



GE VERNOVA

ADMS

16.25.1

Installation Guide

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1. System Overview

This chapter provides a high-level overview of the software and hardware that make up a typical ADMS system. It also introduces key architectural concepts of the ADMS design. People undertaking system integration and maintenance should have a good understanding of these concepts.

This chapter also discusses other related GE products.

1.1. Key Architecture and Design Concepts

ADMS is a suite of applications that has been developed to meet the real-time network management needs of both large and small electric distribution utilities. These applications provide a full range of high-quality network operation functions designed for seamless integration with other systems such as Scada and GIS.

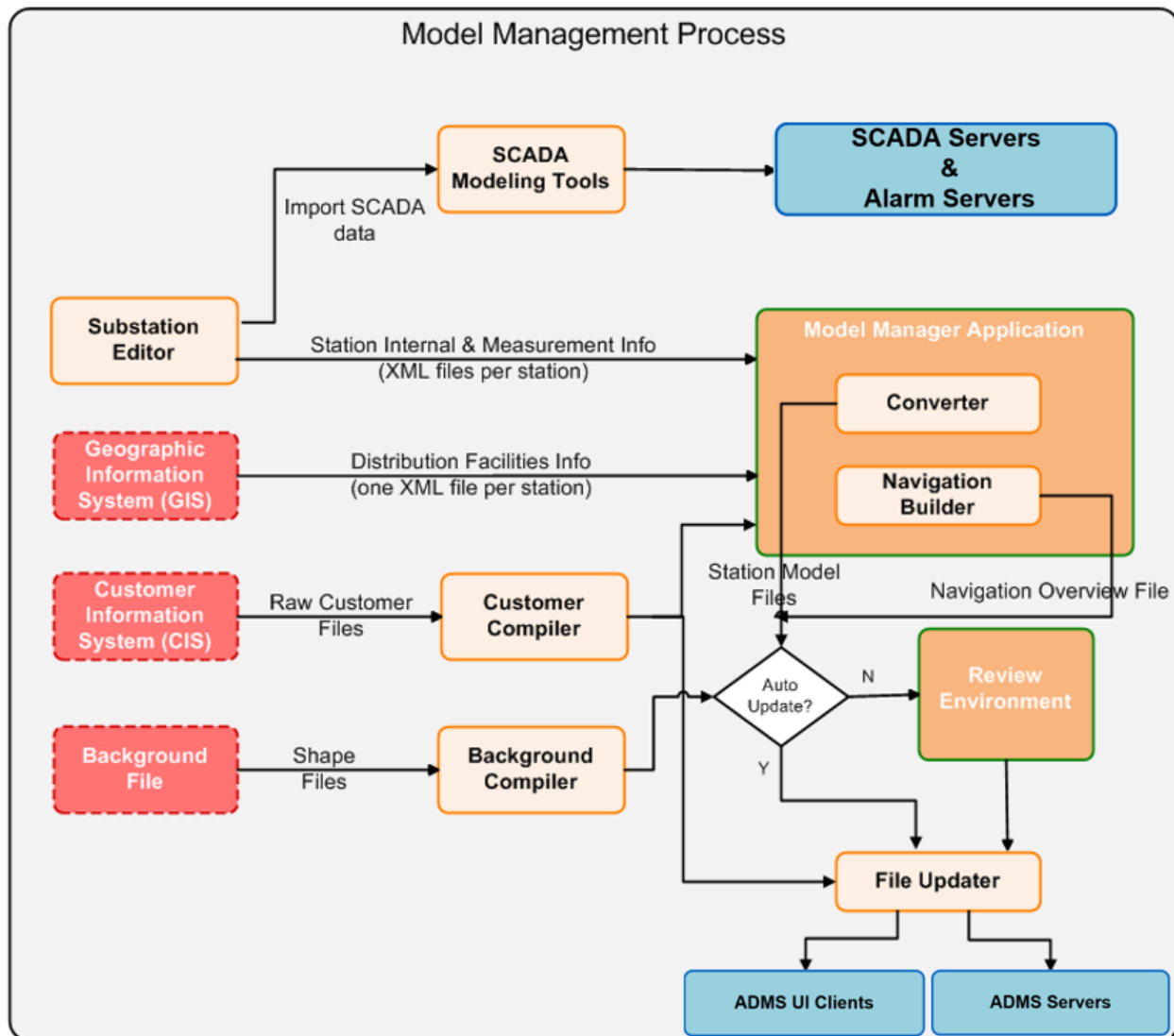


Figure 1. ADMS Modeling Overview

The ADMS software is built around the Distribution Network Object Model (DNOM), which combines the static model data with dynamic data and model updates plus all the other information to support a full suite of advanced network management functions. The model is purposely built to meet the performance and scalability needs of a real-time distribution management system. Its implementation is based on a modern object-oriented technology, which provides special data structures well suited to an integrated, operational system.

The Distribution Network Object Model (DNOM) is at the core of the infrastructure, supporting all real-time distribution operation activities. As such, it includes the description of a comprehensive list of electrical devices and network components, such as switches, lines, cables, transformers, capacitors, regulators, auto-boosters, and more.

The definition of the DNOM includes all necessary structures to support:

- Real-time visualization of network operation events on geographic, schematic, and/or tabular

displays

- High-performance advanced computation performed by applications such as network optimization
- Operation organization by area of responsibility

ADMS populates the DNOM by using data extracted from the utility's asset database, usually a Geographic Information System(GIS). The extracted data includes network device information, connectivity, topology, and nominal status. Non-electrical, "land-based" data such as roads and buildings can also be leveraged from a GIS using the GE background compiler. (Background information is not part of the DNOM.)

This extraction process generates a set of files in the CIM-based format, which are then converted into the GE Distribution Network Object Model format. In this way, the extraction process is kept independent of the real-time network management system. Any GIS model can be extracted to build the DNOM regardless of the supplier or internal schema.

Because changes in the distribution network happen continually, the update process has been designed to allow incremental updates with no need to bring the system down or even fail over to a standby server. The model is built on a per-station basis, and it is fully scalable from a single zone substation to the largest distribution networks. Also, with this approach, all of the data maintenance effort is retained within the GIS, and all parts of the organization are working from the same version of the network information.

The user interface reads this model directly to display geographic and/or schematic views of the network. Special display definitions are not required to show network views. The model supports multiple geographic coordinate sets for each device so that, if available, the network can be displayed in custom geo-schematic formats. The network views can also include various levels of detail depending on the zoom level. Information such as land-based data can also be displayed as required.

The Distribution Network Object Model is a memory-resident, object-oriented model that supports all the data types required for the real-time network management system (see figure below).

- **Static data:** Network equipment electrical parameters, UI display object data
- **Dynamic data:** Temporary modifications, manual entries, abnormal statuses, trouble orders, safety tags, and documents
- **Calculated states:** Power flow results and other output data from network analysis functions

The network operations model is structured on a station basis, that is, the smallest amount of data that is used for a change or update is the portion of the network that is normally supplied from a single substation.

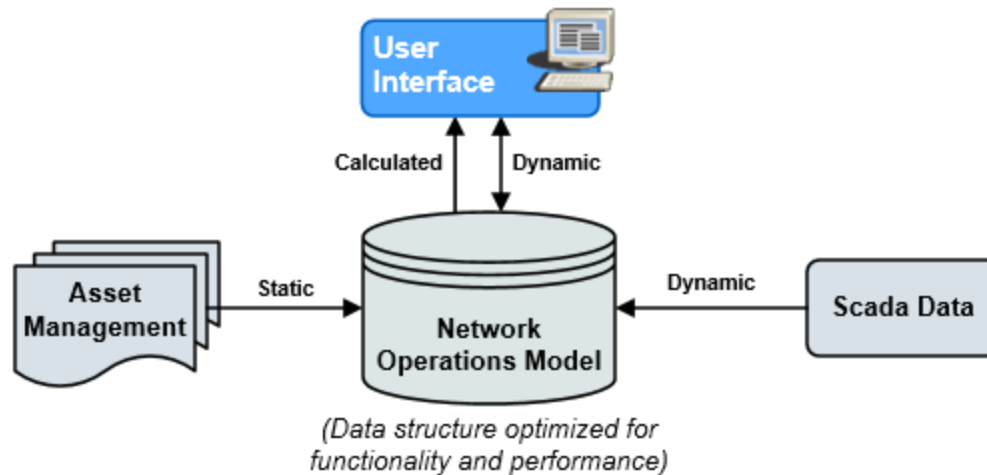


Figure 2. Network Operations Model Data Types

The Distribution Network Object Model includes a comprehensive list of electrical devices and network components as well as some non-electrical objects. Some examples include:

- Substations, which can either be primary stations (HV/MV) or switching stations
- Feeders, which group distribution circuits and components fed from a substation
- Generators, including small customer co-generator plants such as those used as Distributed Energy Resources (DER)
- Line segments, which can be single-phase, two-phase, or three-phase and make up a distribution circuit
- Conductors
- Transformers/regulators, including auto-transformers, two and three winding transformers with constructions, including starstar, deltadelta, stardelta, and several others

The tap changing model includes fixed tap, tap changer, and Load Tap Changer (LTC). It provides the capability to model tap changers on either the HV or LV side of the transformer with the maximum and minimum ratio at either end of the tapping range.

- Capacitors and reactors with various control modes
- Switching devices, including circuit breakers, reclosers, disconnects, sectionalizers, and fuses with a variety of switching schemes, depending on the type of the switch and whether it is remotely controllable
- Grounding devices
- Relays, fault detectors
- Temporary connections to represent jumpers and temporary switches made in the field
- Loads and load categories, which refer to different load shapes and characteristics, including load-to-voltage dependency, power quality requirements, etc.
- Operational limits for components such as lines, transformers, and switching devices

1.1.1. Client-Server Architecture

The design of ADMS is based on a client-server architecture. As in many other similar systems, the client provides the user interface, while the server performs CPU-intensive processing and interfaces with external systems.

Unlike thin clients, the ADMS smart client performs some of the application processing in addition to providing complex and dynamic data presentation. This adds a number of benefits:

- A less-loaded server, which can support a much greater number of clients (i.e., several hundred)
- Light traffic of data between client and server, providing acceptable performance on limited-bandwidth connections
- Some capability for the client to run autonomously and accommodate short disconnections from the server, followed by resynchronization
- Segregated user environments, which allow for stand-alone operation for certain functions such as study mode operations
- Improved performance by leveraging the CPU power from the workstation and performing tasks locally

For example, display call-up is performed by the client and does not impact the server.

The server is responsible for maintaining the current state of the network, based on events received from the users and from Scada. The server is also responsible for running system-wide applications.

There are two types of data that the client and server utilize: static and dynamic.

The term "static data" essentially refers to the description of the distribution network in its normal configuration, including equipment characteristics and connectivity. This data is stored in binary files, also called "station model files" because one file exists per station.

Station model files are maintained on both the ADMS client and ADMS server computers, using the ADMS File Updater, which has client-type components running on both. This utility also has a servertype component running on a computer hosting the current (i.e., "online") versions of the station model files as a reference in a central repository. This repository is typically located on a Web server within the same network as the ADMS system.

The role of the ADMS File Updater utility is to distribute the latest version of the station model files stored in the central location to every computer that needs the station model files. This includes ADMS server and client computers.

Whenever new or modified station model files are detected in the station file central repository, the ADMS File Updater utility automatically updates the files on both servers and clients. The ADMS application code running on both client and server has been designed to accommodate receiving new or modified station model files "on the fly" without interruption to the user.

The term "dynamic data" refers to all actions and events that are applied on top of the static description of the network to reflect the current state of the distribution grid. This includes:

- Device status changes
- SCADA measurements
- Operator actions on non-telemetered devices
- Operator actions related to tags and annotations
- Operator actions related to cuts, jumpers, and grounds

Dynamic data is sent back and forth between the ADMS clients and the ADMS servers using a live connection based on TCP/IP. Dynamic data is also recorded in a journal on both sides for the purpose of synchronization and activity tracking.

Figure below shows the data transfer between client and server. All station model files in the system are present on the server and client computers. The ADMS File Updater utility keeps them in sync.

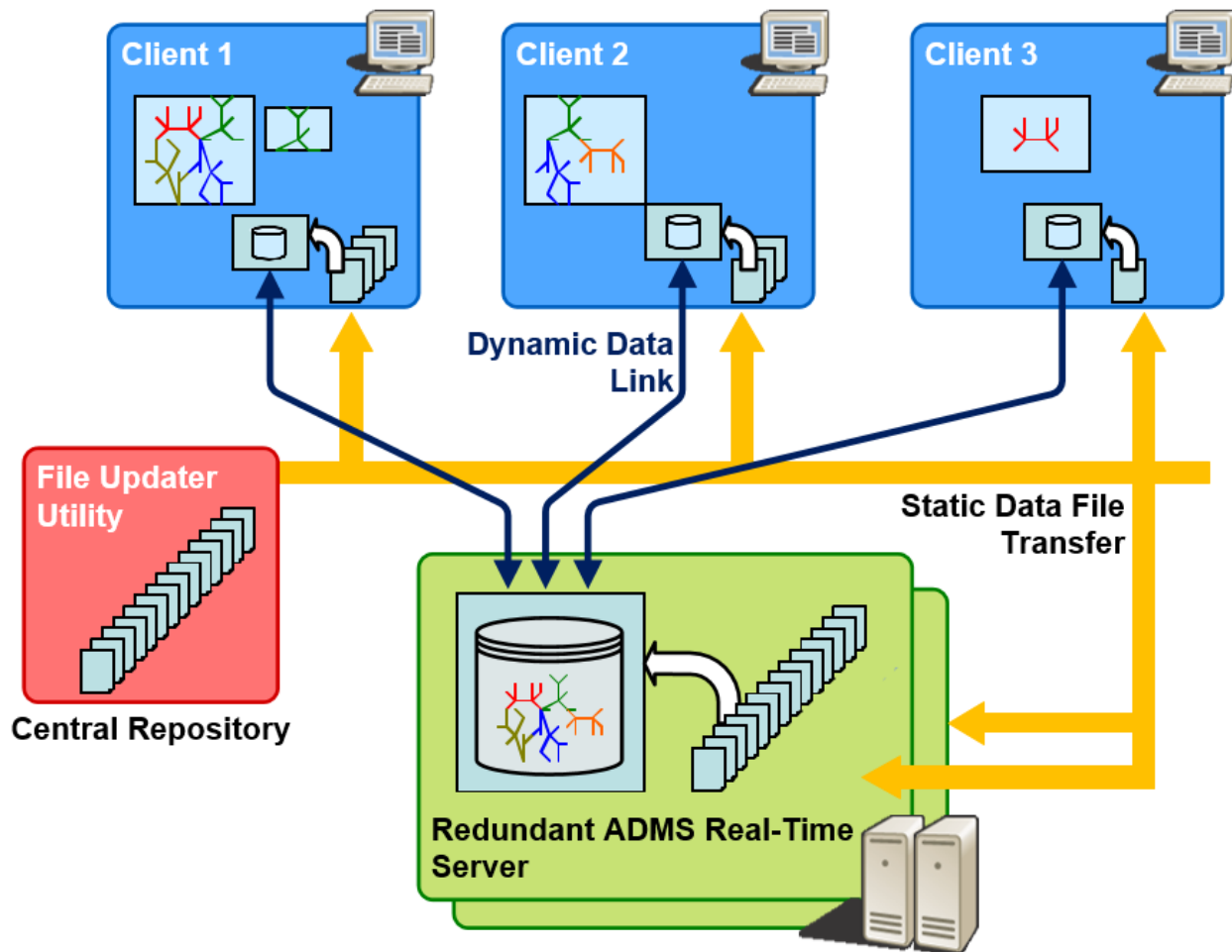


Figure 3. ADMS Client-Server Architecture

The ADMS File Updater components are delivered as part of the suite. The ADMS File Updater transfers model, tabular, picture, configuration, and bitmap files to the ADMS server and client computers. File transfer is based on FTP or SFTP, uses data compression to minimize impact on network bandwidth and uses digital certificates to digitally sign files to ensure files are transferred

securely.

1.1.2. Browser UI Client

The ability to auto-generate distribution network displays from the station model files requires a technology for presenting distribution views that are different from the classic Browser displays created with DisplayBuilder.

However, this ADMS user interface is fully integrated into the Browser client. The code that draws the distribution network and handles user interactions is a .NET component running in the framework of the Browser client. The seamless integration between the two allows the ADMS user interface to leverage the existing feature-rich interface environment of the Browser client as well as to present a common interface for the users.

The following are key features leveraged in this union:

- **Navigation:** An ADMS display can be called up by specifying a URL string. Just as with any Browser display callup, this URL string can be associated with toolbar buttons, pull-down menu entries, navigation points on displays, etc. For example, by specifying the station name within a navigational URL callup string, the user can navigate to a view centered on that station on the geographic display. Additional parameters can be specified in the URL string to modify the resulting display presentation (for example, zoom level and visible layers).
- **Toolbar buttons:** Configurable permission and context-sensitive toolbar buttons.
- **Menus:** Menu-based navigation.
- **Authentication:** Login is handled by the Browser authentication function.
- **Keyboard shortcuts:** Configurable keyboard shortcuts to ease use of highly repeated actions.
- **Rooms:** ADMS displays, specified as URL strings, can be incorporated into Browser rooms in the same fashion as Browser displays. This is done using the traditional Browser room definition functionality. The URLs are produced by the Browser client when the room definition is saved.
- **Printing:** Display hardcopy features for ADMS displays.
- **Operator notifications via popups:** Notifications to the operator via popups are available in the same way as in Browser. For example, when a user action fails, a popup alerts the user.

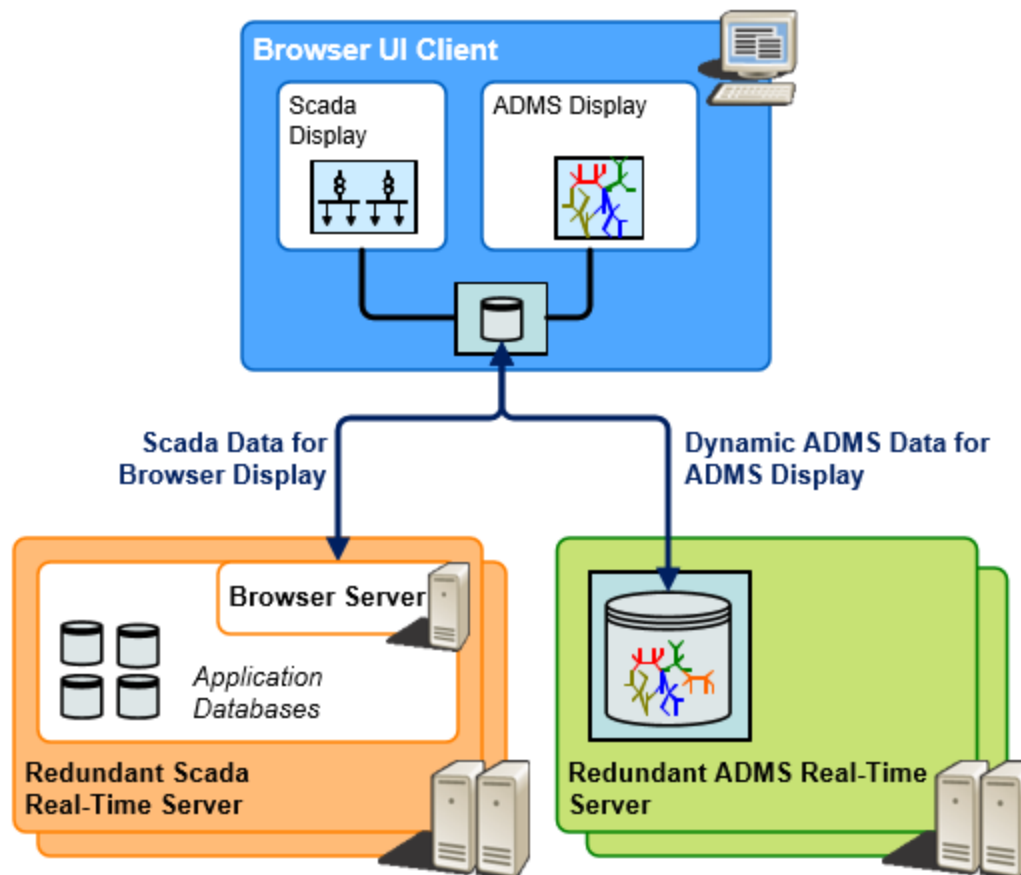


Figure 4. User Interface Client Integration

Note

Browser works as a plug-in for Advanced Presentation Framework. For supported Browser and APF versions, refer to the *ADMS Release Notes*.

1.1.3. Integration with Scada

ADMS provides real-time distribution applications designed to support integration with Scada (SCADA system).

In this integration, the SCADA system brings the data acquisition from the field, including status of telemetered switches, and remote control for devices such as circuit breakers. The SCADA system also provides the alarming and event logging functionality that ADMS applications can use to alert operators about events.

Permission checking and device tagging are two functions shared by both the SCADA and ADMS systems. They are supported by most SCADA systems as part of the core functionality; therefore, ADMS relies on the SCADA implementation and data repository for these two functions.

In a SCADA/ADMS integrated system, both parties need to consider and honor permissions and

tags as they are defined and modified (respectively placed and removed).

To exchange real-time data with the Scada database, ADMS supports the GE InterSite Data (ISD) protocol used with Scada. The main data flow is directed from SCADA to ADMS, as values and quality flags for analog points change.

Controls initiated by the operator from the ADMS user interface are handled as follows:

1. Upon selection of a remote-controlled device (i.e., modeled in SCADA), the ADMS UI invokes the SCADA control popup display by its URL command with the appropriate device context information.
2. The actual control sequence is then handled by SCADA, just like controls issued from SCADA displays.
3. Once the control is executed in the field, the new status is obtained by SCADA and transferred to ADMS via the ISD interface.
4. The DMS server (NetSrv) updates the DNOM model and sends notification of the changes to connected clients (dynamics transfer).

Figure below shows the different types of interactions with SCADA and associated data flows.

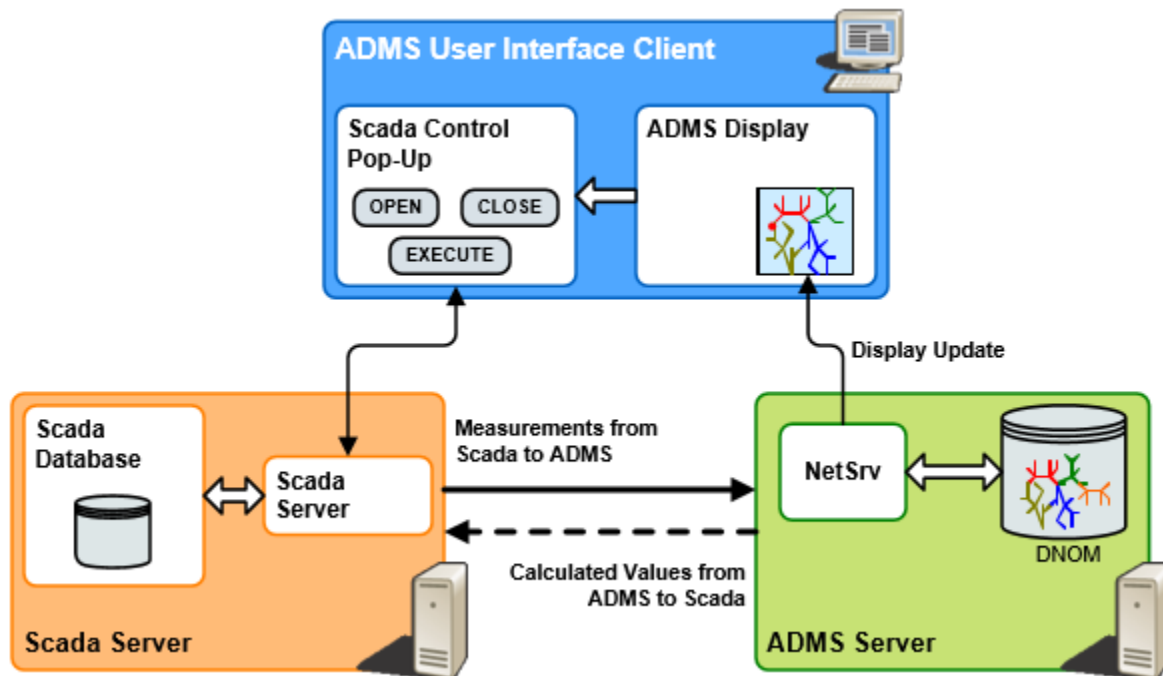


Figure 5. Scada Interaction - Acquisition and Control

Tags are managed by the SCADA server and stored in the Scada database. However, the ADMS user interface needs to show tags placed on distribution devices and to allow distribution operators to place new tags from there.

Existing tags, placed on distribution devices, are transferred from SCADA to ADMS using the InterSite Tagging protocol. An initial download is performed at connection; from there, any

creation, modification, or removal of tags is transferred incrementally. Tag information transfer is always from SCADA to ADMS.

To allow users to place tags from the ADMS UI, an RUSER command is sent by the software to the TAGGING application, which in response opens a tagging popup display. From there, the tag placement sequence is handled the same way as from SCADA displays.

Once the tag is entered, the DMS server receives an update via the InterSite Tagging link. In turn, it updates the DNOM model and refreshes the ADMS view on the client.

For devices that are modeled in SCADA, the tag is placed on the device exactly as it is in the Energy Management System (EMS). For devices that are not modeled in SCADA (most of the distribution devices), tags are placed next to the devices and flagged as DMSONLY tags.

Figure below shows the tag data flow and interactions among the SCADA server, the ADMS server, and the UI client.

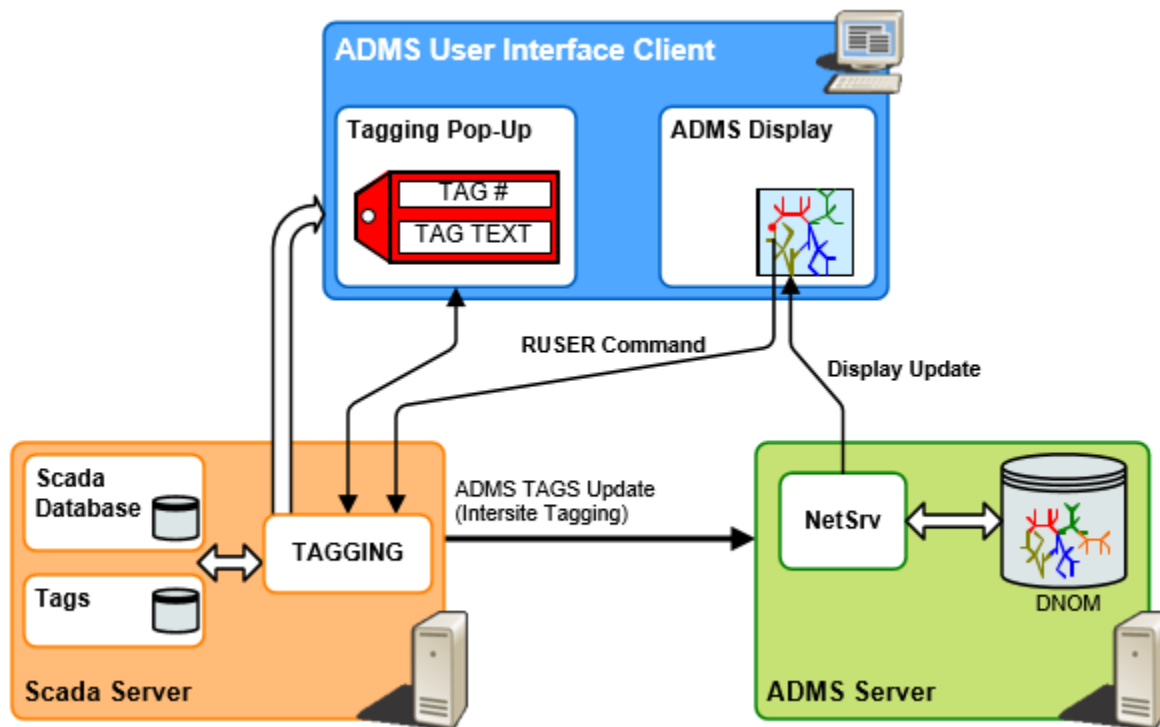


Figure 6. Tagging Interaction

Defining permissions for ADMS is performed in the SCADA environment using the traditional PERMIT application. All permission definitions are transferred to ADMS via a TCP/IP link by the Permission Manager Proxy server (netpermsrv.exe), a Habitatbased application delivered with ADMS that runs on the SCADA/EMS host where PERMIT is running. Upon connection, the content of the PERMIT database is downloaded to ADMS. Periodic updates are sent to make sure that permission modifications are reflected on the ADMS system in real-time fashion.

The DMS server is responsible for enforcing permissions as they are defined in PERMIT. Along with actions performed by users, the DMS server receives user identification from the clients, so that

permissions can be checked and the action request denied if the user is not authorized.

Figure below illustrates permission checking principles. More information about how to install the required Habitatbased NETPERMSRV application is also provided in [Key Architecture and Design Concepts](#).

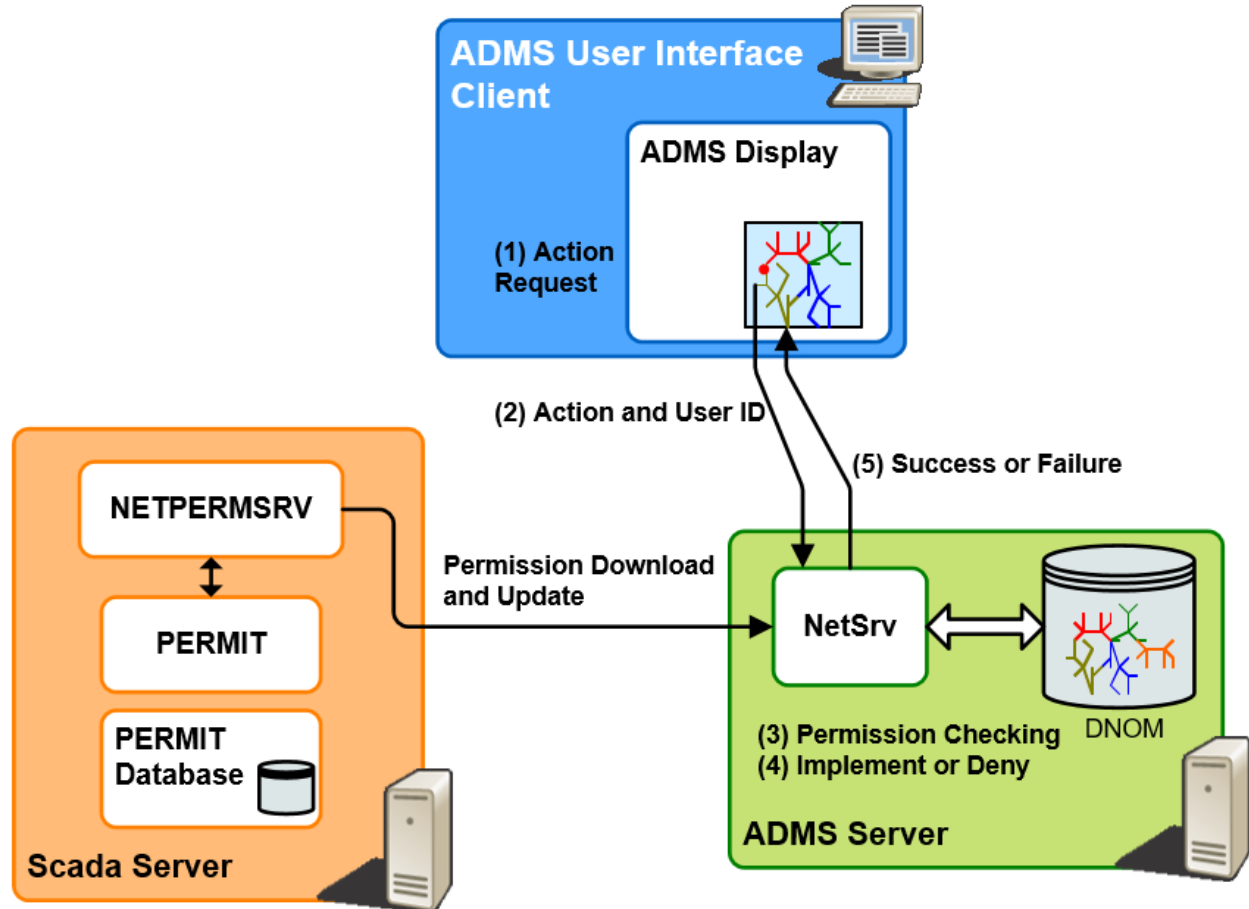


Figure 7. Permission Checking

1.1.4. ADMS Server Redundancy Management/Multi-Host

To ensure the required availability of mission-critical applications, ADMS servers are designed to run in a redundant configuration. The server applications have primary/standby role assignments and automatic failover capabilities. Manually initiated failovers are also useful for orchestrating maintenance activities.

The system can also be set up with a multi-host/multi-site CFGCTRL/CFGMONI/NETIO configuration. Failovers can be initiated to the remote site and within the remote site. The ADMS server applications make no distinction between sites. The ADMS server applications also make no distinction between standby, remote-preferred-standby, and remote-standby. All connections are only open on the primary server. All standby "main" server apps are clients to the primary. For further information about multi-host/multi-site configuration, refer to the *Multiple Standby Host User*

Note

ADMS does not support Scada with the Parallel Site redundancy configuration.

Regardless of whether the ADMS servers run on the same computer as SCADA or on a different computer, they receive their role assignments from the Configuration Manager (CFGMAN) application and the Configuration Manager Proxy server (CfgProxy) application running on the ADMS servers. In cases where ADMS and SCADA servers are separate, the Habitat environment on the ADMS machine runs a minimal set of tasks, including CFGMAN and CfgProxy.

CfgProxy is a Habitatbased application. The purpose of CfgProxy is to communicate the role assigned by CFGMAN to the ADMS server applications.

Details on how to configure CfgProxy can be found in [Integrating with the SCADA System](#).

When assigned the primary role, an ADMS server application listens to and accepts connection requests from clients and attempts to connect to ProxySrv and NETPERMSRV on the primary SCADA server. It also receives a connection request from the standby ADMS server application, which acts like a client to subscribe to dynamic updates. The primary ADMS server application interprets this request as a regular client connection request.

In the standby roles (standby, remote preferred standby, and remote standby), an ADMS server application process is started and loads the station model files into memory, but it does not listen to client connection requests, nor does it attempt to connect to ProxySrv and NETPERMSRV on the SCADA computer. Acting as a client, it receives dynamic updates from the primary server and waits for the primary assignment to take mastership.

In the standby roles, an ADMS server application confirms the role to CFGCTRL as soon as it connects to the primary/enabled server. It then starts copying the dynamic data from the main server. Habitat's Memory Replication Service (MRS) application is not used to copy the dynamic data from the primary/enabled server. The EMS Site Overview display might show that everything is synchronized and it might appear that the system is ready to failover, but this is not correct. This EMS display does not represent the actual synchronization state of ADMS and it must not be used as an indication to decide if the ADMS standby server is ready for failover.

To accurately determine if the servers are ready to be failed over from a client, use the ADMS Site Overview display. This display shows if the enabled and standby server dynamics are in synchronization. For the secondary servers, the display also shows the role and confirmed state and other important information.

In addition, you can open the standby server's log file and look for the following two lines:

```
"StandBy role is announced to main server".  
"dynMsgHelloAck: m_srvSeq = 5000, jst = 6000, main servers srvSeq = 6000, backlog = 0"
```

The value of m_srvSeq represents the dynamic sequence of the standby server. The value of srvSeq is the dynamic sequence of the primary/enabled server. These two values must be equal or nearly

equal.

If the two values are nearly equal, it is possible that a great deal of changes are coming from SCADA because of Fault Indicator chatter and because the standby server is always in a constant trail of synchronizing with the main server. In this case, it is recommended to copy the dynamics file (dynamics.sav) from the enabled machine to a separate location and look at the last few entries using Snapshot Viewer. If the entries on the Sequential View tab are all SCADA related it is okay to failover since these values will again be sent by SCADA upon a failover.

If the OMS is deployed in the customer location, then ADMS Site Overview also shows JournalTime of NetDataReplicator server in the Standby and Enabled System. Before initiating a planned failover to the Standby, the analyst needs to make sure that the JournalTime is equal or nearly equal between Enabled and Standby NetDataReplicators. It is recommended that system stays on the newly Enabled system for at least a day before initiating another failover back to the originally Enabled System.

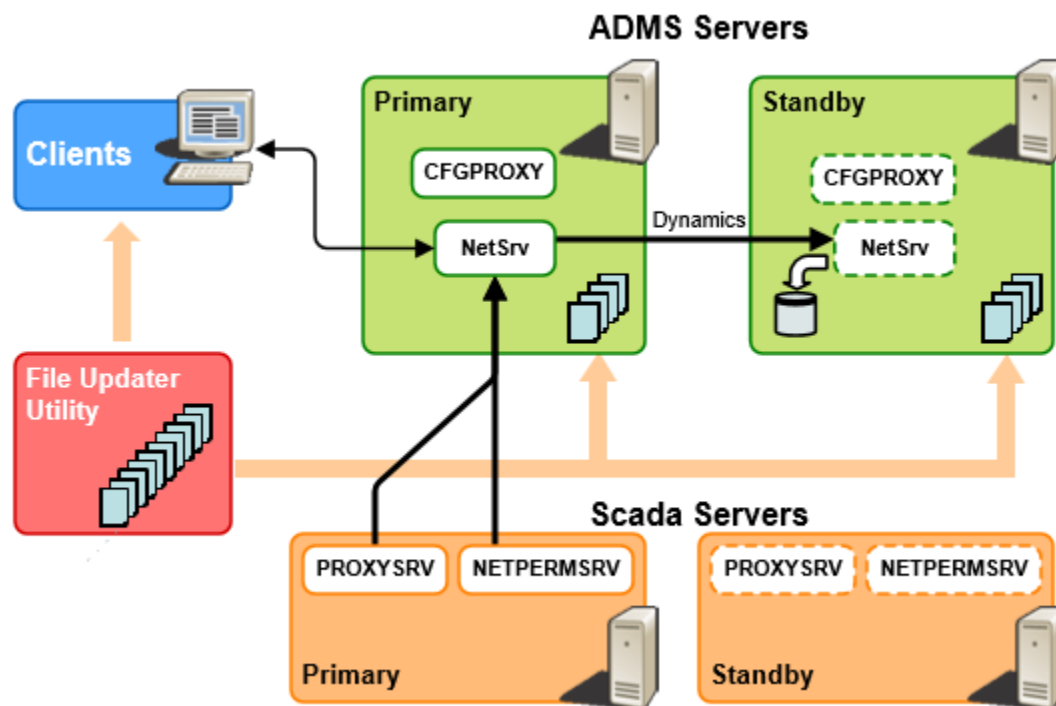


Figure 8. Redundancy Management

On a primary-to-standby transition, an ADMS server application process terminates and restarts in the standby mode.

On a standby-to-primary transition, an ADMS server application process applies dynamics to the station model files, runs the topology processor, opens connection ports for clients, and connects to ProxySrv and NETPERMSRV on the primary SCADA server.

Station model files on both primary and standby ADMS servers are kept up to date by the ADMS File Updater utility.

Note

It is worth mentioning here that, on four-computer configurations (ADMS separate from SCADA), as depicted in figure above, failover can occur on SCADA and ADMS servers independently. A SCADA failover does not cause an ADMS failover (and vice versa).

With combined configurations where ADMS and Scada servers run with the same Habitat on a single computer, failover, by default, causes ADMS and Scada to transition to a new role simultaneously.

To be clear, the NETPERMSRV application must run on the same machine as the SCADA PERMIT server, and the CfgProxy application must run on the same machine as the ADMS applications.

1.1.5. Onlining Distribution Network Model Changes

A station model file contains the static description of all equipment fed by the HV/MV substation in the so-called "normal" configuration of the network. In other words, this includes all distribution feeders of the substation with backbones and laterals from the feeder head down to normally open tie switches or feeder ends. The station model file may also include some of the devices within the substation. This consistent set of data represented by a station file constitutes the minimum increment in terms of a network model change. Models of large distribution systems are potentially made up of hundreds of station model files.

Station model files with a .mod extension are binary files created by the ADMS Converter. They are distributed to ADMS servers and clients by the ADMS File Updater utility.

ADMS servers and clients can accept new or modified station model files while running. When a new or modified station file comes in, the application automatically loads it into memory and applies dynamics onto the static configuration. This process sometimes requires a few seconds to complete before changes are available to the user, as the application ensures that functions using the old version of the station terminate before the switch is performed.

Dynamics contain all changes that need to be applied to the static normal configuration to reflect the current operational state of the network. This includes the switch positions, if different from normal, along with temporary modifications such as cuts, jumpers, grounds, tags, and more. The manner in which dynamics are applied to the station models allows the server and clients to be slightly out of sync and to run on different versions of the same station file. This is to accommodate the time necessary to update all station model files in the system.

Dynamic events are date- and time-stamped. The server maintains a dynamics journal in files that can be archived and used to initialize another environment to the same state. Periodically (typically once a day), the server captures a full picture of dynamics and puts a stamp on it. From there, dynamic changes are computed incrementally based on this snapshot. This is used to re-synchronize clients when they reconnect after being disconnected. Upon connection, the clients send the last timestamp they had before the link went down, which allows the server to update them from the last point.

Multiple benefits are leveraged from this design:

-

The ability to implement model changes, which does not require an administrator to stop/restart the server or even failover

- The ability to support a large number of clients or use low bandwidth communication, by allowing a reasonable time to update the clients and managing the situation where some clients can be "outofsync"
- The ability to accommodate short communication disconnections while display call-up and functions local to the client remain available to the user (for example, tracing)
- Communication between client and server is reduced to a minimum by distinguishing static information from dynamic changes

1.1.6. Outage Management System (Optional Add-On)

The Outage Management System (OMS) is a trouble call and network outage management system that forms an integral part of the ADMS suite. It allows operators to manage unscheduled and scheduled outages from within a unified operating environment that integrates real-time Distribution Management System (DMS), SCADA, crew monitoring, and switching orders.

The Outage Management System provides dispatchers with a means of centrally collecting and coordinating all possible information about network outages. In case of unplanned outages, customer calls, AMI observations, SCADA data, and emergency services calls are all analyzed to assist in determining the most likely point at which a customer supply has been interrupted.

From initial notification of a fault, through prediction, crew assignment, and fault isolation, to return-to-normal switching, the dispatcher is able to work from a single set of network views. All the necessary information for each phase of the job is clearly presented in a way that allows the dispatcher to manage each outage efficiently while also staying aware of other network activity.

The Outage Management System provides the ability to handle major storm situations by providing effective, high-performance tools combined with the ability to easily divide or combine areas of responsibility and therefore balance the dispatcher's workload. A complete record of all actions taken in the restoration process is recorded for each incident. Network reliability and performance indices are automatically calculated.

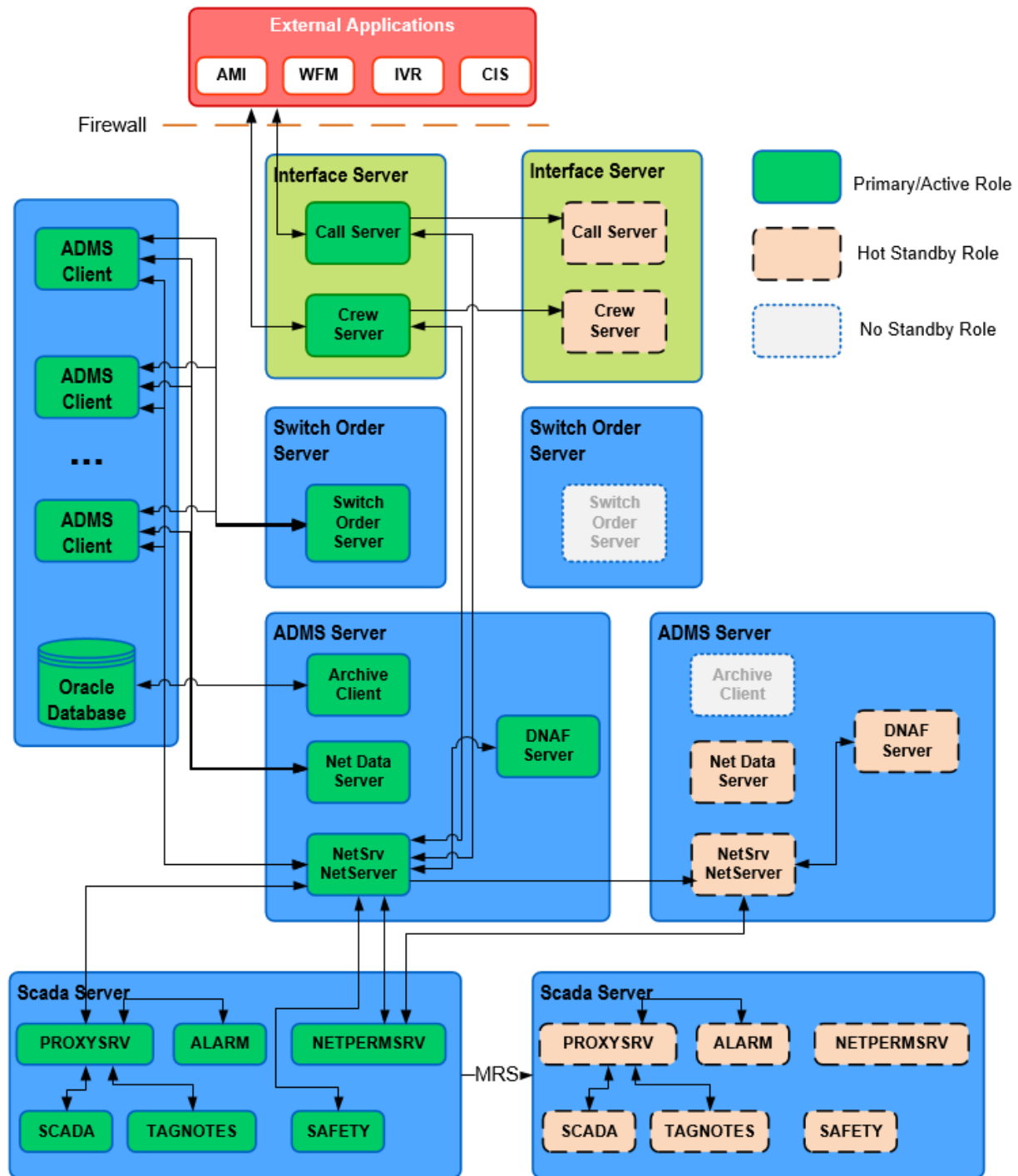


Figure 9. Integrated Distribution Management System

1.2. Supported Hardware Configurations

Two hardware configurations are supported by ADMS for the servers of the main control center:

- The simplified configuration where both Scada and ADMS server applications run on the same pair of computers
- The distributed configuration where Scada and ADMS server applications run on two different pairs of computers

Choosing one or the other is a decision based on the size of the distribution model, the size of the Scada system, the configuration management offered by the Scada system, and the number of clients to support simultaneously. Consideration should also be given to the number and complexity of interfaces to external systems. Also, if an existing Scada system is in operation and ADMS functionality is being added, additional servers are the easiest and recommended approach.

1.2.1. Simplified Hardware Configuration (One Pair of Servers)

In this configuration, the SCADA server software and the ADMS server software run on the same computer.

When integrated with Scada, only one Habitat environment is set up to host all tasks required by SCADA and ADMS.

In this configuration, SCADA and ADMS follow the same failover rules simultaneously.

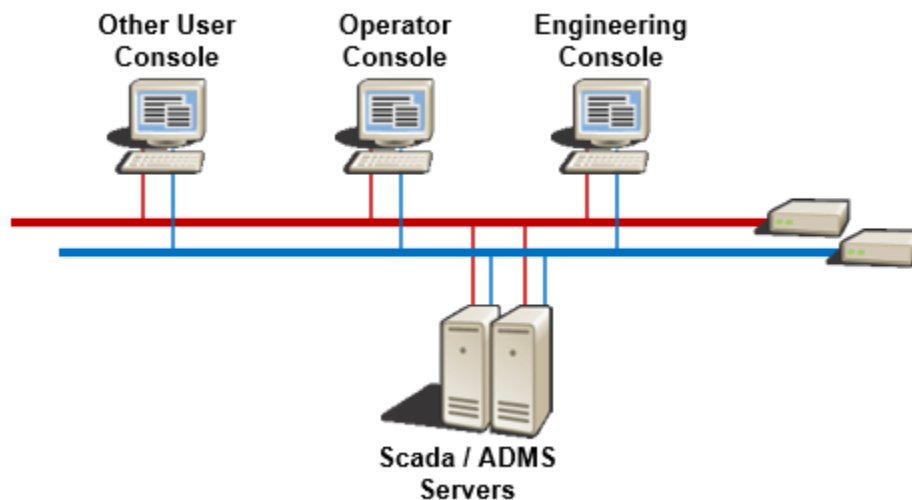


Figure 10. Simplified Hardware Configuration

Figure above depicts the hardware configuration, whereas figure below shows a conceptual view of the software environment. For clarity, both figures only show a limited number of components relevant to ADMS.

i Note

The blue background in figure below denotes the Habitat framework. The ADMS server applications are not native Habitat applications. Also, the TAGGING application is colored differently to highlight the fact that the InterSite Tagging (IST) protocol is used instead of

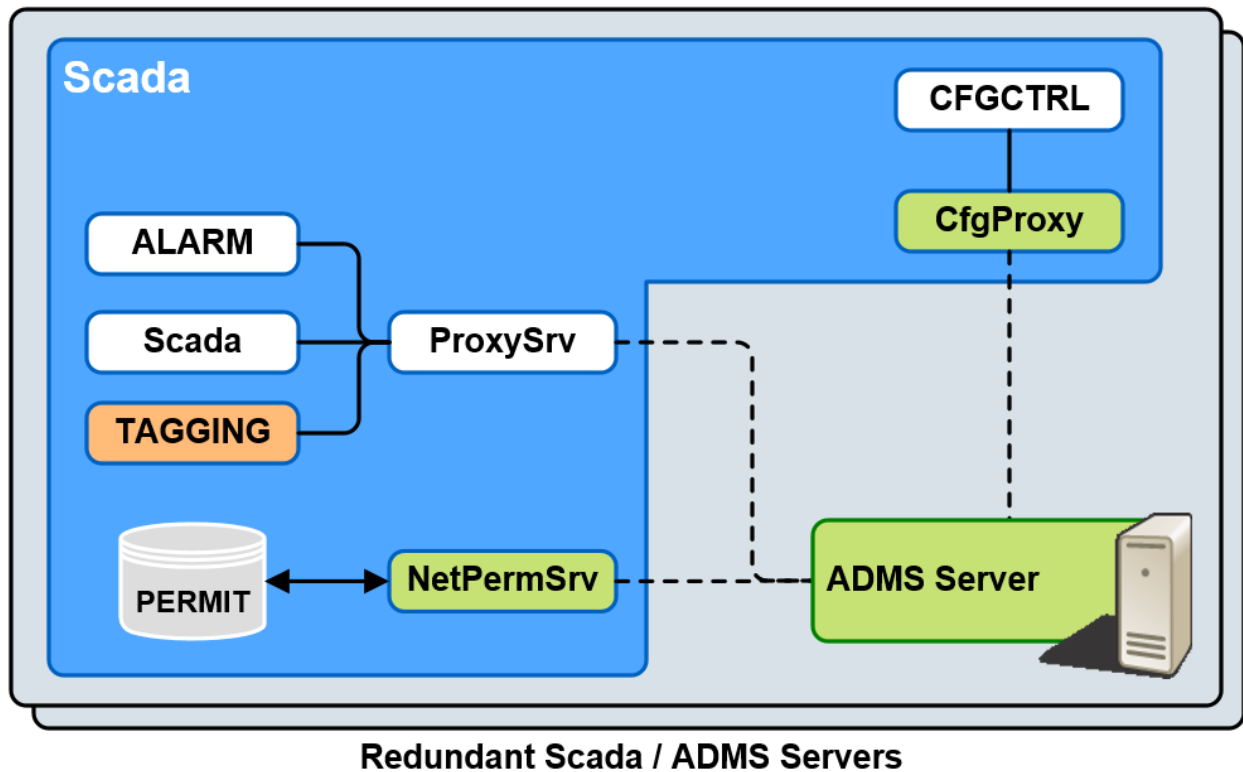


Figure 11. Simplified Software Configuration - Hosting Environment

1.2.2. Distributed Hardware Configuration

In this configuration, SCADA and ADMS run on two separate servers. There are two different Habitat environments: one typical environment on the SCADA server, and a minimal one on the ADMS server.

In this configuration, SCADA and ADMS follow independent failover processes.

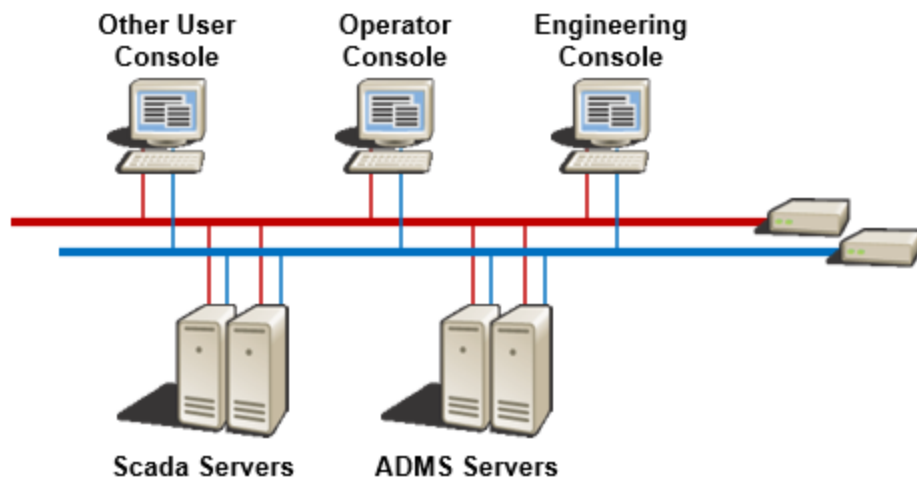


Figure 12. Distributed Hardware Configuration

Figure above depicts the hardware configuration, while figure below shows the conceptual view of the software environment. Again, these are simplified representations of the actual configuration.

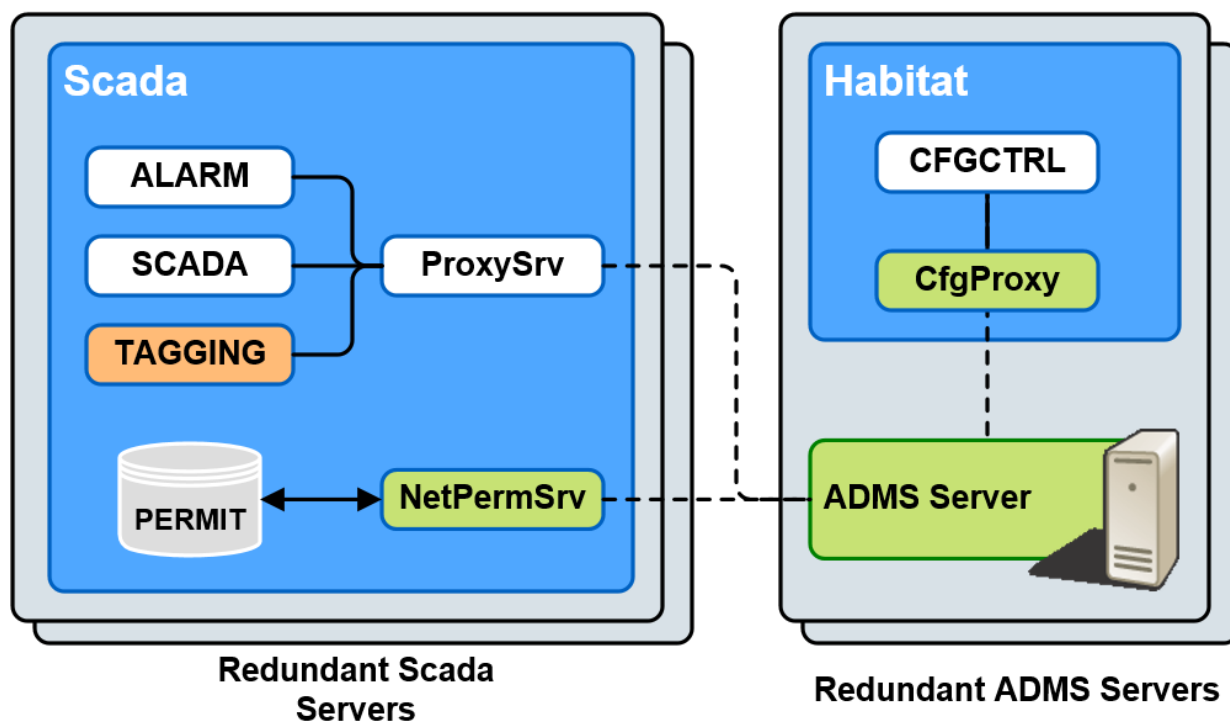


Figure 13. Distributed Software Configuration - Hosting Environment

In this configuration, Habitat is used on the ADMS machines to provide the redundancy management (hence the CfgProxy interface to CFGCTRL).

1.2.3. All-In-One Hardware Configuration (PC for Testing)

For testing, training, configuration work, or demonstrations, it can be useful to have an entire system running on a single computer. The installation kits are designed to work "out of the box" in this monolith-style configuration.

In this configuration, the Browser client, Habitat, Scada, CfgProxy, NetPermSrv, ADMS servers, ADMS clients, and ADMS tools are all installed on the same Windows machine. Again, default configuration files are included to work with this "local host" setup.

Some additional steps and configuration scripts are available to aid in this style of setup, which are explained further in [Monolithic Sample Installation](#).

1.3. Software Overview

This section describes the deliverables and the installed layout of the software. In light of the possible configurations described in previous sections, there are separate individual kits for NetPermSrv and CfgProxy. This takes into consideration the installation needs of either the shared setup (where ADMS and Scada share hardware) or the separated setup.

1.3.1. Software Deliverables

The ADMS software deliverables are listed below. The portable applications are available in Microsoft Windows MSI kits.

Client installation kit (used in conjunction with the Browser client)

- Call Entry (tool for entering outage customer call data)
- DMS Interface server (interface for external applications)
- ADMS client (executable, data, and configuration files)
- Local study server (for offline analysis)
- DOTS: Event Scripter application (operator training tool)
- Localization files
- Additional Browser configuration files
- Sample model files

Server installation kit (portable)

- DMS, OMS, Read-Only, DNAF, Simulator, and Customer-Only server executables and configuration files
- ADMS Switching Order server executable and configuration files
- ADMS supporting tools (distribution operator training simulator, and archive) executables and configuration files

- Net Data server (OMS read-only interface server) executable and configuration files
- Localization files
- Additional Browser configuration files
- Sample model files

Tools installation kit

- ADMS Configuration Tools
 - Coded Translation Editor
 - DNAF Configuration Editor
 - Encryption Tool
 - Grid Tabular Builder
 - Net Dynamics Configuration Editor
 - OMS Client Configuration Editor
 - OMS Server Configuration Editor
 - Resource Editor
 - Snapshot Viewer Configuration Editor
- ADMS Modeling Tools
 - Annotation Importer-Exporter
 - Converter
 - Crew Modeling Tool
 - Customer Compiler
 - DNOM Database Loader
 - DNOM Viewer
 - Geographic Background Compiler Model Manager
 - Model Export
 - Model Quality Stat Logger
 - Navigation Overview Builder
 - Permit Builder
 - Switching Order Server Configuration Editor
 - Switching Order Client Configuration Editor
 - Substation Builder
 - Substation File Builder Lite
 - Substation Editor

File Updater server installation kit (portable)

- File Updater server executables and configuration files to be installed on the file updater server machine (the SFTP host)

File Updater client installation kit (portable)

- File Updater client executables and configuration files to be installed on the ADMS client machines and the ADMS server machines (both need up-to-date model files)

Documentation suite installation kit

- The complete set of documentation supplied with the current release of ADMS

1.3.2. Network Server Configuration Files

The following set of main server configuration files is installed with the ADMS server kit. Their functionalities are explained below.

- **NetServerMainConfig_localhost.txt:** Sets parameters for the ADMS main (DMS) server application that is run by `netsrv.exe`. For more information, see [NetServerMainConfig.txt](#).
- **NetServerOmsConfig_localhost.txt:** Sets parameters for the OMS server application that is run by `NetServer.exe`. For more information, see [NetServerOmsConfig.txt](#).
- **NetServerDnafConfig_localhost.txt:** Sets parameters for the DNAF server application that is run by `DnafSrv.exe`. For more information, see [NetServerDnafConfig.txt](#).
- **NetServerReadOnlyConfig_localhost.txt:** Sets parameters for the Read-Only server application that is run by `RoSrv.exe` (DMS-only) or `NetServerR0.exe` (DMS and OMS). For more information, see [NetServerReadOnlyConfig.txt](#).
- **NetServerSimConfig_localhost.txt:** Sets parameters for the DMS Simulator server application that is run by `SimSrv.exe`. For more information, see [NetServerSimConfig.txt](#).
- **NetServerMainCustomerModeConfig_localhost.txt:** Sets parameters for the Customer Mode server application that is run by `NetServerCM.exe`. For more information, see [NetServerMainCustomerMode.txt](#).

The core distribution management functionality and the advanced applications (DNAF) are written in portable C++. This code can be compiled into an executable named `netsrv.exe` and `DnafSrv.exe`.

The outage management code is Windows-only. It is built on top of the core portable code to create `NetServer.exe`.

1.3.3. Client Configuration Files

The following set of main client configuration files is installed with the ADMS client kit. Their functionalities are explained below.

- **netconf_localhost.xml:** Sets parameters for the ADMS client application that is run by `Apf.exe`. For more information about configuring the client, see [ADMS Client Configuration](#). For more

information about `netconf_localhost.xml` parameters, refer to the *Net Dynamics Configuration Editor User Guide*.

- **NetServerStudyConfig.txt:** Sets parameters for the ADMS main (DMS) server application running in study mode (run by `studysrv.exe`). For more information, see [NetServerMainConfig.txt](#).
- **viewerConfig.xml:** Configures the appearance of the ADMS client displays and network view. For more information, refer to the *Viewer Configuration Editor User Guide*.

2. Software Installation Overview

Regarding operating systems, the ADMS client release media (and the associated tools) are only available as Windows installation kits. The server code is portable and available for Windows. The File Updater software is also portable since it must run on the ADMS client and server machines.

2.1. Supported Operating Systems

For a list of certified and supported operating system configurations, refer to the chapter "Operating System" in the *ADMS Release Notes*.

2.2. Installation Process Overview

Before giving specific instructions for installing and configuring an ADMSbased system, it is useful to discuss the process at an overview level.

The main steps of the ADMS installation and configuration are as follows:

1. Install the server software and start it in a stand-alone fashion.
2. Install the client software and connect it to the server.
3. Configure the DNAF server and start it.
4. Configure the server for outage management. (Optional)
5. Configure the server for high availability in a redundant setup.
6. Configure the server to connect to the SCADA system (configure the ISD and IST links to the SCADA system).
7. Configure the File Updater components (refer to the *File Updater Software Installation and Maintenance Guide*).

The order of steps 3, 4, 5, and 6 is largely up to the integrator.

Note

The ADMS client and server must be installed in the same time zone.

This approach allows for other engineers to start using the basics of the system as integration continues with the subsequent parts. The availability of an electrically meaningful model, the availability of customer data, the availability of workforce data, the availability of redundant hardware, and the availability of a working SCADA system should all factor into the order of integration.

To simplify the integration of product and path updates, the product installation path is defined in the IDMS_ROOT environment variable. The variable is created and defined automatically during system installation and setup.

In Windows, the engineers can set and define the environment variables in the administrative command prompt by using the SETX command or through the environment variable configuration user interface.

To view the list of the existing environment variables, in the administrative command prompt, enter:

```
> SET
```

To verify that the IDMS_ROOT environment is defined and view the build version, enter:

```
> SET IDMS
```

i Note

- The %IDMS_ROOT% environment variable is case sensitive for some ADMS applications and must be in capital letters.
- Localized installation paths are not supported. Only the US-ASCII character encoding can be used; therefore, installation paths and configuration paths must be in English.
- The silent installation does not set the IDMS_ROOT value. The user script must explicitly set the value of IDMS_ROOT.

2.3. Integrating with Scada

Review the *ADMS Release Notes* for more details about the supported and certified configurations, and for potential updates.

For Habitat and Scada installation steps, refer to the appropriate product installation guides.

2.4. WelcomeSetupUI Application

The WelcomeSetupUI application is a setup bootstrapper that helps service the ADMS product installation. It is located in the root of the setup media.

After executing the application, the Quick Setup screen appears, as shown below.

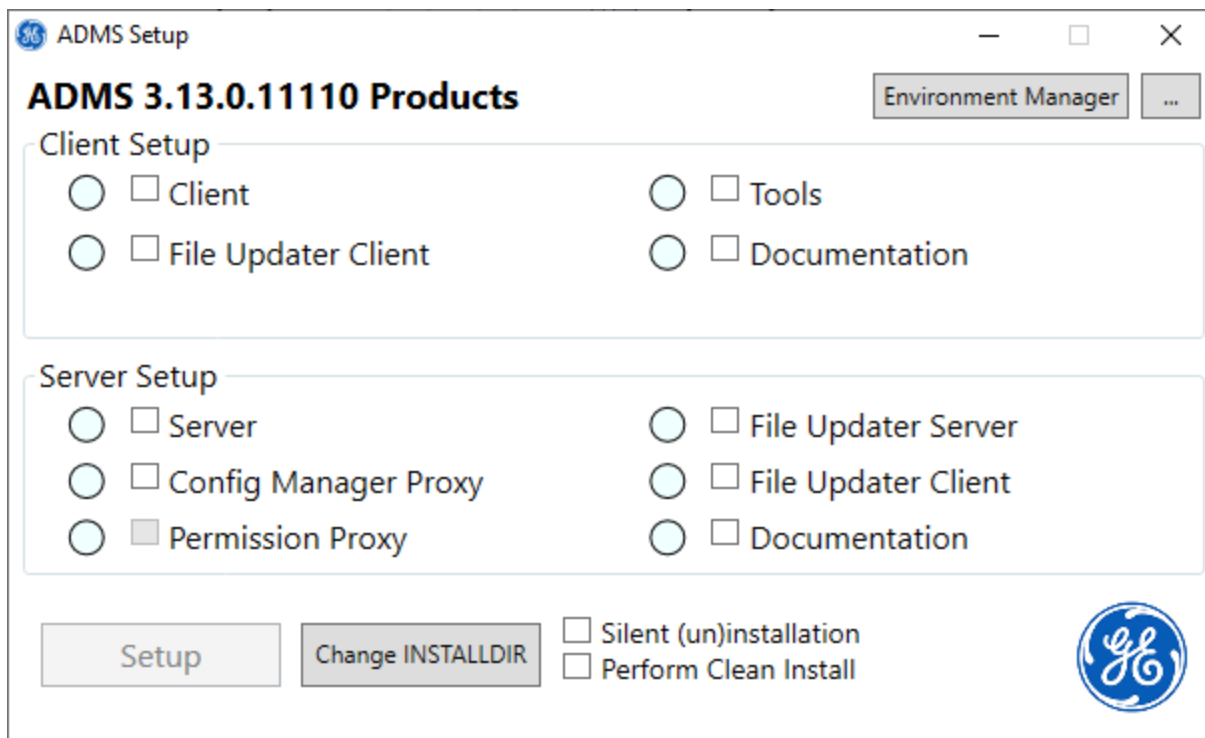


Figure 14. The WelcomeSetupUI Application Quick Setup Screen

ADMS products are grouped by two setup suites: client and server. By selecting the appropriate check box, you can initiate the appropriate servicing procedure for the selected product (Install/Uninstall). Clicking the Setup button initiates the batched maintenance operation for the selected items.

Other options on the Quick Setup screen include:

- **Silent (Un)Installation:** Executes the installation process with default settings and without user interaction.
- **Environment Manager:** Opens a dialog box for configuring the path to the %IDMS_ROOT% environment variable.

IDMS_ROOT Manager [X]

Current IDMS_ROOT value

Per-User:

Per-Machine:

IDMS_ROOT modification

☒ Set IDMS_ROOT from existing installations:

☐ Set IDMS_ROOT manually:

IDMS Root	SCADA Config
☒ Set Per-User IDMSROOT	☐ Enable SCADA Configurations
☐ Set Per-Machine IDMSROOT	☐ SCADA 2.6
	☐ SCADA 3.2

- **Perform Clean Install:** Overrides the idms.appdef file, the Browser toolbar, menu, and rooms, and the environment variable settings.

ADMS Server CurrentDev Setup [Min] [Max] [Close]

Please select additional installation options 

Selecting checkboxes will overwrite existing resources. Grayed out checkbox means that such resource doesn't exist in the system and will be installed

☒ **Perform Clean Installation (all below options will be enabled)**

☐ Backup and overwrite EmpToolbar file

☐ Backup and overwrite WebFGMenus file

☐ Backup and overwrite WebFGRooms file

☐ Backup and overwrite ADMS Application Definition file

☐ Overwrite IDMS_ROOT environment variable

☐ Overwrite ADMS Environment Variables

☐ Overwrite NETPERMSRV_PORT environment variable

- **Ellipsis ("...") (For advanced users):** Opens the ADMS Setup screen to install several instances of the same product by defining the installationID.

State	Product	Action	Properties
☐	☐ Server	Install	ID:
☐	☐ Client	Install	ID:
☐	☐ Tools	Install	ID:
☐	☐ File Updater Server	Install	ID:
☐	☐ File Updater Client	Install	ID:
☐	☐ Documentation	Install	ID:

On this screen, you can manage multi-instance installations. Users should select the products to service, select the service action, and specify an ID.

Clicking the "Process tasks" button starts maintenance for the selected operations. The Quick Instance Setup addition helps to service groups of products based on their instanceID.

In this case, the instanceID is an identifier whose purpose is to help you distinguish between setup products. For instance, you can have three instances of the ADMS server installed with IDs: Production, Test01, and Test02.

2.5. Information about Uninstalling

Because MSI installation kits are used on Windows operating system, each software subset can be easily uninstalled.

From the control panel, choose "Programs and Features" to remove individual install packages. Also, kit maintenance operations can be run through the WelcomeSetupUI application interface described above.

The GE kits follow the standard practice of only removing files that were part of the initial installation. Any new or modified files are left alone. As ADMS does create files (logs, dynamics, etc.), note that files do remain after the un-installation process. In the case of customized configuration files, this feature is useful for preserving that work. If the goal of the uninstall action

is to create a perfectly clean slate from which to start over, some manual cleanup is also required after the programmatic uninstallation is completed.

To ensure an entirely clean uninstall, do the following:

- Remove any remaining directory structures and files.
- Remove clones and savecases (NetPermSrv and CfgProxy).
- Remove modified and customized shortcuts.

2.6. Monolithic Sample Installation

It can be quite useful to install a complete system including Browser, Habitat, Scada, and ADMS on one's personal computer, as well as for stand-alone test environments. This is intended for GE use only, but it can also be useful on a customer project for a Project Development Server (PDS). The default configuration files, savecases, and post-installation scripts are configured to work together for a "localhost" configuration.

The ADMS development and delivery team maintains an up-to-date Quick Install Guide and some associated scripts. Contact the team for more information.

2.7. Configuring the Windows Server

The following changes are required for Windows Server 2019 (for the certified configuration) or Windows Server 2016 (for the supported configuration).

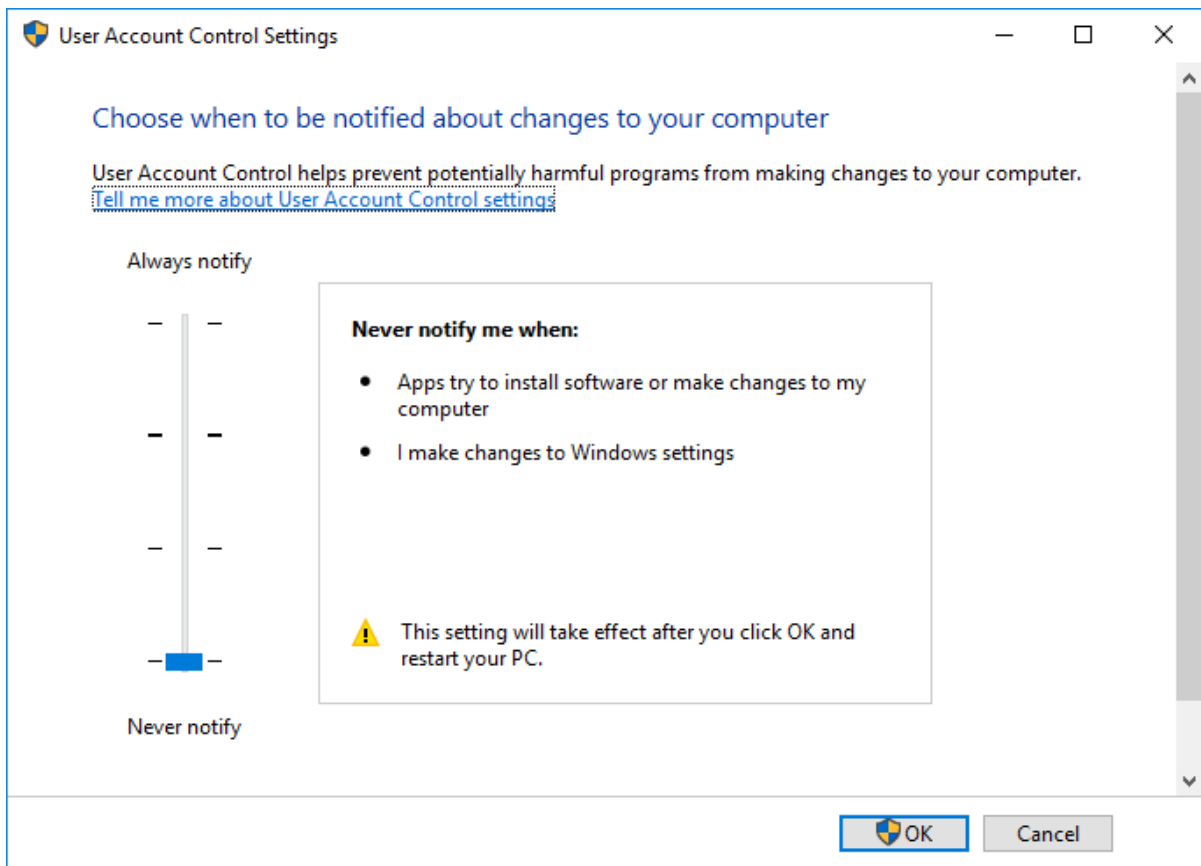
- User Account Control (UAC) must be disabled. This is required to always run the command line in admin mode.
- Windows error reporting service must be disabled and the registry updated.
- The SSL 3.0 protocol must be disabled, for security purposes.

2.7.1. Disabling User Account Control (UAC)

1. Click Start> Control Panel> User Accounts> Change User Account Control Settings.

The User Account Control Settings window appears.

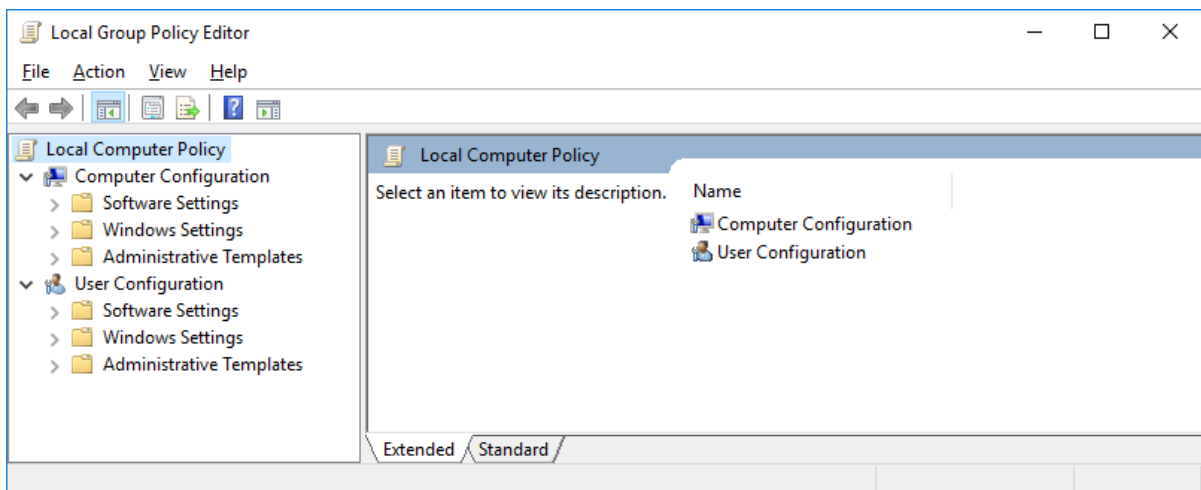
2. Select Never Notify and click OK.



Notifications are now disabled.

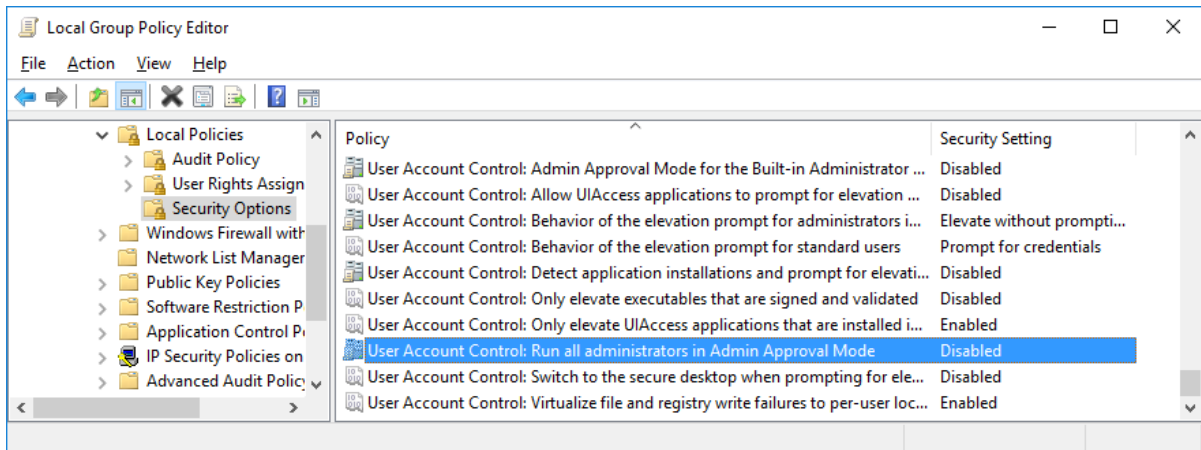
3. From the command line, run `gpedit.msc`.

The Local Group Policy Editor opens.



4. In the left pane, click Local Computer Policy> Computer Configuration> Windows Settings> Security Settings> Local Policies> Security Options.
5. In the right pane, scroll down to the User Account Control: Run all administrators in Admin

Approval Mode feature.



6. Right-click the User Account Control: Run all administrators in Admin Approval Mode feature and select Properties.

7. Select Disabled and click OK.

The feature is now disabled.

2.7.2. Configuring Access Control Lists (ACLs)

1. Create a local group on each ADMS server, for ex. %COMPUTERNAME%\<Custom ADMS Local Group>.

Note

The local group name is provided by the Customer/IT.

2. Add the following groups to the local group created above:

- Network Service
- <ADMS Group> which is modelled in Active Directory. For example, <Customer.com>\<ADMS Group>

ADMS user should be added to the above mentioned active ADMS group, for example, <Customer.com>\<ADMS Group>.

3. Run Windows PowerShell as an administrator.

4. Run the following command:

```
icacls.exe D:\Eterra\distribution /grant "Network Service" : (OI) (CI) (F)
```

5. Run the following command.

```
icacls.exe D:\Eterra\distribution /grant %COMPUTERNAME%\<Custom ADMS Local Group> : (OI) (CI) (F)
```

2.7.3. Disabling Windows Error Reporting using Server Manager

1. In the Windows PowerShell, enter:

```
Disable-WindowsErrorReporting
```

This option disables Windows Error Reporting and prevents it from prompting you to send information about application failures to Microsoft.

2. Make sure that the command result shows "True".

To update the registry:

1. Back up your registry.
2. In the Windows PowerShell, enter "regedit" and click OK to open the Registry Editor window.
3. In the Registry Editor, change the value of HKEY_CURRENT_USER\Software\Microsoft\Windows\Windows Error Reporting\DontShowUI from 0 (Disabled) to 1 (Enabled).

2.7.4. Disabling the SSL 3.0 Protocol

By default, Windows allows web clients to use the SSL 3.0 security protocol, which has vulnerabilities. It is recommended to use TLS 1.2 only.

To disable the SSL 3.0 protocol:

1. Back up your registry.
2. In the Windows Start menu, enter "regedit" and click OK to open the Registry Editor window.
3. In the Registry Editor, navigate to the following registry key:

```
HKey_Local_Machine\System\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\SSL  
3.0\Server
```

Note

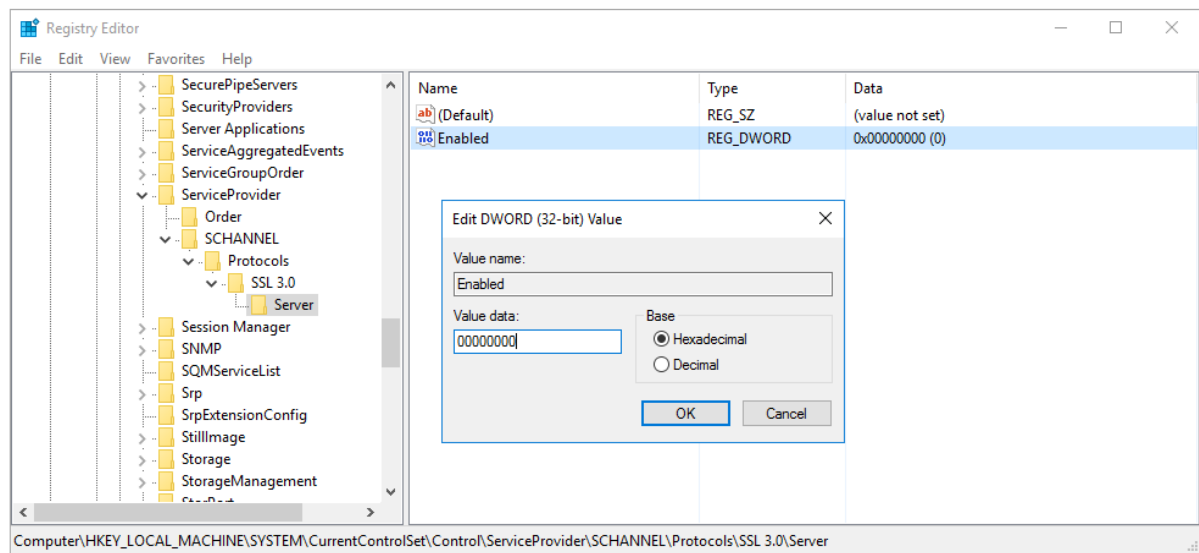
If this registry key path does not exist, create it manually by expanding the available keys and using the New > Key option from the Edit menu.

4. Edit the "Enabled" item in the SSL 3.0\Server registry key.

Note

If the "Enabled" item does not exist, create it manually by using the New > DWORD Value option from the Edit menu.

5. Type 00000000 in Value Data field to set the value of the new key equal to "0".



6. Click OK.
7. Restart the computer.

3. Installing the ADMS Software

This chapter describes prerequisites and installation steps for installing the ADMS server, client, tools, and interfaces software.

3.1. Installing a Real-Time ADMS Server

The ADMS server software kit can only be installed on Windows, as specified in the section "Operating System" in the *ADMS Release Notes*.

Installing an ADMS server as a stand-alone task(i.e., without a SCADA connection and redundancy) is a straightforward process.

Integrating the functionality with a SCADA system does require a more-complex sequence of actions (mostly on the SCADA machine), described in [Integrating with the SCADA System](#).

Important

The PROCMAN run flags (i.e., check-init) variable creation step is based on the REALTIME_FAMILY environment variable. If this variable is not correctly set, you may need to re-create the init variables under the proper family.

3.1.1. ADMS Server Applications

Table below lists applications that are included in the ADMS server installation kit (ADMS_server.msi).

Table 1. ADMS Server Components

Application	Description
ANPS Adapter	The Automated Notification Paging Services (ANPS) Adapter supports sending the notifications about area setting changes and breaker outages to a web client.
Archive	The Archive application reads ADMS data from various sources and stores the data in a relational database.
Call Server	Call server supports communicating with external interfaces (CIS, IVR, and AMI interfaces) via HTTP or TCP.
Crew Server	Crew server supports communicating with an external interface Workforce Management System (WFM) via HTTP or TCP.
Customer Experience Adapter	The Customer Experience (CE) Adapter supports sending notifications about incident and planned outage changes to a web client.
Customer Loader	The Customer Loader tool can archive customer data from a customer model file in the CUSTOMER table in the Archive database.
DMS Main Server	Supports core distribution management system functions, such as managing network model and communicating with clients and SCADA.
DMS Read Only Server	Serves DMS-only data to ADMS read-only clients.

Application	Description
DMS Simulator Server	Supports DMS-only simulation functions.
DNA Server	Supports a modularized architecture for distribution network applications. If enabled without the DNA scheduler, the DNA server runs the Fault Location (FL) and Short Circuit Calculation (SCC) solutions for the DNAF server with a smaller performance impact. If enabled with the DNA scheduler, the DNA server runs the Fault Isolation and Service Restoration (FISR) application that is requested by the DNAF server.
DNAF Server	Supports Distribution Network Analysis Functions (DNAF). The standard DNAF applications (DPF, PRV, FL, and SCC) are used to analyze the system to estimate load, locate faults, estimate bus voltages and losses, etc. The advanced DNAF applications (LVM, FISR, FSD, PLOS, AFR, and FISRCA) find solutions for improving network characteristics, restoring customer energization, isolating a component, etc.
Forecast Server (LVM Analysis Server)	Generates LVM solutions for LVM System Monitor.
Net Data Server	Supports the outage management read-only web service interface.
Net Server Main Customer Mode	Supports the ADMS customer-only mode to work with customer smart meters, visualize the meter data, and use the secondary voltage to refine the system load and voltage characteristics.
Net Server Read Only	Serves DMS and OMS data to ADMS read-only clients.
Net Server Simulator	Supports DMS and OMS simulation functions.
Net Server	Provides functionality for the Outage Management System.
NetData Replicator	Replicates OMS and Switch Order data between servers.
Oms Adapter	Provides a web services portal for sending OMS notifications to web clients and communicating with the GE's Outage Assist solution and other external applications.
Operator Simulator	The operator simulator tool (RoboDispatcher) simulates the control room operators' tasks by automatically assigning crews to the incident, keeping track of the crew work, assessing damage, and clearing incidents.
Process Starter	Allows running ADMS applications without the Habitat's Process Manager (PROCMan).
Relay Server	Simulates the Call server and AMI server external applications (for example, IVR).
Switch Order Server	Supports the Switching Order application.
View Outage Portal Adapter	The View Outage Portal (VOP) Adapter supports sending information from the De-Energized Line Journal to a web client.
DNA Scheduler	Receives the requests from DNAF server and forwards the requests to and idle DNA server.

3.1.2. Prerequisites for ADMS Server Installation

The main ADMS servers (DMS, OMS, and DNAF), as stand-alone tasks, do not impose any prerequisites different from what the client needs. The client installation kit contains a local server

application that can run on the same machine for study purposes. In Windows, this equates to requiring the .NET Framework and the Microsoft Visual C++ run-time files (see below for specific versions).

The actual prerequisites of the server in the context of a project are in fact further driven by the redundancy setup and the SCADA environment targeted for integration. The ADMS server's redundancy is handled by Habitat, so those prerequisites are described in the Habitat documentation.

Note

ADMS does not support Scada with the Parallel Site redundancy configuration.

To check the specific versions (and patches) required for Habitat and Scada compatibility, refer to the *ADMS Release Notes*.

3.1.2.1. Server Hardware Guidelines

There are no strict hardware requirements or configurations. ADMS offers flexible deployment options, such as core applications that can be deployed together on the same machine or deployed separately across several machines. It is recommended that you consult with a solution architect for individual project hardware configurations.

If the ADMS servers are installed on the same machine as the SCADA system, the hardware must meet the requirements of Scada before accommodating the ADMS components.

The ADMS server computers should be sized according to the following guidelines.

Table 2. ADMS Server Computer Hardware Guidelines

Subsystem	Small System (200k customers)			Medium System (500k customers)			Large System (2M customers)			Very Large System (6M customers)		
	Cores	RAM (GB)	HDD (TB)	Cores	RAM (GB)	HDD (TB)	Cores	RAM (GB)	HDD (TB)	Cores	RAM (GB)	HDD (TB)
Distribution Optimization (DMS)	8	32	0.6	10	64	0.6	14	96	0.6	18	128	0.6
Outage Response (OMS)	8	32	0.6	18	64	0.6	32	96	0.6	50	128	0.6
ADMS (DMS/OMS)	10	32	0.6	24	64	0.6	44	128	0.6	64	256	0.6
ADMS Archive RDBMS	4	16	0.6	4	16	0.6	4	16	0.6	4	16	0.6
External Interface	6	16	0.3	8	32	0.6	10	64	0.6	12	64	0.6
ADMS Model Manager	4	32	1.8	4	32	1.8	4	32	1.8	4	32	1.8
DOTS-DMS	8	32	0.6	10	64	0.6	14	96	0.6	18	128	0.6
DOTS-OMS	8	32	0.6	18	64	0.6	32	96	0.6	50	128	0.6
Active Directory	2	8	0.3	2	16	0.3	4	32	0.3	4	32	0.3
Replicated Data Server	8	32	0.6	10	64	0.6	14	96	0.6	18	128	0.6

Because the DMS server application executable runs with the entire distribution model in RAM, the memory must be increased relative to the size of the model. It is recommended to have extra RAM to allow for some model growth over time.

The ADMS server executables are multi-threaded, so more processors equates to a faster system. There can be multiple threads for loading the models into memory upon startup. Also, separate/multiple threads can be configured for handling the dynamics, Network Topology Processing (NTP), interactions with SCADA, and handling queries.

Aside from startup, DNAF is the biggest user of CPU time. Therefore, if a project is using more of the functions and/or running the DNAF more frequently, the number of CPUs should be increased.

There is no difference in the recommendations between physical and virtual server resources.

For new servers/components, it is recommended to start with 4 cores (vCPU) and 16 GB RAM, and then add 2 to 4 cores and 16 GB RAM for each component.

Physical CPU base clock speed should be a minimum of 2.0 GHz. All of the latest generation Intel Xeon CPUs peak around 3.6 to 4.0 GHz turbo speed, so unless you have a particularly CPU-intensive application there is very little difference.

Current hard drives, with several hundred gigabytes of storage capacity, provide plenty of room for storing the history and DNAF results.

Gigabit Ethernet should be used on the server. Depending on the number of clients and the network through which they are connecting, the network infrastructure in general may need to be evaluated.

3.1.2.2. Server Software Requirements

For information about supported operating systems, refer to the section "Operating System" in the *ADMS Release Notes*.

Note

Because of the study server concept, the server software is also supported on the client Windows operating systems in a nonrealtime capacity.

The server software is built on Windows using Microsoft Visual Studio 2019, which drives the prerequisites listed below. The ADMS server application executables and the ADMS File Updater client require the .NET Framework and some C++ libraries.

- Microsoft VisualC++2019 Redistributable Package (x64) of September 2021 or later. It is recommended to download the latest version with all security updates from https://aka.ms/vs/16/release/vc_redist.x64.exe
- Microsoft .NET Framework 4.8 offline installer for Windows <https://support.microsoft.com/en-us/help/4503548/microsoft-net-framework-4-8-offline-installer-for-windows>

These are free installations from Microsoft. Be sure to use the appropriate versions.

If integration with Scada is targeted, refer to the respective documentation for additional software requirements.

DNA Scheduler has the same software requirements as the ADMS Interface applications.

3.1.3. Installing an ADMS Server

There is an installation kit (MSI) for the ADMS server. The following describes what the installation kit does.

Warning

If you are going to install the kits over the network, it is recommended that you copy them locally first. There can be problems with uninstalling MSIs if the installation was performed over a network share and the original cannot be accessed. Also, local kits improve the performance of the installation.

The installation kit creates its directory structure beneath the specified install path. The binary executables are installed by default under:

`D:\Eterra\Distribution\Server\bin`

The default path for data and configuration files is defined in the %IDMS_ROOT% environment variable as the following:

`D:\Eterra\Distribution\FantasyIsland`

To install the ADMS server software:

1. Run the server installation kit (ADMS_server.msi) with Administrator privileges, using one of the following methods:
 - Log on to the Windows machine as "Administrator" and double-click the server MSI file.
 - Press Shift and right-click the ADMS_server.msi file, select "Run as different user", and choose the Administrator user.
 - Right-click the command window shortcut and choose "Run as Administrator". Then execute the server MSI file from the command window.
 - Execute the server MSI file from the command window. For more information, see [Installing an ADMS Server Using Command Line](#).

The ADMS Setup Wizard display appears.

2. Click Next.

The End User License Agreement display appears.

3. Read the license agreement.

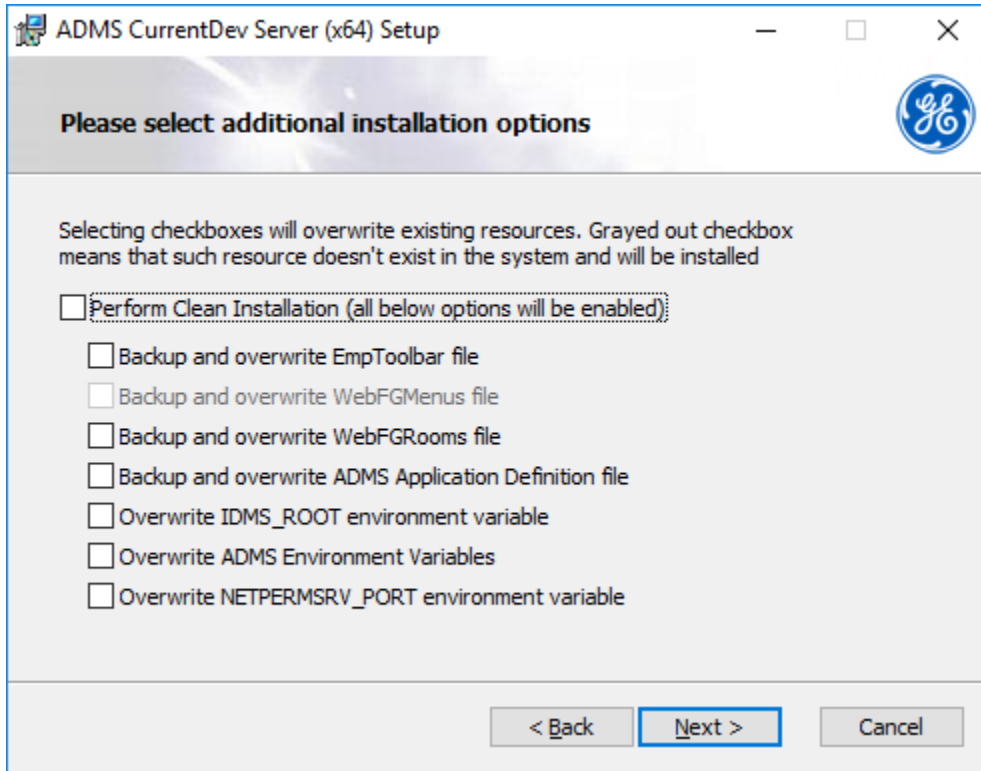
- Click Next to accept the license terms and proceed with the installation.

The Customer Setup display appears with a default setting for features and location of the installation.

- Optionally, adjust the installation path and/or select/deselect features and click Next.

If changed, the dialog box displays the updated path.

- On the Additional Installation Options screen, select any options to overwrite files (for example, the `idms.appdef` file) and environment variables as needed. Click Next.



- On the Ready to Install screen, click Install.

The ADMS server installation begins.

- When the installation finishes, click Finish.

The ADMS Setup Wizard display closes.

- Open a new Command Prompt with administrative privileges and type "set".

All environment variables are displayed.

- Verify that the following variables are all defined:

DMS_BINDIR: Defines the path to the server bin directory. The default path is `D:\Eterra\Distribution\Server\bin\`.

DMSDNAF_CONFIG: Defines the path to the DNAF Server configuration file. The default path is

%IDMS_ROOT%\config\server\NetServerDnafConfig_localhost.txt.

DMSMAIN_CONFIG: Defines the path to the NetSrv configuration file. The default path is %IDMS_ROOT%\config\server\NetServerMainConfig_localhost.txt.

OMSMAIN_CONFIG: Defines the path to the NetServer configuration file. The default path is %IDMS_ROOT%\config\server\NetServerOmsConfig_localhost.txt.

DMS_OMSCALL_CONFIG: Defines the path to the OMS Call server configuration file. The default path is %IDMS_ROOT%\config\server\CallServerConfig.xml.

DMS_OMSCREW_CONFIG: Defines the path to the OMS Crew server configuration file. The default path is %IDMS_ROOT%\config\server\CrewServerConfig.xml.

DMS_OMSRELAY_CONFIG: Defines the path to the OMS Relay server configuration file. The default path is %IDMS_ROOT%\config\server\RelayServerConfig.xml.

DMS_OPERATORSIM_CONFIG: Defines the default Operator Simulator (RoboDispatch) configuration parameters. The default value is /NETCONF
"%IDMS_ROOT%\config\server\NetconfRoboDispatch.xml" /CONFIG
"%IDMS_ROOT%\config\server\RoboDispatchConfig.xml".

DMS_SOBINDIR: Defines the path to the Switch Order server Bin directory. The default path is D:\Eterra\Distribution\Server\bin\.

DMS_SOCONFIG: Defines the path to the Switch Order server configuration file. The default path is %IDMS_ROOT%\config\server\SwitchOrderServerConfig.txt.

IDMS_ROOT: Defines the path for data and configuration files. The default path is D:\Eterra\Distribution\FantasyIsland.

Note

The %IDMS_ROOT% environment variable is case sensitive for some ADMS applications and must be set in all capital letters.

PROCMAN_FLAG_EMS_RUN_DMSMAIN: If set to True, runs the DMS server task in the default Scada PROCMAN model.

PROCMAN_FLAG_EMS_RUN_DMSDNAF: If set to True, runs the DNAF server task in the default Scada PROCMAN model.

Note

The ADMS run flags PROCMAN_FLAG_EMS_RUN_DMSMAIN and PROCMAN_FLAG_EMS_RUN_DMSDNAF are available in the default Scada PROCMAN model.

If you are using a custom PROCMAN model, these tasks are not needed and can be deleted. To learn more about these tasks, refer to the *PROCMAN User Guide* (available in the *Habitat* documentation set).

PROCMAN_FLAG_EMS_START_IDMS: If set to True, starts ADMS tasks in the default Scada PROCMAN model.

PROCMAN_FLAG_EMS_START_OMS: If set to True and PROCMAN_FLAG_EMS_START_IDMS is also set to True, starts OMSrelated tasks in the default Scada PROCMAN model.

11. Run the symbols installation kit (ADMS_symbols_server.msi and ADMS_symbols_client_tools_interface.msi) with administrator privileges and perform the same installation steps as for the server kit. The symbols kit contains PDB files for the server, client, and tools kits. These files contain method signatures that are used for debugging purposes, such as creating crash dumps.

3.1.4. Installing an ADMS Server Using Command Line

You can also install the ADMS server using command line.

The server installation supports the following command-line arguments:

```
INSTALL_<Component_Name>=1
```

Specifies the server component to be installed (ALL, DMS, DNAF, OMS, DOTS, INTERFACE, or CNV). You can use several instances of this argument in a command.

```
INSTALLDIR=<_Installation_Folder_Path_>
```

Specifies the location and name of the customer schema file to use. This argument is required when running the program from the command line.

```
OVERWRITE_ENV_IDMS_ROOT=1
```

Specifies whether the installation overwrites the IDMS_ROOT environment variable.

```
/QB
```

Specifies whether the installation is performed in silent mode (without user interface).

For example:

```
msiexec /i ADMS_server.msi INSTALL_DNAF=1 INSTALL_DMS=1 INSTALLDIR=c:\ADMS OVERWRITE_ENV_IDMS_ROOT=1 /QB
```

3.2. Installing the Client Software

The ADMS client software kit can only be installed on Windows, as specified in the chapter "Operating System" in the *ADMS Release Notes*.

The ADMS client software runs within the Advanced Presentation Framework (APF), which is a framework of the Browser client, so the ADMS installation kit checks for the presence of APF and prevents installation if this prerequisite is not found.

i Note

There is a 256-character limit for the target of a Windows shortcut. Because the ADMS client shortcut must start webfgviewer.exe and pass multiple configuration files, a target with fully qualified paths can be quite long. The client installation kit creates a shortcut that uses some standard Browser environment variables to reduce the length.

However, if you install to a folder with a very long path, if the variables are traded out for full paths, or if the configuration file names used are descriptive and long, there is a chance that the 256character limit may be exceeded. Should this happen, a Windows batch script can be used for launching the client, or some new, more-specific environment variables can be used.

3.2.1. ADMS Client Applications

Table below lists applications that are included in the ADMS client installation kit (ADMS_client.msi).

Table 3. ADMS Client Components

Application	Description
Call Entry	The Call Entry tool allows the direct entry of trouble calls by operators or other staff in cases of emergency, or when an outage is reported directly by authorities such as police. Trouble calls can also be specified by location if there is no direct association with a customer.
Crew Simulator	The Crew Simulator (RoboCrew) tool simulates behavior of field crews. It works together with the Crew server and the operator simulator for automatic changes of crew status, crew position, damage assessment, and fault removal.
DMS Interface Server	The DMS interface server provides a web server interface to allow third party applications to communicate with the DMS Main server (NetSrv).
Event Scripter	Use the Event Scripter to create and play time-based scripts of power system events that are injected into the simulated power system model. Most of these events are common disturbances that occur on the distribution power system. Event scripts allow the instructor to save and repeat complex and/or lengthy training scenarios as required. You can also use scripts for demo purposes.
Local Study Server	Supports the study mode that allows the network view and network analysis functions to be run on a private instance of the network model on the client workstation.
Local Study DNA Server	Supports the study mode for the DNA server, which runs the Fault Location and Short Circuit Calculation solutions for the DNAF Server with a smaller performance impact.

Application	Description
Oms Interface Server	The OMS Interface server provides a standard web server interface to allow third party applications and ADMS services to communicate with the OMS server (NetServer).
Mobile Adapter	The DMS interface server communicates with field mobile devices managed by the Mobile Enterprise solution via the Mobile Adapter application. (This application is not installed by default.)

3.2.2. Prerequisites for Installing the ADMS Client and Tools

The following sections describe the hardware and software requirements for installing and using the ADMS client and tools software.

3.2.2.1. Checking the Browser Client Installation

You need to install Browser and Advanced Presentation Framework prior to installing the ADMS client kit. If they are already installed, note their location.

The following sections assume that Advanced Presentation Framework is installed in the following location (this is not the default location):

D:\Eterra\Presentation Framework\

Most projects follow the paradigm of using the Cdrive for the operating system and the Ddrive for applications. This provides good visibility, and it allows for easier system restoration. Also, as Microsoft continues to improve the security of its systems, use of the Cdrive involves more restrictions.

If your installation is different, you might have to customize some of the ADMS configuration files.

For information about installing Browser, refer to the *Browser Software Installation and Maintenance Guide*.

For information about installing Advanced Presentation Framework, which is required for Browser, refer to the *Advanced Presentation Framework Software Installation and Maintenance Guide*.

3.2.2.2. Hardware Requirements

For the client and tools, the following hardware configuration is recommended:

- 2 @ 2.27 GHz processors with 4 cores each
- 16GB or more of RAM
- 5GB of available disk space
- Due to the intense nature of the displays of the distribution networks, higher-performance video cards that support multiple high-resolution monitors are recommended.

- The certified size scaling for text, applications, and other items on the screen is 100%. Display layout issues may appear at higher size scaling.
- The certified display resolution is 1920x1080. Ultra-high display resolutions (for example, 3840 x 2160) are not certified. Display layout issues may appear.

3.2.2.3. Software Requirements

The ADMS client and tools are certified on the Windows platform, as specified in the section "Operating System" in the *ADMS Release Notes*.

The client and tools software (just like the server software) is built on Windows using Microsoft Visual Studio 2019, which drives the prerequisites listed below.

For Windows, the ADMS client software and the ADMS File Updater client require the .NET Framework and some C++ libraries. You can search online for these free downloads using the following text:

- Microsoft Visual C++ 2019 Redistributable Package (x64) of September 2021 or later. It is recommended to download the latest version with all security updates from https://aka.ms/vs/16/release/vc_redist.x64.exe
- Microsoft .NET Framework 4.8 offline installer for Windows <https://support.microsoft.com/en-us/help/4503548/microsoft-net-framework-4-8-offline-installer-for-windows>

For a complete list of required third-party software, refer to the *ADMS Release Notes*.

3.2.3. ADMS Client Installation Steps

1. Run the client installation kit (ADMS_client.msi) with Administrator privileges, using one of the following methods:
 - Log on to the Windows machine as "Administrator" and double-click the client MSI file.
 - Press Shift and right-click the ADMS_client.msi file, select "Run as different user", and choose the Administrator user.
 - Open a command window by right-clicking the command window shortcut and choosing "Run as Administrator", and then execute the client MSI file from the command window.

The ADMS Setup Wizard display appears.

2. Click Next.

The End User License Agreement display appears.

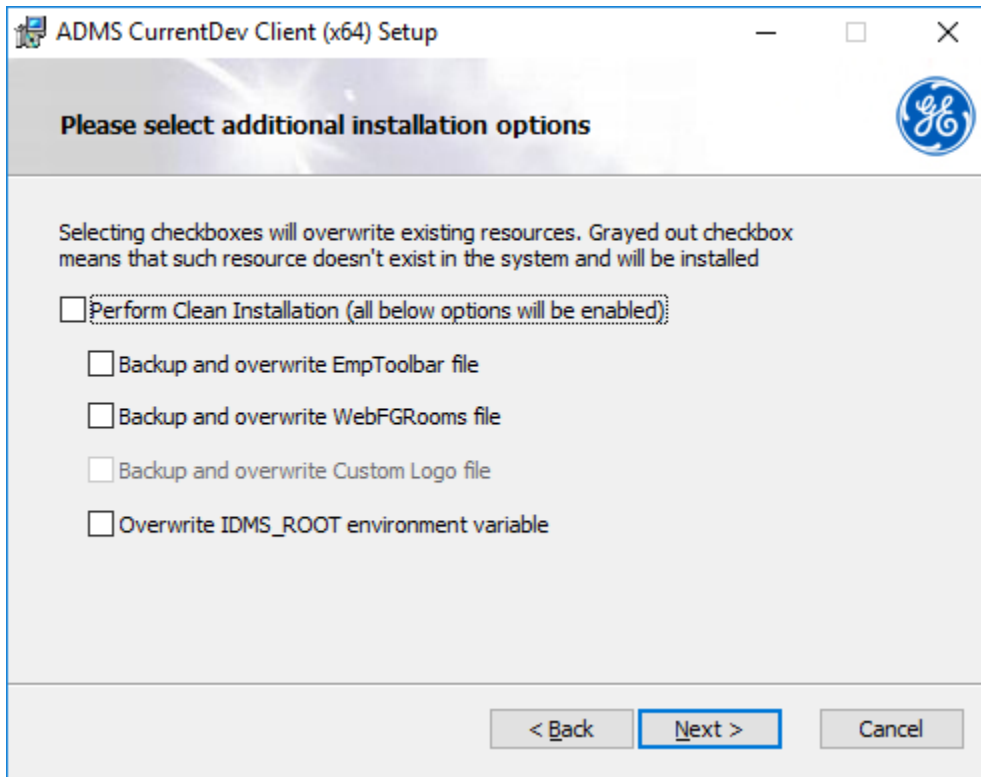
3. Read the license agreement.
4. Click Next to accept the license terms and proceed with the installation.

The Destination Folder and Features List display appears with a default setting for the location.

5. Optionally, adjust the installation path and/or select/deselect features. Click Next.

If changed, the dialog box displays the updated path.

6. On the Additional Installation Options screen, select any options to overwrite files and environment variables as needed, then click Next.



7. On the Ready to Install screen, click Install.

The ADMS client installation begins.

8. When the installation finishes, click Finish.

The ADMS Setup Wizard display closes.

3.2.4. Integration with the Browser Display Server

For systems using Browser, there is always a functional system referred to as "the Browser display server". This can be a system by itself. The Browser display server machine runs IIS or Apache, and serves Browser displays and configuration files for connections, menus, toolbars, and so on. The integration of ADMS within the larger GE system allows for display call-ups and actions to be launched using the standard Browser menu and toolbar mechanisms.

To facilitate a fully functional set of menus and toolbars, the ADMS release contains a menu file and a toolbar file that add the distribution-related items on top of the standard Scada files. The ADMS

server kit installs a WebFGRoot configuration folder (by default, %IDMS_ROOT%\..\FantasyIsland\misc\WebFGRoot) that holds these configuration files. The included menu and toolbar files are supersets of the delivered files included with Scada, so it should be safe to replace the base delivered files with those included in this kit if they have not been customized.

3.2.5. Toolbars for Scada vs. Control SCADA

The delivered ADMS toolbar files were designed to work with either Control (formerly **e-terracontrol** or SMP) SCADA or Scada (formerly **e-terrascada** or EMP) SCADA.

There is a client command-line switch used to set the functionality for integration with the Scada version: "-EMPSCADA". Since this is the normal configuration, the installation kit sets this flag by default.

3.3. Installing the Tools Software

The ADMS tool kit contains a suite of Windows-based programs used for various tasks relating to creating the models and configuring the system.

3.3.1. ADMS Tools Applications

Table below lists applications that are included in the ADMS tools installation kit (ADMS_tools.msi).

Table 4. ADMS Tools

Application	Description
Configuration Tools	
Coded Translation Editor	This tool enables the customer to define a set of bitmaps associated with object types, such as Incident, Outage, and Crew. In addition to defining the bitmaps to be displayed in the ADMS client, this tool allows the user to define the related description and (as needed) the expected behavior when selecting the bitmap (using a URL command).
DNAF Configuration Editor	This tool allows a user to modify the DNAF configuration file, which provides key application data, including convergence tolerances, specific application parameters, and data that is necessary for the optimization applications (LVM, FISR, AFR, POS, and FISRCA), including problem formulation attributes.
Encryption Tool	Encrypts text data, such as passwords and database connection strings in configuration files.
Grid Tabular Builder	Builds the ADMS tabular displays.
Model Manager Configuration Editor	The Model Manager Configuration Editor allows you to set the processing options and modify the locations of the Model Manager's folders, components, and log files.
Net Dynamics Configuration Editor	Used to define and modify the instances of the dynamic data that are used by the ADMS client to connect to the server and to configure other various aspects of the client's behavior.

Application	Description
OMS Client Configuration Editor	The main purpose of this tool is to configure Outage Management System (OMS) clients. It also allows an administrator to configure the appearance and effects of geographic network displays, to meet the needs of individual utilities.
OMS Server Configuration Editor	The main purpose of this tool is to configure the Outage Management System (OMS) server. It also allows an administrator to define estimated time of restoration values, call action classifications, and case note classifications. The appearance and content of the chronology log for outage events are also configured within this tool.
Resource Editor	Facilitates localization of the ADMS components.
Snapshot Viewer	The Snapshot Viewer is an ADMS Outage Management System (OMS) debugging and disaster recovery tool used when all other options of fixing a problem have been exhausted. The snapshot viewer application allows you to view the entries of snapshot and journal files in a grid view, modify them, and save the changes. For NetServer, it is possible to apply journal messages up until a specified point in time and then create a new snapshot file with the saved results.
Switching Order Client Configuration Editor	Used to configure the look of the Switching Order application.
Switching Order Server Configuration Editor	Used to configure the behavior of the Switching Order application.
Viewer Configuration Editor	This editor is used to set up and modify the broad range of functionality that is supported by the network viewer. This tool can be used to change the look and feel of the ADMS network view (map).
Modeling Tools	
Annotation Importer-Exporter	This tool can be used to import and export annotations between binary SAV files and XML files. The Annotation Importer/Exporter application can also be used to view and edit annotation binary files and XML files. It operates between annotation files (binary dynamics files) and XML.
Converter	The converter is the key modeling tool that validates and generates station models used by the ADMS software.
Crew Modeling Tool	Supports creating a crew model and importing and exporting crew data (employees, vehicles, crew areas, employee types, and vehicle types).
Customer List Compiler	This tool generates the customer model used by the Outage Management System (OMS).
DNOM DB Loader	Writes station model data in a relational database.
DNOM Unique ID Checker	Evaluates a set of station model files for unique object IDs to ensure that the DMS system is functioning optimally. When objects IDs are not unique, issues might occur with the dynamics application process.
DNOM Viewer	This tool reads a station model and shows the Distribution Network Object Model (DNOM) contents.
Geographic Background Compiler	This compiler produces background files that are incorporated into the model management process and are used within the ADMS client for displaying roads, buildings, lakes, etc. These background files are generated from GIS data (in particular, from shape files).

Application	Description
Model Manager	The Model Manager tool is a workflow management tool that governs the model creation steps and files. The resulting station model files are then stored in the appropriate folder for further review or transferred into the online ADMS servers.
Model Quality Stat Logger	Reads station models in the Converter's output folder and extracts statistics about the quality of the model and Converter error messages into consolidated log files.
Navigation Builder	Analyzes all the station model files to create an overview.nvw file for the ADMS clients. For more information, see Station Overview Builder .
Permit Builder	The purpose of this tool is to create and maintain a hierarchical representation of the Areas Of Responsibility (AORs) used by the Habitat PERMIT application.
Substation Builder	Creates <Station>_subeditor.xml files that are used as an input to the Substation Editor tool.
Substation File Builder Lite	Updates station externals files <Station>.xml files with the measurement and value placement information from the Scada database. Creates <Station>.xml files that could be for building station models using the Converter or Model Manager.
Substation Editor	Creates SCADA and ADMS displays and portions of the Scada model and the ADMS station models.

3.3.2. ADMS Tools Software Installation

The ADMS tools kit should be installed prior to installing the toolkit.

To install the ADMS tools software:

1. Log on to the target machine and run the `ADMS_tools.msi` installer.
2. At the ADMS Setup Wizard screen, click Next.
3. At the End User License Agreement screen, read the license agreement, and then click Next to accept the license terms and proceed with the installation.
4. At the Destination Folder and Features List screen, adjust the installation path and select/deselect features as needed. Click Next.

If changed, the dialog box displays the updated path.

5. At the Ready to Install screen, click Install.

The software installation begins and the status screen is shown.

6. When the installation finishes, click Finish.

The ADMS Setup Wizard display closes.

7. Verify that the following variable is defined:

DMS_TOOLS_BINDIR: Defines the path to the tools bin directory. The default path is `D:\Eterra\Distribution\Tools\bin\`.

3.4. Installing the Interfaces Software

The ADMS interfaces kit contains a suite of Windows-based programs used for various tasks relating to communicating with external applications.

3.4.1. ADMS Interfaces Applications

Table below lists applications that are included in the ADMS tools installation kit (ADMS_interfaces.msi).

Table 5. ADMS Tools

Application	Description
Call Server	Supports communicating with external interfaces (CIS, IVR, and AMI interfaces) via HTTP or TCP.
Core Notification Server	Supports the Internal Notifications functionality.
Crew Server	Supports communicating with an external Workforce Management System (WFM) via HTTP or TCP.
Data Correction Interface	Supports the Enterprise OMS functionality. Supports the historical data correction function.
DERM Kafka Adapter	Supports communicating with the 2030.5 Gateway.
Failover Broker	Supports the NetViewer Adapter redundancy.
NetData Replicator	Replicates OMS and Switch Order data between servers.
NetViewer Adapter	Provides map data for the web-based network view and provides DMS and OMS data in GraphQL format.
Oms Adapter	Provides a web services portal for sending OMS notifications to web clients and communicating with the GE's Outage Assist solution and other external applications.
OMS Interface Server	Provides a standard web server interface to allow thirdparty applications and ADMS services to communicate with the OMS server (NetServer).
Structure Server	The StructureServer uses the EquipmentAndStructure.index file to return requested information about structures, equipment, and the association between them.
Work Order Adapter	Supports the Enterprise OMS functionality. Reads the Work Model journal files and sends the journals to Work Model-Event Handler to be processed by Work Model Service.
Work Order Archive Adapter	Supports the Enterprise OMS functionality. Writes Work Model journal files for Enterprise OMS.
Work Order Database Sync	Supports the Enterprise OMS functionality. Reads Work Model journal files and writes the data to the Enterprise OMS database.

3.4.2. Prerequisites for Installing the ADMS Interfaces

The ADMS interfaces are certified on the Windows platform, as specified in the section "Operating System" in the *ADMS Release Notes*.

The ADMS interfaces software require the ASP.NET Core Runtime Hosting Bundle and .NET Desktop Runtime bundles, which are part of Microsoft .NET Framework 5. It is recommended to download the latest versions with all security updates from <https://dotnet.microsoft.com/download/dotnet/5.0>

You can search online for these free downloads using the following text:

- ASP.NET Core Runtime Hosting Bundle version 5.0.9 or later.

For example, <https://dotnet.microsoft.com/download/dotnet/thank-you/runtime-aspnetcore-5.0.12-windows-x64-installer>

- .NET Desktop Runtime version 5.0.9 or later.

For example:

<https://dotnet.microsoft.com/download/dotnet/thank-you/runtime-desktop-5.0.12-windows-x64-installer>

For a complete list of required third-party software, refer to the *ADMS Release Notes*.

3.4.3. ADMS Interfaces Software Installation

The ADMS interfaces kit should be installed prior to installing the toolkit.

To install the ADMS interfaces software:

1. Log on to the target machine and run the `ADMS_interfaces.msi` installer.
2. At the ADMS Setup Wizard screen, click Next.
3. At the End User License Agreement screen, read the license agreement, and then click Next to accept the license terms and proceed with the installation.
4. At the Destination Folder and Features List screen, adjust the installation path and select/deselect features as needed. Click Next.

If changed, the dialog box displays the updated path.

5. At the Ready to Install screen, click Install.

The software installation begins, and the status screen is shown.

6. When the installation finishes, click Finish.

The ADMS Setup Wizard display closes.

7. Verify that the following variables are defined:
 - **DMS_INTERFACEAMD_BINDIR:** Defines the path to the interfaces bin directory. The default path is `D:\Eterra\Distribution\Interfaces\bin\`.
 - **DMS_INTERFACE_BINDIR:** Defines the path to the interfaces bincore directory. The default path is `D:\Eterra\Distribution\Interfaces\bincore\`.

4. Initial Configuration

This chapter details the first round of configuration, designed to get the ADMS server and client functioning together.

Warning

Only administrators should have privileges to access server configuration files. It is the responsibility of the customer to harden access for security purposes.

4.1. Configuring the DMS Main Server

This section describes how to configure the DMS Main server (NetSrv), which provides the core functionality of the ADMS system.

The DMS main server application reads a text-based configuration file. This section describes how to get the server up and running using the text-based configuration file, `NetServerMainConfig.txt`. For details about server configuration parameters, see [NetServerMainConfig.txt](#). For a list of all server configuration files that are delivered with the ADMS server, see [Network Server Configuration Files](#).

The following sections discuss the parameters that specify the main configuration of the server.

4.1.1. Configuring the Path Parameters

Verify that the following parameters match the path configuration of your server installation:

```
MODEL_PATH      = %IDMS_ROOT%\models\RT\  
DYNAMIC_PATH    = %IDMS_ROOT%\dynamics\Server\  
SAVECASE_PATH    =   
DNAF_INPUT_PATH = %IDMS_ROOT%\models\DNAFParams\  
LOG_FILE_NAME    = %IDMS_ROOT%\log\netsrv_main.log  
ISD_LOG_FILE_NAME = %IDMS_ROOT%\log\isd_main.log
```

Note

The "SAVECASE_PATH" parameter should be blank on the real-time server (MAIN and DNAF instances). It is only used on the client's local study server.

For more information about the environment variable `IDMS_ROOT`, see [Software Installation Overview](#).

4.1.2. Setting the Server Listening Port

The DMS server applications listen to client connection requests on a specific TCP/IP port, which can be defined in the configuration file using the `SERVER_PORT` parameter. To connect to a server instance, clients must use the same port number for connection requests, which is defined in their `netconf.xml` file (use `NetDynamicsConfigurationEditor.exe` to configure the client settings). For `NetSrv`, either leave this listening port setting as delivered (5025), or change it to fit project needs.

For more information about TCP/IP ports, see [TCP/IP Default Port List](#).

4.1.3. Setting the Thread Parameters

The DMS server application is a multi-threaded application. If you run the server on a multi-CPU (or multi-core) computer, you can improve performance by taking advantage of more than one CPU. Parameters in the `NetServerMainConfig.txt` file are available to specify the number of desired threads by thread activity:

- `THREADS_DYN` (dynamics handling)
- `THREADS_NTP` (network topology processing)
- `THREADS_LOAD` (mod file loader used during startup)
- `THREADS_SCADA` (SCADA communication)
- `THREADS_QUERY` (query management)
- `THREADS_CLIENT` (client communication)

These parameters should not be set to a number greater than the number of processors. Hyper-threading and multiple cores complicate this concept, but ultimately tests should be performed to find the optimal settings. For example, testing the loader thread running on a machine with four hyper-threaded CPUs, the optimal setting might be at six threads (not eight).

Finding the optimal setting is not critical during initial configuration. However, the thread counts should not be set to a number greater than the number of logical processors.

4.1.4. Temporarily Disabling CfgProxy (CFGCTRL Interface)

At this stage of the configuration, the goal is to get the server running and a client connected to it. Later sections detail how to configure the DMS server application to interface with `CFGCTRL` via `CfgProxy`. For now, comment out the redundancy parameters using an apostrophe.



Warning

Do not run redundant DMS servers at the same time with the `CFGCTRL` interface disabled. Only run one DMS server at a time.

Comment out the following lines in the `NetServerMainConfig.txt` file to disable the `CFGCTRL`

interface, allowing the DMS server application to be enabled by default:

```
'DMS Redundancy Parameters
'REDUNDANT_HOST      = mydmsb,mydmsa,mydmse,mydmsd
'REDUNDANT_PORT      = 5025
'CFGPROXY_HOST       = mydmsa
'CFGPROXY_TIMER_PERIOD = 30
'CFGPROXY_PORT       = 5019
```

These settings allow the server to fully start up, become enabled, and accept client connections to truly test the connection to the server.

To disable getting role assignments from CfgProxy, set the parameter:

```
USE_ROLE_FROM_CFGPROXY = 0
```

Later sections provide instructions for completing the redundancy integration.

Warning

After the CfgProxy and redundancy configuration is completed, set `USE_ROLE_FROM_CFGPROXY = 1` for a production system.

4.1.5. Temporarily Disabling Permission Checking (PERMIT Interface)

Again, the goal is to get the server running, get a client connected to it, and ensure that actions and messages can be exchanged. To allow this without a fully integrated set of model files and a matching functional PERMIT definition, make sure that the following parameter is set appropriately.

Make the following change to the `NetServerMainConfig.txt` file to disable permission checking:

```
PERMIT_CHECK_ENABLED = 0
```

Be sure to save your changes. Later sections provide instructions for configuring the permission checking for ADMS.

Warning

After the permission checking configuration is completed, set `PERMIT_CHECK_ENABLED = 1` for a production system.

4.1.6. Temporary Object Auto Correction

As substation model files change over time, it is possible that lines with temporary objects or dynamic state changes may be moved from station to station. This would normally be problematic

in that the target line for the temporary object would not be found in the indicated station model. The DMS server, however, has built-in capabilities to correct for these object shifts through its State Preservation and Orphan Object processing.

All temporary network modifications are stored as dynamic entries along with relevant placement information. This information includes such items as the X and Y coordinates, the geometry used, the target line information, the position as a percentage of the overall line length, etc.

On each station model upload, temporary object processing evaluates each object for correct placement to determine if it should be orphaned or if it can be corrected. This automatic processing may be adjusted using several configuration options in the `NetServerMainConfig.txt` file:

```
CORRECTLY_PLACED_TOLERANCE    = .5
POSITION_UPDATE_TOLERANCE     = 25
```

Additionally, objects that cannot be automatically corrected can be "snapped" to nearby lines by searching within the snap tolerance:

```
SNAP_SEARCH_TOLERANCE         = 5
```

To disable both the state preservation and the orphan object snap correction, make the following change:

```
AUTO_CORRECT_ENABLED          = 0
```

- `CORRECTLY_PLACED_TOLERANCE` is the distance in geometric units that determines whether the temporary object is correctly placed on a line. If the distance from the temporary object indicated position to the object line placement is within this tolerance, the temporary object is assumed to be correctly placed.
- `POSITION_UPDATE_TOLERANCE` is the distance in geometric units that determines whether the temporary object should have its X and Y position automatically updated. If the temporary object indicated position is not correctly placed (above tolerance), but is within the position update tolerance, its X and Y position will be updated to reflect the actual line attachment point. This is helpful if the target line has a slightly changed geometry, but still retains its original phase, ID, and station.

Note

Setting this value to the same value or less than the correctly placed tolerance will effectively disable the position update feature.

- `SNAP_SEARCH_TOLERANCE` is the distance in geometric units in which to search for potential line segments to snap to (lines may be in other stations). If more than one line is found, no assumption is made as to the target position, and the temporary object will become orphaned. If a single line is found within the snap tolerance distance (regardless of which station it is in), the temporary object position will be updated to indicate attachment to that line.

Note that some special handling is required if a substation model is to be removed from the

system. In that instance, the auto correction must be triggered before the actual station removal (see [Temporary Objects Not Auto Corrected on Deleted Substations](#)).

4.1.7. Allow Loading Model Files with Fatal Errors

This setting is used to configure whether station model files with fatal errors can be loaded by the server.

To allow model files to be loaded with fatal errors, make the following change in the `NetServerMainConfig.txt` file:

```
ALLOW_LOAD_FATAL_MODEL = 1
```

Also, make the following change in the `netconf.xml` file:

```
<AllowLoadFatalModel>true</AllowLoadFatalModel>
```

4.1.8. Backing Up Stable Station Models

When station model and overview files are uploaded to the server and the main server safely operates during a specified time period, the models are backed up (note that older backed up models are overridden by newer models).

Model backups are configured using the following parameters:

```
MODEL_PATH_SAFE_MODE = %IDMS_ROOT%\models\RT_SM\  
MODEL_SAFE_MODE_BACKUP_PERIOD = 30
```

Note

By default, model backup parameters are commented out in the configuration file. To enable backing up stable station models, uncomment these parameters.

PROCMan can be configured to run the server in safe mode if the server crashes twice. To enable safe mode, the "-SafeMode" command must be added to the Target field in the Net server shortcut.

When the server is running in safe mode, it loads model files from the backup folder. Model backups are not performed while in safe mode.

4.1.9. Delay freeing DNOM objects

ADMS server has a feature to delay freeing DNOM objects that are created at runtime for live model updating support. By default, this feature is enabled with 10 minutes delay, and it only has effect on unmanaged DNOM objects created at runtime. This feature is controlled by the following option in unmanaged server configuration files:

```
DELETED_OBJECT_RETAIN_MINUTE = 10
```

Setting value of this option to zero will turn off this feature. When this feature is enabled, deleted runtime created DNOM objects will be moved to a deleted object bag for later final deleting by garbage collection process based on the value of the option.

4.1.10. Starting the Server

The server needs at least one model file to read, so make sure that one or more model files are in the models folder.

✓ Tip

- In the absence of a tested and functioning project model, the sample model files delivered with the server and client installation kits are quite useful for getting the initial integration working.
- While initially configuring the server, it is helpful to run NetSrv from a command prompt with a small subset of model files so the station-specific output is minimized. This allows you to focus on the configuration-related messages and keep the startup time short.

Before the server is configured inside of PROCMAN, a shortcut can be used to start the application, or it can be started from a command prompt. To run the server from the command line, mimic the Method1 command below. To use a shortcut, there are two methods: simple, and simple with a little sophistication (i.e., an output log file). This section gives specific examples for Windows.

Method 1: Create a shortcut that contains netsrv.exe in the target as well as the DMS server application's configuration file, shown below:

```
D:\Eterra\Distribution\Server\bin\netsrv.exe /CONFIG  
D:\Eterra\Distribution\Server\config\NetServerMainConfig_localhost.txt
```

i Note

The paths might be different depending on the installation location.

Method 2: Create a batch script that starts the executable and passes the configuration file exactly as shown above. Then create a shortcut that runs the batch script and redirects the output to a named log file. The target should look like this:

```
StartServer.bat > netsrv_output.log 2>&1
```

This approach pipes the general output messages of the server into a text file. The "2>&1" notation on the end is standard syntax for piping the standard output and standard error to the same place.

✓ Tip

Use the command-line method until the initial configuration is done and a client can successfully connect to the server. Seeing the output going directly to the window is the fastest way to troubleshoot initial configuration/connection problems.

4.2. ADMS Client Configuration

4.2.1. Configuring the Client Connection to the DMS Main Server

This section discusses how to use Net Dynamics Configuration Editor to connect the client to the server. For complete details regarding this tool, refer to the *Net Dynamics Configuration Editor User Guide*.

The configuration information that directs the client to the server(s) is stored in the file `netconf.xml`, which is installed by default in the `%IDMS_ROOT%\config\client` folder. This file can be created or modified using the Net Dynamics Configuration Editor.

It is recommended that the file be renamed `netconf_hostname.xml`, where *hostname* is a generic (non-a or -b) variation of the server names on which the ADMS server applications are running.

The Net Dynamics Configuration Editor shortcut is installed with the Tools kit and stored in Start Menu\Programs\Eterra\Distribution\Tools\Configuration Tools\. The shortcut is used to start this application and open the default configuration file. The default target value in the shortcut's properties dialog box is as follows:

```
Target = D:\Eterra\Distribution\Tools\bin\NetDynamicsConfigEditor.exe /DEFAULTFILE  
"%IDMS_ROOT%\config\client\netconf_localhost.xml"
```

To use a custom `netconf.xml` file, update the shortcut's target value accordingly. (To open the shortcut properties window, right-click the shortcut and select Properties.)

To configure the client connection to the server:

1. Start Net Dynamics Configuration Editor.

The General tab is shown by default.

2. Verify that the proper configuration file was opened. Notice that the currently opened file is displayed in the title bar.
3. Select the RT (Real-Time) tab.

Syntax issues are most obvious when you use absolute paths; if environment variables or relative paths are used, check to ensure that the paths are correct for your client installation. Relative paths should be relative to the client's "bin" folder, which houses (for example) `dmsdisplay.dll` and `netdisplay.dll`.

4. Verify that the server IP's parameter (a comma-separated list) is properly defined for your machines, and that the port is defined the same for both the real-time server and the RT client setup.
5. Check that the folder paths match your installation layout.
6. Set the server IP's field to match the server machine that was previously configured.

When listing redundant servers, separate the names with a comma. For study mode or an all-in-one setup, the server IP's field should be set to "localhost", since the network server is running locally.

7. Set the port number to match the "SERVER_PORT" setting in the server's configuration file (NetServerMainConfig.txt).

Note

You can also configure OMS-specific clients to have read-only access to distribution management system data and full access to OMS. For more information, refer to the *Net Dynamics Configuration Editor User Guide*.

4.2.2. Creating Client Shortcuts

Starting the ADMS client is similar to starting a traditional Browser instance; however, additional parameters are passed to webfgviewer.exe.

The syntax for Advanced Presentation Framework and Browser is:

```
<Path>\Presentation Framework\Apf.exe> /NETCONF <Path>\netconf.xml" -EMPSCADA /etb:setting=IDMS  
/roomName:idms_apf /roomType:LocalShared /DataPool:EMS
```

Set the shortcut's Start-In value to the ADMS client's bin directory (for example, D:\Eterra\Distribution\Client\bin\).

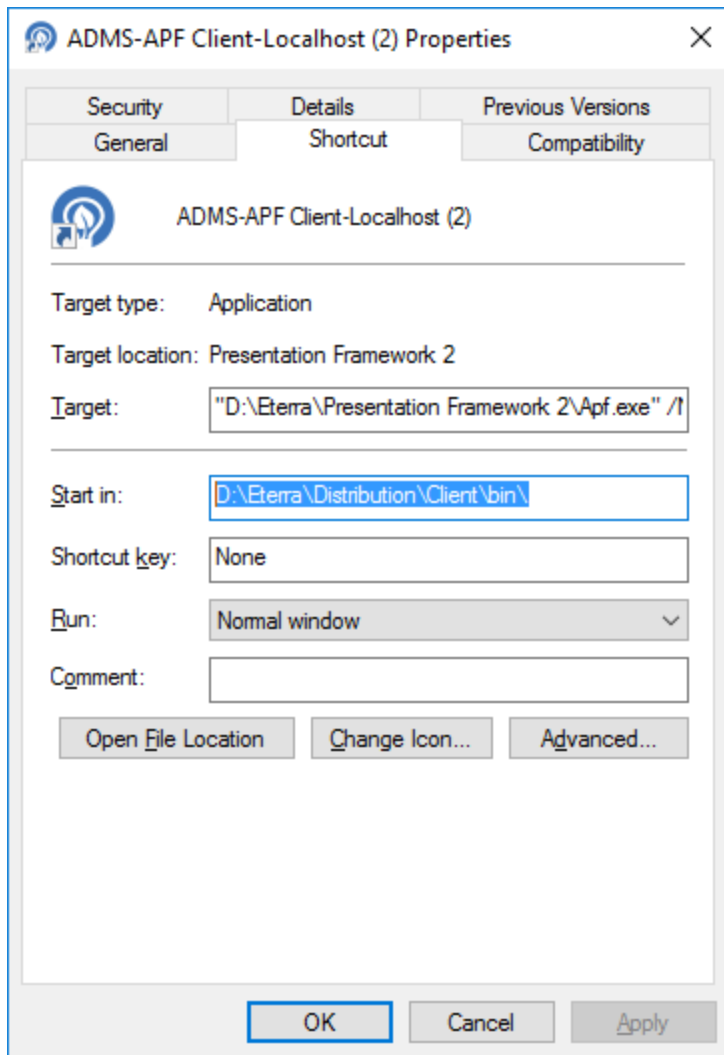


Figure 15. Creating Client Shortcuts

Note

The path for the netconf file can be relative to the shortcut "startin" folder, which must be set to the ADMS client bin directory.

You can also specify the DataPool parameter, which points to the datapool definition as specified in the `DataSourcesConfiguration.xml` APF configuration file. The datapools define the source of the display data (where the Browser can find the Browser Server) and the source of the displays.

You can also specify the client type using the `-CLIENTTYPE "<ClientType>"` parameter. This allows showing different controls and data panels for operators, dispatchers, analysts, and OMS and DNAF users as specified in the client appearance configuration file, `ViewerConfig.xml`.

Additionally, the following troubleshooting parameters can be passed on the command line:

- `-debug`

- -no_exception_handlers
- -DebugMsgSize

The first two flags are passed by default in the delivered shortcut, and it is recommended that they be used during integration and testing activities. The first parameter results in a bit more logging, but it does not have major impact if left on in production.

The third flag, "-DebugMsgSize", creates a "debug" file with information about the sizes of interprocess messages exchanged between the client and the server. This information may be helpful if you need to diagnose network performance issues. The output file is created in the "..\log" (relative to the bin folder) if it exists, or in the .\bin folder otherwise. The file name looks like this:

- Debug_NetClientMsgSizes_RT.log (for real-time messages)
- Debug_NetClientMsgSizes_ST.log (for study messages)
- Debug_NetClientMsgSizes_SIM.log (for simulator messages)

Once your shortcut is ready, start it and try connecting to the server.

4.3. DMS and OMS Tasks

ADMS can be set up with DMS and OMS installations.

4.3.1. DMS Server Production Environment

The following list of tasks is needed for DMS installations:

- CPXNET (CfgProxy for NetSrv)
- CPXDNAF (CfgProxy for DnafSrv)
- CPXREPLICATE (CfgProxy for Net Data Replicator)
- NETPERMSRV (Permission Proxy for NetSrv)
- NETSRV (DMS main server)
- DNAFSRV (DNAF server)
- SWOSERVER (Switching Order server)
- NETDATAREPLICATOR (for replicating Switching Order files)

SWOSERVER does not receive any role assignments from CFGMAN via CfgProxy. Therefore, it should not run in standby mode, but it should be started when the server takes an enabled role using PROCMAN modeling.

4.3.2. DMS and OMS Server Production Environment

The following list of tasks is needed for the DMS and OMS installations:

- CPXNET (CfgProxy for NetSrv)
- CPXOMS (CfgProxy for NetServer)
- CPXDNAF (CfgProxy for DnafSrv)
- CPXNETDATA (CfgProxy for Net Data Server)
- CPXCALL (CfgProxy for CallServer)
- CPXCREW (CfgProxy for CrewServer)
- CPXREPLICATE (CfgProxy for NetDataReplicator)
- NETPERMSRV (Permission Proxy for NetSrv)
- NETSRV (DMS main server)
- NETSERVER (OMS server)
- DNAFSRV (DNAF server)
- SWOSERVER (Switching Order server)
- NETDATAREPLICATOR (Replicates OMS data and Switching Order files)
- NETDATASERVER (For OMS history correction and external interfaces)
- CALLSERVER (Interface server for getting calls and AMI messages)
- RELAYSERVER (Interface server for communicating with Call server, Call Entry application, and external interface applications)
- CREWSERVER (Interface server for getting information about crews)
- ArchiveApp (archives historical data to a relational database)

SWOSERVER does not receive any role assignments from CFGMAN via CfgProxy. Therefore, it should not run in standby mode, but it should be started when the server takes an enabled role using PROCMAN modeling.

5. Advanced Configuration

This chapter details the advanced configuration, designed to expand the functionality of the ADMS suite to provide network analysis and outage management.

5.1. Distribution Network Analysis Functions (DNAF)

ADMS Distribution Network Analysis Functions (DNAF) run as the DNAF server (DnafSrv) using the `NetServerDnafConfig_localhost.txt` configuration file.

To enable DNAF functionality, the following need to occur:

- The `DnafSrv.exe` executable needs to be run.
- The `NetServerDnafConfig_localhost.txt` configuration file needs to be revised depending on project needs.

For more information about configuring server parameters, see [Configuring the DMS Main Server](#). For more information about parameters specific to the DNAF server, see [NetServerDnafConfig.txt](#).

Of particular note are the following parameters in `NetServerDnafConfig.txt`:

- **MAIN_SERVER_HOST:** Specifies the host(s) where the production NetSrv process (DMS Main server) is running.
- **MAIN_SERVER_PORT:** Specifies the TCP/IP listening port for the production system's NetSrv process. This setting must match the `SERVER_PORT` setting in the `NetServerMainConfig.txt` file.
- **SERVER_PORT:** Specifies the TCP/IP listening port number for the DnafSrv process.

For example:

```
MAIN_SERVER_PORT = 5025
MAIN_SERVER_HOST = 127.0.0.1
SERVER_PORT = 5026
```

For more information about DNAF, refer to the following:

- *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide:* A guide for analysts and administrators about configuring the standard Distribution Network Analysis Functions.
- *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide:* A guide for analysts and administrators about configuring the advanced Distribution Network Analysis Functions.
- *Distribution Network Analysis Functions (DNAF) User Guide:* A guide for users, analysts, and administrators about using the Distribution Network Analysis Functions.

Important

The DNAF_MANUAL_TRIGGER parameter must be set to 1 in the main server configuration file, NetServerMainConfig.txt; all other triggers must be set to 0. This redirects manual requests from the main server to the DNAF server.

5.1.1. LVM System Monitor

The Load and Volt/VAR Management (LVM) System Monitor allows users to easily monitor and control LVM execution across the entire system. It also provides the ability to evaluate potential changes to the system from implementing LVM plans, and then easily implement those changes.

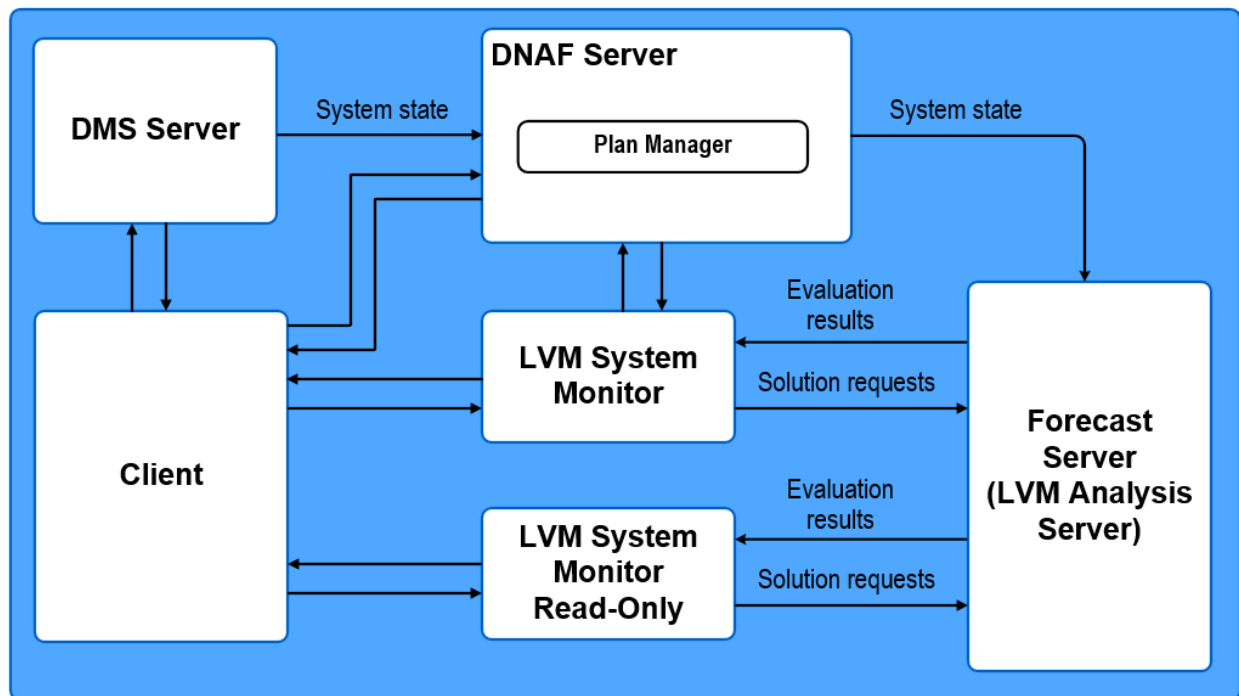


Figure 16. LVM System Monitor Flow

The LVM System Monitor is implemented as a plugin that orchestrates execution and reporting among a number of components. The LVM System Monitor plugin runs on top of the DNAF server and does the core decision making for the LVM System Monitor, including: monitoring the current system state from the DNAF server; requesting LVM solutions as needed from the Forecast server (LVM Analysis server); triggering system-wide closed-loop implementations of LVM plans and making bulk changes to the current LVM problem formulation; and providing data to the user interface so users can monitor the current state of the system and monitor implementation progress, if triggered.

To use the LVM System Monitor, set up the system as follows:

1. Edit NetServerDnafConfig.txt to configure the DNAF server.
2. Set the APP_LVMMONITOR_ENABLED parameter to 1 to enable the LVM System Monitor application.

Alternatively, set the APP_LVMREADONLYMONITOR_ENABLED parameter to 1 to enable the read-only version of the LVM System Monitor application, where you can independently generate and study forecast scenarios, but cannot implement them.

3. Edit `NetServerForecastConfig_localhost.txt` to configure the Forecast server (LVM Analysis server).

5.1.2. Forecast Adapter

This section describes how to configure the Forecast Adapter, which receives dynamic load schedules from the customer's Forecast Engine.

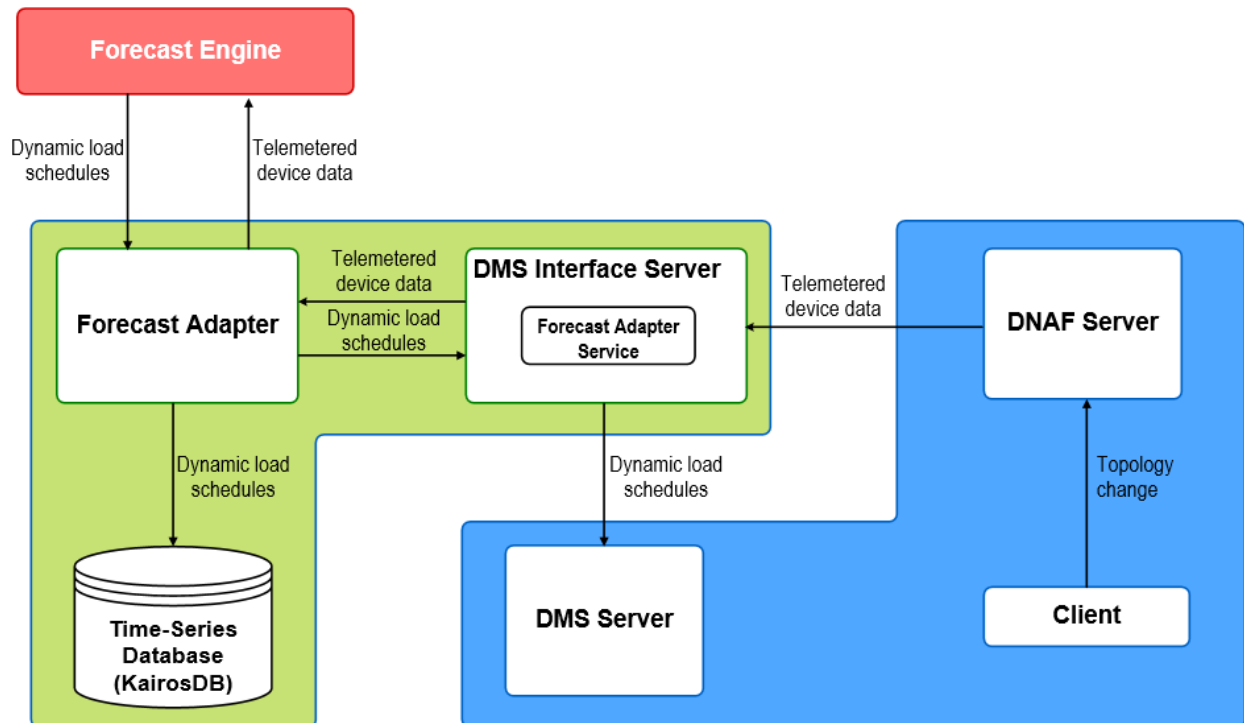


Figure 17. Forecast Adapter Flow

Figure above shows an example of the Forecast Adapter process:

1. A user disconnects a switch.
2. The DNAF server calculates DPF for the switch's substation based on the topology change trigger.
3. The DMS Interface server sends a SwitchStatus notification to the Forecast Adapter.
4. The Forecast Adapter sends a GET call to `http://<DMS Interface server Host:Port>/dms/services/dtl/<Substation>` and receives telemetered device data, which includes device ID, substation, and information for downstream loads.
5. The Forecast Adapter sends telemetered device data to a third-party Forecast Engine, which provides load forecasts.

6. The Forecast Engine sends dynamic schedules for listed loads to the Forecast Adapter.
7. The Forecast Adapter sends dynamic load schedules to a time-series database (KairosDB).
8. The Forecast Adapter sends dynamic load schedules to the DMS Interface server.
9. The DMS Interface server sends dynamic load schedules to the DMS server.

To use the Forecast Adapter, set up the system as follows:

- Deploy the following Docker images: forecast-adapter-service.tar, cassandra.tar, registry.gear.ge.com-pwr-derm-derm-kairosdb-1.0.0.tar, and registry.gear.ge.com-pwr-derm-forecast-timeseries-service-1.0.0.tar.
- Open NetServerDnafConfig.txt to configure the DNAF server.
 - **DYNAMIC_SCHEDULE_PERIOD = 60:** Time period in minutes to create dynamic schedule points.
 - **DYNAMIC_SCHEDULE_QUERY_ENABLED = 1:** Enable dynamic schedule data querying from the main server to synchronize the cached dynamic schedule data with the main server.
- Open DmsInterfaceServiceConfig.xml to configure the DMS Interface server.

For the ForecastAdapter application, you also need to specify a trusted endpoint for the Study Server Adapter in DmsInterfaceServiceConfig.xml to receive DnafStatus notifications. For example:

```
<EnableFAService>true</EnableFAService>
<LogLevel>5</LogLevel>
<TrustedClientList>
<TrustedClient>
<Name>ForecastAdapter</Name>
<Key></Key>
<NotificationBaseURL>http://localhost:8078/api/v1/dms/</NotificationBaseURL>
<NotificationTypes>SWITCH</NotificationTypes>
</TrustedClient>
</TrustedClientList>
<TrustedClientList>
</TrustedClientList>
```

5.1.3. Integration with the 2030.5 Gateway

The 2030.5 Gateway external application supports managing manually-controlled Distributed Energy Resources (DERs). The ADMS system supports communicating with the 2030.5 Gateway using the DERM Kafka Adapter. Both approaches support similar functionality, but use different components.

When integrated with the 2030.5 Gateway, you can send DER controls (LVM plans with manual DER controls or separate controls issued using the DER Manual Dispatch display) from ADMS to the 2030.5 Gateway for implementation.

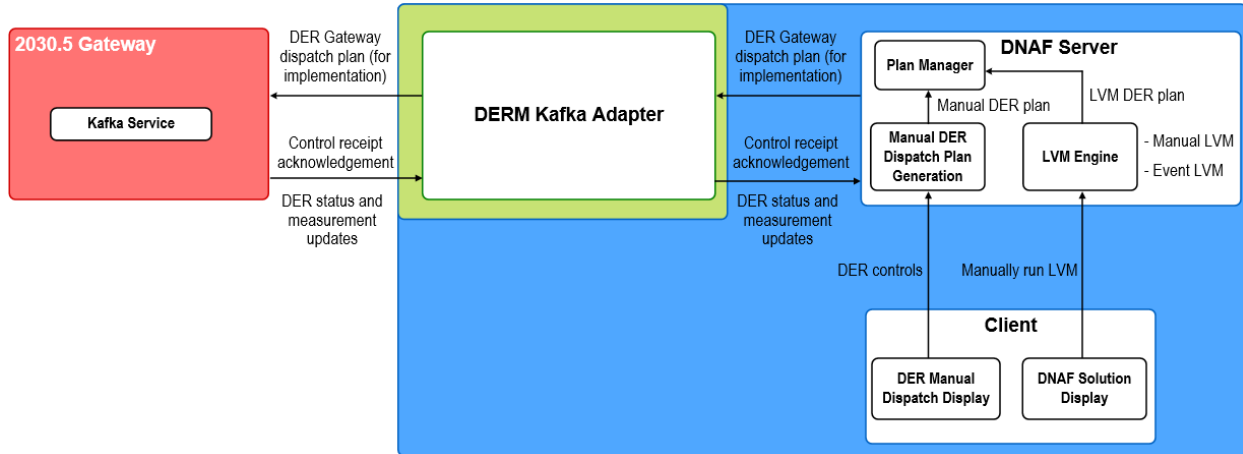


Figure 18. 2030.5 Gateway Process with DERM Kafka Adapter

Figure above shows an example of the process that uses the DERM Kafka Adapter for exchanging DER data between ADMS and the 2030.5 Gateway:

1. DER dispatch plans are generated by LVM or by a user:
 - DER Dispatch plans with control steps for SCADA or gateway-enabled DERs (controllable by the 2030.5 Gateway) are generated as part of an LVM control solution, similar to bank-level plans. (Only PF and kVAR control steps are supported.)
 - A user can issue Manual DER Dispatch plans for DERs using the DER Manual Dispatch display. (PF, kW, kVAR, and Curve control steps are supported.)

Both LVM and user-issued DER dispatch plans appear in the "Advanced Application Plan Status System View" display.

2. Plan Manager receives DER dispatch plans and adds control steps to the implementation list. This list is used for the DERM Kafka Adapter query.

Note

Only one DER dispatch plan (manual or automatic) for a station can be prepared for the DERM Kafka Adapter query. If a newer plan appears for a station, this plan overrides the previous plan.

3. The DERM Kafka Adapter queries the DNAS server for DER dispatch plans (and their control steps lists). Once the control steps are sent to the DERM Kafka Adapter, they are cleared in the DNAS server.
4. The DERM Kafka Adapter receives DER dispatch plans from the DNAS server and formats the data for sending to the 2030.5 Gateway.
5. The DERM Kafka Adapter sends DER dispatch plans to the Kafka Service of the 2030.5 Gateway for implementation.
6. The Kafka Service of the 2030.5 Gateway receives DER dispatch plans and forwards them to the 2030.5 Gateway services for implementation. (The 2030.5 Gateway communicates with third-

party aggregators that implement the recommended controls.)

7. The 2030.5 Gateway sends a control-response status back to the DERM Kafka Adapter, when the 2030.5 Gateway receives a status update from the aggregator (Acknowledged, Received, Rejected, and so on) using the Kafka Service.
8. The DERM Kafka Adapter sends DER updates to the DNAF server.
9. The DNAF server implements DER updates, including measurement and status updates.

The energy resource objects in DNOM are updated with new measurements (status, analog, and alarm) and control statuses. The updated data appears in energy resource Analyst Attribute displays.

To integrate with the 2030.5 Gateway, configure the system as follows:

1. Open `NetServerMainConfig.txt` to configure the DMS server. Configure the `FD_DER_MEASUREMENT_PERIOD` parameter, which defines the time period (in seconds) for querying the DER measurement data.
2. Open `NetServerDnafConfig.txt` to configure the DNAF server.
 - Set the `FD_DER_MEASUREMENT_QUERY_ENABLED` parameter to 1 to enable the DER data query.
3. Open `netconf.xml` to configure the ADMS client.
 - Set the `StateOfChargeQueryEnabled` parameter to true to enable the energy resource state of charge data query.
4. When integrating using the DERM Kafka Adapter, configure `netconf_derm_adapter.xml` as follows:
 - Specify the DNAF server host and port.
5. When integrating using the DERM Kafka Adapter, open `DermKafkaAdapterConfig.xml` to configure the DERM Kafka Adapter.
 - Set the `NetConfigFileName` parameter to the path to the `netconf_derm_adapter.xml` file.
 - Set `BootstrapServers` to the comma-separated list of Kafka Service endpoints (host and port pairs) in the 2030.5 Gateway. If port numbers are not specified, 9092 is used by default.
 - Set topic names in the `DerKafkaTopics` section to match the topic names in the Kafka Service.
 - Set `GroupId` to the Kafka Service group ID.
 - Specify whether a secure connection is used using the `UseSecurityProtocol` parameter.
 - If a secure connection is used, specify the location where the certificate authority is stored using the `CertificateAuthorityLocation` parameter. The format is `"..\\..\\CertFileName"`.

5.1.4. Distribution Network Applications Server

The Distribution Network Applications (DNA) server supports a modularized architecture for

distribution network applications. The new modularized applications are called DNA (Distribution Network Applications) rather than the existing DNAF (Distribution Network Application Functions).

The DNA server is designed to unload the DNAF server. The DNA server has a small memory footprint: it does not load a complete system model and keep it in sync using dynamics. Instead, it runs individual analyses on demand and only loads enough of the system to support that analysis.

The DNA server works as follows:

1. A request to run a network application appears.
2. The DNAF server gathers the current dynamics for the required stations and sends a message to the DNA server to run the analysis.
3. The DNA server runs the analysis and sends a message back to the DNAF server with the results.
4. The DNA server unloads system data and waits for another request.

The Distribution Network Applications (DNA) server runs using the `NetServerDnaconfig_localhost.txt` configuration file.

To enable DNA server functionality, the following need to occur:

- The `DnaSrv.exe` executable needs to be run.
- The `NetServerDnaConfig_localhost.txt` configuration file needs to be revised. For more information, see [Configuring the DMS Main Server](#).

Depending on project needs, configure the DNA server parameters listed in [NetServerDnaConfig.txt](#).

5.1.5. DNA Scheduler

The DNA Scheduler service schedules the analysis requests from the DNAF server to an available (idle) DNA server. The DNA Scheduler works together with the DNAF server and one or multiple DNA servers to distribute the analysis workload across the DNA servers. It receives request messages from the DNAF server and forwards the messages one by one to idle DNA servers.

This service has a configuration file named `DnaSchedulerConfig.xml`. The parameters in this file are described in [DnaSchedulerConfig.xml](#).

5.2. Network Outage Management System (OMS)

This section describes advanced configurations for the ADMS Outage Management System.

5.2.1. Configuring the Network Server for Outage Management

The outage management engine runs separately as NetServer along with the DMS real-time server using the NetServerOmsConfig.txt configuration file.

To enable OMS functionality, the following need to occur:

- The NetServer.exe executable needs to be run.
- The NetServerOmsConfig_localhost.txt configuration file needs to be revised.

For more information about configuring server parameters, see [Configuring the DMS Main Server](#). For more information about parameters specific to the OMS server, see [NetServerOmsConfig.txt](#).

Of particular note are the following parameters in NetServerOmsConfig.txt:

- **MAIN_SERVER_HOST:** Specifies the host(s) where the production NetSrv process (DMS Main server) is running.
- **MAIN_SERVER_PORT:** Specifies the TCP/IP listening port for the production system's NetSrv process. This setting must match the SERVER_PORT setting in the NetServerMainConfig.txt file.
- **SERVER_PORT:** Specifies the TCP/IP listening port number for the NetServer process.

For example:

```
MAIN_SERVER_PORT = 5025
MAIN_SERVER_HOST = 127.0.0.1
SERVER_PORT = 5060
```

5.2.1.1. Configuring the Multiple Prediction Rule Sets

A specific list of feeders can be assigned their own prediction rule sets. These feeder-specific prediction rule sets take precedent over the global rule sets.

Configuration Parameter:

```
MULTIPLE_PREDICTION_RULE_SETS_FILE = ..\..\PredictionRuleSets.csv
```

- **MULTIPLE_PREDICTION_RULE_SETS_FILE (optional):** By default, this option is commented out, and the system will use the global prediction rules.
- When uncommented, the system uses feeder-specific prediction rule sets defined in the PredictionRuleSets.csv file. Refer to the sample rule sets included in the FantasyIsland installation kit for guidance.
- If the specified rule set file is not found, the system falls back to using the global prediction rules defined in NetServerOmsConfig_localhost.txt.
-

If the prediction device real time feeder can't be found in the file, the system falls back to using the global prediction rules defined in NetServerOmsConfig_localhost.txt.

- StationFuse type network protective device always uses global prediction parameters.
- In the feeder specific custom prediction rules configuration file, the header names cannot be localizable.
- For an eligible prediction rule if the value of any network protective device parameter (ex: B1, B2 etc.) is not a number, then the associated default global prediction rule will be used.
- For an eligible prediction rule if the value of any network protective device parameter (ex: B1, B2 etc.) is having more than one decimal, then the default global prediction rule will be used.
- If same feeder from same station is observed again in the prediction ruleset configuration file, then the bottom most added ruleset is considered.

5.2.1.2. Configuring the OMS-Specific Path and File Input/Output Parameters

Verify that the following parameters match the path configuration of your server installation:

```
'-----  
'OMS Server Specific Parameters  
'-----  
OMS_CALL_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\Server\OMSServer\CallJournal\  
OMS_ARCHIVE_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\Server\OMSServer\ArchiveJournal\  
OMS_INCIDENT_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\Server\OMSServer\IncidentJournal\  
OMS_CREW_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\server\OMSServer\CrewJournal\  
OMS_ORDER_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\server\OMSServer\OrderJournal\  
OMS_SNAPSHOT_FILE_PATH = %IDMS_ROOT%\dynamics\Server\OMSServer\Snapshot\  
OMS_DEENERGIZED_LINES_FILE_PATH = %IDMS_ROOT%\dynamics\server\OMSServer\DeenergizedLines\  
OMS_MANAGED_CONFIGFILE = %IDMS_ROOT%\config\server\OmsServerConfig.xml  
OMS_CODED_TRANSLATION_CONFIGFILE = %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml  
'-----  
'OMS Data Files  
'-----  
OMS_POLYGON_GROUPS = %IDMS_ROOT%\models\RT\oparea_polygongroup.xml  
OMS_CUSTOMER_FILE = %IDMS_ROOT%\models\customer\FantasyIslandCustomers.bpcl  
OMS_CREW_EMPLOYEE_FILE = %IDMS_ROOT%\models\crew\FantasyIslandCrewModel.Crew
```

Note

The "OMS Data Files" parameters are files that feed geographic, customer, and crew workforce data for outage management functionality. These parameters should be generated from GIS, CIS, and WFM systems.

5.2.1.3. Configuring the Server Connections to External Interface Applications

The ADMS server applications initiate connection requests on specific TCP/IP ports to the external interface applications. The Call server and Crew server manage communications with enterprise CIS/IVR/AMI and WFM systems, respectively. More detail on integrating these solutions can be found in [Configuring the External Interface Server Applications](#).

```
'Call Server
CALLSERVER_PORT = 5032
CALLSERVER_HOST = localhost
'Crew Server
CREWSERVER_PORT = 5036
CREWSERVER_HOST = localhost
```

The ports and host names are defined in the MAIN mode configuration file using the CALLSERVER_PORT, CALLSERVER_HOST, CREWSERVER_PORT, and CREWSERVER_HOST parameters. To connect to the external interface applications, these parameters need to match the listening parameters for the Call server and Crew server defined below. Either leave the port and host name settings as delivered or change them to fit your project's needs.

5.2.2. Configuring the External Interface Server Applications

Within the ADMS suite, two applications manage the interfaces with CIS, IVR, AMI, and WFM. The following sections describe the configuration of these applications.

In addition to the components that are considered part of ADMS, ADMS also requires integration with a number of external services:

- Workforce Management System (WFM)
- CSS (or CIS: Customer Information System)
- External Interactive Voice Response (IVR)
- Automated Metering Infrastructure (AMI)
- Geographic Information System (GIS)

The interface to the GIS is an extraction process that provides model information about the distribution facilities and land-based information to the ADMS model management system.

Within ADMS, two applications manage the interfaces with the workforce management system, CSS, IVR, and AMI:

- The Crew server (workforce management interface)
- The Call server (CSS, IVR, and AMI interfaces)

Figure below shows the data flow and interaction between the servers and the external services.

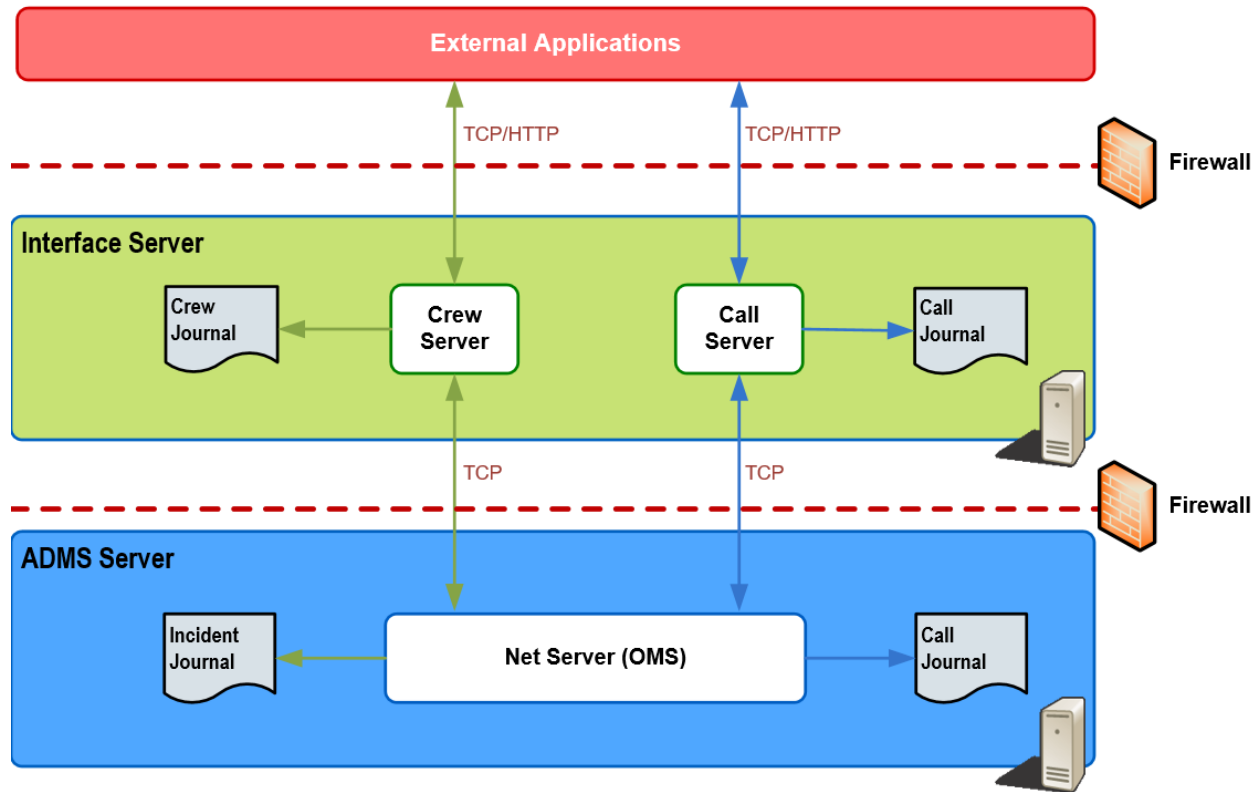


Figure 19. Interface with External Services

The interface between the ADMS channel adapter and the Call server and Crew server is through TCP/IP or HTTP. Each message is defined by an XSD schema. The message structure has five parts:

- A message header that describes the size and framing of the message.
- A security token that is based on the next three bullet items in addition to a secret shared key.

The security token is created by using the SHA256 hashing function, which generates a 256bit digital signature based on the XML message payload, user SID, current UTC time, and secret shared key.

- The XML message payload that the application uses to pass relevant information.
- A user SID.
- The time of the message, expressed in UTC format.

5.2.2.1. Enabling Secure Token Messaging for Crew, Relay, and Call Servers

After the server installation, the optional secure token messaging option can be enabled for the Crew server, Relay server, and Call server tasks. The default path to the task credential and configuration files is %IDMS_ROOT%\config\server.

To enable secure token messaging using registered SIDs and credential files between tasks:

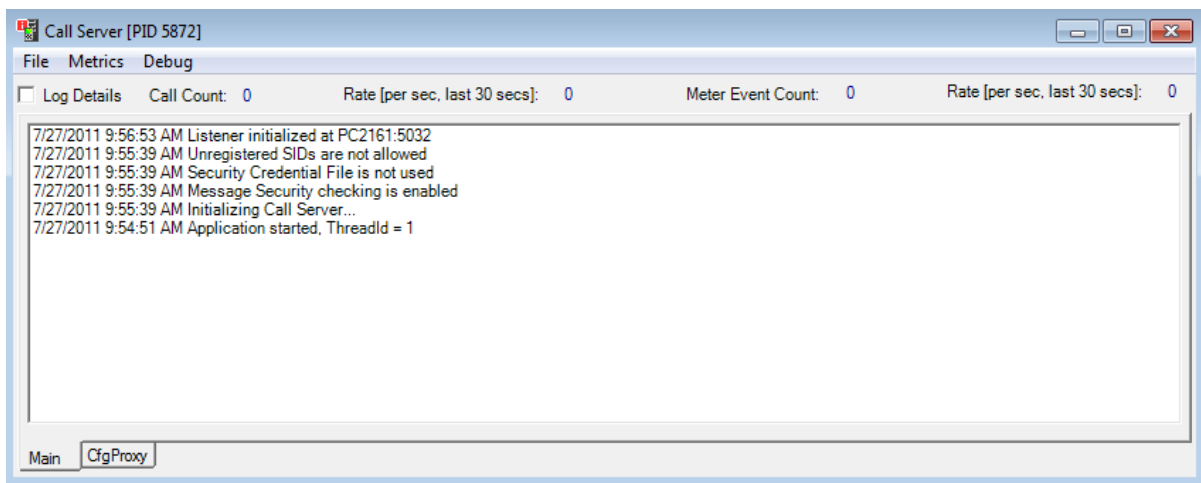
1. Edit the following settings in the each of the task configuration files: `CallServerConfig.xml`, `CrewServerConfig.xml`, and `RelayServerConfig.xml`.

```
<CheckMessageSecurity>true</CheckMessageSecurity>
<CheckMessageSequence>true</CheckMessageSequence>
<UseCredentialFile>true</UseCredentialFile>
<AllowUnregisteredSids>false</AllowUnregisteredSids>
```

Warning

XML files are case-sensitive. Use all lower-case letters when changing the Boolean settings.

2. Start one of the server tasks (for example, the Call server).



3. When the selected server task starts, on the File menu, select Credentials.

The IDDS Credentials display appears.

IDDs Credentials

Current File: D:\Eterra\Distribution\FantasyIsland\config\server\CallServerCi

Valid SIDs: S-1-5-21-1085783642-1965037168-3936496997

Shared Key: *****

Current SID: S-1-5-21-1085783642-1965037168-3936496997

Open Save Save As Close

4. Enter any sequence of characters into the Shared Key field to overwrite the original asterisks. Retain the sequence of shared key characters; it is entered into the IDDS Credentials interface for the remaining server tasks.

Note

The Shared Key and Valid SIDs must be the same.

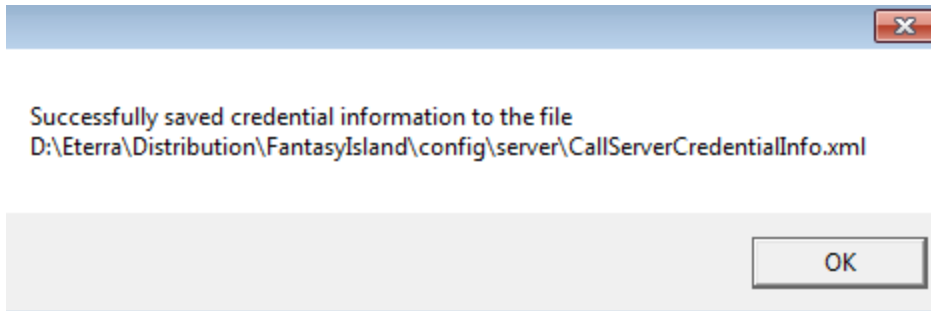
5. Click Save.

The "Do you want to save credential information to the file?" dialog box appears.

6. Click Yes.

The Windows File Browser may open. If so, navigate to the %IDMS_ROOT%\Config\Server folder and select the appropriate credential information file for the server task.

The "Successfully saved credential information to the file" message dialog box appears and displays the full path to the credential file location.



Note

Credential files cannot be copied from one host to another. Create a credential file uniquely on each host where any of the processes run using the `CredentialEditor.exe`.

7. Repeat the same process for the remaining server tasks: Crew server and Relay server. If the Crew Simulator is being used, enable the use of the credential file in the `CrewSimulatorConfig.xml` file and repeat the Credential/Shared Key process for the Crew Simulator.

Note

The secure token messaging process is enabled immediately without restarting the tasks.

Note

All external applications communicating with the ADMS Call server, Relay server, and Crew server tasks need to use secure token messaging by using the same shared key credential information.

If the message validation fails:

1. Set the `ConnectionDebug` parameter to `True` for Call server or Relay server to enable creating a `ConnectionDebug` file in the log folder.
2. Open the `ConnectionDebug` file and check the rows with the text "Using Security" and the rows above to determine what is causing the message validation to fail.

5.2.2.2. Call Server Application (CIS, IVR, and AMI Interfaces)

Call server relays calls from a CIS, IVR, or AMI (a "call server"), or relays calls from a call server (a "relay server") to another call consumer (call server, relay server, or NetServer). It supports the CIS, IVR, and AMI interfaces.

To configure the Call server, revise parameters in `CallServerConfig.xml`. For more details, see [CallServerConfig.xml](#).

Of particular note are the following parameters in `CallServerConfig.xml`:

- **Relay:** Set to true for Relay server or the AMI server. For Call server this parameter must be false.
- **ListeningPort:** TCP/IP port that the application is listening to for incoming messages. Must match CALLSERVER_PORT in the NetServerOmsConfig_localhost.txt file.
- **OmsServerHost:** Hosts where the server will be listening for dual redundant tasks.
- **OmsServerPort:** TCP/IP port used to connect to OMS servers for dual-redundant tasks.
- **CustomerFile:** Relative location (path) and name of the customer file.

5.2.2.3. Relay Server

To configure the Relay server, revise parameters in [RelayServerConfig.xml](#). For more details, see [RelayServerConfig.xml](#).

Of particular note are the following parameters in [RelayServerConfig.xml](#):

- **Relay:** Set to true for Relay server or the AMI server. For Call server this parameter must be false.
- **ListeningPort:** TCP/IP port that the application is listening to for incoming messages.
- **CallServerHost:** Host machine name(s) of the Call server(s).
- **CallServerPort:** TCP/IP listening port number of the Call server. Must match ListeningPort in [CallServerConfig.xml](#).
- **CustomerFile:** Relative location (path) and name of the customer file.

5.2.2.4. AMI Server

To configure the AMI server, revise parameters in [AmiServerConfig.xml](#). For more details, see [AmiServerConfig.xml](#).

Of particular note are the following parameters in [AmiServerConfig.xml](#):

- **Relay:** Set to true for Relay server or the AMI server. For Call server this parameter must be false.
- **CallServerPort:** TCP/IP listening port number of the Call server. Must match ListeningPort in [CallServerConfig.xml](#).
- **CallServerHost:** Host machine name(s) of the Call server(s).
- **CustomerFile:** Relative location (path) and name of the customer file.

5.2.2.5. Crew Server Application (WFM Interface)

Crew server transfers crew and incident information between NetServer and a workforce management system. It implements the workforce management interface to the channel adapter.

To configure the Crew server, revise the parameters in `CrewServerConfig.xml`. For more details, see [CrewServerConfig.xml](#).

Of particular note are the following parameters in `CallServerConfig.xml`:

- **ListeningPort:** TCP/IP port that the application is listening for incoming messages. Must match `CREWSERVER_PORT` in the `NetServerOmsConfig_localhost.txt` file.
- **OmsServerHost:** Hosts where the server will be listening for dual redundant tasks.
- **OmsServerPort:** TCP/IP port used to connect to OMS servers for dal-redundant tasks.

5.2.3. Starting the Net Server

In a similar manner as shown in [Starting the Server](#), it is recommended to start the server manually with the newly configured parameters to ensure that the server operates properly before it is configured inside of PROCMAN.

Note

You do not need to run `NetServer.exe` from a command prompt. You can use the Windows shortcut to start the graphical interface.

Whether the shortcut (method1) or the batch script (method2) was used to start the server, update the target to be `NetServer.exe` instead of `netsrv.exe` and set the configuration file to `NetServerOmsConfig.txt`.

The ADMS client requires extra configuration to connect to the OMS server (NetServer) to leverage Outage Management System functions. You must configure the client connection to the NetServer in the OMS section on the RT tab in the Net Dynamics Configuration Editor. For more details, see [Configuring the Client Connection to the DMS Main Server](#).

After the configuration is completed, close any existing running clients, and reuse the same shortcut created in [Creating Client Shortcuts](#) to start the client, to confirm that connectivity is still functional.

5.2.4. Setting Current Locale

Important

Skip this section for English-based systems.

To use localized client resources and show the time and date in local format on client displays, you must set the locale for the client using the `/CULTURE` command line parameter. The predefined CULTURE names are listed for your reference at the following location:

<https://msdn.microsoft.com/en-us/library/cc233982.aspx>

In addition, use the /LANGUAGE parameter to specify what resource strings to use in the APF UI. Use the culture code for the language to use (for example, "fr" = French). Matching satellite resource DLLs must exist in the APF installation folder.

The example below demonstrates the command line syntax for setting a French locale:

```
"D:\Eterra\Presentation Framework\Apf.exe" /NETCONF "%IDMS_ROOT%\config\client\netconf_localhost.xml"
-EMPSCADA -no_exception_handlers /etb:setting=IDMS /roomName:idms_apf /roomType:LocalShared /CULTURE
fr-FR /LANGUAGE:fr
```

Right-to-Left Culture

For Windows locales with the right-to-left (RTL) culture, all display menus and input fields have right-to-left cursor direction.

You can enable the RTL mirroring for client displays using the /RIGHTTOLEFTLAYOUT client command line parameter.

The example below demonstrates the command line syntax to be added to the Target field of the client shortcut for using the client display mirroring:

```
"C:\Program Files\Eterra\e_terrabrowser\WebFGViewer.exe"
"%WEBFGROOT%\WebFGCache\config\IDMS_localhost_standalone_client_config.txt" /NETCONF
"%IDMS_ROOT%\config\client\netconf_localhost.xml" -EMPSCADA -debug -no_exception_handlers /CULTURE he-
IL /RIGHTTOLEFTLAYOUT
```

In addition, you can toggle the text entry mode (left-to-right or right-to-left) for text fields using the Left Ctrl + Left Shift or Right Ctrl + Right Shift keyboard combinations, correspondingly.

5.2.5. Setting Prediction Rules

The OMS can predict which protective device (network protector, distribution transformer, fuse, sectionalizer, recloser, breaker, or station fuse) might be the cause of trouble, based on calls and AMI events from downstream customers. All protective devices are eligible for prediction unless the Block Prediction flag is set for the device.

Customers and downstream devices are either directly connected to a protective device or cascaded. For example, a recloser may have a directly connected fuse that has distribution transformers connected downstream. The distribution transformers are directly connected to the fuse and cascaded from the recloser. A recloser might also have directly connected distribution transformers.

5.2.5.1. Prediction Rules

Each type of protective device has its own set of prediction rules which require input values to be entered in the NetServer configuration file. The input values from the configuration file are used in the prediction rule equations to vary the behavior of the prediction engine.

A rule set defines the basic behavior of the prediction engine. Each protective device type has four available rule sets, and the value of the <protective device>_PREDICTION_RULE_SET parameter determines which algorithm will be used.

5.2.5.1.1. Network Protector Rule Sets

Note

For the purpose of prediction, a call is either a customer call of an outage type as defined in the Call Action Matrix or an AMI event.

Available network protector rule sets:

- NETPROTECT_PREDICTION_RULE_SET = 1
Uses the absolute number of calls.
- NETPROTECT_PREDICTION_RULE_SET = 2
Uses the percentage of calls.
- NETPROTECT_PREDICTION_RULE_SET = 3
Uses the absolute number of calls *or* the percentage of calls.
- NETPROTECT_PREDICTION_RULE_SET = 4
Uses the absolute number of calls *and* the percentage of calls.

5.2.5.1.2. Distribution Transformer Rule Sets

Note

For the purpose of prediction, a call is either a customer call of an outage type as defined in the Call Action Matrix or an AMI event.

Available distribution transformer rule sets:

- DISTTF_PREDICTION_RULE_SET = 1
Uses the absolute number of calls.
- DISTTF_PREDICTION_RULE_SET = 2
Uses the percentage of calls.
- DISTTF_PREDICTION_RULE_SET = 3
Uses the absolute number of calls *or* the percentage of calls.
- DISTTF_PREDICTION_RULE_SET = 4

Uses the absolute number of calls *and* the percentage of calls.

5.2.5.1.3. Fuse Rule Sets

Note

For the purpose of prediction, distribution transformers and fuses are the downstream protective devices considered for fuses.

Available fuse rule sets:

- FUSE_PREDICTION_RULE_SET = 1
Uses the absolute number of downstream predicted out protective devices and customer calls.
- FUSE_PREDICTION_RULE_SET = 2
Uses the percentage of downstream predicted out protective devices and customer calls.
- FUSE_PREDICTION_RULE_SET = 3
Uses the absolute number *or* percentage of downstream predicted out devices and customer calls.
- FUSE_PREDICTION_RULE_SET = 4
Uses the absolute number *and* the percentage of downstream predicted out devices and customer calls.

5.2.5.1.4. Sectionalizer Rule Sets

Note

For the purpose of prediction, fuses and distribution transformers are the downstream protective devices considered for sectionalizers.

Available sectionalizer rule sets:

- SECTIONALIZER_PREDICTION_RULE_SET = 1
Uses the absolute number of downstream predicted out protective devices and customer calls.
- SECTIONALIZER_PREDICTION_RULE_SET = 2
Uses the percentage of downstream predicted out protective devices and customer calls.
- SECTIONALIZER_PREDICTION_RULE_SET = 3
Uses the absolute number *or* the percentage of downstream predicted out devices and customer calls.

- SECTIONALIZER_PREDICTION_RULE_SET = 4

Uses the absolute number *and* the percentage of downstream predicted out devices and customer calls.

5.2.5.1.5. Recloser Rule Sets

Note

For the purpose of prediction, reclosers, sectionalizers, fuses, and distribution transformers are the downstream protective devices considered for reclosers.

Available recloser rule sets:

- RECLOSER_PREDICTION_RULE_SET = 1

Uses the absolute number of downstream predicted out devices and customer calls.

- RECLOSER_PREDICTION_RULE_SET = 2

Uses the percentage of downstream predicted out devices and customer calls.

- RECLOSER_PREDICTION_RULE_SET = 3

Uses the absolute number *or* the percentage of downstream predicted out devices and customer calls.

- RECLOSER_PREDICTION_RULE_SET = 4

Uses the absolute number *and* the percentage of downstream predicted out devices and customer calls.

5.2.5.1.6. Breaker Rule Sets

Note

For the purpose of prediction, reclosers, sectionalizers, fuses, and distribution transformers are the downstream protective devices considered for breakers.

Available breaker rule sets:

- BREAKER_PREDICTION_RULE_SET = 1

Uses the absolute number of downstream predicted out protective devices and customer calls.

- BREAKER_PREDICTION_RULE_SET = 2

Uses the percentage of downstream predicted out protective devices and customer calls.

- BREAKER_PREDICTION_RULE_SET = 3

Uses the absolute number OR the percentage of downstream predicted out devices and customer calls.

- BREAKER_PREDICTION_RULE_SET = 4

Uses the absolute number AND the percentage of downstream predicted out devices and customer calls.

5.2.5.1.7. Station Fuse Rule Sets

Note

For the purpose of prediction, breakers, reclosers, sectionalizers, fuses, and distribution transformers are the downstream protective devices considered for station fuses.

Available rule sets for fuses located in the station internals:

- STATIONFUSE_PREDICTION_RULE_SET = 1

Uses the absolute number of downstream predicted out protective devices and customer calls.

- STATIONFUSE_PREDICTION_RULE_SET = 2

Uses the percentage of downstream predicted out protective devices and customer calls.

- STATIONFUSE_PREDICTION_RULE_SET = 3

Uses the absolute number OR the percentage of downstream predicted out devices and customer calls.

- STATIONFUSE_PREDICTION_RULE_SET = 4

Uses the absolute number AND the percentage of downstream predicted out devices and customer calls.

5.2.5.1.8. Special Prediction Out Cases

Some special prediction-out cases are described below:

1. If all direct downstream devices are predicted out, the device will always be predicted out.

Example: A fuse has one distribution transformer directly connected, which has only one directly connected customer. When the customer calls, the fuse will be predicted out, regardless of the rule set.

2. If a device has more than one directly connected downstream device or customer, a minimum of two devices or customers must be predicted out before the upstream device is predicted out, regardless of the rule set.

Example: A transformer has three customers connected. The prediction rules are set such that

0.75 customers must call to predict out the transformer. One customer calls and a single customer incident is created after the simmer timer has expired. When a second customer connected to the distribution transformer calls in, the device will predict out.

3. If a device has more than one directly connected downstream device, the device will not be predicted out if all the prediction calls are from only one predicted out directly connected downstream device.

Example: Five calls are received under one fuse, which has three directly connected distribution transformers. All five calls are from customers connected to a single distribution transformer. The fuse will not be predicted out regardless of the rule set; only the transformer will be predicted out (if the number of calls warrants the prediction for the distribution transformer rule set).

Example: A recloser has two directly connected downstream fuses and all calls are from customers connected under only one of the fuses. Only the fuse will predict out (if the number of calls warrants the prediction for the fuse rule set); the recloser will not be predicted out regardless of the rule set.

5.2.5.2. Protective Device Parameter Sets

Each type of protective device has a set of weight parameters named B1-B5 and a set of threshold value parameters named T1-T5 plus T2AMI that can be specified by the system administrator in the NetServer configuration file. Each of these parameters influences the prediction engine behavior in conjunction with the prediction rule set selected for the device type.

Typically, weight values B1, B2, B4, and B5 are zero or one, but they can be set to any higher positive integer value to give more weight to components of the prediction calculation. B3 is not currently used and does not need to exist in the configuration file.

Threshold values T1, T3, and T5 should be integer or float values, typically in the range of 2-10. Threshold values T2, T2AMI, and T4 are percentages expressed as decimals (for example, T4 = 0.14 sets the T4 value to 14%). These parameters can be adjusted during prediction by using the REDUCTION parameters.

Impact of the B1-B5 values:

- <protective device type>_PREDICT_B1 is multiplied by the number of calls and AMI events from directly connected customers. Setting the B1 value to 0 removes the absolute number of directly connected calls and AMI events from the prediction calculation.
- Set the B1 value to 1 to enable calls and AMI events from directly connected customers. Set the B1 value to a whole number greater than 1 to give greater significance to the number of directly connected customer calls and AMI events.
- <protective device type>_PREDICT_B2 is multiplied by the number of calls and AMI events from cascaded customers. Setting the B2 value to 0 removes the absolute number of cascaded calls and AMI events from the prediction calculation.
- Set the B2 value to 1 to enable calls and AMI events from cascaded customers. Set the B2 value

to a whole number greater than 1 to give greater significance to the number of cascaded customer calls and AMI events.

- <protective device type> **_PREDICT_B3** is currently unused.
- <protective device type> **_PREDICT_B4** is multiplied by the number of downstream directly connected predicted out protective devices. Setting the B4 value to 0 removes the absolute number of downstream directly connected predicted out protective devices from the prediction calculation.
- Set the B4 value to 1 to enable downstream directly connected predicted out protective devices. Set the B4 value to a whole number greater than 1 to give greater significance to the number of downstream directly connected predicted out protective devices.
- <protective device type> **_PREDICT_B5** is multiplied by the number of downstream cascaded predicted out protective devices. Setting the B5 value to 0 removes the absolute number of cascaded predicted out protective devices from the prediction calculation.
- Set the B5 value to 1 to enable downstream cascaded predicted out protective devices. Set the B5 value to a whole number greater than 1 to give greater significance to the number of downstream cascaded predicted out protective devices.

Impact of the T1-T5 and T2 AMI threshold values:

- <protective device type> **_PREDICT_T1** is the absolute number of customer calls and AMI events. The B1 and B2 parameters can be used to ignore, enable, or differently weight directly connected or cascaded calls and AMI events.
- <protective device type> **_PREDICT_T2** is the percentage of customer calls compared to the total number of customers downstream of the device. The B1 and B2 parameters can be used to ignore, enable, or differently weight directly connected or cascaded calls and AMI events.
- <protective device type> **_PREDICT_T2AMI** is the percentage of AMI meters reporting out compared to the total number of all meters downstream of the device. The B1 and B2 parameters can be used to ignore, enable, or differently weight directly connected or cascaded calls and AMI events.
- <protective device type> **_PREDICT_T3** is the absolute number of downstream protective devices predicted out. The B4 and B5 parameters can be used to ignore, enable, or differently weight directly connected or cascaded devices.
- <protective device type> **_PREDICT_T4** is the percentage of downstream protective devices predicted out compared to the total number of downstream protective devices. The B4 and B5 parameters can be used to ignore, enable, or differently weight directly connected or cascaded devices.
- <protective device type> **_PREDICT_T5** is the sum of calls/AMI events and downstream protective devices predicted out. The B4 and B5 parameters can be used to ignore, enable, or differently weight directly connected or cascaded devices. The B1 and B2 parameters can be used to ignore, enable, or differently weight directly connected or cascaded calls and AMI events.

The above threshold values are applied on a per-phase basis to determine if a given device phase is to be predicted out. For example, if the downstream device count (T3) were set to 2, and a three-

phase fuse had 2 A-phase distribution transformers predicted out, the A phase of the fuse would predict out. If the two downstream predicted devices were on different phases, the fuse would not be predicted out since none of the phases had 2 downstream predicted devices.

When one or more phases of a device is predicted out, the threshold values can be reduced by applying a "reduction" parameter to all of the threshold values. This will cause the per phase prediction to roll up in a more balanced way to upstream devices. For example, consider the above three-phase fuse. When 2 downstream A-phase transformers are predicted out, the A-phase of the fuse will also predict out. With no reduction, it would require 2 downstream B-phase transformers be predicted out for the B-phase of the fuse to also be predicted. However, with a reduction parameter of 0.5 applied to the fuse, the B and C phases of the fuse will require only 1 downstream B or C transformer be predicted out for the fuse B or C phase to be predicted.

- `<protective device type>_PREDICT_REDUCTION` is the scaling factor applied to all threshold values (T1, T2, T3, etc.) once any phase of a device is predicted out. If set to 1.0 (the default), no scaling is done and the threshold values remain as they are set. If set to 0.5 (50%), all threshold values are reduced by half once any phase of a device is predicted out.

5.2.5.3. Rule Set Passes

Each rule set runs two passes for prediction: a call/AMI pass and a device pass.

- Rule set 1:
 - Pass 1: If $(B1 * \text{the number of directly connected calls and AMI events}) + (B2 * \text{the absolute number of cascaded calls and AMI events}) \geq T1$, then predict the device out.
 - Pass 2: If $(B4 * \text{the number of directly connected predicted out downstream protective devices}) + (B5 * \text{the number of cascaded predicted out downstream protective devices}) \geq T3$, then predict the device out.
- Rule set 2:
 - Pass 1: If $B1 * \text{the number of directly connected calls} + (B2 * \text{the absolute number of cascaded calls divided by the number of downstream directly connected + cascaded customers}) \geq T2$ OR if $B1 * \text{the number of directly connected AMI events} + (B2 * \text{the absolute number of cascaded AMI events divided by the number of downstream directly connected + cascaded customers}) \geq T2_{AMI}$, then predict the device out.
 - Pass 2: If $\text{the number of directly connected predicted out downstream devices divided by the number of directly connected protective devices} \geq T4$, then predict the device out.
- Rule set 3:
 - Pass 1: If $(B1 * \text{the number of directly connected calls and AMI events}) + (B2 * \text{the absolute number of cascaded calls and AMI events}) \geq T1$ OR if $B1 * \text{the number of directly connected calls} + (B2 * \text{the absolute number of cascaded calls divided by the number of downstream directly connected + cascaded customers}) \geq T2$ OR if $B1 * \text{the number of directly connected AMI events} + (B2 * \text{the absolute number of cascaded AMI events divided by the number of downstream directly connected + cascaded customers}) \geq T2_{AMI}$, then predict the device out.

- Pass 2: If $(B4 * \text{the number of directly connected predicted out downstream protective devices}) + (B5 * \text{the number of cascaded predicted out downstream protective devices}) \geq T3$ OR If the number of directly connected predicted out downstream protective devices divided by the number of directly connected downstream protective devices $\geq T4$ OR if $(B1 * \text{the number of directly connected calls}) + (B2 * \text{the number of cascaded calls}) + (B4 * \text{the number of directly connected predicted out downstream protective devices}) + (B5 * \text{the number of cascaded predicted out downstream protective devices}) \geq T5$, then predict the device out.
- Rule set 4:
 - Pass 1: If $(B1 * \text{the number of directly connected calls and AMI events}) + (B2 * \text{the absolute number of cascaded calls and AMI events}) \geq T1$ AND if $B1 * \text{the number of directly connected calls}) + (B2 * \text{the absolute number of cascaded calls divided by the number of downstream directly connected + cascaded customers}) \geq T2$ AND if $B1 * \text{the number of directly connected AMI events}) + (B2 * \text{the absolute number of cascaded AMI events divided by the number of downstream directly connected + cascaded customers}) \geq T2_{AMI}$, then predict the device out.
 - Pass 2: If $(B4 * \text{the number of directly connected predicted out downstream protective devices}) + (B5 * \text{the number of cascaded predicted out downstream protective devices}) \geq T3$ AND if the number of directly connected predicted out downstream protective devices divided by the number of directly connected downstream protective devices $\geq T4$, then predict the device out.

5.2.5.4. Examples of Prediction Rule Settings

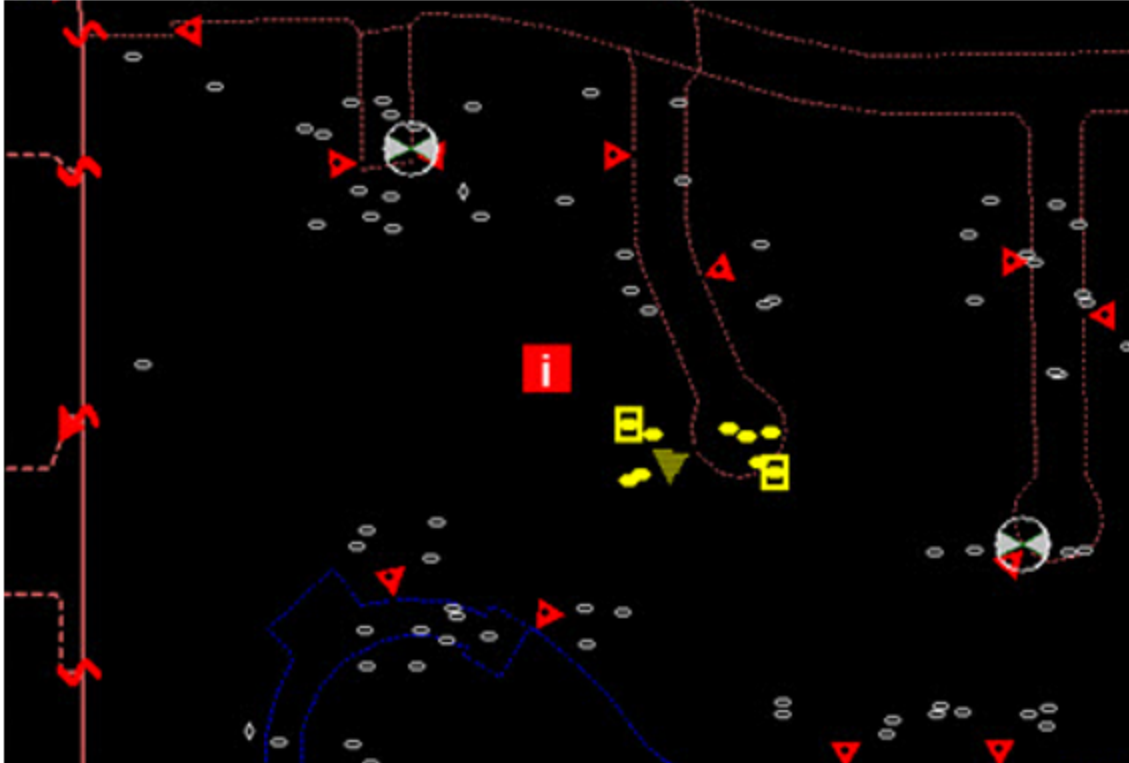
5.2.5.4.1. Example #1

Distribution transformers will be predicted out if two directly connected customers call or send AMI events.

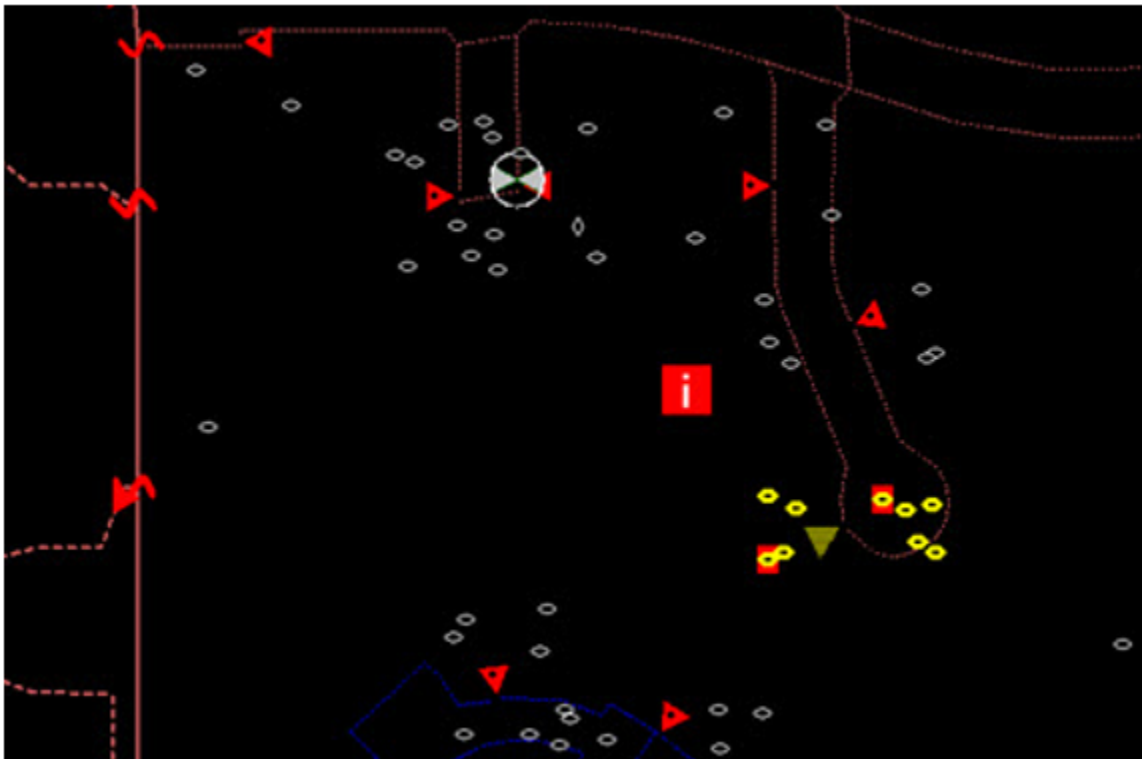
`DISTTF_PREDICTION_RULE_SET = 1`

`DISTTF_PREDICT_B1 = 1`

`DISTTF_PREDICT_T1 = 2`



Two out of nine customers have phoned in. The distribution transformer and all the connected customers are displayed as predicted out and a predicted incident is created.



Two out of nine meters have sent AMI Power Out events. The distribution transformer predicts out and a predicted incident is created.

5.2.5.4.2. Example #2

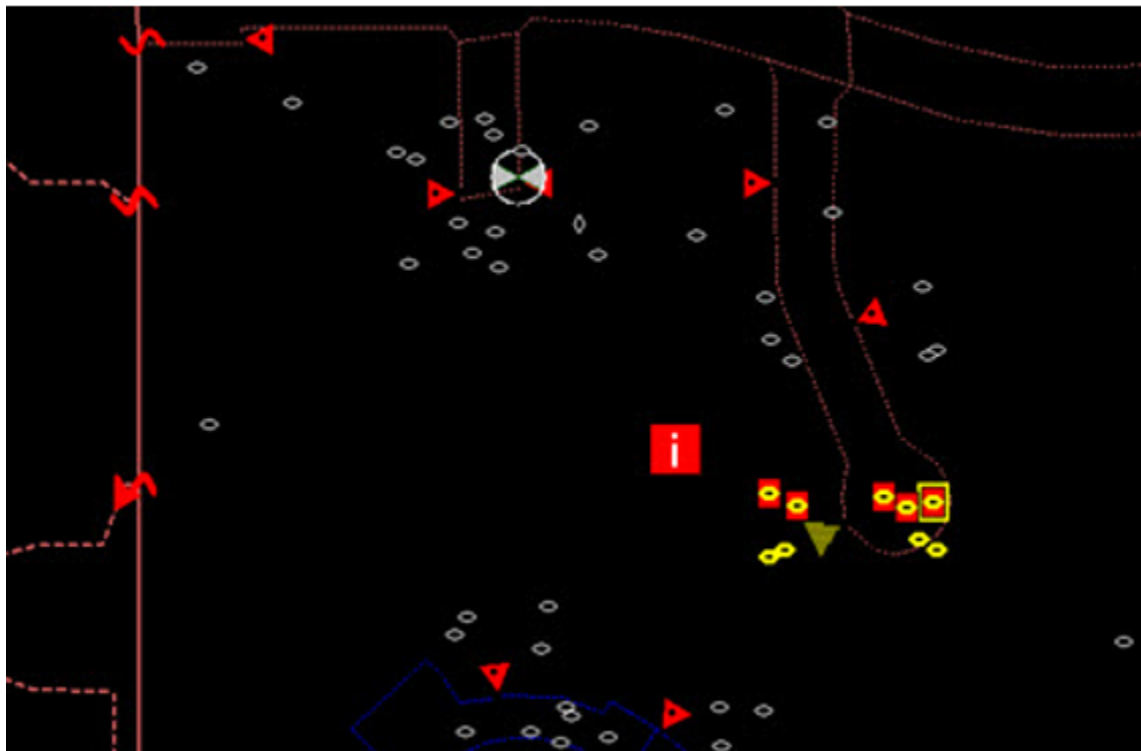
Distribution transformers will be predicted out if 30% of the directly connected customers call OR if 50% of the meters send AMI events.

DISTTF_PREDICTION_RULE_SET = 2

DISTTF_PREDICT_B1 = 1

DISTTF_PREDICT_T2 = 0.30

DISTTF_PREDICT_T2AMI = 0.50



One customer called, and then five meters sent AMI Power Out events, including the meter of the customer who called in. The distribution transformer predicts out and a single predicted incident is created.

5.2.5.4.3. Example #3

Distribution transformers will be predicted out if 2 directly connected customers call or send AMI events.

DISTTF_PREDICTION_RULE_SET = 1

DISTTF_PREDICT_B1 = 1

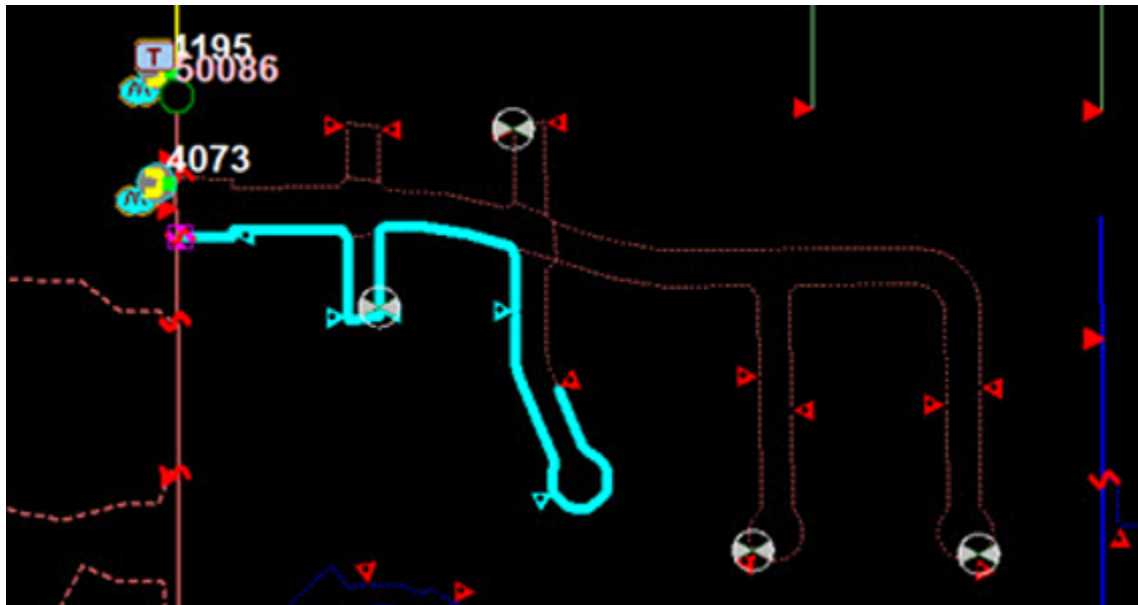
DISTTF_PREDICT_T1 = 2

Fuses will be predicted out if 10% of the directly connected downstream protective devices (distribution transformers in the case of fuses) are predicted out.

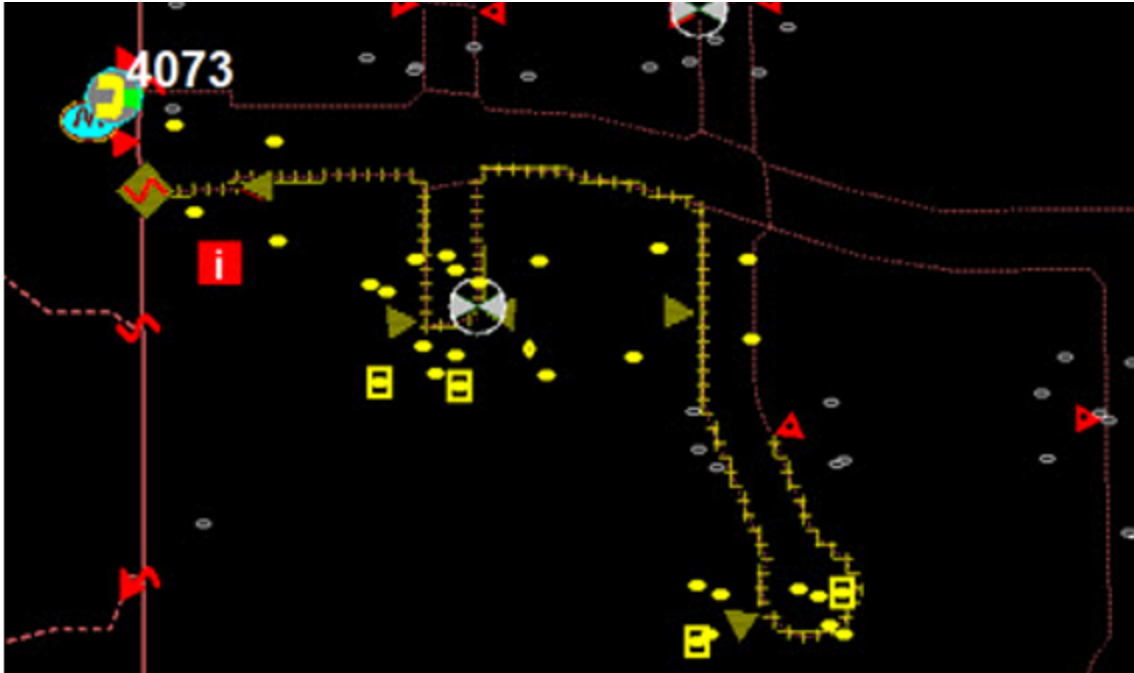
FUSE_PREDICTION_RULE_SET = 2

FUSE_PREDICT_B4 = 1

FUSE_PREDICT_T4 = 0.10



Fuse has five transformers directly connected downstream.



Two customers called on each of two transformers. The fuse predicts out and a single predicted incident is created for the fuse.

5.2.5.4.4. Example #4

Distribution transformers will be predicted out if two directly connected customers call or send AMI events.

`DISTTF_PREDICTION_RULE_SET = 1`

`DISTTF_PREDICT_B1 = 1`

`DISTTF_PREDICT_T1 = 2`

Fuses will be predicted out if 10% of the directly connected downstream protective devices (distribution transformers in the case of fuses) are predicted out.

`FUSE_PREDICTION_RULE_SET = 2`

`FUSE_PREDICT_B4 = 1`

`FUSE_PREDICT_T4 = 0.10`

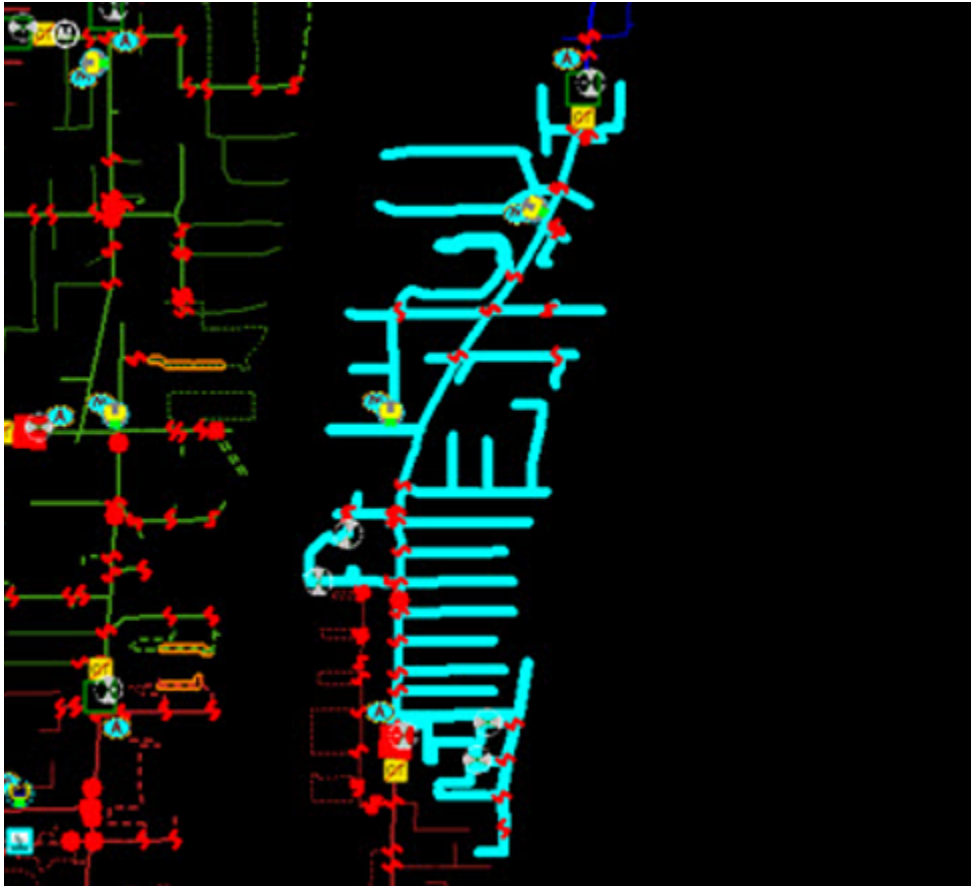
Reclosers will be predicted out if 20% of the directly connected and cascaded downstream protective devices (fuses and distribution transformers in the case of reclosers) are predicted out.

`RECLOSER_PREDICTION_RULE_SET = 2`

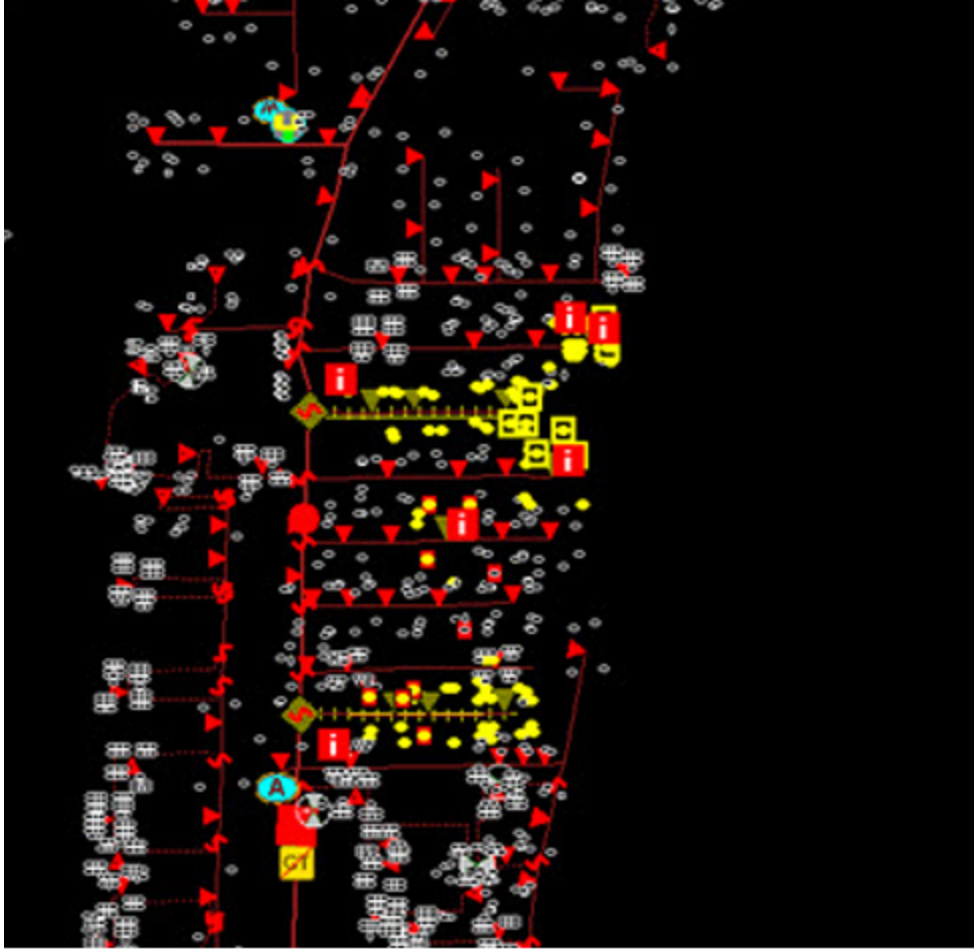
`RECLOSER_PREDICT_B4 = 1`

`RECLOSER_PREDICT_B5 = 1`

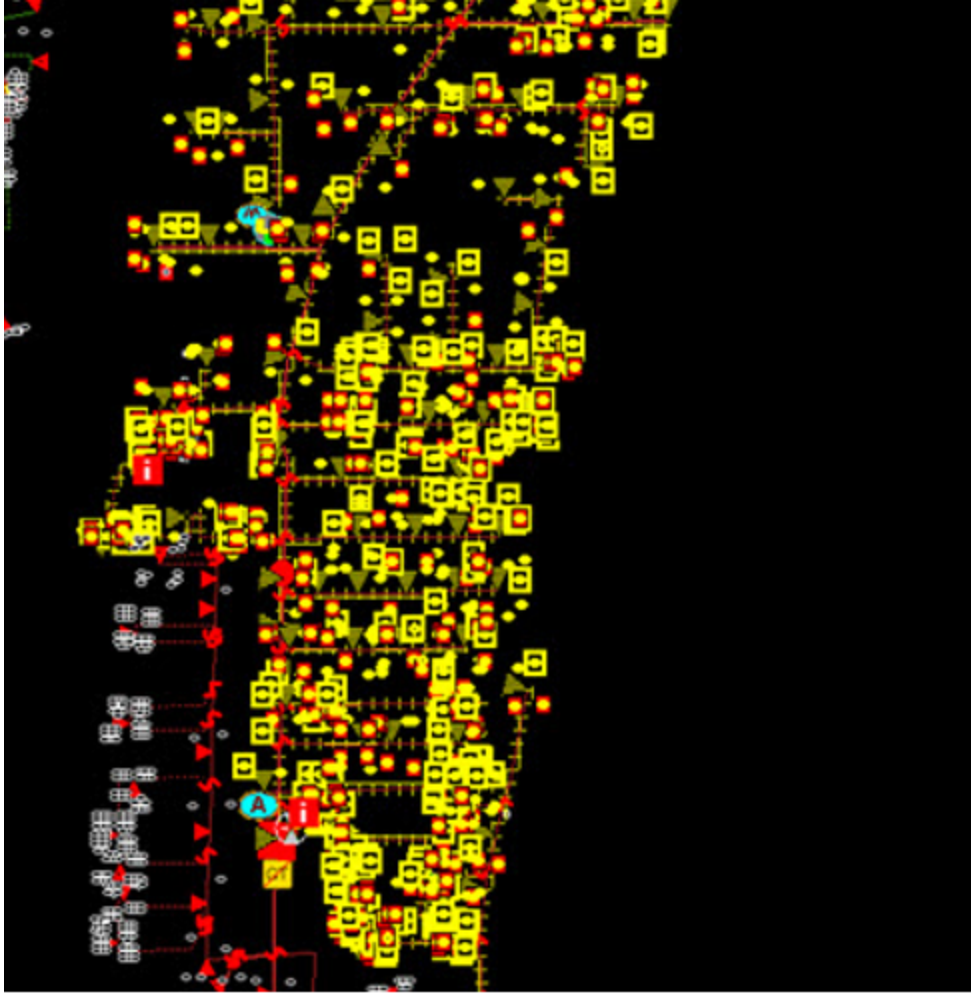
RECLOSER_PREDICT_T4 = 0.20



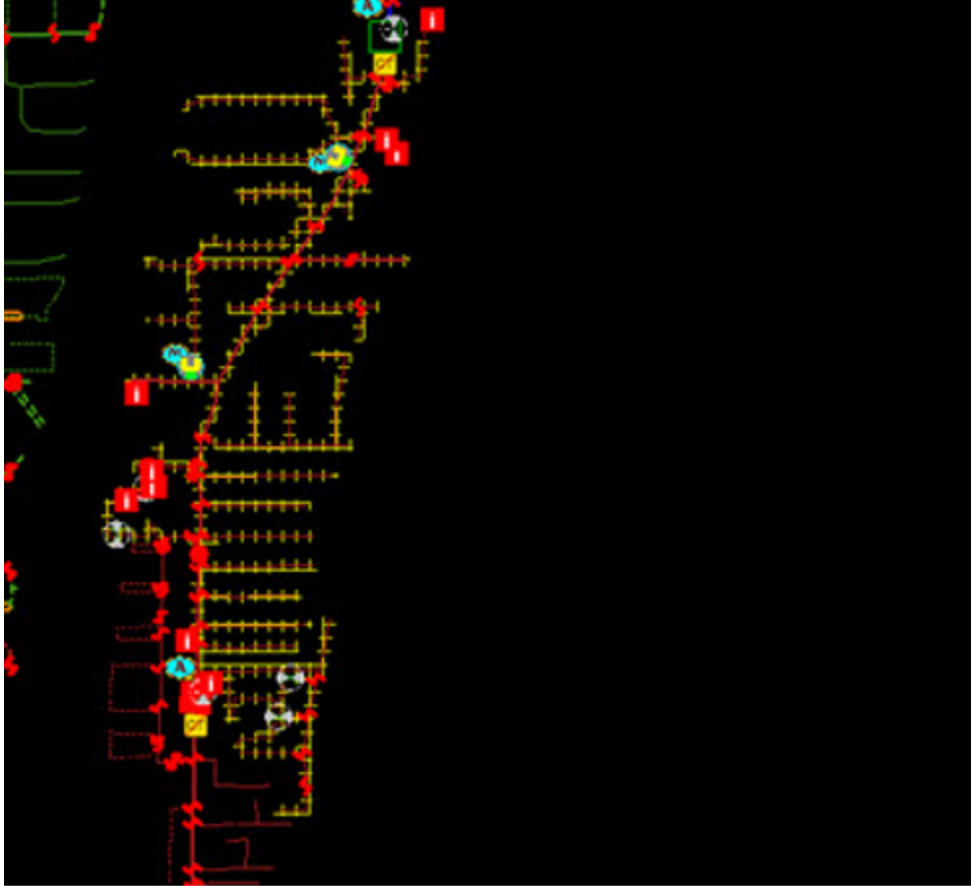
142 transformers and approximately 50 fuses are downstream of the selected recloser.



As customer calls and AMI Power Out events from meters begin to come in, transformers and fuses begin to predict out downstream of the recloser.



Eventually enough downstream devices predict out and all the predicted incidents roll up into a single predicted incident at the recloser.



View with customers turned off. The entire feeder section from the predicted recloser to the normally open is predicted out. If there are mismatch issues in the customer or electrical model, some single customer incidents may also appear.

5.2.6. Custom Outage Management Configuration

To meet the needs of individual utilities, the OMS server Configuration Editor and OMS Client Configuration Editor tools allow customization of outage management related functionality and behavior.

For more information about ADMS Outage Management, refer to the following:

- *Outage Management System Server Configuration Editor User Guide*: A guide for analysts and administrators in the configuration of the OMS server functionality.
- *Outage Management System Client Configuration Editor User Guide*: A guide for analysts and administrators in the configuration of the OMS client functionality.
- *Outage Management System User Guide*: A guide for operators in the use of the outage management system.

5.2.7. Modeling the Management Areas Hierarchy

In **e-terra** *distribution* 3.7 and earlier, the management areas hierarchy only included Company, Division, OpCenter, Station, and Feeder levels. In ADMS 3.8.6 and later, management areas hierarchy is configurable.

There are two group types in the management areas hierarchy:

- **Logical Group:** A Logical group is created in a parent-child relationship. Any child can belong to only one parent. You can create custom groups with names and IDs, where the ID must be unique across all hierarchy groups, including the Electrical group.
- **Electrical Group:** An Electrical group includes Station and Feeder groups. The Station group is just below the lowest level of the Logical group, and the Feeder group is below the Station group. The Feeder group is the lowest level of the hierarchy. The Electrical group cannot be modified.

The Logical group and Electrical group are tied together using a "service center" concept. A service center is the lowest level polygon for logical operating boundaries in a utility; therefore, the service center corresponds to the lowest level in the Logical group hierarchy.

All devices and station data in the model must have the ServiceCenter field populated. In addition, every customer in the customer file must be associated with a service center using the ServiceCenter field.

The ServiceCenter field must be created in the customer model based on the OperatingCenter field. However, the OperatingCenter field is still supported for backward compatibility.

To model and configure the management areas hierarchy:

1. Using the Customer File Compiler tool, on the Region Schema tab, model the management areas hierarchy in a customer schema and rebuild the customer model file.
2. Generate scripts to load the management areas hierarchy from a schema file into the Archive database using the Customer File Compiler tool. Update the Archive database.
3. Using the Crew Modeling tool, on the Crew Areas tab, associate crew areas with service centers in the crew model file. These relationships can be one-to-one or one-to-many, which means that one crew area can be associated with one or more service centers.
4. Associate each device in the station model with an AOR using the PermitAreaId field. The PermitAreaId fields of substations and feeders can be defined using the Substation Editor tool. All PermitAreaId fields must be modeled in the PERMIT system (note that the field length is limited to 8 characters). Also, all IDs defined in the management areas hierarchy must be present in the PERMIT model for the Navigation Overview display to work properly.
5. Using the OMS Client Configuration Editor tool, configure the management areas hierarchy in the `OmsDisplayPluginConfiguration.xml` file.

The number of hierarchy groups on the OMS Display plug-in configuration must match the management areas hierarchy modeled in the customer file. The lowest logical hierarchy level (service center) is queried using the ORGLEVEL1NAME field. The higher hierarchy levels are

queried using the ORGLEVEL2NAME-ORGLEVEL9NAME fields. For a default configuration of 7 management areas hierarchy levels (5 Logical groups and 2 Electrical groups), you should have a query of ORGLEVEL5NAME, ORGLEVEL4NAME, ORGLEVEL3NAME, ORGLEVEL2NAME, ORGLEVEL1NAME, STATION, and FEEDER in the hierarchy group of `OmsDisplayPluginConfig.xml`.

Note

In ADMS 3.8.6, the HistoryGroup query was renamed to HierarchyGroup. The HistoryGroup query is still supported for backward compatibility, but it is recommended to use HierarchyGroup instead.

6. Using the Coded Translation Editor tool, update the ManagementAreasHierarchy coded translation item.
7. Using the Grid Tabular Builder tool, create and update OMS displays for the new hierarchy.

5.3. Tuning/Configuring ADMS Server Parameters

The ADMS servers are multi-threaded applications with the option of allowing an analyst to configure the number of threads that each module can have. The parameters are defined as:

- THREADS_DYN = 2
- THREADS_NTP = 2
- THREADS_DNAF = 1 (applicable in DNAF deployment only)
- THREADS_LOAD = 2
- THREADS_SCADA = 2
- THREADS_CLIENT = 2
- THREADS_DNAF_CBO = 2 (applicable in DNAF deployment only)
- THREADS_QUERY = 2
- THREADS_OMS = 2 (Applicable in OMS deployment only)
- OMS_QUERY_THREADS = 4 (Applicable in OMS deployment only)

Configuring the appropriate number of threads for each module might need some experimentation. With many cores per CPU, a general recommendation is to set the number of threads equal to the number of cores (hyper-threaded, in case hyper-threading is enabled).

OMS uses Windows Threadpool for most of its heavy-duty tasks, which reconfigures itself based on the number of cores in a server.

5.4. Configuring Net Data Replicator

The Net Data Replicator application copies DMS and OMS data (snapshots, journal files, and switching order files) from the enabled ADMS system host to the standby and read-only hosts. Net

Data Replicator ensures that data created or updated on the enabled server is replicated to the standby servers, so that the standby servers can quickly obtain the enabled role without loss of data in the event of a failover. For more information on Net Data Replicator, refer to the *ADMS Architecture Overview*.

The Net Data Replicator application is configured using the `NetDataReplicatorConfig.xml` configuration file.

To enable data replication, the following need to occur:

- The Net Data Replicator application needs to be run on the enabled server host and on the standby server host.
- The `NetDataReplicatorConfig.txt` configuration file needs to be revised on the enabled server host and on the standby server host.

For more information about parameters specific to the Net Data Replicator application, see [NetDataReplicatorConfig.xml](#).

Of particular note are the following parameters in `NetDataReplicatorConfig.xml`:

- **UseRoleFromCfgProxy:** Indicates whether to obtain role (Enabled, Standby, or Disabled) from CfgProxy server.

The enabled and standby Net Data Replicator applications use the CfgProxy connection, therefore CfgProxy should be enabled.

- **LocalListenerAddress:** TCP/IP address, in <Host>:<Port> format, at which the local Net Data Replicator is listening for replication queries from the remote standby Net Data Replicator when the local replicator's role is enabled (Primary).
- **RemoteListenerAddresses:** Comma-separated list of TCP/IP addresses, in <Host>:<Port> format, to which the local OMS replicator sends replication queries when its role is standby.
- **ReadOnlyMode:** Set to False for Net Data Replicator on the ADMS enabled and standby servers.
- **DmsOnlyMode:** If DmsOnlyMode is set to true, Net Data Replicator supports copying only Switch Order files. If OMS replication is needed for the ADMS server, set DmsOnlyMode to False.
- **<ReplicationDataType>Path:** Path to folders with files to be replicated (for example, Incident journal files).
- **Replicate<ReplicationDataType>:** Indicates whether to replicate specific files (for example, Incident journal files). Set to true for all types of files that need to be replicated.

For example, for the DMS and OMS setup:

```
<UseRoleFromCfgProxy>true</UseRoleFromCfgProxy>
<CfgProxyHost>localhost</CfgProxyHost>
<CfgProxyPort>5020</CfgProxyPort>
<CfgmanProcessName>REPLICATESRV</CfgmanProcessName>
<RemoteListenerAddresses>192.168.2.11:8201</RemoteListenerAddresses>
<OmsJournalFilePath>%IDMS_ROOT%\dynamics\Server\OMSServer</OmsJournalFilePath>
<ReplicateCallJournal>true</ReplicateCallJournal>
<ReplicateIncidentJournal>true</ReplicateIncidentJournal>
```



```

<ReplicateCrewJournal>true</ReplicateCrewJournal>
<ReplicateArchiveJournal>true</ReplicateArchiveJournal>
<ReplicateDailySnapshots>true</ReplicateDailySnapshots>
<ReplicateSwitchOrderServerFiles>true</ReplicateSwitchOrderServerFiles>
<SwitchOrderServerFilesPath>%IDMS_ROOT%\dynamics\server\SwitchOrderServer\</SwitchOrderServerFilesPath>
>
<DmsOnlyMode>false</DmsOnlyMode>
<ReadOnlyMode>false</ReadOnlyMode>

```

For more information about NetDataReplicatorConfig.xml parameters, see [NetDataReplicatorConfig.xml](#).

5.5. Configuring Net Data Server

Net Data Server supports reviewing and correcting cleared and canceled incidents in the Outage Management System. Net Data Server also supports read-only access to the current OMS data and recent history for users inside and outside of the control room environment via web services.

Net Data Server is a web service with interfaces available for OMS. Net Data Server reads Call, Incident, Crew, and Real Time Incident journal files and creates daily snapshot files, which contain the data for the previous day consolidated in a single file.

Net Data Server also reads and writes an Archive Journal for history correction purposes. It does not archive, but instead creates the Archive Journal for the Archive client application. The Archive client application replicates the archive journals into an Oracle or SQL database instance.

The Net Data Server process configuration file NetDataServerConfig.txt is typically located in the %IDMS_ROOT%\config\server subdirectory. Of particular note are the following Net Data Server parameters:

- **USE_ROLE_FROM_CFGPROXY:** The Net Data Server application should have a CfgProxy connection, therefore the USE_ROLE_FROM_CFGPROXY should be set to 1.
- **WEBSERVICE_SERVER_NAME:** A host where web services are installed and listening to other interfaces.
- **WEBSERVICE_PORT_NUMBER:** TCP/IP port used for web services.
- **OMS_ENABLE_DASHBOARD_SUPPORT:** Set to "true" to enable the dashboard and history correction support.

For example:

```

USE_ROLE_FROM_CFGPROXY      = 1
CFGPROXY_HOST               = 127.0.0.1
CFGPROXY_PORT               = 5018
CFGMAN_PROCESS_NAME        = NETDATASRV
WEBSERVICE_SERVER_NAME    = localhost
WEBSERVICE_PORT_NUMBER    = 8800
OMS_ENABLE_DASHBOARD_SUPPORT = true

```

For more information about NetDataServerConfig.txt parameters, see [NetDataServerConfig.txt](#).

5.5.1. Adding an HTTP URL Namespace Reservation

⚠ Caution

The URL namespace reservation must correspond to the Net Data server HTTP/HTTPS configuration. If the Net Data server is configured for a secure connection (HTTPS), use "https" instead of "http" when reserving a namespace.

Net Data Server implements the `http://localhost:8800/Oms` endpoint. By default, listening at a particular HTTP address requires administrator privileges. You may receive the error "HTTP could not register URL `http://+:8800/Oms`". There are two ways to avoid this error:

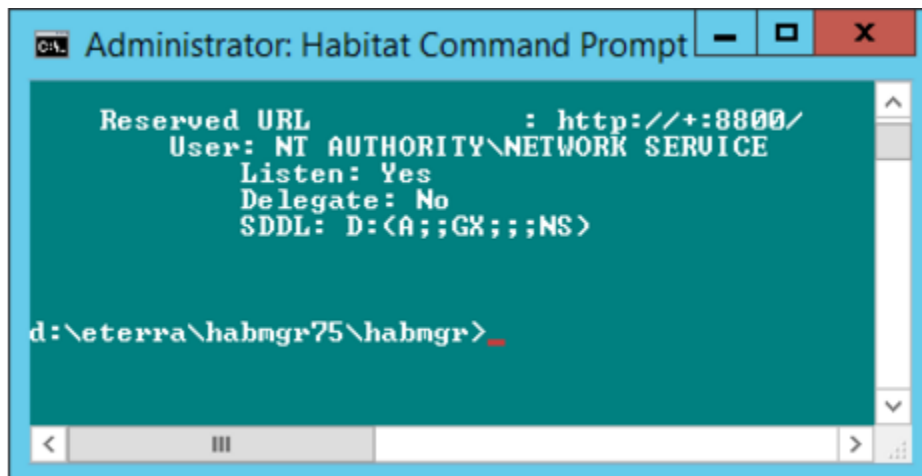
- Run Net Data Server as an administrator.
- Use `Netsh.exe` to give the account permissions to reserve the URL.

To use `Netsh.exe`, open a command prompt with administrator privileges and enter the following command:

```
netsh http add urlacl url=http://+:8800/ user=machine\username
```

✓ Tip

To allow access for any valid user with appropriate permissions, set the `machine\user` to "NETWORK SERVICE". For example, if Habitat is running as a service, access is allowed for the user that starts the Habitat service.



To delete the reservation, enter:

```
netsh http delete urlacl url=http://+:8800/
```

5.6. Installing the Browser Web Server

The Browser web server primary function is to serve the Browser display files (Scada and Habitat displays) from a central location to ADMS clients. The web server can be installed on the production system server, or it can be installed on a separate server.

Note

Browser web server can only serve data to the clients in the same security zone. For clients in another security zone, another Browser web server must be set up.

5.6.1. Copying the Production ADMS Browser Displays

The Browser display files that are used by the ADMS clients to view display data replicated from the production system by RdServer must be copied onto the server where the IIS web server is installed. These files can be copied from the Browser display file web server used by the production ADMS system. Normally display files are stored in the %HABITAT_WEBROOT%\displays folder.

5.6.2. Configuring Internet Information Services (IIS)

To confirm that IIS is installed on the server where you plan to set up the Browser web server:

1. From the Windows Start menu, open server Manager.

The "Server Manager" window opens.

2. In the Roles and Features grid you should see "Web Server (IIS) Tools" listed.

If you find that IIS is not installed, read the "configure Internet Information Services (IIS) on Windows Server 2016" in the *System Security Guide for Windows Server 2016* for information related to installing and configuring IIS.

Microsoft Windows IIS must be specifically configured to serve the Browser display files. For specific instructions, refer to the following chapters in the *System Security Guide for Windows Server 2016*:

- "Configure Internet Information Services (IIS) on Windows Server 2016"
- "Configure IIS for Browser on Windows Server 2016"

Be sure to add the appropriate IIS roles, including:

- Web Server (IIS)
- CGI
- ISAPI Extensions
- Management Services
- IIS 6 Metabase Compatibility

- IIS 6 Management Console

as instructed in the Procedures in the *System Security Guide for Windows Server 2016*.

After you finish configuring IIS, edit the `DataSourcesConfiguration.xml` file in the Browser display server directory structure so that it references the WfgServer process that is configured on the ADMS server. The file is located in the `D:\Eterra\APF\DataContext` folder within the Browser display server directory structure.

Configure either the default datapool, or a secondary datapool to reference the WfgServer process on the ADMS server. Details concerning the `DataSourcesConfiguration.xml` file, as well as additional information about configuring the Browser display web server, can be found in the "Configuring and Maintaining the Web Server" section of the *Browser Software Installation and Maintenance Guide*. Information about configuring the WfgServer process is available in the "Browser Data Server Configuration" section of the *Habitat Software Installation and Maintenance Guide*.

5.7. Managing Sensitive Information

Specifying certain information in clear text may be a security risk. The ADMS system supports encryption for text strings, text files, journal files, and station, customer, and crew models.

The Archive, DNOM Database Loader, and Customer Loader applications can use database passwords stored in the CyberArk Enterprise Password Vault.

5.7.1. Encrypting Text Data

The ADMS system uses the Encryption tool to encrypt text data, such as passwords and database connection strings in configuration files.

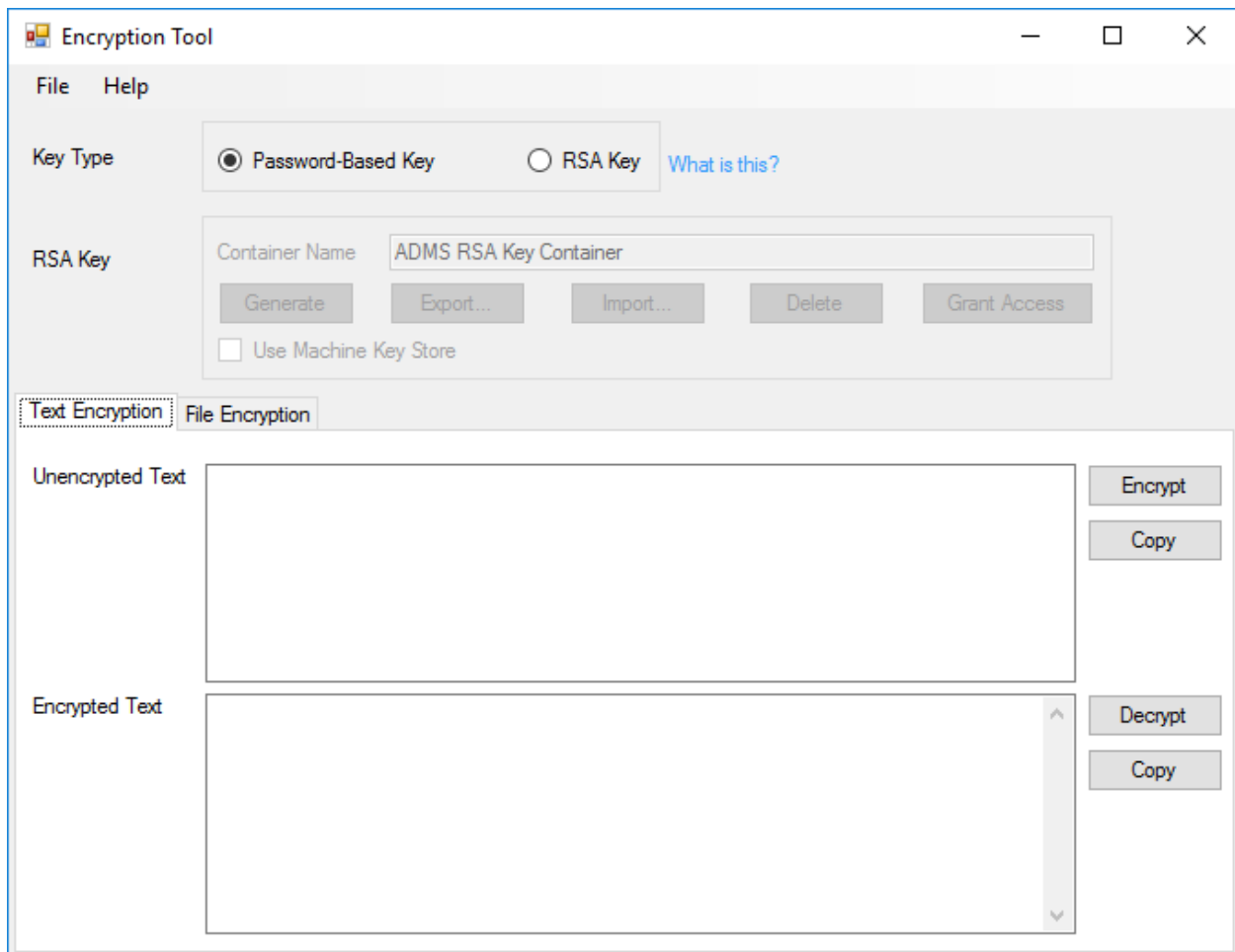


Figure 20. Encryption Tool

Encryption Tool Fields and Controls

The Encryption tool includes the following fields and controls:

- **Key Type:** The Encryption Tool supports two Key Type options: Password-Based Key and RSA Key.

If Password-Based Key is selected, the Windows Data Protection API (DPAPI) is used to encrypt and decrypt text with a key derived from the user's logon credentials. Text encrypted with DPAPI can only be decrypted on the computer on which it was encrypted, or by the same user account on another computer in the same domain (with a roaming profile).

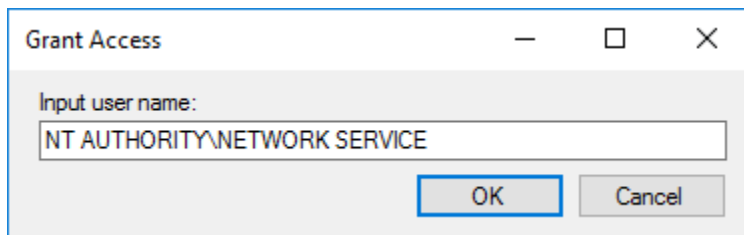
Thus, if a configuration file contains text encrypted with DPAPI, the configuration file must be changed on each computer to which it is copied. In a high-availability configuration with a primary server and a standby server, the configuration file on the standby server cannot be a copy of the configuration file on the primary server. (Encrypted text may be extracted to a separate file so that the configuration file on the standby server may be a copy of the configuration file on the primary server, but the separate file containing the encrypted text on the standby server may not be a copy of the file on the primary server.) For more information about DPAPI, refer to: <http://msdn.microsoft.com/en-us/library/ms995355.aspx>.

If RSA Key is selected, the RSA cryptography algorithm is used to encrypt and decrypt text with an asymmetric public/private key pair. The RSA key may be exported from the encrypting computer (for example, the primary server) and imported into another computer (for example, the standby server), thereby enabling decryption on a computer other than the encrypting computer.

! Important

Exporting an RSA key creates a potential security risk. After the key is imported into the other computer, the key file should be deleted.

- **RSA Key options:** RSA key options are only enabled if RSA Key is selected for the Key Type.
 - **Container Name:** Key container name.
 - **Generate:** Generates a key container.
 - **Export:** Exports the key container to a file.
 - **Import:** Use the Encryption Tool to import an RSA key container on any other computer that needs to decrypt the text.
 - **Delete:** Deletes an RSA key container.
 - **Grant Access:** By default, a generated key can only be used by the user who generated it. If a generated key needs to be used by another user, select the Use Machine Key Store option, and then grant access for that user using the Grant Access option. Use this option when running the Archive application from PROCMAN.



- **Text Encryption:** Options to encrypt, decrypt, and copy text strings.

Text Encryption File Encryption

Unencrypted Text: Some text

Encrypted Text: AQAANCMnd8BFdERjHoAwE/CI+sBAAAAe3GR3f/zBE2eH4ofUPxafQAAAAACAAAAAAQZgAAAAEAACAAAAAGHYbxaVAkR PQ9pY1JsNOeVnyEFOv9EY/7LGHc08TrewAAAAAOgAAAAIAACAAADcRQH/p63JAReeaX a8N7epYy2K5KhyxVH7/A8JF8f2AhAAAACYvVmtMfGkSvzSUWpzF2gKQAAAAFo +fwOc0PaQK66EP5EqRc4p1vjXtQ4kvTV3LsU9ukDr0IQ4k6H4ow50WzSnskjW8PpsEHXUdv V8vFw7FN8A=

Buttons: Encrypt, Copy, Decrypt, Copy

- **File Encryption:** Options to encrypt and decrypt text in a file.

Text Encryption File Encryption

Encryption

Input file: d:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\File.txt

Output file: d:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\File_Encrypted.txt

Decryption

Input file: d:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\File_Encrypted.txt

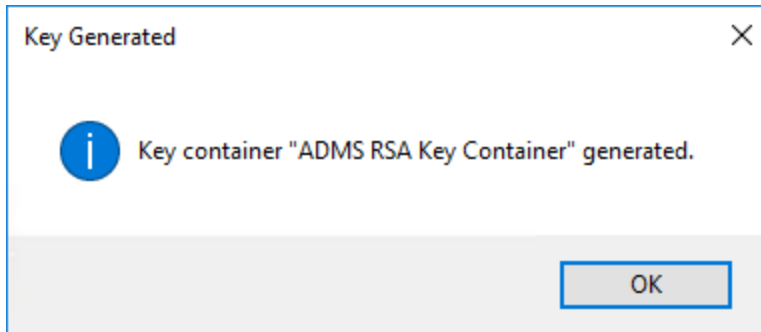
Output file: d:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\File_Decrypted.txt

Buttons: Encrypt, Decrypt

Example: Encrypting an Archive Database Connection String and Using it on Another Machine

The following example describes how to use the Encryption Tool to encrypt an Archive database connection string and enable using the connection string on another machine.

1. On a machine that hosts the Archive application (hereinafter, Machine 1), open the Encryption Tool (click All Programs> Eterra> Distribution> Tools> Configuration Tools> Encryption Tool).
2. Select the RSA Key option.
3. Select the Use Machine Key Store option.
4. Click Generate.



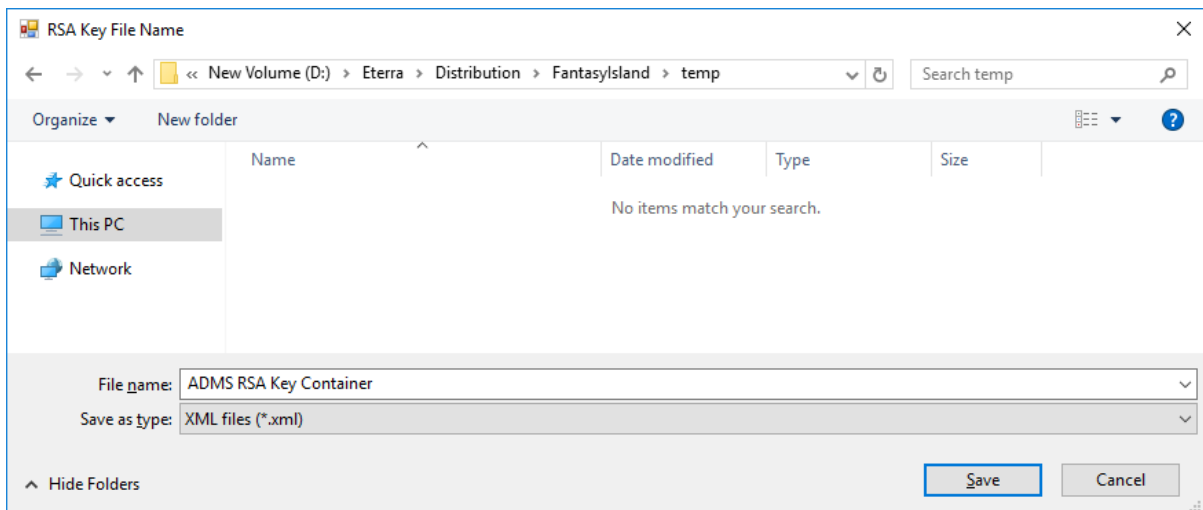
5. Click Grant Access.

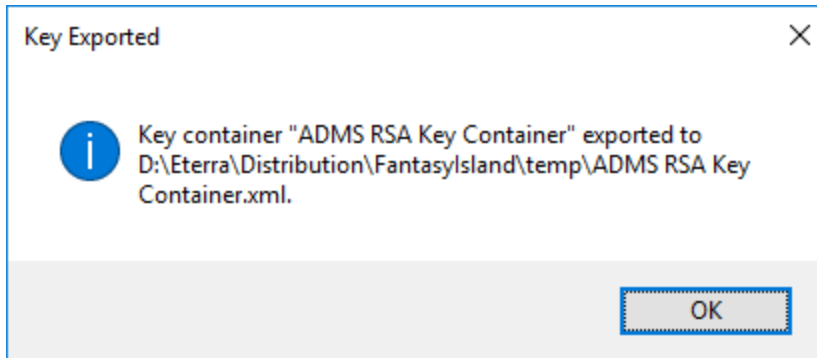
6. Enter the default user name (NT AUTHORITY\NETWORK SERVICE) and click OK.



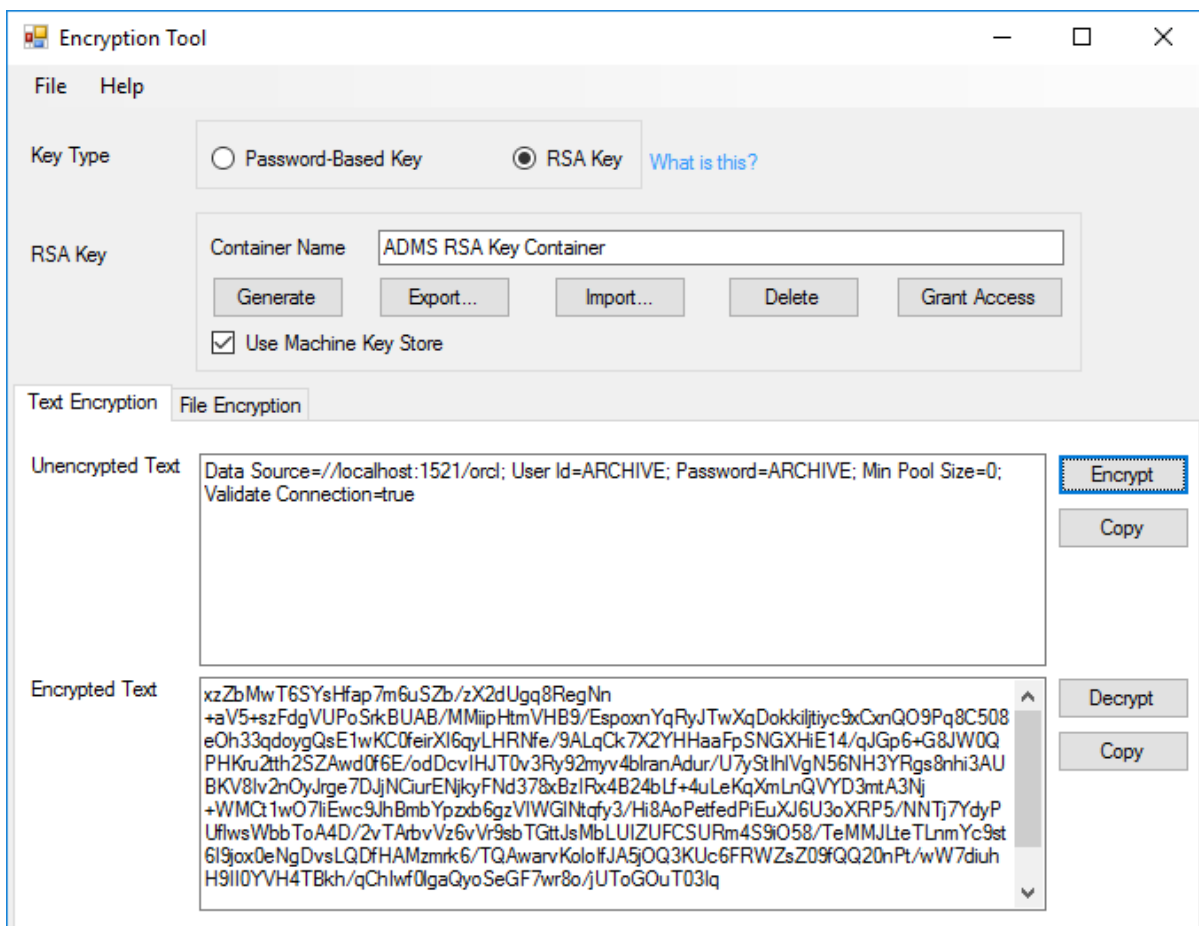
7. Click Export.

8. Select a folder and enter a file name (for example, ADMS RSA Key Container), and click Save.





9. In the Unencrypted Text field, paste an unencrypted SqlConnectionString connection string value from the ArchiveAppConfig.xml file (for example, Data Source=//localhost:1521/orcl; User Id=ARCHIVE; Password=ARCHIVE; Min Pool Size=0; Validate Connection=true).
10. Click Encrypt.



11. Copy the ADMS RSA Key Container file to another machine where the Archive application needs to be run (hereinafter, Machine 2).
12. Run the Encryption Tool on Machine 2.
13. Select the RSA Key option.

14. Select the Use Machine Key Store option.
15. Click Import.
16. Open the copied ADMS RSA Key Container file.
17. Click Grant Access and grant access to the NT AUTHORITY\NETWORK SERVICE user.
18. On Machine 1, copy the encrypted connection string from the Encrypted Text field of the Encryption Tool.
19. On Machine 2, modify the DbConnectionString parameter in ArchiveAppConfig.xml file:
 - Use the encrypted connection string into the "value" attribute.
 - Set the "encrypted" attribute to "true".
 - Add the "keyContainerName" attribute: keyContainerName="ADMS RSA Key Container".

If RSA Key is selected for the Key Type, set the DbConnectionString configuration parameter's "keyContainerName" attribute to the value specified in the RSA Key Container Name field. The "keyContainerName" attribute is not required for a Password-Based Key.

If the connection string is encrypted, the "encrypted" attribute of the DbConnectionString element must be set to "true". This setting indicates to the Archive application that it must decrypt the connection string before it attempts to connect to the database. If the connection string is not encrypted, the "encrypted" attribute must be set to "false".

If the connection string is encrypted with a password-based key, the "keyContainerName" attribute of the DbConnectionString element must either be deleted or set to an empty string. If the connection string is encrypted with an RSA Key, the "keyContainerName" attribute of the DbConnectionString element must be set to the RSA Key Container Name specified in the Encryption Tool.

Note that the Archive application interprets the existence of a "keyContainerName" attribute as an indication that the connection string must be decrypted with that key, so it is important not to specify a "keyContainerName" attribute if a Password-Based Key was used for encryption.

20. Save changes to the ArchiveAppConfig.xml file.

```

7      <DataProviderName>Oracle.ManagedDataAccess.Client</DataProviderName>
8      <!-- <DataProviderName>Oracle.DataAccess.Client</DataProviderName> -->
9      <!-- <DataProviderName>System.Data.SqlClient</DataProviderName> -->
10
11      <!-- Database connection string -->
12      <DbConnectionString value=
13      "xzZbMwT6SYsHfap7m6uSZb/zX2dUgq8RegNn+aV5+szFdgVUPoSrKBUAB/MMiipHtmVHB9/EspoxnYqRyJTwXqDokkiljt
14      iyo9xCxnQ09Pq8C508eOh33qdoygQsE1wKC0feirXI6qyLHRNfe/9ALqCk7X2YHHaaFpSNGXHIE14/qJGp6+G8JW0QPHKru
15      2tth2SZAWd0f6E/odDovIHJT0v3Ry92myv4blranAdur/U7yStIhlVgN56NH3YRgs8nhi3AUBKV8Iv2nOyJrge7DjJNCiur
16      ENjkyFNd378xBzIRx4B24bLf+4uLeKqXmLnQVYD3mtA3Nj+WMct1w07IiEwc9JhBmbYpzb6gzVIWGlntqfy3/Hi8AoPetf
17      edPiEuXJG6U3oXRP5/NNTj7YdyPUflwsWbbToA4D/2vtArbvVz6vVr9sbTGttJsMbLUIZUPCSURm4S9iO58/TeM4JLteTLnm
18      Yc9st6I9jox0eNgDvsLQDFHAMzmrk6/TQAwArvKoloIfJA5jOQ3KUc6FRWzsZ09fQQ20nPt/wW7diuhH9II0YVH4TBkh/qC
19      hIwf0lgaQyoSeGF7wr8o/jUToGOuT03Iq+vbBrSq21w7uIxx7ulqxxbtOGPc3GzTsp8feEAQEKVnmR0WE/W7IkJzstyQ1/4
20      yUWbrdzVYgvwZHswtHY=" encrypted="true" keyContainerName="ADMS RSA Key Container" />
21
22      <!-- <DbConnectionString value="Data Source=server\instance,port; Database=database;
23      Integrated Security=true; MultipleActiveResultSets=true" encrypted="false" /> -->

```

Now you can run the Archive application by the current user, as well as by the Task Scheduler

(by NETWORK SERVICE).

5.7.2. Encrypting Journals and Models

ADMS can encrypt journals files and customer, crew, and station models.

The data is encrypted using AES-256 in CBC mode. Advanced Encryption Standard (AES) is a specification of encryption from the US National Institute of Standards and Technology (NIST). AES uses a block size of 256 bits, and a symmetric encryption key of size 128 bits. Cipher Block Chaining (CBC) is a mode of operation in which the result of encrypting a block depends not only on the encryption key, but also on the immediately previous block.

The 256-bit key used to encrypt journal and model files can itself be encrypted using the Microsoft Data Protection API (DPAPI) encryption method. For enhanced security, it is recommended to encrypt the key value. The key value encryption can be done by the AesKeyMgr tool.

The encryption functionality uses the following properties:

- **Encryption Key Scope:** The scope property enables using different key names for different applications, subsystems (for example, ADMS, Modeling, Test) or different sites (for example, GEADMS, TESTADMS, PROJECTADMS). The scope value format is an upper-case text string with maximum length of 11 characters.
- **Encryption Key Version:** The version property enables changing the encryption key to be used while maintaining the same EncryptionKeyScope value. It is assumed that the version is incremented periodically for the user to switch to a new key value. The version value format is a short integer value (any value between 0 and the 32767).

The <EncryptionKeyScope>_<EncryptionKeyVersion> value is used as the entry to look up the actual value of the encryption key in a key storage. This value is also stored in the header of encrypted files, so that the value can be used to look for the encryption key when a file needs to be decrypted.

- **Encryption Key Storage Type:** The storage type property designates where the application looks for the key (Default, EnvironmentVariable, or EnvironmentVariableDpApi). If the EncryptionKeyStorageType parameter is not defined in the application configuration or the system, the application uses the EnvironmentVariableDpApi storage type to look for the key.
 - **Default:** The Default value internally resolves to EnvironmentVariableDpApi.
 - **EnvironmentVariable:** Specifies that the encryption key is defined in the system using an environment variable and the key value is not encrypted.
 - **EnvironmentVariableDpApi:** Specifies that the encryption key is defined in the system using an environment variable and the key value is encrypted using the DPAPI encryption method.
- **Encryption Key Value:** The key value is the AES key, which the application uses to encrypt and decrypt files. The key value format is a free-form text string comprised of 64 Hexadecimal characters (0...9; A...F).

You must define an encryption key in the system using an environment variable. The environment variable format defines the key storage type.

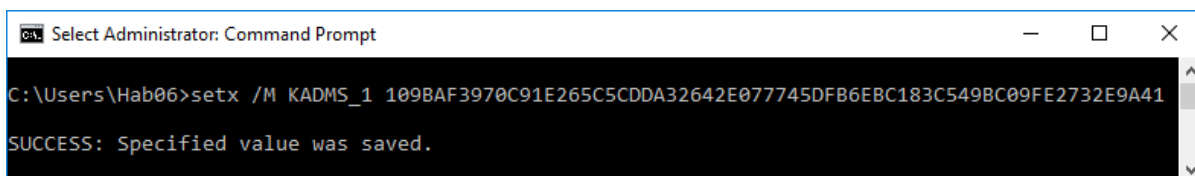
- To store an encryption key in the EnvironmentVariable storage type, the environment variable format is "<EncryptionKeyScope>_<EncryptionKeyVersion> <Key_Value>".
- To store an encryption key in the EnvironmentVariableDpApi storage type, the environment variable format is "<EncryptionKeyScope>_<EncryptionKeyVersion>_DPAPI <Encrypted_Key_Value>". For more details about encrypting the key value, see [Encrypting the AES Key using DPAPI](#).

Examples:

To define an encryption key in the EnvironmentVariable storage type (machine-wide):

1. Open a command prompt.
2. Execute the following command:

```
setx /M KADMS_1 109BAF3970C91E265C5CDDA32642E077745DFB6EBC183C549BC09FE2732E9A41
```

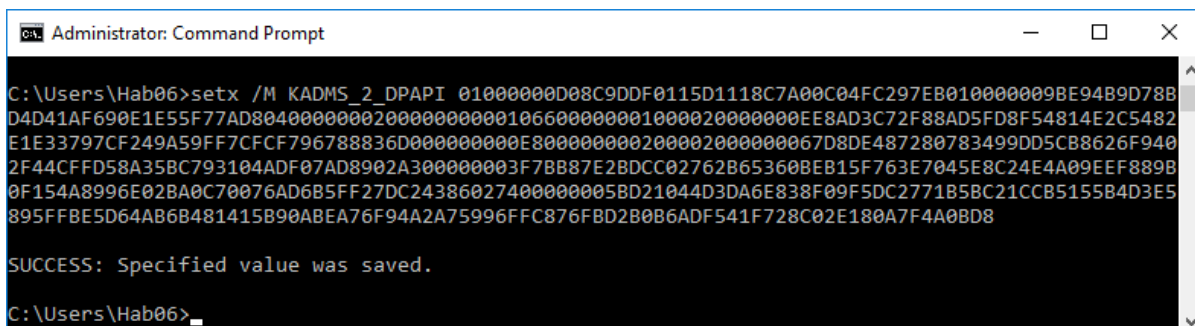


This adds the key to the EnvironmentVariable key storage.

To define an encryption key in the EnvironmentVariableDpApi storage type (machine-wide):

1. Open a command prompt.
2. Execute the following command:

```
setx /M KADMS_2_DPAPI 01000000D08C9DDF0115D1118C7A00C04FC297EB010000009BE94B9D78BD4D41AF690E1E55F77AD8040000002000000000106600000001000020000000EE8AD3C72F88AD5FD8F54814E2C5482E1E33797CF249A59FF7CF796788836D00000000E800000000200002000000067D8DE487280783499DD5CB8626F9402F44CFFD58A35BC793104ADF07AD8902A300000003F7BB87E2BDCC02762B65360BEB15F763E7045E8C24E4A09EEF889B0F154A8996E02BA0C70076AD6B5FF27DC24386027400000005BD21044D3DA6E838F09F5DC2771B5BC21CCB5155B4D3E5895FFBE5D64AB6B481415B90ABEA76F94A2A75996FFC876FBD2B0B6ADF541F728C02E180A7F4A0BD8
```



This adds the key to the EnvironmentVariableDpApi key storage.

After you define an encryption key, you can use the key to encrypt and decrypt journals and models. You can use multiple encryption keys.

Journal file encryption is configured using the associated server configuration files (for example, use the OMS_ENCRYPTION_KEY_SCOPE, OMS_ENCRYPTION_KEY_VERSION, and OMS_ENCRYPT_ALL_JOURNALS parameters in NetServerOmsConfig.txt to encrypt OMS journal files). Customer, crew, and station model encryption is performed by the associated tools. For more information, refer to the following:

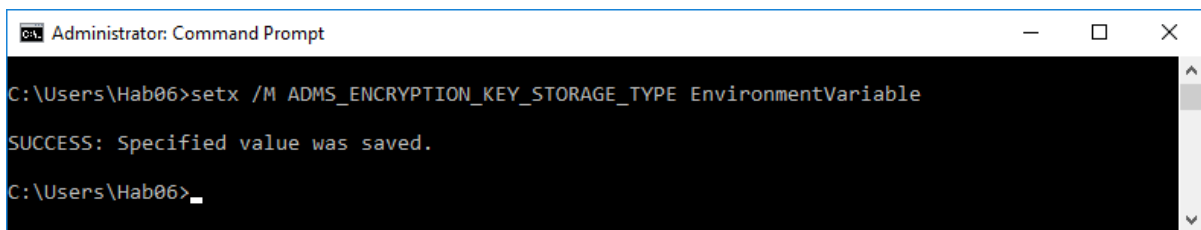
- *Customer List Compiler User Guide*
- *Crew Modeling User Guide*
- *ADMS Modeling Overview and Converter User Guide*

Most applications do not have the encryption storage type configuration and use the EnvironmentVariableDpApi storage type by default (for example, NetServer, which encrypts journal files; DNOM Viewer, which reads model files; and Crew Server, which reads crew model files). To globally define another storage type for applications that do not have the storage type configuration, use the ADMS_ENCRYPTION_KEY_STORAGE_TYPE environment variable.

For example, to define that applications look for an encryption key in the EnvironmentVariable storage type (machine-wide):

1. Open a command prompt.
2. Execute the following command:

```
setx /M ADMS_ENCRYPTION_KEY_STORAGE_TYPE EnvironmentVariable
```



This specifies that the EnvironmentVariable storage type is used for applications that do not have the ENCRYPTION_KEY_STORAGE_TYPE (EncryptionKeyStorageType) configuration.

Note

The key must be defined for users that encrypt files and for users that run applications which use the encrypted files. If an application that creates encrypted files cannot obtain the encryption key, the application writes unencrypted files. If an application that reads encrypted files cannot retrieve the encryption key, the application cannot decrypt the data and read the file.

5.7.2.1. Encrypting the AES Key using DPAPI

This section describes how to encrypt the AES key that is used to encrypt journal and model files using the Microsoft Data Protection API (DPAPI) encryption method. The key value encryption can be done by the AesKeyMgr tool, located in the Distribution\Tools\bin folder.

The AesKeyMgr tool works in command-line mode. Below is a list of command-line arguments that the AesKeyMgr tools supports.

```
genRandomKey
```

Generates and prints a random AES256 key (a string composed of 64 hexadecimal characters).

```
dpApiUserEncrypt <AES Key>
```

Encrypts the AES key using DPAPI and prints the encrypted value. The key applies to the current user only.

```
dpApiMachineEncrypt <AES Key>
```

Encrypts the AES key using DPAPI and prints the encrypted value. The key applies machine-wide.

```
genDefHardcodedKey
```

Prints a hardcoded AES256 key, which is always the same.

```
userConfigDefKeyHardcoded
```

Defines the KADMS_1 and KADMS_1_DPAPI environment variables with the default hardcoded key values (non-encrypted and encrypted, correspondingly) for the current user.

```
machineConfigDefKeyHardcoded
```

Defines the KADMS_1 and KADMS_1_DPAPI environment variables with the default hardcoded key values (non-encrypted and encrypted, correspondingly) machine-wide.

```
version
```

Prints the AesKeyMgr version.

```
help
```

Outputs descriptions of the available command-line arguments, as shown in the following image.

```
Administrator: C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

D:\Eterra\Distribution\Tools\bin>AesKeyMgr help

AesKeyMgr (Vers: 02-NOV-2020-a) Usage:

    (1) > AesKeyMgr genRandomKey
           Generates and prints a random AES256 key

    (2) > AesKeyMgr dpApiUserEncrypt 64-hex-character-string
           Encrypts (AES256) 64-hex-character-string passed using DPAPI (*USER* based) and prints encrypted value

    (3) > AesKeyMgr dpApiMachineEncrypt 64-hex-character-string
```

To generate a DPAPI-encrypted value for an encryption key (machine-wide):

1. Open a command prompt.
2. Navigate to the folder with AesKeyMgr.exe.
3. Execute the following command:

```
AesKeyMgr dpApiMachineEncrypt 109BAF3970C91E265C5CDDA32642E077745DFB6EBC183C549BC09FE2732E9A41
```

```
Administrator: C:\Windows\System32\cmd.exe

D:\Eterra\Distribution\Tools\bin>AesKeyMgr dpApiMachineEncrypt 109BAF3970C91E265C5CDDA32642E077745DFB6EBC183C549BC09FE2732E9A41
Encrypted value: 01000000D08C9DDF0115D1118C7A00C04FC297EB010000009BE94B9D78BD4D41AF690E1E55F77AD804000000200000000001066000000010000200000000EE8AD3C72F88AD5FD8F54814E2C5482E1E33797CF249A59FF7CFCF79678836D000000000E800000000200002000000067D8DE487280783499DD5CB8626F9402F44CFD58A35BC793104ADF07AD8902A300000003F7BB87E2BDCC02762B653608EB15F763E7045E8C24E4A09EEF889B0F154A8996E02BA0C70076AD6B5FF27DC24386027400000005BD21044D3DA6E838F09F5DC2771B5BC21CCB5155B4D3E5895FFBE5D64AB6B481415B90ABEA76F94A2A75996FFC876FBD2B0B6ADF541F728C02E180A7F4A0BD8
```

This generates a DPAPI-encrypted value for the specified key value.

5.7.3. Storing Secret Values in CyberArk

ADMS applications can retrieve relational database passwords, trusted user IDs, and UAA secret values from the centralized protected vault, CyberArk Enterprise Password Vault, so that you do not need to provide sensitive information in the application configuration files as a plain text or encrypted string.

ADMS applications can retrieve secret values from CyberArk as follows:

- The Archive, DNOM Database Loader, and Customer Loader applications can retrieve relational database passwords from CyberArk.
- The Certificate Loader application can retrieve certificate passwords from CyberArk.
- The DMS server, OMS server, DNAF server, OMS Interface server, OMS Adapter, DMS Interface server, NetViewer Adapter, Read-Only server, Study server, Forecast server, and Simulator

server can retrieve trusted user IDs and UAA secrets from CyberArk.

CyberArk Enterprise Password Vault is a component of the CyberArk Privileged Access Manager (PAM) solution, which is designed to discover, secure, rotate and control access to privileged account passwords used to access systems throughout the enterprise IT environment.

The retrieval of secrets from the CyberArk Enterprise Password Vault requires the client machine to have CyberArk Credential Provider installed. The Credential Provider installation includes the following libraries:

- **NetPasswordSDK.dll:** Provides APIs for secret retrieval from the vault for C#/.NET framework based applications.
- **NetStandardPasswordSDK.dll:** Provides APIs for secret retrieval from the vault for C#/.NET Core based applications.
- **CPasswordSDK64.dll:** Provides APIs for secret retrieval from the vault for C++ based applications.

On the ADMS side, the following libraries are used to work with CyberArk Credential Provider libraries:

- **CyberArkSecret.dll:** ADMS-side wrapper that interfaces with NetPasswordSDK.dll. CyberArkSecret.dll implements the CyberArkSecretStore class that is accessed by the iSecretStore class that retrieves the secrets stored in the CyberArk Enterprise Password Vault. By default, this file is delivered with the ADMS_Interfaces_SDK.msi kit and stored in D:\Eterra\Distribution\IdmsInterfaceSDK\SDKSamples\CyberArkSecret\bin\.
- **NetCoreCyberArkSecret.dll:** ADMS-side wrapper that interfaces with NetStandardPasswordSDK.dll. NetCoreCyberArkSecret implements the CyberArkSecretStore class that is accessed by the iSecretStore class that retrieves the secrets stored in the CyberArk Enterprise Password Vault. By default, this file is delivered with the ADMS_Interfaces_SDK.msi kit and stored in D:\Eterra\Distribution\IdmsInterfaceSDK\SDKSamples\NetCoreCyberArkSecret\bin\.

For using CyberArk for C++ based applications, such as NetServer, the Credential Provider installation must be stored in an environment variable CYBERARK_CP_HOME on all clients that need to communicate with the CyberArk Enterprise Password Vault server to retrieve secrets.

To define the CYBERARK_CP_HOME environment variable:

1. Open a command prompt.
2. Execute the following command:

```
setx CYBERARK_CP_HOME <Full path of the Credential Provider installation>
```

For using CyberArk for C#/.NET, C#/.NET Core, and C++ based ADMS applications, NetPasswordSDK.dll, NetStandardPasswordSDK.dll, CPasswordSDK64.dll, CyberArkSecret.dll, and NetCoreCyberArkSecret.dll must exist in the Distribution\Server\bin folder.

Note

For details about installing CyberArk Enterprise Password Vault and CyberArk Credential Provider, refer to the CyberArk documentation.

5.7.3.1. CyberArk Configuration Format

For C#/ .NET applications (for example, `ArchiveAppConfig.xml`), the configuration format is as follows:

```
<CustomSecretStore>
  <AssemblyName>cyberArkSecret.dll</AssemblyName>
  <ClassName>CyberArkSecret.CyberArkSecretStore</ClassName>
  <Supported Object Keys>
</CustomSecretStore>
```

For C#/ .NET Core applications (for example, `CoreNotificationServerConfig.xml`), the configuration format is as follows:

```
<CustomSecretStore>
  <AssemblyName>NetCoreCyberArkSecret.dll</AssemblyName>
  <ClassName>CyberArkSecret.CyberArkSecretStore</ClassName>
  <OtherAssemblyNames>NetStandardPasswordSDK.dll</OtherAssemblyNames>
  <Supported Object Keys>
</CustomSecretStore>
```

For C++ applications (for example, `NetServer`), the configuration format is as follows:

```
SECRET_STORE_ENABLED = TRUE
SECRET_STORE_DLL_PATH = %CYBERARK_CP_HOME%\ApplicationPasswordSdk\CPasswordSDK64.dll
<Supported Object Keys>
```

5.7.3.2. Retrieving Passwords

Use the following parameters in configurations files for C#/ .NET applications (`ArchiveAppConfig.xml`, `CustomerLoaderConfig.xml`, `DnomDBLoaderConfig.xml`, and `CertificateLoader.xml`) to use passwords stored in CyberArk:

- **AssemblyName:** Name of the assembly (DLL) containing a class that implements the CyberArk secret store.
- **ClassName:** Name of the class that implements the CyberArk secret store. This class is implemented based on the CyberArk software development kit (SDK) and supports connecting to CyberArk API.
- **ObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the password defined in the Enterprise Vault Server. The format is `<DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>`.

For example, the default configuration in `OmsInterfaceServerConfig.xml` is as follows:

```
<CustomSecretStore>
  <AssemblyName>cyberArkSecret.dll</AssemblyName>
  <ClassName>CyberArkSecret.CyberArkSecretStore</ClassName>
  <ObjectKey>ETD,ETD_Databases,root,Application-CyberArkPTA-User-ARCHIVE-ARCHIVE</ObjectKey>
</CustomSecretStore>
```

Use the following parameters in configurations files for C#/.NET Core applications (`CoreNotificationServerConfig.xml`) to use passwords stored in CyberArk:

- **AssemblyName:** Name of the assembly (DLL) containing a class that implements the CyberArk secret store.
- **ClassName:** Name of the class that implements the CyberArk secret store. This class is implemented based on the CyberArk software development kit (SDK) and supports connecting to CyberArk API.
- **ObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the password defined in the Enterprise Vault Server. The format is `<DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>`.
- **OtherAssemblyNames:** Name of the assembly (DLL) that implements the CyberArk secret store for the C#/.NET core based applications.

For example, the default configuration in `CoreNotificationServerConfig.xml` is as follows:

```
<CustomSecretStore>
  <AssemblyName>NetCoreCyberArkSecret.dll</AssemblyName>
  <ClassName>CyberArkSecret.CyberArkSecretStore</ClassName>
  <OtherAssemblyNames>NetStandardPasswordSDK.dll</OtherAssemblyNames>
  <ObjectKey>ETD,ETD_Databases,root,Application-CyberArkPTA-User-ARCHIVE-ARCHIVE</ObjectKey>
</CustomSecretStore>
```

Note

When you provide the relational database password both in the database connection string and the Custom Secret Store node, the application (for example, Archive) first tries to retrieve the password from the connection string, and then from CyberArk.

5.7.3.3. Retrieving Trusted User IDs and UAA Secrets

Use the following parameters in configurations files for C#/.NET applications (`OmsInterfaceServerConfig.xml`, `OmsAdapterConfig.xml`, and `netconf_dataservice.xml`) to use secret values stored in CyberArk:

- **AssemblyName:** Name of the assembly (DLL) containing a class that implements the CyberArk secret store.

- **ClassName:** Name of the class that implements the CyberArk secret store. This class is implemented based on the CyberArk software development kit (SDK) and supports connecting to CyberArk API.
- **OtherAssemblyNames:** Name of the assembly (DLL) that implements the CyberArk secret store for the C#/.NET core based applications. This parameter appears in configuration files for C#/.NET Core based applications, such as NetviewerAdapterConfig.xml.
- **ObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the trusted user ID defined in the Enterprise Vault Server. The format is <DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>.
- **UaaIdObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the UAA ID defined in the Enterprise Vault Server.
- **UaaSecretObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the UAA secret defined in the Enterprise Vault Server.

For example, the default configuration in OmsInterfaceServerConfig.xml is as follows:

```
<CustomSecretStore>
  <AssemblyName>CyberArkSecret.dll</AssemblyName>
  <ClassName>CyberArkSecret.CyberArkSecretStore</ClassName>
  <ObjectKey>ETD,ETD_Databases,root,Application-CyberArkPTA-OIS-OIS</ObjectKey>
  <UaaIdObjectKey>ETD,ETD_Databases,root,Application-CyberArkPTA-OIS_UAA_ID-
OIS_UAA_ID</UaaIdObjectKey>
  <UaaSecretObjectKey>ETD,ETD_Databases,root,Application-CyberArkPTA-OIS_UAA_SECRET-
OIS_UAA_SECRET</UaaSecretObjectKey>
</CustomSecretStore>
```

Use the following parameters in configurations files for C#/.NET Core applications (NetviewerAdapterConfig.xml) to use secret values stored in CyberArk:

- **AssemblyName:** Name of the assembly (DLL) containing a class that implements the CyberArk secret store.
- **ClassName:** Name of the class that implements the CyberArk secret store. This class is implemented based on the CyberArk software development kit (SDK) and supports connecting to CyberArk API.
- **OtherAssemblyNames:** Name of the assembly (DLL) that implements the CyberArk secret store for the C#/.NET core based applications.
- **ObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the trusted user ID defined in the Enterprise Vault Server. The format is <DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>.
- **UaaIdObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the UAA ID defined in the Enterprise Vault Server.
-

UaaSecretObjectKey: CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the UAA secret defined in the Enterprise Vault Server.

For example, the default configuration in `NetviewerAdapterConfig.xml` is as follows:

```
<CustomSecretStore>
  <AssemblyName>NetCoreCyberArkSecret.dll</AssemblyName>
  <ClassName>CyberArkSecret.CyberArkSecretStore</ClassName>
  <OtherAssemblyNames>NetStandardPasswordSDK.dll</OtherAssemblyNames>
  <UaaIdObjectKey>ETD,ETD_Databases,root,Application-CyberArkPTA-NVA_UAA_ID-
NVA_UAA_ID</UaaIdObjectKey>
  <UaaSecretObjectKey>ETD,ETD_Databases,root,Application-CyberArkPTA-NVA_UAA_SECRET-
NVA_UAA_SECRET</UaaSecretObjectKey>
</CustomSecretStore>
```

Use the following parameters in configurations files for C++ applications (for example, `NetServerMainConfig.txt`, `NetServerOmsConfig.txt`, and `NetServerDnafConfig.txt`) files to use secret values stored in CyberArk:

- **SECRET_STORE_ENABLED:** Specifies whether using values from the CyberArk secret store is enabled.
- **SECRET_STORE_DLL_PATH:** Full path to the assembly (DLL) containing a class that implements the CyberArk secret store.
- **SECRET_STORE_TRUSTED_USER_DIS_OBJECT_KEY:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the DMS Interface Server trusted user ID defined in the Enterprise Vault Server. The format is
<DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUserName>.
- **SECRET_STORE_TRUSTED_USER_DFWK_OBJECT_KEY:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the DMS Framework trusted user ID defined in the Enterprise Vault Server.
- **SECRET_STORE_TRUSTED_USER_OIS_OBJECT_KEY:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the OMS Interface Server trusted user ID defined in the Enterprise Vault Server.
- **SECRET_STORE_TRUSTED_USER_WFM_OBJECT_KEY:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the Workforce Management System trusted user ID defined in the Enterprise Vault Server.
- **SECRET_STORE_UAA_ID_OBJECT_KEY:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the UAA ID defined in the Enterprise Vault Server.
- **SECRET_STORE_UAA_SECRET_OBJECT_KEY:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the UAA secret defined in the Enterprise Vault Server.

For example, the default configuration in `NetServerMainConfig.txt` is as follows:

```
SECRET_STORE_ENABLED = FALSE
SECRET_STORE_DLL_PATH = %CYBERARK_CP_HOME%\ApplicationPasswordSdk\CPasswordSDK64.dll
SECRET_STORE_TRUSTED_USER_DIS_OBJECT_KEY = ETD,ETD_Databases,root,Application-CyberArkPTA-DIS-DIS
SECRET_STORE_TRUSTED_USER_DFWK_OBJECT_KEY = ETD,ETD_Databases,root,Application-CyberArkPTA-DFWK-DFWK
SECRET_STORE_UAA_ID_OBJECT_KEY = ETD,ETD_Databases,root,Application-CyberArkPTA-OMS_UAA_ID-OMS_UAA_ID
SECRET_STORE_UAA_SECRET_OBJECT_KEY = ETD,ETD_Databases,root,Application-CyberArkPTA-OMS_UAA_SECRET-OMS_UAA_SECRET
```

5.7.4. Managing Certificates

The Certificate Loader application is designed to simplify the distribution and installation of digital certificates for ADMS systems.

The Certificate Loader application adds and installs digital certificates on ADMS hosts. The application monitors specific folders for new digital certificate files and adds those certificates to the host's Windows certificate store. The Certificate Loader also binds newly added certificates to specific server ports based on how the certificate is defined in the Certificate Loader configuration file. ADMS applications retrieve digital certificates from the host's certificate store on startup.

The Certificate Loader application should run on each host (node) in an ADMS system that requires digital certificates. For example, on hosts with ADMS services that must support the HTTPS protocol or where transport layer security (TLS) is required for connections between servers and clients.

The Certificate Loader process can be modeled in PROCMAN as a "passive task" because it does not interact with PROCMAN or Configuration Manager.

Digital certificates can be distributed in several formats and use different schemes for managing certificate usage and lifetime. Therefore, each customer may have different use cases and requirements for their certificate management.

To cover various needs, the application supports TLS for TCP/IP, and dual SSL for REST services (HTTPS).

To support different use cases, the following assumptions were considered when developing the Certificate Loader.

- Each ADMS application host that uses certificates will have an instance of the ADMS Certificate Loader locally.
- The Certificate Loader configuration will be driven by the Certificate Common Name (SubjectName) defined at the time the certificate is issued (<CertSubjectName> in the Certificate Loader configuration file).
- Certificates that do not have a matching Subject Name in the configuration will not be added and installed.
- Certificate files created for ADMS applications should be in the p12 (PKCS#12) file format and contain only one certificate per file.
- Root Certificate Authority certificates will be installed and updated manually.

- Intermediate client and server certificates will be added to the Windows certificate store by the Certificate Loader instance running on each node.
- Applications that support HTTPS and TLS (TCP/IP) endpoints will need to restart to reload a new certificate.
- After TCP/IP TLS and HTTPS interfaces are configured, the system administrator will need to restart the affected applications according to the runbook sequence (for example, using PROCMAN to stop and start processes).

It is expected that in a dual-redundant ADMS configuration, the following will occur when new certificates are issued:

- The Certificate Loader application running on the enabled and standby servers will deploy new certificates to their local Windows Certificate Store.
- The system administrator will verify that the new certificates have been added to the required Windows Certificate Store by verifying that the following log message is written in the CertificateLoaderApp log (%idms_root%\log\CertificateLoaderApp_YYYYMMDD_HHMMSS.log):

```
"CertificateLoaderPlugin: Success Adding VALID cert: 'cert-file-path\cert-file-name.p12-subject-name',"
```

- The processes on the standby machine are restarted to use the new certificates.
- The failover is initiated.
- The processes on the new standby machine are restarted to use new certificates.

5.7.4.1. Certificate Loader Configuration

The Certificate Loader application is installed with the Server kit. The application is configured using the `CertificateLoaderAppConfig.xml` configuration file. By default, this file is stored in `%IDMS_ROOT%\config\server`.

The `CertificateTypes` definitions determine how certificate files are managed. The `CertPath` folder path defines where the newly issued certificate files are expected to be copied. The `PortBindings` definitions determine the list of ports to which the certificate is bound.

The Certificate Loader application monitors the certificate folder for new files. When a new file appears, the Certificate Loader retrieves all files with names that match the `CertFileName` template. Each file that matches the template is opened to retrieve the certificate's "subject name / CN".

- If a match is found between the file's certificate name and the config `<CertSubjectName>` value, the certificate is added to the local Windows Certificate Store at `<CertStore>` and location `<CertStoreLocation>`. For example, `TrustedRoot/LocalMachine`.
- If the file certificate is already defined in the host's Windows Certificate Store, the file is not added (the certificate's "thumbprint" is used as the certificate's unique identifier when searching in the store).
- If the `<CertPort>` element is defined, the Certificate Loader binds the new certificate to the IP

Port defined.

The default CertificateLoaderAppConfig.xml contents are shows below:

```
<?xml version="1.0" encoding="utf-8"?>
<CertificateLoaderAppConfig xmlns:json='http://james.newtonking.com/projects/json'>
  <EnableDebugLog>false</EnableDebugLog>
  <LogFile>%IDMS_ROOT%\log\CertificateLoaderApp.log</LogFile>
  <MaxMessagesInLogFile>10000</MaxMessagesInLogFile>

  <PluginApiConfig>
    <Plugins>

      <!-- Certificate Loader Plugin -->
      <!-- Monitor and Process X509 Certificate Files -->
      <PluginDefinition>
        <Enabled>true</Enabled>
        <Type>Plugin</Type>
        <Name>CertificateLoaderPlugin</Name>
        <Config>
          <Name>CertificateLoaderPlugin</Name>
          <Description>Certificate Loader OpenApi Plugin</Description>
          <VerboseLogging>false</VerboseLogging>

          <CertFileAccessPasswordCfg>testpassword</CertFileAccessPasswordCfg>

          <!-- Sample CyberArk secret store configuration -->
          <CustomSecretStore>
            <AssemblyName>cyberArkSecret.dll</AssemblyName>
            <ClassName>CyberArkSecret.CyberArkSecretStore</ClassName>
            <ObjectKey>ETD,ETD_Databases,root,Application-CyberArkPTA-CertPassword</ObjectKey>
          </CustomSecretStore>
          -->

          <Certificates>

            <!-- CertificateTypes define the attributes and actions for a particular type of
certificate file. -->
            <!-- The order in which the CertificateType definitions appear in this config file is
important, they should be defined in dependency order.-->
            <!-- If a certificate depends on another for validation that other should be added first,
e.g.-->
            <!-- 1. Any "root" certificate should be added first. -->
            <!-- 2. Any "intermediate" certificates should be added next. -->
            <!-- 3. Then "client" certificates. -->
            <!-- 4. Then "server" certificates. -->

            <!-- Root certificate type definition. -->
            <CertificateTypes json:Array='true'>
              <!-- The path to monitor for new certificate files. -->
              <CertPath>C:\Certificates</CertPath>
              <!-- The filename template to monitor. -->
              <CertFileName>*root*.crt</CertFileName>
            </CertificateTypes>
          </Certificates>
        </Config>
      </PluginDefinition>
    </Plugins>
  </PluginApiConfig>
</CertificateLoaderAppConfig>
```

```

    <!-- CertStore Choices are: Personal, TrustedRoot, Intermediate. -->
    <CertStore>TrustedRoot</CertStore>
    <!-- CertStoreLocation Choices are: LocalMachine or CurrentUser. -->
    <CertStoreLocation>LocalMachine</CertStoreLocation>
    <!-- The "Subject Name" that identifies the certificate's purpose and role. This value
is used -->
    <!-- as the key when matching the certificate file to the correct CertificateType
configuration. -->
    <CertSubjectName>ADMS_Root_CA</CertSubjectName>
  </CertificateTypes>

  <!-- Intermediate certificate type definition. -->
  <CertificateTypes json:Array='true'>
    <!-- The path to monitor for new certificate files. -->
    <CertPath>C:\Certificates</CertPath>
    <!-- The filename template to monitor. -->
    <CertFileName>*intermed*.p12</CertFileName>
    <!-- CertStore Choices are: Personal, TrustedRoot, Intermediate -->
    <CertStore>Intermediate</CertStore>
    <!-- CertStoreLocation Choices are: LocalMachine or CurrentUser -->
    <CertStoreLocation>LocalMachine</CertStoreLocation>
    <!-- The "Subject Name" that identifies the certificate's purpose and role. This value
is used -->
    <!-- as the key when matching the certificate file to the correct CertificateType
configuration. -->
    <CertSubjectName>ADMS_Intermediate_CA</CertSubjectName>
  </CertificateTypes>

  <!-- Client certificate type definition. -->
  <CertificateTypes json:Array='true'>
    <!-- The path to monitor for new certificate files. -->
    <CertPath>C:\Certificates</CertPath>
    <!-- The filename template to monitor. -->
    <CertFileName>*client*.p12</CertFileName>
    <!-- CertStore Choices are: Personal, TrustedRoot, Intermediate. -->
    <CertStore>Personal</CertStore>
    <!-- CertStoreLocation Choices are: LocalMachine or CurrentUser. -->
    <CertStoreLocation>LocalMachine</CertStoreLocation>
    <!-- The "Subject Name" that identifies the certificate's purpose and role. This value
is used -->
    <!-- as the key when matching the certificate file to the correct CertificateType
configuration. -->
    <CertSubjectName>ADMS_Client</CertSubjectName>
  </CertificateTypes>

  <!-- Server certificate type definition. -->
  <CertificateTypes json:Array='true'>
    <!-- The path to monitor for new certificate files. -->
    <CertPath>C:\Certificates</CertPath>
    <!-- The filename template to monitor. -->
    <CertFileName>*server*.p12</CertFileName>
    <!-- CertStore Choices are: Personal, TrustedRoot, Intermediate -->
    <CertStore>Personal</CertStore>

```



```

    <!-- CertStoreLocation Choices are: LocalMachine or CurrentUser -->
    <CertStoreLocation>LocalMachine</CertStoreLocation>
    <!-- The "Subject Name" that identifies the certificate's purpose and role. This value
is used -->
    <!-- as the key when matching the certificate file to the correct CertificateType
configuration. -->
    <CertSubjectName>ADMS_Server</CertSubjectName>
    <!-- The IP port bindings that should be added for this certificate. -->
    <PortBindings>
      <Binding json:Array='true'>
        <!-- The IP port the certificate should be bound to. -->
        <Port>0.0.0.0:5050</Port>
        <!-- Should the support dual (server & client side) authentication. -->
        <EnableMutualAuthentication>true</EnableMutualAuthentication>
      </Binding>
      <Binding json:Array='true'>
        <!-- The IP port the certificate should be bound to. -->
        <Port>0.0.0.0:5051</Port>
        <!-- Should the support dual (server & client side) authentication. -->
        <EnableMutualAuthentication>false</EnableMutualAuthentication>
      </Binding>
    </PortBindings>
  </CertificateTypes>

</Certificates>
</Config>
</PluginDefinition>
</Plugins>
</PluginApiConfig>
</CertificateLoaderAppConfig>

```

6. Integrating with the SCADA System

i Note

This entire chapter specifically refers to integration with a GE Scada system.

To integrate ADMS with Scada, Habitat includes the following applications:

- NetPermSrv (Permission Proxy server), which allows the ADMS servers to access permission definitions stored in the PERMIT database. This application runs on the SCADA server.

i Note

Habitat installation includes NetPermSrv. There is no separate ADMS kit for NetPermSrv since **e-terra*distribution 3.6*.

- CfgProxy (Configuration Manager Proxy server), which allows the ADMS server applications to receive individual roles from CFGMAN and act accordingly. This application runs on the ADMS server.

For the purpose of integrating the ADMS server applications with a SCADA system, you need to perform the tasks described in this chapter so that the ADMS server applications:

- Receive role assignments from CFGMAN (running on the same machine)
- Receive real-time data from SCADA through an InterSite Data(ISD) link
- Can access permission definitions maintained by the PERMIT application on the SCADA system
- Can send events and alarms to the ALARM application on the SCADA system
- Can create, modify, and delete tags using the TAGGING application on the SCADA server via InterSite Tagging (IST)

i Note

ADMS does not support the Smart Tagging and Group Tagging features when integrated with **e-terrascada** 3.0 and later.

- Can create, modify, and delete safety documents using the SAFETY application on the SCADA server

Figure below shows the main communication links and components involved in the SCADA/ADMS integration.

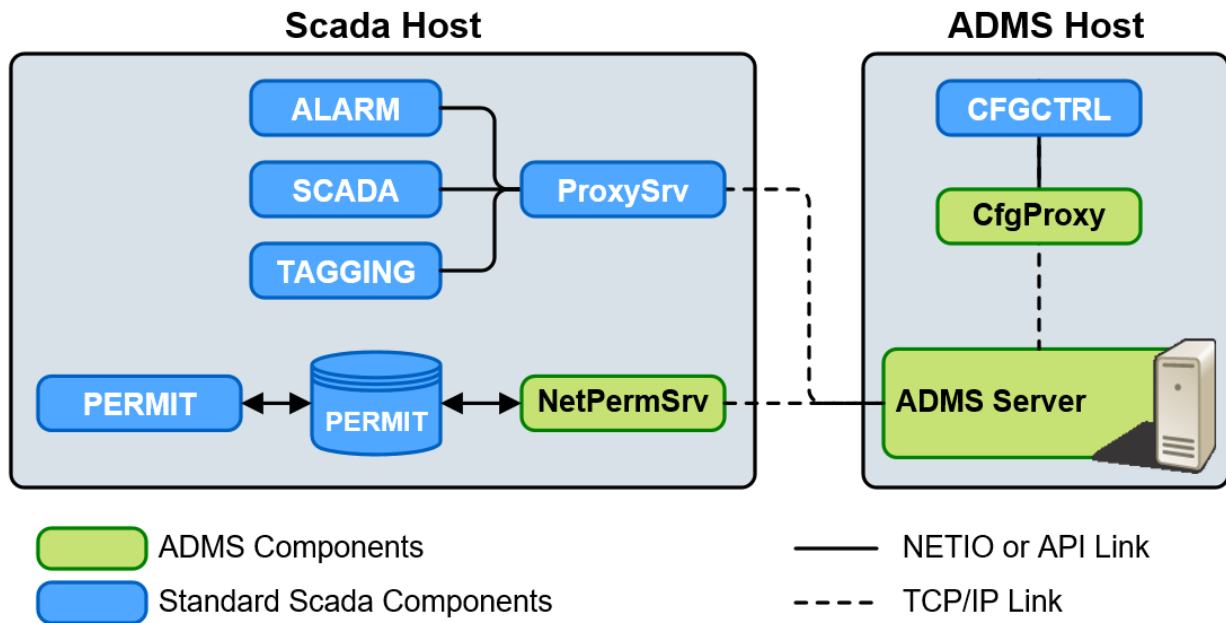


Figure 21. Scada/ADMS Communication

This figure is appropriate for configurations where SCADA and ADMS are on the same machine and on separate machines. The CfgProxy application passes CFGCTRL role assignments from the CFGCTRL application to the nonHabitatbased ADMS server applications. This application runs on the ADMS server machine. Follow the steps below to integrate this software.

The NetPermSrv application (running on the SCADA server machine) passes permission data from the PERMIT database hosted by the SCADA system to the ADMS server applications. Follow the steps below to integrate this software.

The NetPermSrv and CfgProxy source files are located in the %HABITAT_SRCDIR%\dms\ folder. They can be used to rebuild applications as appropriate.

Note

Habitat is leveraged on the ADMS server machines for the redundancy features it provides. Therefore, if the ADMS server applications are running on a separate machine than SCADA, install the appropriate version of Habitat on the ADMS servers before you proceed.

6.1. Configuration Manager Proxy Server

Note

Habitat installation already includes CfgProxy. CfgProxy, supplied with **e-terrah**abitat 5.10, only supports the DMS configuration. To support DMS and OMS configurations on ADMS with **e-terrah**abitat 5.10, you need to install CfgProxy patch supplied with **e-terrah**abitat 5.10. For more details, contact your GE project team.

The CfgProxy that runs on the ADMS host communicates between the ADMS server applications and CFGCTRL. Messages between CfgProxy and the server applications occur via TCP/IP. Messages between CfgProxy and CFGCTRL occur via the CFGMAN API.

Currently, the following ADMS applications interface with CfgProxy:

- CallServer
- CrewServer
- DnafSrv
- Net Data Server
- NetSrv
- NetServer
- Net Data Replicator

Because CfgProxy is a Habitatbased application, it is installed at %HABITAT_SRCDIR%\dms\cfgproxy\, and the user is not asked for an installation location. The installer then performs a loadhab operation, creates a clone, retrieves the default savecase, and sets a PROCMAN run flag (PROCMAN_FLAG_EMS_RUN_CFGPROXY).

Important

The clone creation and PROCMAN run flag (i.e., check-init) variable creation steps are based on the REALTIME_FAMILY environment variable. If this variable is not correctly set, you may need to recreate the clone and the PROCMAN variables under the proper family.

The default Scada PROCMAN model contains tasks for NetPermSrv, CfgProxy, and NetSrv (the DMS server application). These tasks do not have their init flags selected by default, but are configured to read the run flags.

For the CfgProxy application, the following additional operating system environment variable can be defined:

- CFGPROXY_PORT: If set, the application overwrites its default listening ports of 50155021 to the values of this parameter.

Note

This environment variable is retained only for legacy purposes. The ports should be defined in the CfgProxy clone as described below.

Based on the configuration, you may or may not need to define this environment variable.

Note

The listening port must match what is defined in the NetServerMainConfig.txt file (also called "CFGPROXY_PORT") and in the configuration files for CallServer. In a DMS/OMS installation, this port is also defined in the configuration files for CallServer, CrewServer, DnafSrv, Net Data

Server, NetServer, and Net Data Replicator.

CfgProxy also contains the following environment variable options:

- **CfgProxy_LOGDEBUG:** If set, the application logs additional debugging information.
- **CfgProxy_LOGTOCONSOLE:** If set, the application outputs all messages to the console and corresponding .out files.

For debugging purposes, it is recommended to set both of these environment variables.

The CfgProxy DBDEF file includes listening port numbers for each application that connects to CfgProxy. These ports are modeled in the CfgProxy clone in the CfgProxy database. The port field names are defined as follows:

Table 6. Application Port Field Names

Application	Port Field Names
Call Server	CALLPORT_ITEMS
Crew Server	CREWPORT_ITEMS
DNAF Server	DNAFPORT_ITEMS
Net Data Server	NETDPORT_ITEMS
DMS Main Server	NETSPORT_ITEMS
Net Server (OMS Server)	OMSPORT_ITEMS
Net Data Replicator	REPLPORT_ITEMS

Warning

The listening port field names are not configurable. Only the port numbers are configurable.

The default CfgProxy savecase contains listening port numbers for each application that connects to CfgProxy.

Table 7. Application Executable Files, Configuration Files, and Port Numbers

Application	Executable	Configuration File	Port Number
Call Server	CallServer.exe	CallServerConfig.xml	5015
Crew Server	CrewServer.exe	CrewServerConfig.xml	5016
DNAF Server	DnafSrv.exe	NetServerDnafConfig_localhost.txt	5017
Net Data Server	NetDataServer.exe	NetDataServerConfig_localhost.txt	5018
DMS Main Server	NetSrv.exe	NetServerMainConfig_localhost.txt	5019
Net Server (OMS Server)	NetServer.exe	NetServerOmsConfig_localhost.txt	5021
Net Data Replicator	NetDataReplicator.exe	NetDataReplicatorConfig.xml	5020

Warning

The listening port numbers modeled in the CfgProxy clone must match the corresponding port numbers defined in the configuration file for each server application.

6.1.1. Updating the Clones.dat File

You may need to update your clones.dat file to describe the appropriate savecases for CfgProxy. As you modify or create additional savecases, update the clones.dat file to reflect those changes as well.

Modify your clones.dat file to include the following:

```
! -----  
! ADMS : DMS  
! -----  
CfgProxy %RT_FAMILY% case_CfgProxy_CfgProxy.dms_default
```

6.2. Permission Proxy Server

Note

Habitat installation already includes NetPermSrv. NetPermSrv must not be installed separately for ADMS.

For the NetPermSrv application, the following additional operating system environment variables can be defined:

- **NETPERMSRV_USE_LOGFILES:** If set, the application logs additional output messages into the NetPermSrvPorts_<family_name>.out file.
- **NETPERMSRV_NOCFGMAN:** If set, the application does not request role assignments from CFGMAN. This variable is used to run NetPermSrv in a standalone mode.
- **NETPERMSRV_PORT:** If set, the application overwrites the default listening port of 5345 to the value of this parameter.

Based on the configuration, the engineer may or may not need to define these environment variables.

Note

The listening port must match what is defined in the NetServerMainConfig.txt file (labeled "PERMIT_SERVER_PORT").

6.3. Modifying the CFGCTRL Model



Important

The default CFGCTRL savecase (case_cfgctrl_cfgctrl.esca_emp_sample) delivered with Scada already includes PROCESS records for NetPermSrv and CfgProxy. The information in this section only describes how to manually re-create the configurations as in the default savecase.

Note

The ADMS servers are **not** modeled in CFGCTRL as a PROCESS record because they are not Habitatbased applications. The purpose of CfgProxy is to be a liaison between CFGCTRL and NetSrv (in the case of a DMS-only installation), or several ADMS applications including CallServer, CrewServer, DnafSrv, Net Data Server, NetSrv, NetServer, and Net Data Replicator (in the case of the DMS/OMS installation).

As with the PROCMAN model, depending on your hardware configuration i.e., simplified vs. distributed (for details, see [Supported Hardware Configurations](#)) you may have one or two CFGCTRL models to modify.

The following sections explicitly indicate which CFGCTRL models to modify (SCADA server or ADMS server). If you have a simplified configuration (one pair of servers only), you need to perform all the CFGCTRL model modifications on your combined SCADA/ADMS server.

6.3.1. Adding NETPERMSRV to the CFGCTRL Model (SCADA Server)

On the SCADA server, the NETPERMSRV process is added to the CFGCTRL model.

Add a process for the NETPERMSRV task, as shown in figure below.

The screenshot shows the configuration window for the NETPERMSRV process. The title bar reads "Process (52) : NETPERMSRV". The window is divided into several sections:

- Owned By:** A radio button is selected for "Application Set" with the value "1" and the label "EMSHOST". The "Local System" option is unselected.
- Timeouts:** Three input fields are present: "Disable Role Timeout:" with the value "30", "Stall Check Interval Timeout:" with the value "0", and "Other Role Confirm Timeout:" with the value "30".
- Participates in Process Set:** An unchecked checkbox.
- Enabled role should be resent:** An unchecked checkbox.
- Crash System on Failure:** An unchecked checkbox.
- Contributes To Status of Owner:** An unchecked checkbox.

Figure 22. NETPERMSRV CFGCTRL Process

6.3.2. Adding CfgProxy to the CFGCTRL Model (ADMS Server)

When a server application connects to CfgProxy, the first message it sends is an "identify" message. The identify message includes the process name that CfgProxy registers with CFGMAN. Each server application uses a unique process name, which is modeled in the corresponding server application configuration file. The process names are also modeled in the CFGCTRL database.

For each of the CfgProxy-related processes on your system, create a corresponding process flag

and node records as shown below.

The default CFGMAN process names are defined as follows:

Table 8. Application CFGMAN Process Names

Application	CFGMAN Process Name
CallServer	CALLSERVER
CrewServer	CREWSERVER
DnafSrv	DNAFSERVER
NetDataServer	NETDATASRV
NetSrv	NETSERVER
NetServer	OMSSERVER
NetDataReplicator	REPLICATESRV

 Warning

The CFGMAN process names in the server application configuration files must match those modeled in the CFGCTRL database.

CFCTRL contains timeouts for the role confirmation messages. If the process does not confirm its role before the timeout expires, the CFGCTRL stops the process. There are separate role confirmation timeouts for DISABLED role assignments and other role assignments (for example, ENABLED and STANDBY). For more details, refer to the *Configuration Manager User Guide* in the Habitat documentation set.

On the ADMS server, the CfgProxy processes are added to the CFGCTRL model. Add a process for each of the CfgProxy tasks, as shown in figure below.

Configuration Control Definition

Dataset Validation

Process (17) :

NETSERVER

Owned By

Application Set

1

EMSHOST

Local System

Disable Role Timeout:

300

Stall Check Interval Timeout:

0

Other Role Confirm Timeout:

300

Participates in Process Set:

Crash System on Failure:

Enabled role should be resent:

Contributes To Status of Owner:

Process (18) :

REPLICATESRV

Owned By

Application Set

1

EMSHOST

Local System

Disable Role Timeout:

300

Stall Check Interval Timeout:

0

Other Role Confirm Timeout:

300

Participates in Process Set:

Crash System on Failure:

Enabled role should be resent:

Contributes To Status of Owner:

Process (19) :

NETPERMSRV

Owned By

Application Set

1

EMSHOST

Local System

Disable Role Timeout:

30

Stall Check Interval Timeout:

0

Other Role Confirm Timeout:

30

Participates in Process Set:

Crash System on Failure:

Enabled role should be resent:

Contributes To Status of Owner:

Process (20) :

OMSSERVER

Owned By

Application Set

1

EMSHOST

Local System

Disable Role Timeout:

120

Stall Check Interval Timeout:

0

Other Role Confirm Timeout:

120

Participates in Process Set:

Crash System on Failure:

Enabled role should be resent:

Contributes To Status of Owner:

Figure 23. Adding CfgProxy CFGCTRL Processes

6.3.3. Using CFGMONI between SCADA and the ADMS Server

This section only applies to the distributed configuration (two pairs of servers).

CFGMONI can be used to propagate CFGMAN system information from the ADMS server to the SCADA server. In other words, the main EMS site overview display can monitor the health of the ADMS servers as well.

There is nothing specific to ADMS in this process. It is similar to bringing back Comm CFGMAN data. For more information, refer to the CFGCTRL/CFGMONI documentation, including the *CFGTEAM User Guide* (included in the Habitat documentation set).

6.3.4. Configuring CFGCTRL Timers for a Multi-Host ADMS System

This section describes how to configure the Other Role Confirm Timeout and Disable Role Timeout CFGCTRL timers that influence the failover processes on a multi-host ADMS system. If these times

are set to low values, issue may occur after failover, such as the following: server processes may not stop, duplicate connections may appear, or CFGCTRL may terminate a process that not yet received an Enabled or Standby role from CfgProxy.

- **Other Role Confirm Timeout:** Role confirmation timeout (in seconds) when the enabled or standby role is assigned to the process. If the process does not confirm the new role in the specified amount of time, CFGCTRL ends the process and lets the system handle that event as it normally does. If the value is zero, CFGCTRL gives the process 30 seconds to confirm its role.

The Other Role Confirm Timeout timer should be larger than CFGPROXY timers defined using the CFGPROXY_TIMER_PERIOD and CfgProxyTimerPeriod parameters in server configuration files. The recommended value is 300 seconds.

- **Disable Role Timeout:** Role confirmation timeout (in seconds) when the Disable role is assigned to the process.

The Disable Role Timeout timer should be larger than CFGPROXY timers defined using the CFGPROXY_TIMER_PERIOD and CfgProxyTimerPeriod parameters in server configuration files. The recommended value is 300 seconds.

Set the Other Role Confirm Timeout and Disable Role Timeout timers to 300 seconds for the following processes:

- CALLSERVER
- CREWSERVER
- OMSSERVER
- NETSERVER
- NETDATASRV
- DNAFSERVER
- REPLICATESRV

CFGCTRL timers, including the Other Role Confirm Timeout and Disable Role Timeout timers, can be changed using the CFGCTRL Definition display or using a command prompt.

To update CFGCTRL timers from the CFGCTRL Definition display:

1. Open a command prompt.
2. Enter the following commands:

```
tview r hdbserver hdb hdb  
tview r procman procman habitat
```

3. Run the client.
4. Navigate to HABITAT Applications > Configuration Control > CFGCTRL Master.
5. Navigate to Analyst Displays > CFGCTRL Model Definition > CFGCTRL Definition.
6. Update timers for the specified processes (CALLSERVER, OMSSERVER, etc.)

Configuration Control Definition

[Dataset Validation](#)

Process (3) : **CALLSERVER**

Owned By

- ☒ Application Set **1** **EMSHOST**
- ☐ Local System

Disable Role Timeout: **300**

Stall Check Interval Timeout: **0**

Other Role Confirm Timeout: **300**

Participates in Process Set: ☐

Crash System on Failure: ☐

Enabled role should be resent: ☐

Contributes To Status of Owner: ☐

Essential to Site-State change: ☐

Do not terminate after Site-State change timeout: ☐

Response to Request Site State Change Timeout: **0**

7. Validate the CFGCTRL dataset (click Dataset Validation > Run Verify).

To update CFGCTRL timers from a command prompt:

1. Open a command prompt.
2. Enter the following commands to update the CFGCTRL timers for the CALLSERVER process:

```
tview r hdbserver hdb hdb
Context cfgctrl
hdbrio cfgctrl
p process
find process=callserver
/rolecfm=300 {'Other Role Confirm Timeout' timer}
/discfrm=300 {'Disable Role Timeout' timer}
```

3. Repeat step 2 to update the CFGCTRL timers for the other processes.
4. Validate the CFGCTRL dataset (cfgctrl_verify).

For more details about configuring CFGCTRL, refer to the *Habitat Configuration Manager User Guide*.

6.4. Configuring the ADMS/SCADA ISD Link

To allow ADMS/SCADA ISD communication, an ISD site record must be added in the SCADA database. The parameter values used to define this site need to match the ISD parameters located in the `NetServerMainConfig.txt` configuration file. The main parameters are presented in the following sections.

In the sample SCADA savecase (case_scadamdl_ade.fantasyisland), the ISD links are all properly configured to work with the example configuration files for the ADMS servers.

i Note

For ADMS to retain SCADA time using ISD, it is necessary to set `ISEXTEND_ITEMS(1) = True`.

6.4.1. Adding the ISD Site for ADMS to Scada

There are a few key parameters that need to be properly set up for the ISD link to work. In particular, the NETIO objects for InterSite Data, InterSite Control, and InterSite TAGGING need to match the NETIO objects contained in NetServerMainConfig.txt (see [Setting Up the ADMS Server ISD Parameters](#)).

! Important

The "SMP Type" parameter must be set so the ISD link goes through ProxySrv, rather than using a direct NETIO link.

Figure below shows an example of an ISD site record for ADMS. Specifically, this is how the example SCADA savecase is configured. The two sites "NS" and "NR" stand for Simulator server and RealTime DMS server, respectively.

Figure 24. ISD Site for ADMS As Seen with the SCADA Online Editor

i Note

It is recommended that you name the DMS site "DMS" instead of using the "NR" convention from the Fantasy Island example.

6.4.2. Setting Up the ADMS Server ISD Parameters

To configure InterSite Data, TAGGING, and Control communications paths with SCADA, set the ISD parameters within NetServerMainConfig.txt. For more information about the parameters, see

INTERSITE DATA (SCADA CONNECTION) in [NetServerMainConfig.txt](#) Parameters.

For running a server in study mode, the Alarm parameters, ISD parameters, and Permit parameters are not used. The server port must be identical to what was defined in `netconf.xml`, and `MODEL_PATH` indicates where the `.mod` files are stored that get used for the study mode. The `DYNAMIC_PATH` and the `DYNAMIC_NAME` indicate the directory and the name of the dynamic file.

6.4.3. Additional Tagging Communication Configuration

Tags placed on DMS devices are stored with other SCADA tags in the SCADA system. The DMS server application receives the DMS tags from SCADA using an InterSite Tagging link.

To enable the communication link, the "InterSite Tagging Enabled" parameter of the DMS ISD site must be set to True in the SCADA database (see figure above) and the main SCADA site (`SYSTEM_SITE` in SCADAMOM).

The DMS server uses the InterSite Tagging (IST) protocol version 2.1.1, which means that on the Scada system, the following environment variable definition is needed to force the IST communication protocol to 2.1.1:

```
TAGNOTES_22_USING_211_IST
TAGNOTES_22_USING_211_DMS_IST
```

Both of these environment variables should be defined as True.

Since InterSite Tagging uses NETIO, there must be NETIO system records that match the SCADA site record system names. The system records must be modeled under the local SCADA system's NETIO site.

Figure 25. NETIO Configuration Example for InterSite Tagging

6.5. Tagging Configuration

In addition to the ISD/IST configuration described in the previous section, the following configuration tasks must be performed to get tagging fully working for your system.

6.5.1. Selecting/Configuring Tag Types for Use in ADMS

The ADMS devices that are also modeled in SCADA are handled by the SCADA-based tagging as if ADMS was not involved. Contrarily, the ADMS-only (i.e., DNOM-only) devices are tagged as "non-modeled" tags within the SCADA-based Tagging application. These non-modeled tags are stored as substation tags.

Listed below are the possible tag types in SCADA using HDBRIO:

- context scada ems
- hdbrio scadamom
- pos tagtyp
- /id(*)

For certain tag-types such as STICKY, NOSHED, and PREVENT_SETPOINT, it does not make sense to make them available for use on DNOM-only devices like manual fuses. For the tag types that do make sense for DNOM-only devices, make sure those records have the SUBSTN_TAGTYP field set to TRUE.

For example:

- find tagtyp=info
- /substn=true

If the ANOTEDMS_TAGTYP flag is set to True for a specific tag type in the SCADAMOM database, the tag will be treated as an annotation instead of a tag and will not be displayed on the ADMS geographic display or on any tag lists. Set ANOTEDMS_TAGTYP to False for the tags to be correctly displayed on the geographic and all tag lists.

You must stop and restart the tagging application to test these changes. When these changes are tested and agreed upon, be sure they make it into the proper savecase or SCADA modeling environment.

For more details about configuring and validating the TAGGING database, refer to the *Scada System Installation and Maintenance Guide*.

6.5.2. Tagging Configuration in NetServerMainConfig.txt

The following lines reflect a potential setup from the NetServer configuration file. This example uses the default "NR" site name.

```
'-----  
InterSite Data (SCADA Connection) Parameters  
'-----  
ISD_LOCAL_SITE           = NR  
ISD_LOCAL_SCANNER_PATH   = NR.SCADA_NR_ISD  
ISD_LOCAL_CONTROL_PATH   = NR.SCADA_NR_CTRL  
ISD_LOCAL_TAGNOTES_PATH  = NR.TAGGING_NR_TGDATA  
ISD_REMOTE_SITE          = EMS  
ISD_REMOTE_SCANNER_PATH  = EMS.SCADA_EMS_ISD  
ISD_REMOTE_CONTROL_PATH  = EMS.SCADA_EMS_CTRL  
ISD_REMOTE_TAGNOTES_PATH = EMS.TAGGING_EMS_TGDATA  
ISD_REMOTE_HOST1         = 127.0.0.1  
'ISD_REMOTE_HOST2       = EMSB  
'ISD_REMOTE_HOST3       =  
'ISD_REMOTE_HOST4       =  
ISD_REMOTE_PORT          = 5001  
ISD_REMOTE_POLLRATE      = 2  
ISD_REMOTE_INTEGRITYRATE = 900  
ISD_MAX_DOWNLOAD_TIME    = 300  
ISD_KEEPALIVE_TIMEOUT    = 300  
'ISD_DETAILED_MESSAGES  = 1  
'ISD_SKIP_MESSAGES_PATH_DOWN = 1  
ISD_ANALOG_VALUE_MESSAGES = FALSE  
'ISD_DEFAULT_SITE       =  
USE_NEW_ISD_LOGIC        = 1  
ISD_RECONNECT_TIMEOUT    = 5  
'ISD_DUMP_SOCKET_TRAFFIC = 1  
'ISD_CTRL_STATUS_ON_OFF =
```



```
12:Closed#CLOSE:Open#TRIP,14:On#CPON:Off#CPOF,111:Enabled#ENABLE:Disabled#DISABLE  
'ISD_HIGHEST_PROTOCOL = 7
```

6.5.3. Tagging Configuration in ViewerConfig.xml

In cases where multiple tags are applied to a single device, the normal display mode is to show a single tag symbol with a plus sign ("+"), indicating that multiple tags have been applied. The color of the tag symbol is set to the highest-priority tag that is currently applied. This color priority is defined in the Viewer Configuration Editor.

The `ViewerConfig.xml` file (edited using the Viewer Configuration Editor) contains all commands associated with toolbar buttons and right-click menus. The commands for tagging are therefore defined within this file.

By default, the commands are targeted for TAGGING running under the EMS family. If this is not the case for your system, you need to modify the file accordingly.

Although use of the editor is strongly encouraged, in this instance, doing a search and replace of `SCADA_EMS` with `SCADA_{YourFamily}` is also effective.

Also, the bitmaps associated with the tags (specifically the colors) must be configured on the Bitmaps tab within the Viewer Configuration Editor.

To configure the bitmaps:

1. Start the Viewer Configuration Editor and open the appropriate `viewerconfig.xml` file.
2. Select the Bitmaps tab.
3. On the DMS Objects combo-box, select `dmsTag`.
4. Use the Subset combo-box and the main form to adjust the first 1 through `n` tags to match your project set up in terms of label and bitmap (as seen/configured in the Bitmap Preview column).
5. Clear the labels on any unused tag types to reduce possible confusion in the future.

Note

The tags in SCADAMOM under the TAGTYP records may not be defined in order of priority.

For example, when tag types are derived via the `/ID_TAGTYP(*)` command and copied into Viewer Configuration Editor, the tag list may be defined as shown below:

```
TAGTYP(1) ---- PRIOR1_TAGTYP = TRUE  
TAGTYP(2) ---- PRIOR3_TAGTYP = TRUE  
TAGTYP(3) ---- PRIOR5_TAGTYP = TRUE  
TAGTYP(4) ---- PRIOR4_TAGTYP = TRUE  
TAGTYP(5) ---- PRIOR2_TAGTYP = TRUE  
TAGTYP(6) ---- PRIOR6_TAGTYP = TRUE  
TAGTYP(7) ---- PRIOR7_TAGTYP = TRUE
```

Note that the tag types are not defined in order of priority. In this case, the Viewer Configuration Editor definitions need to be re-ordered to match.

6.5.4. Additional Tagging Items

For tagging on the ADMS to function, permissions must be enabled on the ADMS. For a detailed description, see [Configuring the Link to PERMIT \(ADMS Host\)](#).

Once permissions are enabled, the AORs in DNOM (or assigned to DNOM objects) and the rights of the Browser user must match for the user to be allowed to place a tag. To help with troubleshooting, use the Browser message window to see the commands and proxy-resolutions that are being sent from the tagging commands. The AREA value (i.e., AOR) should be passed in the command.

For help with tagging, a troubleshooting-only ADMS tabular display is provided. This display lists the tags that the DMS server application knows about because they have been successfully transmitted from the SCADA Tagging application. Use the following syntax to manually call up this display:

```
Netview://rt/griddtabular/tagsummary_griddisplay
```

To test the integration incrementally, once everything has been configured perform the following steps:

1. To ensure that all configuration/modeling changes are current, stop and restart the TAGGING application and restart the ADMS system.
2. Verify that the tagging connection was properly formed by reviewing the proxysrv/log display in Browser. The following PXS004 messages should be present (in addition to the "opening" PXS004):

NETIO path opened from DMS.TAGGING_DMS_TGDATA to EMS.TAGGING_EMS_TGDATA

Note

Your specific TAGGING path names may be different.

If this does not work, troubleshoot your ISD/SITE/NETIO configuration; otherwise, proceed to the next step.

3. Select a SCADA breaker that is correctly modeled between SCADA and DNOM.
4. Open two Browser viewports: one for the SCADA oneline and another for the ADMS display showing the SCADA device (for example, station internal display).
5. On the SCADA oneline display, toggle or operate the breaker. If the corresponding device changes on the ADMS display, the modeling is correct and you may proceed with tagging it; if not, you need to fix the model or find a device that is modeled properly before proceeding to the next step.

6. Tag the properly modeled device from the SCADA oneline.
 - If the tag appears on the ADMS display's device, proceed to the next numbered step.
 - If the tag does not appear on the SCADA oneline, you need to fix SCADA Tagging first.
 - If a "beachball" symbol is presented on the ADMS display instead of a colored tag bitmap, the tagging communication is thus far correct but the Viewer Configuration settings on the client are incorrect.
 - If nothing appears, the IST connection between ADMS and SCADA is not working.
7. Select a DNOM-only switch on the geographic display and tag it.
 - If the appropriate tag types do not appear on the ADMS display, check the DMS Main server log for error messages.
 - If the appropriate tag types are not all presented on the tagging pop-up, revisit setting the SUBSTN_TAGTYP fields in SCADAMOM. See [Selecting/Configuring Tag Types for Use in ADMS](#).
 - If you get a TAGNOPERM error message using the Browser Messages tab, assess what AOR is getting passed in the command to the SCADA server. If the AOR does not exist in PERMIT, or if you simply do not have permissions for that AOR, or if the AOR is blank, that must be corrected before tagging will work.
 - If a "beachball" symbol is presented on the ADMS display instead of a colored tag bitmap, the tagging communication is thus far correct but the Viewer Configuration settings on the client are incorrect.
8. Once tags are properly showing up on the geographic display with their proper coloring, place one of each type individually and verify that all open/close-action restrictions function properly.

6.5.5. Specific Configurations for e-terra_scada_Versions 3.0 and Later

Compared to **e-terrascada_version 2.6**, versions 3.0 and later contains certain differences. For example, the TAGNOTES application name in **e-terra_scada 2.6** changed to TAGGING in **e-terrascada 3.0**. Therefore, some applications may not be completely functional when using earlier configurations.

6.5.5.1. Configuring ADMS Server

For configurations with **e-terra_scada_versions 3.0** and later, make the following changes:

1. Replace the DmsProxy.dll file in %HABUSER_BINDIR% with the DmsProxy.dll file that is supplied with the ADMS server and located in %DMS_BINDIR%. This allows the **e-terra_scada_versions 3.0** and later Tagging application to retrieve DNOM information directly, without using the RUSER parameter, which is necessary for interacting with ADMS.

When running Scada on the Linux platform, use the Linux version of the DMS Proxy file, libDmsProxy.so. For more information, contact your GE project team.

2. In the NetServerMainConfig.txt file, uncomment the service module listening socket and socket parameters:

```
SERVICE_PORT = 6032
SERVICE_SOCKET_OUTPUTBUF_MAX_SIZE = 1000000
SERVICE_SOCKET_WOULDBLOCK_RETRY = 2
SERVICE_SOCKET_MAX_CLIENT_MES_SIZE = 640000
```

The service module is used by the *e-terra*_scada_versions 3.0 and later Tagging and Safety Documents applications to retrieve data from ADMS.

3. In the NetServerMainConfig.txt file, replace EMS.TAGGING_NR_TGDATA with NR.TAGGING_NR_TGDATA:

```
ISD_LOCAL_TAGNOTES_PATH = NR.TAGGING_NR_TGDATA
```

4. Replace SCADA displays with the displays that are built specifically for **e-terra**_scada_versions 3.0 and later.

Replace files in %HABITAT_WEBROOT%\displays\scada\ with files from %IDMS_ROOT%\..\FantasyIsland\misc\WebFGRoot\displays\scada31\.

5. Navigate to the IST Site Config tab of the Tagging Control Panel display (SCADA Applications menu > Tagging > Tagging Configuration > IST Site Config).
6. Edit the IST Site "NR" entry:
 - Enable the Connection Through the Proxy Server option if available. (Alternatively, set the THROUGHPROXY_ISTSITE record to True in the TAGGING database using the Habitat HDBRIO tool.)
7. When integrating with Scada 3.4, uncomment the following command in the NetServerMainConfig.txt file to enable the Scada ISD protocol version 7 and support the related features, such as the Scada load shed bit for DNOM objects.

ISD_HIGHEST_PROTOCOL = 7

6.5.5.2. Configuring Host Names

Note

This section applies only to Scada 3.4 or later.

When using Scada 3.4 or later, perform the following steps:

1. Navigate to the IST Site Config tab of the Tagging Control Panel display (SCADA Applications menu > Tagging > Tagging Configuration > IST Site Config).
2. Edit the IST Site "NR" entry:
 - Enable the Connection Through the Proxy Server option. (Alternatively, set the

THROUGHPROXY_ISTSITE record to True in the TAGGING database using the Habitat HDBRIO tool.)

- When Tagging and the DMS server are deployed on the same machine, set the First Host Id to 0.0.0.0 or 127.0.0.1 instead of LOCALHOST.

The screenshot shows the ProxySrv Configuration window. At the top, there are fields for 'IST Site: NR', 'Conn. Status: UP' (highlighted in green), and 'System Name: NR'. Below these are several configuration options: 'Path Definition: %SYSTEM%.TAGGING_%ID%.TGDATA', 'Communications Path: NR.TAGGING_NR.TGDATA', 'Protocol Version: 21', and 'Controlling flags: SEND: [checked], RECEIVE: [checked], FORWARD: [unchecked]'. There are also checkboxes for 'NIS: [unchecked]', 'Inhibit: [unchecked]', and 'FOWARDED: [unchecked]'. The 'Connection through proxy server:' checkbox is checked. Under 'Type: DMS', there are four host ID fields: 'First Host ID: 0.0.0.0', 'Second Host ID: [empty]', 'Third Host ID: [empty]', and 'Fourth Host ID: [empty]'. A 'Description:' field is at the bottom.

- When Tagging and the DMS server are deployed on different machines and you want to use DMS server host names instead of IP addresses, select the Get Host Name option in the ProxySrv Configuration display (To show this display, navigate to SCADA Applications menu > SCADA > Substation Directory; Analyst Displays menu > ProxySrv Configuration > Get Host Name)

Alternatively, set the GETHOSTNAME_ITEMS record to True in the PROXYSRV database using the Habitat HDBRIO tool.

This screenshot is identical to the one above, showing the ProxySrv Configuration window with the same settings: 'IST Site: NR', 'Conn. Status: UP', 'System Name: NR', 'Path Definition: %SYSTEM%.TAGGING_%ID%.TGDATA', 'Communications Path: NR.TAGGING_NR.TGDATA', 'Protocol Version: 21', 'Controlling flags: SEND: [checked], RECEIVE: [checked], FORWARD: [unchecked]', 'NIS: [unchecked]', 'Inhibit: [unchecked]', 'FOWARDED: [unchecked]', 'Connection through proxy server: [checked]', 'Type: DMS', and host IDs: 'First Host ID: 0.0.0.0', 'Second Host ID: [empty]', 'Third Host ID: [empty]', 'Fourth Host ID: [empty]'. A 'Description:' field is at the bottom.

The screenshot shows the 'PROXYSRV Configuration' window. The title bar is blue with 'PROXYSRV' in a white box and 'Configuration' in white text. The main content area has a dark background. It features a section titled 'Get Host Name' with a checkbox 'Get Host Name: [checked]'. Below this is a text block: 'Info: if this flag is FALSE, all logs will display <NOT RESOLVED> for the node instead of computer name. Use this setting ONLY if your DNS server or host file is not accessible in order to prevent PROXYSRV from hanging.' At the bottom, there is a warning: 'Warning: This flag should always be TRUE if SCADA SITE configuration uses host name instead of IP address for the valid IP addresses'.

6.5.5.3. Configuring the ISTSITE Records

The Inter-Site Tagging (IST) connection between DMS and TAGGING requires two IST records in the TAGGING database to be configured one for the remote (DMS) IST connection and another for the local (EMS/SCADA) IST connection.

The remote DMS IST connection can be configured from the TAGGING IST configuration display.

For configurations with **e-terra**_scada_versions 3.0 and later, make the following changes:

1. On the Configuration tab of the Tagging Control Panel display (SCADA Applications menu > Tagging > Tagging Configuration), set the DMS Synchronization parameters, where the Port value equals the SERVICE_PORT value in the NetServerMainConfig.txt file.

The screenshot shows the 'DMS Synchronization' configuration window. It contains several input fields for configuring the DMS connection. The 'Interval (seconds)' is set to 0, and 'Max Substations to process per synch' is also set to 0. The 'Synchronize Unlinked' checkbox is unchecked. The 'Retry Time(Seconds)' is set to 10, and the 'Timeout(ms)' is set to 120000. There are four sections for DMS Servers. The first section, 'DMS Server #1:', has 'Host Name' set to 'LOCALHOST' and 'Port' set to '6032'. The other three sections, 'DMS Server #2:', 'DMS Server #3:', and 'DMS Server #4:', each have 'Host Name' and 'Port' fields that are currently empty.

Parameter	Value
Interval (seconds)	0
Max Substations to process per synch	0
Synchronize Unlinked	☐
Retry Time(Seconds)	10
Timeout(ms)	120000
DMS Server #1: Host Name	LOCALHOST
DMS Server #1: Port	6032
DMS Server #2: Host Name	
DMS Server #2: Port	
DMS Server #3: Host Name	
DMS Server #3: Port	
DMS Server #4: Host Name	
DMS Server #4: Port	

Set other Tagging configuration parameters as needed.

TAGGING

Control Panel

Modifications only take effect after TAGGING is restarted.

Status

Configuration

IST Site Config

Default Tag Types Configuration

Name of Default Tag Type:

INFO

Name of Default Tag Group Type:

DEFAULT

Alarm Configuration

Permit Areas

Location for Group:

TSTSCADA

Default Area for Group:

EAST

Location for Tag:

TSTSCADA

Default Area for Tagging:

EAST

Category for Tagging:

TAGGING

Default Area for Test Mode:

TESTMODE

Restrictions

Logging Flags

Require Permit Checks:

☒

Function Trace:

☐

Require Execute Privilege:

☐

Debug Log:

☒

Require Analyst for Removal:

☐

API Log:

☐

Legacy Console Support:

☐

API Dump:

☐

Allow 'Created By' modification:

☐

IST Log:

☒

Allow 'Created For' modification:

☐

IST Dump:

☐

Allow Tag Type changes:

☐

RUSER Log:

☐

Disable Archive on Sync:

☐

DMS Dump:

☒

Other General Configuration

Autogroup Assignment:

☐

Max Comment Lines:

0

IMS Resync Time:

0

IMS Request Timeout:

1

IMS Max Sync Objects:

0

API Principal:

API_PASSWORD

Functional Permission Checking

Enable Functional Permission Checking:

☐

Switching:

No Checking:

2. On the IST Site Config tab of the Tagging Control Panel display (SCADA Applications menu > Tagging > Tagging Configuration), add a node for the NR site. Set the node properties as follows:

- Local Site: EMS

This is the name of the local IST TAGGING site. This value must match the corresponding SCADA ISD local SITE record ID. Configuration modeling requirements for the local IST sites are similar to those for the remote IST sites. However, the current versions of the TAGGING IST configuration display do not allow you to enter the required values. Therefore, you must use Habitat HDBRIO tool to enter the values for the local IST site.

- IST Site: NR

This is the remote IST site for the DMS system. The "IST Site" field value must match the

corresponding SCADA Inter-Site Data (ISD) SITE record ID used to configure the connection between Scada and the DMS system.

- SystemName: NR
- Protocol version: 21
- Path Definition: %SYSTEM%.TAGGING_%ID%_TGDATA
- Communications Path: NR.TAGGING_NR_TGDATA (populated automatically)

The Path Definition field value must match the value specified by the ISD_LOCAL_TAGNOTES_PATH parameter in the NetServerMainConfig.txt configuration file (for example, "ISD_LOCAL_TAGNOTES_PATH = NR.TAGGING_NR_TGDATA").

- Type: DMS (use the right-click menu to select DMS)
- First Host ID: Localhost

The four Host ID fields are used to specify the IP addresses of up to four DMS system servers that can connect to TAGGING.

- Control flags: Send/Receive
- Host IDs: Similar to host IDs in the NetServerMainConfig.txt.

3. After adding the NR node, click Validate Model.

4. Click Apply Model.

5. Restart the Tagging application.

The screenshot shows the 'TAGGING Control Panel' with a dark blue header. Below the header are two buttons: 'Validate Model' and 'Apply Model'. The main area has three tabs: 'Status', 'Configuration', and 'IST Site Config' (which is selected). Under the 'IST Site Config' tab, the following fields are visible:

- Local Site: EMS
- System Name: EMS
- Description: ENERGY CONTROL CENTER
- IST Site: NR
- Conn. Status: UP (indicated by a green bar)
- System Name: NR
- Path Definition: %SYSTEM%.TAGGING_%ID%_TGDATA
- Communications Path: NR.TAGGING_NR_TGDATA
- Protocol Version: 21
- Controlling flags: SEND: ☒ RECEIVE: ☒ FORWARD: ☐
- NIS: ☐ Inhibit: ☐
- Type: ▴ DMS
- First Host ID: LOCALHOST
- Second Host ID: (empty field)
- Third Host ID: (empty field)
- Fourth Host ID: (empty field)
- Description: (empty field)

6.5.5.3.1. Configuring the LOCAL_ISTSITE Record using HDBRIO

Open the TAGGING database in HDBRIO and position on the ISTSITE record that has the local IST site ID value (in the current example, "EM") in the ID_ISTSITE field.

The example below shows a list of the field's values for the local IST site. The important field values are:

- LOCAL_ISTSITE: This field must be set to "True".
- ID_ISTSITE: This field must be set to a value that matches the local ID_SITE record value in SCADA.
- OBJDEF_ISTSITE and OBJECT_ISTSITE: These field values must match the value specified by the ISD_REMOTE_TAGNOTES_PATH parameter in the NetServerMainConfig.txt configuration file (for example, "ISD_REMOTE_TAGNOTES_PATH = EM.SCADA_EM_TGDATA").

```
ITAGGING.TAGGING.EMS I
ISTSITE<2>> read
Display of record ISTSITE<2>
=====
BitContainer  FLAGS_ISTSITE = B*4
  Bit: 0      NIS_ISTSITE = False
  Bit: 1      INHIBIT_ISTSITE = False
Integer*4     PROTOCOL_ISTSITE = 0
BitContainer  TYPE_ISTSITE = B*4
  Bit: 0      LOCAL_ISTSITE = True
  Bit: 1      PEER_ISTSITE = False
  Bit: 2      PSC_ISTSITE = False
  Bit: 3      DMS_ISTSITE = False
BitContainer  USAGE_ISTSITE = B*4
  Bit: 0      SENDDATA_ISTSITE = False
  Bit: 1      RECUDATA_ISTSITE = False
  Bit: 2      FWDALL_ISTSITE = False
Char*80       DESC_ISTSITE = ENERGY CONTROL CENTER <EMP>
Char*20       HOSTID01_ISTSITE =
Char*20       HOSTID02_ISTSITE =
Char*20       HOSTID03_ISTSITE =
Char*20       HOSTID04_ISTSITE =
Char*8        ID_ISTSITE = EM
Char*56       OBJDEF_ISTSITE = EM.SCADA_EM_TGDATA
Char*52       OBJECT_ISTSITE = EM.SCADA_EM_TGDATA
Char*16       SYSTEM_ISTSITE = EMS
Integer*8     OID_ISTSITE = 4685444189003545688
BitContainer  COMSTATE_ISTSITE = B*4
  Bit: 0      UNINIT_ISTSITE = True
  Bit: 1      CONNPEND_ISTSITE = False
  Bit: 2      CONN_ISTSITE = False
  Bit: 3      FAIL_ISTSITE = False
Char*15       CONNECTIONSTATUS_ISTSITE = UNKNOWN
```

Figure 26. Configuring the LOCAL_ISTSITE Record using HDBRIO

The TAGGING database must be re-validated for any changes to the TAGGING database records and fields referenced above.

To validate the TAGGING database, do one of the following:

- If TAGGING is running, click the Validate Model button on the TAGGING IST Configuration display.
- If TAGGING is not running, use the tagging_utility.exe command line tool to validate the

tagging database. For example, execute the following commands:

```
Context tagging ems
tagging_utility.exe -validate
```

6.5.5.3.2. Configuring the DMS DATASRC Record using HDBRIO

The DMS DATASRC record must be correctly configured in the TAGGING database.

The example below shows a list of the field's values for the DMS DATASRC record. The important field values are:

- DMS_DATASRC: The value of this field must be set to "True".
- APP_DATASRC and DB_DATASRC: The value of these fields must be set to "DMS".
- ID_DATASRC: The value of this field must be set to "Hdb-DNOM".
- HOSTx_DATASRC: There are four of these fields, which are used to specify the IP addresses of each DMS server that will connect to TAGGING.

```
[TAGGING.TAGGING.EMS]
DATASRC<2>> read
Display of record DATASRC<2>
=====
BitContainer  FLAGS_DATASRC = B*4
  Bit: 0      VERIFIED_DATASRC = False
Integer*4     NUMRECS_DATASRC = 8681
BitContainer  OPTIONS_DATASRC = B*4
  Bit: 0      USEALIAS_DATASRC = False
Integer*4     RETRY_DATASRC = 60
Integer*4     TIMEOUT_DATASRC = 10000
BitContainer  TYPE_DATASRC = B*4
  Bit: 0      SCADA_DATASRC = False
  Bit: 1      DMS_DATASRC = True
  Bit: 2      PSA_DATASRC = False
  Bit: 3      HABITAT_DATASRC = False
Char*80       ALIASFORM_DATASRC =
Char*8        APP_DATASRC = DMS
Char*20       DB_DATASRC = DMS
Char*80       DESC_DATASRC = Default DMS Data Source
Char*8        FAM_DATASRC = EMS
Char*48       HOST1_DATASRC = .112
Char*48       HOST2_DATASRC =
Char*48       HOST3_DATASRC =
Char*48       HOST4_DATASRC =
Char*8        PORT1_DATASRC = 5040
Char*8        PORT2_DATASRC =
Char*8        PORT3_DATASRC =
Char*8        PORT4_DATASRC =
Char*30       ID_DATASRC = Hdb-DNOM
Integer*8     OID_DATASRC = 4463375692544990108
BitContainer  COMSTATE_DATASRC = B*4
  Bit: 0      UNINIT_DATASRC = True
  Bit: 1      CONNPEND_DATASRC = False
  Bit: 2      CONN_DATASRC = False
  Bit: 3      FAIL_DATASRC = False
```

Figure 27. Configuring the DMS DATASRC Record using HDBRIO

The TAGGING database must be re-validated for any changes to the TAGGING database records and fields referenced above.

6.5.5.3.3. Configuring the DSRECTYP DMS DNOM Records using HDBRIO

A list of DMS Distribution Network Operating Model (DNOM) record types that the TAGGING application can process and manage tags for must be specified in the TAGGING database DSRECTYP table. The DSRECTYP record is a hierarchical child record of the DATASRC record.

Use HDBRIO to see the list of DSRECTYP records under the DMS DATASRC record in the TAGGING database. Position HDBRIO to the DMS DATASRC record and issue the "where -s" command.

```

[TAGGING.TAGGING.EMS]
DATASRC(2)> where -s
    DSRECTYP(14) = X
    DSRECTYP(15) = IMPSWT
        DSTTYREF(59) = CONTROL INHIBIT
        DSTTYREF(60) = TRIP ONLY
        DSTTYREF(61) = INFO
        DSTTYREF(62) = MAINTENANCE
        DSTTYREF(63) = STICKY
        DSTTYREF(64) = ANNOTATION
        DSTTYREF(65) = LD CLEARANCE
        DSTTYREF(66) = STICKY_TYP
    DSRECTYP(16) = IF
        DSTTYREF(67) = CONTROL INHIBIT
        DSTTYREF(68) = TRIP ONLY
        DSTTYREF(69) = INFO
        DSTTYREF(70) = MAINTENANCE
        DSTTYREF(71) = STICKY
        DSTTYREF(72) = ANNOTATION
        DSTTYREF(73) = LD CLEARANCE
        DSTTYREF(74) = STICKY_TYP
    DSRECTYP(17) = SHIMP
        DSTTYREF(75) = CONTROL INHIBIT
        DSTTYREF(76) = TRIP ONLY
        DSTTYREF(77) = INFO
        DSTTYREF(78) = MAINTENANCE
        DSTTYREF(79) = STICKY
        DSTTYREF(80) = ANNOTATION
        DSTTYREF(81) = LD CLEARANCE
        DSTTYREF(82) = STICKY_TYP
    DSRECTYP(18) = GEN
        DSTTYREF(83) = CONTROL INHIBIT
        DSTTYREF(84) = TRIP ONLY
        DSTTYREF(85) = INFO
        DSTTYREF(86) = MAINTENANCE
        DSTTYREF(87) = STICKY
        DSTTYREF(88) = ANNOTATION
        DSTTYREF(89) = LD CLEARANCE
        DSTTYREF(90) = STICKY_TYP
    DSRECTYP(19) = ENERGVRES
        DSTTYREF(91) = CONTROL INHIBIT
        DSTTYREF(92) = TRIP ONLY
        DSTTYREF(93) = INFO
        DSTTYREF(94) = MAINTENANCE
        DSTTYREF(95) = STICKY
        DSTTYREF(96) = ANNOTATION
        DSTTYREF(97) = LD CLEARANCE
        DSTTYREF(98) = STICKY_TYP
    DSRECTYP(20) = CUT
        DSTTYREF(99) = CONTROL INHIBIT
        DSTTYREF(100) = TRIP ONLY
        DSTTYREF(101) = INFO
        DSTTYREF(102) = MAINTENANCE
        DSTTYREF(103) = STICKY
        DSTTYREF(104) = ANNOTATION
        DSTTYREF(105) = LD CLEARANCE
        DSTTYREF(106) = STICKY_TYP
    DSRECTYP(21) = SWT
        DSTTYREF(107) = CONTROL INHIBIT
        DSTTYREF(108) = TRIP ONLY
        DSTTYREF(109) = INFO
        DSTTYREF(110) = MAINTENANCE
        DSTTYREF(111) = STICKY
        DSTTYREF(112) = ANNOTATION
        DSTTYREF(113) = LD CLEARANCE
        DSTTYREF(114) = STICKY_TYP
    DSRECTYP(22) = COMPSW
        DSTTYREF(115) = CONTROL INHIBIT
        DSTTYREF(116) = TRIP ONLY
        DSTTYREF(117) = INFO
        DSTTYREF(118) = MAINTENANCE
        DSTTYREF(119) = STICKY
        DSTTYREF(120) = ANNOTATION
        DSTTYREF(121) = LD CLEARANCE
        DSTTYREF(122) = STICKY_TYP
    DSRECTYP(23) = SUBSTN
        DSTTYREF(123) = TRIP ONLY
        DSTTYREF(124) = STICKY
        DSTTYREF(125) = MAINTENANCE
        DSTTYREF(126) = CONTROL INHIBIT
        DSTTYREF(127) = STICKY_TYP
        DSTTYREF(128) = INFO
        DSTTYREF(129) = ANNOTATION
        DSTTYREF(130) = LD CLEARANCE
    DSRECTYP(24) = DEVICE
        DSTTYREF(131) = TRIP ONLY
        DSTTYREF(132) = STICKY
        DSTTYREF(133) = MAINTENANCE
        DSTTYREF(134) = CONTROL INHIBIT
        DSTTYREF(135) = STICKY_TYP
        DSTTYREF(136) = XFR AUTH INACTIU
        DSTTYREF(137) = INFO
        DSTTYREF(138) = XFR AUTH ACTIVE
    DSRECTYP(25) = ANALOG
        DSTTYREF(139) = STICKY
        DSTTYREF(140) = MAINTENANCE
        DSTTYREF(141) = STICKY_TYP
        DSTTYREF(142) = INFO
    DSRECTYP(26) = COUNT
        DSTTYREF(143) = STICKY
        DSTTYREF(144) = MAINTENANCE
        DSTTYREF(145) = STICKY_TYP
        DSTTYREF(146) = INFO
    DSRECTYP(27) = POINT
        DSTTYREF(147) = TRIP ONLY
        DSTTYREF(148) = STICKY
        DSTTYREF(149) = MAINTENANCE
        DSTTYREF(150) = CONTROL INHIBIT
        DSTTYREF(151) = STICKY_TYP
        DSTTYREF(152) = XFR AUTH INACTIU
        DSTTYREF(153) = INFO
        DSTTYREF(154) = XFR AUTH ACTIVE

```

Figure 28. Configuring the DSRECTYP Records using HDBRIO

The DSRECTYP records above highlighted in yellow are unique to the DMS DNOM database and must be specified for the TAGGING application to process tags for those DMS devices. Also note the DSTTYREF child records under each DSRECTYP record. The DSTTYREF records specify the tag types that can be processed for each DSRECTYP.

The TAGGING database must be re-validated for any changes to the TAGGING database records

and fields referenced above.

6.5.5.4. Configuring ADMS Clients

For configurations with *e-terra*_scada_versions 3.0 and later, make the following changes:

1. In the grid tabular display files, replace TAGNOTES_SCADA_EMS with TAGGING_TAGGING_EMS. This change is required for the following files:
 - Abnormal_GridDisplay.xml
 - AbnormalFdrRF_GridDisplay.xml
 - OrphanTagSummary_GridDisplay.xml
2. In SwitchOrderClientConfig.xml, replace TAGNOTES_SCADA_EMS with TAGGING_TAGGING_EMS.
3. In ViewerConfig.xml, replace TAGNOTES_SCADA_EMS with TAGGING_TAGGING_EMS.

6.6. Configuring the Link to PERMIT (ADMS Host)

A handful of parameters in the ADMS server configuration files define the connection to PERMIT (via NetPermSrv) on the SCADA server. Open a server configuration file (for example, NetServerMainConfig.txt) on the ADMS server to check the following settings:

```
'-----  
'Habitat PERMIT Server Parameters  
'-----  
  
PERMIT_CHECK_ENABLED = 1  
PERMIT_CROSS_ZONES_MSGS_ENABLED = 0  
PERMIT_SERVER_IP = 127.0.0.1  
PERMIT_SERVER_PORT = 5345  
PERMIT_SERVER_TIMER_PERIOD = 9  
PERMIT_LOG_ENABLED = 0  
PERMIT_JUMPER_OR_MODE = 0  
PERMISSION_PER_STATION = 0  
PERMIT_CACHE_DUMP_TRIGGER_FILENAME = %IDMS_ROOT%\log\trigger_permit_cache_dump.txt
```

For more information about the parameters, see Habitat PERMIT Server in [NetServerMainConfig.txt](#) Parameters.

Important

The PERMISSION_PER_STATION parameter is deprecated. It defaults to zero and is used for legacy purposes.

For the entire permission system to function (to allow a user to operate a switch), the following must work together.

- There must be a user modeled in PERMIT that has MODES giving it access to an AREA record in

PERMIT (AOR) that matches the AOR for a specific substation.

- The model files must be created to have AORs defined per substation.

If you are unsure of the AOR defined for a specific station, enable permission checking and attempt to toggle a switch. This should result in a permission denied message that includes the AOR. If the error message is blank, that is also indicative of a problem.

- The ADMS server applications and NetPermSrv must be running, enabled, and configured to communicate using the same port.

For an ADMS server application, the port is listed in the PERMIT server parameters (PERMIT_SERVER_PORT) section in the NetServer configuration file. Default port number is 5345.

6.6.1. Defining Permissions for ADMS Users

Some parts of the permission scheme in ADMS differ from the permission scheme for traditional Browser displays. For traditional displays, specific objects on a display (or entire displays) can be blocked/hidden using AORs or application filtering. For ADMS, if the user has a navigational path to the display, it will be rendered. However, when a user tries to act on the system, SCADA AORs are checked prior to allowing actions such as switching operations, tagging, and creating temporary devices.

The following PERMIT AREA records are currently supported for functional permission:

- **ANNO:** For placing/removing annotations.
- **POWERAPP:** For manually running power flow applications and editing system-level manual parameters for the real-time server.
- **CASENOTE:** For creating/modifying a case note.
- **AREANOTE:** For creating/modifying an area setting.
- **UNLKINCD:** For unlocking a locked incident.

These are the specific strings used in the code, and must be modeled exactly.

Optionally, an administrator could model AREA records such as:

- DMS
- DNAF
- OMS

These could be used for filtering the available navigation options in the Browser toolbars and menus. For example, the RT OMS button could be hidden by adding the `<AreaPermission>OMS</AreaPermission>` line in the `emp_toolbar.xml` configuration.

6.6.1.1. Service Center Based Permissions

For a single customer call, the OMS can be configured to predict out the service point (service drop) instead of predicting out the distribution transformer. To enable this feature, set the following parameter in the NetServerOmsConfig.txt file to 1.

```
'Determines whether to use service point name for the name of device of record for single customer incident.  
USE_SERVICE_POINT_FOR_SINGLE_CUSTOMER_INCIDENT = 0
```

These service-point based, single customer incidents are hidden or shown based on the Service Center. Therefore, the Service Centers (as defined in the customer model) need to be modeled as AREA records in PERMIT.

Note

The AREA record is char*8, so the Service Centers must have names of eight characters or less.

6.6.1.2. Optional Permissions

Some optional permission settings can be used to allow more granular control for less privileged dispatchers or "techs". By default, this additional granularity is disabled. These optional settings must be enabled within the NetServerMainConfig.txt and NetServerOmsConfig.txt configuration files, as shown below:

```
'-----  
Permission Settings  
'-----  
  
REQUIRE_SWITCHING_AND_GEOGRAPHIC_PERMISSION = 0  
SWITCHING_FUNCTION_AOR_NAME = SWITCHOP  
REQUIRE_DISPATCH_AND_GEOGRAPHIC_PERMISSION = 0  
DISPATCH_FUNCTION_AOR_NAME = DISPATCH  
OUTAGE_PLANNING_FUNCTION_AOR_NAME = PLOUTAGE  
ADHOC_CREW_FUNCTION_AOR_NAME = ISADHOC  
OUTAGE_INCIDENT_FUNCTIONAL_PERMISSION_NAME = OUTAGE  
HISTORY_CORRECTION_FUNCTION_AOR_NAME = HISTCOR  
CUSTOMER_MOVE_FUNCTION_AOR_NAME = MOVECUST  
INCIDENT_EDIT_ETR_AOR_NAME = EDITETR  
REQUIRE_FUTURE_COMMISSIONING_PERMISSION = 0  
PROJECT_COMMISSIONING_PERMIT_AOR = COMMIS
```

Although the AOR names are configurable, it is recommended to leave them set to their default names to help with troubleshooting, should there be any permission-related confusion. Also, if the AOR name is changed, the new name must fit within the char*8 restriction of the PERMIT AREA record.

- **SWITCHOP:** For changing a device's state or attaching a tag.

Changing a device state or adding a tag requires geographic write permission for the device or

SCADA point, and a "Switching" functional permission.

- **DISPATCH:** For confirming a device to another state from the current state.

Confirming a device requires write access for the geographic permission for the device or SCADA point, and a "Dispatch" functional permission.

To confirm a state for any non-SCADA switching device and for customers, write access to the Dispatch functional permission is required. This does not apply to the Study mode.

- **PLOUTAGE:** For creating, updating, or cancelling planned outages.
- **ISADHOC:** For creating crew, employee, and vehicle records from the OMS Display plugin.
- **OUTAGE:** For creating, updating, or cancelling outage incidents and for operating protective devices.
- **HISTCOR:** For creating, updating, merging, or splitting historical incidents.
- **MOVECUST:** For dynamically relocating customers on the geographic display and cancelling the customer move operations.
- **EDITETR:** For setting and updating an incident's Estimated Time for Restoration (ETR).
- **COMMIS:** For updating the commissioning stage and state (active, future, future removed, or removed) for network components.

Note

To disable the permission checking for an operation, leave the value blank. For example, set: HISTORY_CORRECTION_FUNCTION_AOR_NAME =

Incidents where the associated station is missing and AOR cannot be determined are assigned to DEFPAREA. Therefore, you need to model the DEFPAREA mode in Permit and assign this mode to operators to see incidents with DEFPAREA on OMS displays.

In addition, permissions for managing follow-up work orders and non-outage incidents can be configured using the OMS server Configuration Editor. By default, the following permissions are defined:

- **ARBOR:** For modifying specific types of follow-up work orders.
- **EDISON:** For modifying specific types of follow-up work orders and for modifying specific types of non-outage incidents.

6.7. Configuring the Link to CFGCTRL

The ADMS server applications communicate with CfgProxy via a TCP/IP link, which can be defined as follows:

- The CFGPROXY_PORT environment variable defines the ports that the CfgProxy application uses to communicate with the ADMS server applications. If this variable is not defined, CfgProxy

defaults to using ports 5015-5020.

Note

The CfgProxy_PORT environment variable is retained for legacy purposes only. The port numbers are defined in the CfgProxy clone (as described in [Configuration Manager Proxy](#)). Use the default port to decrease the risk of port conflicts.

- The CFGPROXY_PORT and CfgProxyPort parameters within a server configuration file (NetServerMainConfig.txt, NetServerOmsConfig.txt, CallServerConfig.xml, and so on) defines the port that the ADMS server application uses to communicate with CfgProxy.

For the communication link to be established, the port numbers defined in the CfgProxy application and in an ADMS server configuration must match.

Configuring CfgProxy is another instance where running NetSrv from the command line is useful for initial troubleshooting. The application outputs a message stating whether it connects to CfgProxy. From there, if NetSrv loads the model files but does not proceed any further (i.e., applying dynamics and running NTP), it is clear that it has not received the enabled role from CfgProxy.

6.8. Configuring the Link to ALARM

To configure alarms to be sent to SCADA, set ALARM parameters in the NetServerMainConfig.txt file. For more details about the parameters, see "HABITAT ALARM SERVER" in [NetServerMainConfig.txt Parameters](#).

To test the configuration, call up the alarm/alarm_system_activity display in one viewport, and an ADMS geographic display in another viewport. Operator actions that modify the state of the electrical network are logged so that, if you toggle a switch, it should show up on the system activity display.

6.9. Defining Alarms and Events for the ADMS System

The ADMS server applications route the alarm and event messages to the ALARM system on the EMS SCADA server through ProxySrv.

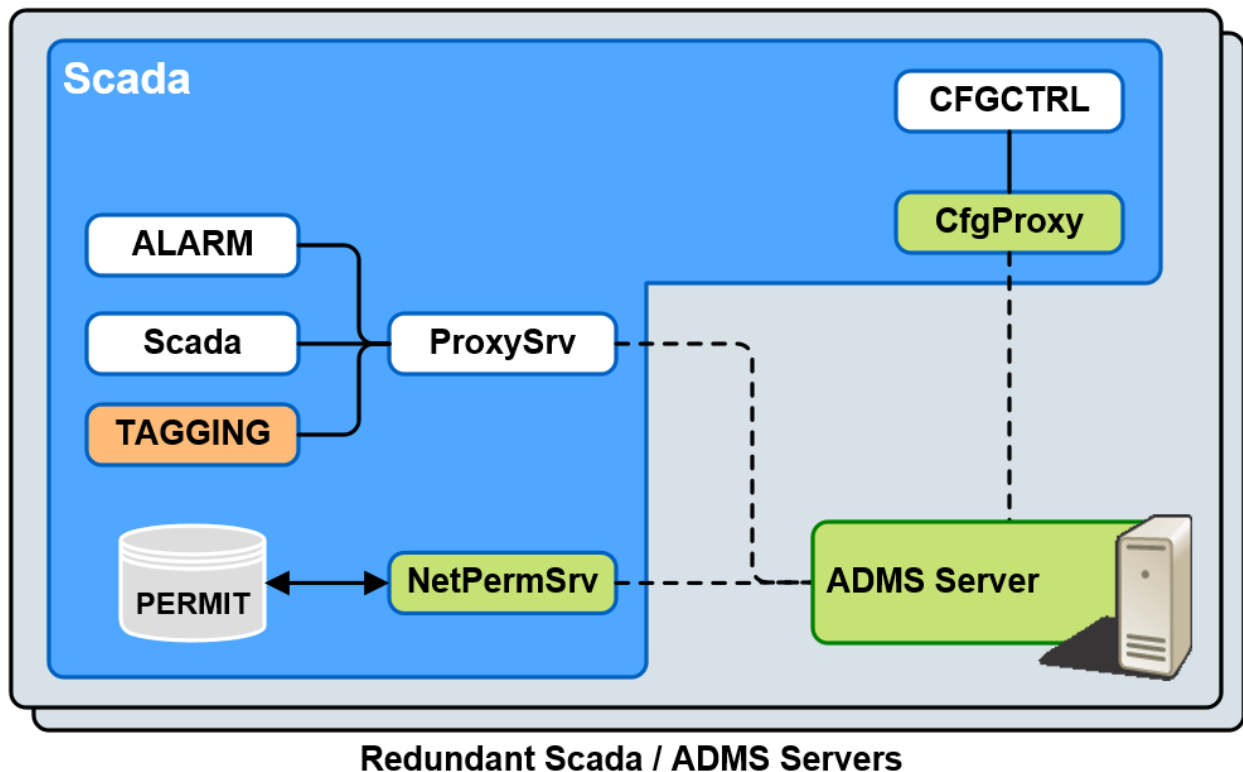


Figure 29. Simplified Software Configuration - Hosting Environment

The following new categories, objects, and exceptions need to be defined in the ALARM database model under the application ProxySrv. The texts of the messages themselves are embedded within the ProxySrv.ALMSG file.

- APP = ProxySrv
- CAT = DMSANA
- CAT = DMSOPER
- CAT = DMSSYS
- CAT = DNAF_F1
- CAT = DNAF_F2
- CAT = DNAF_F3
- CAT = DNAF_F4
- CAT = DNAF_F5
- CAT = DNAF_FI
- CAT = DNAF_HV3
- CAT = DNAF_LV1
- CAT = DNAF_LV2
- CAT = DNAF_LV3

- CAT = DNAF_LV4
- CAT = DNAF_LV5
- CAT = DNAF_SC
- CAT = DNAF_VI
- CAT = DMS_CAP
- CAT = OMSANA
- CAT = OMSOPER
- CAT = OMSSYS
- CAT = OMSCREW
- CAT = OMSINCD
- CAT = OMSMISC
- CAT = FE
- CAT = FE_COMM
- CAT = FE_DBUPD
- CAT = FE_RTU
- CAT = PX
- CAT = RF
- CAT = DNAF_BF
- OBJDEF = FEPAIR
 - EXCDEF = FEDBUPD
- OBJDEF = FESOBJ
 - EXCDEF = FEALM
- OBJDEF = DMSOPER
 - EXCDEF = DMSOP001 // Device commanded to a new state
 - EXCDEF = DMSOP002 // Permanent status changed for a switch
 - EXCDEF = DMSOP003 // Insert or remove a jumper
 - EXCDEF = DMSOP004 // Insert or remove a cut
 - EXCDEF = DMSOP005 // Apply or remove temporary ground
 - EXCDEF = DMSOP006 // RAL, GRAL, and GRAM switches commanded to a new state
 - EXCDEF = DMSOP007 // RA-21, RA-32, and SRAM-33 switches commanded to a new state
 - EXCDEF = DMSOP008 // A client dynamic is not in sync for some period of time
 - EXCDEF = DMSOP009 // A server thread has backlog over a value for some period of time
 - EXCDEF = DMSOP010 // G Switch (three-phase gang-operated switch in URD transformer)
 - EXCDEF = DMSOP011 // Regulators and LTC transformers: block/unblock regulators

- EXCDEF = DMSOP012 // Regulators and LTC transformers: block/unblock station
- EXCDEF = DMSOP013 // Regulator device: remote/local
- EXCDEF = DMSOP014 // Regulator device: tap step change
- EXCDEF = DMSOP015 // Regulator device: supervisory cutout
- EXCDEF = DMSOP028 // Capacitor banks: On/Off
- EXCDEF = DMSOP034 // Fault Indicators: Set/Reset
- EXCDEF = DMSOP048 // Insert or remove a phase jumper (deprecated)
- EXCDEF = DMSOP049 // Device commanded to a new state
- EXCDEF = DMSOP050 // Event is fired when standby processing is complete on the server
- EXCDEF = DMSOP052 // Apply or remove temporary ground
- EXCDEF = DMSOP054 // Capacitor Banks: Set Inoperable/Operable
- EXCDEF = DMSOP055 // Station Model file was updated, but failed to load successfully
- EXCDEF = DMSOP056 // Distribution transformer: connected/disconnected
- EXCDEF = DMSOP057 // Distribution transformer: confirm connected/disconnected
- EXCDEF = DMSOP060 // DNAF application configuration was changed
- EXCDEF = DMSOP061 // Device confirmed open/closed
- EXCDEF = DMSOP062 // Device phase confirmed open/closed
- EXCDEF = DMSOP063 // Standby connect to enabled server status change
- EXCDEF = DMSOP064 // Switching order Mobile alarm 1 (for example, "SOS message was accepted by field employee. Employee Id: FI2945")
- EXCDEF = DMSOP065 // Switching order Mobile alarm 2 (for example, "Failed to Deliver SwitchOrder to Mobile server. SWO: SO1216815")
- EXCDEF = DMSOP066 // DMS Interface server failed to send a DER control plan to the 2030.5 Gateway
- EXCDEF = DMSOP067 // DER control plan status change
- EXCDEF = DMSOP068 // Standby OMS server connected/disconnected from a DMS Follower server
- EXCDEF = DMSOP069 // Untrusted user connection rejected for a DMS Follower server
- EXCDEF = DMSOP070 // DMS server rejected connection from a blacklisted user.
- EXCDEF = DMSOP071 // DMS server rejected connection from a user with an excessive rate of connection attempts.
- EXCDEF = DMSOP072 // Stage 1 FSD application triggered for a transformer.
- EXCDEF = DMSOP073 // Stage 1 FSD application completed.
- EXCDEF = DMSOP074 // Stage 1 FSD application completed with no result.
- EXCDEF = DMSOP075 // Stage 1 FSD application control failure.

- EXCDEF = DMSOP076 // Stage 2 FSD application triggered for a transformer.
- EXCDEF = DMSOP077 // Stage 2 FSD application completed.
- EXCDEF = DMSOP078 // Stage 2 FSD application completed with no result.
- EXCDEF = DMSOP079 // Stage 2 FSD application control failure.
- EXCDEF = DMSOP080 // Web API request to create a hot line hold document.
- EXCDEF = DMSOP081 // Switching device specific open state classification change to Temporary Open, Bad Open, Normal Open, or Normal.
- OBJDEF = OMSOPER
 - EXCDEF = OMSOP041 // NetServer Overloaded
 - EXCDEF = OMSOP042 // Unacknowledged momentary incident created
 - EXCDEF = OMSOP043 // Unacknowledged sustained incident created
 - EXCDEF = OMSOP044 // Incident affects critical customers
 - EXCDEF = OMSOP045 // Emergency call received
 - EXCDEF = OMSOP046 // Customer suspended
 - EXCDEF = OMSOP047 // Customer unsuspended
 - EXCDEF = OMSOP048 // Undo confirmation was performed for an incident
 - EXCDEF = OMSOP049 // Block/Unblock prediction on a device
 - EXCDEF = OMSOP050 // Failure of a user-requested AMI operation
 - EXCDEF = OMSOP051 // Area setting ETR expired
 - EXCDEF = OMSOP052 // Planned outage not converted to an incident
 - EXCDEF = OMSOP053 // OMS server rejected connection from a blacklisted user
 - EXCDEF = OMSOP054 // OMS server rejected connection from a user with an excessive rate of connection attempts
 - EXCDEF = OMSOP055 // Non-outage incident originated
 - EXCDEF = OMSOP056 // Area setting expired
- OBJDEF = DMSSYS // DNAF Application Status
 - EXCDEF = AF111 // AFR could not complete due to voltage or loading violation
 - EXCDEF = AF112 // AFR completed
 - EXCDEF = AF113 // AFR failed to complete
 - EXCDEF = AF115 // AFR is starting
 - EXCDEF = AF122 // AFR plan implementation ends
 - EXCDEF = AF123 // AFR plan implementation failed on device
 - EXCDEF = AF125 // AFR plan implementation starts
 - EXCDEF = AF128 // AFR plan implementation rejected

- EXCDEF = AF129 // AFR plan implementation approved
- EXCDEF = AF158 // AFR plan canceled or execution mode changed
- EXCDEF = AF217 // AFR is disabled for feeder
- EXCDEF = AF237 // AFR is disabled for feeder by SCADA DA status measurement
- EXCDEF = AF238 // AFR enable/disabled for feeder by SCADA DA status measurement not changed
- EXCDEF = AF2310 // AFR is enabled for feeder by SCADA DA status measurement
- EXCDEF = AF241 // AFR plan found, but violations cannot be addressed with LVM
- EXCDEF = AF242 // AFR plan found with violations that can be addressed with LVM
- EXCDEF = AF313 // Overload solution for device NOT found by AFR
- EXCDEF = AF314 // Overload solution for device found by AFR
- EXCDEF = AF323 // Device failed to complete AFR command
- EXCDEF = AF326 // Switch has low current flow, which may indicate a failure to close.
- EXCDEF = TEXTONLY
- EXCDEF = BADALARM
- EXCDEF = DUPPATH
- EXCDEF = FI111 // FISR could not complete due to voltage or loading violation
- EXCDEF = FI112 // FISR completed
- EXCDEF = FI113 // FISR failed to complete
- EXCDEF = FI115 // FISR is starting
- EXCDEF = FI1111 // FISR is not triggered
- EXCDEF = FI122 // FISR plan implementation ends
- EXCDEF = FI123 // FISR plan implementation failed
- EXCDEF = FI125 // FISR plan implementation started
- EXCDEF = FI128 // FISR plan implementation rejected
- EXCDEF = FI129 // FISR plan implementation approved
- EXCDEF = FI158 // FISR plan canceled or execution mode changed
- EXCDEF = FI217 // FISR is disabled for feeder
- EXCDEF = FI237 // FISR is disabled for feeder by SCADA DA status measurement
- EXCDEF = FI238 // FISR enable/disabled for feeder by SCADA DA status measurement not changed
- EXCDEF = FI241 // FISR plan found, but violations cannot be addressed with LVM
- EXCDEF = FI242 // FISR plan found with violations that can be addressed with LVM
- EXCDEF = FI2310 // FISR is enabled for feeder by SCADA DA status measurement

- EXCDEF = FI323 // Device failed to complete FISR command
- EXCDEF = FI326 // Switch has low current flow, which may indicate a failure to close.
- EXCDEF = FL112 // Fault location calculation completed successfully (is shown only when fault location event happens)
- EXCDEF = FL113 // Fault location failed to complete (is shown only when fault location event happens). Should display failure reason.
- EXCDEF = FL115 // Fault location started (is shown only when a fault location event happens)
- EXCDEF = PF112 // DPF completed
- EXCDEF = PF113 // DPF failed to complete
- EXCDEF = PF696 // DPF has not been solved recently
- EXCDEF = PV112 // PRV completed
- EXCDEF = PV113 // PRV failed to complete
- EXCDEF = VV112 // LVM completed
- EXCDEF = VV113 // LVM failed to complete
- EXCDEF = VV122 // Implementable LVM plans generated
- EXCDEF = VV126 // No implementable LVM plans generated in XX hours
- EXCDEF = VV136 // Too many infeasible solutions in a row
- EXCDEF = VV146 // Too many infeasible solutions in a day
- EXCDEF = VV156 // Too many discarded solutions in a row
- EXCDEF = VV166 // Too many discarded solutions in a day
- EXCDEF = VV1106 // LVM station too many "controls not available" in a day
- EXCDEF = VV1130 // LVM distribution automation (DA) support mode turned off after FISR/AFR plan without violations
- EXCDEF = VV1135 // LVM distribution automation (DA) support mode turned on to reduce violations in FISR/AFR plans
- EXCDEF = VV1140 // Energized Sensor Zero Voltage Measurement Return To Normal
- EXCDEF = VV1143 // Energized Sensor Zero Voltage Measurement
- EXCDEF = VV1162// LVM failsafe conditions ended for a station
- EXCDEF = VV1165// LVM failsafe conditions started for a station
- EXCDEF = VV1175// LVM retriggers due to a plan execution failure
- EXCDEF = VV1183// LVM plan pre-implementation device state check failed
- EXCDEF = VV1198// LVM plan canceled
- EXCDEF = VV217 // LVM is disabled for feeder
- EXCDEF = VV277 // LVM is off - too many controllers out of service
- EXCDEF = VV2106 //Too many "controls not available" in a day

- EXCDEF = VV2120 // LVM problem formulation returned to normal
- EXCDEF = VV2122 // LVM problem formulation changed
- EXCDEF = VV2162 // LVM failsafe conditions ended for a feeder
- EXCDEF = VV2165 // LVM failsafe conditions started for a feeder
- EXCDEF = VV473 // LVM Capacitor switch failed
- EXCDEF = VV490 // Actual Delta-V mismatch at capacitor controller (Return to Normal)
- EXCDEF = VV491 // Actual Delta-V mismatch at capacitor controller
- EXCDEF = VV4106 // Too many capacitor controls issued in XX hours
- EXCDEF = VV4116 // Too many capacitor controls issued in a day
- EXCDEF = VV573 // LVM tap change failed
- EXCDEF = VV578 // Tap changer move cancelled
- EXCDEF = VV780 // Regulator group voltage imbalance (return to normal)
- EXCDEF = VV781 // Regulator group voltage imbalance
- EXCDEF = VV5106 // Too many tap changes issued in XX hours
- EXCDEF = VV5116 // Too many tap changes issued in a day
- EXCDEF = VV5162 // LVM failsafe conditions ended for a transformer
- EXCDEF = VV5165 // LVM failsafe conditions started for a transformer
- EXCDEF = VV61212// Optimization group LVM system implementation percentage complete
- EXCDEF = VV6122// Optimization group LVM system implementation ended
- EXCDEF = VV6125// Optimization group LVM system implementation started
- EXCDEF = VV6128// Optimization group LVM system implementation stopped
- EXCDEF = VV6150// Optimization group LVM throttling (return to normal)
- EXCDEF = VV6154// Optimization group LVM throttling requested an execution pause
- EXCDEF = VV6155// Optimization group LVM throttling started
- OBJDEF = VIOLFI
 - EXCDEF = PF150 // Imbalance flow violation for Station (RETURN TO NORMAL)
 - EXCDEF = PF151 // Imbalance flow violation for Station
 - EXCDEF = PF250 // Flow imbalance violation for feeder (RETURN TO NORMAL)
 - EXCDEF = PF251 // Imbalance flow violation for feeder
- OBJDEF = VIOLFL
 - EXCDEF = PF140 // DNAF overloaded for station (RETURN TO NORMAL)
 - EXCDEF = PF141 // DNAF overloaded for station
 - EXCDEF = PF240 // DNAF overloaded for feeder (RETURN TO NORMAL)

- EXCDEF = PF241 // DNAF overloaded for feeder
- OBJDEF = VIOLHL
 - EXCDEF = PF130 // DNAF high voltage violation for station (RETURN TO NORMAL)
 - EXCDEF = PF131 // DNAF high voltage violation for station
 - EXCDEF = PF230 // DNAF high voltage violation for feeder (RETURN TO NORMAL)
 - EXCDEF = PF231 // DNAF high voltage violation for feeder
- OBJDEF = VIOLLV
 - EXCDEF = PF120 // DNAF low voltage violation for station (RETURN TO NORMAL)
 - EXCDEF = PF121 // DNAF low voltage Violation for station
 - EXCDEF = PF220 // DNAF low voltage violation for feeder (RETURN TO NORMAL)
 - EXCDEF = PF221 // DNAF low voltage violation for feeder
- OBJDEF = VIOLSC
 - EXCDEF = PV370 // Protection zone violation returned to normal (this is detected by the PRV application, device is now in a proper state due to a change in the network or a change in the protection settings)
 - EXCDEF = PV371 // Device has abnormal protection validation settings
 - EXCDEF = PV326 // Protection setting group mismatch for device
- OBJDEF = VIOLVI
 - EXCDEF = PF160 // Imbalance voltage violation for station (RETURN TO NORMAL)
 - EXCDEF = PF161 // Imbalance voltage violation for station
 - EXCDEF = PF261 // Imbalance voltage violation for feeder
 - EXCDEF = PF260 // Voltage imbalance violation for feeder (RETURN TO NORMAL)
 - EXCDEF = PF1130// AMI voltage mismatch violation for Station (RETURN TO NORMAL)
 - EXCDEF = PF1131// AMI voltage mismatch violation for station
- OBJDEF = VIOCAP
 - EXCDEF = PF280 // Capacitor position validation for feeder (RETURN TO NORMAL)
 - EXCDEF = PF281 // Capacitor position validation for feeder (failed)
 - EXCDEF = PF3106 // Unexpected voltage measurement from energized device
 - EXCDEF = PF5116 // LTC/Regulator source voltage adjusted for tap position estimation
- OBJDEF = BCKFED
 - EXCDEF = PF2121 // Backfeed state on feeder
 - EXCDEF = PF2120 // Backfeed state on feeder (RETURN TO NORMAL)

For more information about alarms, refer to the *ADMS Alarmable Events* document.

Note

The PROXYSRV.ALMMMSG file that is supplied with Scada does not contain all ADMS ALARM message definitions.

If the PROXYSRV application definition PROXYSRV.ALMMMSG in your current ALARM database does not already include the record definitions shown above, the record definitions must be added to the ALARM database.

To add ADMS ALARM record definitions:

1. Create a backup savecase of the current ALARM database. Also, make backup copies of your current proxysrv.exe file in the %HABUSER_SRCDIR%\scada\proxysrv directory and habalmmmsg.dll files in the %HABUSER_BINDIR% directory.
2. Execute the following commands:

```
getenv proxysrv
tview r hdbserver hdb hdb
hdbimport -d alarm -a alarm -f ems -overrideoid -s
%IDMS_ROOT%\..\FantasyIsland\misc\dms\QuickInstall_hab511\alarm\idms_alarm_hdbimport.txt
```

This merges the ADMS ALARM message definitions into your current version of the PROXYSRV.ALMMMSG file. For importing ADMS ALARM records and the related field data into the ALARM database, these commands use the ADMS HdbImport file named "idms_alarm_hdbimport.txt" from the %IDMS_ROOT%\..\FantasyIsland\misc\dms\QuickInstall_hab511\alarm directory.

Note

For clean installations, alternatively, you may replace your current PROXYSRV.ALMMMSG file in %HABUSER_SRCDIR%\scada\proxysrv with the PROXYSRV.ALMMMSG file in the %IDMS_ROOT%\..\FantasyIsland\misc\dms\QuickInstall_hab511\alarm folder.

3. After the new records are added, validate the ALARM database and check the ALARM message log to make sure there are no validation error messages.
4. Rebuild ProxySrv:

```
buildlst -almmmsg
```

(If the "buildlst -almmmsg" command does not work, execute "build -almmmsg" or "build *.almmmsg")

This should update the habalmmmsg.dll file in the %HABUSER_BINDIR% directory to include the new ADMS ALARM messages.

Note

"Bad string" messages might appear until the upgrade is finished.

5. To test these changes to ALARM and ProxySrv, open a line switch or place a cut on a line from one of the ADMS geographic displays. When you do this, you should see corresponding event messages appear on the ALARM System Activity display (ALARM_SYSTEM_ACTIVITY). For example:

 **RAVEN** **51999** **SET TO OPEN By DMSALL**

6. After you confirm that the changes to ALARM and ProxySrv are working properly, propagate the changes to your other test and production ADMS SCADA systems.

7. Switching Order Configuration

The ADMS switching operations module is an optional component in the greater ADMS suite. This module allows for the creation, storage, and execution of switching orders. The server component is a simple application that manages the saved switching orders in a centralized location under a group of organizing folders.

7.1. Switching Order Server Configuration

The Switching Order server is delivered as an optional component of the Server kit. By default, the Switching Order server is installed under the base server folder: D:\Eterra\Distribution\Server\bin.

The syntax for starting the server is as follows:

```
EXE /CONFIG Config.txt  
%DMS_BINDIR%\SwitchOrderServer.exe /CONFIG %IDMS_ROOT%\Config\Server\SwitchOrderServerConfig.txt
```

The Switching Order server also reads the SwitchOrderServerConfig.xml configuration file, from which the Switching Order client requests the Switching Order server configurations when connection is established.

For more information about SwitchOrderServerConfig.txt and SwitchOrderServerConfig.xml configuration parameters, refer to the *Switching Order Server Configuration Editor User Guide*.

Caution

The Switching Order server does not integrate with CFGCTRL and therefore does not support a redundant configuration.

7.1.1. Configuring File Folders for the Switching Order Server

The Switching Order server configuration file contains the FilesFolderPath parameter. This path specifies the base directory where the switching order files will be stored. The directory contains subfolders for approved, completed, and saved switching orders, daily logs, and templates.

If the FilesFolderPath parameter is empty, the Switching Order server uses the "files" folder in place of where the Switching Order server configuration file is situated.

7.2. Switching Order Client Configuration

The ADMS switching operations component of the client software has its own configuration file (SwitchOrderClientConfig.xml) that resides in the same configuration folder as the client's main netconf_hostname.xml file. Once the Switching Order server application is configured and running, the host name and targeted port number need to be updated in the client's configuration file. To

define this information, use the Switching Order Client Configuration Editor delivered with the Tools kit. Be sure to open the appropriate file. For the Device and State tab in the tool, the default settings should be a good starting point, unless Scada has been significantly modified.

While using the Switching Order Client Configuration Editor, there is an easy way to get the tag types from your SCADA environment to add to the TagTypes tab in the editor. From a command prompt on your SCADA system:

```
context scada %realtime_family%  
hdbrio -c "/id_tagtyp(*)" scadamom
```

To test your changes, restart the Browser/ADMS client application.

8. Configuring Web Services

8.1. Running Web Services from PROCMAN

Several web services in the ADMS suite need specific ports open for HTTP traffic. The Net Data Server process and the MultiSpeak-mode CallServer and RelayServer all fall into this category. During initial integration, applications might be run interactively using a normal Windows shortcut. This method uses the logged in user's rights and will likely work fine. However, once these tasks are configured in PROCMAN, and once PROCMAN is configured to run as a service (LOCALSERVICE), the permission set becomes restricted and the application will not be able to open the listening ports.

Windows provides a mechanism for granting privileges to specific user accounts to open specific HTTP ports. This section describes how to use the "netsh" command for this purpose.

8.1.1. Configuring the DMS Interface Server Process

Caution

The URL namespace reservation must correspond to the DMS Interface server HTTP/HTTPS configuration. If DMS Interface server is configured for a secure connection (HTTPS), use "https" instead of "http" when reserve a namespace.

In the default configuration, DMS Interface server listens at `http://+:8801/dms`. By default, listening at a particular HTTP address requires administrator privileges. Without the proper rights, you might receive the error "HTTP could not register URL `http://+:8801/dms`". There are two ways to avoid this error:

- Run DMS Interface server as an administrator.
- Use `Netsh.exe` to give the account permissions to reserve the URL.

To use `Netsh.exe`, open a command prompt with administrator privileges and enter the following command:

```
netsh http add urlacl url=http://+:8801/dms user=DOMAIN\username
```

For the situation when PROCMAN is starting this task, and PROCMAN is running as a service (the official production setup), the "user" specification should be set as follows:

```
user="NT AUTHORITY\NETWORK SERVICE"
```

To delete the reservation, enter:

```
netsh http delete urlacl url=http://+:8801/dms
```

To view existing reservations, enter:

```
netsh http show urlacl
```

8.1.2. Configuring CallServer and RelayServer for MultiSpeak

The CallServer and RelayServer can use the MultiSpeak protocol for interfacing with a phone call system (IVR) and smart meters (AMI). These MultiSpeak web services must be set up similarly to the Net Data Server as described above.

The <MultiSpeakSettings> section in the CallServerConfig.xml and RelayServerConfig.xml files specifies the various web service endpoints.

You must add permissions for the <RelayServerODServerEndPoint> and the <CallServerOAServerEndPoint> on their respective machines. Depending on your needs, the Relay server and Call server might be on the same machine or on separate machines.

```
netsh http add urlacl url=http://+:6090/MultiSpeakWebServices/OA_Server.svc user="NT AUTHORITY\NETWORK SERVICE"
netsh http add urlacl url=http://+:6091/MultiSpeakWebServices/OD_Server.svc user="NT AUTHORITY\NETWORK SERVICE"
```

To verify if the process is actually listening, you can use the command:

```
netstat -ano | findstr 6090
```

Note

The MultiSpeak web services run under PID=4.

8.2. Configuring the Switching Order Web Service

The Switching Order web service is implemented based on the DMS Interface server using Microsoft Windows Communication Framework (WCF), which uses the open data protocol (OData) to expose services. The data exchange format supported by the Switching Order web service is XML or JSON.

The DMS Interface server contains three specific modules to support the Switching Order web service:

- **Web Service:** This module is responsible for instantiating the web service and intercepting any incoming web service requests from external systems.
- **SWO Interface Agent:** This module is responsible for business logic, which includes getting a switching request to formulate a Switching Order Request and then handing it over to the Communication client module to send it to the Switch Order server.

- **Communication Client:** This module is responsible for implementing the communication interface with the Switch Order server.

The following DMS Interface server configuration properties relate to the Switching Order web service:

- **WebServiceBaseURL:** This configures the base URL for trusted applications. Note that the trailing "/" is needed at the end. The default value is `http://localhost:8801/dms/`.

Each web service application adds a specific ending to this base. For example, the URL for the Switching Order web service is `http://localhost:8801/dms/SWO`.

- **SecureWebService:** This configures whether the secure web service connection will be used.
- **TrustedClientList and TrustedClient:** Used to set up trusted application lists. Each application can have the following settings:
 - **Name:** Used to identify the application.
 - **Key:** Used to calculate the hash value for the authorization header.
 - **NotificationBaseURL:** The base URL for trusted applications to receive notifications. The default value is `http://localhost:9090/dms/event/`.

The DMS Interface server uses this base URL as an address where the POST web-request for the notification will be sent. For example, the POST web-request for the Switching Order status notification will be sent to `http://localhost:9090/dms/event/SWO/StatusNotification`.

You may need to reserve the URL for the notifications (for instructions, refer to [Configuring the DMS Interface Server Process](#)).

- **NotificationTypes:** The notification types that the application is interested in. The currently valid values are SWITCH, DNAF, SWO-ALL, SWO-STATUS, SWO-STATUS-EXT, SWO-STEP, and SWO-PDF. If not specified, the default value is ALL.
- **EnableSwitchingOrderWebServices:** If true, the Switching Order web service is enabled.
- **ListenerHost:** Name or IP address of the host machine on which the Switching Order server is running.
- **ListenerPort:** TCP/IP port number on which the Switching Order server is listening for client connections.
- **TimeToLive:** The number of seconds before received messages are considered expired.
- **CompressMessage:** If set to True, Switching Order server communication client messages are sent compressed. This decreases message size and improves performance. This parameter must be set with the same value with the CompressMessage, OMS_SEND_COMPRESSED_MESSAGE, and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
- **EnableTcpiTls:** If set to true, the TLS protocol is enabled for the TCP/IP connection.
- **HelloIntervalSeconds:** Interval between periodic hello messages to check that the connection is still open.

The following is a sample DmsInterfaceServerConfig.xml configuration:

```
<WebServiceBaseURL>http://localhost:8801/dms/</WebServiceBaseURL>
<SecureWebService>false</SecureWebService>
<TrustedClientList>
<TrustedClient>
<Name>DmsISClient</Name>
<Key>Test123</Key>
<NotificationBaseURL>>http://localhost:9090/dms/event/</NotificationBaseURL>
<NotificationTypes>SWO-ALL</NotificationTypes>
</TrustedClient>
</TrustedClientList>
<EnableSwitchingOrderWebServices>true</EnableSwitchingOrderWebServices>
<SwitchingOrderServerCommConfig>
<ListenerHost>localhost</ListenerHost>
<ListenerPort>5044</ListenerPort>
<TimeToLive>300</TimeToLive>
<CompressMessage>true</CompressMessage>
<EnableTcpipTls>false</EnableTcpipTls>
<HelloIntervalSeconds>30</HelloIntervalSeconds>
</SwitchingOrderServerCommConfig>
```

Also, you need to enable the Switching Order web service notification system via the SwitchOrderServerConfig.txt file:

```
EnableNotificationSystem = true
```

In addition, you can enable the DMS Interface server's Workforce Management System (WFM) switching order interface web service client plugin to exchange messages with an external web client. For more details, refer to the *DMS Interface Server Programmer Guide*.

8.3. Configuring HTTPS for ADMS Interface Servers

ADMS supports using a secure HTTPS encrypted connection with TLS termination for interface server web services (Call Server, Crew Server, AMI Server, OMS Interface Server, DMS Interface Server, Net Data Server, OMS Adapter, CE Adapter, ANPS Adapter, and VOP Adapter).

To enable HTTPS for an ADMS web service or web service client, you need to bind an SSL certificate to the web service port, and then enable HTTPS configuration in the associated configuration file.

For production, it is recommended to use SSL certificates issued by a certificate authority. For testing and demo purposes, you can use selfsigned certificates.

If using self-signed certificates, you need the following certification files for ADMS interface servers:

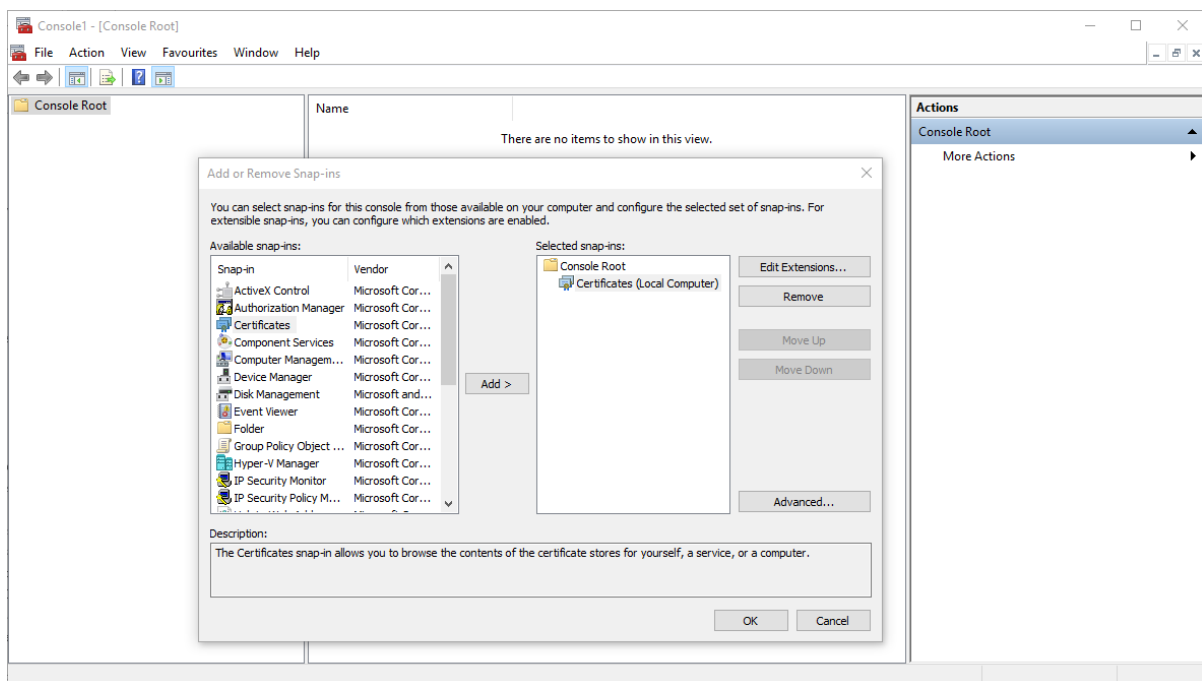
- Root certificate authority certificate. This certificate is used to issue and sign certificates. (For example, file name = ca.p12; CN = ADMS CA).
- ADMS interface server certificate. This certificate is used for secure communication between ADMS interface servers and clients. (For example, file name = keystore.p12; CN = ADMS)

8.3.1. Importing Certificates to the ADMS Hosts

When using self-signed certificates, you need to apply ADMS certificates on each host where ADMS interface servers are running.

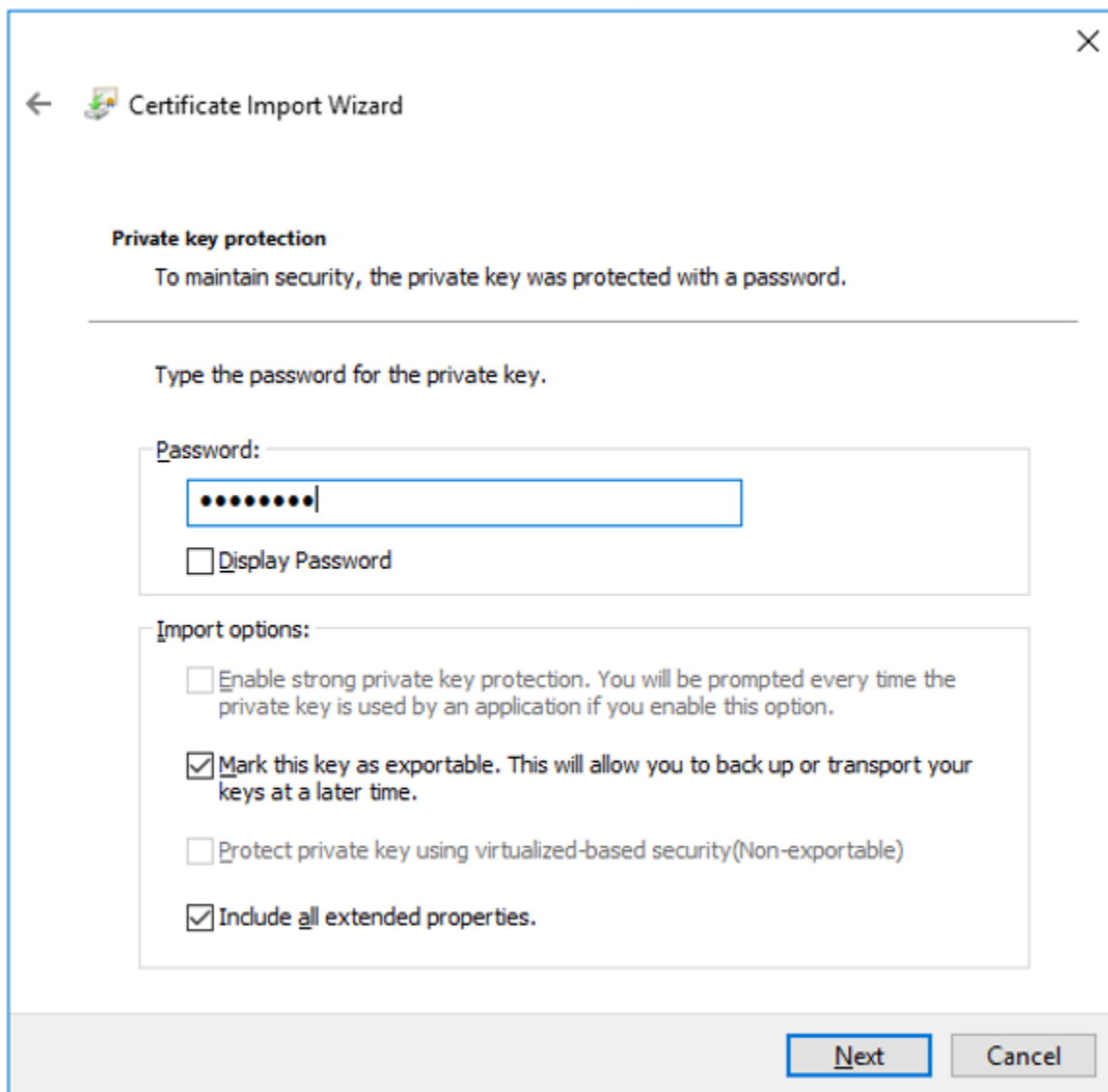
To import certificates to an ADMS interface server host:

1. Copy the Root certificate authority certificate (ca.p12) and the ADMS interface server certificate (keystore.p12) to the ADMS interface server host.
2. Open the Microsoft Management console.
3. From the File menu, select Add/Remove Snap-in.
4. From the available snap-ins, select Certificates and click Add.
5. Select the Computer Account option and click Next.
6. Select the Local Computer option and click Finish.



7. Click OK.
8. In Console Root, expand the Certificates (Local Computer) list.
9. Right-click Trusted Root Certification Authorities, select All Tasks, and click Import.
The Certificate Import wizard appears.
10. Verify that the Store location is Local Machine and click Next.
11. Click Browse, select the ca.p12 file to import, and click Next.
12. Enter the certificate password.
13. Disable the "Strong Private Key Protection" option.

14. Enable the "Mark this key as exportable" option, and click Next.

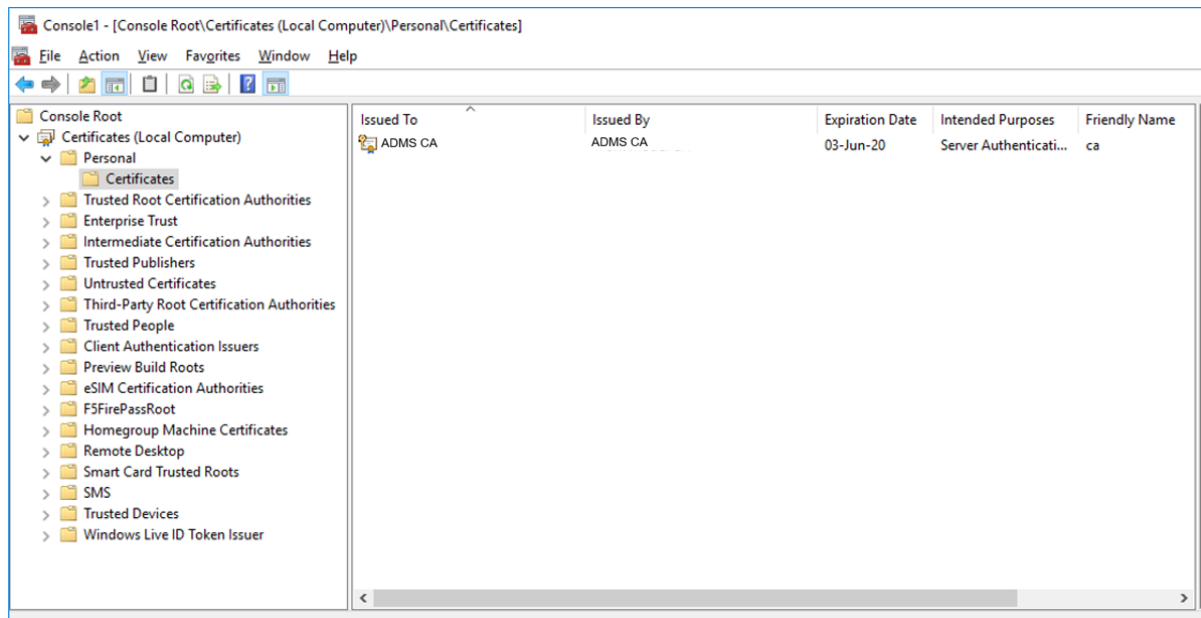


The image shows the 'Certificate Import Wizard' dialog box. At the top, there is a back arrow and the title 'Certificate Import Wizard'. Below this, the section 'Private key protection' is displayed, with the text 'To maintain security, the private key was protected with a password.' followed by a horizontal line. Underneath, it says 'Type the password for the private key.' There is a 'Password:' label followed by a text box containing ten dots. Below the text box is a checkbox labeled 'Display Password'. The 'Import options:' section contains four checkboxes: 'Enable strong private key protection. You will be prompted every time the private key is used by an application if you enable this option.' (unchecked), 'Mark this key as exportable. This will allow you to back up or transport your keys at a later time.' (checked), 'Protect private key using virtualized-based security(Non-exportable)' (unchecked), and 'Include all extended properties.' (checked). At the bottom right, there are 'Next' and 'Cancel' buttons.

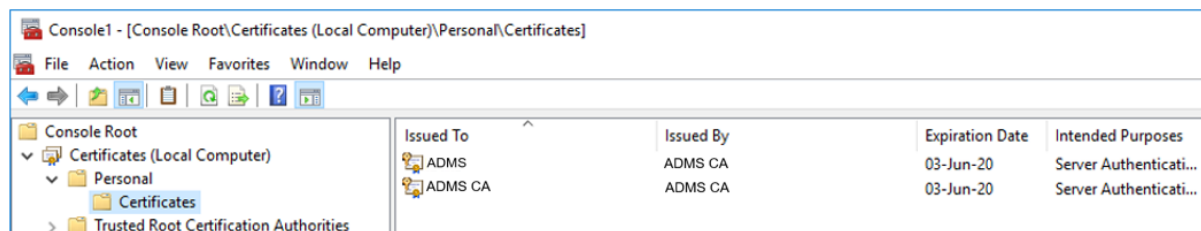
15. Select the Place All Certificates in the Following Store option, select Personal, and click Next.

16. Click Finish.

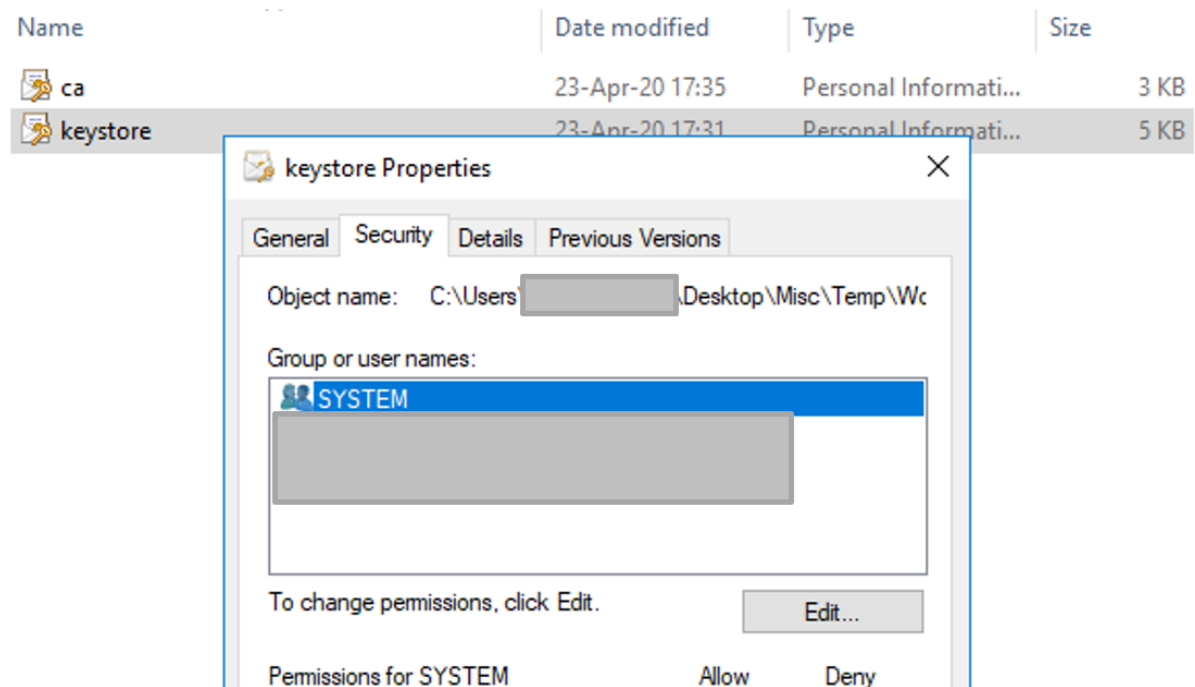
The root certificate appears in the Certificates list.



17. Perform the same operations to import the keystore.p12.
18. Verify that the Root certificate authority certificate (ADMS Root CA) and the ADMS server certificate (ADMS) are listed in the Personal folder under Certificates (Local Computer).



19. Right-click the ADMS server certificate (ADMS), point to All Tasks, and click Manage Private Keys. Alternatively, open the Security tab of the certificate Properties display.

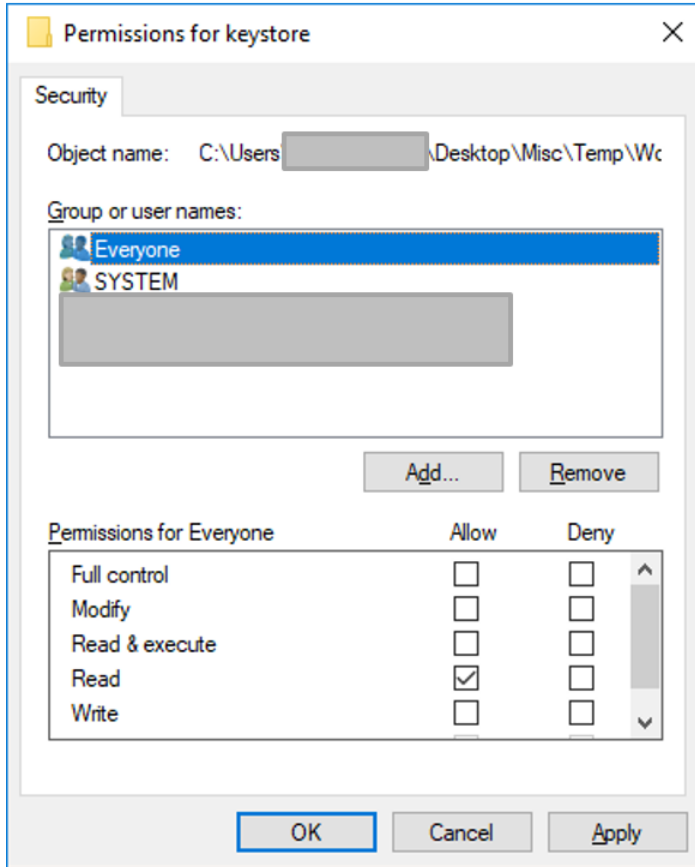


20. Add the user or service account that is used to run ADMS servers (or select Everyone, which is less secure) and give Read permissions.

To read from Trusted Root Certification Authorities, permission must be granted for users that do not have Read permissions.

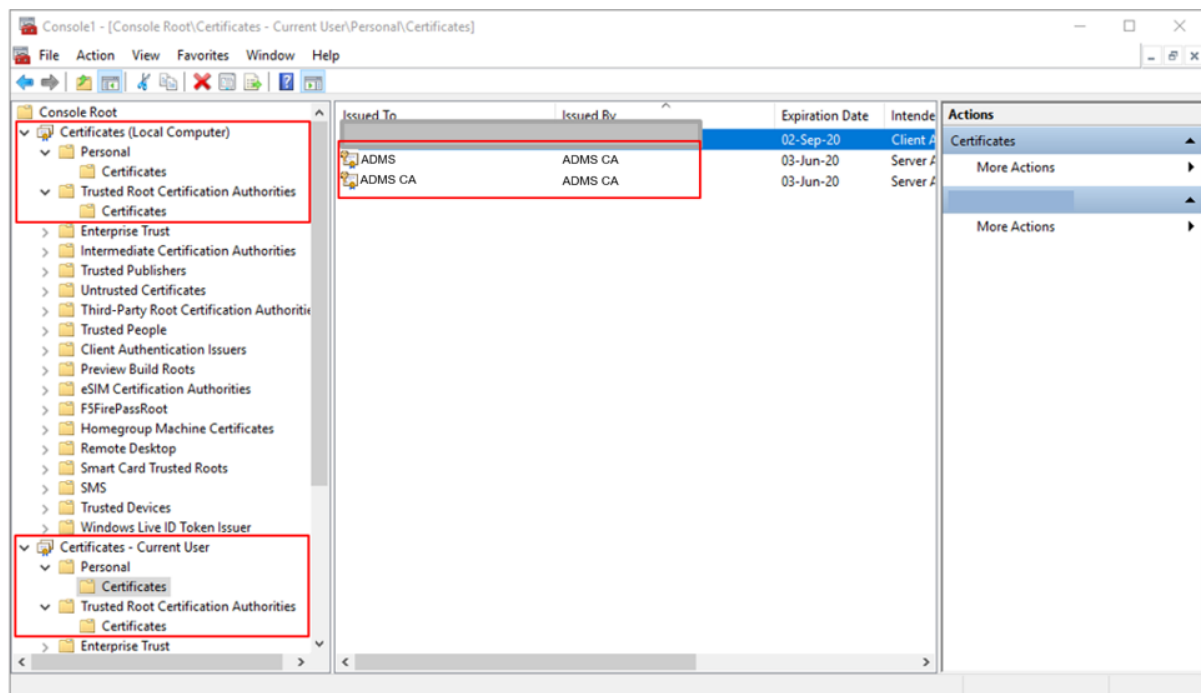
Important

Ensure that you give permissions to the private key, not the public key.



This allows access to the private key of the imported certificate to the user (service account) that is used to run ADMS servers. Without this, the servers cannot serve data in HTTPS mode.

21. Copy the ADMS CA and ADMS certificates using the context menu.
22. Navigate to the Certificates > Trusted Root Certification Authorities > Certificates folder.
23. Paste the ADMS and ADMS CA certificates from the Certificates folder context menu.
24. Add or copy the ADMS and ADMS CA certificates to the Certificates - Current User store (you may need to create the Current User store similar to the Certificates (Local Computer) store).



8.3.2. Binding Certificates to ADMS Interface Server Ports

For ADMS interface server HTTPS operation, you must bind certificates to all ADMS interface server listening ports (when running as web services) on each host where the interface servers are running.

Note

You can bind one certificate to all server ports.

To bind certificates to ADMS interface server ports:

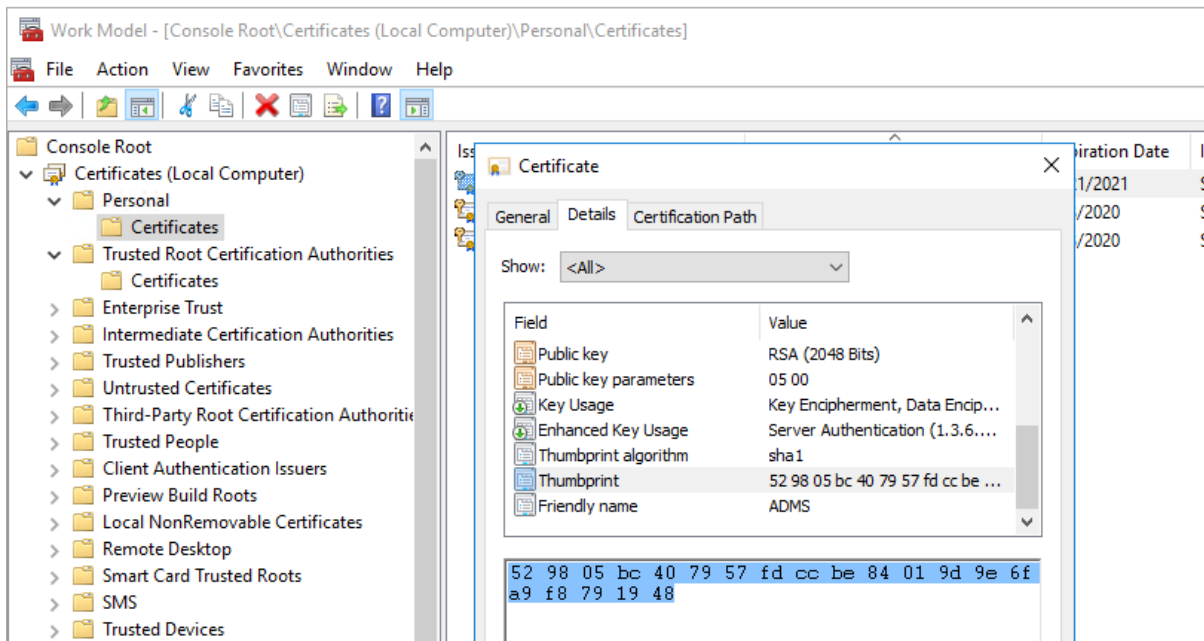
1. In the Microsoft Management Console, double-click a certificate (for example, ADMS).

The certificate properties window appears.

2. On the Details tab, copy the Thumbprint value.

Important

Delete all spaces in the copied string, if any.



3. Run Windows PowerShell as an administrator.
4. Run the following command.

```
netsh
```

5. Run the following command.

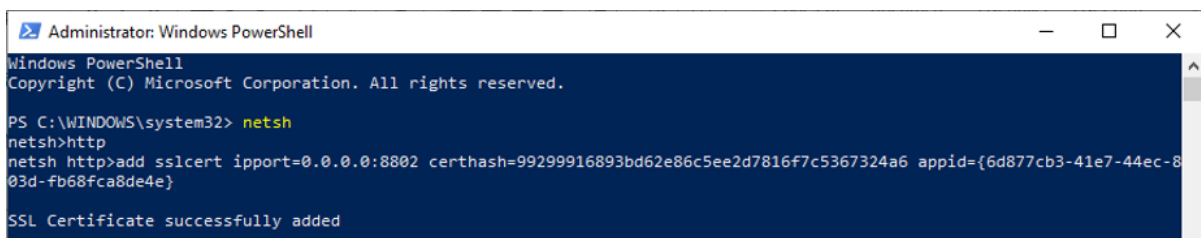
```
http
```

6. Run the following command to bind a certificate to a server port:

```
netsh http> add sslcert ipport=0.0.0.0:<Port> certhash=<Cert_Thumbprint> appid=<Application_GUID>
```

If you configure mutual client-server authentication, add the `clientcertnegotiation=enable` argument in the command. For example:

```
netsh http> add sslcert ipport=0.0.0.0:<Port> certhash=<Cert_Thumbprint> appid=<Application_GUID>  
clientcertnegotiation=enable
```



Note

This action fails if a certificate is already bound to the specified port.

7. Verify the certificate port binding configuration using the following command:

```
netsh http> show sslcert
```

8. Similarly, bind a certificate to the other ADMS interface server ports.

8.3.3. Configuring ADMS Interface Servers

You can configure secure communication using interface server configuration files (for example, CallServerConfig.xml).

To modify parameters in the interface server configuration files:

1. Change all "http" occurrences in parameter values to "https". For example:

```
<UriScheme>https</UriScheme>
<BaseUri>https://localhost:8443/api/adms/relay/call/v2</BaseUri>
<BaseUri>https://localhost:8443/api/adms/relay/ami/v2</BaseUri>
<Field name="OmsServiceURL">https://localhost:8802/Oms/services/</Field>
```

2. Enable secure web services. For example:

```
<SecureWebService>true</SecureWebService>
<Field name="SecureWebService">true</Field>
WEBSERVICE_SECURE = true
```

This enables HTTPS mode for ADMS interfaces web services.

3. Enable a web socket security web service:

```
<WSSecurityEnabled>true</WSSecurityEnabled>
```

4. Optionally, you can enable a two-way SSL authentication (this enables the server to request the client certificate).

In a client configuration file, set EnableDualSSL to true and specify the certificate store type and subject value. For example:

```
<EnableDualSSL>true</EnableDualSSL>
<CertStore>Personal</CertStore>
<CertSubjectName>ADMS</CertSubjectName>
```

This enables ADMS interfaces that are running as web service clients to authenticate servers.

8.3.4. Verifying HTTPS Configuration

To verify HTTPS configuration for a server:

1. Open a web browser.
2. Navigate to a server web interface URL that supports a GET call. For example:

For Call Server: `https://localhost:8446/api/adms/call/v2/Calls?from=0001-01-01T00:00:00.0000000`

For OMS Interface Server: `https://localhost:8802/oms/services/rest/GetAllCrews`

For the OMS Interface Server web service:

`https://localhost:18802/api/adms/oms/v2/Regions?groupType=STATION&id=EQSI`

3. Check that the web page loads and shows data.

To verify HTTPS configuration for a client:

1. Configure HTTPS for the Call Server and Relay Server.
2. Check that the Call Server and Relay Server successfully exchange HTTPS messages in the server logs.

8.3.5. Configuring Clients

Configuring HTTPS on the external application client side is optional. If HTTPS is configured only on the server side, clients will access the web service using the "https" scheme and the server will authenticate the clients. If HTTPS is configured on both the server and client sides, both the client and the server authenticate each other.

To enable HTTPS on the client side:

1. Install the client and server certificates in the client and server trusted root stores.
2. Use the `clientcertnegotiation=enable` argument when binding certificates to ports. This enables the Client Certificate Negotiation option, which is disabled by default.
3. Modify the client code to set the certificate when initializing the https connection. (In ADMS, clients are implemented using the .Net Framework.)

To modify the client code:

- Implement `MyHttpsRequestHandler` based on `System.Net.Http.WebRequestHandler`:
`MyHttpsRequestHandler.ClientCertificates.Add(clientCertificate)`, where the `clientCertificate` is an `X509Certificate2` object retrieved from either the Personal or TrustedRoot store.
- Implement `MyHttpsMessageHandler` based on `System.Net.Http.DelegatingHandler`:
initialize the base class using `MyHttpsRequestHandler`.
- Instantiate `System.Net.Http.HttpClient` with `MyHttpsMessageHandler`.
- Call `System.Net.Http.HttpClient.PostAsJsonAsync` to send message.

8.4. Configuring UAA for ADMS Interface Servers

User Account and Authentication (UAA) is another layer of security that can be applied with or without HTTPS. To use UAA, there must be a UAA web service to provide and authenticate UAA tokens. (The ADMS server-side implementation expects "password-grant" tokens.)

Note

ADMS does not provide a UAA server or a token service LDAP server. The UAA server must be implemented on the customer side. For more information about UAA, see <https://docs.cloudfoundry.org/concepts/architecture/uaa.html>

The UAA server is used for centralized identity management. The UAA server is an OAuth2 server, which issues and validates JWT (JSON Web Token) formatted access tokens for ADMS interface servers, such as OMS Interface server.

Access tokens are used in token-based authentication to allow an application to access an API. A token is verified each time when an application attempts to access an API. The passed token informs the API that the bearer of the token has been authorized to access the API.

The UAA workflow is as follows:

- ADMS interface clients retrieve an access token from the UAA server's URL (for example, <https://localhost/oauth/token>)
- ADMS interface servers validate clients' tokens at the UAA server's URL (for example, <https://localhost/uaa/introspect>)

Active (valid) tokens are those issued by the UAA server, not revoked by the resource owner, and issued within the validity time period (12 hours).

The client-side UAA authentication is implemented as follows:

- If there is a valid token cached, that token is added to the authorization header of the outgoing HTTPS requests.
- If there is no token cached, before sending any outgoing requests, the client application requests an access token (JWT) from a UAA server or a token service LDAP server.

The response from UAA has information about the token expiration time. The client keeps this data and requests a new token shortly before the expiration time.

The server-side UAA authentication is implemented as follows:

- The server intercepts incoming requests. The token is extracted from the authorization header of the request. This token is sent to the UAA server to check whether the token is valid. If the token is valid, the request is processed. Otherwise, the 401 error (unauthorized) is returned.

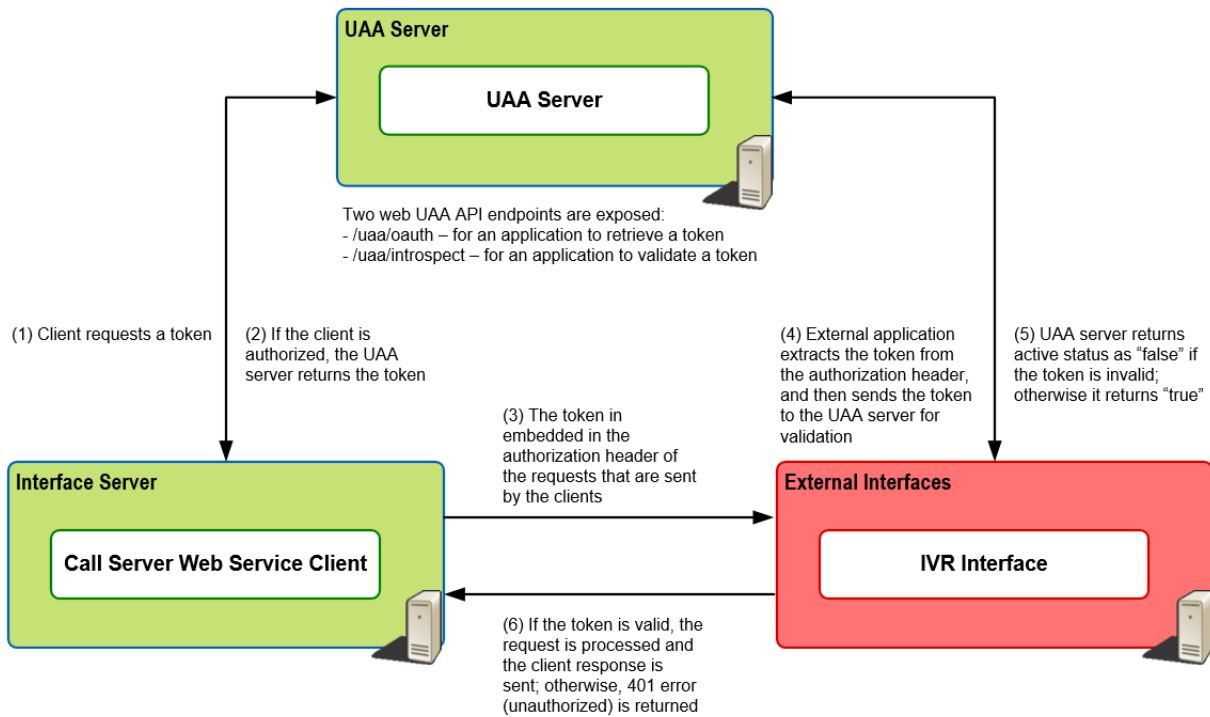


Figure 30. UAA Workflow Example

8.4.1. Configuring UAA Authentication

Web service clients use the UAA web service to get UAA tokens. Web services use the UAA web service to authenticate the incoming token in the client requests. Therefore, different endpoints need to be configured for getting tokens and for verifying tokens.

To set up UAA authentication:

1. For web services, enable the functionality to verify UAA tokens and specify the associated UAA server endpoint using the `UseUAAToken`, `UseUAATokenForIntrospect` and `UaaIntrospectEndPoint` parameters in ADMS server configuration files.

Note

Web services that are initialized by plugin definition use the `UseUAAToken` parameter to enable the functionality to verify UAA tokens. The other web services use the `UseUAATokenForIntrospect` parameter.

For example:

```
<UseUAAToken>true</UseUAAToken>
<UaaIntrospectEndPoint>https://localhost/uaa/introspect</UaaIntrospectEndPoint>
```

2. For web service clients, enable the functionality to get UAA tokens and specify the associated

UAA server endpoint using the UseUAAForGettingToken, UseUAAToken, and UaaEndPoint parameters in ADMS server configuration files.

Note

Web service clients that are initialized by plugin definition use the UseUAAToken parameter to enable the functionality to get UAA tokens. The other web services use the UseUAAForGettingToken parameter.

For example:

```
<UseUAAToken>true</UseUAAToken>
<UaaEndpoint>https://localhost/uaa/oauth/token</UaaEndpoint>
#Zitael
<UaaEndpoint>https://localhost/oauth/v2/token</UaaEndpoint>
```

3. When using the UAA server that is supplied with the GE's Outage Assist solution, on web service and web service client hosts, add system environment variables which are used to access UAA APIs. The format of the environment variables is the following:

```
<service/client name>_UAA_AUTH_PROJECID = <project-id>
<service/client name>_UAA_AUTH_CLIENTID = <id>
<service/client name>_UAA_AUTH_CLIENTSECRET = <secret>
```

Note

The defined environment variables determine the grant type that is used for an application to authenticate with the token server. If only <Application>_UAA_AUTH_CLIENTID and <Application>_UAA_AUTH_CLIENTSECRET are defined, the "client credentials" grant type is used (recommended approach). If <Application>_UAA_AUTH_PASSWORD and <Application>_UAA_AUTH_USERNAME are also defined, the "password" grant type is used (legacy approach).

Optional environment variables:

```
<service/client name>_UAA_AUTH_USERNAME = <Client_Username>(not needed for servers)
<service/client name>_UAA_AUTH_PASSWORD = <Client_Password>(not needed for servers)
```

The system gets the information from the environment variables and use them to send requests to the UAA web service.

Note

For web services and web service clients that are initialized by plugin definition, the name is defined in the Name field in the Configuration section in the configuration file.

For example, execute the following commands to add environment variables for OMS Interface Server:

```
SETX OMSINTERFACESERVER_UAA_AUTH_PROJECID 12345624233243
SETX OMSINTERFACESERVER_UAA_AUTH_CLIENTID oms-interface-service-client-id
SETX OMSINTERFACESERVER_UAA_AUTH_CLIENTSECRET oms-interface-service-client-secret
```

Optional environment variables:

```
SETX OMSINTERFACESERVER_UAA_AUTH_USERNAME service.account.etd
SETX OMSINTERFACESERVER_UAA_AUTH_PASSWORD Northstar#1
```

8.4.2. Verifying UAA Configuration

To verify UAA configuration for web service clients:

1. Open an API development tool (for example, Postman).
2. Send a token request (POST) of the Basic Authorization type to the UAA server to get the token.
For example:

Token get URL: <https://localhost/oauth/token>

Request Format:

```
POST /oauth/token HTTP/1.1
Content-Type: application/x-www-form-urlencoded
Accept: application/json
Host: localhost
client_id=<id>&client_secret=<secret>&grant_type=password&username=<username>&password=<password>
```

The UAA server returns the new token in the "access_token" section of the response. For example:

```
{
  "access_token": "<returned access token here>",
  "token_type": "bearer",
  "expires_in": 43199,      // unit is second, so this is 12hrs.
  "scope": "uaa.none",
  "jti": "a782cc5df49840dab9e434b87ee29e81"
}
```

3. Send a GET request of the Bearer Token type with the new token in the request header to an ADMS interface server.

The ADMS interface server returns a correct response.

To verify UAA configuration for web services:

1. Open an API development tool (for example, Postman).
2. Send a token request (POST) of the Bearer Token type with the new token in the request header to the UAA server to verify the token. For example:

Token verify URL: <https://localhost/uaa/introspect>

Request Format:

```
POST /introspect HTTP/1.1
Authorization: bearer 21d1530d20414379997461200b3a4232
Host: localhost
Content-Type: application/x-www-form-urlencoded

token=<access token here>
```

The UAA server returns the token status in the response. Active token is a token that was issued by this authorization server, was not revoked by the resource owner, and is not expired.

For example:

```
{
  "revocable": false,
  "active": true
}
```

9. Configuring Web Maps

The ADMS system supports web-based map technology, allowing you to choose among backgrounds such as street, aerial, clouds, and radar using the Common toolbar button on the network view. For web maps, you need to pre-download map tiles or have access to a map web service to provide background tiles for ADMS.

To add web maps to the network view and configure the toolbar menu to manipulate the map's layers:

1. Using the Converter tool, process the latitude/longitude coordinates to convert into x,y coordinates.

For security purposes, an offset might be created to hide the actual coordinates of the devices. To show the maps, your GE project team will need to implement a converter to transpose x,y coordinates to lat-long.

The converter must implement the interface `ICoordinateConverter` from `NetUtil.dll`.

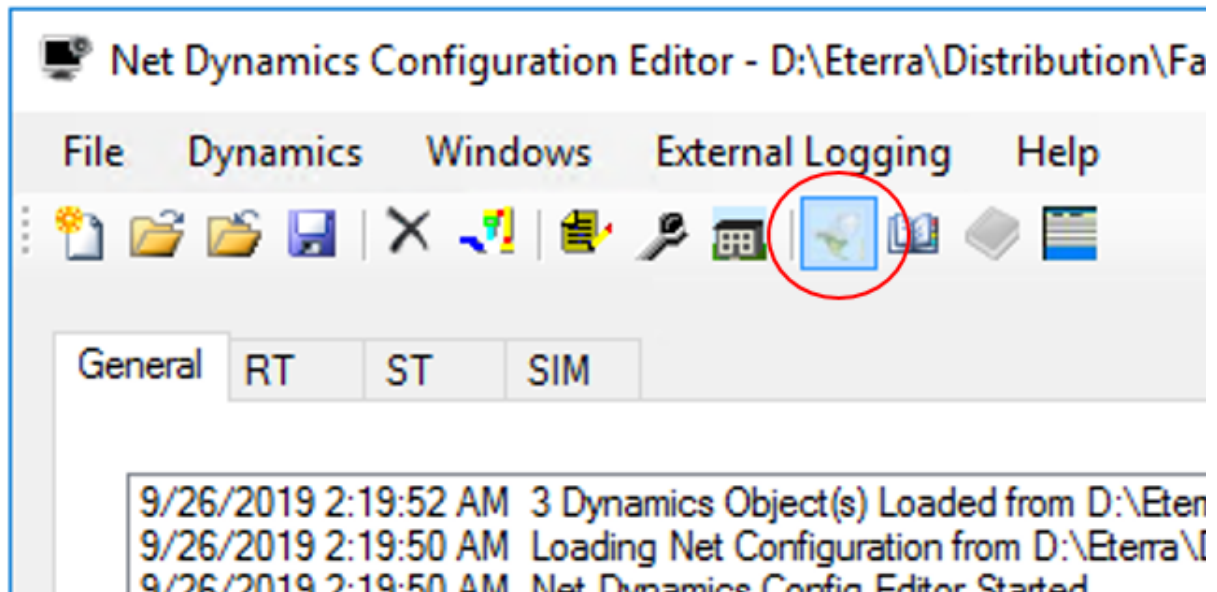
The signature of the interface is as follows:

```
public interface ICoordinatesCovertor
{
    void ToXY(double Latitude, double Longitude, ref double X, ref double Y);
    void ToLatitudeLongitude(ref double Latitude, ref double Longitude,
        double X, double Y);
    bool GetURI(string inputString, ref string URI);
}
```

2. For web map purposes, only the function `ToLatitudeLongitude` is needed. Contact GE to obtain a starter project for `sampleconverter`.
3. Enable a map web service, such as Map Proxy.

Alternatively, download tiles from your choice of map tile providers (consistent with the provider's license terms). OpenStreetMaps is one of many choices. GE does not provide tiles in the standard kit. Contact GE for guidance on how to select and download tiles from a provider, if needed.

4. Open the Net Dynamics Configuration Editor tool.
5. Click the Web Map Providers button.

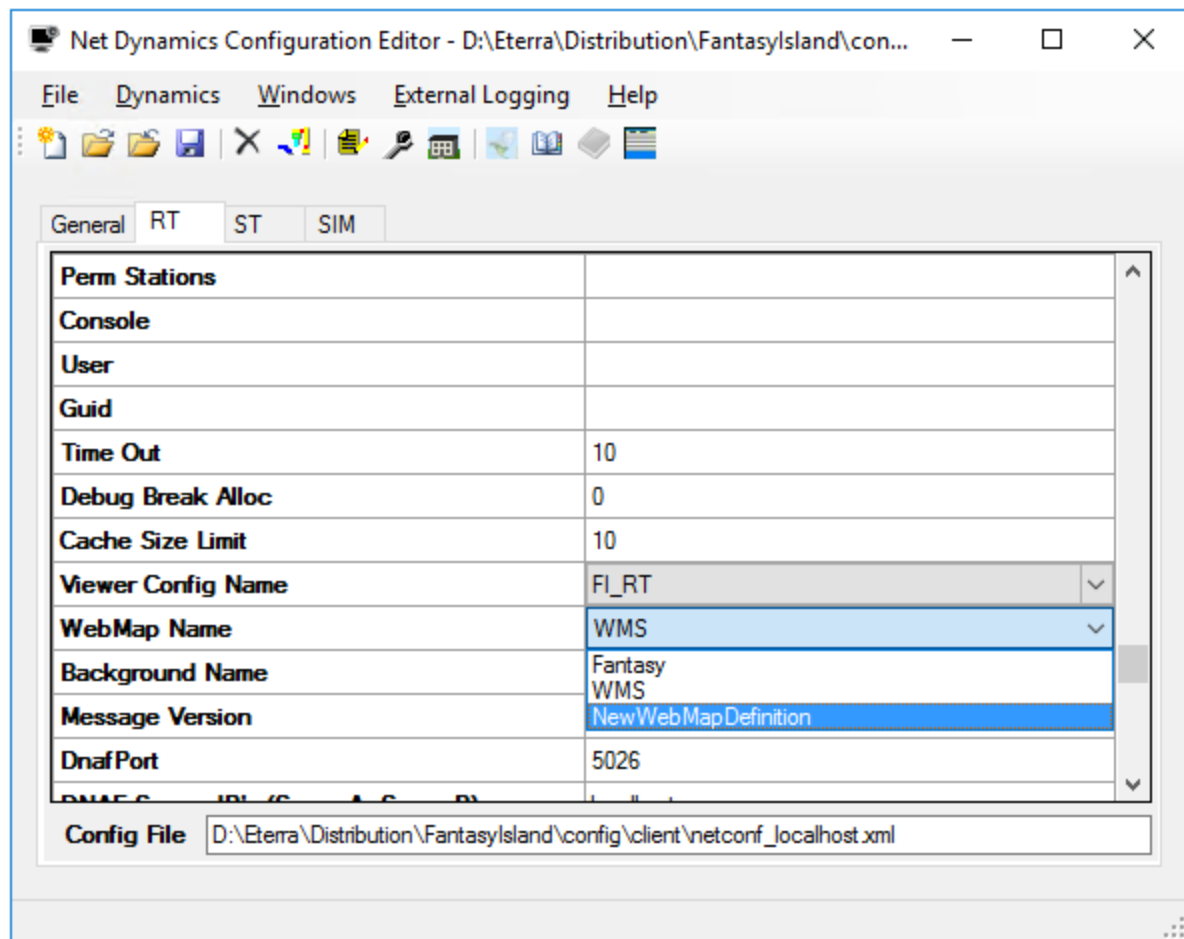


6. Create a new Web Map Provider object by entering the following fields:

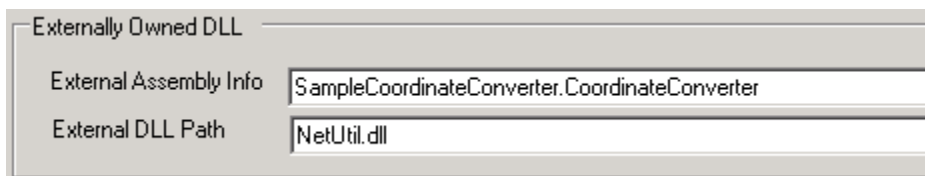
Web Map Name	Local Image Path	WMS Providers	Tile Name Pattern	Max Map Zoom Level
Fantasy	%IDMS_ROOT%\models\...		{0}\{1}\{2}.tile	18
WMS	http://<MapProxyHost>:8080			22
NewWebMapDefinition	http://localhost:8080			0

- **Web Map Name:** Web map provider name.
- **Local Image Path:**
 - For using pre-downloaded map tiles: Path to a folder with map tiles.
 - **For using a map web service:** URL of the map web service.
- **WMS Providers:** This field is for future use.
- **Tile Name Pattern:** Three components uniquely specify a tile: map zoom level, column, and row. Entering {0}\{1}\{2}.tile in this field indicates that map tiles are stored with a hierarchy. This improves performance when large numbers of map tiles are stored in a folder. This field is required when using pre-downloaded map tiles only.
- **Max Map Zoom Level:** Maximum map zoom level, from 1 to 23, where 23 indicates best resolution. Depending on which map provider is used, this parameter can change.

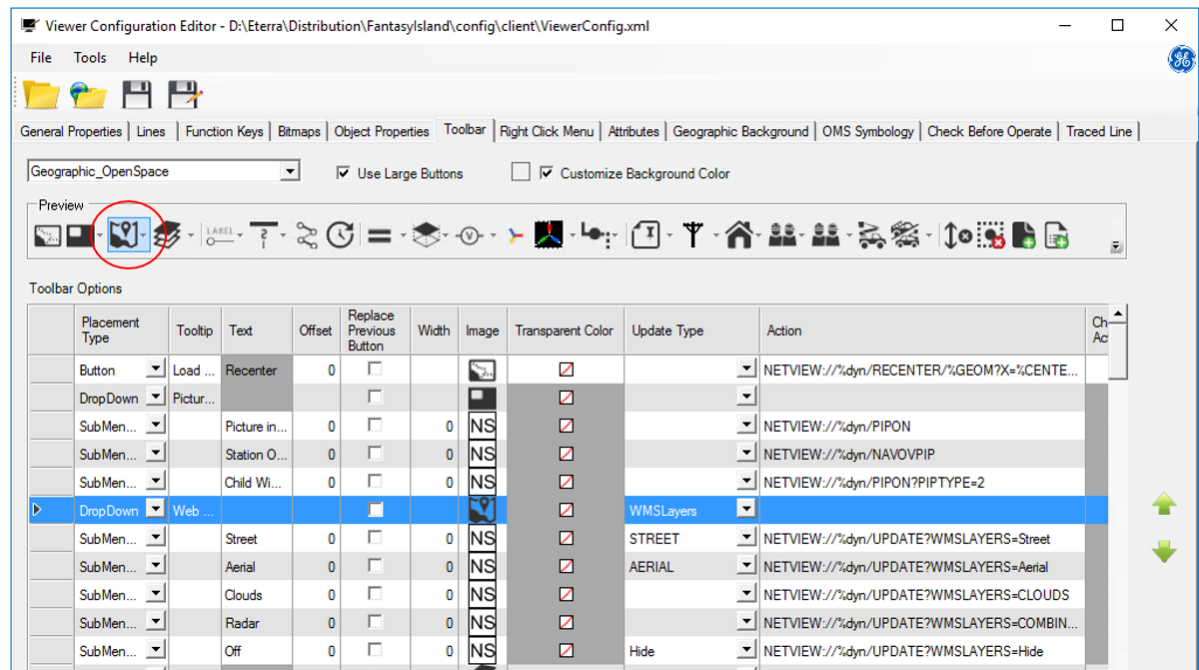
7. Click OK. The Web Map Name menu is updated with the new web map definition for each dynamics instance.



8. Save your changes and close the Net Dynamics Configuration Editor.
9. Open the Viewer Configuration Editor.
10. On the General Tab, specify the Converter DLL and the full class name in the Externally Owned DLL group.



11. On the Toolbar tab, adjust the configuration of the toolbar button to show/hide Webmaps layers. The Fantasy Island kit has five menu options configured: Street, Aerial, Clouds, Radar, and Off.



For more information about how to configure toolbar buttons and layers, refer to the "Toolbar" chapter in the *Viewer Configuration Editor User Guide*.

10. Read-Only Server Configuration

This chapter describes the read-only system configuration and the installation tasks required to set up this configuration.

10.1. System Overview

The ADMS read-only system configuration allows you to replicate a copy of your production DMS/OMS system data so that users who require read-only access to the data can view it without interfacing directly with the production system. Typically, the ADMS read-only system is placed in a Production Control System Demilitarized Zone (DMZ) located outside the production system's network firewall. The ADMS read-only system's primary subsystems are:

- Habitat
- ADMS read-only server components
- Web server
- ADMS client

The following figures shows the read-only ADMS client/server system for the DMS-only and DMS and OMS production ADMS systems.

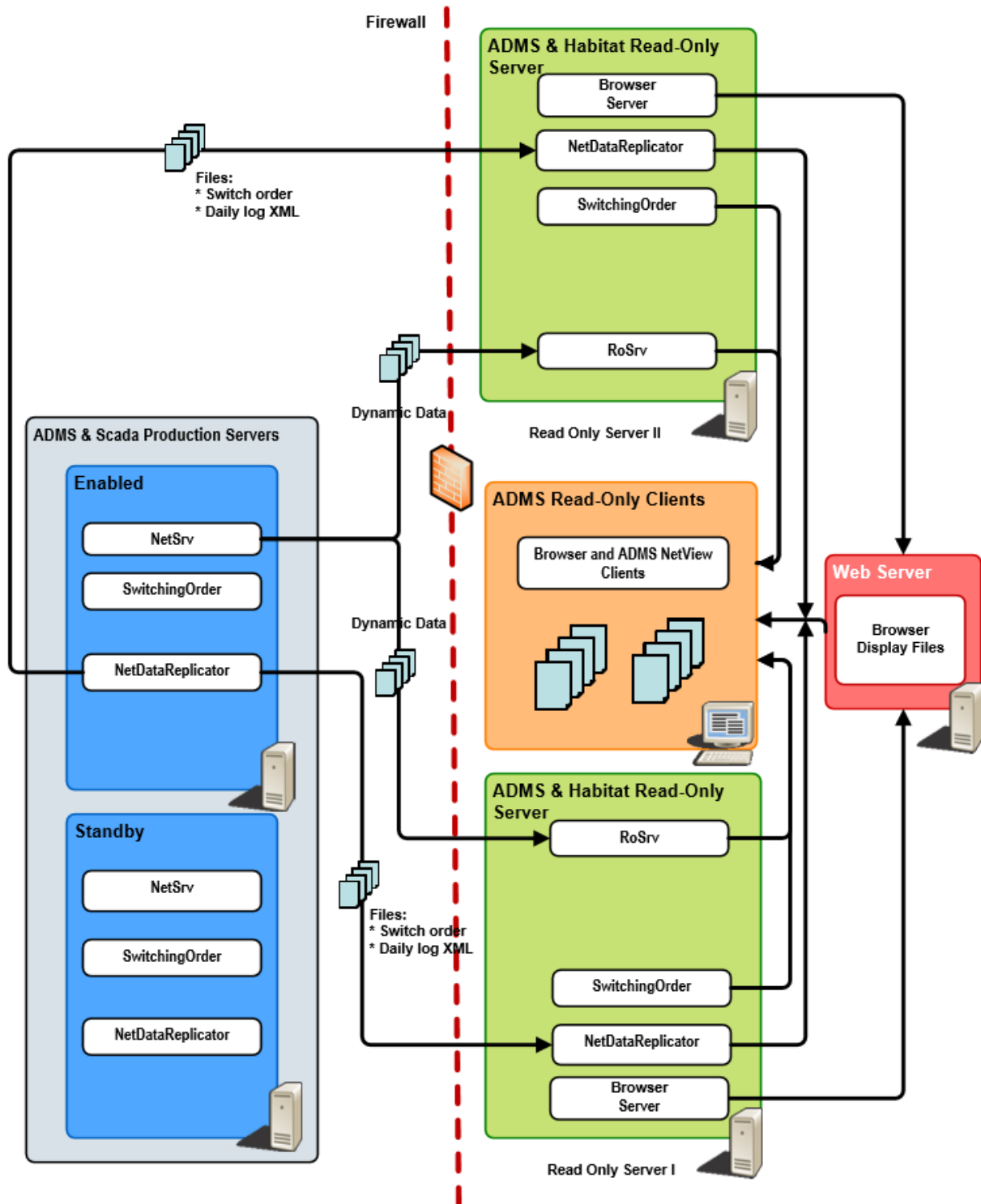


Figure 31. Read-Only System (DMS Only) Configuration Overview

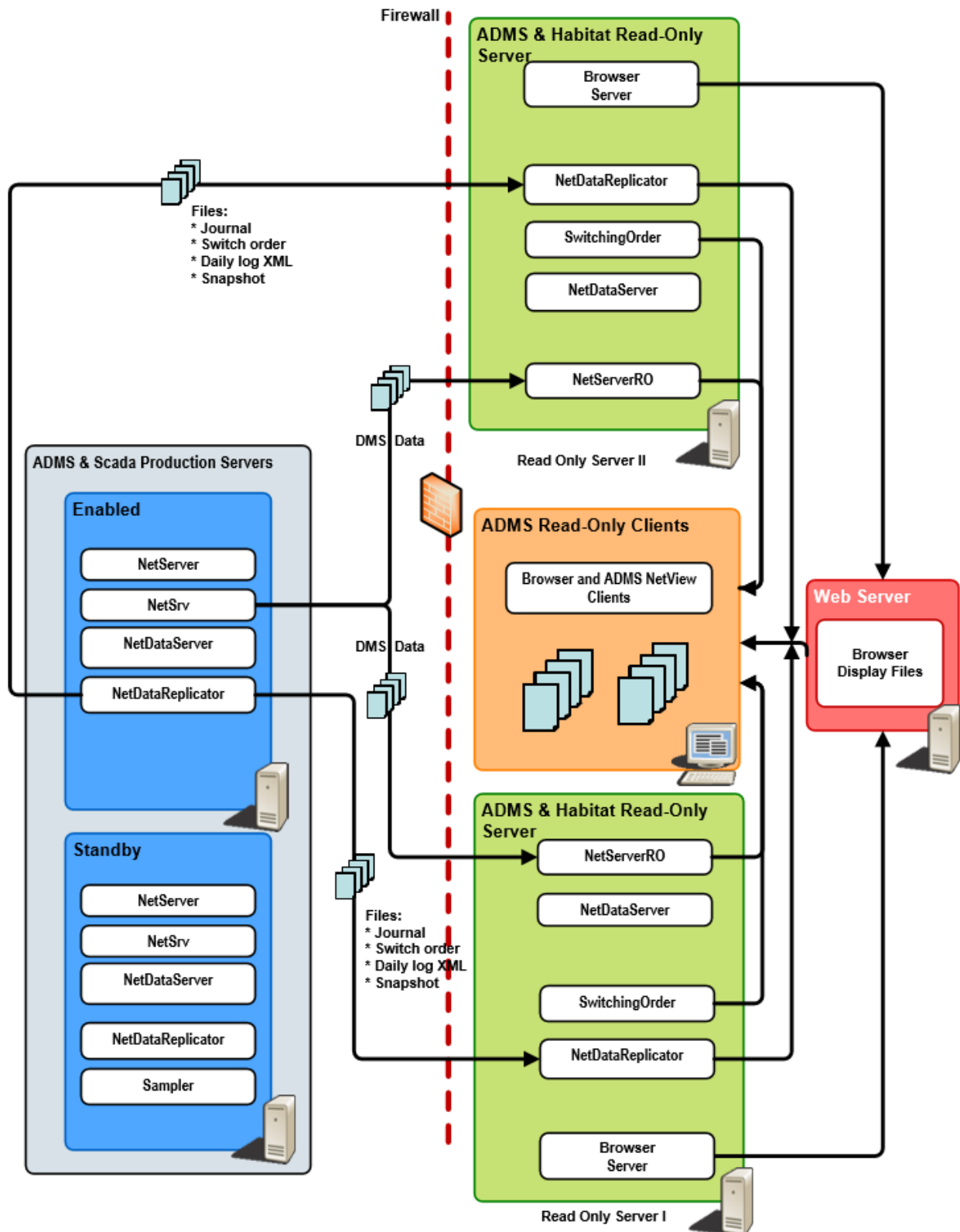


Figure 32. Read-Only System (DMS and OMS) Configuration Overview

Note that the direction of the arrows in the diagram illustrates the "one-way" flow of data from the production ADMS system to the read-only ADMS system and its clients. Basic descriptions of each read-only system component and their functions are provided below.

Read-Only RoSrv/NetServerRO

On the production DMS-only system, the NetSrv server process running on the read-only server connects to an instance of the NetSrv process on the production server.

On the production DMS and OMS system, the NetServerRO server process running on the read-only server connects to an instance of the NetSrv process on the production server.

Read-only RoSrv/NetServerRO process queries the production NetSrv process for data and replicates it onto the ADMS read-only system. It also serves the replicated DMS/OMS data to the read-only system clients. The read-only RoSrv/NetServerRO process prevents the read-only clients from executing DMS/OMS operations (for example, opening and closing switches) and making data changes. For more information on NetSrv, NetServer, and NetServerRO, refer to the *ADMS Architecture Overview*.

The read-only RoSrv/NetServerRO process can replicate the following categories of DMS/OMS data from the production system:

- Regular dynamics
 - Switches
 - Cuts
 - Temporary switches
 - Faults
 - Tags
 - Capacitors (if enabled)
 - Tap changers
 - Composite switches
 - Decorations (moveable objects)
- Mobile dynamics
 - Scada annotations
 - Vehicles (not in use)
- Manual entry dynamics
 - Manual entry
- Annotation Dynamics
- Fast Dynamics
 - Power flow
 - Fault locations

- SCADA measurement qualities
- SCADA measurement qualities and values
- DNAF Input Data

Net Data Server

The read-only Net Data Server processes replicated OMS data from the production ADMS system. For more information about Net Data Server, refer to the *ADMS Architecture Overview*.

Net Data Replicator

The read-only Net Data Replicator processes replicated DMS and OMS data served by the enabled Net Data Replicator on the production ADMS system. The Net Data Replicator application copies journal files, and switch order and daily log XML files, from the enabled machine to the standby machines. It can run in DMSONLY and READONLY modes. In DMSONLY mode, defined by the Net Data Replicator configuration file, it copies only the switch order and daily log XML files from the enabled system. For more information on Net Data Replicator, refer to the *ADMS Architecture Overview*.

RdServer

RdServer is an application delivered with the Windows version of Habitat. It connects to the enabled Habitat Sampler application running on the production ADMS system and replicates Browser display data, which is written into HABITAT/SCADA application container clones on the read-only ADMS system.

Browser Server

The Browser data server (WfgServer) is a Habitat process. On the ADMS read-only system, WfgServer interacts with the ADMS read-only clients and queries the HABITAT/SCADA application clones that contain replicated Browser display data from RdServer. The replicated display data is then served to the ADMS read-only clients by WfgServer. WfgServer also interfaces with the Permit application in Habitat and serves the Browser display user permissions to the ADMS read-only clients.

Web Server

The web server provides the following functions on the read-only ADMS system:

- Serves the compiled Browser display files to the ADMS read-only clients.
- Serves required configuration and infrastructure files to the ADMS read-only clients.
- Helps initiate and facilitate the network connection between the ADMS read-only clients and the WfgServer process on the ADMS read-only server.

ADMS Read-Only Client

Architecturally, the ADMS read-only client is essentially the same as a real-time, production ADMS client. Its two main components are the Browser client and the ADMS NetView client. The primary difference is that the read-only client is configured specifically to only query for the ADMS data and Browser display data served by the read-only server.

It is important to note that the ADMS read-only system can be deployed in a variety of ways. Configuration options are not limited to the example system overview schematic shown above. For example, you may want to deploy the read-only RoSrv/NetServerRO, Net Data Server, and Net Data Replicator processes on one server and the RdServer process on another server. Conversely, you may want to consolidate all ADMS read-only server processes, as well as the web server, on a single server. Finally, you can use the Habitat CFGMAN subsystem to configure a dual-redundant ADMS read-only system.

Important

If the read-only RoSrv/NetServerRO application cannot communicate with the production NetSrv application, the clients will not be able to connect, or will become disconnected. This is intentional so that read-only users are not misinformed by old data and they must contact a system administrator to figure out what is wrong.

10.2. System Requirements

Note

Refer to the *ADMS Release Notes* for the latest information about supported software versions.

Supported Operating Systems:

- Microsoft Windows 10 64-bit (ADMS clients and tools)
- Microsoft Windows Server 2019 64-bit (ADMS servers and ADMS clients)
- Microsoft Windows Server 2016 64-bit (ADMS servers and ADMS clients)

Required GE Software:

- Habitat
- ADMS
- Browser
- Advanced Presentation Framework

Web Server Software:

- Microsoft Internet Information Services (IIS) 7.0 or later

For more information about system software and hardware requirements, consult the following documents for the supported versions:

- *Habitat Release Notes*
- *Browser Release Notes*
- *ADMS Release Notes*
- *Advanced Presentation Framework Release Notes*

10.3. Installing the ADMS Read-Only System

This section describes the basic ADMS read-only system installation steps. As stated earlier, the system can be deployed and configured in a number of different ways, and it is assumed that you have at least a basic working knowledge of how to install and configure Habitat, ADMS, APF, Browser, and IIS. In general, only information specific to configuring the ADMS read-only system is covered. The instructions are oriented towards deploying a read-only system configuration similar to the diagrams shown in [System Overview](#).

10.3.1. Installing Habitat

Habitat is the first ADMS read-only system server component that must be installed. Refer to the *Habitat Release Notes* and the *Habitat Software Installation and Maintenance Guide* for a list of Habitat system requirements and software prerequisites.

Install the software prerequisites and then install the default Habitat configuration. This typically includes installing the APF and the Browser client. Refer to the *Advanced Presentation Framework Release Notes* and the *Advanced Presentation Framework Software Installation and Maintenance Guide* for a list of Advanced Presentation Framework software prerequisites. Refer to the *Browser Release Notes* and the *Browser Software Installation and Maintenance Guide* for a list of Browser software prerequisites.

After you install APF, Browser, and Habitat, make sure:

- Habitat starts successfully.
- The Browser client starts and successfully connects to the Habitat WfgServer process.
- The Browser client can call up Habitat displays.

10.3.1.1. Configuring Permit

After you confirm that Habitat and the Browser client are functioning properly, you can configure the Habitat Permit application to include your read-only users. Refer to the *PERMIT User Guide* (available in the *Habitat* documentation set) for information about modeling Permit users and permissions.

On the ADMS read-only server system, make sure that the read-only users in Permit are configured to only have read permissions. This helps ensure that the read-only users are not allowed to change data or perform any Habitat application functions on the read-only server.

10.3.1.2. Configuring RdServer

RdServer is part of the Windows-only Habitat delivery. Within the context of the ADMS read-only system, RdServer is used to replicate HABITAT and SCADA application display data from the production ADMS system to the ADMS read-only server. RdServer accomplishes this by connecting

to a Sampler application process running on the production ADMS system (for more information about Sampler, refer to the *HABConnect User Guide*). RdServer writes the display data it receives from Sampler into container application clones on the ADMS read-only server. The ADMS read-only clients are configured to query for the Browser display data in the container application clones on the ADMS read-only server.

Information related to installing and configuring the ADMS read-only system web server and ADMS read-only clients is provided later in this document. Specific information about configuring RdServer can be found in the "Configuring and Maintaining a Replicated Data server" section of the *Habitat Software Installation and Maintenance Guide*. As stated in that document, a copy of the compiled Browser HABITAT and SCADA application display file directories from the production ADMS are required in order to configure RdServer. These compiled display directories should be copied from the production ADMS system's Browser display web server.

10.3.2. Installing the DMS Server (DMS Only)

Refer to the *ADMS Release Notes* for a list of ADMS server software prerequisites. Install the software prerequisites and then install the default ADMS configuration on the ADMS read-only server.

For DMS only, the two DMS server processes that are used in the DMS read-only system are NetSrv and Net Data Replicator. You do not need to install the Net Data Server if you are running a read-only, DMS-only system; if so, skip [Configuring the Read-Only Net Data Server for OMS](#).

10.3.3. Installing the DMS Server and OMS Server (DMS and OMS)

Refer to the *ADMS Release Notes* for a list of ADMS server software prerequisites. Install the software prerequisites and then install the default ADMS configuration on the ADMS read-only server.

For ADMS, which includes DMS and OMS functionality, the ADMS server processes that are used in the ADMS read-only system are NetSrv, NetServer, Net Data Server, and Net Data Replicator.

ADMS read-only system configuration information related to these processes appears in the following sections.

10.3.4. Configuring the Read-Only RoSrv/NetServerRO

A read-only RoSrv/NetServerRO process configuration file (`NetServerReadOnlyConfig.txt`) is typically located in the `%IDMS_ROOT%\config\server` folder. For more information, see [NetServerReadOnlyConfig.txt](#).

An example of the most relevant read-only configuration parameters in this file are shown below:

```
DYNAMIC_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly\  
REALTIME      = 0
```

```

FD_POWERFLOW_QUERY_ENABLED = 1
FD_MEASUREMENT_QUERY_ENABLED = 1
FD_MEASUREMENT_QUALITY_QUERY_ENABLED = 1
FD_FAULT_LOCATION_QUERY_ENABLED = 1

DNAF_RESULTS_QUERY_ENABLED = 1

MAIN_SERVER_HOST = mydmsa,mydmsb,mydmsc,mydmsd
MAIN_SERVER_PORT = 5025
DNAF_SERVER_HOST = mydnafa,mydnafb,mydnafc,mydnafd
DNAF_SERVER_PORT = 5026
SERVER_PORT      = 5055
OMS_SERVER_PORT  = 5051

```

Of particular note are the following parameters:

- **MAIN_SERVER_HOST:** Specifies the host(s) where the production NetSrv process is running.
- **MAIN_SERVER_PORT:** Specifies the TCP/IP listening port number for the production system's NetSrv process.
- **SERVER_PORT:** Specifies the TCP/IP listening port number for the read-only RoSrv/NetServerRO process.
- **DNAF_SERVER_HOST:** Host name of the DNAF server. This parameter is used for querying DNAF results from the DNAF server.
- **DNAF_SERVER_PORT:** TCP/IP port used by the DNAF server. This parameter is used for querying DNAF results from the DNAF server.
- **OMS_SERVER_PORT:** The TCP/IP port that the NetServerRO listens on to support client OMS capabilities. (Managed connection port.) This is the port to which the clients connect, such as OMS Interface server, Netviewer Adapter, and Net Data Replicator.
- **DNAF_RESULTS_QUERY_ENABLED=1:** When set to 1, enables replicating DNAF results to the read-only server.

To configure the read-only RoSrv/NetServerRO process:

1. Create a read-only dynamics file subdirectory on the read-only server.

This subdirectory is typically created below the %IDMS_ROOT%\dynamics subdirectory. The DYNAMIC_PATH parameter in the configuration file should be set to point to this directory.

2. Copy the model files from the production ADMS system into the corresponding model files directory on the ADMS read-only server.

Typically, this is the %IDMS_ROOT%\models\RT subdirectory. The MODEL_PATH parameter in the configuration file should be set to point to this directory.

3. Integrate a HABITAT startup task for the read-only RoSrv/NetServerRO process into the PROCMAN model on the ADMS read-only server.

Figure below shows examples of the TASK, FLAG, NODE, PATH, and TFLAG records that must be added to the PROCMAN model. Note that the "Parameters" display field for the TASK record

entry references an environment variable string (%ROSRV_CONFIG%). The environment variable points to the directory path and RoSrv/NetServerRO configuration file name. This environment variable must be defined and set for this PROCMAN TASK record modeling convention to work.

4. Add the ROSRV entry to the idms.appdef file to allow PROCMAN to start the read-only RoSrv/NetServerRO process.

The screenshot displays the HABITAT configuration interface with the following settings:

- Task (1):** ROSRV. Application: IDMS. Family: . Taskset: 0. Appmgr ID: ROSRV. Test Sim Time: 1. Stall Checking: 0. Start Timeout: . Stop Timeout: 30. Parameters: %ROSRV_CONFIG%. Down Alarm: . Passive: ☒. Task Must Stop: . Test Mode: . Retry Alarm: . Restart Logic... Hang Alarm: . Startup: . Timeout Alarm: . Log: . Quota...
- Flag (1):** RUN_ROSRV. Check Env: . Init State: ☒. Reset: . Set Alarm: . Reset Alarm: . Log: ☒.
- Node (1):** Name: ST_ROSRV. Max. Counter: 0. Log: . Start: ☒. Exit: . WaitExit: . Junction: . Parameters: .
- Path (1):** ROSRV. To Node: 1 ROSRV. ST_ROSRV 1. Check Count: . Update Count: . Reset Count: . Auto Wait: ☒. Parallel: ☒. Log: . Open: .
- Flag Condition (1):** IF RUN_ROSRV 1.

Figure 33. Example HABITAT Startup Task, Read-Only RoSrv/NetServerRO Process

10.3.5. Configuring the Read-Only Net Data Server for OMS

Read-only Net Data Server supports reviewing cleared and canceled incidents in the Outage Management System. Net Data Server also supports read-only access to the current OMS data and recent history for users via web services.

The read-only Net Data Server configuration file NetDataServerReadOnlyConfig.txt is typically located in the %IDMS_ROOT%\config\server subdirectory. Of particular note are the following Net Data Server parameters:

- **USE_ROLE_FROM_CFGPROXY:** The read-only Net Data Server application will not have a CfgProxy connection, therefore the USE_ROLE_FROM_CFGPROXY should be set to false or zero.
- **SERVER_PORT:** Specifies the TCP/IP listening port number for the read-only Net Data Server process.
- **WEBSERVICE_SERVER_NAME:** A host where web services are installed and listening to other interfaces.
- **WEBSERVICE_PORT_NUMBER:** TCP/IP port used for web services.
- **OMS_ENABLE_DASHBOARD_SUPPORT:** Set to "true" to enable the dashboard and history correction support.
- **OMS_POLYGON_GROUPS:** The file referenced by the OMS_POLYGON_GROUPS parameter must

be copied to the ADMS read-only server from the ADMS production system. This file is typically placed in the %IDMS_ROOT%\models\RT directory. The OMS_POLYGON_GROUPS parameter must be configured to reference the file.

For example:

```
USE_ROLE_FROM_CFGPROXY      = 0
SERVER_PORT = 6096
WEBSERVICE_SERVER_NAME = localhost
WEBSERVICE_PORT_NUMBER = 9900
OMS_ENABLE_DASHBOARD_SUPPORT = true
OMS_POLYGON_GROUPS          = %IDMS_ROOT%\models\oparea_polygongroup.xml
```

For more information about NetDataServerReadOnlyConfig.txt parameters, see [NetDataServerConfig.txt](#).

A HABITAT startup task for the read-only Net Data Server process must be integrated into the PROCMAN model on the ADMS read-only server.

Examples of the TASK, FLAG, NODE, PATH, and TFLAG records that must be added to the PROCMAN model are shown in figure below. Note that the "Parameters" display field for the TASK record entry references an environment variable string (%RODATASRV_CONFIG%). The environment variable points to the directory path and Net Data Server configuration file name. This environment variable must be defined and set for this PROCMAN TASK record modeling convention to work.

The RODATASRV entry must also be added to the idms.appdef file to allow PROCMAN to start the read-only Net Data Server process.

The screenshot displays the PROCMAN configuration interface with five records:

- Task (2):** RODATASRV. Application: IDMS. Family: (empty). Taskset: 0. Appmgr ID: RODATASRV. Test Sim Time: 1. Stall Checking: 0. Start Timeout: (empty). Stop Timeout: (empty). Parameters: %RODATASRV_CONFIG%. Down Alarm: (empty). Passive: (checked). Task Must Stop: (empty). Test Mode: (empty). Retry Alarm: (empty). Restart Logic... (button). Hang Alarm: (empty). Startup: (empty). Timeout Alarm: (empty). Log: (checked). Quota... (button).
- Flag (2):** RUN_RODATASRV. Check Env: (empty). Init State: (checked). Reset: (empty). Set Alarm: (empty). Reset Alarm: (empty). Log: (checked).
- Node (2):** Name: ST_RODATASRV. Max. Counter: 0. Log: (empty). Start: (checked). Exit: (empty). WaitExit: (empty). Junction: (empty). Parameters: (empty).
- Path (2):** RODATASRV. To Node: 2. RODATASRV. ST_RODATASRV. Check Count: (empty). Update Count: (empty). Reset Count: (empty). Auto Wait: (checked). Parallel: (checked). Log: (empty). Open: (empty).
- Flag Condition (2):** IF. RUN_RODATASRV. 2.

Figure 34. Example HABITAT Startup Task, Read-Only Net Data Server Process

Also, you need to add an HTTP URL namespace reservation for the read-only Net Data Server web service URL. For more details, see [Adding an HTTP URL Namespace Reservation](#).

10.3.6. Configuring the Read-Only Net Data Replicator

The Net Data Replicator application copies journal files and switch order XML files from the enabled machine to the standby machines. It can run in DmsOnly and ReadOnly modes. In DmsOnly and ReadOnly mode, defined by the Net Data Replicator configuration file, it copies just the switch order and daily log XML files by connecting to the enabled machine and copies the data to the read-only server. The application may also interface with CFGMAN via CfgProxy, which allows it to behave differently in case role assignments change. The interface to CFGMAN via CfgProxy is not enabled when ReadOnly mode is defined.

A read-only Net Data Replicator configuration file (NetDataReplicatorReadOnlyConfig.xml) is typically located in the %IDMS_ROOT%\config\server subdirectory.

Net Data Replicator replicates OMS data and Switching Order files, and acts as an external interface.

The most relevant configuration parameters used to configure Net Data Replicator on the ADMS read-only server are:

- **UseRoleFromCfgProxy:** Indicates whether to obtain role (Enabled, Standby, or Disabled) from CfgProxy server.

The read-only ADMS server applications do not use the CfgProxy connection, therefore CfgProxy should be disabled. In addition, Net Data Replicator has no interface to CfgProxy and starts in standby mode to get data from the enabled real-time ADMS system.

- **ReadOnlyMode:** Set to True for Net Data Replicator on the ADMS read-only server.
- **DmsOnlyMode:** If OMS replication is needed for the ADMS read-only server, set DmsOnlyMode to False. If DmsOnlyMode and ReadOnlyMode are set to true, Net Data Replicator supports copying Switch Order files and does not wait for the snapshot files before it confirms standby role.
- **RemoteListenerAddresses:** Comma-separated list of TCP/IP addresses, in <Host>:<Port> format, to which the local OMS replicator sends replication queries when its role is standby.

For example, for the DMS and OMS setup:

```
<UseRoleFromCfgProxy>false</UseRoleFromCfgProxy>
<DmsOnlyMode>false</DmsOnlyMode>
<ReadOnlyMode>true</ReadOnlyMode>
<RemoteListenerAddresses>192.168.2.11:8200</RemoteListenerAddresses>
```

For more information about NetDataReplicatorConfig.xml parameters, see [NetDataReplicatorConfig.xml](#).

Finally, a HABITAT startup task for the read-only Net Data Replicator process must be integrated into the PROCMAN model on the ADMS read-only server. Examples of the TASK, FLAG, NODE, PATH, and TFLAG records that must be added to the PROCMAN model appear below.

Note that the "Parameters" display field for the TASK record entry references an environment

variable string (%NETDATAREPLIC_CONFIG%). The environment variable points to the directory path and Net Data Replicator configuration file name. This environment variable must be defined and set for this PROCMAN TASK record modeling convention to work.

The screenshot displays the HABITAT configuration interface for the NETDATAREP process. It is organized into five main sections:

- Task (69):**
 - Appmgr ID: NETDATAREP
 - Application: IDMS
 - Family: (empty)
 - Taskset: 0
 - Test Sim Time: 1
 - Stall Checking: 0
 - Start Timeout: (empty)
 - Stop Timeout: 30
 - Parameters: /CONFIG %NETDATAREPLIC_CONFIG%
 - Down Alarm: ☐ Passive: ☒ Task Must Stop: ☐ Test Mode: ☐ Retry Alarm: ☐ Restart Logic... (button)
 - Hang Alarm: ☐ Startup: ☐ Timeout Alarm: ☐ Log: ☒ Quota... (button)
- Flag (133):**
 - Check Env: ☒ Init State: ☒ Reset: ☐ Set Alarm: ☐ Reset Alarm: ☐ Log: ☒
- Node (156):**
 - Name: ST_NETDATARP
 - Max. Counter: 0
 - Log: ☒ Start: ☒ Exit: ☐ WaitExit: ☐ Junction: ☐
 - Parameters: (empty)
- Path (33):**
 - To Node: 156 NETDATAREP ST_NETDATARP
 - Check Count: ☐ Update Count: ☐ Reset Count: ☐ Auto Wait: ☒ Parallel: ☒ Log: ☐ Open: ☐
- Flag Condition (31):**
 - IF RUN_NETDATAREP 133

Figure 35. Example HABITAT Startup Task, Read-Only Net Data Replicator Process

After you finish configuring the RoSrv/NetServerRO, Net Data Server, and Net Data Replicator processes and integrating them into the PROCMAN model on the ADMS read-only data server, start PROCMAN and confirm that the RoSrv/NetServerRO, Net Data Server, and Net Data Replicator processes start and remain up. Note that it takes several seconds for the Net Data Server process to start.

In the Windows task bar on the ADMS read-only server you should see icons for the RoSrv/NetServerRO, Net Data Server, and Net Data Replicator processes. Maximize these task bar icons and check the log messages on the corresponding windows for any problems. You can also check the RoSrv/NetServerRO and Net Data Server log files for possible issues. The LOG_FILE_NAME parameter in the RoSrv/NetServerRO and Net Data Server configuration files indicates the location of each log file.

10.4. Installing the Browser Web Server

For read-only clients that are located outside of the production security zone, a separate Browser web server must be set up to serve Browser display files. For more information, see [Installing the Browser Web Server](#).

10.5. Installing the ADMS

As mentioned earlier in the [System Overview](#), the ADMS read-only client structure is essentially the same as a real-time, production ADMS client. Its two main components are the Browser client and the ADMS NetView client. The primary difference is that the read-only client is configured specifically to only query for the DMS/OMS data and Browser display data served by the ADMS read-only server.

A good approach is to initially install and configure the ADMS read-only client on the read-only server where Habitat and the ADMS RoSrv/NetServerRO and Net Data Server components are installed. It is also helpful to base the initial read-only client configuration on the current production ADMS client configuration and then tailor it to meet the specific requirements of the read-only users. Once the client configuration on the ADMS read-only server is working, you can use it as an example "template" for the remaining read-only client installations.

The first step in the process is to install the APF and Browser client software. Refer to the *Browser Release Notes* and the *Browser Software Installation and Maintenance Guide* for a list of Browser software prerequisites. Refer to the *Advanced Presentation Framework Release Notes* and the *Advanced Presentation Framework Software Installation and Maintenance Guide* for a list of Advanced Presentation Framework software prerequisites.

The next step is to install the ADMS client software. Refer to the *ADMS Release Notes* for a list of software prerequisites.

After the APF, Browser, and ADMS client software is initially installed, both client components must be configured to work with your ADMS read-only system.

10.5.1. Configure Browser

Specific information about configuring the Browser client software can be found in the *Browser Software Installation and Maintenance Guide*. Information about integrating the Browser client software with the ADMS client software can be found in [Installing the Client Software](#). The following content highlights Browser configuration activities that are specific to adapting the client to your read-only configuration.

On the "Settings" tab of the Browser WebFGViewer main status and configuration window, or in the corresponding WebFGViewer configuration file:

- Make sure the "Top Level Display URL 1" parameter (file parameter "TopLevelUrl1") references the URL for the Browser "displays" subdirectory on the ADMS read-only system web server (for example, <http://dmsRO/web/displays>).
- Make sure the following parameters point to the correct display file web server URL paths: "Automated Script URL", "Room URL", and "Tool Bar URL".
- Make sure the "DMS bin directory" parameter points to the bin directory for the ADMS client installation (for example, <D:\Eterra\Distribution\Client\bin>).
- Make sure the "Tool Bar Name" references the correct custom toolbar configuration file title

that should be used by the ADMS read-only client configuration.

- If in [Configuring Internet Information Services \(IIS\)](#) you configured a secondary datapool in the `DataSourcesConfiguration.xml` file that references the `WfgServer` process running on the ADMS read-only server, make sure the "DataPool" Browser parameter references this datapool name. Otherwise, if you configured the default datapool in the `DataSourcesConfiguration.xml` file to reference the `WfgServer` process running on the ADMS read-only server, make sure the "DataPool" Browser parameter is blank.
- Make sure the "Startup Room" parameter references a valid room file name in the directory path specified by the "Room URL" parameter.

10.5.2. Configure the ADMS Client Software

Specific Information about configuring the ADMS NetView client software can be found in [Installing the Client Software](#). The remaining content in this section highlights the ADMS client configuration activities that are specific to adapting the client to your read-only configuration.

The model files referenced by the `RoSrv/NetServerRO` configuration on the ADMS read-only server should be copied into a corresponding ADMS client directory. It is good practice to keep a separate set of models for each client data context (i.e., real-time, study, simulated). With this in mind, it is recommended that you create a read-only model subdirectory (for example, "RO") below the ADMS NetView client main model subdirectory, and then copy the model files into the new read-only model directory.

You should also create a new read-only dynamics subdirectory (for example, "RO") for the ADMS read-only client dynamics files. A logical location for this new subdirectory would be just below the `%IDMS_ROOT%\dynamics\client` directory.

The read-only ADMS NetView configuration file should be modified to reference the new dynamics and model file paths. The settings in the file for:

```
<DynPath>./DynPath>  
<ModelPath>./ModelPath>
```

are used to reference the model and dynamics file directory paths for the read-only client.

Also of note are the following parameter settings:

```
<Port>5055</Port>  
<ServerIPs>dmsro</ServerIPs>
```

The Port parameter setting indicates the listening port number used by the `RoSrv/NetServerRO` process on the ADMS read-only server. The `serverIPs` parameter setting specifies the host name or IP address for the server where the read-only `RoSrv/NetServerRO` process is running.

```
<ReadOnlyColor>661328</ReadOnlyColor>  
<BackgroundColor>661328</BackgroundColor>
```

These settings specify a dark blue background color for the ADMS geographic displays when viewed by the read-only client configuration. Configuring a specific background color for the read-only client data context helps inform the user that the ADMS geographic display data is from the read-only system.

10.5.3. Optional Read-Only Client Configuration Changes

There are other ADMS read-only system client configuration changes that you may want to consider implementing. Some of these are:

- Modifying the custom Browser toolbar configuration to restrict the user to the read-only data context. Removing toolbar buttons and functions related to ADMS, SCADA, and Habitat write and execute operations.
- Modifying the Browser pull-down menu scripts to restrict the user to the read-only data context. Removing pull-down menus and functions related to ADMS, SCADA, and Habitat write and execute operations.
- Only including a read-only data context section in the NetView configuration file used by the ADMS read-only clients.

After you finish the initial ADMS read-only client installation and configuration activities, start up the client and make sure it connects to the ADMS read-only server processes. Verify that you can view the ADMS, SCADA, and Habitat displays in a read-only mode and confirm that you cannot enter data or execute ADMS, SCADA, and Habitat operations from the displays.

Make any necessary corrections and changes to your client configuration. Once you have the initial client configuration working to your satisfaction, propagate similar configurations to your read-only user workstations.

10.5.4. Optional Switch Order Configuration Changes

If you decide to view switch orders and daily logs with your ADMS read-only system, you need to run the Net Data Replicator to replicate the switch order and daily logs from the real-time ADMS system to the ADMS read-only system.

In addition, you also need to make client configuration changes in the SwitchOrderClientConfig file. The switch order configuration file is typically located in the %IDMS_ROOT%\config\client subdirectory. The most relevant read-only configuration parameters in this file are:

```
<RequirePermitLogin>>false</RequirePermitLogin>  
<NeedDailyLog>>false</NeedDailyLog>
```

The RequirePermitLogin parameter should be changed from true to false, and the NeedDailyLog parameter can be changed from true to false depending on your preference.

11. Specific System Configurations

11.1. Configuring Customer-Only Mode

The ADMS customer-only mode is required to work with customer smart meters, to visualize the meter data, and for using the secondary voltage to refine the Integrated Volt VAR Control (IVVC). This mode requires the user's customer file (for more details about building a customer file, refer to the *Customer List Compiler User Guide*).

As opposed to the Outage Management System, the customer-only mode does not provide functionality to work with incidents.

Figure below shows the high-level message flow in customer-only mode.

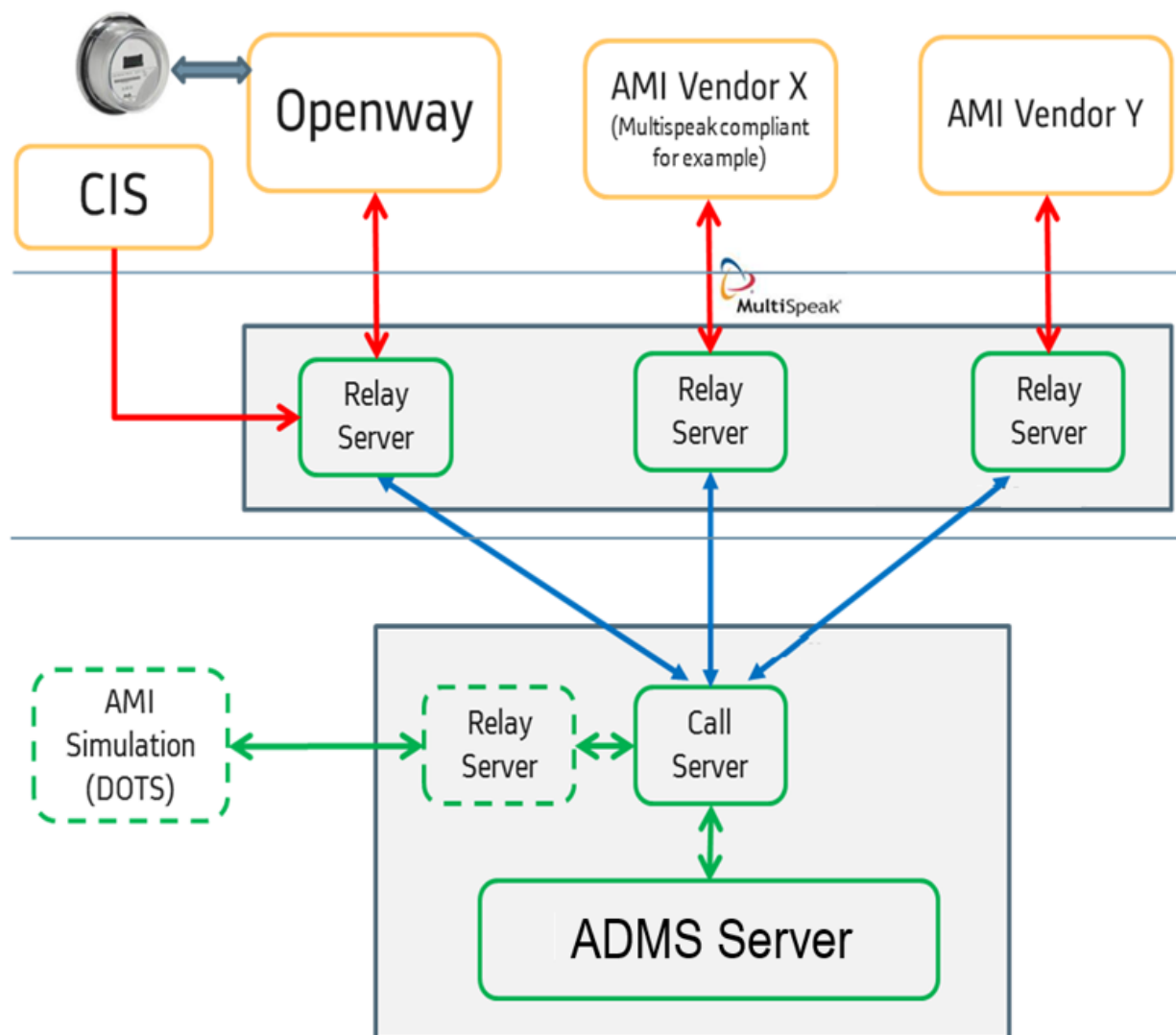


Figure 36. Customer-Only Mode Message Flow

When Call server receives a message from the AMI or CIS system via Relay server, it journals the message in Call Journal and then forwards the message to the ADMS server for further processing. The ADMS server maintains a list of events according to the customer-only mode configuration.

To run ADMS in customer-only mode, ensure that the parameters in the server and client configuration files are set as follows:

- **NetServerMainCustomerModeConfig.txt:** Configures the Customer Mode server. For more information, see [NetServerMainCustomerMode.txt](#).
 - **CUSTOMERONLY_MODE = 1:** If set to 1, the real-time server runs in customer-only mode.
 - **OMS_SEND_COMPRESSED_MESSAGE = 0:** If set to 0, Net Server messages are sent not compressed.
 - **OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION = 0:** If set to 0, Net Server messages for managed connections are sent not compressed.
 - **OMS_METER_VOLT_QUERY_INTERVAL_SECONDS = 60:** Period of time to query for the meter voltage, in seconds.
 - **METER_BASE_SECONDARY_VOLTAGE = 110:** Meter base secondary voltage, in volts. This parameter is used together with the threshold value for setting the meter voltage status.
 - **METER_SECONDARY_VOLTAGE_VIOLATION_THRESHOLD = 0.8:** Meter secondary voltage violation threshold. This parameter is used with the base secondary voltage for setting the meter voltage status.
 - **CALL_STATUSES_TIMEOUT = 120:** The number of seconds for the customer calls to be shown on the display.
 - **CIS_IVR_INTERFACE_ENABLED = 1:** If set to 1, interface to call server is enabled. Verify that this parameter exists and is set to 1.
 - **FD_POWER_MEASUREMENT_QUERY_ENABLED = 1:** Set to 1 to enable the query to get the customer meter power measurement data from the main server.
 - **MAIN_SERVER_HOST:** Specifies the host(s) where the production NetSrv process is running. This parameter is also required to establish connection to the Switch Order server.
- **NetServerDnafCustomerModeConfig.txt:** Configures the DNAF server in customer-only mode.
 - **FD_POWER_MEASUREMENT_QUERY_ENABLED = 1:** Set to 1 to enable the query to get the customer meter power measurement data from the main server.
 - **FD_POWER_MEASUREMENT_PERIOD = 30:** Time period in seconds for DNAF server to query the customer meter power measurement data from the main server.
- **NetServerSimCustomerModeConfig.txt:** Configures the DMS Simulator server in customer-only mode.
 - **CUSTOMERONLY_MODE = 1:** If set to 1, the simulator server runs in customer-only mode.
 - **OMS_SEND_COMPRESSED_MESSAGE = 0:** If set to 0, Net Server messages are sent not compressed.

- **OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION = 0:** If set to 0, Net Server messages for managed connections are sent not compressed.
- **OMS_METER_VOLT_QUERY_INTERVAL_SECONDS = 60:** Period of time to query for the meter voltage, in seconds.
- **METER_BASE_SECONDARY_VOLTAGE = 110:** Meter base secondary voltage, in volts. This parameter is used with the threshold value for setting the meter voltage status.
- **METER_SECONDARY_VOLTAGE_VIOLATION_THRESHOLD = 0.8:** Meter secondary voltage violation threshold. This parameter is used with the base secondary voltage for setting the meter voltage status.
- **CALL_STATUSES_TIMEOUT = 120:** The number of seconds for customer calls to be shown on the display.
- **CIS_IVR_INTERFACE_ENABLED = 1:** If set to 1, the interface to call server is enabled. Verify that this parameter exists and is set to 1.
- **OmsServerSimCustomerModeConfig.txt:** Configures the OMS Simulator server in customer-only mode.
 - **CUSTOMERONLY_MODE = 1:** If set to 1, the simulator server runs in customer-only mode.
 - **OMS_METER_VOLT_QUERY_INTERVAL_SECONDS = 60:** Period of time to query for the meter voltage, in seconds.
 - **METER_BASE_SECONDARY_VOLTAGE = 110:** Meter base secondary voltage, in volts. This parameter is used with the threshold value for setting the meter voltage status.
 - **METER_SECONDARY_VOLTAGE_VIOLATION_THRESHOLD = 0.8:** Meter secondary voltage violation threshold. This parameter is used with the base secondary voltage for setting the meter voltage status.
- **NetConf.xml:** Client configuration file.
 - **CustomerOnlyMode:** Set each <CustomerOnlyMode> node (for RT, ST, and SIM) to True to run the client in customeronly mode. In this mode the client does not send redundant queries to the real time NetServer.
- **CallServerCustomerModeConfig.xml:** Call Server (AMI Server) configuration file in customer-only mode.
 - **CustomerOnlyMode:** If true, the Call Server runs in customeronly mode.
Add <CustomerOnlyMode>true</CustomerOnlyMode> between the </ApplicationParameters> and <RemoteProcessCapabilitiesList> nodes in CallServerConfig.xml.
 - **CompressMessage:** If set to false, Call Server messages are sent not compressed.
- Use the following parameters in CallServerCustomerModeConfig.xml and RelayServerCustomerModeConfig.xml to enable data exchange between CallServer and RelayServer in customer-only mode. Configure Relay server similarly to Call server.
 -

RemoteProcessCapabilitiesList: This section configures communications with remote servers.

- **RemoteProcessCapabilities:** This section configures communications with the particular remote server.
- **ID:** The ID of the remote server (for example, Relay server).
- **Capabilities:** This section configures the communication capabilities.
- **SupportsCaseNoteQuery:** Set to True to enable the case note queries.
- **SupportsMeterEventQuery:** Set to True, to enable the meter event queries.
- **SupportsMeterVoltEventQuery:** Set to True to enable the meter voltage event queries.
- **SupportsMeterPingRequest:** Set to True to enable the meter ping requests.
- **SupportsMeterVoltPingRequest:** Set to True to enable the meter voltage ping requests.
- **SupportsCallQuery:** Deprecated functionality. Set to False to avoid having duplicate call queries (over TCP and HTTP).
- **CompressMessage:** If set to false, Relay Server messages are sent not compressed.

The following is an example Relay Server configuration:

```
<RemoteProcessCapabilitiesList>
  <RemoteProcessCapabilities>
    <Id>Call Server</Id>
    <Capabilities>
      <SupportsCaseNoteQuery>>false</SupportsCaseNoteQuery>
      <SupportsMeterEventQuery>true</SupportsMeterEventQuery>
      <SupportsMeterPingRequest>true</SupportsMeterPingRequest>
      <SupportsCallQuery>>false</SupportsCallQuery>
      <SupportsAutomatedCallbackRequest>true</SupportsAutomatedCallbackRequest>
      <SupportsManualCallbackRequest>true</SupportsManualCallbackRequest>
    </Capabilities>
  </RemoteProcessCapabilities>
</RemoteProcessCapabilitiesList>
<CompressMessage>>false</CompressMessage>
```

- Update the APF shortcut ADMS-APF Client-Localhost.Ink to use the `netconfCustomerMode.xml` file.
- Update the Net Server Simulator shortcut Net Server Simulator.Ink to use the `NetServerSimCustomerModeConfig.txt` file.
- Update the DNAF Server shortcut DNAF Server.Ink to use the `NetServerDnafCustomerModeConfig.txt` file.
- Update the Call Server shortcut Call Server.Ink to use the `CallServerCustomerModeConfig.xml` file.
- Update the Relay Server shortcut Relay Server.Ink to use the `RelayServerCustomerModeConfig.xml` file.

11.2. Configuring Follower Mode

The Leader-Follower configuration allows more than one ADMS system to stay enabled concurrently. There are two types of follower modes: Independent and Dependent. The main difference between these two modes is that the Dependent Follower mode forwards all manual actions to the Leader and dynamics are always in sync with the Leader, whereas Independent Follower mode's dynamics are not in sync with the Leader.

A DMS server can run in follower mode to the enabled DMS server. In Independent Follower mode, follower servers receive changes from the enabled server and independently implement DMS commands from the follower server clients. In Dependent Follower mode, follower servers receive changes from the enabled server and send requests to the enabled server to implement DMS commands from the follower server clients. An OMS server can receive permissions from a DMS server in follower mode. Multiple DMS follower servers can run simultaneously. Dependent Follower servers do not support redundancy. Independent Follower servers support redundancy.

11.2.1. Setting Up Dependent Follower Mode (OMS)

The DMS server leader-follower configuration can be used to deploy DMS servers and OMS servers in different security zones. This allows you to prevent the OMS servers, which are located in a less secure "Corporate" zone, from connecting directly to the DMS and SCADA servers located in the secure "Production Control" zone, as shown in figure below.

Note

The ADMS UI clients (full functionality in the Secure zone and OMS-only in the Corporate zone) need to connect to the NetPermSrv/Permit host for authentication. This requires an OMS-only client from the less secure Corporate zone to cross through the Demilitarized zone to the secure Production Control zone.

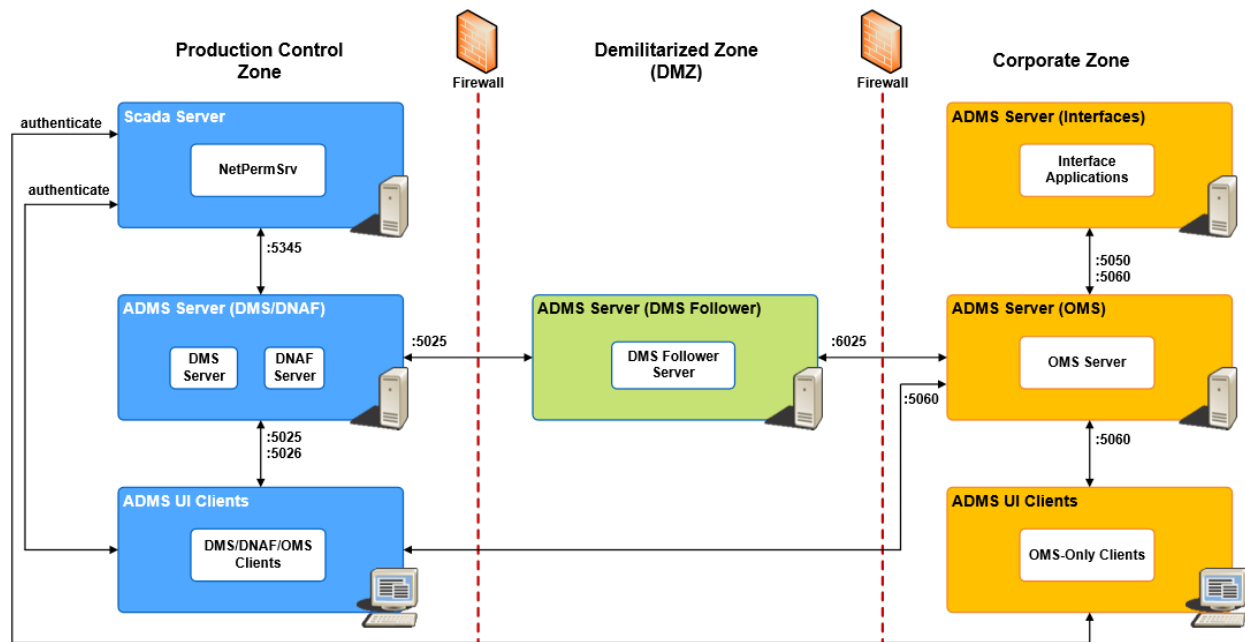


Figure 37. Dependent Follower Mode Message Flow

The DMS Follower Server consist of a single NetSrv task that runs with the `NetServerFollowerConfig.txt` file. Follower servers do not support the Enabled/Standby functionality, therefore they are not subject to multi-host server configuration limits.

To set up the DMS server leader-follower configuration, ensure that the parameters in the server configuration files are set as follows. (This example demonstrates the DMS server dependent follower configuration.)

- **NetServerMainConfig.txt:** Configures the DMS server (leader). For more information, see [NetServerMainConfig.txt](#).
 - **PERMIT_CHECK_ENABLED = 1:** If set to 1, the permission checking for server queries is enabled.
- **NetServerFollowerConfig.txt:** Configures the DMS server (follower).
 - **PERMIT_CHECK_ENABLED = 1:** If set to 1, the permission checking for server queries is enabled.
 - **PERMIT_DATA_OPTION = 1:** The allowed values are: 0 = No Permit data; 1 = Get Permit data from the (leader) Main server; 2 = Connect to directly Permit server to get Permit data (PERMIT_SERVER_IP and PERMIT_SERVER_PORT below must also be provided). The default value is 0. Dependent follower must use the option value of 1, getting permit data from leader Main Server and not connect directly to permit server itself.
 - **MAIN_SERVER_PORT = 5025:** The TCP/IP port that the DMS server (leader) listens on. This is the port to which the DMS server (follower) connects.
 - **SERVER_PORT = 6025:** The TCP/IP port that the DMS server (follower) listens on. This is the port to which the OMS server connects.

- **FOLLOWER_MODE = 2:** Set to 2 for DMS servers running in the Dependent Follower mode to the main DMS server. Dependent follower servers must not connect to SCADA directly.
- **FD_MEASUREMENT_QUERY_ENABLED = 1:** Since dependent followers are not connected directly to SCADA, they must subscribe to Leader Main server for measurement dynamics by enabling this option.
- These two options must be commented out, or set to value of zero (default), so the dependent followers not to ignore these types of dynamics from Leader Main Server.

'DMS_IGNORE_DNAF_CREATED_DYN_FROM_LEADER = 1

'DMS_IGNORE_SCADA_FI_DYN_FROM_LEADER = 1

- **RETURN_TO_NORMAL_BY_RESTART = 0:** Optionally, add the RETURN_TO_NORMAL_BY_RESTART = 0 entry to NetServerFollowerConfig.txt to prevent the Follower server to restart at midnight after the dynamic compression. The Follower server restart may cause the OMS server to reconnect to another instance of the Follower server in a redundant (multiple host) system configuration.
- **NetServerOmsConfig.txt:** Configures the OMS server.
 - **PERMIT_CHECK_ENABLED = 1:** If set to 1, the permission checking for server queries is enabled.
 - **PERMIT_DATA_OPTION = 2:** The allowed values are: 0 = No Permit data; 1 = Get Permit data from the Main server; 2 = Connect Permit server to get Permit data (PERMIT_SERVER_IP and PERMIT_SERVER_PORT below must also be provided). The default value is 0.
 - **PERMIT_SERVER_IP = [null]:** Clear this value, because the OMS server receives permissions from the DMS server (NetSrv) running in the Follower mode, not from NetPermSrv.
 - **PERMIT_SERVER_PORT = 0:** Set to 0, because the OMS server receives permissions from the DMS server (NetSrv) running in the Follower mode, not from NetPermSrv.
 - **MAIN_SERVER_PORT = 6025:** The TCP/IP port that the DMS server (follower) listens on. This is the port to which the OMS server connects.
 - **SERVER_PORT=5060:** The TCP/IP port that the OMS server listens on. This is the port to which the clients connect. (Unmanaged connection port.)
- **OMS_SERVER_PORT=5050:** The TCP/IP port that the OMS server listens on. (Managed connection port.) This is the port to which the clients connect, such as OMS Interface server and Net Data server. This port is also required for the Site Overview display. (The clients and servers must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
- **NetViewerAdapterConfig.xml:** Configures the NetViewer Adapter. (Used for the Group Control application and Outage Assist solution.)

```
<OmsConnectionConfig>
  <OmsServerIPs>192.150.1.1,192.150.1.3</OmsServerIPs>
  <OmsServerPort>5050</OmsServerPort>
</OmsConnectionConfig>
```

```
<HostName>192.150.1.1,192.150.1.3</HostName>  
<HostPort>5060</HostPort>
```

The Netviewer Adapter configuration is referenced in this section because this file has DMS SERVER and OMS server configuration sections. In a non-follower configuration, the NetViewer Adapter would likely connect directly to the DMS directly over a different port (5025). However, in a follower configuration, the NetViewer Adapter can avoid referencing the DMS server directly and can connect to OMS server (5060).

For the other interface servers, such as OMS Interface Server and OMS Adapter, common configurations can be used.

11.2.2. Setting Up Independent Follower Mode (DNAF)

The independent follower configuration can be used by utilities to try newer features in the Follower subsystem without making any changes to the production system software.

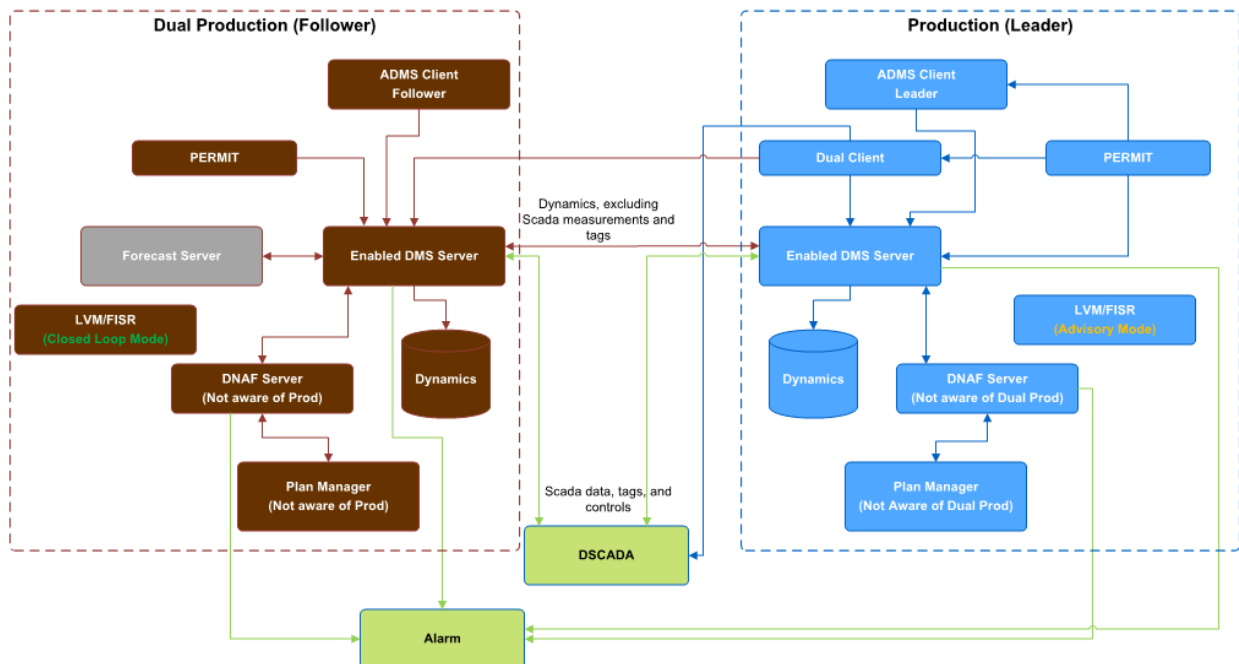


Figure 38. Independent Follower Mode Message Flow

Production (Leader) Subsystem

In figure above, the production subsystem (at the right) acts as a Leader. The enabled DMS server running as a Leader is configured to know about the Follower connection. The DMS server leader sends dynamics (excluding SCADA measurements, dynamics midnight compression data, and some manual dynamics) to the Follower DMS server. All SCADA measurements are sent to the Independent Follower DMS server via the SCADA Inter-Site Data (ISD) connection.

The ADMS client leader is a regular DMS client that is connected to the DMS, DNAF, and OMS

servers. This client can show and operate devices modeled in the DMS server leader. No configuration change is needed for the client to run in this mode.

The follower's client is also a regular DMS client that connects to the follower DMS, DNAF, and OMS servers. It is recommended to configure the network view and grid tabular display background colors to distinguish whether the client is connected to the Production (Leader) subsystem or Dual Production (Follower) subsystem.

It is recommended to connect these Leader and Follower subsystems to two different PERMIT systems, so that their AORs can be controlled differently. Devices that are common to both subsystems should be given the read-only permission in the Follower subsystem. If you choose to have just one PERMIT system for both Leader and Follower subsystems, operators must be trained to not operate devices from the Follower network view. Currently, there are no warnings or errors to prevent such an action from the operator.

The independent Follower system is a fully functional DMS system that can work with the DNAF, OMS, and Study servers.

No configuration change is needed for the DNAF server that runs on the Leader subsystem. There are a few extra configuration changes that are required for the DMS and DNAF Followers to work independently from the DMS and DNAF Leaders which is detailed below.

Dual Production (Follower) Subsystem

The Dual Production (Dual Prod) subsystem is typically where a utility would use newer features of the product. The Dual Prod subsystem has model files associated with the newer version of ADMS. Therefore, there must be two different modeling systems feeding models to the Production and Dual Prod subsystems. The Dual Prod subsystem supports hot standby so it can failover independently of the Production subsystem.

A DMS Follower can be run in two modes: Dependent and Independent.

A dependent DMS Follower behaves like a read-only DMS server. It synchronizes dynamics with the Leader DMS server. The main difference is that the DMS follower will forward device change actions from its clients to the Leader DMS server. So, the change will be made if the DMS follower is given permissions in the Leader DMS server to make the change. A Follower DNAF with an dependent DMS follower is not supported.

An independent DMS follower works just like a normal enabled DMS server in addition to accepting normal dynamics and some manual dynamics. It does not synchronize dynamics with the Leader, and it only merges dynamics received from the Leader. So, this means that the Follower server will see more dynamics than the Leader. The dynamic sequences on the Leader and Follower are independent. An Independent Follower will never send dynamics information back to the Leader server.

With an independent DMS Follower, you can also configure a follower DNAF (Follower DNAF) server to run with it that is independent of the DNAF (Leader DNAF) server that is working with the DMS leader.

Two Dual Production Systems

You can have two Dual Production subsystems: one with a DNAF server running FISR, and another with a DNAF server running LVM. Only these two types of the follower subsystems are supported to keep the minimum complexity of the system. However, it is recommended to use just one Dual Prod subsystem. When two Dual Prod subsystems are used, they must have the same ADMS version.

A Dual Prod system can have its own set of Enabled/Standby systems using the CFGMAN interface. This interface is independent from the production system. This interface gets the Enabled/Standby roles and can failover independently of the production subsystem.

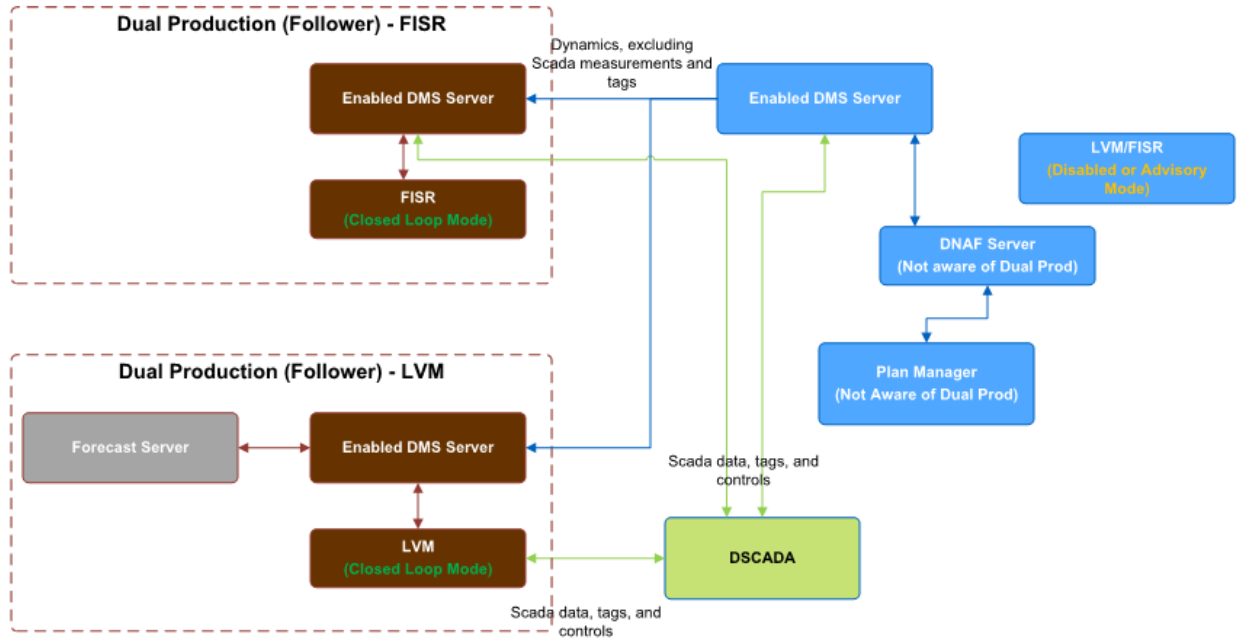


Figure 39. Independent Follower Mode - Two Dual Production Systems

Manuals Dynamics

Additional options are available in the independent Follower DMS server that controls whether the Follower servers should accept manual dynamics from the Leader server. There are two kinds of manual dynamic entries:

- Entries that are created from the DNAF System Configuration (DSC) and LVM Profile UI – these entries have a field name that is prefixed with one of the following: DNAF, LVM, FISR, AFR, SLS, or FSD
- Remaining entries that are created from DataEntry popup UI from geographic viewer or the Manual Entry Parameter tabular display.

By default, the independent Follower server accepts entries from the leader that are created by the DataEntry popup UI or the Manual Entry Parameter tabular display and rejects entries from the leader that are created from the DSC and LVM profile display. The DMS follower has three options to control acceptance of these manual dynamics. The first of these three options is to help decide the category of each dynamic entry. The remaining three options are used to control the acceptance in the DMS Follower. The values listed here are the default values.

```
DMS_MANUAL_DYN_DSC_LVMPROFILE_FIELD_NAME_PREFIXES    = DNAF,LVM,FISR,AFR,SLS,FSD
DMS_MANUAL_DYN_KEY_CONTAINS_STR = /RFBSIF=,/FLTRIGGER=
DMS_IGNORE_MANUAL_DYN_DSC_LVMPROFILE_FROM_LEADER     = 1
DMS_IGNORE_MANUAL_DYN_FROM_LEADER                    = 0
```

SCADA System Interaction

Leader and Follower subsystems connect to SCADA and receive and send measurements. SCADA supports sending the same measurements to many different systems. The DNAF server Leader sends DPF-calculated aggregation values to SCADA. The master of any Status point for FISR and LVM controls is SCADA. The Leader and Follower subsystems can issue controls for those Status points. However, it is the utility's responsibility to not enable FISR and LVM for the same feeders to avoid conflicts. A single instance of the SCADA database can be used without any change in the configuration.

Tagging Interaction

The Tagging application, running in the SCADA subsystem, only connects to the Production subsystem to place tags. All tags placed in SCADA are sent to the Dual Prod subsystem by the Inter-Site Tagging protocol.

Read Only Clients

Read-only clients cannot connect to both Production and Dual Production subsystems. You must set up read-only clients separately for each subsystem.

OMS Support

OMS is not supported in the Independent Follower mode. Use the Dependent Follower mode for OMS support.

Assumptions

- The operator can use the Dual client to view DNAF data from the Dual Prod or Production subsystems. The selection is similar to selecting between the RT and SIM modes.
- If the enabled Leader server becomes unavailable, the Follower connects to the next available Leader predefined from the four-node follower configuration.
- The operator can adjust FISR and LVM execution and alarm data from the DNAF System Configuration display. For example, the operator can enable or disable a feeder for closed loop FISR for the Follower DNAF using the Follower client's DNAF System Configuration display.

Configuration

For the DNAF dual production environment setup, most of the configuration settings are made on the Follower subsystem. The following example demonstrates the DMS server independent follower configuration. Note that we recommend that Follower servers be configured to run on a different machine than the Leader Servers even though it could be setup to run on the same machine. If you decide to run both servers on the same machine, then it is recommended to change the result

output and log path to avoid interference. The same applies to the client log file names.

- **NetServerDnafConfig.txt:** Configures the DNAF server (leader). For more information, see [NetServerDnafConfig.txt](#).

```
PLAN_MANAGER_LOG_FILE = PlanManager
```

- **NetServerDnafFollowerConfig.txt:** Configures the DNAF server (follower). For more information, see [NetServerDnafConfig.txt](#).

```
MAIN_SERVER_HOST      = localhost ' This must point to the DMS Follower
MAIN_SERVER_PORT      = 6025      ' This must be the part of the DMS Follower

SERVER_PORT           = 6026

REDUNDANT_HOST        = localhost
REDUNDANT_PORT        = 6025

USE_ROLE_FROM_CFGPROXY = 1 ' Optional
CFGPROXY_HOST         = localhost ' Only applicable if USE_ROLE_FROM_CFGPROXY is enabled
CFGPROXY_PORT         = 6020
```

The following changes are required to distinguish DNAF server (follower) outputs from the DNAF server (leader).

```
DNAF_SERVER_RESULTS_PATH = %IDMS_ROOT%\dynamics\Server\DNAF_Follower\
DNAF_OUTPUT_PATH         = %IDMS_ROOT%\log\DNAF_Follower\
DYNAMIC_PATH             = %IDMS_ROOT%\dynamics\Server\DNAF_Follower\
LOG_FILE_NAME            = %IDMS_ROOT%\log\netsrv_dnaf_follower.log
DYNAMIC_SCHEDULE_PATH    = %IDMS_ROOT%\dynamics\Server\DNAF_Follower\DynSchedules
APP_LVMMONITOR_LOG_FILE  = LVMSysSystemMonitor_Follower
PLAN_MANAGER_LOG_FILE    = PlanManager_Follower
```

Comment out the ISD_LOG_FILE_NAME parameter.

```
'ISD_LOG_FILE_NAME = %IDMS_ROOT%\log\isd_dnaf.log
```

- **NetServerFollowerConfig.txt:** Configures the DMS server (follower). For more information, see [NetServerMainConfig.txt](#).

DMS follower must be an independent follower for the DNAF follower to be able to add or change device statuses to run FISR and LVM.

```
FOLLOWER_MODE      = 1

SERVER_PORT        = 6025
MAIN_SERVER_HOST    = localhost
MAIN_SERVER_PORT    = 5025

LEADER_DYNAMIC_NAME = leader_dynamics.sav
```

```

USE_ROLE_FROM_CFGPROXY = 1 ' Optional
CFGPROXY_HOST          = 127.0.0.1    ' Only applicable if USE_ROLE_FROM_CFGPROXY is enabled
CFGPROXY_PORT          = 5019

SERVICE_PORT = 7032

```

You must also properly setup a different local ISD site for the DMS follower server to be different from that of the DMS server. The connecting ISD site (for example, FEP_TEST) must be properly setup on the SCADA service.

```

ISD_LOCAL_SITE          = FEP_TEST
ISD_LOCAL_SCANNER_PATH  = FEP_TEST.SCADA_FEP_TEST_ISD
ISD_LOCAL_CONTROL_PATH  = FEP_TEST.SCADA_FEP_TEST_CTRL
ISD_LOCAL_TAGNOTES_PATH = FEP_TEST.TAGGING_FEP_TEST_TGDATA

```

The following changes are needed to skip some dynamics that are received from the DMS server (leader). These are all default values.

```

DMS_IGNORE_DNAF_CREATED_FAULT_DYN_FROM_LEADER = 1
DMS_IGNORE_SCADA_FI_DYN_FROM_LEADER           = 1
DMS_IGNORE_MANUAL_DYN_FROM_LEADER             = 1

```

The following changes are required to prevent changes to the DMS and OMS servers on the Production subsystem.

```

READ_ONLY_DMS_SERVER_MODE = 1
READ_ONLY_OMS_SERVER_MODE = 1

```

The following changes are required to use different output log file and dynamics folder.

```

DYNAMIC_PATH = %IDMS_ROOT%\dynamics\Server\Follower
LOG_FILE_NAME = %IDMS_ROOT%\log\netsrv_follower.log
ISD_LOG_FILE_NAME = %IDMS_ROOT%\log\isd_follower.log

```

- **netconf_follower_localhost.xml:** Configures the follower client. For more information, refer to the *Net Dynamics Configuration Editor User Guide*.

```
<DnafPort>6026</DnafPort>
```

The following changes are required to use different log files and working folders from the regular client:

```

<LogFile>%IDMS_ROOT%\log\NetClientDebug_follower.txt</LogFile>
<DynPath>%IDMS_ROOT%\dynamics\Client\RT_Follower\</DynPath>

```


12. Upgrading to ADMS 16.X.X

This chapter describes the process for upgrading real-time production systems and demonstration systems from an earlier version to ADMS 16.X.X.

12.1. Project Development Server (PDS) Upgrade

The procedures in this section describe the PDS upgrade to ADMS 16.X.X

1. Review the *ADMS Release Notes* for a list of supported operating systems.

If a newer version of ADMS requires another operation system, refer to [Switching to Another Operating System](#) for upgrade details.

2. Review the *ADMS Release Notes* for a list of compatible GE software.

If a newer version of ADMS requires a different version of Habitat, Scada, or Browser, upgrade the required products. For details, refer to each product's release notes or contact your GE project team.

3. When upgrading to a system with **e-terrascada** versions 3.0 and later, review [pecific Configurations for e-terra_scada_Versions 3.0 and Later](#).

4. Shut down the ADMS servers, tools, and client.

5. Back up the system files and dynamics (for Fantasy Island users).

On both the client and server machines, back up all files stored in Eterra\Distribution\FantasyIsland as necessary. Ensure that this backup is recent to include the latest "dynamics" (that is, current non-nominal device states).

Note

This step is only required when the %IDMS_ROOT% environment variable points to the Eterra\Distribution\FantasyIsland directory. In this case, the Fantasy Island data is typically overwritten by the installation and uninstallation processes. Skip this step if %IDMS_ROOT% points to another directory.

6. Upgrade or install any prerequisite software. Refer to the *ADMS Release Notes* for a list of software prerequisites.

For any software required by ADMS that is not included in its installation kits, upgrade or install as indicated in that product's instructions.

7. Optionally, uninstall the previous version of ADMS using the Windows Add/Remove Programs utility.

Note

You may skip this step because the ADMS installation kits apply the Major Upgrade feature,

which means that the old version of a component (for example, the client) is automatically uninstalled during the installation of a newer version.

8. Install the current release version of ADMS.

For more information about installing the ADMS software, refer to [Software Installation Overview](#).

Note

When installing the File Updater server and client applications, select the Migrate Existing Default File Updater Service option to migrate the default server and client services. All non-default services are migrated automatically.

Note

Do not overwrite the IDMS_ROOT environment variable during installation.

9. Update the previous configuration as described in [Updating Configuration Files](#).

10. Make management areas hierarchy updates as described in [Modeling the Management Areas Hierarchy](#).

Important

This procedure is required only when upgrading **e-terrardistribution 3.7** and earlier.

11. If you are using the Fantasy Island expansion pack, copy your saved Fantasy Island directories (Dynamics, Logs, ToolsWorkspace, and Models). Perform an update and merge of the configuration files as described in [Updating Configuration Files](#), as necessary.

12. Perform other configuration changes.

Perform updates to accommodate any Alarm Message format changes, and also update the proxysrv.almmmsg file, the SCADA alarm, the Habitat ALARM savecase, and the habalarmmsg.dll message library.

Refer to the *ADMS Alarmable Events* table and update your Habitat Alarm savecase as needed to accommodate any customizations you have made. The Fantasy Island Expansion Pack includes an example savecase.

PERMIT definitions might need to be updated to allow creating planned outages, working with historical incidents, moving customers, modifying area settings, case notes, and DNAF calculations, and using the "Confirm" OMS functionality.

SCADA and TAGGING configurations need to be updated when migrating to another version of Scada.

13. Upgrade the Archive database.

Existing Archive databases must be upgraded to preserve the data. Upgrading Archive

databases is described in the *ADMS Relational Database Software Installation and Maintenance Guide*.

14. Rebuild the model files.

Unless otherwise noted, all binary models must be rebuilt, including DNOM and Customer.

15. Re-create the localized resource files.

16. Start the ADMS servers, tools, and client.

17. Confirm the successful update.

Verify that the installation process completed successfully, the servers and the client started successfully, and the dynamics and configuration files have been picked up.

For information about inter-version file compatibility, see [Inter-Version Compatibility Matrix](#).

i Note

- 3.6 and earlier clients are not compatible with 16.X.X servers because of server separation, which occurred in version 3.7.
- 3.7 and later clients can work with 3.14 or later servers (except Switching Order).
- Using 3.14 or later clients to connect to the older server (version 3.13 and earlier) is not supported. The recommended procedure is to always upgrade the server before upgrading the clients.

12.2. Production System Planned Failover Upgrade

This section describes the failover upgrade process to ADMS16.X.X.

The procedures in this section use the ADMS server replication and redundancy mechanism to ensure no downtime during the production environment upgrade. For more information, refer to [ADMS Server Redundancy Management/Multi-Host](#).

12.2.1. Prerequisites

Before upgrading an earlier version of ADMS to version 16.X.X, do the following:

1. Ensure that all IDs are unique across all model inputs using the DNOM Unique ID Checker tool, as documented in the *ADMS Modeling Overview and Converter User Guide*. Failure to do this will cause issues during and after the upgrade, including operational issues.
2. Verify all new software, functionality, and configurations on the PDS demo system, including:
 - Expansion pack upgrades.
 - Configuration files that might be company-specific (for example, DNAF_PARAMS_DNAF.xml, Tftype.dat, so on).

- Rebuilding geographic background shapes.
- Making management areas hierarchy updates. (This procedure is required only when upgrading **e-terra**distribution 3.7 and earlier.)
- Rebuilding the customer and crew models.

Note

Crew models are essentially journal files. Crew models that are written with legacy journal processing functionality (version 1) cannot be read with enhanced journal processing functionality (version 2, implemented in ADMS 3.9 and enabled by default in ADMS versions 3.11 and later). To convert a crew model version 1 into a crew model version 2, it is required to resave the crew model version 1 using the Crew Modeling tool.

- Rebuilding models with the new version of Converter.
 - Re-creating localized resource files from new resources using the Resource Editor.
3. Test the upgrade procedure on a Quality Assurance (QA) environment to verify, for example, that old dynamics are properly imported.

12.2.2. Upgrade Procedure

Warning

The upgrade process must be tested on a test system before upgrading the production system.

Note that interface integrations (for example, Call server, Crew server) remain live through this upgrade process and are connected to the primary machine.

1. Review the *ADMS Release Notes* for a list of supported operating systems.

If a newer version of ADMS requires another operation system, refer to [Switching to Another Operating System](#) for upgrade details.

2. Review the *ADMS Release Notes* for a list of compatible GE software.

If a newer version of ADMS requires a different version of Habitat, Scada, or Browser, upgrade the required products. For details, refer to each product's release notes or contact your GE project team.

3. When upgrading to a system with *e-terra*_scada_versions 3.0 and later, review [Specific Configurations for e-terra-scada Versions 3.0 and Later](#).
4. Prepare to bring down the standby servers by saving dynamics from the primary (enabled) and standby servers.

Back up the dynamics stored at %IDMS_ROOT%\dynamics\server (dynamics.sav, manuals.sav, preservedstates.sti, dynamic schedules, CallServer and RelayServer journals, DNAF dynamics, station measurements, DMS server, OMS server, and SimServer journals and snapshots,

Switching Order files, and annotation dynamics).

For example, create a zip package with the dynamics by using the `idmsSystemSnapshot.bat` batch file located in `%IDMS_ROOT%\..\FantasyIslandmisc\demo\Scripts`.

5. Stop the current standby servers (for example, by using the Task Viewer (tview) or the Process Starter tool).

Note

As appropriate, also bring down the DNAF standby servers at this time.

6. Pause any model updates (both DNOM and customer), and pause the File Updater utility.
7. Upgrade the current standby server based on the PDS configuration. For more details, see [Project Development Server \(PDS\) Upgrade](#).
8. Upgrade the Model Management environment.

Note

There is a chance that a fix or a new feature in the converter/model might not be compatible with an older server or client; this should be tested in advance on the PDS demo system.

If any issues occur with inter-version model compatibility, keep the new Model Management environment offline/paused.

Note

Models and configuration files are not backward compatible. Models compiled with a newer version of the Converter are not usable in an older version of the server or client.

9. Update integrations with SCADA, Browser, and Habitat.

- Update NetPermSrv and CfgProxy.

The new version of the Configuration Manager Proxy server must be installed for Habitat for the DMS and OMS configuration only. (CfgProxy, supplied with Habitat, only supports the DMS configuration.)

- Update any PERMIT, ALARM, or SCADA models.
- Update TAGGING configurations.
- Update the NetPermSrv and CfgProxy Appdef files as needed. (In **e-terrahabitat** 5.9 and later, the default location of the NetPermSrv and CfgProxy executables changed from `%HABUSER_BINDIR%` to `%HABITAT_BINDIR%`.)

10. When upgrading **e-terrardistribution** 3.6 and earlier, update your configurations according to the modularization approach that was introduced in **e-terrardistribution** 3.7.

In **e-terrardistribution** 3.7, DMS server, OMS server, DNAF server, Read-Only server, Simulator

server, and Customer-Only Mode server processes were split from the NetServer/NetSrv application. The new server executables are: NetServerRO, NetServerSim, SimSrv, NetServerCM, and Studysrv. **e-terr**adistribution 3.7 clients maintain separate connections to these servers to obtain the required data. However, **e-terr**adistribution 3.6 and earlier clients are no longer fully functional with ADMS 16.X.X servers.

- Prepare the PROCMAN, CFGCTRL, and CfgProxy models.
- Update idms.appdef to allow PROCMAN to start the renamed DMS and OMS servers.
- Update your server shortcuts.

To support the new NetServer (OMS server) process, do the following:

- Install CfgProxy using the installer.
 - Rebuild the CfgProxy clone. For example, use CfgProxy source files from the QuickInstall_hab511 or QuickInstall_hab510 folder in %IDMS_ROOT%\..\FantasyIsland\misc\dms.
 - Add the CPXOMS task to the PROCMAN model.
 - Add OMSSERVER to the CFGCTRL model.
11. Delete the dynamics files from the standby servers using %IDMS_ROOT%\..\FantasyIsland\misc\demo\scripts\clear_dynamics.bat. All files under %IDMS_ROOT%\dynamics and its subdirectories should now be deleted.
 12. Prepare the new standby systems based on the PDS configuration. This includes the server and tools kits.
 13. Update the configuration files as described in [Updating Configuration Files](#).
 14. Resume model updates and enable the File Updater. Make sure that the following models are synchronized between the enabled and the standby servers and correctly applied on the standby server.
 - Station model files
 - Customer model
 - Crew model
 - Geographic background models
 15. Ensure that the dynamics on the primary and standby server are synchronized and the system is ready for the failover. For this purpose, use the DMS/ADMS Site Overview displays that can be accessed from the RTDMS or OMS menu.
 16. When upgrading from versions 3.6 and earlier, copy the .sav files from the DMS Server (NetSrv) dynamics folder to the OMS Server (NetServer) dynamics folder to preserve data about split incidents and Block Prediction conditions. Otherwise, NetServer will create its own dynamics from the current system state and re-create incidents with new names for open devices and customer calls.
 17. Bring up the new standby systems.

At this point, the new standby servers and the standby DNAF server will get the current dynamics from the primary server. The standby server will obtain the "Standby" role. The Net Data Replicator will sync the switch order and other files from the primary to the standby server.

 Caution

If the new standby servers are not automatically getting the current dynamics from the primary server, manually copy the dynamics from the enabled server, as described in section [Manually Transferring Dynamic Files](#).

 Warning

Before failing over to the standby server, make sure that the standby server's dynamic data is up-to-date with the primary server. For details about how to ensure that the server is ready to failover, see [ADMS Server Redundancy Management/Multi-Host](#). If you failover to the standby server before the dynamic data is synchronized, errors may occur and you may lose the dynamics information entered in the primary system.

18. Failover to the new standby server by using CFGMan displays to disable the primary server.
19. Simultaneously with the failover or before the failover, install the new ADMS client on all consoles.

For each client:

- Stop the client.
- Remove dynamics.
- Upgrade to the software tested on the PDS. This includes updating the client configuration files.

If disk space is a concern, files with the extension ".cri" can be deleted. Files with the extension ".ncri" should be retained.

- Install newer versions of APF and Browser
- Start the client.

 Note

If 3.6 and earlier clients are connected to the 16.X.X server, which just became the primary, these clients will not support complete functionality (DMS + OMS + DNAF) because of the server separation. By default, the 3.6 and earlier clients connect to NetSrv and only support DMS functions. Alternatively, you may configure 3.6 and earlier clients to connect to NetServer or DnafSrv to only support OMS or DNAF functions. If 3.13 and earlier clients are connected to the 16.X.X server, planned outage functionality of the OMS is not fully supported.

20. Wait until the newly enabled machine is up, running, and stable.

- Verify that the Archiving application and database are working properly.
- Verify that interfaces are connected.
- Verify that clients are working properly in this configuration.

In addition, you can use the ADMS Site Overview display to check the state of secondary servers and connected clients (if enabled).

21. Bring down the current standby (the old enabled server), and upgrade it following the procedures described earlier in this section.
22. Shut down the read-only server.

Note

Read-only clients are out of service at this point.

23. Upgrade and enable the read-only server.
24. Update any remaining clients, including the File Updater client software.
25. Upgrade the Archive Database as described in the *ADMS Relational Database Software Installation and Maintenance Guide*.
26. Update the ADMS Reports.

Note

Several issues were observed during the system upgrade. For more information, see [Known Issues](#).

12.3. Updating Configuration Files

The old server, client, and tool configuration files must be merged with the new configuration files to make sure that any new or changed items get picked up. In addition, configuration files must be updated with new paths for components that were moved to other folders.

For more information about configuration changes implemented in this release, see [Configuration Changes in ADMS](#). For more information about configuration changes implemented in previous releases, see the associated Release Notes documents.

Tip

The Configuration Accelerator tool is designed to facilitate identifying changes in the ViewerConfig.xml file. For more details, see [Configuration Accelerator Tool](#).

The configuration must be done before the upgrade.

If there is a configuration utility for a file (i.e., the Viewer Configuration Editor), it is recommended that you load the old configuration into the utility, and then save it to pick up any "schema"

changes. For configuration files with no configuration utility, you need to use a comparison tool to detect and make changes.

For a planned failover upgrade, do the following:

1. Copy all configuration files from the older server to the newer server. The default location is %IDMS_ROOT%\config.
2. Merge the old configuration files with the new configuration files as described above.
3. In the configuration files, replace the servers' host names and IPs to those of the new servers, as necessary.
4. In the configuration files, update paths to ADMS components, as necessary. (For example, components may be moved to other folders in the new release.)
5. Apply the new configuration files.

! Important

In **e-terr**adistribution 3.7, OMS server settings moved from NetServerMainConfig.txt to a new configuration file, NetServerOmsConfig.txt. Therefore, when upgrading from **e-terr**adistribution 3.6 and earlier, you must modify these files and transfer your OMS settings to NetServerOmsConfig.txt.

12.3.1. Upgrading Shared Configuration Files

You must update the CodedTranslationConfig.xml and DefaultServerLocalization.txt files. Otherwise, some new features might not work, such as text and images on OMS displays.

For example, in ADMS 3.12, the ManagementAreasHierarchy coded translation item replaced the OrganizationalHierarchy coded translation item. The following example shows the default configuration. For more information about using the Coded Translation Editor, refer to the *Coded Translation Editor User Guide*.

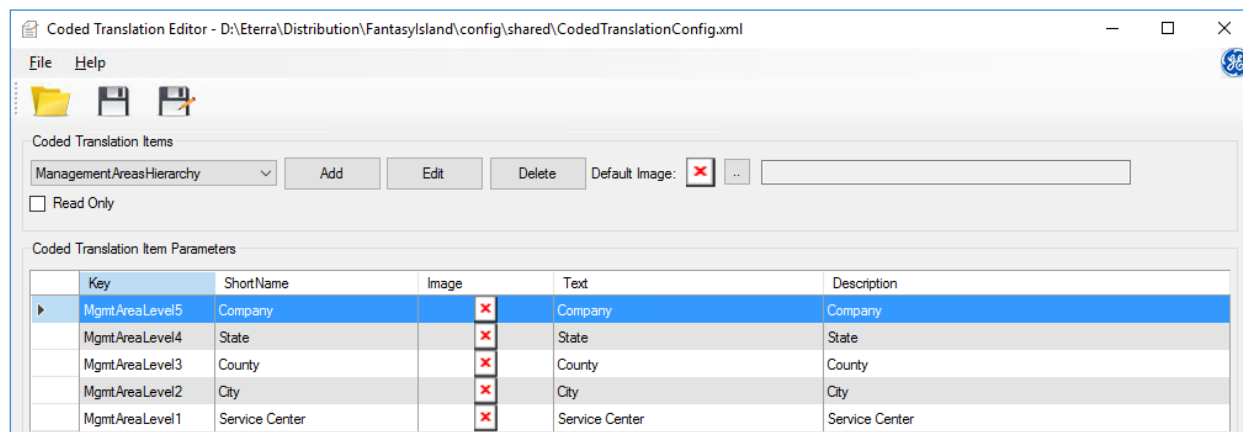


Figure 40. ManagementAreasHierarchy Coded Translation Item

12.4. Switching to Another Operating System

Upgrading to a newer version of ADMS might require an operating system upgrade (for example, from Windows Server 2008 R2 SP1 to Windows Server 2016). In this case, the upgrade process includes the following steps:

1. Deploying the new system on new machines with the required version of the operating system. The installation and configuration details are described earlier in this document. For a list of software prerequisites, refer to the *ADMS Release Notes*.
2. Transferring and applying data and configurations from the older system.
3. Bringing up the new system (as a PDS or as a standby system).
4. Failing over to the new system (for the planned failover upgrade).

For details about bringing up the new system and failing over to the new system, see [Project Development Server \(PDS\) Upgrade](#) and [Production System Planned Failover Upgrade](#).

12.5. Inter-Version Compatibility Matrix

The following tables show how components in earlier versions of ADMS are supported in ADMS 16.X.X.

Table 9. Inter-Version Component Compatibility

Component/Configuration	From 3.15	From 3.14	From 3.13	From 3.12	From 3.11	Comments
File Updater						
Can File Updater server configuration files be loaded in File Updater server version 16.X.X?	Yes	Yes	Yes	Yes	Yes	
Can File Updater client configuration files be loaded in File Updater client version 16.X.X?	Yes	Yes	Yes	Yes	Yes	
Customer Model (.bpcl)						
Can a generated BPCL be loaded in the 16.X.X server?	Yes	Yes	Yes	Yes	Yes	Due to changes in the functionality for detecting whether a file is encrypted, encrypted customer model files that were built using an older version of the Customer List Compiler need to be rebuilt using the new version of the tool.
Can a generated BPCL be loaded in the 16.X.X Customer Compiler?	Yes	Yes	Yes	Yes	Yes	
Crew Model (.crew)						

Component/Configuration	From 3.15	From 3.14	From 3.13	From 3.12	From 3.11	Comments
Can a generated crew file be loaded in the 16.X.X server?	Yes	Yes	Yes	Yes	Yes	Crew models that are written with legacy journal processing functionality (version 1) cannot be read with enhanced journal processing functionality (version 2, implemented in ADMS 3.9 and enabled by default in ADMS versions 3.11 and later). To convert a crew model version 1 into a crew model version 2, it is required to resave the crew model version 1 using the Crew Modeling tool.
Can a generated crew file be loaded in the 16.X.X Crew Modeling Tool?	Yes	Yes	Yes	Yes	Yes	
Station Models (.mod)						
Can generated .mod files be loaded into 16.X.X servers and clients?	Yes	Yes	Yes	Yes	Yes	Models that are built by the Converter version 3.11 and later cannot be loaded into the servers and clients, whose versions are 3.10 and earlier, because 3.11 supports the int64 serialization.
Structures						
Can structure files be used for 16.X.X?	Yes	Yes	Yes	No	No	
Alarms						
Can alarm configurations be used for 16.X.X?	Yes	Yes	Yes, partially	Yes, partially	Yes, partially	Alarms that were added in ADMS3.15 or later will not be available.// REVIEW COMMENT, 2023-06-07 [Hultstrand, Alyda (GE Vernova)]: When adding the 3.16 column, I have updated this from saying "3.15" to "3.15 or later". It looks like changes related to alarms happened in 3.15. So I assume the statement is applicable not just for 3.15 and not just for current release, but 3.15 and later. @SME, please confirm that this is correct.
Configuration files						

Component/Configuration	From 3.15	From 3.14	From 3.13	From 3.12	From 3.11	Comments
Can server configuration files be used by a 16.X.X server?	No	No	No	No	No	Existing configurations must be carefully transferred to the new configuration files.
Can client configuration files be used by a 16.X.X client?	No	No	No	No	No	Existing configurations must be carefully transferred to the new configuration files.
Can tools configuration files be used by 16.X.X tools?	Yes, partially	Yes, partially	Yes, partially	Yes, partially	Yes, partially	Older configuration files should first be opened with newer versions of the tools and then saved. This adds any new configuration items that are required.
Other files						
Can generated Switching Order application files be used by the 16.X.X Switching Order application?	Yes	Yes	Yes	Yes	Yes	
Can localized resource files be used by 16.X.X applications?	No	No	No	No	No	

12.6. Known Issues in Upgrade Process

This section describes known issues related to upgrading to ADMS 16.X.X.

12.6.1. Upgrade Known Issues (DMS)

Upgrading Structures

Version 3.12 and earlier clients do not support structure models supplied with ADMS version 16.X.X. To use 3.12 clients with 16.X.X server (limited functionality), remove structure model files from FantasyIsland\models\structures. Optionally, you can back up the 16.X.X models and then restore the backup when the client is upgraded to 16.X.X.

Upgrading Server Configuration Files

e-terrdistribution 3.7 split the DMS server, OMS server, DNAF server, Read-Only server, Simulator server, and Customer-Only Mode server processes from the NetServer and NetSrv processes. The **e-terr**distribution 3.7 release also added the NetServerR0.exe, NetServerSim.exe, SimSrv.exe, NetServerCM.exe, and Studysrv.exe server executables as well as a new NetServerOmsConfig.txt server configuration file. Therefore, the PROCMAN model and server shortcuts must be updated, and the new server configuration files must be adjusted when upgrading from **e-terr**distribution 3.6 and earlier.

Upgrading the Archive Database

The Archive database in MS SQL server is upgraded incrementally by executing `upgrade_to_3.X.sql` scripts and `ARCHIVE.dacpac` files.

For more information, refer to the *ADMS Relational Database Software Installation and Maintenance Guide*.

12.6.2. Upgrade Known Issues (OMS)

Non-Outage Incidents

When upgrading from **e-terr***adistribution* 3.7 and earlier to ADMS 16.X.X, "Wants Callback" flags are not transferred for calls that are associated with non-outage incidents. (ETD-37666)

When upgrading from **e-terr***adistribution* 3.7 and earlier to ADMS 16.X.X, "Call Type" values for non-outage incidents are shown as "None" on the Incident Update display. (ETD-37663)

Area Settings

When upgrading from **e-terr***adistribution* 3.5 and earlier to ADMS 16.X.X, area settings are not transferred to the new system and must be re-applied manually. (ETD-35501)

When upgrading from ADMS 3.8.6 and earlier to ADMS 16.X.X, area settings are not shown on client displays until the old OMS display plugin configuration is merged with the new configuration. (ETD-45529)

Upgrading the Management Areas Hierarchy Model

In **e-terr***adistribution* 3.7 and earlier, the management areas hierarchy only included Company, Division, OpCenter, Station, and Feeder levels. In ADMS 3.8.6, the management areas hierarchy model was redesigned to support customization and multiple hierarchy levels.

Therefore, when upgrading from **e-terr***adistribution* 3.7 and earlier to ADMS 16.X.X, you must configure your management areas hierarchy model. Otherwise, some OMS data might not be shown. For more information, see [Modeling the Management Areas Hierarchy](#).

Incident Detail Display

When upgrading from **e-terr***adistribution* 3.7 and earlier to ADMS 16.X.X, the Station and Feeder columns in the Incident Detail display are empty. (ETD-40150)

Incident Summary Displays

When upgrading from **e-terr***adistribution* 3.7 and earlier to ADMS 16.X.X, incident summary displays (for example, Company Summary and Service Center Summary) are empty. (ETD-40163)

Manually Confirmed Single Customer Incidents

When upgrading from **e-terr***adistribution* 3.6 and earlier to ADMS 16.X.X, manually confirmed

incidents with one affected customer (for example, a customer split to a separate incident) where the device of record is not a service point are merged to an incident on the associated transformer. (ETD-45452)

13. Maintenance and Troubleshooting

13.1. Installation and Configuration

- In the absence of a tested and functioning customer project model, the sample model files delivered with the server and client installation kits are quite useful for getting the initial integration working.
- While initially configuring the server, it is helpful to run NetSrv from a command prompt with a small subset of model files, so the output is minimized and the startup time is kept short.
- In the client kit, model files are delivered only to the RT folder. If you want to use the ST configuration, either use the study-case setup feature, or copy the files from RT to the ST folder.
- Use the command-line method of running an ADMS server application until the initial configuration is done and a client can successfully connect to the server. Seeing the output going directly to the window is the fastest way to troubleshoot initial problems. This technique can be used for troubleshooting at any time.
- Using the default ports decreases the chance of having port conflicts.
- The targets of Windows shortcuts have a 256character limit. Because the ADMS client shortcut must start `webfgviewer.exe` and pass multiple configuration files, a target with fully qualified paths can be quite long. The client installation kit creates a shortcut that uses standard Browser environment variables to reduce the length. However, if you install to a folder with a very long path, if the variables are traded out for full paths, or if the configuration file names used are descriptive and long, the 256character limit may be exceeded. Should this happen, a Windows batch script can be used for launching the client, or some new more-specific environment variables can be used.

13.2. System/Software Maintenance

Maintaining an ADMS system is similar to other real-time systems.

- Keep the hardware up to date with firmware updates.
- Patch the operating system (after getting approval from GE).

As part of maintenance agreements, GE does operating system patch testing and provides the results.

- Monitor and manage disk space.

In particular, keep an eye on the log files (especially the `isd.log`) and the dynamics backups on the ADMS server machine.

- Perform regular backups of pertinent data.
- If appropriate, periodically defragment the disk (only in the standby mode).

Most of these are standard system administration tasks that do not need to be detailed here. Tasks particular to an ADMS system are described in the following sections.

13.2.1. Files to Back Up

The ADMS server applications generate a series of dynamics files that get check-pointed and written out to a timestamp-named file every 30minutes (configurable by the `DYNAMICS_BACKUP_MINUTES` parameter). All the dynamics files that are generated by the server are stored in the dynamics folder. These files should be backed up on a periodic basis.

Should all the real-time servers suffer a catastrophic failure, a new server can be created and one of the saved dynamics files can be used to initialize the server to almost the exact state prior to the catastrophe. Also, these files can be used to go back in time and see the state of the electrical network at 30-minute intervals.

13.2.2. Files to Clean Up

The dynamics backup files and the log files are the main areas of automated file creation that need to be systematically cleaned on a periodic basis.

Given that the appropriate files are being backed up, the real-time server should only have the time-stamped backup files for the last two weeks, or something close that makes sense for a specific organization. That said, backed-up files older than the designated period should be removed from the dynamics folder and subfolders.

Keep in mind that files cannot be blindly deleted by date. The annotations functionality generates a dynamics file per annotation type. Given that "storm" is a likely annotation type, a `storm.sav` dynamics file exists to contain all the storm-related annotations. If the last storm was three months ago, most of the annotations were cleaned up within a week, and no activity has been done on the remaining annotations; that `storm.sav` file could be three months old.

Therefore, only delete ".sav" files that have a date in the file name, that have been properly backed up, and are older than the designated purge time frame (two weeks, for example).

13.2.3. Security Management

This section introduces the security management aspects.

13.2.3.1. ADMS System Security Maintenance

Security management should include change control, testing, and auditing. Change control includes the application updates.

GE notifies customers via email when a security patch is available. Details about the security

patches can be found on GE's online customer support web portal (<https://gegrid.force.com/sws>), in the Security Bulletins area of the Security Corner.

It is important to conduct a post-installation security test before deploying a system in production. The following sections provide the initial set of tasks required for ongoing system security maintenance. Once the system is deployed, proper maintenance is critical for keeping the security posture of the system and for continued compliance.

13.2.3.2. Security Management and Auditing for Compliance

Security management means change control and auditing. Change control includes the workflow and the approval processes for managing the changes; deploying the patches, new releases, and configuration updates; and change auditing.

There are many automatic change auditing solutions available on the market that can help you monitor changes and generate audit reports automatically at certain levels.

Contact GE if you need help designing and deploying automatic change control and auditing on your system.

13.2.3.3. Anti-Virus Software Installation and Signature Updates

For a secure Windows installation, a suitable anti-virus package must be installed as early in the setup process as possible. Other products and procedures can apply to project integration and customer environments, which mandate other anti-virus products.

Install the latest anti-virus signatures (virus updates available from the vendor) along with the product and use them to scan all local drives.

GE has determined that system code builds can take significantly longer when on-demand scanning is enabled with some anti-virus products. GE has also found that the un-installation process for MSI kits can also take significantly longer. Some customers may consider disabling on-demand scanning in their host anti-virus product immediately before system builds and product uninstalls. You should then re-enable on-demand scanning after system builds and software uninstallations.

There are security implications when you disable on-demand scanning and the tradeoff between speed and protection must be a customer decision per your corporate security policy. Customers should contact their anti-virus software vendor to determine the best method for avoiding performance issues. If the decision is to turn off the anti-virus software, GE recommends that the customer run a full virus scan immediately following the system build or uninstallation process.

GE has had no reports or experiences of anti-virus software impacting performance during production.

An anti-virus signature is an algorithm or hash that uniquely identifies a specific virus. Depending on the anti-virus vendor, a signature can be referred to as a "signature", a "definition file", or a "DAT file".

A new signature must be created each time a new virus is discovered that is not detectable by an existing signature, or that may be detectable but cannot be properly removed because its behavior is not totally consistent with previously known threats. When a new signature is available, the anti-virus vendors usually issue a signature update.

To maintain the highest level of protection, we recommend that you configure your anti-virus software to check for updates as often as the software allows. Keeping the signatures up to date does not guarantee that a new virus never slips through, but it does make it far less likely.

13.3. Tools and Techniques

13.3.1. Application and Server Logs

The application logs can be used to obtain additional information to investigate and resolve problems.

The ADMS server applications create human-readable log files, the content of which can be used for troubleshooting:

- NetServer.log
- netsrv.log
- isd.log
- SwitchOrderServer.log
- RelayServer.log
- NetDataServer.log
- DMSInterfaceServer.txt
- OMSInterfaceServer.txt
- CrewServer.log
- CallServer.log

Unless a different working directory is specified, these files are created in the server's log directory, which by default is %IDMS_ROOT%\log.

The level of debugging information produced can be adjusted using parameters of the NetServerMainConfig.txt file (for descriptions, see [NetServerMainConfig.txt](#)). Keep in mind that, at the maximum level, the size of these files can increase rapidly and can require a significant amount of disk space.

While running, an ADMS server application also sends informational messages to the standard output. Such messages are displayed to the console window, if running interactively, or stored in a file (netsrv_netsrv_<fam>.out) in the HABITAT_LOGDIR directory, if running from PROCMAN.

13.3.1.1. Enhanced Logging

The enhanced logging features are supported by most of the ADMS servers, such as NetSrv, NetServer, DMS Interface server, OMS Interface server, Crew server, and Switching Order server.

The enhanced logging consists of additional values following the message text. These tab-separated values enable log files to be opened in a tool such as Microsoft Excel to facilitate log message filtering and review.

The first field in each log message is the message text, prefixed with a timestamp. The new fields are Severity, Category, StationName, DeviceName, KeyId, and KeyName. Severity and Category are specified for all enhanced log messages. The other fields are optional since they do not apply to all log messages.

The enhanced logging fields are described below:

- **Severity:** This field shows the message severity. The following table describes the predefined Severity values.

Severity	Description
UNKNOWN	Message severity is unknown
INFOALWAYS	Informational message that is always logged, even if the configuration parameter OMS_SERVER_LOG_LEVEL is set to a higher level
FATAL	Fatal error message that causes the application to shut down
ERROR	Error message that does not shut down the application
WARNING	Warning message
INFO	Informational message
DEBUG	Detailed debug message
DEBUG1	More detailed debug message
DEBUG2	Most detailed debug message
ALL	All messages

Filtering on Severity helps you focus on important messages. Another approach is to log only messages of a specified level. For more information, refer to the OMS_SERVER_LOG_LEVEL parameter, described in [NetServerOmsConfig.txt](#).

- **Category:** This field shows the message category. The following table describes the predefined category values.

Category	Description
None	Not specified
General	General application processing, and for messages that do not fit into one of the other categories
InternalState	Internal application state, typically to describe an invalid condition that represents an application programming error

Category	Description
Configuration	Configuration, such as values of configuration parameters specified in the configuration file and used by the application
Role	High-availability role (for example, Enabled, Standby)
Communications	Message-based communications for interfaces between ADMS components or between ADMS and external applications
Performance	Application performance
Security	Authentication or authorization (user permissions)

- **StationName and DeviceName:** These fields contain the name of a station (substation) and device, respectively.
- **KeyId and KeyName:** These fields contain the identifier and name, respectively, of a specific application entity such as an incident or switching order.

13.3.2. Capturing URL Commands

The following information might be helpful for troubleshooting URL commands, or for sending an exact display call-up to someone else to help with any other problems.

On a NetView display, "Ctrl+C" copies the current URL to the clipboard. Then you can paste it into any other text box (using "Ctrl+V", for example). This technique is highly recommended for writing Software Problem Reports (SPRs).

13.3.3. XML Comparer

The XML Comparer tool facilitates the migration of ADMS configurations by allowing you to review the XML configuration files of your current version and the new version when you upgrade an ADMS product. It displays a side-by-side comparison of a specified XML configuration file from one version to the another, so that you can easily read through the elements and values that have changed and update your configurations accordingly.

13.3.3.1. Comparing Versions

Caution

Every time a new comparison is initiated, the previous comparison is deleted regardless of whether it was saved or not.

To execute the "Compare files" function:

1. Select the two versions of XML files to be compared. To select a file there are three different methods:
 -

Through the File option.

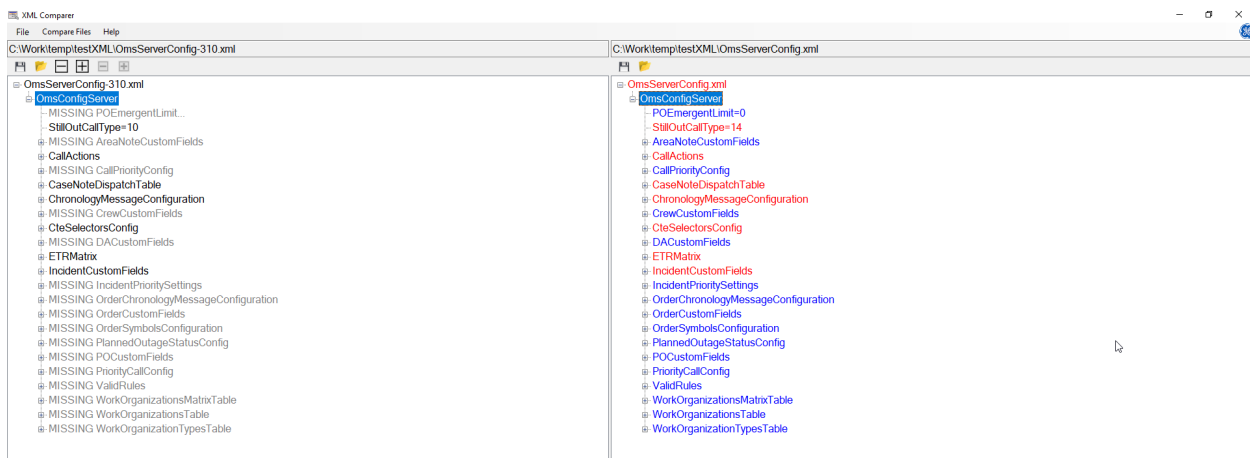
- Select the File option in the upper-left corner.
 - Click "Select left file" and open the configuration file for the current version.
 - Click "Select right file" and open the configuration file for the new version.
 - Using the folder icon.
 - Click the folder (Select left file) button on the tool bar above the left panel and open the configuration file for the current version.
 - Click the folder (Select right file) button on the tool bar above the right panel and open the configuration file for the new version.
2. Select the Compare Files option on the menu bar to compare the two files.

The two files will be displayed in their corresponding panels.

Note

This option is only enabled after selecting the files to compare.

The XML Comparer tool presents the files content in a two-panel display, as shown in the figure below.



13.3.3.1.1. Display

The content of the XML files is displayed as elements arranged in alphabetical order. The files may contain the following types of values:

- **Leaf nodes:** Elements or nodes with only single values.
- **Branches:** Elements or nodes with one or more values and/or branches.

Note

The XML Comparer tool only affects the arrangement of the values, nodes, and branches while

doing the comparison. Files will maintain their original arrangement once saved.

13.3.3.2. Color Codes

The "Compare files" function uses color codes to identify the different scenarios in the comparison. For more details, see the below table.

Scenario	Color Code	Description
Elements and values are the same	Black	The values in both panels are marked with black font.
Values of an element are different	Red	The value in the panel for the new version file is marked with red font.
Element is missing in one file	Gray	The file with the missing value has the value marked with the text MISSING prior to the value name and a gray font. For example: MISSING DisableHistoryCorrection
	Blue	The file with the additional value has the value marked in blue font.
Element and/or value has been copied	Green	The values that have been copied from one file to another are marked in green font.

13.3.3.3. Viewing Detailed Information

The XML Comparer tool allows you to review detailed information related to changes per branch like the following:

- The leaf nodes values that have been changed, added, or deleted.
- The branch elements added or deleted.

You may review the nodes content in the following ways:

- Expand all nodes: On the menu bar select the option Expand all nodes .

You may reclose the nodes by selecting "Collapse all nodes" .

- Expand selected nodes: Select the nodes to be expand, right-click on the selection, and select the option "Expand selection" of the pop-up menu.

You may reclose the nodes by following the same steps and selecting "Collapse selection"

To review the detailed information, hover over a value marked in blue or red, a tooltip will be displayed next to the cursor with the following information:

- **Changed values:** Total amount of values that are different between files.
- **Deleted values:** Total amount of values that exist only in the configuration file of the old version.
- **New values:** Total amount of values that exist only in the configuration file of the new version.

- **Deleted branches:** Total amount of branches that exist only in the configuration file of the old version.
- **New branches:** Total amount of branches that exist only in the configuration file of the new version.

13.3.3.4. Copying Values

To facilitate the customization of new configuration files, the XML Comparer tool allows you to copy elements and branches marked in blue or red from the new configuration version to the current configuration version and vice versa.

To copy an element from one file to the other:

1. Select the value or element on the corresponding panel by right-clicking on it.

Note

You can copy an element from left-to-right or right-to-left when they are marked in blue. You cannot copy elements that appear in gray to the other file because this would be a delete operation, which is not currently supported.

2. Select the "Copy value to right" or "Copy value to left" option accordingly.

The element and value in both panels will be marked in green font indicating that the copy has been completed.

To copy a branch from one file to the other:

1. Select the branch on the corresponding panel by right-clicking on it.

Note

You can copy a branch from left-to-right or right-to-left when they are marked in red. You cannot copy elements that appear in gray to the other file because this would be a delete operation, which is not currently supported.

2. Select the "Copy branch to right" or "Copy branch to left" option accordingly.

The element and value in both panels will be marked in green font indicating that the copy has been completed.

Note

The "Copy value to right", "Copy value to left", "Copy branch to right", and "Copy branch to left" options will be available only if a leaf or branch value has not yet been copied.

Note

The "Copy value to left", "Copy value to right", "Copy branch to left", and "Copy branch to

right" functionalities only apply for elements marked in blue or red.

13.3.3.5. Undoing Changes

For modified values for elements (marked in green font) you can undo the changes if needed. When you do so, the XML Comparer tool will recover the prior value and re-establish the color code correspondingly.

To undo a change, right-click an element that appears in green text and select Undo. The element changes back to the previous values and color coding.

13.3.3.6. Saving Files

You can save the changes done to either the older or newer version of the file. When saving a file, the XML Comparer tool saves the changes directly in the corresponding nodes without affecting their original order.

After completing the changes to the configuration files:

1. Use the "Save left file" or "Save right file" option to save the XML configuration files with the migration changes

Click the save button above the panel of the file to be saved.

2. Enter the name of the new file.

To replace the version used for comparison, enter the same name and select to replace it in the pop-up window.

Note

The XML Comparer tool automatically add the prefix "New_" prior to the file name.

For example: *New_OMSDisplayPluginConfig*

13.3.4. Configuration Accelerator Tool

The Configuration Accelerator tool is designed to facilitate identifying changes in the `ViewerConfig.xml` configuration file. This tool extracts values of selected elements in the input `ViewerConfig.xml` file and save those values in an output csv file. In this file:

- Values of the same element are converted to a tab-separated line.
- For elements that are not direct children of the selected element, path relative to the selected element is prepend to the value with a vertical broken line for separation.

You can compare Configuration Accelerator outputs for several `ViewerConfig.xml` file versions in a text comparison tool to identify changes between those `ViewerConfig.xml` files.

- **Selection:** A specific element to convert.
- **Path:** Path of the selected element relative to document root.
- **Omission:** Paths of elements to omit, separated by commas. Each path is relative to selected element.
 - **Other:** The "Other" element is a pseudo element that does not exist in the source `ViewerConfig.xml`. It is used to exclude multiple specific VCE entries. It represents all elements that are both leaf nodes and direct children of document root, including but not limited to: Version, LineBkgdColor, LineHaloTypeLoop, ScaleX, ScaleY, DecorationsMinZoom, Label, and Find.

13.3.5. Disabling Enhanced Journal Processing

The enhanced journal processing feature (version 2) was implemented in ADMS 3.9 to increase performance and decrease memory footprint. In ADMS 3.11 and later, the enhanced journal processing functionality is enabled by default.

In some cases, you may need to disable the enhanced journal processing functionality and use the legacy journal processing functionality (version 1).

To disable the enhanced journal processing functionality:

Important

Perform this operation on all hosts where the ADMS system is running.

1. Open a command prompt.
2. Execute the following command:

```
setx NET_JOURNAL_VERSION = 0
```

This command adds the `NET_JOURNAL_VERSION = 0` environment variable to disable enhanced journal processing functionality for all server journals.

13.3.6. Troubleshooting Tagging

There are multiple aspects of the overall system and multiple configuration entries that all have to work together for a tag to show up on the NetView (i.e., Geographic) display.

First, verify that the two tagging-specific environment variables are set on the SCADA host where the TAGGING application is running. If these are not set, refer to [Additional Tagging Communication Configuration](#).

Next, if the initial configuration of ISD and IST was done correctly, there should be a NETIO path formed for tagging. Verify this.

1. Call up the ProxySrv/LOG display.
2. In another viewport, stop and restart SCADA and TAGGING from the PROCMAN master display.
3. Watch the ProxySrv/LOG display to ensure that the data("ISD") and control ("CTRL") NETIO paths are opened (msg PXS004).
4. Also, make sure that the NETIO path for tagging data ("TGDATA") is opened (also msg PXS004).

If this is all in place, try to place a tag on a DMS-only switch from the geographic display. From the tag entry screen, write down the TAGID, and enter text to make the tag unique and noticeable.

If no error messages are presented, make sure that the tag was created in the SCADA system. For example:

```
context scada ems
hdbrio scadamom
find tag = _TAGID_
```

If you are unable to find the tag in SCADA, use the SCADA/Tagging logs to troubleshoot the issue.

If you can find the tag in SCADA, check to see if the tag moved from SCADA to the ADMS server by calling up the following display on the DMS:

```
netview://<mode>/gridtabular/TagSummary_GridDisplay
```

If the tag shows up in this tabular, but not on the geographic display, you need to investigate the UI configuration in the Viewer Configuration Editor against the tag definitions in the NetserverMainConfig.txt file.

If the tag is not in the tabular, check the netsrv_main log file to see if the tag was sent to DMS and identify any issues with processing.

13.3.7. Maintaining Orphan Dynamics Entries

This chapter describes how to maintain/handle the dynamics entries for orphan objects, such as the following:

- **Orphaned Tag:** A tag can occasionally become "orphaned" in two cases:
 - When a DMS tag is not linked to a valid EMS Tagging/TAGNOTES tag.
 - When an EMS tag marked as DMS-only does not have a valid link to any DMS tag.
- **Orphaned Device:** A DMS device that is mapped to the EMS device which no longer exists.
- **Orphaned Entry:** A DMS dynamic entry that is related to a static DMS device which no longer exists or has changed position. For example, a cut on a line that was removed from the model file, or an abnormal state for a switch that was removed from the model file.

For more information, refer to [Temporary Object Auto Correction](#).

13.3.7.1. Orphan Tag (Otags) Utility

The Orphan tags (Otags) utility is a command-line executable program designed to locate "orphaned" tags, devices, and entries.

During the Otags installation, the utility's executable is copied to the HABITAT_BINDIR directory. Therefore, the working directory location does not matter.

To allow Otags queries, you need to enable the "Service Port" parameters in the NetServerMainConfig.txt configuration file:

To enable the "Service Port" parameters:

1. Locate the NetServerMainConfig.txt configuration file. The default location is %IDMS_ROOT%\config\server.
2. Locate the "Service module listening socket and socket parameters" line and uncomment the following parameters:

```
SERVICE_PORT = 6032
SERVICE_SOCKET_OUTPUTBUF_MAX_SIZE = 1000000
SERVICE_SOCKET_WOULDBLOCK_RETRY = 2
SERVICE_SOCKET_MAX_CLIENT_MES_SIZE = 640000
```

3. Save the changes and close the configuration file.

To start the Otags application and perform the queries:

1. Start the DMS and EMS systems.
2. Start the command prompt.
3. Set the EMS context that contains tags (SCADA/TAGGING). To set the SCADA context, type the following command:

```
context scada ems
```

4. Start the Otags application by typing the command:

```
otags [-args]
```

where [-args] is the otags command-line arguments, described in table below.

The following example demonstrates the required arguments needed for the Otags utility to start queries, including the EMS application name, the EMS family where the tags reside, the host address, and the DMS service port number. The final argument enables the creation of a CSV file containing the list of orphan tags, if any are found.

```
> otags -a scada -f ems -h localhost -p 5032 -c
```

Table below describes the available Otags command-line arguments.

Table 10. Otags Command-Line Arguments

Argument	Description
-h	The DMS host address, which can be either a host name (for example, localhost) or an IP address.
-p	The DMS service port number.
-a	The EMS application name (SCADA/TAGGING) of the clone where the tags reside.
-f	The EMS family name of the clone where the tags reside.
-s	Report the substation name for each orphan tag.
-d	Set the debug flag for the entire application.
-c	Enables creation of the otags.csv file.

For performance enhancement purposes, the Otags application queries the DMS in two steps:

1. Query the DMS for all stations that have any tags.
2. Query each station for all tags.

The list of orphan tags is saved into the otags.csv file, which can be imported into a spreadsheet or a database. The CSV file is not created when no orphan tags are found.

To locate the CSV file, at the command prompt, type "dir *.csv".

	A	B	C	D	E
DMS Heading	1 DMS ID	TAGSTATIONNAME	TAGCOMPID		
First DMS Tag Orphan	2 TEMS3CA52F50	3LIONS	22240300		
	3 TEMS3CA52F3D	PEBBLE BEACH	131109215		
	4 TEMS3CA52DDC	RAVEN	22233017		
	5 TEMS3CA52DDC	RAVEN	22233017		
Tagnotes/Tagging Heading	6 TAG ID	SUBSTN	RECCKEY		
First Tagnotes/Tagging Orphan	7 TEMS3CA52F36	PBLBEACH	PEBBLE BEACH.F6532.SWT.131109215		
	8				
	9				
	10				

Figure 42. Otags Output Data in an Excel Spreadsheet

13.3.7.2. Retaining Orphan Dynamics

Retaining orphan dynamics is controlled using the ORPHAN_DYN_RETAIN_DAYS parameter in the NetServerMainConfig.txt configuration file.

The ORPHAN_DYN_RETAIN_DAYS option sets orphan dynamics cleanup periodicity in days. Cleanup occurs during midnight dynamics compression.

- If ORPHAN_DYN_RETAIN_DAYS is set to zero, the cleanup feature is disabled.
- If ORPHAN_DYN_RETAIN_DAYS is set to a positive value, all orphan dynamics with timestamps are retained until they are older than the ORPHAN_DYN_RETAIN_DAYS value in days.

All orphan dynamics with no creation time are removed during midnight compression.

In addition to the server configuration file option, the following command can be used to control

orphan dynamics cleanup:

```
netview://rt/orphanDyn?rd=0&compressDyn=1
```

- The *rd* and *compressDyn* parameters are optional. The default value for both parameters is zero, which indicates that no dynamics compression is set to occur and the system only logs all orphan dynamics.
- *rd* can be set to a positive, zero, or a negative integer.

A negative integer indicates seconds, while a positive integer sets the parameter in days.

- *compressDyn* can be set to 1, 0, true, or false.

To report orphan dynamics without changing the dynamics.sav file, use the following command:

```
netview://rt/orphandyn
```

During compression, a log file, `orphandynamics_yyyymmdd_hhmm.txt`, is created to log all orphaned dynamics regardless of the `ORPHAN_DYN_RETAIN_DAYS` parameter value. The log file is stored in the same location as the `dynamics.sav` file:

```
%IDMS_ROOT%\dynamics\server\orphandynamics_yyyymmdd_hhmm.txt
```

The log file records the orphan dynamics in a plain text format with one orphaned dynamics record per line. The recorded data includes the associated component type, ID, substation, dynamics timestamp, and the purge reason. Dynamic data types and order may slightly differ.

Dynamics can be purged due to the following reasons:

- The associated component does not exist anymore. In this case of true orphaned dynamics, the purge reason indicates "Orphaned".
- The associated components are in the nominal state, and there is no need to have the dynamics. In this case, the logged purge reason is "Dev nominal".

Currently, only two types of dynamics are checked for the nominal state: switch and fault indicator (CDmsFaultInd and CDmsSwitch DNAF objects).

When purging is enabled, the log file records only purged dynamics.

The example below demonstrates a log file output:

```
## Orphan Dynamics of 9/24/2012 10:07:17 Pacific Daylight Time ####

SWT: 3234511, Sub1: RAVEN, Sub2: , CreateTime: 9/24/2012:10:03:27, PendPerm: 0, Purge reason: Dev
nominal
SWT: 3234523, Sub1: RAVEN, Sub2: , CreateTime: 9/24/2012:10:03:27, PendPerm: 0, Purge reason: Orphaned
FI: 155439001XFD, Sub: OLYMPIC, Purge reason: Dev nominal
FI: 155439002XFD, Sub: OLYMPIC, Purge reason: Dev nominal

##### End of 9/24/2012 10:07:17 Pacific Daylight Time #####
```

13.3.8. Forcing Garbage Collection

For debug purposes, you may need to force garbage collection for the server or client.

Client Garbage Collection

To force garbage collection for the client, use the following command:

```
NETVIEW://RT/DEBUG/GC/<Generation Level>
```

Where Generation Level is an optional parameter to force garbage collection for a specific generation level. Supported generation levels are 0, 1, and 2.

For example:

```
NETVIEW://RT/DEBUG/GC/1
```

Server Garbage Collection

To force garbage collection for the server, use the following command:

```
NETVIEW://RT/SERVERDEBUG/GC?CLIENT=<Collect for Client?>&AGE=<Object Age>
```

Where:

- CLIENT is an optional parameter, which defines whether to also perform garbage collection for the client. Supported values are 0 (disabled) or 1 (enabled).
- AGE is an optional parameter, which defines that the garbage collection will include objects that have been deleted more than the specified number of seconds ago.

For example:

```
NETVIEW://RT/SERVERDEBUG/GC?CLIENT=1&AGE=1
```

13.3.9. ADMS Debugging Pop-Up Display

Use the ADMS Debugging pop-up display to view and monitor the following parameters and configurations:

- Both default and updated server configuration parameters.
- All message threads in the server.
- All clients currently connected to the server.
- Current statistics for the current client.
- Parameters specified in the corresponding dynamics configuration file.

Use the following commands to call up the display for NetSrv, NetServer, and DnafSrv, respectively:

```

NETVIEW://%dyn/POPUP/DEBUG
NETVIEW://%dyn/POPUP/OMSDEBUG
NETVIEW://%dyn/POPUP/DNAFDEBUG

```

Where "%dyn" is the name of the dynamics, for example, "RT", "SIM", or "ST".

The display includes five tabs: Client Statistics, Message Statistics, server Configuration, My Statistics, and My Configuration. The tabs are described in the sections below. The three statistics tabs refresh every four seconds, but the configuration tabs do not refresh after the application starts.

13.3.9.1. Client Statistics Tab

The Client Statistics tab shows all clients that are currently connected to the server. The GUID of a client matches the GUID found on the ACTIVE SESSIONS display in PERMIT. The GUID of an individual client can also be seen on the My Statistics tab.

Select a client to display the messages for that client in the right side, starting on the next refresh cycle.

Guid	App Type	IP Address	Port	Dynamics Sequence	Number R
{22931A5E-1F11-46FF-8940-QA380C2}	NetClient	127.0.0.1	51934	171	694893
bb7c893b-0fd3-4...	DNAF	127.0.0.1	55928	171	8707
fc1e8f31-c1eb-4...	OMS Main	127.0.0.1	5025	171	0

Message	Read	Sent
dynMsgHelloAck	0	127
dynMsgQuery	55	0
dynMsgQueryAck	0	55
dynMsgExecuteA...	0	1943014
dynMsgResult	0	1943014
dynMsgCancel	55	0
dynBufferUpdate	0	441
dynMsgGUID	6	0
dynRequestSessi...	6	0
dynSessionData	0	6
dynFastDynamics	20380	674
dynManagedQue...	525914	0
dynManagedResp	0	1
dynManagedQue...	0	314512
dynRequestOms...	1	0
dynManagedDat...	21	0
dynManagedDat...	0	21

The columns for each client include:

- Guid: The GUID supplied by Browser, and used by PERMIT.
- App Type: The client application type.
-

IP Address: The IP address of the client.

- Port: The IP port (from the server's point of view) for the socket connection to the client. This corresponds to information that the Windows NETSTAT command will return.
- Dynamics Sequence: The sequence number acknowledged by the client. In a well-behaving system, the sequence number of all clients will be the same, and will match the sequence if the dynamics file (`dynamics.sav`) is viewed in the Snapshot Viewer application.
- Number Read: The number of messages received from the client.
- Number Sent: The number of messages sent to the client.
- Bytes Read: The number of bytes corresponding to the messages read.
- Bytes Sent: The number of bytes corresponding to the messages sent.
- Total Bytes Read: The total number of bytes read from the client. The client/server protocol frames messages in 1024-byte buffers, so the total bytes of a 50-byte message will be 1024 bytes, the total number of bytes for a 1900-byte message is 2048, etc. The total bytes read will always be larger than the bytes read.
- Total Bytes Sent: the total number of bytes sent to the client, reflecting the 1024-byte framing.
- Average Number Read: The average number of messages received from the client.
- Average Number Sent: The average number of messages sent to the client.
- Average Bytes Read: The average number of bytes corresponding to the messages read.
- Average Bytes Sent: The average number of bytes corresponding to the messages sent.
- Average Total Read: The average total number of bytes read from the client.
- Average Total Sent: The average total number of bytes sent to the client.

13.3.9.2. Message Statistics Tab

The Message Statistics tab displays the number of internal "dynamics" messages read from and sent to the client. The tab shows all the threads in the server (except for the main dispatcher, which can be inferred from the sum of the dispatch threads listed, and the managed OMS threads in a NetServer environment).

Select a thread in the list on the left to display the thread's messages on the right, starting in the next refresh cycle.

The screenshot shows a 'Debugging Popup' window with two main panes. The left pane, titled 'Message Statistics', contains a table of thread statistics. The right pane, titled 'NTP Dispatcher Messages', contains a table of message backlogs. At the bottom, there is a checkbox 'Highlight item with backlog over:' followed by a text input field containing the value '5'.

ThreadName	Number Received	Number Processed	Received Backlog	Processed Backlog
NTP Dispatcher	614	6394	0	0
NTP Thread0	304	960	0	0
NTP Thread1	280	878	0	0
NTP Timer	37	37	0	0
NtpCBO Thread	1	0	0	0
DYNsrv Disp	5111012	826803	0	0
DYNsrv Thd0	764658	686574	0	0
DYNsrv Thd1	141019	121045	0	0
DYNsrv SOCK	4771246	1159773	0	2
DYNsrv Journal	28906	91115	0	0
DYNsrv Permit	0	4	0	0
DYNsrv Timer	1	44475	0	0
DYNsrv w/RwR	415442	14421	0	0
DYNsrv Ann	280643	34	0	0
DYNsrv TT	5	6	0	0

Message	Received	Processed	Received Backlog
msgLog	0	5730	0
msgDynBoundSa...	0	2	0
msgDynamicsAp...	1	0	0
msgNtpSubstatio...	0	580	0
msgNtpSchedule...	9	0	0
msgNtpDoneIniti...	0	3	0
msgNtpEventsSo...	0	42	0
msgNtpSwitchLo...	37	37	0
msgStrnMap	1	0	0
msgDynamicRest...	35	0	0
msgNtpPrediction...	530	0	0
msgNtpPrediction...	0	532	0
msgDynMobileSt...	0	2	0
msgNtpSchedule	1	0	0

☒ Highlight item with backlog over: 5

The server is architected with several modules, each with a dispatcher and one or more generic or special worker threads. Each thread includes an input and output queue, where the input queue contains messages added by the threads dispatcher (or the main dispatcher in the case of a dispatcher thread), and the output queue contains messages added by the thread to be read by the dispatcher thread.

The list of threads includes these columns:

- Thread name: The thread name includes the module of the thread; for example, DYNsrv indicates the Dynamics server module.
- Number Received: The number of messages read by the thread.
- Number Processed: The number of messages sent to the dispatcher from the thread.
- Received Backlog: The difference between the number of messages sent by the dispatcher and the number of messages read by the thread. In a well behaving system, this number should be zero. Temporarily this number might be greater than zero for dynamics threads, during a large burst of activity.
- Processed Backlog: The difference between the number of messages sent by the thread and the number of messages read by the dispatcher. This number should be zero, except that temporarily it might be one depending on the query cycle.

For the messages for each thread, these columns are included:

- Message: The internal message name.
- Received: The number of this message type received by the thread.
- Processed: The number of this message type sent to the dispatcher from this thread.
- Received Backlog: The difference between the number sent by the dispatcher and the number read by the thread, for this message type.

- **Processed Backlog:** The difference between the number sent by the thread and the number read by the dispatcher, for this message type.

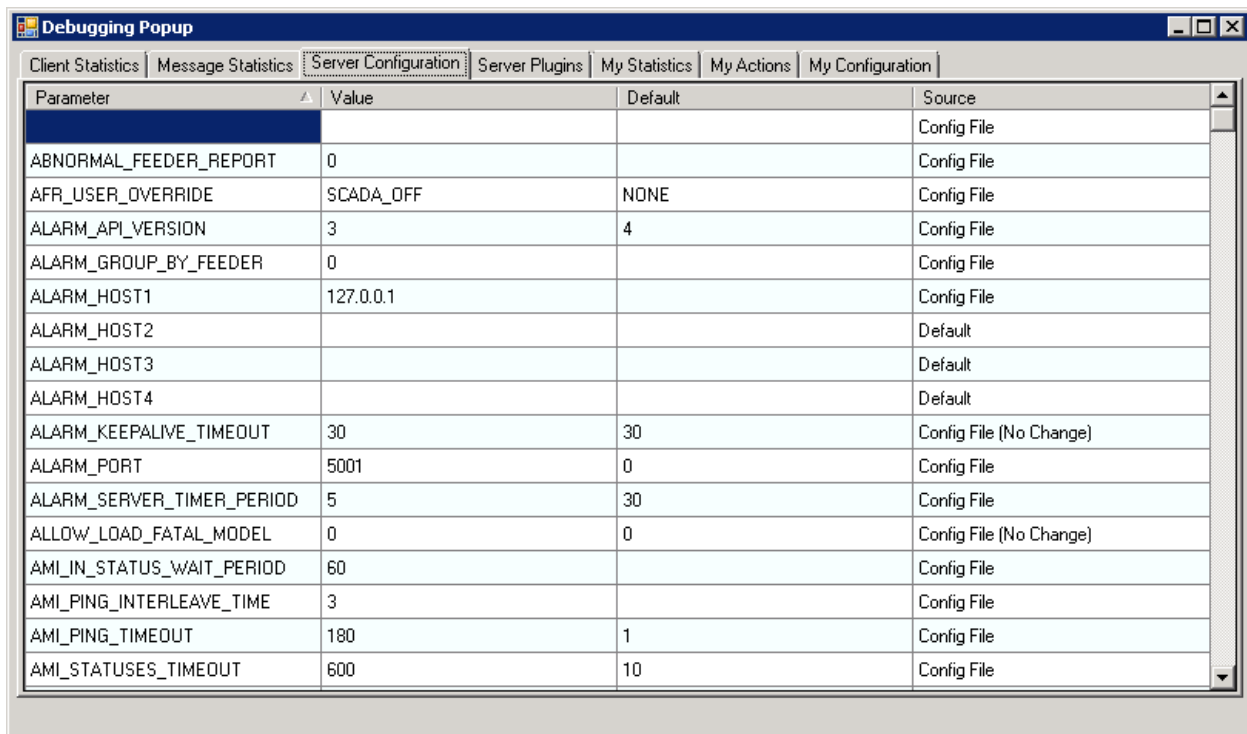
Generic worker threads are typically named "Thread*n*:" where *n* is a number 0, 1, 2, etc. The number of generic worker threads is specified in the server configuration; for example, "THREADS_NTP = 2" will cause the server to create two worker threads. Generic worker threads perform some work that can be done in parallel. For example, NTP and DNAF worker threads solve a single station, and Dynamics Worker and Query threads resolve a single request.

The Debugging pop-up display itself is populated using "fast dynamics" requests and responses, so the data is supplied by a generic Dynamics server worker thread.

Nongeneric worker or dispatcher threads are special threads that perform a specific function.

13.3.9.3. Server Configuration Tab

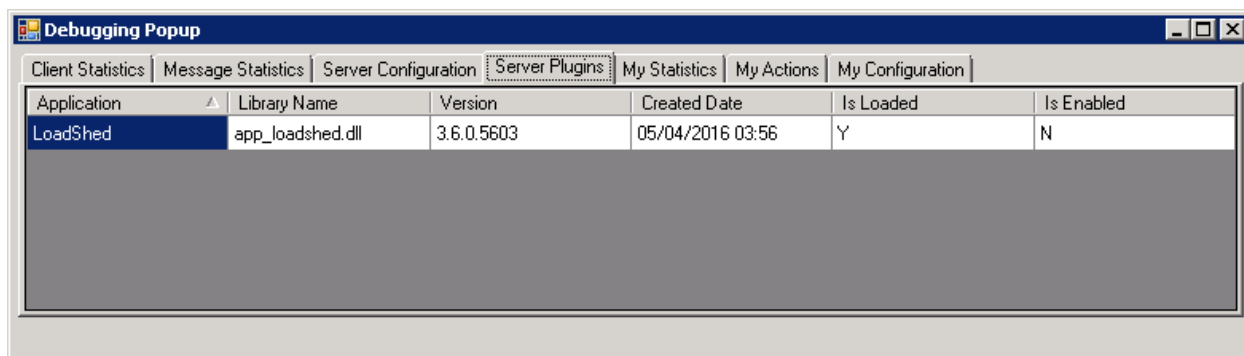
The server Configuration tab includes all the server parameters, their values, the default value, and whether the value has been changed. You can sort on the Changed column to verify which parameters have been changed from their default value. The purpose of this tab is to display the server's configuration without directly accessing the server.



Parameter	Value	Default	Source
			Config File
ABNORMAL_FEEDER_REPORT	0		Config File
AFR_USER_OVERRIDE	SCADA_OFF	NONE	Config File
ALARM_API_VERSION	3	4	Config File
ALARM_GROUP_BY_FEEDER	0		Config File
ALARM_HOST1	127.0.0.1		Config File
ALARM_HOST2			Default
ALARM_HOST3			Default
ALARM_HOST4			Default
ALARM_KEEPAIVE_TIMEOUT	30	30	Config File (No Change)
ALARM_PORT	5001	0	Config File
ALARM_SERVER_TIMER_PERIOD	5	30	Config File
ALLOW_LOAD_FATAL_MODEL	0	0	Config File (No Change)
AMI_IN_STATUS_WAIT_PERIOD	60		Config File
AMI_PING_INTERLEAVE_TIME	3		Config File
AMI_PING_TIMEOUT	180	1	Config File
AMI_STATUSES_TIMEOUT	600	10	Config File

13.3.9.4. Server Plugins Tab

The server Plugins tab shows information for connected server plug-ins.

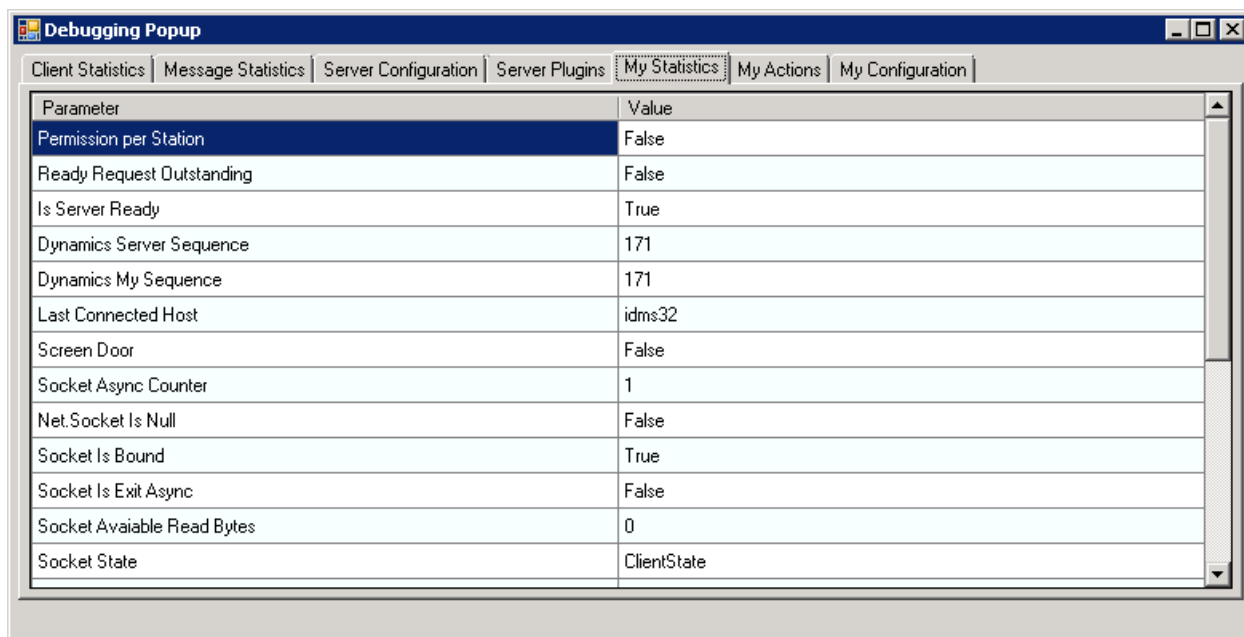


The columns on this tab are as follows:

- Application: Plug-in application name.
- Library Name: Plug-in library name.
- Version: Version details.
- Created Date: Plug-in build date and time.
- Is Loaded: Indicates if the DLL file exists and loaded.
- Is Enabled: Indicates if the application is enabled.

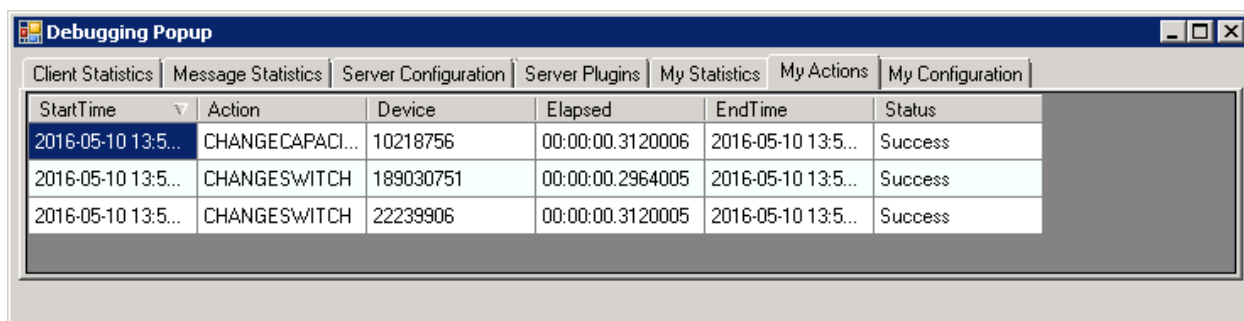
13.3.9.5. My Statistics Tab

The My Statistics tab shows current statistics for the current client. Some of the parameters are based on the connection to the server, and the dynamics sequence corresponds to dynamics sequence for this server connection. Some parameters are process-wide and include all ADMS and Browser viewports, such as the memory statistics and the handle count.



13.3.9.6. My Actions Tab

The My Statistics tab shows the recent user actions, such as toggling devices.

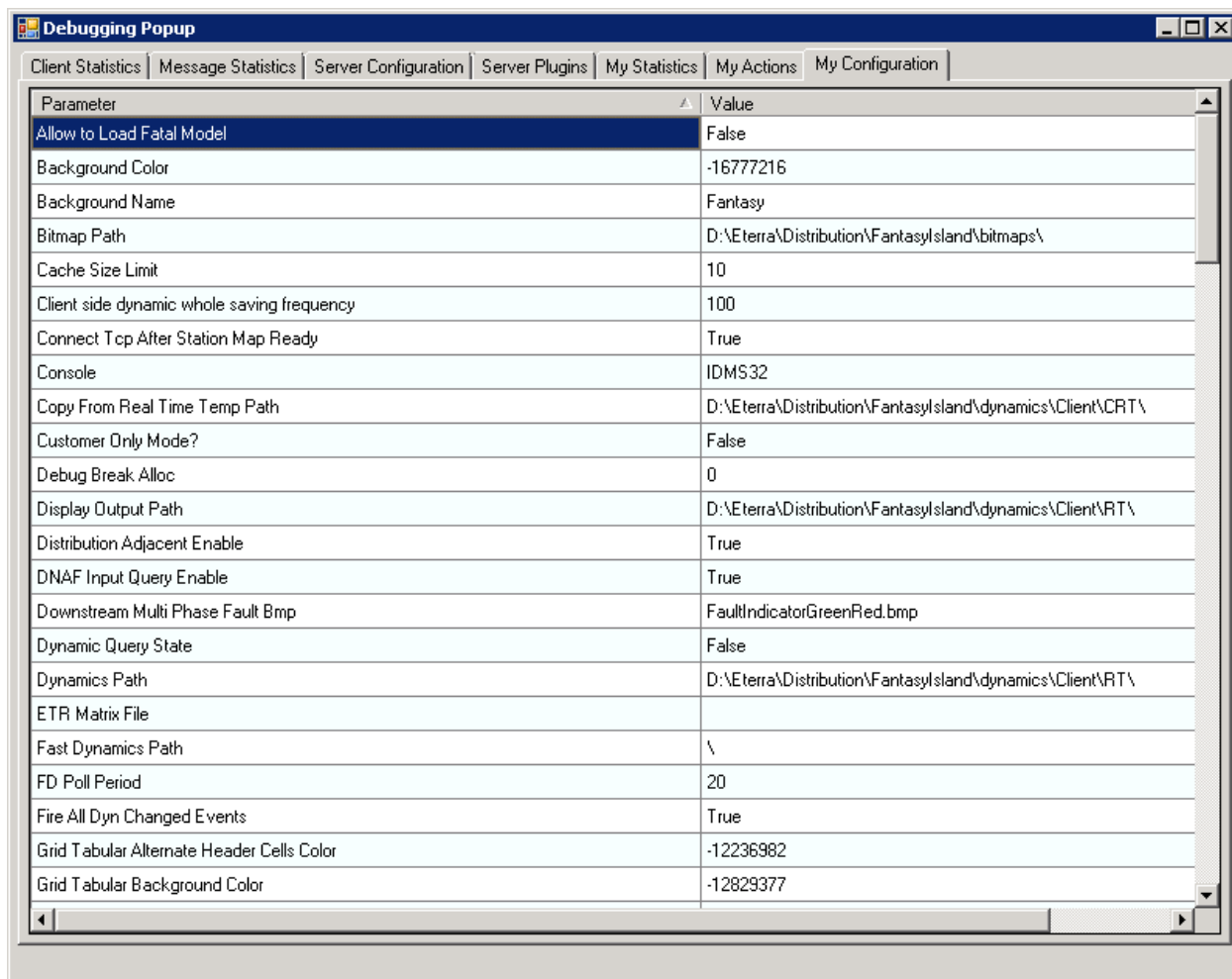


The screenshot shows the 'Debugging Popup' window with the 'My Actions' tab selected. The tab displays a table of recent user actions. The table has six columns: StartTime, Action, Device, Elapsed, EndTime, and Status. There are three rows of data, all showing 'Success' status.

StartTime	Action	Device	Elapsed	EndTime	Status
2016-05-10 13:5...	CHANGECAPACI...	10218756	00:00:00.3120006	2016-05-10 13:5...	Success
2016-05-10 13:5...	CHANGESWITCH	189030751	00:00:00.2964005	2016-05-10 13:5...	Success
2016-05-10 13:5...	CHANGESWITCH	22239906	00:00:00.3120005	2016-05-10 13:5...	Success

13.3.9.7. My Configuration Tab

The My Configuration tab lists parameters specified in the corresponding dynamics configuration file (for example, `netconf_localhost.xml`). It is also possible to see these values in the Net Dynamics Configuration Editor tool or by directly reading the XML file.



The screenshot shows the 'Debugging Popup' window with the 'My Configuration' tab selected. The tab displays a table of configuration parameters and their values. The table has two columns: Parameter and Value. There are 25 rows of data, including parameters like 'Allow to Load Fatal Model', 'Background Color', 'Background Name', 'Bitmap Path', 'Cache Size Limit', 'Client side dynamic whole saving frequency', 'Connect Top After Station Map Ready', 'Console', 'Copy From Real Time Temp Path', 'Customer Only Mode?', 'Debug Break Alloc', 'Display Output Path', 'Distribution Adjacent Enable', 'DNAF Input Query Enable', 'Downstream Multi Phase Fault Bmp', 'Dynamic Query State', 'Dynamics Path', 'ETR Matrix File', 'Fast Dynamics Path', 'FD Poll Period', 'Fire All Dyn Changed Events', 'Grid Tabular Alternate Header Cells Color', and 'Grid Tabular Background Color'.

Parameter	Value
Allow to Load Fatal Model	False
Background Color	-16777216
Background Name	Fantasy
Bitmap Path	D:\Eterra\Distribution\FantasyIsland\bitmaps\
Cache Size Limit	10
Client side dynamic whole saving frequency	100
Connect Top After Station Map Ready	True
Console	IDMS32
Copy From Real Time Temp Path	D:\Eterra\Distribution\FantasyIsland\dynamics\Client\CRT\
Customer Only Mode?	False
Debug Break Alloc	0
Display Output Path	D:\Eterra\Distribution\FantasyIsland\dynamics\Client\RT\
Distribution Adjacent Enable	True
DNAF Input Query Enable	True
Downstream Multi Phase Fault Bmp	FaultIndicatorGreenRed.bmp
Dynamic Query State	False
Dynamics Path	D:\Eterra\Distribution\FantasyIsland\dynamics\Client\RT\
ETR Matrix File	
Fast Dynamics Path	\
FD Poll Period	20
Fire All Dyn Changed Events	True
Grid Tabular Alternate Header Cells Color	-12236982
Grid Tabular Background Color	-12829377

13.3.10. Station Overview Builder

The Navigation Builder utility is used to build the Station Overview display, which shows an overview of all electrical substations and allows navigation to any substation by simply clicking on the display.

The screenshot below is an example of a navigational overview display.

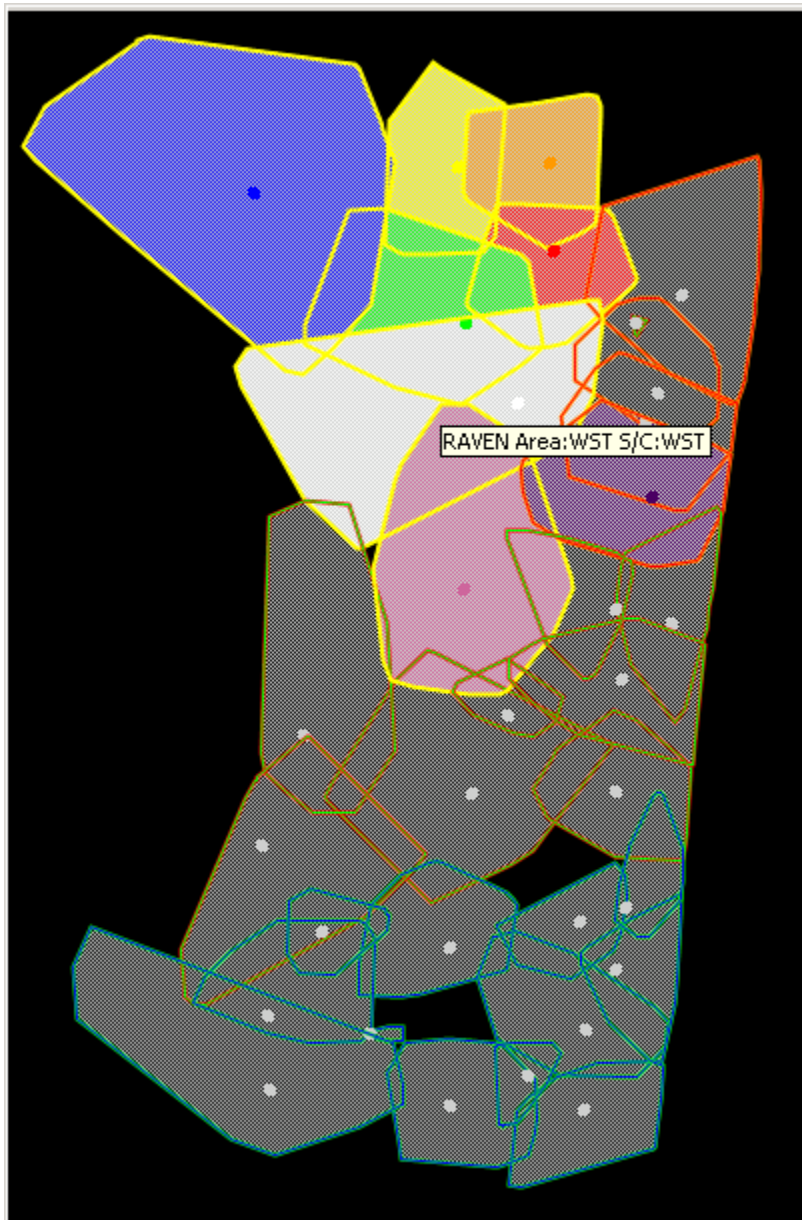


Figure 43. Example of a Navigational Overview Display

The navigational overview file (overview.nvw) is generated from the .MOD files using the NavBuilder program.

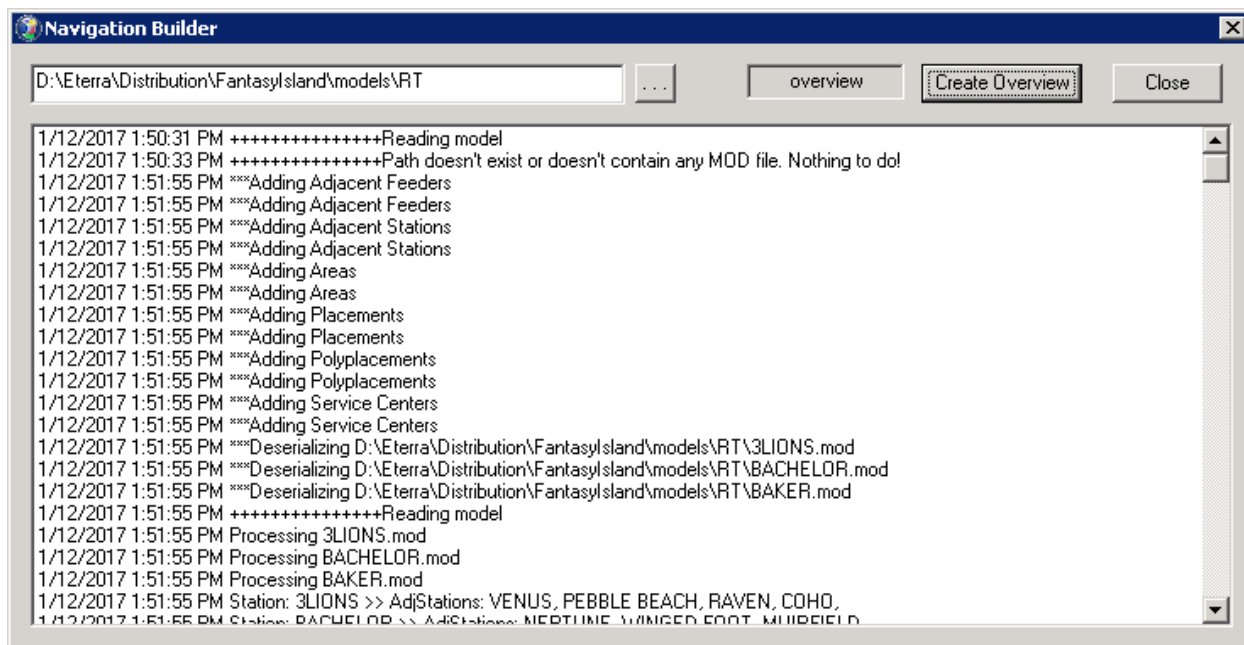


Figure 44. NavBuilder

To generate the navigational overview file:

1. Start NavBuilder.exe.
2. In the first text box, enter the directory containing all the .MOD files (or Browse to the correct folder).
3. Click the Create Overview button.

The "overview.nvw" file is created in the same folder that contains the .MOD files.

The overview.nvw file must be kept with the .MOD files. This enables the overview functionality of the clients and allows adjacent stations to be called up around the selected station.

This application can also be run from the command line for batch processing capabilities.

```
Navbuilder.exe /PATH: D:\ThePathWithTheModelFiles /FILE: NameOfOverviewFileWithoutExtention
```

```
Navbuilder.exe /PATH: D:\working\mods /FILE: overview
```

The "/LOG:" argument can be used to specify the application log file name (the default value is NavBuilder.log).

Notice that the file name "overview" does not include the file extension or the absolute path.

13.4. Performance

13.4.1. Performance Statistics

The performance statistics gathering capability of the ADMS client allows users to measure and analyze the duration of typical user activities. The gathered data can be used for diagnostic purposes in case the durations are longer than expected.

The performance statistics gathering can work for a stand-alone client or for a Citrix-based client.

The following user actions are measured and recorded to the statistics file.

- **Geographic display call-up:** Measures the time from receiving a parsed command, until the display is ready to be drawn to the screen.
- **Changing the status of a device:** Measures the time from when a request is made for a device change, until the client has received a dynamics change and completed the topology processing for the change. The following devices are measured for changes:
 - Changing a switch state (both all phases and individual phase operation)
 - Adding, removing, or changing the state of a cut
 - Adding, removing, or changing the state of a temporary jumper
- **Calling up an attribute or output pop-up dialog:** Measures the time from opening the pop-up until the query is received from the server.
- **Performing a device search:** Measures the time from a search being initiated until the query results are received and ready to be displayed.
- **Performing a tabular display call-up:** Measures the time from when a parsed command is received until the query results are received and the display is ready to be drawn.

To enable statistics at startup, the `InitializeStatisticsLogging` parameter in the `netconfig.xml` file must be set to `True`. By default, it is `False`.

Statistics can be manually enabled or disabled using the following commands, where `%dyn` is the dynamics name (for example, RT).

```
NETVIEW://%dyn/STATS/ON
NETVIEW://%dyn//STATS/OFF
```

Note

When enabled, the statistics are recorded for all dynamics.

If statistics are enabled, a log file will be produced in the log directory, specified by the `StatisticsLogDir` parameter in the dynamics configuration file `netconf.xml`. By default, the `%IDMS_ROOT%\Log\` directory is set.

The file has the following naming convention, where `pid` is the process ID:

The statistics log file is a text file where the fields are tab-delimited, so that is easy to write a script to parse the file, or to bring the file into Excel to gather the statistics.

The file format is a self-describing file where each record type has a header field, followed by a list of records. The second column is used to differentiate header rows (=0) that specify which columns the record supports from the data rows (=1), as shown in the example below. The durations are measured in milliseconds.

Table 11. Example Statistics Log File Details

GEOCALLUP	0	ELAPSEDMSEC	TIME	STATIONCOUNT	URL
GEOCALLUP	1	1591	July 21, 2016 10:20:05 AM	8	NETVIEW://RT/VIEW/%GEOM/RAVEN
TABULAR	0	ELAPSEDMSEC	TIME	ROWCOUNT	URL
TABULAR	1	904	July 21, 2016 10:20:21 AM	33	NETVIEW://RT/GRIDTABULAR/AREALIST _GRIDDISPLAY
ATTRIBUTEPOPUP	0	ELAPSEDMSEC	TIME	NOTUSED	URL
ATTRIBUTEPOPUP	1	31	July 21, 2016 10:19:12 AM		NETVIEW://SIM/VIEW/%GEOM/RAVEN

Note

When statistics are enabled, a statistics file is produced and updated until the client is shut down or restarted.

13.4.2. Performance Metrics

This section provides performance metrics for some common system activities.

13.4.2.1. DNAF Application Solution Times

Hardware Configuration: 1x6 core CPU @ 3.2 GHz and 64 GB of memory (RAM).

Table below shows DNAF application solution times. Testing was based on the Fantasy Island demo model with default parameter values in the DNAF Configuration Editor.

Table 12. DNAF Application Performance

Application	Scenario	Solution time, seconds
FISR (Single-phase fault)	Station: BETHPAGE Line: 22276020 Place single-phase BG fault on line in SIM mode.	1.927
FISR (Three-phase fault)	Station: BETHPAGE Line: 339967940 Place three-phase ABCG fault on line in SIM mode.	24.153
FISR (Feeder Loss of Voltage)	Station: BETHPAGE Line: 22336220 Place cut on line in SIM mode.	9.37
AFR (Line Overload)	Station: ROYAL TROON Line: 327959850212.981 Set line flow limit for the all phases to [15] Amps.	212.697
AFR (Transformer Overload)	Station: MUIRFIELD Set transformer flow limit for the all phases to [4.5] MVA.	180.694
DPF (manual execution for stations in the demo model)	Station: N/A (average for all stations)	2.3
DPF (manual execution for a station with full secondary network)	Station: Custom	21
DPF (manual execution for a station with no secondary network, with lumped load at the service transformer)	Station: Custom	5
LVM (manual execution with large number of DERs)	Station: SPYGLASS Manual execution with "Reactive Support" problem formulation.	12.203
PRV (manual execution with large number of DERs)	Station: SPYGLASS Manual execution.	10.504
PLOS	Station: RAVEN Line: 329340298 Manual execution in Study mode.	58.088
FISRCA (manual execution for stations in the demo model)	Station: N/A (average for all stations) Problem formulation: Minimum Unserved kW (Auto + Manual)	16.9 seconds per switchable section 37.3 minutes per station

Note

DNAF application solution performance is impacted by the size and complexity of the network

model, the quality of the network model data, the power flow convergence, and the application settings in the DNAF Configuration Editor.

Important

GE has no control over the switching reconfiguration plan execution time. This depends on the number of steps, the ability to implement switching steps in parallel, the presence of failed controls, and the performance of field device communications networks.

13.4.2.2. OMS Processing Times

General Configuration:

- Model size is 1.8 million customers
- Call Server and NetServer are configured on two different machines
- Network bandwidth between machines is 1Gbps. The average and peak network consumptions in case of a storm are 50 Kbps and 5Mbps respectively.

Scenario 1: 500,000 AMI meter out messages in 1 hour.

- **Hardware Configuration:** 4x10 core CPU @ 2.6 GHz and 128 GB of memory (RAM).
- **Result:** NetServer gets in sync with CallServer in near-real time with 2-3 sec delay intermittently.

Scenario 2: 100,000 AMI meter restoration messages in a burst of 180 seconds.

- **Hardware Configuration:** 4x10 core CPU @ 2.6 GHz and 128 GB of memory (RAM).
- **Result:** NetServer gets in sync with CallServer in about 130 seconds.

Scenario 3: 60,000 calls per hour for 4 hours.

- **Hardware Configuration:** 4x10 core CPU @ 2.6 GHz and 128 GB of memory (RAM).
- **Result:** The system handles 60,000 calls per hour for 4 hours without any issue.

Scenario 4: The response time for the Incident Update and Order Worksheet displays must be within 5 seconds (worst case) in a normal situation and 10 seconds (worst case) in a storm situation.

- **Hardware Configuration:** 4x10 core CPU @ 2.6 GHz and 128 GB of memory (RAM).
- **Result:** The response time is within the requirement.

Scenario 5: The time Difference between call entries in the call journal and call entries in the Archive database must be within 30 seconds in a normal situation and 300 seconds in a storm situation.

- **Hardware Configuration:** 2x8 core CPU @ 3.0 GHz and 64 GB of memory (RAM). MS SQL Server 2015.
- **Result:** In a normal situation, the replication time for 100% of the records is less than 30 seconds. In a storm situation, the replication time for 92% of the records is 300 seconds.

13.5. Problems and Solutions

13.5.1. Using File Transfer Tools with Model Files

Symptom: The model (*.mod) file is corrupted. It cannot be read by an ADMS server application.

The model files (.mod) are not always interpreted correctly by FTP or SFTP applications as being binary. If using FTP/SFTP to transfer individual files, be sure to enter the "binary" command. Using zip files to transfer large groups of model files is faster (since the files compress well), and this also retains the binary settings. However, if you are using InfoZip's unzip to unzip the package, do not use the "-a" flag because it incorrectly auto-converts the models into ASCII.

13.5.2. Cannot Overwrite a Model File

Symptom: Cannot overwrite a model file, such as a customer model.

When overwriting model files, it is recommended that you rename the original file, and then copy the new file.

13.5.3. Communications Failure between the Client and the DMS Server

Symptom: The ADMS viewer client does not communicate with the DMS server application.

Symptom: Not able to toggle switch. When trying to toggle a switch, a popup box appears with the error message "no connection to server".

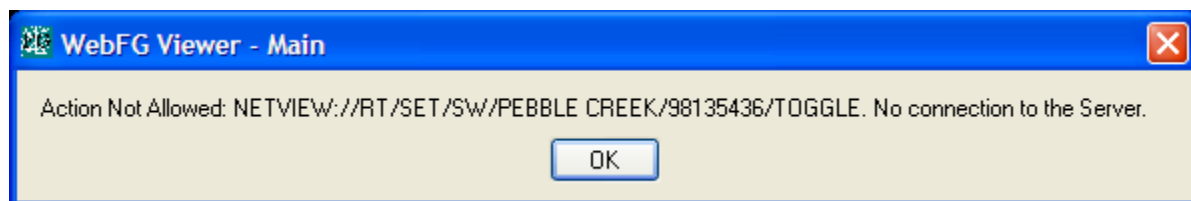


Figure 45. Sample Message Dialog Box

User Action	System Response
Check to see if NetSrv is running.	If not, start NetSrv by providing the appropriate configuration file on the target machine.
Make sure that "Netconf.xml" has the parameter <Port> set to the same number as the parameter SERVER_PORT in NetsrvConfig.txt.	

Note

The default port number on the client should be 5025. This should be exactly the same for the parameter `SERVER_PORT` in `NetServerMainConfig.txt`.

If the client is still not connected to the server, check for the "Online" flag in the `NetConf.xml` file, which must be set to True.

Another way to troubleshoot communication between the client and the DMS server is to check that firewall rules have been implemented by trying to telnet from client to server on the 5025 port.

13.5.4. ADMS Displays, Bitmaps, or Pictures Not Found

Symptom: Displays, bitmaps, or pictures do not show up.

User Action	System Response
Open " <code>Netconf.xml</code> " and make sure that the paths for model files, pictures, and so on are correct.	Close the Browser viewer and relaunch.

13.5.5. Cannot Uninstall (Missing MSI)

Symptom: The Windows uninstall process does not complete because it cannot find the MSI kit/information needed.

The uninstall process fails, but it might not provide a helpful error message. Try using the repair feature from Add/Remove Programs, which should also fail, but hopefully provides a "browse" window showing the location where it is trying to find the MSI. If the original MSI is no longer there and cannot be found, use a comparable kit to do the repair. This should update the Global Unique Identifier (GUID) information, and subsequently allow for the uninstall process to function.

Note

A "comparable kit" means that, if you are trying to uninstall kit 3.8.x, do not try a 3.8.y kit. Make sure that the version is the same even if the build date is not.

Another "trick" is to copy the original or a similar MSI into the `C:\Windows\Installer` folder, and then rerun the uninstall process.

13.5.6. MSI Fails to Install (Needs Windows MSI Hotfix)

Symptom: Certain MSI-based installation kits can fail due to a problem with the Microsoft installer. This is most noticeable on new installations of Microsoft Windows7 that have missed a particular update. The error message is not very clear: "Number: -214682823 Description: Invalid procedure call or argument".

Run Microsoft Windows Update, and click the link to check online for updates from Microsoft Update. Not all WSUS configurations include this particular update.

13.5.7. Install Fails via WelcomeSetupUI

Symptom: On a few computers, the installation kits experience problems when run via the WelcomeSetupUI.

The solution is to run the installation kits directly. Be sure to read [MSI Fails to Install \(Needs Windows MSI Hotfix\)](#) as well.

13.5.8. Error in ViewControl/OpenPluginControl

Symptom: When starting up the client shortcut for Browser plus ADMS, the following error message is displayed:

```
Error in ViewControl.OpenPluginControl: Could not load file or assembly "NetDisplay, Version=1.0.3735.38035, Culture=neutral, PublicKeyToken=null" or one of its dependencies. An attempt was made to load a program with an incorrect format.
```

Most likely, the problem is that there is a 32bit vs. 64bit mismatch between the Browser and the ADMS installation kits. Be sure that the 64bit versions of both kits were used.

13.5.9. Study Mode Initialization Fails

Symptom: After study mode initialization, power flow fails to solve. The client reports missing DAT files.

The User Access Control (UAC) permissions in Windows7 require that the user running the application have write permissions to the target folder. In this case, the target folder is the "CRT" folder under dynamics.

There are several possibilities for resolving this issue:

- Restart the application with elevated permissions
- Grant additional permissions to the folder
- It is recommended that the software be installed on the D drive (or another drive other than C:), since Windows has more stringent security measures on the Cdrive.

13.5.10. Target of an Invocation

Symptom: When starting up the client or when calling up a DMS display for the first time, an error message is shown that states, "Exception has been thrown by the target of an invocation".

This might occur if the `Netconf.xml` file cannot be found by the client software. Check the shortcut to see which file is being passed. Try passing the file name (as written in the shortcut) to Notepad with the RUN command to verify that the file exists and is specified correctly.

If that piece is correct, there could also be a problem where the folders specified within the `Netconf.xml` file are not available (for example, if the client cannot find the Models folder specified in `Netconf.xml`).

13.5.11. Scroll-Zoom Not Working with My Mouse

Symptom: If the PC has a custom third-party mouse driver, there is a chance that scroll-zoom might not work (this was seen with a Lenovo mouse). Normally, the scroll wheel can be used to zoom the geographic display in and out.

To resolve this, use the custom mouse control software delivered from the vendor, and set the mouse wheel to "Office97 scrolling emulation only".

For other specialty mice, the solution may vary slightly.

13.5.12. Cannot Start DMSMAIN from TVIEW

Symptom: Tried to start "dmsmain idms ems" from TVIEW. Although the task appears in the task manager, the clients are not connecting and the PROCMAN Master display is not showing the task as running.

Resolution: Aside from `CfgProxy` and `NetPermSrv`, the ADMS programs are not native Habitat applications; therefore, they do not have a `PROCMANAPI`. In `PROCMAN` terms, these are passive tasks, which means that `PROCMAN` must be used to start the task for `PROCMAN` to know about it and to control it properly. Therefore, do not try to use TVIEW to start tasks such as "DMSMAIN".

13.5.13. Simulator ISD Connection is not formed with SCADA

Symptom: After starting the NetServer simulator, an ISD connection is not formed with SCADA, measurements are incorrect, the site communication status shows "failed", and the `proxysrv` log shows error `PSX821`.

Resolution:

1. In `proxysrv netiopxytcp.h`, change:

```
#define MAX_BACKLOG 2000
```

to:

```
#define MAX_BACKLOG 4000
```

2. Rebuild `proxysrv.lib` and `proxysrv.exe`.

13.5.14. CfgProxy Aborts/Restarts and ADMS Tasks Do Not Take the Enabled Role

Symptom: After starting the ADMS tasks, none of the tasks become enabled. Entering the "tview list" command shows that all CfgProxy tasks are restarting. Looking at the corresponding .OUT file in the LOGD folder shows the CClLogger error below:

```
0000000003745B29 0000000000128920 sal_1_1::LoggerFactory::GetRootLogger + 3E79
0000000003741C78 0000000000128970 sal_1_1::LoggerFactory::CreateLogger + 38
00000000033665B8 0000000000128B00 Eterra::Cls::CClLogger::GetLogger + 258
(e:\cls_main_teambuild\cls\cls_21_spr_build\sources\clspublic\clscore\cclogger.cpp:56)
0000000003389A16 0000000000129050 ConvertUTF8toUTF16 + 86
(e:\cls_main_teambuild\cls\cls_21_spr_build\sources\clspublic\clscore\convertutf.cpp:617)
000000000337E4A5 0000000000129190 Eterra::Cls::Utility::FromUtf8 + 145
(e:\cls_main_teambuild\cls\cls_21_spr_build\sources\clspublic\clscore\clscommon.cpp:288)
00000000035B4CEF 0000000000129640 cSetApplicationId + AF
(e:\cls_main_teambuild\cls\cls_21_spr_build\sources\clspublic\clsappender\clsappender_c.cpp:79)
000000000671204 000000000012AA60 MlfLogDllCfgapiCfga01 + 204
(D:\eterra\habdata18\tmp\src\mlf_dll_cfgapi.c:139)
00000000067BB5B05 000000000012C8D0 MlfLogCfgapiCfga01 + 125
(D:\eterra\habsrc18\habitat\rcse\cfgapi\workspace\cfgapi_lib\mlf_cfgapi.c:196)
00000000067BB4199 000000000012C950 CfgmanCProcess::SendRegistrationMessage + A9
(D:\eterra\habsrc18\habitat\rcse\cfgapi\cfgapi_process.cxx:627)
```

Resolution: Check the PATH environment variable and make sure that the Browser path is not listed before the Habitat path. In the command prompt, PATH sometimes truncates values if they are too long; therefore, it is better to look at the path for the user as well as the system using the Windows user interface.

13.5.15. Annotations Are Not Getting Copied to the Study Server

Symptom: After adding an annotation in the real-time server and clicking "Initialize to Real-Time" from the Study Case Setup pop-up, annotations are not appearing in the study mode of the client.

Resolution: Make sure that the real-time server configuration file parameter FD_DNAF_INPUTS_FILES has the correct path set. For example, to copy Annotation_KeyCustomer.sav, the path should be set to \$DYNAMIC_PATH\$Annotation_KeyCustomer.sav.

13.5.16. Study Environment Cannot be Initialized

Symptom: Everything appears to be working except that when clicking the initialization button to bring up the study pop-up, a pop-up appears on the client with the title "Error Initializing Study Case Setup" and message "Remote Study servers not supported. Please try using Local Study

server".

The error pop-up indicates it is trying to connect to a remote StudyServer, but the Browser session and the StudyServer log show that it is forming the connection with itself.

Resolution 1: The host file has an incorrect IP address for its own machine name, for example:

```
Client1 172.1.1.168
```

Should be:

```
Client1 10.1.1.168
```

Resolution 2: Ensure that your workstation is correctly listed within the etc\hosts file.

For example, the etc\hosts originally had a working entry defined as:

```
10.10.33.27 Workstation27
```

But later the subnet was changed to:

```
10.10.55.x
```

The host checking done by the client would block the study case setup and present the aforementioned message. The incorrect entry must either be corrected or removed. This is true even if you are pointing the ST dynamic to "localhost" or "127.0.0.1".

13.5.17. Font Related Client Crash

Symptom: Placing cuts or jumpers crashes Browser, but other ADMS functionality seems unaffected.

Resolution: Review the managed Browser crash logs (for example, WebFGViewer_.managed.log). If a line similar to the following appears, review the fonts installed on the server.

```
InnerException = System.ArgumentException: Font 'MS Reference Sans Serif' does not support style 'Bold'
```

You may need to locate the missing font and install it on the affected machine(s).

13.5.18. %HABITAT_LOGDIR% Folder Files Contain Zero Bytes While Server Is Running

Symptom: Some or all the files with server output in the %HABITAT_LOGDIR% folder contain zero bytes while the server is running.

Resolution: A disconnect may occur between Windows and the output files generated by PROCMAN-initiated tasks. It might appear in the %HABITAT_LOGDIR% folder that some or all the files have zero bytes. However, the files have content, are usable, and can be opened for viewing.

Opening the files using one of the methods below causes the system to checkpoint or recognize the file contents. After opening, the file sizes are properly displayed.

- Open the files with Notepad.
- Type "tail -f" to view the content in the command prompt.

13.5.19. Temporary Objects Not Auto Corrected on Deleted Substations

Symptom: Temporary objects (jumpers, cuts, temporary grounds) are not auto corrected on a substation that is combined with another station.

Resolution: If an entire substation is being removed (such as when its network components are being combined with another station), the auto correction feature must be "triggered" for the deleted station so that the temporary objects that exist there can be associated with the new substation model. Simply removing the model file will not accomplish this because the system has no way of distinguishing whether the model was removed prior to a replacement, or if the model is being permanently removed.

To trigger the auto correction on the deleted station, a "pseudo" station of the same name but having only a single line, must be created. When the new "combined" station model is uploaded, this pseudo station must also be uploaded. This will trigger the auto correction for all temporary objects in the pseudo/deleted station. Once all the temporary objects have been corrected for the new combined station, the pseudo station can be removed.

The general steps are as follows:

- Create a pseudo station with the same name as the station to be deleted, using only a single line.
- Upload the pseudo station along with other stations that may be changing.
- Correct or delete any orphaned entries for the pseudo station.
- Delete the pseudo station.

13.5.20. Foreign Key Constraint Violation in Archive Database

To guarantee referential integrity, the Archive database contains many foreign key constraints. For example, a switching step cannot reference a switching order that does not exist. In some circumstances, including a NetServer restart, NetServer may write data to the OMS archive journal out of order. For example, a switching step may be written to the archive journal before the switching order that it belongs to is written to the archive journal. In this case, the Archive

application will attempt to store the switching step to the Archive database before it stores its parent switching order, leading to a foreign key constraint violation, which causes the Archive application to shut down.

A temporary workaround is to disable the constraint, with a SQL statement like the following:

```
ALTER TABLE SWITCHING_STEP DISABLE CONSTRAINT FK_SWITCHING_STEP_ORDER;
```

The orphan switching steps problem may correct itself, since the switching order will be written to the OMS archive journal (and hence to the Archive database) when the switching order is completed. Otherwise, you may need to manually insert the missing record before re-enabling the constraint.

13.5.21. Manually Transferring Dynamic Files

If you need to manually transfer the dynamic files from the enabled server to a standby server, follow the procedures described in the table below.

Table 13. Data Type and Procedures

Data Type	Procedure
OMS dynamics, including crew assignments and temporary crews	Copy the latest regular snapshots, crew snapshots, and journal files (Call journals, Incident journals, Call server journals, and Relay server journals) from the enabled server to the standby server. Also copy the dynamics.sav and preservedstates.sti files. The default file locations are: • %IDMS_ROOT%\dynamics\server\OMSServer • %IDMS_ROOT%\dynamics\server\OMSServer\Snapshot • %IDMS_ROOT%\dynamics\server\OMSServer\CallJournal • %IDMS_ROOT%\dynamics\server\OMSServer\CrewJournal • %IDMS_ROOT%\dynamics\server\OMSServer\IncidentJournal • %IDMS_ROOT%\dynamics\server\RelayServer
OMS history documents	Copy the archive journals from the enabled server to the standby server. The default file location is: • %IDMS_ROOT%\dynamics\server\OMSServer\ArchiveJournal
Switching Order documents	Copy all Switching Order documents from the enabled server to the standby server. The default location of the Switching Order documents is: • %IDMS_ROOT%\dynamics\server\SwitchOrderServer
Dynamic schedules	Copy all dynamic schedules from the enabled server to the standby server. The default location is: • %IDMS_ROOT%\dynamics\server\DynSchedules

Data Type	Procedure
DMS dynamics	Copy the <code>dynamics.sav</code> and the <code>preservedstates.sti</code> files from the enabled server to the new server. The default location is: • <code>%IDMS_ROOT%\dynamics\server</code>
Annotations	Copy all annotation dynamics from the enabled server to the standby server. The default location of the annotation dynamics is: • <code>%IDMS_ROOT%\dynamics\server</code>
Study savecases	1. Copy all study savecases from the enabled server to the standby server. 2. Retrieve the savecases on the standby server.
Fast dynamics	Copy the <code>manual.sav</code> file from the enabled server to the standby server. The default location is: • <code>%IDMS_ROOT%\dynamics\server</code>
DNAF dynamics	Copy all DNAF dynamics from the enabled server to the standby server. The default location of the DNAF dynamics is: • <code>%IDMS_ROOT%\dynamics\server\DNAF</code>

13.5.22. Simulate-Only Measurements

Symptom: Although a point measurement quality in SCADA and SIM is shown as "Good", the same point measurement quality in RT is shown as "Suspect".

Solution: Set the "Simulate Only" flag for these measurements to "False" in the Substation Editor and rebuild the station model.

13.5.23. Synchronizing Client and Server Time

Symptom: OMS issues appear on the client (for example, incident details on OMS displays are not updating). Performance of the client significantly decreases when used with the Outage Management System.

Solution: Synchronize the system time on both the client and the server. Alternatively, set the `UsePreComputeTimes` parameter to `False` in the `netconf.xml` client configuration file.

13.5.24. The Properties Tab in the Update Incident Display is Empty

Symptom: On the Update Incident display, the Properties tab is empty while the other tabs are populated correctly.

Solution: Set the Browser client or APF shortcut's Start-In value to the ADMS client's bin directory, such as D:\Eterra\Distribution\Client\bin\.

13.5.25. Permission Changes are Not Applied

Symptom: After applying permission changes, the PERMIT model is not updated for the current session. The changes are applied only after restarting NetPermSrv.

Solution: On the machine where NETPERMSERVER is running, add the NETPERMSRV_FORCEFULLPERMUPDATE environment variable, set its value to 1, and then restart NetPermSrv.

13.5.26. Cannot Uninstall ADMS Server

Symptom: When trying to uninstall ADMS server, the error message "The specified account already exists" appears and the uninstallation process rolls back.

Solution: Run the ADMS server installer and click Repair. After ADMS server is repaired, it can be successfully uninstalled.

13.5.27. Client Exception Occurs when Initializing Client UI Components

Symptom: A client exception occurs when initializing client UI components (error in the InitializeComponent function) after upgrading to a newer version of ADMS. For example, an exception occurs when navigating from a grid tabular display to a geographic view.

Solution: Update the client's localized resource files for the new version of resources.

1. Clear the contents of the folder with localized resource files under Client\bin (for example, Client\bin\fr-FR).

Warning

Do not remove the DefaultResources folder.

2. Using the Resource Editor, create localized resources for the newer version of the ADMS client. For more information, refer to the *Resource Editor User Guide*.
3. Copy the created resources to the folder with localized resource files under Client\bin.

13.5.28. Interface Server Cannot Run

Problem: An interface server (for example, Net Data server or OMS Interface server) cannot start

on the assigned port and may be crashing.

Solution: Check and update the system HTTP URL namespace reservations.

Important

You must not have two different namespace reservations on the same port. For example, if you already have an HTTP reservation, you must delete this reservation to make an HTTPS reservation.

To check and update the system HTTP URL namespace reservations:

1. Run Windows PowerShell as Administrator.
2. Run the "netsh" command.
3. Run the "http" command.
4. Run one of the following commands to check the current URL assignments in the system:

```
netsh http show urlacl
netsh http show urlacl | findstr /I 880
```

5. Run the following command to remove an unwanted URL assignment:

```
netsh http delete urlacl URL=<URL>
```

For example:

```
netsh http delete urlacl URL=http://+:8800/
```

6. Run the following command to add a new URL assignment:

```
netsh http add urlacl URL=<URL>
```

For example:

```
netsh http add urlacl URL=https://+:8800/ user="Network Service"
```

13.5.29. Toolbar Changes are not Applied

Problem: Toolbar changes are not applied in the ADMS client.

Solution: Click the Purge Browser Cache button on the Browser toolbar. If the problem persists, clear the internet cache in the Internet Explorer and restart the client.

13.5.30. OMS Server Disconnects from the DMS Server and Cannot Reconnect

Problem: During high load conditions with a big number of queued messages, the OMS server disconnects from the DMS server and cannot reconnect. The following error message appears in the `omsserver_main.log` file: "DYNC_RW: Disconnected from Main", and "*** Too many queued send, disconnecting, queued:" messages.

Solution: Increase the maximum number of queued messages that can be processed by the DMS server using the `SOCKET_SEND_MAX_BACKUP` parameter. For example:

1. Add the `SOCKET_SEND_MAX_BACKUP = 30000` parameter in `NetServerMainConfig.txt` configuration file.
2. Restart the DMS server.

13.5.31. Cannot Update Permission Modes in Permission Management Plug-in

Problem: Permission mode changes done in the Permission Management plug-In are not applied.

Reason: For the default Fantasy Island permit model, the Hierarchical Mode functionality is not compatible with Enterprise OMS organizations membership switching functionality, developed as part of the Enterprise OMS organizations functionality in the ADMS 3.11 release. In the default Fantasy Island permit model, the Hierarchical Mode functionality is disabled.

Solution 1: Modify the Permit model to support both Hierarchical Mode functionality and Enterprise OMS organizations functionality. For more information, contact your GE project team.

Solution 2: Enable the Hierarchical Mode functionality and restart PERMSRV.

To enable the Hierarchical Mode functionality:

1. From the Browser menu, navigate to HABITAT Applications> Permit> Mode Management.
2. The Analyst Displays option appears on the Browser menu.
3. From the Browser menu, navigate to HABITAT Applications> Analyst Displays > Server Parameters.
4. Enable the Enabled option in the Hierarchical Mode group.

Permit Server Parameters

Changes made on this display require a restart of the Permit Server

Alarm

Alarm Location: LOCAL

Alarm Area: SYSTEM

Critical Area Category: COVERAGE

Critical Area Exceptions: COVERAGE

Log On Category: LOGIN

Log On Exception: LOGIN

Mode Management Category: MODE

Mode Management Exception: MODEMGMT

Guest Account

Guest WorkStation: GUEST

Options

Operational Mode: ☒ Dynamic ☐ Static

Hierarchical Mode

Enabled: ☒

Don't Use Default HMODE ☐

User Account Propagation

Timestamp:

Check interval: Seconds

Environment:

Target:

13.5.32. Grid Tabular Displays Are Not Working as Expected

Problem: DMS tabular summaries are not providing expected information or they work slowly.

Solution: Due to server modularization activities implemented in **e-terrardistribution 3.7**, all DMS-based queries are routed to the DNAF server first. If the DNAF server is down, the queries are then routed to the DMS server. This affects some of the tabular summaries, such as Abnormal Summary, Feeder Exception Summary, and Find Results. When a slowness issue appears, this may be because the DNAF server is running slowly.

To identify DNAF slowness:

1. Open the DNAF Debug display (called by NETVIEW://RT/POPUP/DNAFDEBUG).
2. Check whether there is a backlog in DNAF query threads.

For DMS tabular displays that do not contain any DNAF data, execute the following steps to always redirect the related queries to the DMS server:

1. Open the `netconf_localhost.xml` file in `%IDMS_ROOT%\config\client\` for editing.
2. Navigate to the `TabularQueriesFromDms` parameter. By default:

```
TabularQueriesFromDms>System.Collections.Generic.HashSet`1+Enumerator[System.String]</TabularQueriesFromDms>
```

3. Update the parameter value to a comma-separated list of grid tabular displays that should be resolved on the DMS server side. For example, to apply the configuration for `Abnormal_GridDisplay.xml` and `AnnoSummary_GridDisplay.xml`:


```
<TabularQueriesFromDms>Abnormal,AnnoSummary</TabularQueriesFromDms>
```

Note

You must use the internal tabular name (typically, remove "_GridDisplay.xml" from the file name) when you specify a tabular name in the list.

13.5.33. Cannot Acquire the Incident Lock and Update an Incident

Problem: The Update Incident display is inactive or read-only. The user cannot acquire the incident lock and update the incident.

Reason: The Permission Manager Proxy server (netpermsrv.exe) connection is down or the logged-in user does not have AORs for modifying the incident.

Troubleshooting steps:

- Make sure that this is a permission issue (to check whether the issue is resolved when permission checking is disabled, set PERMIT_CHECK_ENABLED =0. After checking, set PERMIT_CHECK_ENABLED =1).
- Make sure that the OUTAGE AOR is modeled in the Permit database and the user has read and write access.
- Make sure that the user has the read and write access to the AOR that is associated with the incident (for example, use the Analyst Query display to check the AOR of the incident).
- Make sure that the port used by the ADMS server to communicate with the EMS/SCADA server where the Permit runs is open.
- Make sure that the NETPERMSRV_FULLFORCEPERMUPDATE environment variable is set to 1.

13.5.34. DMS Server Does Not Attempt to Connect to ISD/Tagging

Problem: Tagging is started but the DMS Server does not attempt to connect.

Solution: Perform the following steps:

1. Initiate an ISD/Tagging connection manually using the following command:

```
netview://rt/serverdebug/onlinepath
```

2. Check the NetSrv_main.log file for an ISD/Tagging connection status. For example:

```
"*** CIsdThread processed a CMsgForceIsdPathOnline at ...."
```

13.5.35. Vehicle Location DLL Blocked

Symptom: Vehicle location updates fail to reach the Crew Server.

Problem: A custom DLL used to convert latitude and longitude into X and Y coordinates is blocked by Windows. Windows can automatically block DLLs downloaded from external sources, preventing applications from using them.

Resolution: Perform the following steps:

- Locate the DLL (or ZIP archive if downloaded).
- Right-click the file and select Properties.
- On the General tab, click Unblock (or check the Unblock checkbox).
- Click Apply and then OK.
- Vehicle location updates can be retried after unblocking the DLL.

All assemblies downloaded from external sources should be unblocked before use.

14. Configuring and Using mTLS (mutual Transport Layer Security)

14.1. How mTLS works?

Mutual transport layer Security mandates both the client and server authenticate each other with digital certificates. Like other secure communications mTLS also uses private and public keys. The below steps give high level overview of mTLS process.

- First client connects to the server.
- Server sends its transport layer security certificate.
- Client verifies the certificate sent by server.
- Next client sends its transport layer security certificate.
- Server verifies the certificate sent by client.
- Server grants permission.
- Client and server exchange information over encrypted connection.

14.2. Configuration file NetServerMtls.xml

This file is in the same directory where the `NetServerMainConfig` is located. The below three flags should be enabled for mTLS (Mutual Transport Layer Security).

```
<conf:sslSettings
  clientAllowClientAuth="true"
  clientSendClientAuth="true"
  serverRequireClientAuth="true" >
```

14.3. Adding Certificates for mTLS

To generate and configure certificates on Windows, perform the following steps as a user with administrator privileges:

1. Verify that the Microsoft certificate tools (`certutil.exe` and `certreq.exe`) exist in the `C:\Windows\System32` directory.
2. Start a Habitat command prompt as an administrator and navigate to `C:\Eterra\habitat<NN>\habitat\cfg\SSL\bin`.

Where `<NN>` is the group under which Habitat is installed.

3. Generate a certificate by executing the following command.

powershell ./certtool.ps1

This generates a CA signed certificate for which an Active Directory server needs to be configured and respective permissions need to be set.

To generate a self-signed certificate that is used for testing purposes only (as it is not recommended due to its lower level of security compliance), enter:

powershell ./certtool.ps1 -selfsigned

Once the certificate is generated, it is placed in the local machine's personal certificate store.

14.3.1. Generate Certificate

1. Check if the certificate (.cer) is generated at C:\Eterra\habitat<NN>\habitat\cfg\SSL\bin
2. Open Microsoft Management Console as an administrator (type mmc in the Habitat command prompt).
3. On the File menu, click Add/Remove Snap-in.
4. Add the Certificates snap-in for a local Computer account and click Ok.
5. Verify that a certificate is added to Certificates > Personal > Certificates.
6. Verify that a certificate is added to Certificates > Trusted Root Certification Authorities > Certificates.

Important commands

```
Certmgr.msc
```

This command will navigate to user certificates store.



```
Certlm.msc
```

This command will navigate to local computer certificates store



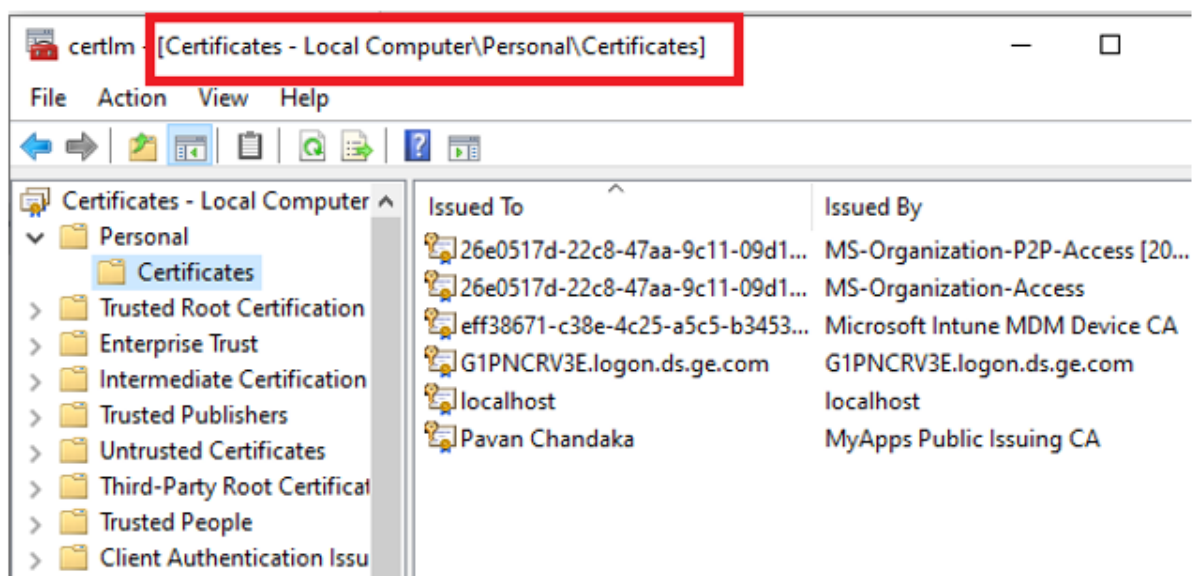
14.3.2. Read Access to the Private Key of the Generated Certificate for the User

The following steps explain how to provide access to the private key:

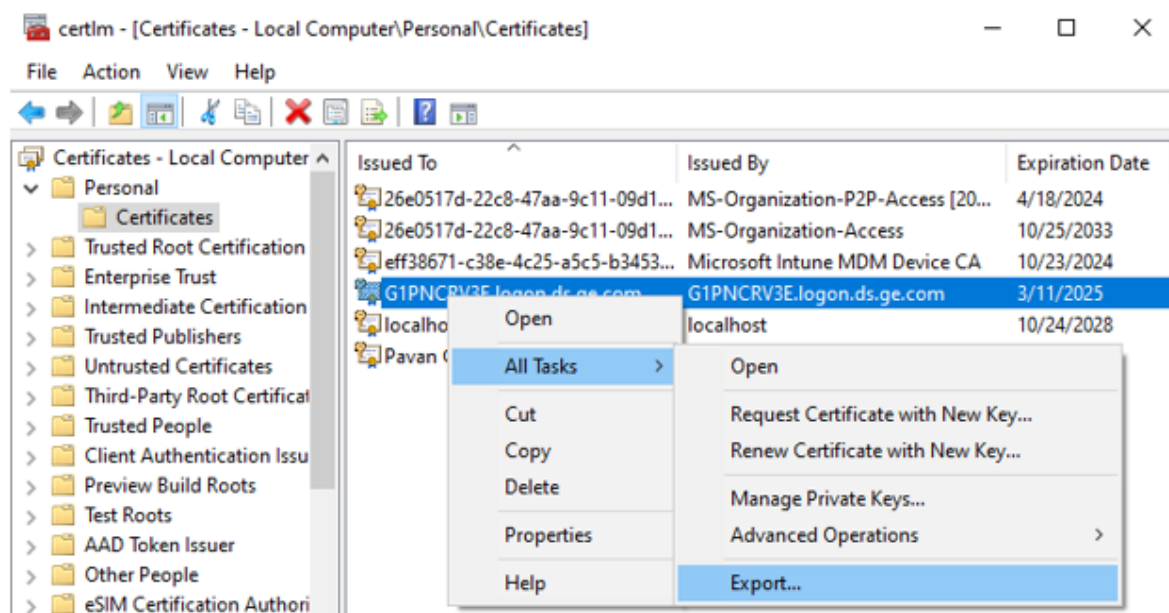
1. Open Microsoft Management Console as administrator (type mmc in the Habitat command prompt).
2. On the File menu, click Add/Remove Snap-in.
3. Add the Certificates snap-in for a local Computer account and click Ok.
4. Right-click the generated certificate and select All Tasks > Manage Private Keys.
5. Add the current user with "Read" permissions and apply the changes.

14.3.3. Export the certificate

1. Run certlm.msc and go to the certificate that needs to be exported.



2. Right-click the certificate and click All Tasks > Export.

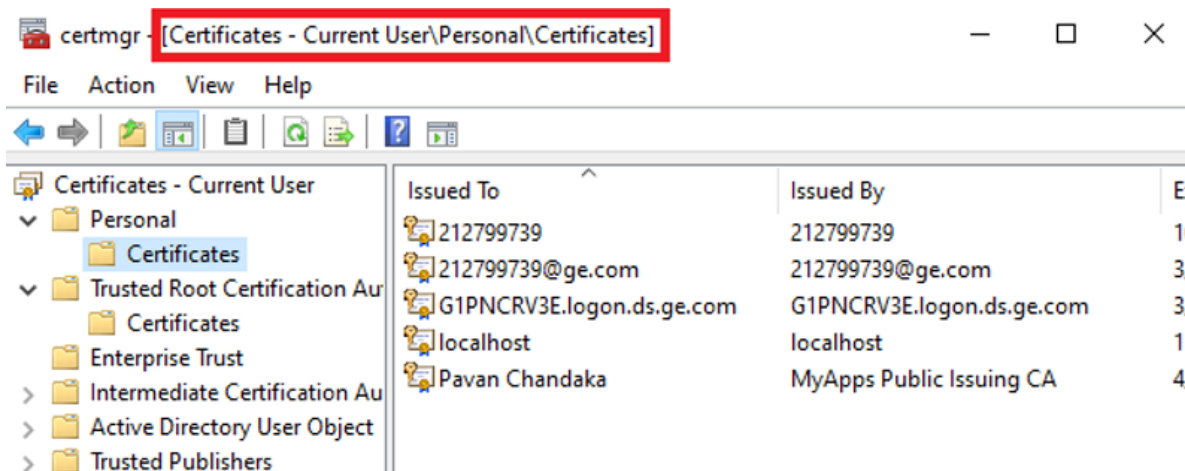


3. Select the format to export and click Next.
4. Browse to the location to store the file, click Next, and then click Finish.

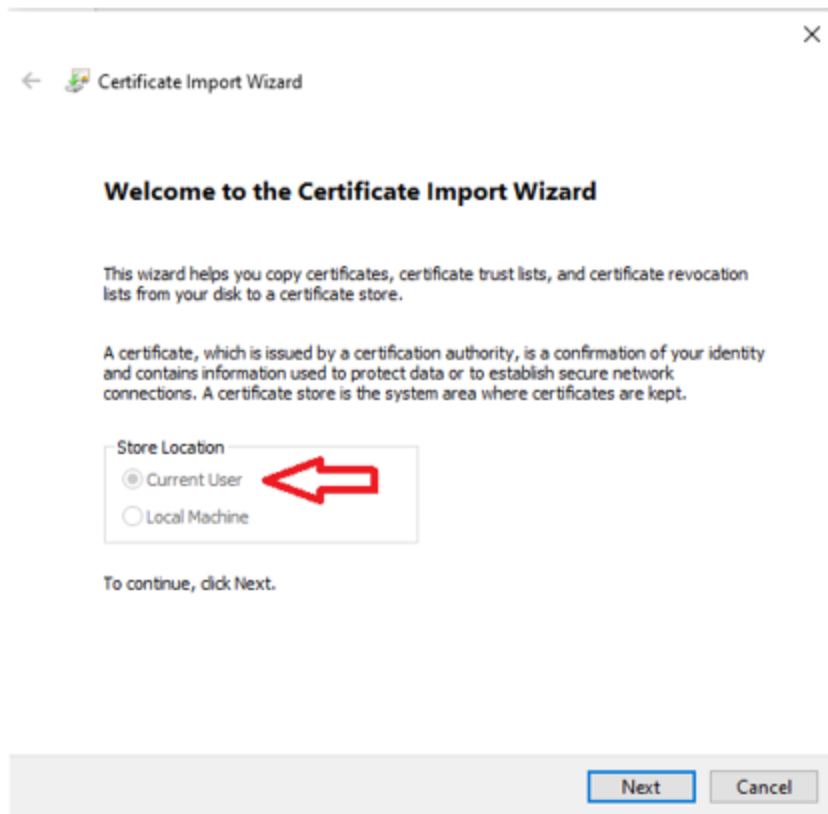
In mTLS, for the client-side authentication, the application looks for the certificate in user store, not in the local computer store.

14.3.4. Import Certificate in User Store

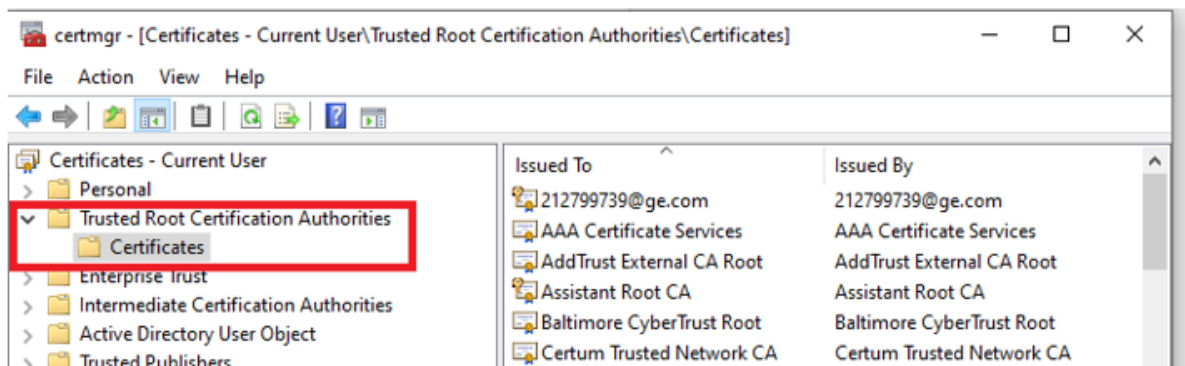
1. Run `certmgr.msc` and go to the certificates (Personal).



2. Right-click the certificate and click All Tasks > Import. In the wizard select Current User. Select next. A window to browse the certificate should appear.



3. Browse to the location of the file, select the certificate that was exported in earlier steps.
4. Click Next, and then click Finish. After the import completes, copy the imported certificate, and place it in Trusted Root certification Authorities.



14.4. Required Application Specific Configuration Changes

Details about the mentioned below configuration variables explained in the [NetServerMainConfig.txt](#) parameters.

NetServerMainConfig	ProxySrv (SCADA) Set remote host. It should be the fully qualified name. ISD_REMOTE_HOST1 = G1PNXXXXXX.logon.ds.ge.com Enable MTLS flag. ISD_REMOTE_USE_MTLS = 1 Permit Server Set remote host. It should be the fully qualified name. PERMIT_SERVER_IP = G1PNXXXXXX.logon.ds.ge.com Enable MTLS flag. PERMIT_SERVER_USE_MTLS = 1 ProxySrv (Alarm) Set remote host. It should be the fully qualified name. ALARM_HOST1 = G1PNXXXXXX.logon.ds.ge.com Enable MTLS flag. ALARM_SERVER_USE_MTLS = 1 CfgProxy Set remote host. It should be the fully qualified name. REDUNDANT_HOST = G1PNXXXXXX.logon.ds.ge.com CFGPROXY_HOST = G1PNXXXXXX.logon.ds.ge.com Enable MTLS flag. CFGPROXY_USE_MTLS = 1
NetServerDnafConfig	Set SITEOV_DISABLE_HOSTNAME_QUERY_FROM_IP = 1 to use fully qualified name, instead of IP address in the application. ProxySrv (Alarm) Set remote host. It should be the fully qualified name. ALARM_HOST1 = G1PNXXXXXX.logon.ds.ge.com Enable MTLS flag. ALARM_SERVER_USE_MTLS = 1 CfgProxy Set remote host. It should be the fully qualified name. REDUNDANT_HOST = G1PNXXXXXX.logon.ds.ge.com CFGPROXY_HOST = G1PNXXXXXX.logon.ds.ge.com Enable MTLS flag. CFGPROXY_USE_MTLS = 1
NetServerSimConfig	Set SITEOV_DISABLE_HOSTNAME_QUERY_FROM_IP = 1 to use fully qualified name, instead of IP address in the application. ProxySrv (SCADA) Set remote host. It should be the fully qualified name. ISD_REMOTE_HOST1 = G1PNXXXXXX.logon.ds.ge.com Enable MTLS flag. ISD_REMOTE_USE_MTLS = 1

15. TCP/IP Default Port List

This chapter lists the default ports used by the ADMS software.

These ports are listed in their respective sections of this document. If you want ports other than the defaults for a particular implementation, refer to those respective sections for instructions that describe all the places that need updating.

Note

It is the responsibility of the utility to whitelist all incoming connections to the listening TCP/IP ports by ADMS applications.

Important

Applications and endpoints need to be configured to use unique port values.

DMS-Only Configuration

- ADMS server machine
 - DMS Main server application (NetSrv): 5025
 - DNAF server application (DnafSrv): 5026
 - CfgProxy
 - DnafSrv: 5017
 - NetSrv: 5019
 - Net Data Replicator: 5020
 - DnafSrv Follower: 6020
 - Switching Order server: 5044
 - Net Data Replicator (Local): 8200
 - Net Data Replicator (Remote): 8201
- ADMS read-only server machine
 - DMS Main server read-only application (RoSrv): 5055
 - Net Data Replicator (Local): 8200
 - Net Data Replicator (Remote): 8201
- ADMS follower server machine
 - DMS Main server application in Follower mode (NetSrv): 6025
 - CfgProxy
 - DnafSrv Follower: 6020
- ADMS simulator server machine

- DMS simulator server (SimSrv): 5029
- SCADA/PERMIT server machine
 - NetPermSrv: 5345
 - Proxysrv: 5001
- ADMS client machine
 - Local Study server application: 5027 (this is listed here for completeness, but since it is on the client, port conflicts are not a big concern)
- ADMS Interface server (for communicating data using web services)
 - DMS Interface server: 8801
- Tools
 - Model Manager Service: 8085
 - Model Manager Web Service: 8090

DMS+OMS Configuration

- DMS server machine
 - DMS Main server application (NetSrv): 5025
 - Dnaf server application (DnafSrv): 5026
 - CfgProxy
 - CallServer: 5015
 - CrewServer: 5016
 - DnafSrv: 5017
 - Net Data Server: 5018
 - NetSrv: 5019
 - NetServer: 5021
 - Net Data Replicator: 5020
 - Switching Order server: 5044
- OMS server machine
 - OMS server application (NetServer): 5060
 - OMS server application managed connections (NetServer): 5050
 - Net Data server: 6096
 - Net Data server web service: 8800
- ADMS Interface server (for communicating data using web services)
 - Call server: 5032
 - Crew server: 5036

- Relay server: 5033
- Net Data server: 6096
- Net Data server web service: 8800
- DMS Interface server: 8801
- Crew server web service: 8802
- OMS Interface server - without UAA: 8802
- OMS Interface server - with UAA: 8812
- Call server, Relay server, AMI server, Crew server and Mobile Switching Order web services: 8443
- OMS Adapter: 8803
- ANPS Adapter and CE Adapter: 8443
- VOP Adapter: 8447
- NetViewer Adapter: 9991
- ADMS Interface server web service plugins:
 - Call server web service: 8443
 - Call server Call interface web service client: 8443
 - Call server AMI interface web service client: 8443
 - Crew server web service: 8443
 - Crew Modeling web service: 8443
 - Crew Simulator Crew web service: 8445
 - Crew Simulator Switching Order web service: 8445
 - Crew server WMS web client: 8445
 - Crew server Modeling web client: 8445
 - Customer Experience Adapter web service client: 8447
 - OMS Interface server web service - without UAA: 18802
 - OMS Interface server web service - with UAA: 18812
 - Netviewer Adapter: 24468
 - Structure Server: 17642
- ADMS read-only server machine
 - DMS Main server read-only application (NetServerRO): 5055
 - OMS server port on NetServerRO: 5099
 - OMS server application managed connections (NetServerRO): 5051
 - Net Data Replicator (Local): 8200

- Net Data Replicator (Remote): 8201
- Net Data server: 6096
- Net Data server web service: 9900
- Switching Order server: 5044
- ADMS follower server machine
 - DMS Main server application in Follower mode (NetSrv): 6025
 - OMS server application in Follower mode (NetServer): 6060
- ADMS simulator server machine
 - DMS+OMS simulator server (NetServerSim): 5029
- SCADA/PERMIT server machine
 - NetPermSrv: 5345
 - Proxysrv: 5001
- ADMS client machine
 - Local Study server application: 5027

This application is listed here for completeness. Since it is on the client, port conflicts typically can be skipped.
- Tools
 - Model Manager Service: 8085
 - Model Manager Web Service: 8090

To find the related sections within this document, search using the port numbers.

15.1. Default Port Configuration

Table 14. TCP/IP Default Port List

	List ning	NETS RV	NetS erver	Dnaf Srv	Simul ator	Study	CallS erver	Crew Serve r	NetD ataSe rver	NetD ataR eplic ator	RELAY *	Proxy Srv	CfgPr oxy
NETSRV	5025											5001	5019
NetServer	5060						5032	5036				5001	5021
DNAFSRV	5026												5017
Simulator	5029										5033		
NetClient (RT)	NA	5025	5060	5026									
NetClient (SIM)	NA				5029								
NetClient (ST)	NA					5027							

	Liste ning	NETS RV	NetS erver	Dnaf Srv	Simul ator	Study	CallS erver	Crew Serve r	NetD ataSe rver	NetD ataR eplic ator	RELAY *	Proxy Srv	CfgPr oxy
NetDataReplicator													5020
NetDataServer													5018
CallServer	5032												5015
CrewServer	5036												5016
CrewSimulator	NA				5029			5036					
RELAY*	5033						5032						
NETPERMSRV	5345												
ProxySrv	5001												
CfgProxy	5014	5019	5021	5017			5015	5016	5018	5020			

15.2. Advanced Port Configuration

Table 15. TCP/IP Advanced Port List

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
DMS					
22	File Updater		File Updater Servers	DMS Servers Read-Only Servers OMS Servers Clients	N/A
80	WebServers IIS	http://<webserver>/<IIS default site>	Web Servers OMS Servers	DMS Servers Read-Only Servers OMS Servers Clients	N/A
443	Remote Desktop SSL		RDS Servers to/from Clients		N/A
445	Standard Network Drive Mapping Port		RDS Servers DMS Servers OMS Servers	DMS Servers OMS Server Support Clients	N/A
	Note: Opening this port between servers is optional.				

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
1521	Oracle database server		Oracle Server	Archive Servers Source Servers Clients	ArchiveAppConfig.xml: DbConnectionString
3389	Remote Desktop Standard (RDS)		DMS Servers Read-Only Servers OMS Servers Modeling Servers Web Query Servers	Support Clients	N/A
5001	ISD / ProxySrv / ALARM		SCADA Servers	DMS Servers OMS Servers	NetServerMainConfig.txt: ISD_REMOTE_PORT , ALARM_PORT NetServerDnafConfig.txt: ALARM_PORT NetServerSimConfig.txt: ISD_REMOTE_PORT
5014	CfgProxy		SCADA Servers	Read-Only Servers Clients	NetServerReadOnlyConfig_localhost.txt NetServerStudyConfig.txt: CFGPROXY_PORT
5017	DNAF Server CfgProxy		Internal		NetServerDnafConfig.txt: CFGPROXY_PORT
5018	Net Data Server CfgProxy		Internal		NetDataServerConfig.txt: CFGPROXY_PORT
5019	DMS Server CfgProxy		Internal		NetServerMainConfig.txt: CFGPROXY_PORT
5020	Net Data Replicator CfgProxy		Internal		NetDataReplicatorConfig.xml: <CfgProxyPort>

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
5025	DMS Server (NetSrv)		DMS Servers	DMS Servers Read-Only Servers OMS Servers Web Query Servers Clients	netconf.xml: <Port> NetServerDnafConfig: REDUNDANT_PORT, MAIN_SERVER_PORT NetServerMainConfig: SERVER_PORT, REDUNDANT_PORT NetServerOmsConfig: MAIN_SERVER_PORT NetServerReadOnlyConfig: MAIN_SERVER_PORT
5026	DNAF Server		DNAF Servers	Read-Only Servers Clients	netconfig_dnaf.xml : <Port> netconf.xml: <DnafPort> netconf_follower.xml: <DnafPort> NetServerDnafConfig.txt: SERVER_PORT NetServerDnaConfig.txt: SERVER_PORT NetServerForecastConfig.txt: DNAF_SERVER_PORT

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
5028	DNA Scheduler Server		DNA Scheduler Servers	Clients	DnaSchedulerConfig.xml: <AppPort> NetServerDnafConfig.txt: DNA_SCHEDULER_PORT NetServerDnaConfig.txt: DNA_SCHEDULER_PORT
5044	Switching Order Server Communication Client		DMS Servers	Clients	SwitchOrderServerConfig.txt: Port
5345	Permit and NetPermSrv		PERMIT Servers	DMS Servers OMS Servers	NetServerOmsConfig_localhost.txt: PERMIT_SERVER_PORT NetServerMainConfig_localhost.txt: PERMIT_SERVER_PORT NetServerDnafCustomerModeConfig_localhost.txt: PERMIT_SERVER_PORT NetServerFollowerConfig_localhost.txt: PERMIT_SERVER_PORT
6025	DMS Follower		DMS Follower Servers	OMS Only Clients	netconf_follower.xml: <Port> NetServerFollowerConfig.txt: SERVER_PORT, REDUNDANT_PORT

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
6032	DMS Proxy Orphaned Tag (Otags Utility) DNOM2HDB Service		SCADA Servers DMS Servers	DMS Servers	NetServerMain Config_localhost.txt: SERVICE_PORT
6060	DMS Follower		OMS Servers	OMS Only Clients	netconf_follower.xml: <OmsPort>
7032	DMS Follower service model		DMS Follower Servers		NetServerFollower Config.txt: SERVICE_PORT
7168	DNA Scheduler Server		DNA Scheduler Servers	Clients	DnaScheduler Config.xml: <RESTfulAppPort>
8200	Net Data Replicator Local		DMS Servers OMS Servers	DMS Servers OMS Servers	NetDataReplicator Config.xml: <LocalListenerAddress>
8201	Net Data Replicator Remote		DMS Servers OMS Servers Read-Only Servers	DMS Servers OMS Servers Read-Only Servers	NetDataReplicator Config.xml: <RemoteListenerAddress>
9222	Web Remote Debugging Port		DMS Servers	Clients	netconf_localhost.xml netconf_oms_localhost.xml netconf_ReadOnly_localhost.xml netconf_follower_localhost.xml: <WebRemoteDebuggingPort>
9400	NIOARC (9400 + Hab Grp # in Hex converted to Decimal) If connected to EMS/SCADA use EMS Group #, else DMS Group #		DMS Servers or EMS Servers		N/A

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
35347*	Datapool Ports calculated based on WebFG defined PORT for various data source. For example, port 5002 has hex value 138A, invert the two bytes to 8A13 and convert back to decimal which is 35347.		Datapool Servers	Clients Support Clients	N/A
Note: DBSEQ datapool firewalls only need to be opened to the Support Clients					
PORT FIREWALL					
5002 35347					
5010 37395					
5011 37651					
5020 39955					
5090 57875					
Interfaces - DMS					
8801	Telemetered Data to External system from DNAF DMS Interface Server DMS Data Service Switching Order Web Service DMS Control Service Annotation Web Service Schedules Web Service Forecast Adapter Web Service	http://localhost:8801/dms/data http://localhost:8801/dms/SWO http://localhost:8801/dms/data/\$metadata http://localhost:8801/dms/services http://localhost:8801/dms/services/annotation http://localhost:8801/dms/services/schedules http://localhost:8801/dms/services/dtl	DMS Servers	External Applications	DmsInterfaceServerConfig.xml:WebServiceBaseURL OmsInterfaceServerConfig.xml:DMSInterfaceServerURL
8025	Notification Publisher	http://localhost:8025/DISEvents/	External Applications	DMS Servers	DmsInterfaceServerConfig.xml:NotificationBaseURL

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
9090	Notification Base URL for trusted applications Switching Order Status Notification	http://localhost:9090/dms/event/ http://localhost:9090/dms/event/SWO/StatusNotification	External Applications External Notifications	DMS Servers Interface Servers	
8180	Mobile Application Notifications	http://mobilityserver:9180/event	External Applications	DMS Servers	DmsInterfaceServerConfig.xml: NotificationBaseURL
OMS					
5015	Call Server CfgProxy		Internal		CallServerConfig.xml: <CfgProxyPort>
5016	Crew Server CfgProxy		Internal		CrewServerConfig.xml: <CfgProxyPort>
5021	OMS Server CfgProxy		Internal		NetServerOmsConfig.xml: CFGPROXY_PORT
5032	Call Server		Call Servers	OMS Servers	NetServerMainCustomerModeConfig_localhost.txt: CALLSERVER_PORT AmiServerConfig.xml: <CallServerPort> CallServerConfig.xml: <ListeningPort> NetServerOmsConfig.txt: CALLSERVER_PORT RelayServerConfig.xml: <CallServerPort>

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
5036	Crew Server		Crew Servers	OMS Servers WFM System Crew Simulator	NetServerMainCustomer ModeConfig_localhost.txt: CREWSERVER_PORT CrewServerConfig.xml: <ListeningPort> CrewSimulatorConfig.xml: <CrewServerPort> NetServerOmsConfig.txt: CREWSERVER_PORT
5050	OMS Server		OMS Servers	Call Servers Crew Servers Clients	netconf.xml: <ManagedConnectionPort> netconfig_follower.xml: <ManagedConnectionPort> CallServerConfig.xml: <OmsServerPort> CrewServerConfig.xml: <OmsServerPort> NetDataReplicatorConfig.xml: <OmsServerPort> NetDataServerConfig.txt: OMS_SERVER_PORT NetServerOmsConfig.txt: OMS_SERVER_PORT SwitchOrderServerConfig.txt: OMS_SERVER_PORT

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
5060	OMS Server		OMS Servers	Clients	netconf.xml: <OmsPort> netconf_oms.xml: <Port> NetServerOmsConfig.txt: SERVER_PORT
6090	MultiSpeak Web Services		Call Servers Relay Servers	IVR and/or AMI	
6091	MultiSpeak Web Services		Call Servers Relay Servers	IVR and/or AMI	
Interfaces - OMS					
8800	NetData Server (Historical Correction) Read-Only Web Service	http://localhost:8800/oms	OMS Servers Read-Only Server	Clients Ext App	NetDataServerConfig_localhost.txt WEBSERVICE_PORT_NUMBER
8802	Crew Server Web Services Base Url	http://localhost:8802/crew	Crew Servers	OMS Servers	CrewServerConfig.xml: WebServiceBaseURL
8802	OMS Interface Server - without UAA	http://localhost:8802/oms/services/	Interface Servers	OMS Servers	OmsInterfaceServerConfig.xml: WebServiceBaseURL
8803	OMS Adapter	http://localhost:8803/OmsAdapter/CrewService	OMS Servers Interface Servers	Call Servers Crew Servers Interface Servers Kubernetes Cluster External Notification	OmsAdapterConfig.xml: CrewServiceURL
8812	OMS Interface Server - with UAA	http://localhost:8812/oms/services/	Interface Servers	OMS Servers	OmsInterfaceServerConfig_UAA.xml: WebServiceBaseURL
8443	ANPS Adapter	http://localhost:8443/api/ext/ANPS/v2/ANPSNotification	Interface Servers	External App	ANPSAdapterConfig.xml: ANPSEndPoint

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
8447	View Outage Portal Adapter	http://localhost:8447/ api/ext/ANPS/v2/ViewOutagePortalNotification	Interface Servers	External App	ViewOutagePortalAdapterConfig.xml: ViewOutagePortalEndPoint
9090	Crew Notification Netviewer Adapter Tile Update Notification	http://localhost:9090/ dms/event/Crew/ http://localhost:9090/notify	External Applications External Notifications	OMS Servers Interface Servers	CrewServerConfig.xml: NotificationBaseUrl NetviewerAdapterConfig.xml: CnfUrl
Interface Plugin Web Services - OMS					
8443	Call Server Web Service	http://localhost:8443/ api/adms/call/v2	Interface Servers	External App	CallServerConfig.xml: UriPort
8443	Call Server Call Interface Web Service Client Call Server AMI Interface Web Service Client Relay Server Call Web Service Relay Server AMI Web Service	http://localhost:8443/ api/adms/relay/call/v2 http://localhost:8443/ api/adms/relay/ami/v2	External Notification External App	Interface Servers	CallServerConfig.xml: BaseUri RelayServerConfig.xml: BaseUri
8443	Crew Server Web Service	http://localhost:8443/ api/adms/crew/v2	Interface Servers	External App	CrewServerConfig.xml: UriPort
8443	Crew Modeling Web Service	http://localhost:8443/ api/adms/crew/model/v2	Interface Servers	External App	CrewServerConfig.xml: UriPort
8445	Crew Simulator Switching Order Web Service	http://localhost:8445/api/ext/swo/v2	Interface Servers	Crew Simulator WFM System	CrewSimulatorConfig.xml: UriPort DmsInterfaceServerConfig.xml: BaseUri

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
8445	Crew Server WMS Web Client Crew Server Modeling Web Client Crew Simulator Crew Web Service	http://localhost:8445/api/ext/wms/v2	Interface Servers	Crew Simulator WFM System	CrewServerConfig.xml: BaseUri CrewSimulatorConfig.xml: UriPort
8447	Customer Experience Adapter Web Service Client	http://localhost:8447/api/ext/ce/v2	External Notification External App	Interface Servers	CustomerExperienceAdapterConfig.xml: BaseUri
18802	OMS Interface Server Web Service - without UAA	http://localhost:18802/api/adms/oms/v2	Interface Servers	Interface Servers	OmsInterfaceServerConfig.xml: UriPort CoreNotificationServerConfig.xml: OmsInterfaceServerURL ANPSAdapterConfig.xml: BaseUri CrewServerConfig.xml: BaseUri CustomerExperienceAdapterConfig.xml: BaseUri OmsAdapterConfig.xml: BaseUri
18802	OMS Interface Server Web Service - with UAA	http://localhost:18812/api/adms/oms/v2	Interface Servers	Interface Servers	OmsInterfaceServerConfig_UAA.xml: UriPort
Read-Only					
5051	OMS Interface Server		OMS Read Only Servers	OMS Interface Server	NetServerReadOnlyConfig.txt: OMS_SERVER_PORT
5055	RoSrv or NetServerRO		Read Only Servers	Read Only Clients Support Clients	netconf_ReadOnly.xml: <Port> NetServerReadOnlyConfig.txt: SERVER_PORT

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
5099	NetServerRO		Read Only Servers	OMS Interface Server (optionally, when served from NetServerRO)	OmsInterfaceServerConfig.xml: <OmsServerPort >
8200	Net Data Replicator Local		OMS Servers Read-Only Server	OMS Servers Read-Only Server	NetDataReplicatorReadOnlyConfig.xml: <LocalListenerAddress>
8201	Net Data Replicator Remote		OMS Servers Read-Only Server	OMS Servers Read-Only Server	NetDataReplicatorReadOnlyConfig.xml: <RemoteListenerAddress>
6096	Net Data Server		OMS Servers Read-Only Server	Clients	NetDataServerReadOnlyConfig_localhost.txt SERVER_PORT
9900	Net Data Server (Historical Correction) Read-Only Web Service	http://localhost:9900/oms	OMS Servers Read-Only Server	Clients External Applications	NetDataServerReadOnlyConfig_localhost.txt WEBSERVICE_PORT_NUMBER
Study					
5027	Local Study Server		DMS Servers	Clients	netconf_dnaf.xml: <Port> netconf.xml: <Port> netconf_follower.xml: <Port> netconf_oms.xml: <Port> netconf_ReadOnly.xml: <Port> NetServerStudyConfig.txt: SERVER_PORT
5030	DMS Citrix Client Study		Citrix Servers	DMS Servers	

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
5039	Highest Allowed Citrix Study		DMS Servers to/from Citrix Servers		
Distribution Operator Training Simulator					
5029	DMS Simulator		DMS Servers	Support Clients	netconf_dnaf.xml: <Port> netconf.xml: <Port> netconf_follower: <Port> netconf_oms.xml: <Port> netconf_ReadOnly. xml: <Port> NetSServerSimCon fig.xml: SERVER_PORT
5033	OMS Simulator		Relay Servers	Relay Servers DMS Servers OMS Servers	NetServerSimConfi g: CALLSERVER_PORT RelayServerConfig: <ListeningPort>
Modeling					
8085	Model Manager Service	http://localhost:80 85/ ModelManagerSer vice	Model Managers	Model Managers Support Clients	
8090	Model Manager Web Service	http://localhost:80 90/ MmaWebService.s vc	Source	Model Managers	
8041-8044	Sampler		Source	Model Manager Servers	
Group Control					
8800	Net Data Server	http://localhost:88 00/oms	Interface Server	Client	NetDataServer Config_localhost.tx t WEBSERVICE_PORT _NUMBER
9990	Apache HTTP Server		Interface Server	Client	httpd.conf: <Listen>

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
9991	Tile Service		Interface Server	Client	NetviewerAdapter Config.xml: <TileServiceUrl>
9991	Netviewer Adapter	ws://localhost:9991/tilecacheservice	Interface Server	Client	httpd.conf: <ProxyPass>, <ProxyPassReverse>
9993	PostgreSQL		Interface Server	Client	httpd.conf: <ProxyPass>, <ProxyPassReverse>
9994	Preference Service		Interface Server	Client	httpd.conf: <ProxyPass>, <ProxyPassReverse>
9997	GraphQL Service		Interface Server	Client	httpd.conf: <ProxyPass>, <ProxyPassReverse>
Integrations - Mobile Enterprise - Switching Order					
8443	Switching Order Mobile Web Service	http://localhost:8443/api/adms/MobileSwo	Interface Server	Interface Server	DmsInterfaceServerConfig.xml: MobileUrl MobilityGatewayAppConfig.xml: MobileSwitchOrderWebServiceBaseUri
Integrations - DERM 2030.5 Gateway					
8888	DERM 2030.5 Gateway Web Service	http://localhost:8888/DERCONTROLS/dispatch	DMS Servers	DERM 2030.5 Gateway Web Service	DmsInterfaceServerConfig.xml: DermsGatewayDispatchUrl
Integrations - Forecasting					
8078	Forecast Adapter	http://<DockerHost>:8078/api/v1/dms/			N/A
Integrations - Outage Assist					

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
5051	Outage Assist		OMS Read Only Servers	Net Data Replicator Netviewer Adapter	NetViewerAdapter.xml: <OmsServerPort> NetDataReplicator Config.xml: <OmsServerPort> NetServerReadOnlyConfig.txt: OMS_SERVER_PORT
8088	OMS Adapter to Outage Assist		OMS Servers		N/A
9090	OMS Adapter to Outage Assist	http://localhost:9090/service/soap/	Outage Assist ES Adapter	OMS Servers	OmsAdapterConfig.xml: OmsEsAdapterURL
24467	Outage Assist	ws://localhost:24467/tilecacheservice	Websocket TileService	Netviewer Adapter	NetviewerAdapter Config.xml: <TileServiceUrl>
24468	Outage Assist		Netviewer Adapter	Outage Assist	NetviewerAdapter Config.xml: <Port>
Integrations - Enterprise OMS					
5432	PostgreSQL DB Connection Port		Work Model Server or Kubernetes cluster	Interface Servers	N/A
8802	Work Model Event Handler Service connection to OMS Interface Server	http://localhost:8802/	Interface Servers	Enterprise OMS Server or Kubernetes cluster	manifest.yml: externalPort OmsInterfaceServerConfig.xml: WebServiceBaseURL
8083	Work Order Database Sync and NetServer connection to Work Model Notifications Service	ws://localhost:8083 / oms/api/v1/events	Enterprise OMS Server or Kubernetes cluster	OMS Servers Interface Servers	NetServerOmsConfig_localhost.txt: WORKMODEL_EVENT_API WorkOrderArchive Adapter Config.xml: WMNotificationServer

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
8080	Connection to Work Model Event Handler Service from Work Order Adapter, NetServer, APF, Browser	http://localhost:8080 http://localhost:8080/oms/api/v1/	Enterprise OMS Server or Kubernetes cluster	OMS Servers Interface Servers Clients	NetServerOmsConfig_localhost.txt: WORKMODEL_SERVICE_API OmsDisplayPluginConfig.xml: WorkModelServiceApi WorkOrderAdapterConfig.xml: WorkOrderServiceUrl
30001	Work Order Adapter connection to PostgreSQL DB		Enterprise OMS Server or Kubernetes cluster	Interface Servers	manifest.yml: port_external
Integrations Mobile Enterprise					
8080	Mobility Gateway App (Work Model API Client)	http://localhost:8080	Work Model Event Handler Service	EOMS Servers	MobilityGatewayAppConfig.xml
8083	Mobility Gateway App (Mobility Gateway WorkOrder Events Processor)	ws://localhost:8083 / oms/api/v1/events	Work Model Event Handler Service	EOMS Servers	MobilityGatewayAppConfig.xml
8801	Mobility Gateway App (DMS Interface Server Switching Order Web Service Client)	http://localhost:8801/dms/SWO<	OMS Interface Server	Interface Servers	MobilityGatewayAppConfig.xml
8802	Mobility Gateway App (OMS Interface V1 Web Service Client)	http://localhost:8802/ oms/services/rest	OMS Interface Server	Interface Servers	MobilityGatewayAppConfig.xml
8443	Mobility Gateway App (Crew Server Web Service Client)	http://localhost:8443/ api/adms/crew/v2	Crew Server	Interface Servers	MobilityGatewayAppConfig.xml

Port	Usage	URL	To (destination server where the port is open)	From (source server/client that connects via the port)	Configuration file / Parameter
8443	Mobility Gateway App (Mobile Gateway Switch Order Plugin Web Service)	http://localhost:8443/ api/adms/MobileSwo	DMS Interface Server	Interface Servers	MobilityGatewayAppConfig.xml
8445	Mobility Gateway App (Mobile Enterprise Web Service Client)	http://localhost:8445/ api/ext/MobileEnterprise/v1	Mobile Enterprise Web Service Client	Interface Servers	MobilityGatewayAppConfig.xml
18802	Mobility Gateway App (OMS Interface Query Service)	http://localhost:18802/ api/adms/oms/v2	OMS Interface Server	Interface Servers	MobilityGatewayAppConfig.xml
38669	Mobility Gateway App (Tagging Web Service Client)	http://localhost:38669/tagrest/api	Tagging Service	Scada Server	MobilityGatewayAppConfig.xml

16. Configuration Files

16.1. NetServerMainConfig.txt

The following table describes the parameters that are specific to the ADMS Main (DMS) server.

Note

Parameters followed by an asterisk (*) are for developer use only.

Note

Boolean parameters in NetServer configuration files can be set to 1/0 or True/False (case insensitive). If a parameter is removed or the parameter value is not defined, the default parameter value is used.

Table 16. NetServerMainConfig.txt Parameters

PARAMETER	VALUE
REAL-TIME SERVER MODE	
REALTIME	This flag is used to distinguish between real-time server mode and study mode. The difference is in how the ADMS server application obtains energization information. In real-time mode, it uses SCADA. In study mode, it creates a fictitious high-side energy source for each station. For a real-time server or simulator server, set the value to 1. For a study server, set the value to 0.
SERVER_PORT	The TCP/IP port that NetSrv listens on. This is the port to which the client connects.
READ-ONLY SERVER MODE	
READ_ONLY_DMS_SERVER_MODE	This flag is used to distinguish between real-time server mode and read-only mode. For a read-only DMS server, set the value to 1.
READ_ONLY_OMS_SERVER_MODE	This flag is used to distinguish between real-time server mode and read-only mode. For a read-only OMS server, set the value to 1.
MAIN_SERVER_HOST	The host(s) where the production NetSrv process is running.
MAIN_SERVER_PORT	The TCP/IP listening port for the production system's NetSrv process.
MAIN PATHS	
MODEL_PATH	The path to the directory where the model files are placed.
DYNAMIC_PATH	The path to the directory where the dynamic data are placed. For example, %IDMS_ROOT%\dynamics\Server.
SAVECASE_PATH	The path to the directory where the savecases are placed. Used on the study server only.
DNAF_INPUT_PATH	The path to the directory where the DNAF parameter files are placed.

PARAMETER	VALUE
LICENSES_PATH	The path to the directory where the license files are placed. (Deprecated functionality.)
MTLS_PATH	This is full path to the mTLS configuration file. The default file is NetServerMtls.xml.
DEFAULT_MTLSAUTH_TIMEOUT	The default time out for MTLS authentication.
INPUT/OUTPUT FILES	
EMS_DMS_TAG_MAP	The full path to the file that is used for EMS and DMS tagging.
STATION_AREA_MAP	Deprecated.
DYNAMIC_NAME	The name of the file that holds the dynamic data. Default is dynamics.sav.
LOGGING	
LOG_ENABLED	A flag to enable (=1) or disable (=0) NetSrv logging. This option is enabled (=1) by default.
LOG_SIZE_LIMIT	A numeric value specified in kilobytes. If LOG_SIZE_LIMIT is 0, the file size is not limited. If a numeric number is greater than zero, when the current log file reaches the specified size, the file is renamed to <current log file name>_bak_N.log (where N is a number incremented for each new file). A new log file is created. For example, when a log file netsrv_main.log reaches the size limit, it is renamed to "netsrv_main_bak_1.log", and a new empty "netsrv_main.log" is created. Each log file is limited in size and set in the LOG_SIZE_LIMIT value. When the LOG_DAILY option is enabled, the log file has the same behavior, except the file name also contains a date. The default value is 100000.
LOG_DAILY	A flag to enable (=1) or disable (=0) NetSrv to create a separate log file for each day. For example, the LOG_FILE_NAME defines the log name as "netsrv_main.log" and the LOG_DAILY flag is enabled. The log file name is netsrv_main_20140708.log (date appended). When a new day begins, a new log file is created. This option is disabled (=0) by default.
LOG_ADD_FILENAME_PREFIX	A flag to enable (=1) or disable (=0) adding the server host and role prefix to the server log file name. If enabled, a new log file is created each time the server role changes, such as from Enabled to Standby. The name format is <HostName>_<Role>_<Server>_<Time Stamp>.log (for example, idms32_Enabled_NetRTServer_20161228_014732.log). Adding the prefix to the name is configurable.
LOG_FILE_NAME	The full path to the file that is used for NetSrv logging purposes.
ISD_LOG_FILE_NAME	The full path to the file that is used for ISD logging purposes.

PARAMETER	VALUE
ISD_LOG_ENABLED	A flag to enable (=1) or disable (=0) logging for the InterSite Control communications. This option is enabled (=1) by default. This parameter can be changed at run-time via the following URL command on the client side: NETVIEW://RT/SERVERDEBUG/ISDLOG?ENABLED=1
MTLS_LOG_FILE_NAME	The full path to the file that is used for mTLS logging purposes.
MTLS_LOG_ENABLED	A flag to enable (=1) or disable (=0) logging for the mTLS communication. This option is disabled by default. When enabled, it logs very high-level information about security context initialization, authentication, send and receive details.
MTLS_LOG_ERROR	A flag to enable (=1) or disable (=0) logging for the errors during mTLS communication. The errors contain both application-level errors and windows API errors.
MTLS_LOG_TRACE	A flag to enable (=1) or disable (=0) logging for the function level trace during mTLS communication.
MTLS_LOG_ALL	A flag to enable (=1) or disable (=0) logging for all the messages, errors and function level trace during mTLS communication.
SERVICE MODULE LISTENING SOCKET	
SERVICE_PORT	TCP/IP port that is used for the DMS service module. The DMS service module on the DMS server is used by the *e-terra*scada versions 3.0 and later Tagging and Safety Documents applications to retrieve data from ADMS. The DMS service module is supported for DMS, DNAF, Study, Read-Only, Simulator, and Customer-Only servers.
SERVICE_SOCKET_OUTPUTBUF_MAX_SIZE	The maximum output buffer size. The default value is 6000000.
SERVICE_SOCKET_WOULDBLOCK_RETRY	The default value is 2.
SERVICE_SOCKET_MAX_CLIENT_MES_SIZE	The maximum client message size in bytes. The default value is 640000.
DUMP_DYN_SOCKET_TRAFFIC	If enabled (=1), detailed information about sockets is logged to separate files in the log folder. The log files are updated dynamically. Default is disabled (=0). Note This parameter is absent in the NetServer configuration file and can be added manually, as needed. This parameter can be changed at run-time via the following URL command on the client side: NETVIEW://RT/SERVERDEBUG/DYNSOCKLOG?ENABLED=1

PARAMETER	VALUE
SOCKET_NO_SLEEP_AFTER_SEND (*)	This option should be enabled (=1) if DNAF server communication errors are observed when connecting read-only servers that obtain DNAF results via TCP. This option is disabled (=0) by default. Note: This parameter is absent in the NetServer configuration file and can be added manually, as needed.
SOCKET_SEND_MAX_BACKUP	The maximum number of queued messages that can be sent from a client (for example, ADMS client or OMS server) to the DMS server. If the number of queued sent messages is exceeded, the DMS server disconnects this client. A big number of queued messages may appear at high system load or when a client is disconnected or is behind the DMS server for a long time. The default value is 3000. Note: This parameter is absent in the NetServer configuration file and can be added manually, as needed.
QUERY MAX ROW RESULT	
QUERY_MAX_RESULT_ROW_COUNT	The maximum number of rows in the server query result. This number affects the maximum number of rows in grid tabular displays and the maximum number of output sent to the web services (via the DMS interface server). The default value is 30000.
DEBUG	
TRACE_LEVEL	This parameter specifies the level of detail that is shown on the Analyst Message tab of the Net server RT window. The following levels are currently supported: • 0 = some server messages • 1 = triggering messages • 2 = summary iteration count and alarms • 3 = internal DNAF performance • 4 = additional, less important internal DNAF performance messages • 5 = configurator messages • 6 = show all messages Note Each trace level shows messages for lower levels as well (for example, level 2 shows messages for 0, 1, and 2 levels). This parameter can be changed at run-time on the client side via the following URL command: NETVIEW://RT/SERVERDEBUG/LOG?LEVEL=5 The value of -1 is ignored as it is reserved by the system. Other values beyond the 0-6 range result in an error message.

PARAMETER	VALUE
TRACE_TYPE	This parameter specifies the type of messages that are shown on the Analyst Message tab of the Net server RT window. The following debug types are currently supported: • 0 = TT_NONE • 1 = TT_REGULAR • 6 = TT_DNAF Types 2, 3, 4, and 5 can be entered, but no message types are currently assigned to them. This parameter can be changed at run-time on the client side via a URL command. To define several trace types, separate them with commas. NETVIEW://RT/SERVERDEBUG/LOG?TYPES=1,6 In the command above, the value of -1 is ignored as it is reserved by the system. Other values beyond the 0-6 range result in an error message.
TRACE_TO_LOG_FILE	A flag to enable (=1) or disable (=0) storing portable code processing details in the log file. This option should always be set to enabled (=1) on the product environment for debugging purposes. This option is enabled (=1) by default.
CLIENT CONNECTION	
CLIENT_HELLO_TIMEOUT	The number of seconds to wait before disconnecting when Hello messages are not responded to. The default value is 180seconds.
HELLO_SEND_PERIOD	The frequency (in seconds) at which Hello messages are sent.
SITEOV_DEAD_ENTRY_STAY_TIME	Time period (in seconds) since the last status update before an application is considered "dead" and the application connection entry is removed from the ADMS Site Overview grid tabular display.
FAST DYNAMICS	
FD_FLOW_INFO_PERIOD	Time period in seconds for querying NTP flow direction changes. A value of zero means this feature is disabled.
FD_MEASUREMENT_PERIOD	Time period in seconds for querying measurements changes from SCADA. A value of zero means this feature is disabled.
FD_POWER_MEASUREMENT_PERIOD	Time period in seconds for querying the customer meter power measurement data. A value of zero means this feature is disabled.
FD_DER_MEASUREMENT_PERIOD	Time period in seconds for querying DER measurement data. A value of zero means this feature is disabled.

PARAMETER	VALUE
FD_DNAF_INPUTS_FILES	This parameter specifies files (with paths), separated by commas, which are needed when initializing a study from real time. The default files are: • \$DYNAMIC_PATH\$manuals.sav • \$DNAF_INPUT_PATH\$DNAF_PARAMS_DNAF.xml • \$DNAF_INPUT_PATH\$Manual_Entry.dat • \$DNAF_INPUT_PATH\$tftype.dat • \$DNAF_INPUT_PATH\$alg.dat • \$DNAF_INPUT_PATH\$optalg.dat • \$DNAF_INPUT_PATH\$msgdef.dat • \$DNAF_INPUT_PATH\$phid.dat The values inside the "\$" symbols specify paths that are provided by other configuration parameters.
FD_POWERFLOW_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the DPF power flow data query. This option is enabled (=1) by default.
FD_MEASUREMENT_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the measurement data query. This option is enabled (=1) by default.
FD_FLOW_INFORMATION_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the NTP flow direction data query. The flow information is maintained in the all_flow.fldr file. The data is used in case of a server restart or failover. This option is enabled (=1) by default.
FD_VALUE_PLACEMENT_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the placement value data query. This option is enabled (=1) by default.
FD_HMS_EVENT_DATA_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the HMS event area fast dynamics data query. Enable this option on secondary servers that need Active Hazard Event Visualization data from the main server. This option is disabled (=0) by default.
FD_POWER_MEASUREMENT_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the customer meter power measurement query. This option is enabled (=1) by default.
FD_TROUBLE_TICKET_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the trouble ticket data query. This option is enabled (=1) by default.
THREAD POOLS	
THREADS_DYN	The number of worker threads in a thread pool in the dynamics module.
THREADS_NTP	The number of worker threads in a thread pool in the NTP module.
THREADS_LOAD	The number of worker threads in a thread pool in the loader module.
THREADS_SCADA	The number of worker threads in a thread pool in the SCADA module.
THREADS_QUERY	The number of worker threads in a thread pool in the query module.

PARAMETER	VALUE
THREADS_CLIENT	The number of worker threads in a thread pool in the client module (it handles primary-standby server communication).
THREADS_PLUGIN	Number of worker threads in a thread pool in the plug-in module. The plug-in module is used for pluggable application DLLs, such as the Load Shed application. Note For DNAF server plug-ins (LVM Monitor), only one thread is allowed. Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
THREADS_DNAF_PRIORITY (*)	Operating system priority of the DNAF worker thread. Values can range from -2 (highest priority) to 2 (lowest priority). Default is zero. This is supported on the Windows platform only. This parameter is especially useful on small demo machines where the CPU may get overburdened with DNAF processing.
ENABLE_PROACTIVE_WORKER_THREAD	If enabled (=1), the Query module worker threads actively pull messages from the Dispatcher module. If disabled (=0), the Query module worker threads passively wait for the Dispatcher module to post messages to the Query module. Enabling this option evenly distributes tasks to each available worker thread, reduces the task queue, and improves system performance.
INTERSITE DATA (SCADA CONNECTION)	
ISD_LOCAL_SITE	Set to match the SYSTEM_SITE field within SCADAMOM representing the ADMS site. Although not used directly by the ADMS server application, this can be a useful indication of which site represents the ADMS system at the remote.
ISD_LOCAL_SCANNER_PATH	Set to match the OBJECT_SITE field in SCADAMOM representing the ADMS site; it is used for InterSite Data. If this parameter is not set, no communications on the path are attempted.
ISD_LOCAL_CONTROL_PATH	Set to match the CBJECT_SITE field in SCADAMOM representing the ADMS site; it is used for InterSite Control. If this parameter is not set, no communications on the path are attempted.
ISD_LOCAL_TAGNOTES_PATH	Set to match the TGOBJECT_SITE field in SCADAMOM representing the ADMS site; it is used for InterSite TAGGING. If this parameter is not set, no communications on the path are attempted.
ISD_REMOTE_SITE	Set to match the SYSTEM_SITE field within SCADAMOM representing the remote Scada site. Although not utilized directly by the ADMS server application, this can be a useful indication of the remote site identifier.
ISD_REMOTE_SCANNER_PATH	Set to match the OBJECT_SITE field in SCADAMOM representing the Scada site; it is used for InterSite Data. If this parameter is not set, no communications on the path are attempted.
ISD_REMOTE_CONTROL_PATH	Set to match the CBJECT_SITE field in SCADAMOM representing the Scada site; it is used for InterSite Control. If this parameter is not set, no communications on the path are attempted.
ISD_REMOTE_TAGNOTES_PATH	Set to match the TGOBJECT_SITE field in SCADAMOM representing the Scada site; it is used for InterSite TAGGING. If this parameter is not set, no communications on the path are attempted.

PARAMETER	VALUE
ISD_REMOTE_HOST1	Set equal to the first EMS (SCADA) host name or IP. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
ISD_REMOTE_HOST2	Set equal to the second EMS (SCADA) host name or IP. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
ISD_REMOTE_HOST3	Set equal to the third EMS (SCADA) host name or IP, if applicable. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
ISD_REMOTE_HOST4	Set equal to the fourth EMS (SCADA) host name or IP, if applicable. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
ISD_REMOTE_PORT	Set to match the port used for communications on the remote machines. It should be set to equal the DEFPORT_ITEMS field within the ProxySrv database on the remote machine.
ISD_REMOTE_POLLRATE	Time period in seconds between periodic polls to the remote site. This parameter is only used for the InterSite Data protocol.
ISD_REMOTE_USE_MTLS	If enabled (=1), the network communication uses mutual authentication (Mutual Transport Layer Security). By default, disabled.
ISD_REMOTE_INTEGRITYRATE	Time period in seconds the ADMS server application uses to send all current data values to the remote site. This parameter is only used for the InterSite Data protocol.
ISD_MAX_DOWNLOAD_TIME	Set to the desired period, in seconds, the ADMS server application should wait before disconnecting if it has not received any message from SCADA. This could happen if SCADA is running slowly.
ISD_KEEPAIVE_TIMEOUT	Time period in seconds the ADMS server application waits without receiving an appropriate message from the remote before disconnecting. Setting this value to zero disables keep-alive logic from NetSrv for InterSite protocols.
ISD_DETAILED_MESSAGES	If enabled (=1) and other non-zero values evaluate to True, verbose message logging relating to InterSite Protocol processing is produced. The default file is placed in the working directory of the ADMS server application and is called "isdlog.txt". Note that warning and error messages are sent to this log file regardless of the setting of the ISD_DETAILED_MESSAGES parameter. This parameter can be changed at run-time via the following URL command on the client side: NETVIEW://RT/SERVERDEBUG/ISDLOG?DETAILED=1

PARAMETER	VALUE
ISD_SKIP_MESSAGES_PATH_DOWN	If enabled (=1), ISD data is not sent to SCADA when the connection is down. This parameter is used to resolve an intermittent issue where DMS cannot connect to SCADA after failover.
ISD_ANALOG_VALUE_MESSAGES (*)	If enabled (=1), ISD logs analog measurements received and their value and quality. The data is logged to the file defined in ISD_LOG_FILE_NAME. The default value is Disabled (=0).
ISD_PARTIAL_MEAS_DOWNLOAD	If enabled (=1), after a model push, the DMS server requests and re-applies the measurement data from SCADA only if the measurement hash sum differs from the previous value. If the measurement hash sum is the same, the DMS server applies measurement values from the measurement file. The default value is Enabled (=1).  Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
ISD_DEFAULT_SITE	Set to match the DEFAULT_SITE field in SCADAMOM representing the ADMS site; it is used for InterSite Data. If this parameter is not set, no communications on the path are attempted.
USE_NEW_ISD_LOGIC	A flag to enable (=1) or disable (=0) the new InterSite Data logic. This option is enabled (=1) by default.
ISD_RECONNECT_TIMEOUT	Time period in seconds for ISD reconnection.
ISD_DUMP_SOCKET_TRAFFIC (*)	If enabled (=1), ISD logs all ISD protocol socket traffic (message size and other lower-level information) to the file defined with the configuration parameter ISD_LOG_FILE_NAME. Data is presented in a raw little-endian format. The messages are output to the isdlog.txt file in the working directory of the NetSrv application. The default value is enabled (=1). This parameter can be changed at run-time via the following URL command on the client side: NETVIEW://RT/SERVERDEBUG/ISDLOG?DUMPTRAFFIC=1
ISD_CTRL_STATUS_ON_OFF	The list of comma-separated mappings between the device On/Off status measurements and device On/Off status names. Used for the group control feature. The mapping format is: <MeasTypeEnumeration>:<OnDisplayNameString>#<OnInternalNameString>:<OffDisplayNameString>#<OffInternalNameString> For example, the mapping value for the SwitchStatusMeasurementType measurement is 12:Closed#CLOSE:Open#TRIP
ISD_HIGHEST_PROTOCOL	The highest Scada ISD protocol version that is supported. Set to 7 when integrating with Scada 3.4 to support the ISD protocol version 7. This is required to recognize when a device is operated due to a Scada rotating load shed event via the SWMASK2 query field of the SWT object.

PARAMETER	VALUE
ISD_EXCLUDE_OUTPUT_MEAS_TYPES	This parameter specifies a list of comma-separated output measurement types, which are excluded in the ISD message when sending DMS-to-SCADA output measurements. Each output measurement type in the list is an integer value. For the mapping between measurement type strings and integer values, see the Measurement Types sheet in the <i>Modeling Reference Tables</i>. For example, to skip sending TrueLoadPOutput (270) and DERProductionPOutput (272) measurements: ISD_EXCLUDE_OUTPUT_MEAS_TYPES = 270, 272 When the parameter value is empty, DMS sends all output measurements to SCADA. This parameter is designed for DMS servers in the leader-follower configuration.

MEASUREMENTS

MEASUREMENT_INVERT_PNTNAM	This parameter specifies the measurement point types whose Scada values need to be inverted in ADMS (for example, when showing values on ADMS grid tabular displays). The parameter value is a list of measurement point types, separated by commas. Example: To invert measurement values for the point types MAN, SC, HT, and A: MEASUREMENT_INVERT_PNTNAM = MAN,SC,HT,A The default value is ARE,GTE.
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Note

This parameter does not apply to the simulate-only status measurements (for example, a low-battery alarm). However, it does apply to SCADA controls sent for these simulate-only measurements. This parameter value must match the parameter values in NetServerOmsConfig.txt, NetServerSimConfig.txt, and NetServerStudyConfig.txt.

TAGGING

TAG_LOCAL	If enabled (=1), local (non-SCADA) tags are enabled. If disabled (=0), the Tagging function from EMS is used. Set to 1 for study mode.
TAGTYPE_COUNT	The number of tag types. The default value is 14.
TAGTYPE_ID_1=DO NOT OPERATE TAGTYPE_PRIORITY_1=1	If TAG_LOCAL is set to 1, this set of configuration entries is used to configure tag types.
TAGTYPE_CONTROLRESTRICTIONS_1 = 2,8,32,64	
TAGTYPE_ID	Label of the tag type, such as DO NOT OPERATE, HOLD OPEN, or INFO.
TAGTYPE_PRIORITY	The priority value of the tag type.
TAGTYPE_CONTROLRESTRICTIONS	The control restrictions applied by the tag type. To use several restrictions, separate them with commas.

PARAMETER	VALUE
SHOW_TAG_FOR_MEAS_TYPES	Specifies the enumerations of Scada measurement types for which tags are shown in the ADMS Tag Summary display. By default, only tags for the following measurement types are shown: • CapMeasurementType • XftapMeasurementType • LtcCapRemoteMeasurementType • LtcCapBlockMeasurementType • SwitchStatusMeasurementType • CapStatus • DeviceReclosingEnabled • GroundTripEnabled • SimulateSwitchStatus • DeviceAutoManual • FaultStatus For example, to show tags for the AnalogActiveRegulationMode measurement type, set SHOW_TAG_FOR_MEAS_TYPES = 164.

Note

This parameter is absent in the NetServer configuration file and can be added manually, as needed.

DERM Kafka Adapter Server	
DMS_SSL_CERTIFICATE_FILE	File path for GRPC SSL secure connection certificate, empty for insecure connection
DERM_ADAPTER_HOST	Host name or ID address of the DERM Kafka Adapter.
DERM_ADAPTER_PORT	TCP/IP port used by the DERM Kafka Adapter.
DERM_SEND_STATIONDERID_INTERVAL	Time period in seconds at which the ADMS main server sends a DER Id and station name to the DERM Kafka Adapter.
DERM_DER_ENERGIZATION_CHECK_ENABLED	A flag to enable (=1) or disable (=0) the automatic disconnect feature.
DERM_DER_ENERGIZATION_CHECK_DELAY	The wait time in seconds for checking if a DER energization status change occurred due to a feeder backbone or lateral topology change.
DERM_CHECK_LOCKOUT_PERIOD	The periodic interval in seconds to check for a new lockout status measurement.
DERM_LOCKOUT_TIME_TOLERANCE	This is the time interval in seconds that is used to determine whether a topology change was the result of a recent breaker or recloser trip and lockout event. A topology change that occurs within this interval is considered to be related to the protection device lockout event.

HABITAT PERMIT SERVER

PARAMETER	VALUE
PERMIT_CHECK_ENABLED	A flag to enable (=1) or disable (=0) checking permissions when queries are made. If enabled (=1), NetPermSrv parameters need to be defined. Setting PERMIT_CHECK_ENABLED to 0 not only prevents an ADMS server from connecting to the SCADA server to obtain permission information, but it also disables permission enforcement on the DMS system. This is helpful during initial integration and for certain troubleshooting. However, this should be set to 1 for real-time operations. Also, note that even during integration, tagging cannot function without permissions.
PERMIT_SERVER_IP	The IP address of the permit server. The PERMIT_SERVER_IP parameter must point to the SCADA servers where PERMIT is running. Host names or IP addresses are acceptable. If using host names, ensure that the names resolve successfully using ping before running the ADMS server application. Also, the PERMIT_SERVER_PORT must be set to match the port used by NetPermSrv running on the specified SCADA servers. The default port is 5345, but it must be defined in this file. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
PERMIT_SERVER_PORT	The port the permit server is running on.
PERMIT_SERVER_TIMER_PERIOD	Time period in seconds the ADMS server application checks the connection with the Permit server and request permit information.
PERMIT_SERVER_USE_MTLS	If enabled (=1), the network communication uses mutual authentication (Mutual Transport Layer Security). By default, disabled.
PERMIT_LOG_ENABLED	A flag to enable (=1) or disable (=0) Permit server logging so that cause PERMIT-related messages are written to the standard ADMS server log. This information is useful for integration and troubleshooting. However, this parameter should be set to 0 during normal operations.
PERMIT_JUMPER_OR_MODE	If enabled (=1), placing a jumper is allowed when write permissions for either (at least) of the AORs associated with the jumper are enabled. If disabled (=0), placing a jumper is allowed when write permissions for both AORs associated with the jumper are enabled.
PERMISSION_PER_STATION	Deprecated. This parameter defaults to zero and is used for legacy purposes.
PERMIT_CACHE_DUMP_TRIGGER_FILENAME	The full path to the file that triggers dumping internal permission cache to the server's log file. To trigger the cache dumping, rename or remove the file. This option is used for debugging the permission functionality.
TRUSTED USERS	
SECRET_STORE_ENABLED	Specifies whether using secret values from the CyberArk secret store is enabled. For more details, see Storing Secret Values in CyberArk .
SECRET_STORE_DLL_PATH	Full path to the assembly (DLL) containing a class that implements the CyberArk secret store.

PARAMETER	VALUE
SECRET_STORE_TRUSTED_USER_DIS_OBJECT_KEY	CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the DMS Interface Server trusted user ID defined in the Enterprise Vault Server. The format is <DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>.
SECRET_STORE_TRUSTED_USER_DFWK_OBJECT_KEY	CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the DMS Framework trusted user ID defined in the Enterprise Vault Server.
SECRET_STORE_UAA_ID_OBJECT_KEY	CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the UAA ID defined in the Enterprise Vault Server.
SECRET_STORE_UAA_SECRET_OBJECT_KEY	CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the UAA secret defined in the Enterprise Vault Server.
TRUSTED_USER_ID_ROBO	Trusted user ID for Robodispatch.
TRUSTED_USER_NAME_ROBO	Trusted user name for Robodispatch.
TRUSTED_USER_ID_DIS	Trusted user ID for the DMS Interface server.
TRUSTED_USER_NAME_DIS	Trusted user name for the DMS Interface server.
TRUSTED_USER_ID_DFWK	Trusted user ID for the DMS Framework. Required for TAGGING to work properly.
TRUSTED_USER_NAME_DFWK	Trusted user name for the DMS Framework. Required for TAGGING to work properly.
TRUSTED_USER_ID_WFM	Trusted user name for the Workforce Management System. Required for implementing device state updates from WFM.
TRUSTED_USER_NAME_WFM	Trusted user name for the Workforce Management System. Required for implementing device state updates from WFM.
TRUSTED_USER_ID_DFOLLOWER	Trusted user ID for the DMS server running in the Dependent Follower mode. Required for implementing DMS commands from this server.
TRUSTED_USER_NAME_DFOLLOWER	Trusted user name for the DMS server running in the Dependent Follower mode. Required for implementing DMS commands from this server.
TRUSTED_USER_ID_DCMFOLLOWER	Trusted user ID for the DMS server running in the Customer-Only Dependent Follower mode. Required for implementing DMS commands from this server.
TRUSTED_USER_NAME_DCMFOLLOWER	Trusted user name for the DMS server running in the Customer-Only Dependent Follower mode. Required for implementing DMS commands from this server.
TRUSTED_USER_ID_OMSSERVER	Trusted user ID for the OMS server running in the Dependent Follower mode. Required for implementing OMS commands from this server and receiving permissions from the DMS server running in the Dependent Follower mode.
TRUSTED_USER_NAME_OMSSERVER	Trusted user name for the OMS server running in the Dependent Follower mode. Required for implementing OMS commands from this server and receiving permissions from the DMS server running in the Dependent Follower mode.
TRUSTED_USER_ID_DERMKAFKAADAPTER	Trusted user ID for the DERM Kafka Adapter. Required for implementing distributed energy resource device measurement updates from DERMS.
TRUSTED_USER_NAME_DERMKAFKAADAPTER	Trusted user name for the DERM Kafka Adapter. Required for implementing distributed energy resource device measurement updates from DERMS.
DEFAULT_OMS_SYSTEM_USERID	Default user ID for the OMS server.
AOR PERMISSION MANAGER	

PARAMETER	VALUE
PERMISSION_UAA_END_POINT	The UAA server endpoint. NetServer needs to access the UAA server to communicate with the AOR Manager application in the Outage Assist's Kubernetes cluster to get AOR data for Outage Assist. For NetServer to access the UAA server in the Kubernetes cluster, you also need to add environment variables that define service connection credentials. For more information, see Configuring ADMS Interface Servers. For example, to use the "client credentials" grant type: <pre>SETX NETSERVER_UAA_AUTH_CLIENTID netserver-service-client-id</pre> <pre>SETX NETSERVER_UAA_AUTH_CLIENTSECRET netserver-service-client-secret</pre>
PERMISSION_READ_VALUE	The name of the user permission to read data from NetServer (by default, oms.Read).
PERMISSION_WRITE_VALUE	The name of the user permission to write data to NetServer (by default, oms.Write).
PERMISSION_API_VERSION	The AOR Manager API version.
PERMISSION_REFRESH_INTERVAL	Time period for refreshing user AOR data, in minutes.
EXCESSIVE RATE OF CONNECTION ATTEMPTS (DENIAL OF SERVICE) PARAMETERS	
CONNECT_ATTEMPT_RATE_LIMIT	The threshold number of server listening socket connection attempts per minute to detect an excessive rate of connection attempt event. If set to 0, excessive rate of connection attempt events are not detected.
RATE_LIMIT_VIOLATION_MAX	The number of excessive rate of connection attempt events to wait before triggering a protective action specified by the CONNECT_ATTEMPT_DISABLE_SOCKET parameter.
CONNECT_ATTEMPT_DISABLE_SOCKET	Specifies a protective action to trigger when the number of excessive rate of connection attempt events specified by the RATE_LIMIT_VIOLATION_MAX parameter is reached. If enabled (=1), the server listening socket will be disabled: all client connections will be terminated and no more connections will be accepted until the server is restarted. If disabled (=0), clients that triggered the protective action will be added to a black list. Their connection will not be accepted by the server. To reset a black list for the DMS server, run the following command: Netview://RT/ServerDebug/RESETBL?TYPE=1 To reset a black list for the OMS server: Netview://RT/ServerDebug/RESETBL?TYPE=1&DMSOMS=1 Alternatively, to reset a black list for the OMS server, from the Net server application display, click Debug > Reset Blacklist. To restart the listening socket for a server, restart the server. Alternatively, to restart the listening socket for the OMS server, from the Net server application display, click Debug > Reset Listener.
PERMISSIONS	
REQUIRE_SWITCHING_AND_GEOGRAPHIC_PERMISSION	If enabled (=1), the user must have both Switching functional permissions and Geographic AOR permissions to confirm a device to a new state.

PARAMETER	VALUE
SWITCHING_FUNCTION_AOR_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is SWITCHOP. If it is not defined, permissions will go back to the default Geographic AOR mode.
REQUIRE_FUTURE_COMMISSIONING_PERMISSION	If enabled (=1), the user must have functional permissions to modify the commissioning stage and state (active, future, future removed, or removed) for network components.
PROJECT_COMMISSIONING_PERMIT_AOR	Set to match the name of the commissioning functional AOR defined in PERMIT. The default name is COMMIS.
COMMISSIONING_LOG_ENABLED	If enabled (=1), additional logging for commissioning activities is enabled.
HABITAT ALARM SERVER	
ALARM_HOST1	Set equal to the first EMS (SCADA) host name or IP. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
ALARM_HOST2	Set equal to the second EMS (SCADA) host name or IP. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
ALARM_HOST3	Set equal to the third EMS (SCADA) host name or IP. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
ALARM_HOST4	Set equal to the fourth EMS (SCADA) host name or IP. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
ALARM_PORT	Set to match the TCP/IP port used by Proxysrv on the EMS (SCADA) servers (DEFPORT_ITEMS field within the ProxySrv database).
ALARM_API_VERSION	Set to 3 (the only supported Alarm API).
ALARM_KEEPALIVE_TIMEOUT	Time period in seconds to wait before disconnecting when Hello messages are not responded to. The default value is 30seconds.
ALARM_SERVER_TIMER_PERIOD	The frequency (in seconds) at which Hello messages are sent.
ALARM_GROUP_BY_FEEDER	A flag to enable (=1) or disable (=0) grouping alarms by substation feeder. If disabled (default), alarms are grouped by substation.
ALARM_SERVER_USE_MTLS	If enabled (=1), the network communication uses mutual authentication (Mutual Transport Layer Security). By default, disabled.
ALARMS_WITH_SUPPRESSED_TONE	Specifies alarms groups or specific alarms for which sound is disabled. To disable sound for all DMS, OMS, and DNAF alarms: ALARMS_WITH_SUPPRESSED_TONE = DNAFALL,DMSALL,OMSALL To disable sound for DMSOP070 and OMSOP041 alarms: ALARMS_WITH_SUPPRESSED_TONE = DMSOP070,OMSOP041
ALARMS	

PARAMETER	VALUE
MONITOR_CONNECTION_STATUS_OF_CLIENTS	This parameter specifies types of clients, separated by commas, for which the connection status to the enabled server is monitored. When a client is connected or disconnected, an alarm is displayed. The available types of clients are: • DMS: DMS (Main) servers • DMF: DMS servers in Follower Mode • OMS: Enabled OMS server • MR: Read-only DMS servers • DE: Enabled DNAF server • DIS: DMS interface servers • OMR: Read-only OMS servers • VC: Common clients • VIPC: Important clients. To mark a client as important, set ApplicationType = VIPC in the client configuration file netcoft.xml.  Note The connection status of the standby DMS server is always monitored, regardless of the configuration. Monitoring the connection status of standby DNAF and OMS servers is not supported.
MONITOR_DYN_SYCH_STATUS_OF_CLIENTS	This parameter specifies types of clients, separated by commas, for which the synchronization status with the enabled server is monitored. When the server and client are out of synchronization, an alarm is shown. The available types of clients are: • MR: Read-only servers • DE: DNAF servers • DIS: DMS interface servers • VC: Clients • VIPC: Important clients  Note The synchronization status of the standby server is always monitored, regardless of the configuration.
ALARM_DYNAMIC_NOT_SYNC_OVER_PERIOD	Time period in seconds for the client dynamic synchronization status monitoring. This value must be over 0 to enable monitoring. Every two seconds, the DMS Main server checks if the dynamics are synchronized with each connected client. If the client dynamics have not been synchronized with the server over the specified period, the server issues the DMSOP008 alarm.  Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.

PARAMETER	VALUE
ALARM_ENABLE_MONITOR_THREAD_BACKLOG	A flag to enable (=1) or disable (=0) the server thread backlog status monitoring. Every second, the DMS Main server scans the threads. To issue the DMS OP009 alarm, two other parameters must be set: ALARM_QUEUE_BACKLOG_THRESHOLD ALARM_BACKLOG_OVER_THRESHOLD_OVER_PERIOD Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
ALARM_QUEUE_BACKLOG_THRESHOLD	The minimum value of backlog for a thread message queue to be considered not in good shape. This backlog threshold limit (in seconds) triggers the DMSOP009 alarm, which is used for server thread queue backlog monitoring. Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
ALARM_BACKLOG_OVER_THRESHOLD_OVER_PERIOD	Minimum length of period a thread has been persistently in bad shape to trigger the DMSOP009 alarm. Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
ALARM_ONCE_PER_DYN_SYCH_VIOLATION	If enabled (=1), the alarm that signals a client is not synchronized with the server is shown only once. Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
DMS REDUNDANCY	
USE_ROLE_FROM_CFGPROXY	Specifies whether the server gets role assignments from CfgProxy.
REDUNDANT_HOST	Host names of the ADMS servers (up to four redundant server names, separated by commas). For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
REDUNDANT_PORT	The TCP/IP port used for redundancy communication. Note The REDUNDANT_PORT in the DMS main server is used only by the Follower Standby.

PARAMETER	VALUE
CFGPROXY_HOST	The list of hosts (separated by commas) where CfgProxy runs. For mTLS communication, the fully qualified name is required. IP address will not work. Ex: G1PNXXXXXX.logon.ds.ge.com
CFGPROXY_TIMER_PERIOD	Time period in seconds for CfgProxy to confirm a process role (Enabled or Standby) before rejecting the process role assignment.
CFGPROXY_PORT	The ADMS server application communicates with CfgProxy via a TCP/IP link. This parameter specifies the TCP/IP port that is used by CfgProxy. This must match the value of the listening port that is defined in the CfgProxy clone.
CFGPROXY_USE_MTLS	If enabled (=1), the network communication uses mutual authentication (Mutual Transport Layer Security). By default, disabled.
CFGMAN_PROCESS_NAME	Configuration Manager process name.
DYNAMICS_BACKUP_MINUTES	Time period in minutes for dynamic data backup.
DEVICE STATE PRESERVATION	
PRESERVE_STATE_ENABLED	A flag to enable (=1) or disable (=0) preserving normal states. This option is enabled (=1) by default.
PRESERVE_STATE_NAME	The name of the file storing the preserved normal states. This file is created in the dynamics folder when dynamics are compressed.
ANNOTATIONS	
ANNOTATIONS_COUNT	The number of annotation types. Default value is 13.
ANNOTATION_TYPE_1 = 0	The set of configuration entries that is used to configure annotation types.
ANNOTATION_NAME_1 = KeyCustomer	
ANNOTATION_MODE_1 = journal	
ANNOTATION_FILE_1 = Annotation_KeyCustomer.sav	
ANNOTATION_TYPE	Subscript of the annotation type. By default, set as "Annotation type - 1".
ANNOTATION_NAME	Label of the annotation type.
ANNOTATION_MODE (*)	For developer use only. The default value is "journal" and it should not be changed.
ANNOTATION_FILE	File name where the corresponding annotations are saved.
TEMP OBJECT AUTO-CORRECTION (ORPHANED OBJECTS)	
CORRECTLY_PLACED_TOLERANCE	Distance in geometric units that determines whether the temporary object is correctly placed on a line. If the distance from the temporary object indicated position to the object line placement is within this tolerance, the temporary object is assumed to be correctly placed.
POSITION_UPDATE_TOLERANCE	Distance in geometric units that determines whether the temporary object should have its X and Y position automatically updated. If the temporary object indicated position is not correctly placed (above correctly placed tolerance), but is within the position update tolerance, its X and Y position will be updated to reflect the actual line attachment point.

PARAMETER	VALUE
AUTO_CORRECT_ENABLED	A flag to enable (=1) or disable (=0) orphan object autocorrection. This option is enabled (=1) by default. If auto correction is enabled, the system attempts to correct any orphaned objects. The position of the original object is used with Snap Search Tolerance to search within a given area.
SNAP_SEARCH_TOLERANCE	Distance in geometric units in which to search for potential line segments to snap to (lines may be in other stations). If more than one line is found, no assumption is made as to the target position, and the temporary object will become orphaned. If a single line is found within the snap tolerance distance (regardless of which station it is in), the temporary object position will be updated to indicate an attachment to that line.
ORPHAN_DYN_RETAIN_DAYS	Time period in days for Orphan dynamics cleanup. The cleanup occurs during midnight dynamics compression. To disable the Orphan dynamics cleanup, set to 0.

SCADA OR DER DEVICE COLLECTIONS

SCADA_OR_DER_COLLECTION	The list of real device collections that are considered for the SCADA or DER device lookup. (By default, LOAD,GEN,ENERGYRES,SWT,TF,SHIMP,EXTBRC,RFBSIF,SV,SERIES,DISTTF.) These collections can be queried using the SCADA_OR_DER virtual object collection, which supports the same queries as TOPOCOMP. Used for the SCADA/DER Devices Tree View display.
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Note

This parameter value must match the parameter value in NetServerDnafConfig.txt.

LOCALIZATION

LOCALE_LANGUAGES_FILE	Path to the Languages file.
LOCALE_COUNTRIES_FILE	Path to the Countries file.
ISD_REMOTE_ENCODING	InterSite Data remote encoding.
ISD_REMOTE_LOCALE	Set to match the language code selected for the operating system.
PERMIT_REMOTE_ENCODING	Permit remote encoding.
PERMIT_REMOTE_LOCALE	Set to match the language code selected for the operating system.
LOCALIZATION_FILE	Full path to the server localization file.

REGULATOR BYPASS BEHAVIOR PARAMETERS

PARAMETER	VALUE
REG_BLOCKED_BYPASS	This parameter determines the regulator bypass behavior when the regulator operation is blocked. 0: The regulator is bypassed using an internally created jumper when the regulator is blocked, even if the regulator group has an explicit bypass switch modeled externally. 1: The regulator is bypassed using an internally created jumper when the regulator is blocked only if the regulator group does not have any bypass switch modeled externally. 2: When the regulator is blocked, no internal jumper is created. Any existing internally created jumper across the regulator will be removed.  Note Tap changing power transformers (LTCs) cannot be bypassed, regardless of the REG_BLOCKED_BYPASS setting.
REG_BLOCKED_TAP	This parameter determines the tap behavior when the LTC/regulator operation is blocked. 0: The tap position is set to nominal when an LTC/regulator is blocked, even if there is a bypass switch. 1: The tap position is set to nominal when an LTC/regulator is blocked only if there is a bypass switch; otherwise, the tap position is not changed. 2 = The tap position is not changed when an LTC/regulator is blocked, even if there is a bypass switch.  Note If a valid tap position SCADA measurement exists when a blocking action happens, the tap position is set to the measured value, regardless of the REG_BLOCKED_TAP setting. The measurement change after the blocking action has no impact on the blocking behavior.
FL HOLD FAULT INDICATION PARAMETERS	
FAULT_INDICATION_HELD_NORMAL_DURATION	Time period in minutes during which the fault indicator halo is held if the associated fault indication SCADA measurement is reset. The time period starts when the breaker is re-energized or the fault indication status is reset (the latest of these events). Applies to all fault indicators that are either associated with a breaker capable of holding fault indications, or associated with a switch that is downstream of a breaker capable of holding fault indications. This capability is defined in the static model for breakers using the HoldFaultIndicationCapable field.
FAULT_INDICATION_HELD_MAXIMUM_DURATION	Time period in minutes during which the fault indicator halo is held after the associated SCADA measurement is reset (the associated breaker may still be open). Applies to all fault indicators that are either associated with a breaker capable of holding fault indications, or associated with a switch that is downstream of a breaker capable of holding fault indications. This capability is defined in the static model for breakers using the HoldFaultIndicationCapable field.
OTHER PARAMETERS	
MEAS_PROT_SETTINGS_TOLERANCE	Tolerance that is used when converting a received MeasProtectionSettingGroup analog measurement value to a protection setting group value.

PARAMETER	VALUE
FAILOVER_DYN_FILE	The replicated dynamic file standby that the ADMS server application uses for failover.  Note This parameter is used in the study server configuration file.
ALLOW_LOAD_FATAL_MODEL	A flag to enable (=1) or disable (=0) loading models with fatal errors.
AOR_CONFIG_FILE	Path to the Permit model input file in XML. This file will be used to extract management areas hierarchy information for the AOR API used by the Operations Summary application.
ABNORMAL_FEEDER_REPORT	A flag to enable (=1) or disable (=0) writing feeder abnormal customer count reports.  Note Do not set this parameter to 1 in the DNAF configuration.
STATION_MAP_PUBLISH_PERIOD	Time period in seconds for loading model files after new or removed files are detected. The reason for having this parameter is to avoid server distress with many small changes. The default value is 30seconds. All values less than 15 are interpreted as 15, which means the minimum value for this parameter is 15seconds.
DNAF_CHECK_DPF_STATUS	A flag to enable (=1) or disable (=0) issuing an alarm if the Distribution Power Flow (DPF) has not been solved in the last DNAF_CHECK_DPF_STATUS_INTERVAL minutes. This check is done at the top of every hour. This alarm is issued by division/area, which is defined in the static DMS model. This option is disabled (=0) by default.
DNAF_CHECK_DPF_STATUS_INTERVAL	Time period in minutes used for checking if the DNAF_CHECK_DPF_STATUS alarm is issued.
DNAF_SET_RLD_SUSPECT_ON_DNAF_DISCONNECT	A flag to enable (=1) or disable (=0) marking the RLD SCADA measurement quality as suspect when the DNAF server is disconnected. The default value is enabled (=1).
EFFECTIVE_ZERO_VOLTAGE_MEASUREMENT_PERCENTAGE	Defines the threshold percentage for considering the voltage measurement at the interface bus as effective zero voltage. If an interface bus voltage measurement value is below the valid interface bus voltage measurement value multiplied by this threshold percentage, the interface bus voltage is considered effective zero voltage. When zero voltage is identified, the system opens the interface bus switch. This only applies to switches that were automatically inserted in the station model by the Converter tool. For more information, refer to section "Reference Bus Energization" in the <i>ADMS Modeling Overview and Converter User Guide</i>.
MODEL_PATH_SAFEMODE	The path to the directory where the system backs up stable station models.  Note The folder must be different than the folder specified by the MODEL_PATH parameter.

PARAMETER	VALUE
MODEL_SAFEMODE_BACKUP_PERIOD	Time period in minutes before backing up station models. The system performs Cyclic Redundancy Checks (CRC) for model files and overrides them only if a CRC is different. When the server is running in safe mode, backups are not performed.
FOLLOWER SERVER	
FOLLOWER_MODE	Set to 1 for DMS servers running in the Independent Follower mode to the enabled DMS server. In this mode, the follower servers receive changes from the enabled server and independently implement DMS commands from the follower server clients. Note The dynamics between the independent follower server and the enabled server are not synchronized but merged from the enabled to the follower. The follower server merges changes from the enabled server when changes occur and during the midnight dynamics compression. Therefore, errors appear when similar operations (for example, device status changes) are performed from the both servers. Set to 2 for DMS servers running in the Dependent Follower mode to the main DMS server. In this mode, the follower servers receive changes from the enabled server and send requests to the enabled server to implement DMS commands from the follower server clients. Note SCADA is disabled for dependent follower servers.
FOLLOWER_PRESEVE_LEADER_DYNAMICS	If enabled (=1), the follower DMS server preserves the enabled DMS server's dynamics.
FOLLOWER_LEADER_DYNAMIC_NAME	The name of the enabled DMS server's dynamics.
FOLLOWER_LEADER_DYNAMIC_MAN_NAME	The name to be used for the enabled DMS server's dynamics.
DISCONNECT_UNTRUSTED_USER	If enabled (=1), the follower DMS server disconnects clients (for example, OMS server) that do not have trusted user ID.
RETURN_TO_NORMAL_BY_RESTART	A flag to enable (=1) or disable (=0) the follower DMS server to re-initialize itself (restart) after the midnight dynamics compression. Note This parameter is absent in the server configuration file but can be added manually, as needed. The default value is enabled (=1).
DMS_IGNORE_DNAF_CREATED_FAULT_DYN_FROM_LEADER	If set to true, the DMS server (follower) does not merge the dynamics created by DNAF server (leader) while performing DNAF functions, such as fault placement. This option applies to the independent follower mode.

PARAMETER	VALUE
DMS_IGNORE_SCADA_FI_DYN_FROM_LEADER	If set to true, the DMS server (follower) does not merge fault indicator dynamics that are created from SCADA measurements. The DMS server (follower) receives all those dynamics directly from SCADA. This option applies to the independent follower mode.
DNAF_SKIP_SAVE_DPF_RESULTS	If set to true, the DNAF server will not write DPF solution result into file system (.LST), it will improve DPF solving performance and DNAF start/restart time. If disabled, DNAF will write DPF solution result into file <substation>_DPF_DNAFSRV.LST. When DNAF server start/restart, it will also try to restore the result from the files. The default is disabled (=0).
DNAF_LOG_CLEANUP_PERIOD_MINUTES	When DNAF server is configured to "Use Unique File Name" for DPF log file, it will append a time stamp to the new log file name. In the production environment, this could produce a lots of log file as DNAF server runs for long period of time. Based on the setting of another DNAF option ("Maximum Number of DPF result files Created for Any One Substation"), DNAF server has to cleanup periodically the old files to keep the total number of DPF result log files per substation in the limit. The value of this option is the frequency in minutes that DNAF server will check and perform the cleanup. The default value is 120 (2 hours), the minimum possible value is 10 minutes.
READ_ONLY_DMS_SERVER_MODE	If set to true, the DMS server (follower) is set to read-only.
READ_ONLY_OMS_SERVER_MODE	If set to true, the OMS server on the Follower subsystem is set to read-only.

16.2. NetServerOmsConfig.txt

Table below describes the parameters that are specific to the Outage Management System (OMS) server. All other parameters are identical to the ADMS main server parameters described in the preceding table.

Note

Parameters followed by an asterisk (*) are for developer use only.

Table 17. NetServerOmsConfig.txt Parameters

PARAMETER	VALUE
REAL-TIME SERVER MODE	
REALTIME	This flag is used to distinguish between real-time server mode and study mode. The difference is in how the ADMS server application obtains energization information. In real-time mode, it uses SCADA. In study mode, it creates a fictitious high-side energy source for each station. For a real-time server or simulator server, set the value to1. For a study server, set the value to0.
MAIN_SERVER_HOST	The host(s) where the production NetSrv process is running.
MAIN_SERVER_PORT	The TCP/IP listening port for the production system's NetSrv process.
SERVER_PORT	The TCP/IP port that NetServer listens on. This is the port to which the client connects.

PARAMETER	VALUE
OMS_ENGINE_V2_VERSION*	Specifies whether NetServer uses the Version 2 processing (future functionality). This parameters is used for development and testing purposes. This parameter must always be commented out or disabled in production environment.
MAIN PATHS	
MODEL_PATH	The path to the directory where the model files are placed.
STRUCTURES_PATH	The path to the directory where the structure files are placed.
DYNAMIC_PATH	The path to the directory where the dynamic data are placed. For example, %IDMS_ROOT%\dynamics\Server\OMSServer.
THREAD POOLS	
THREADS_OMS	The number of worker threads in a thread pool in the OMS module.
PARAMETERS TO DEFER PROCESSING NTP REQUESTS	
OMS_NTP_DEFER_AFTER_DYNCREATE_SECONDS	Time period (in seconds) to defer NTP processing after dynamics are re-created. Deferring NTP processing is required to avoid re-creating incidents after re-creating dynamics during the midnight dynamics compression when a big number of outages exist (for example, during storm conditions).
OMS_NTP_DEFER_AFTER_DYNCREATE_INCIDENTS	Time period (in seconds) to defer NTP processing after the specified number of incidents are created.
PERMISSIONS	
REQUIRE_DISPATCH_AND_GEOGRAPHIC_PERMISSION	If enabled (=1), the user must have both DISPATCH functional permissions and Geographic AOR permissions to confirm a device to a new state.
DISPATCH_FUNCTION_AOR_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is DISPATCH. If it is not defined, the permissions will go back to the default Geographic AOR mode.
REQUIRE_SWITCHING_AND_GEOGRAPHIC_PERMISSION	If enabled (=1), the user must have both Switching functional permissions and Geographic AOR permissions to confirm a device to a new state. This parameter is used to enable the OMS Interface server to operate switching devices.
SWITCHING_FUNCTION_AOR_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is SWITCHOP. If it is not defined, permissions will go back to the default Geographic AOR mode. This parameter is used to enable the OMS Interface server to operate switching devices.
OUTAGE_PLANNING_FUNCTION_AOR_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is PLOUTAGE. Leave this value blank to disable permission checking for creating planned outages.
ADHOC_CREW_FUNCTION_AOR_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is ISADHOC. Leave this value blank to disable permission checking for creating crew, employee, and vehicle records from the OMS Display plugin.

PARAMETER	VALUE
OUTAGE_INCIDENT_FUNCTIONAL_PERMISSION_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is OUTAGE. Leave this value blank to disable permission checking for managing outage incidents and operating protective devices.
HISTORY_CORRECTION_FUNCTION_AOR_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is HISTCOR. Leave this value blank to disable permission checking for incident history correction operations.
CUSTOMER_MOVE_FUNCTION_AOR_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is MOVECUST. Leave this value blank to disable permission checking for moving customers and canceling customer move operations.
INCIDENT_EDIT_ETR_AOR_NAME	Set to match the name of the functional AOR defined in PERMIT. The default name is EDITETR. Leave this value blank to disable permission checking for setting and updating an incident's Estimated Time for Restoration (ETR).
ALLOW_MANUAL_DEVICES_SWITCHING	If enabled (=1), OMS server supports toggling nonprotective devices, such as manual switches and elbows. The devices can be toggled by OMS-only clients and from the Update Incident display (using the Confirm Device In button).
DEFAULT_ORDER_ORGANIZATION_NAME	The name of the work organization that new incidents and orders are assigned to, if an organization cannot be determined using rules specified using the OMS Server Configuration Editor.
TRUSTED USERS	
SECRET_STORE_TRUSTED_USER_OIS_OBJECT_KEY	CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the OMS Interface Server trusted user ID defined in the Enterprise Vault Server. The format is <DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>.
SECRET_STORE_TRUSTED_USER_WFM_OBJECT_KEY	CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the Workforce Management System trusted user ID defined in the Enterprise Vault Server. The format is <DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>.
TRUSTED_USER_ID_OIS	Trusted user ID for the OMS Interface server. Required for implementing OMS commands via the OMS Interface server.
TRUSTED_USER_NAME_OIS	Trusted user name for the OMS Interface server. Required for implementing OMS commands via the OMS Interface server.
DEFAULT_OMS_SYSTEM_USERID	Default user ID for the OMS server.
AMI PARAMETERS	

PARAMETER	VALUE
AMI_STATUSES_TIMEOUT	Time period, in seconds, after the last AMI communication time to reset AMI statuses, such as AMI Ping In, AMI Ping Out, and AMI Meter Low Voltage.  Important AMI statuses are not reset until the associated incident is cleared by the operator. This parameter only applies if the incident is immediately cleared by the operator after the origination.  Note The AMI Power In status is reset by timing out in Customer-Only mode or when there is an incident associated with a customer. The AMI Power Out status is reset by timing out only in the customer-only mode.
AMI_IN_STATUS_WAIT_PERIOD	Time period, in seconds, to wait for AMI In messages after clearing the incident. For example, there is a fuse outage with 10 customers affected, and of these customers, two requested callbacks. When the fuse is energized, the system immediately calls back the two customers who requested callback. After that, the system waits for AMI_IN_STATUS_WAIT_PERIOD for any AMI restore messages. If three AMIs sent AMI In messages, the system will ping five ($10 - 2 - 3 = 5$) meters and wait for AMI_PING_TIMEOUT time for getting ping success messages. If four ping success messages from these meters are received, the system will then call back one ($5 - 4 = 1$) of these customers to make sure that the customer has power. If the callback verified that customers are de-energized, single customer incidents are created for them. Therefore, the process verifies step by step the restoration for all the customers.
AMI_PING_TIMEOUT	Time period, in seconds, to wait for a ping response before declaring that a ping failed.
AMI_PING_INTERLEAVE_TIME	Time period, in seconds, to wait for an inter ping response before declaring that a ping failed.
RETRY_PING_WAIT_PERIOD	Time period in seconds to wait before performing a meter ping attempt after the previous ping attempt, if a meter failed to respond. This value must be less than OMS_PENDING_CALLBACK_PERIOD.
MAX_METER_PING_RETRY_COUNT	The number of times to retry meter pinging if the meter fails to respond (the meter status is NotTried). If multiple retries need to be performed, adjust the OMS_PENDING_CALLBACK_PERIOD parameter accordingly.  Note This number of retries does not include the original try.
MAX_METER_PING_TO_SEND	The number of AMI Meter Ping requests to send at a time. If set to 0, the number of requests is not limited.

PARAMETER	VALUE
POWEROUT_METER_ON_VOLTAGE_PING_RESPONSE	If enabled (=1), a meter voltage below the threshold specified by the MIN_METER_VOLTAGE parameter is considered a meter power out event and an incident is created for the associated customer. If disabled (=0), meter voltage below the threshold is not considered a meter power out event.
MIN_METER_VOLTAGE	Meter voltage threshold for detecting a meter power out event and creating an incident.
OMS SERVER LISTENING SOCKET	
CIS_IVR_INTERFACE_ENABLED	A flag to enable (=1) or disable (=0) the interface to the Call server.
CALLSERVER_PORT	The TCP/IP port that the Call server listens on. This is the port to which the Call server connects.
CALLSERVER_HOST	The host where the Call server is installed and listening to other interfaces.
OMS_NET_QUERY_INTERVAL_SECONDS	Time period in seconds at which NetServer queries the Call server for new data.
CALLSERVER_CHECK_MESSAGE_SEQUENCE	If enabled (=1), checking for the sequence number of Call server messages is enabled.
CALLSERVER_CHECK_MESSAGE_SECURITY	If enabled (=1), checking for the Call server security token authenticity/expiration is enabled.
CALLSERVER_USE_CREDENTIAL_FILE	If enabled (=1), using the Credential file for defining the private key and user SIDs for the Call server is enabled.
CALLSERVER_CREDENTIAL_INFO_FILE	The file name and path of the Credential Info file that stores the valid SIDs and secret shared key.
CALLSERVER_USE_TLS	If enabled (=1), the TLS protocol is enabled for the Call server TCP/IP connection.
CALLSERVER_TIME_TO_LIVE	Time period, in seconds, for Call server messages to live on the network.
WFM_INTERFACE_ENABLED	A flag to enable (=1) or disable (=0) a WFM interface to the Crew server.
WFM_INTERACTION_DISABLED_FOR_MANUALLY_ASSIGNED_CREWS	A flag to enable(=0) or disable(=1) WFM interaction for manually assigned Crews.
CREWSERVER_PORT	The TCP/IP port that the Crew server listens on. This is the port to which the Crew server connects.
CREWSERVER_HOST	The host where the Crew server is installed and listening to other interfaces.
CREWSERVER_HOST_QUERY_PERIOD	Time period in seconds at which the Crew server queries the host for new data.
CREWSERVER_CHECK_MESSAGE_SEQUENCE	If enabled (=1), checking for the sequence number of Crew server messages is enabled.
CREWSERVER_CHECK_MESSAGE_SECURITY	If enabled (=1), checking for the Crew server security token authenticity/expiration is enabled.
CREWSERVER_USE_CREDENTIAL_FILE	If enabled (=1), using the Credential file for defining the private key and user SIDs for the Crew server is enabled.
CREWSERVER_USE_TLS	If enabled (=1), the TLS protocol is enabled for the Crew server TCP/IP connection.
CREWSERVER_CREDENTIAL_INFO_FILE	The file name and path of the Credential Info file that stores the valid SIDs and secret shared key.

PARAMETER	VALUE
CREWSERVER_TIME_TO_LIVE	The time period, in seconds, for Crew server messages to live on the network.
WFM_DONT_APPLY_VEHICLE_RADIO_CREW	If disabled (=0), the system applies a vehicle radio number, received in an UpdateVehicle message to a crew that is associated with the vehicle in the crew model. Both the crew and the vehicle radio numbers are updated in the OMS displays. If enabled (=1), vehicle radio number updates received in UpdateVehicle messages are not applied to the crews that are associated with the vehicles.
WFM_DONT_APPLY_VEHICLE_RADIO_EMPLOYEE	If disabled (=0), the system applies a vehicle radio number, received in an UpdateVehicle message to an employee that is associated with the vehicle in the crew model. Both the employee and the vehicle radio numbers are updated in the OMS displays. If enabled (=1), vehicle radio number updates received in UpdateVehicle messages are not applied to the employees that are associated with the vehicles.
WFM_CUSTOMER_COUNT_FOR_INCIDENT_UPDATE	The maximum number of customer records to be included in the IncidentUpdate messages sent to the WFM system. The customer records are sorted by customer call priority and call time.
WFM_NONOUTAGE_INCLUDE_PRIORITY_AND_CUSTOMER	If enabled (=1), the IncidentUpdate messages for non-outage incidents include all customer data (CustomerList array) and the priority data.
WFM_ORG_LEVEL_FOR_OP_CENTER	Controls the org level that is to be communicated as the "OpCenter" in the IncidentUpdate message of the Workforce Management System Interface. An integer (n) from 1 to 9 specifying the ORGLEVELnNAME field to be passed as the value for OpCenter. If unspecified or invalid, the default is 1.
OMS SERVER	
OMS_QUERY_THREADS	The number of query threads that OMS runs for resolving client queries.
OMS_ARCHIVE_CUSTOMER_COUNTS	A flag to enable (=1) or disable (=0) extracting and writing ARCHIVECUSTOMERCOUNT messages with customer counts by feeder, station, service center, and city to the archive and incident journals. The messages are only written after a new DNOM station model file or customer file is loaded. The messages are written once, at 1:00 AM. If enabled, the Archive application also stores ARCHIVECUSTOMERCOUNT in the Archive database.
OMS_ARCHIVE_FEEDER_UPDATES	A flag to enable (=1) or disable (=0) extracting and writing ARCHIVEFEEDERUPDATE messages with feeder updates (total kVA served values by feeders) to the archive and incident journals. The messages are only written after a new DNOM station model file or customer file is loaded. The messages are written once, at 1:00 AM. If enabled, the Archive application also stores ARCHIVEFEEDERUPDATE in the Archive database.
OMS_SNAPSHOT_ENTRIES_THRESHOLD	Defines the maximum number of journal entries in the snapshot file. When a snapshot is written and the file reaches the maximum number of entries, the file is saved with the .snapshot_Part<N> suffix, and the next entries are written to the new file with the .snapshot_Part<N+1> suffix. The snapshot chunking threshold should be set to a reasonably high value to avoid overburdening the system with too many chunks at normal conditions and only use snapshot chunking when a very large number of incidents appear in the system (for example, during a storm). The default value is 500,000, which is equal to around 30,000 new incidents on the demo model (Fantasy Island).

PARAMETER	VALUE
OMS_NUMBER_OF_DAILY_SNAPSHOTS_REQUIRED	The number of daily snapshots required by the system. This number is equal to the number of days that history data is kept in the system and can be used for history correction. This number should be the same in all server configurations. Snapshot creation can be disabled using the OMS_DISABLE_SNAPSHOT_CREATION configuration parameter in the NetDataServerConfig.txt file.
OMS_ENABLE_HISTORY_GROUPS	If enabled (=1), management areas hierarchy group statistics are collected for each area defined in the region schema, as well as for each station and feeder. One hierarchy group includes all outage device types. Hierarchy groups are used for summary tables (for example, Service Center Summary and Station Summary) on the OMS Display plugin. If the summary tables are not used, disabling the hierarchy group calculation decreases the amount of data sent to the client.
OMS_IGNORE_AREA_NOTE_START_TIME	If enabled (=1), an area setting affects incidents that existed before the area setting's start time. If disabled (=0), an area setting does not affect incidents that existed before the area setting's start time.
DELETE_AREA_SETTINGS_ON_EXPIRATION	If enabled (=1), area settings are automatically deleted after the specified expiration time. If disabled (=0), area settings are preserved. For compatibility with *e-terra*_distribution_3.7 and earlier, set this parameter to enabled (=1).
UPDATE_INCIDENT_ETR_WHEN_AREA_NOTE_DELETED	If enabled (=1), the incident's Area Settings ETR is deleted and the ETR is updated based on the current conditions when the area setting that affected the incident is deleted. If disabled (=0), the incident's Area Settings ETR and ETR are preserved after the area settings is deleted. The default value is disabled (=0).  Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
MIN_DISTANCE_FOR_VEHICLE_LOCATION_UPDATES	The minimum distance for vehicle location updates to be stored in the OMS database and written to the crew journal. The unit of measure depends on the GIS. The default value is 500.
WFM_AVL_CAPABLE	If enabled (=1), the WFM system can provide crew location.
WFM_CANCEL_ONSITE	If enabled (=1), an incident is cancelled in the WFM system if an operator cancels the incident in ADMS while there are crews on site.

PARAMETER	VALUE
WFM_UPDATE_FIELDS	Lists the fields from the Incident journal whose manual update triggers sending an IncidentUpdateWfmV1 message from the Crew Server to the WFM system (applies to incidents that have Work Order ID and incidents in the Pending Dispatched, Complete or Sent To WFM statuses). For example, enter "ETR,DISPATCHERCOMMENTS" to issue an IncidentUpdateWfm message when an operator changes incident ETR or adds comments.  Tip To find the required field name, you can make the associated incident update and look for the field name in the Incident Update journal message in the Incident journal.  Note Not all incident fields are supported. Incident changes can also trigger a CrewUpdateWfmV1 message to each assigned MDT crew. This option is configured by the WFM_CREWUPDATE_FOR_INCIDENT_CHANGE parameter.
WFM_CREWUPDATE_FOR_INCIDENT_CHANGE	If enabled (=1), a CrewUpdateWfmV1 message is sent to each assigned MDT crew along with the IncidentUpdateWfmV1 for any incident changes that are specified by the WFM_UPDATE_FIELDS parameter. If disabled (=0), only IncidentUpdateWfm is sent.
WFM_SEND_UPDATES_IF_WORKORDERID_IS_SET	If enabled (=1), the Crew Server sends IncidentUpdateWfmV1 messages to WFM for incidents that already have a Work Order ID, but are not yet assigned to a crew. If disabled (=0), the Crew Server does not send IncidentUpdate messages for such incidents. The WFM_UPDATE_FIELDS parameter configures the list of fields whose manual update triggers sending an IncidentUpdateWfmV1 message.
WFM_CREWUPDATE_USE_CREW_ID	If disabled (=0), the CrewId value of the CrewUpdateWfmV1 message is set to AutomaticDispatchId. If enabled (=1), the CrewId value of the CrewUpdateWfmV1 message is set to CrewId.
RELEASE_ALL_WORK_FROM_CREW	If disabled (=0), the assigned crew's work will not be released automatically. If enabled (=1), the assigned crew's work will be released automatically.
RELEASE_ALL_WORK_FROM_CREW_STATUS_CODES	This Configuration is a list of Short Names taken from coded translation editor's ARMSStatus tab. The comma-separated values are the Short Names for which all the work for a crew can be released. The default values are Unavailable,SignOff.  Note If RELEASE_ALL_WORK_FROM_CREW is enabled and the current crew status matches any of the short names listed in RELEASE_ALL_WORK_FROM_CREW_STATUS_CODES, then all work assigned to that crew will be released.

PARAMETER	VALUE
OVERRIDE_CREW_STATUSES _ON_MODEL_UPDATE	If disabled (=0), the existing crew statuses are retained when updating the .crew file. If enabled (=1), the existing crew statuses are overridden when updating the .crew file. The default value is 0.
OMS_SNAPSHOT_TIMER_INTERVAL	Time period in minutes for creating snapshot files.
OMS_SERVER_PORT	The TCP/IP port that the OMS server listens on. (Managed connection port.) This is the port to which the clients connect, such as OMS Interface server and Net Data server. This port is also required for the Site Overview display. (Servers and clients must connect to the OMS server (NetServer) managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
OMS_SERVER_USE_STREAMS	If enabled (=1), the OMS server uses the stream OmsConnection interface. (This setting must be set to the same value in all configuration files.)
OMS_SERVER_USE_TLS	If enabled (=1), the OMS server uses the TLS security protocol. (This setting must be set to the same value in all configuration files.)
OMS_SERVER_TLS_SUBJECTNAME	The TLS certificate subject name's CN value.
OMS_SERVER_CHECK_MESSAGE_SECURITY	If enabled (=1), the OMS server checks for security token authenticity/expiration. (This setting must be set to the same value in all configuration files.)
OMS_SERVER_CHECK_MESSAGE_SEQUENCE	If enabled (=1), the OMS server checks message sequence numbers. (This setting must be set to the same value in all configuration files.)
OMS_SERVER_USE_CREDENTIAL_FILE	If enabled (=1), the OMS server uses a credential file for defining the private key and user SIDs. (This setting must be set to the same value in all configuration files.)
OMS_SERVER_CREDENTIAL_INFO_FILE	Path to the credential file. (This setting must be set to the same value in all configuration files.)
OMS_SERVER_TIME_TO_LIVE	Time to live, in seconds, for OMS server connections.
SITE_OVERVIEW_WITH_VIEWCLIENT	A flag to enable (=1) or disable (=0) showing clients connected to the server through the NetDynCtrl object on the Site Overview display.

 **Note**

This parameter is absent in the NetServer configuration file and can be added manually, as needed.

You can enable showing these clients by adding the "WITH_NETCLIENT = 1" filter to the "Where" content query field of the grid display file. For example:

```

SELECT APPID, APPTYPE, PORT, DYNSEQ, DYNGUID, SYNCAGE, CONFIRMED, PMV,
MISMATCHMOD, LASTMSGTIME, ALIVE, USEPROXY, ROLE, INCIDENTJT, CALLJT, CREWJT,
CALLSERVERJT

FROM RT.SiteOV

WHERE WITH_NETCLIENT = 1

```

If this filter is specified in the grid display file, it overrides the server settings.

PARAMETER	VALUE
OMS_DEENERGIZED_LINES_COLLECTION_ENABLED	A flag to enable (=1) or disable (=0) writing journal files which include data about de-energized lines, predicted out lines, and cuts by substation to a location specified by the OMS_DEENERGIZED_LINES_FILE_PATH parameter.
	<i>i</i> Note De-energized lines without affected customers are not written to the journal.
OMS_DEENERGIZED_LINES_COLLECTION_INTERVAL_MS	Specifies the time period (in milliseconds) before creating a new journal file with information about currently de-energized lines.
OMS_DEENERGIZED_LINES_FILE_TIMEOUT_MIN	Specifies the time period (in minutes) for keeping old journal files with information about de-energized lines.
OMS_DISABLE_LOADING_OF_NEW_DAILY_SNAPSHOT_FILE	When the value is set to true, NetServer Standby will not load the daily snapshot file.

OMS JOURNAL AND CONFIGURATION FILES

OMS_CALL_JOURNAL_FILE_PATH	The path to the directory where CallJournal files are placed.
OMS_ARCHIVE_JOURNAL_FILE_PATH	The path to the directory where ArchiveJournal files are placed.
OMS_INCIDENT_JOURNAL_FILE_PATH	The path to the directory where IncidentJournal files are placed.
OMS_CREW_JOURNAL_FILE_PATH	The path to the directory where CrewJournal files are placed.
OMS_ORDER_JOURNAL_FILE_PATH	The path to the directory where OrderJournal files for follow-up work orders are placed.
OMS_SNAPSHOT_FILE_PATH	The path to the directory where snapshot files are placed.
OMS_DEENERGIZED_LINES_FILE_PATH	The path to the directory where DeenergizedLineJournal files with information about de-energized lines are placed.
OMS_MANAGED_CONFIGFILE	The full path to the OMS server configuration file.
OMS_CODED_TRANSLATION_CONFIGFILE	The full path to the Coded Translation configuration file.
OMS_REALTIME_INCIDENT_JOURNAL_FILE_PATH	The path to the directory where Real Time Incident Journal files with Near Real-Time (NRT) data are placed.
	<i>i</i> Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
OMS_WRITE_REALTIME_INCIDENT_JOURNAL	A flag to enable (=1) or disable (=0) writing RealTimeIncidentJournal files. This option is disabled (=0) by default.
	<i>i</i> Note This parameter is absent in the NetServer configuration file and can be added manually, as needed. It is recommended that Real Time Incident Journal files are written by Net Data server.

OMS JOURNAL RETIREMENT

OMS_RETIREMENT_JOURNAL_AND_SNAPSHOT_FILE_PATH	Path to the folder where retired journals and snapshots are stored.
---	---

PARAMETER	VALUE
OMS_RETIREMENT_AGE _DAYS_FOR_JOURNAL_AND _SNAPSHOT_FILES	Period, in days, at which the file retirement is delayed after the files are not used by the system.
OMS JOURNAL FLUSHING	
OMS_JOURNAL_FLUSHING _INTERVAL	Specifies the time period (in milliseconds) for flushing buffered journal file contents to disk.
	<i>i</i> Note Depending on the operating system, flushing may occur more frequently than the specified time period.
OMS JOURNAL INCIDENT LOCK	
OMS_SPECIAL_LOCK_PROCESSING	If disabled (=0), incident locking information is written to the Incident journal. If enabled (=1), incident locking information is only stored in memory.
OMS DATA FILES	
OMS_POLYGON_GROUPS	The full path to the polygon groups file. This file defines the boundaries of cities, service centers, zip code areas, roads, cities, and other shapes. This file is typically placed in the %IDMS_ROOT%\models\RT folder. These polygons are required to set the City and Service Center values for unassociated calls. If the polygons are not defined, unassociated calls are not placed within the hierarchy. The file referenced by this parameter also must be copied to the ADMS read-only server from the ADMS production system.
OMS_CUSTOMER_FILE	The full path to the customer data file.
OMS_CREW_EMPLOYEE_FILE	The full path to the crew data file.
OMS LOGGING	
OMS_SERVER_LOG_ENABLED	A flag to enable (=1) or disable (=0) logging for the OMS server. This option is enabled (=1) by default.
OMS_SERVER_LOG_ ANALYST_MESSAGES_ENABLED	A flag to enable (=1) or disable (=0) logging for the OMS server analyst messages. <i>i</i> Note This option must be disabled (=0) in the production system.
OMS_SERVER_LOGFILENAME	The full path to the file that is used for OMS server logging purposes, such as %IDMS_ROOT%\log\NetOMSServer.log.
OMS_SERVER_LOG_DAILY	If enabled (=1), OMS server creates a separate log file each day. For example, the OMS_SERVER_LOGFILENAME defines the log name as "NetOMSServer.log" and the LOG_DAILY flag is enabled. The log file name is NetOMSServer_20140708.log (date appended). When a new day begins, a new log file is created. This option is enabled (=1) by default.

PARAMETER	VALUE																											
OMS_SERVER_LOGSIZE_LIMIT	A numeric value specified in kilobytes. If set to 0, the file size is not limited. If a numeric number is greater than zero, when the current log file reaches the specified size, the file is renamed to <current log file name>_bak_N.log (where N is a number incremented for each new file) and a new log file is created. For example, when a log file NetOMSServer.log reaches the size limit, it is renamed to "NetOMSServer_bak_1.log" and a new empty "NetOMSServer.log" is created. Each log file is limited in size by the value specified by the OMS_SERVER_LOGSIZE_LIMIT parameter. When OMS_SERVER_LOG_DAILY is set to 1, the log file has the same behavior, except the file name also contains a date. The default value is 100000.																											
OMS_SERVER_LOG_LEVEL	An integer value in the range 0-90 that controls the types of log messages written to the log. Higher values result in more detailed log messages. The values correspond to Common Logging Service (CLS) Level values, even though CLS is not currently used for logging in ADMS. Each log message has a level corresponding to its severity, as indicated by the following table: <table><tr><td>INFOALWAYS</td><td>=</td><td>10</td></tr><tr><td>FATAL</td><td>=</td><td>11-20</td></tr><tr><td>ERROR</td><td>=</td><td>21-30</td></tr><tr><td>WARNING</td><td>=</td><td>31-40</td></tr><tr><td>INFO</td><td>=</td><td>41-50</td></tr><tr><td>DEBUG</td><td>=</td><td>51-60</td></tr><tr><td>DEBUG1</td><td>=</td><td>61-70</td></tr><tr><td>DEBUG2</td><td>=</td><td>71-80</td></tr><tr><td>ALL</td><td>=</td><td>90</td></tr></table> For higher debug levels, the lower level messages are also written to the log. The default value is 50. The server log level may be changed at run time either via the Logging/Set Log Level menu item or via a command request in the following format: NETVIEW://OMS.RT/COMMAND/CONFIGURE?OMS_SERVER_LOG_LEVEL=<level> For example, the following command will cause only messages of severity INFOALWAYS, FATAL, or ERROR to be logged: NETVIEW://OMS.RT/COMMAND/CONFIGURE?OMS_SERVER_LOG_LEVEL=30	INFOALWAYS	=	10	FATAL	=	11-20	ERROR	=	21-30	WARNING	=	31-40	INFO	=	41-50	DEBUG	=	51-60	DEBUG1	=	61-70	DEBUG2	=	71-80	ALL	=	90
INFOALWAYS	=	10																										
FATAL	=	11-20																										
ERROR	=	21-30																										
WARNING	=	31-40																										
INFO	=	41-50																										
DEBUG	=	51-60																										
DEBUG1	=	61-70																										
DEBUG2	=	71-80																										
ALL	=	90																										

PARAMETER	VALUE
OMS_SERVER_SHOW_LOG_ITEMS	A flag to enable (=1) or disable (=0) the display of log items on the Net server RT log tabs. This option is disabled (=0) by default.

Note

You can toggle displaying the log messages at runtime by selecting the Show Log Items option from the Logging menu of Net server.

JOURNAL ENCRYPTION

OMS_ENCRYPT_ALL_JOURNALS	Specifies whether all OMS journals (Incident, Crew, Call, Order, RealTimeIncident, Archive, and Deenergized Line journals) are encrypted using the encryption key specified by the key scope and version parameters.
OMS_ENCRYPT_INCIDENT_JOURNAL	If enabled (=1), Incident journals are encrypted.
OMS_ENCRYPT_CREW_JOURNAL	If enabled (=1), Crew journals are encrypted.
OMS_ENCRYPT_CALL_JOURNAL	If enabled (=1), Call journals are encrypted.
OMS_ENCRYPT_ORDER_JOURNAL	If enabled (=1), Order journals are encrypted.
OMS_ENCRYPT_REALTIME_INCIDENT_JOURNAL	If enabled (=1), Real Time Incident journals are encrypted. Note This option applies only if NetServer is configured to write Real Time Incident journals using the OMS_REALTIME_INCIDENT_JOURNAL_FILE_PATH and OMS_WRITE_REALTIME_INCIDENT_JOURNAL parameters. (By default, real-time incident journals are written by Net Data Server and the encryption is configured by OMS_ENCRYPT_REALTIMEINCIDENT_JOURNAL in NetDataServerConfig.txt.)
OMS_ENCRYPT_ARCHIVE_JOURNAL	If enabled (=1), Incident journals are encrypted.
OMS_ENCRYPT_DEENERGIZED_LINE_JOURNAL	If enabled (=1), Deenergized Line journals are encrypted.
OMS_ENCRYPTION_KEY_SCOPE	Encryption key scope.
OMS_ENCRYPTION_KEY_VERSION	Encryption key version.
OMS_ENCRYPTION_KEY_STORAGE_TYPE	Encryption key storage type. Note This parameter is absent in the NetServer configuration file and must be added manually. If not defined, the system uses the EnvironmentVariableDpApi storage type.

OMS DEBUG

CALLSERVER_CONNECTION_DEBUG	A flag to enable (=1) or disable (=0) Call server connection debug logging. This option is disabled (=0) by default.
OMS_SERVER_LOG_ANALYST_DEBUG_INCIDENT_PROCESSING_ENABLED	A flag to enable (=1) or disable (=0) the extra debug logging related to incident processing. This option is disabled (=0) by default.

PARAMETER	VALUE
OMS DATABASE	
OMS_DB_EXPORT_PATH	The path to the directory where the OMS database will be exported if triggered.
OMS_DB_EXPORT_MAX_ENTRIES_PER_FILE	The number of entries written to each OMS database export file. Must be over 1000.
OMS_DB_EXPORT_ON_STARTUP	A flag to enable (=1) or disable (=0) the OMS database export when OMS starts. This option is disabled (=0) by default.
OMS_DB_EXPORT_WHEN_GOING_ENABLED	A flag to enable (=1) or disable (=0) the OMS database export when OMS is enabled. This option is disabled (=0) by default.
OMS_DB_EXPORT_WHEN_NEW_CUSTOMERFILE_IS_LOADED	A flag to enable (=1) or disable (=0) the OMS database export when a new customer file is loaded. This option is disabled (=0) by default.
OMS_DB_EXPORT_ON_SHUTDOWN	A flag to enable (=1) or disable (=0) the OMS database export when OMS is shut down. This option is disabled (=0) by default.
MESSAGE COMPRESSING	
OMS_SEND_COMPRESSED_MESSAGE	If enabled (=1), Net server messages are sent compressed. This decreases message size and improves performance. This parameter must be set with the same value with the CompressMessage and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION	If enabled (=1), Net server messages for managed connections are sent compressed. This decreases message size and improves performance. This parameter must be set with the same value with the CompressMessage and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
OMS_ISOLATED_INCIDENT_INVALID_CUSTOMER_CALL	If enabled (=1), the system creates an isolated outage incident for an outage call from an invalid or unlinked customer account. The system also ignores an AMI outage notification from an invalid or unlinked customer account. If disabled (=0), the system creates an unassociated call for a non-priority outage call and creates a non-outage isolated incident for a priority outage call from an invalid or unlinked customer account. The system also creates a single customer incident for an AMI outage notification from an invalid or unlinked customer account.
 Note The system always creates an unassociated call for a non-priority unassociated call and creates a non-outage isolated incident for a priority unassociated call from any customer account.	
OMS ALARMS	
OMS_ALARM_UNACKNOWLEDGED_MOMENTARY_INCIDENTS	If enabled (=1), Alarm messages are created for unacknowledged momentary incidents (incidents of type Momentary that have not been acknowledged by an operator, either explicitly or implicitly, and were created more than OMS_ALARM_INCIDENT_ACKNOWLEDGEMENT_WAIT_SECONDS ago).

PARAMETER	VALUE
OMS_ALARM_UNACKNOWLEDGED_SUSTAINED_INCIDENTS	If enabled (=1), Alarm messages are created for unacknowledged sustained incidents (incidents of type other than Momentary, Planned, Confirmed, or ManuallyConfirmed that have not been acknowledged by an operator, either explicitly or implicitly, and were created more than OMS_ALARM_INCIDENT_ACKNOWLEDGEMENT_WAIT_SECONDS ago).
OMS_ALARM_INCIDENT_ACKNOWLEDGEMENT_WAIT_SECONDS	The number of seconds to wait for incident acknowledgement before creating Alarm messages for unacknowledged incidents. Incidents may be acknowledged either explicitly by selecting the Acknowledge menu item from the context menu on the OMS incident display, or implicitly by updating an incident (for example, assigning a crew).
OMS_ALARM_INCIDENTS_AFFECTING_CRITICAL_CUSTOMERS	If enabled (=1), Alarm messages are created for incidents affecting critical customers. These Alarm messages are additional to Alarm messages for unacknowledged incidents.
OMS_ALARM_NON_OUTAGE_INCIDENTS	If enabled (=1), Alarm messages are created for nonoutage incidents.
OMS_ALARM_EMERGENCY_CALLS	If enabled (=1), Alarm messages are created for emergency (911) calls.
OMS_ALARM_CUSTOMER_SUSPENDED_OR_UNSUSPENDED	If enabled (=1), Alarm messages are created for suspending or unsuspending customers.
OMS_ALARM_UNDO_CONFIRMATION_PERFORMED_FOR_INCIDENT	If enabled (=1), Alarm messages are created for undoing an outage confirmation.
OMS_ALARM_AMI_OPERATION_ERROR	If enabled (=1), Alarm messages are created when user-requested AMI operation fail.
OMS_ALARM_IN_STORM_MODE	Specifies the types of OMS Alarm messages to create even in storm mode. Types are UnacknowledgedMomentaryIncidents, UnacknowledgedSustainedIncidents, IncidentsAffectingCriticalCustomers, EmergencyCalls, CustomersSuspendedOrUnsuspended, and UndoConfirmationPerformedForIncident. For this purpose, "storm mode" includes both "Storm" and "MajorStorm" modes, by service center.
OMS PREDICTION	
AUTO_RESET_BLOCK_PREDICTION	If enabled (=1), resets any Block Prediction settings if no downstream devices are predicted out and if no calls or AMI out messages exist on any downstream customers that are not confirmed as de-energized.
USE_PER_PHASE_PREDICTION	If disabled (=0), prediction rules will be applied to devices without regard to the phases of the customers or devices. This mimics old, pre-3.4 prediction behavior. The default value is enabled (=1).
AUTO_ROLLUP_OF_ALL_DOWNSTREAM_OUT	If enabled (=1), the system considers only the number of connected customers downstream of a device when calculating outage prediction. If disabled (=0), the system considers the total number of customers (both connected and disconnected) when calculating outage prediction. For example, if disabled, the system does not predict an outage for a fuse with two distribution transformers downstream, one of which is disconnected and another is predicted out. The default value is enabled (=1).

PARAMETER	VALUE
NO_PREDICTION_OF_GOOD_QUAL_DEVICES	If enabled (=1), only SCADA devices with Suspect or Bad quality measurement data can be predicted out. If disabled(=0), SCADA devices with Good, Suspect, or Bad quality measurement data, or Replaced data can be predicted out. The default value is enabled (=1).
<PROTECTIVE_DEVICE>_PREDICTION_RULE_SET (for example, BREAKER_PREDICTION_RULE_SET)	Prediction rule sets for breakers, reclosers, fuses, distribution transformers, and network protectors that define the basic behavior of the prediction engine. For more information, see Prediction Rules.
<PROTECTIVE_DEVICE>_PREDICT_REDUCTION (for example, BREAKER_PREDICT_REDUCTION)	Prediction reduction scaling factor that is applied to all threshold values once any phase of a device is predicted out. For more information, see Protective Device Parameter Sets.
<PROTECTIVE_DEVICE>_PREDICT_<PARAMETER> (for example, BREAKER_PREDICT_B1)	Weight and threshold values that configure the prediction engine behavior for a device type. For more information, see Protective Device Parameter Sets.
DO_ROLL_UP_IF_ALL_THE_CASCADE_CALLS_ARE_BENEATH_A_SINGLE_DEVICE	If disabled (=0), when a prediction rule is satisfied, prediction does not roll up to the next upstream protection device if all customer calls or AMI outage events appear under only one of multiple downstream protection devices. For example, prediction does not roll up to a recloser when all customer calls appear under one of several downstream fuses. If enabled (=1), when a prediction rule is satisfied, prediction rolls up to the next upstream protection device if all customer calls or AMI outage events appear under a single protection device. Note This parameter is absent in the NetServer configuration file and must be added manually. The default value is disabled (=0).
MERGING INCIDENTS	
OMS_ENABLE_INCIDENT_HOLD_ON_DISPATCH	If enabled (=1), a predicted incident or a single customer incident is "frozen" as long a crew is assigned to it. This does not allow the prediction to change to other devices in the network as other customer calls are received. In this case, if there are more calls to predict an upstream device out, then the OMS engine creates a new outage island for the device, which excludes the outage island for the "dispatched" (original) device. If disabled (=0), the OMS engine merges the original incident to a new incident with a new device of record, and re-assigns the crew to the new incident. Note Single customer incidents with assigned crews participate in outage prediction if OMS_AUTO_MERGE_XFR_ENABLE is enabled (=1).
OMS_DONOT_ALLOW_MERGING_TO_PENDING_CLEAR_INCIDENT	A flag to disable (=1) or enable (=0) the ability to merge incidents to an incident in the Pending Clear state.

PARAMETER	VALUE
OMS_ENABLE_AUTO_MERGE_COMMON_DEVICE_INCIDENTS	If enabled (=1), all predicted incidents on a device are automatically merged to a confirmed incident on the same device. For example, if a confirmed incident exists on phase A of a recloser and then a predicted incident appears on phase B of the recloser, these incidents will merge. In the resulting incident, separate restoration steps are created for the merged predicted incidents.
OMS_AUTO_MERGE_XFR_ENABLE	If enabled (=1), predicted stable and assigned single customer incidents are merged to a predicted or confirmed incident on an upstream distribution transformer. In addition, predicted stable and assigned single customer incidents still participate in the outage prediction for the upstream distribution transformer. In the resulting incident, separate restoration steps are created for the merged single customer incidents.
OMS_MERGE_LIMIT_WFM_ENABLED	If enabled (=1), the following specific incident merge and rollout behavior is applied when merging incidents assigned to WFM: • When merging incidents assigned to WFM, the master incident will have only one crew and one work order assigned (when merging, the master incident's crew and work order have priority over the child incident crews and work orders). • When several predicted incidents roll up into an incident on an upstream device, the work order and crew of the earliest merged incident will be assigned to the resulting incident (if the earliest incident does not have a work order and a crew, a single crew and work order from any of the other merged incidents will be assigned). • When predicted incidents are merged or rolled up, Crew Update and Incident Update messages are sent for all deleted incidents. • When prediction is blocked, the Incident Update message of the CancelIncident type is sent to WFM for the deleted incident before creating downstream incidents. If disabled (=0), the common incident merge and rollout behavior is applied: child crews and work orders are merged to the parent incident.
ALLOW_NON_OUTAGE_MERGE_WITH_PREDICTED	If enabled (=1), predicted incidents can be merged as a parent when merging with non-outage incidents. If disabled (=0), predicted incidents cannot be merged as a parent.
CHANGE_PREDICTED_INCIDENT_TO_PREDICTED_STABLE_AT_MERGING	If enabled (=1), predicted incidents are set to predicted stable after merging with non-outage incidents.
OMS_MERGE_PREDICTED_INCIDENTS	If enabled (=1), single customer and predicted incidents (excluding predicted stable ones) are merged to an incident on the upstream SCADA protective device after the incident changes from momentary to confirmed. If disabled (=0), single customer and predicted incidents become predicted stable when an incident on the upstream SCADA protective device changes from momentary to confirmed.
 Note This parameter is absent in the NetServer configuration file and must be added manually.	
The default value is disabled (=0).	

PARAMETER	VALUE
CLEARING INCIDENTS	
OMS_AUTOCLEAR_MOMENTARY_INCIDENTS	If enabled (=1), a restored momentary incident is auto-cleared after the time period specified by the OMS_MOMENTARY_OUTAGE_DURATION parameter. If disabled (=0), the restored momentary incident status changes to Pending Clear after the time period specified by the OMS_MOMENTARY_OUTAGE_DURATION parameter.
OMS_CLEAR_PREDICTED_MUST_CONFIRM	If enabled (=1), a predicted incident must be confirmed before it can be cleared. An incident is not changed to Pending Clear, and the Clear button is disabled until the device of record has been confirmed out, and then confirmed in.

PARAMETER	VALUE
FIELD_CLEAR_DEVICE_LEVEL	This option allows crews equipped with a Mobile Data Terminal (MDT) to toggle outage devices and clear incidents directly from the field. This option is not applicable to devices with SCADA status measurements. For enabled device types, this option works as follows: • The outage device state changes to Open or Disconnected after an MDT crew sends the Confirm Open message via the Workforce Management System. This option is not applicable to single customer incidents. • The outage device state changes to Closed or Connected and the incident status changes to Pending Clear after an MDT crew sends the Completed message (reports the power restoration) via the Workforce Management System. • The incident clears after an MDT crew sends the Cleared message via the Workforce Management System. The Confirm Device In operation is not required in this case. For non-outage incidents, the incident clears after an MDT crew sends the Completed and Cleared messages. The parameter value can be one of the following: • None: No devices can be toggled or cleared from the field. • Nonoutage: Non-outage incidents can be cleared from the field. • Customer: Non-outage and single customer incidents can be cleared from the field. • Transformer: Distribution transformer, single customer, and non-outage incidents can be cleared from the field. • Fuse: Fuse, transformer, single customer, and non-outage incidents can be cleared from the field. • Sectionalizer: Sectionalizer, fuse, transformer, single customer, and non-outage incidents can be cleared from the field. • Recloser: Recloser, sectionalizer, fuse, transformer, single customer, and non-outage incidents can be cleared from the field. In addition, when device incidents are enabled for field clearing, predicted incidents on devices change to Pending Clear when AMI-In signals are received from any downstream service point meters for a device that is predicted out. This option overrides the OMS_CLEAR_PREDICTED_MUST_CONFIRM setting. This setting may be changed at run time via a command request in the following format: <pre>NETVIEW://PLUGIN.RT/OMSDISPLAYPLUGIN/COMMAND/CONFIGURE?FIELD_CLEAR_DEVICE_LEVEL=<DeviceLevel></pre>

CONFIRMED INCIDENT TIMERS	
OMS_BREAKER_LOCKOUT_DURATION	Time period in seconds after which an unrestored momentary incident becomes a confirmed incident.
OMS_SINGLE_CUSTOMER_SIMMER_PERIOD	Time period in seconds for the call to wait before it becomes a single customer incident.
OMS_AUTO_CONFIRM_TIMER	Time period in seconds for a predicted or a single customer incident to wait before it becomes confirmed.

PARAMETER	VALUE
OMS_FIX_RECLOSER_INCIDENT	If enabled (=1), a breaker incident upstream of a recloser incident is automatically cleared and the switching and restoration steps with customers from the breaker incident are moved to the recloser incident after the breaker is re-energized. This may occur if both a recloser and a feeder breaker trip because of a fault, and the breaker recloses while the recloser remains open.  Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
MOMENTARY INCIDENT TIMERS	
OMS_CLEAR_OUTAGES_CALLS_FOR_MOMENTARY	If enabled (=1), the OMS Engine removes outage calls associated with a momentary incident when the incident is removed.
OMS_INCLUDE_MOMENTARY_INCIDENT	If enabled (=1), momentary incidents are shown on the Incident Detail tab.
OMS_MOMENTARY_OUTAGE_DURATION	Momentary outage duration that is used for performance indices calculation.
PREDICTED STABLE INCIDENT TIMERS	
OMS_PREDICTED_STABLE_TIMER_SINGLE_CUSTOMER_INCIDENT	Time period in minutes before moving a Single Customer incident to the Predicted Stable state.
OMS_PREDICTED_STABLE_TIMER_PREDICTED_TF_INCIDENT	Time period in minutes before moving a predicted incident on a distribution transformer or transformer bank to the Predicted Stable state.
OMS_PREDICTED_STABLE_TIMER_PREDICTED_FUSE_INCIDENT	Time period in minutes before moving a predicted incident on a fuse to the Predicted Stable state.
OMS_PREDICTED_STABLE_TIMER_PREDICTED_RECLOSER_INCIDENT	Time period in minutes before moving a predicted incident on a recloser to the Predicted Stable state.
OMS_PREDICTED_STABLE_TIMER_PREDICTED_SECTIONALIZER_INCIDENT	Time period in minutes before moving a predicted incident on a sectionalizer to the Predicted Stable state.
OMS_PREDICTED_STABLE_TIMER_PREDICTED_BREAKER_INCIDENT	Time period in minutes before moving a predicted incident on a breaker to the Predicted Stable state.
CUSTOMER CALL TIMERS	
OMS_PENDING_CALLBACK_PERIOD	Time period in seconds for an incident to be in the pending callback state. • To enable calling back all customers where the ping has failed and AMI In has not been received to verify that all customers are restored after a restoration step, enter a value greater than AMI_IN_STATUS_WAIT_PERIOD + AMI_PING_TIMEOUT. • To enable calling back only customers that have requested callback, enter a value less than AMI_IN_STATUS_WAIT_PERIOD + AMI_PING_TIMEOUT.

PARAMETER	VALUE
OMS_INCIDENT_CALLBACK_WAIT_DURATION	Time period in seconds before starting the callback process after a restoration step.  Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
OMS_AUTOPING_METER_WHEN_CUSTOMER_CALL	If enabled (=1), the customer meter is automatically pinged when the customer call is received. The ping status is shown on the network view and on the Customers tab of the Update Incident display. If disabled (=0), the customer meter is not pinged when the customer call is received.
OMS_NEW_INCIDENT_DELAY_AFTER_ENERGIZATION	During the time period (in seconds) after an incident goes pending clear, any new downstream calls recorded are associated with the original incident without triggering any change in its status. These associated calls are simply closed when the incident is cleared. If no new calls are recorded within this time period, the existing calls that are associated with the pending clear incident are cleared from the network view and from the incident. (The incident call count is preserved.) New calls that occur after this time period are recorded in new single customer or predicted incidents. When set to "0", all customer calls are always recorded in the original incident if the incident has not been cleared. The customer calls are not cleared from the network view and from the incident.  Note For pending clear incidents, all customer calls are cleared after the timeout if no new calls are received. Therefore, it is recommended to clear incidents within the timeout period to prevent losing call information.
OMS_INCIDENT_PROPERTY_UPDATE_INTERVAL	Time period (in seconds) for the OmsEngine to update incident properties, including call count data and CMI. Used for performance purposes. If set to "-1", incident properties are updated at the regular OmsEngine incident timer.
AUTO_CLEAR_SINGLE_CUSTOMER_INCIDENTS_ON_AMI_IN	If enabled (=1), single-customer incidents originated from an AMI-Out signal are automatically cleared after an AMI-In signal is received if no crews are assigned to the incidents. Incidents with assigned crews are set to Pending Clear. If disabled (=0), single-customer incidents are set to Pending Clear after an AMI-In signal is received.
AUTO_CLEAR_SINGLE_CUSTOMER_INCIDENTS_ON_CANCEL_CALL	If enabled (=1), single-customer incidents originated from a customer call are automatically cleared after the customer call is canceled. If disabled (=0), single-customer incidents are set to Pending Clear after customer calls are canceled.

PARAMETER	VALUE
AUTO_CLEAR_PROTECTED_DEVICE_ON_AMI_IN	If enabled (=1), predicted incidents on devices that are available for field clearing using the FIELD_CLEAR_DEVICE_LEVEL parameter are automatically deleted after AMI-In signals are received from all downstream service point meters if no crews are assigned to the incidents. Incidents with assigned crews and incidents on devices above the device specified by FIELD_CLEAR_DEVICE_LEVEL are set to Pending Clear. If set to False (=0), predicted incidents on devices are set to Pending Clear after AMI-In signals are received from all downstream service point meters.
INCIDENT_PENDING_CLEAR_EXCLUDE_INACTIVE_CUSTOMERS	If set to 1, when incident CurrentCustomer (which is active customer count) becomes 0, the incident becomes pending clear. If set to 0, when incident TotalCurrentCustomer (which includes both active and inactive customers) becomes 0, the incident becomes pending clear. Default value is 1.
UPDATE INCIDENT DISPLAY	
OMS_ETR_EXPIRY_TIME	Time period in minutes before the ETR end time, when the ETR field background color changes to the "Expiring" color (default is yellow).
OMS_ETR_EXPIRY_SOON	Time period in minutes before the ETR end time, when the ETR field background color changes to the "Expiring Soon" color (default is orange).
OMS_INCIDENT_LOCKING_TIMEOUT	Time period in minutes that is used to compare to the time span since the Update Incident display was opened.
OMS_INCIDENT_LOCKING_IDLETIMEOUT	Time period in minutes that is used to compare to the time span since the Update Incident display was last updated.
FOLLOW-UP WORK ORDERS	
OMS_DO_EXPIRE_OLD_ORDERS	If enabled (=1), follow-up work orders are automatically removed by the system after the specified number of days.
OMS_ORDERS_EXPIRATION_LIMIT	Specifies the number of days before removing follow-up work orders.
FOLLOW_UP_ORDER_COMPLETED_STATUS_CODE	The OrderStatus coded translation item key that defines the completed status for follow-up work orders. Used for archiving purposes.
WORK ORDERS	
WORKMODEL_WEBDB_ENABLED	If enabled (=1), NetServer queries the Enterprise OMS REST API for the order data.
WORKMODEL_SERVICE_API	Enterprise OMS REST API endpoint to get work order data to populate the Orders tab in the OMS display plugin.
WORKMODEL_EVENT_API	Event API for listening to work order updates.
WORKMODEL_TLS_ENABLED	Specifies whether the HTTPS TLS security is enabled for the Work Model Event Handler service connection for web clients and web socket clients.
WORKMODEL_TLS_SUBJECTNAME	The certificate subject name that is used when searching for client certificates to send to the Work Model Event Handler service.
WORKMODEL_TLS_IGNORE_REMOTENAME_MISMATCH	Specifies whether the remote server certificate name mismatch is ignored. If set to True, the remote certificate name can be different from the URL host name that is used to connect to the server.
WORKMODEL_GET_ORDERS_PAGE_SIZE	The number of rows in pages with order query results.
OTHER PARAMETERS	

PARAMETER	VALUE
USE_SERVICE_POINT_FOR_SINGLE_CUSTOMER_INCIDENT	If enabled (=1), the service point name is used as the name of the device of record for a single customer incident. If disabled (=0), the transformer name is used as the name of the device of record for a single customer incident. In addition, the AOR of the incident is set to the AOR of the transformer that supplies the customer.
RECALCULATE_ETR_AFTER_CREW_ASSIGNMENT	If enabled (=1), the incident's Estimated Time for Restoration (ETR) is recalculated when a crew is assigned to the incident.
SET_EARLIEST_ETR_FOR_MERGED_INCIDENTS	If enabled (=1), the earliest ETR among the merged incidents is assigned to the parent incident. If disabled (=0), the parent incident's ETR is kept when merging incidents.
ETR_CHANGE_THRESHOLD_FOR_CALLBACK_ENABLED	If enabled (=1), the server issues automatic callbacks to customers who requested callback if an ETR change exceeds the threshold specified by the ETR_CHANGE_THRESHOLD_FOR_CALLBACK parameter. Applies only to ETR changes due to incident merges.
ETR_CHANGE_THRESHOLD_FOR_CALLBACK	ETR change threshold value (in minutes) for the server to issue automatic callbacks to customers who requested callback. Applies only to ETR changes due to incident merges.
OMS_CLEAR_EXISTING_INCIDENTS_ITR_ON_AREA_NOTE_ITR_DISABLED	If enabled (=1), the system clears ITRs for incidents that already exist in an area where an area setting with the ITR Disable option is created. If disabled (=0), the ITRs are preserved.
AUTOMATICALLY_CONVERT_PLANNED_OUTAGE_TO_INCIDENT	If enabled (=1), the system and turns the planned outage into a Planned Outage incident at the scheduled start time. If disabled (=0), the system the does nothing at the scheduled start time. You can automatically turn a planned outage into an incident from the planned outage context menu. Note that the AUTOMATICALLY_CONVERT_PLANNED_OUTAGE_TO_INCIDENT_WHEN_ASSOCIATED_DEVICE_IS_OPEN value must be able to do so.
AUTOMATICALLY_CONVERTED_PLANNED_OUTAGE_SHOULD_NOT_OPEN_DEVICE	If enabled (=1), when automatically turning the planned outage into a Planned Outage incident at the scheduled start time (configured by AUTOMATICALLY_CONVERT_PLANNED_OUTAGE_TO_INCIDENT), the system does not open the device of record. If disabled (=0), when automatically turning the planned outage into a Planned Outage incident at the scheduled start time, the system opens the device of record.
AUTOMATICALLY_CONVERT_PLANNED_OUTAGE_TO_INCIDENT_WHEN_ASSOCIATED_DEVICE_IS_OPEN	Must be enable (=1) when disabling the AUTOMATICALLY_CONVERT_PLANNED_OUTAGE_TO_INCIDENT to automatically turn a planned outage into an incident from the planned outage context menu.
ALLOWED_NUMBER_OF_PLANNED_OUTAGES	Specifies the allowed number of planned outages for a given period for a customer. The default value is 1. If the value set to -1, the validation check is disabled.

 **Note**
 Ensure that ALLOWED_NUMBER_OF_PLANNED_OUTAGES is set to same value as in [NetDataServerConfig.txt](#).

PARAMETER	VALUE
NUMBER_OF_DAYS_PRIOR_TO_PLANNED_OUTAGE_CREATION	Specifies the number of days prior to the start of a planned outage creation. The default value is 30 days. Note Ensure that NUMBER_OF_DAYS_PRIOR_TO_PLANNED_OUTAGE_CREATION is set to same value as in NetDataServerConfig.txt.
UPDATE_PLANNED_OUTAGE_CUSTOMER_LIST_ON_LOAD_OF_CUSTOMER_FILE	If enabled (=1), when a new customer file is loaded and the file includes customer updates downstream of the planned outage device, the planned outage customer list is updated to the nominal list of customers downstream of the planned outage device. The exception is a planned outage where the operator manually modified the customer list or changed the device of record. If disabled (=0), the planned outage customer list is not updated when a new customer file is loaded.
OMS_IGNORE_DEVICE_OPERATION_TIME_THRESHOLD	Specifies the threshold (in minutes) for ignoring device operation times reported by the dynamics file. If the dynamics report indicates an earlier or later device operation time, and the difference exceeds this threshold, then OMS will disregard the reported time and retain the current time as the incident start time. If this value is not set or is less than or equal to 0, OMS will always use the device operation time from the dynamics report. Default Value: 30 minutes This option helps to avoid setting the incident start time to incorrect times in the past, which may result in having high CMI values.
OMS_OBJECT_NOT_OFFSET	Specifies the distance between an OMS icon (for example, an incident or planned outage icon) and the associated devices in pixels. The distance is calculated as 125 multiplied by the specified value.
CUSTOMER_PHONENUMBER_FORMAT	Specifies the customer phone number format. The format can include the AREACODE, PHONENUMBER, PHONEPREFIX, and PHONESUFFIX query fields. The default format is (%AREACODE)%PHONEPREFIX-%PHONESUFFIX. For example, (555)589-0773. The formatted phone number can be queried using the AREACODEPHONENUMBER query field.
OMS_CREATE_INCIDENTS_WHEN_NO_CUST_OUT	If enabled (=1), the system creates a non-outage isolated incident when opening a device de-energizes a part of the network, but does not result in a customer outage. Also, when selecting the Create Predicted Incident option for a protective device without downstream customers, the prediction halo is shown but the predicted incident is not created. If disabled (=0), the system does not create an incident when opening a device de-energizes a part of the network, but does not result in a customer outage. Also, when selecting the Create Predicted Incident option for a protective device without downstream customers, an error is shown.
CLIENT CONNECTION	
CLIENT_HELLO_TIMEOUT	The number of seconds to wait before disconnecting when Hello messages are not responded to. The default value is 180seconds.
HELLO_SEND_PERIOD	The frequency (in seconds) at which Hello messages are sent.

PARAMETER	VALUE
SITEOV_DEAD_ENTRY_STAY_TIME	Time period (in seconds) since the last status update before an application is considered "dead" and the application connection entry is removed from the ADMS Site Overview grid tabular display.
SITE_OVERVIEW_UPDATE_PERIOD	Time period (in seconds) between application status info requests. This setting determines the frequency of data updates for the ADMS Site Overview grid tabular display.
FAST DYNAMICS	
DYNUPDATE_CACHE_SIZE	Dynamics update cache size.
BULK CLOSE	
OMS_BULKCLOSE_RETRY_THRESHOLD_COUNT	Specifies the maximum number of times the Bulk Close Engine will check whether the Task status has been updated and closed. The maximum allowed value is 15.
OMS_BULKCLOSE_MAXBACKOFF_TASKUPDATE	Defines the maximum wait duration is calculated by multiplying OMS_BULKCLOSE_MAXBACKOFF_TASKUPDATE by OMS_BULKCLOSE_RETRY_THRESHOLD_COUNT. Ensure the value does not exceed the limit. The maximum allowed value for OMS_BULKCLOSE_MAXBACKOFF_TASKUPDATE is 30 seconds.
OMS_BULKCLOSE_MAXCONCURRENT_BULKCLEARJOBS	Indicates the number of concurrent bulk close jobs that can run at any given time. The default value is 5.
OMS_BULKCLOSE_PARALLEL_PROCESS_THREAD_COUNT	Indicates the number of concurrent orders processed per request. The value depends on CPU capacity and the available core. The default value is 10.
OMS_BULKCLOSE_FAILURE_RETRY_THRESHOLD_COUNT	Indicates the number of times the bulk close process will retry closing orders that were not cleared due to delay in task updates. The default value is 1, and the maximum allowed value of 3.
OMS_BULKCLOSE_FAILURE_RETRY_STATIC_DELAY	Specifies the minimum delay before re-attempting to close orders that were not cleared due to task update delays. The default value is 5 seconds, and the maximum allowed value is less than 10 seconds.

16.3. NetServerDnafConfig.txt

The following table describes the parameters that are specific to the Distribution Network Analysis Functions (DNAF). All other parameters are identical to the ADMS main server parameters described in [NetServerMainConfig.txt](#).

Note

Parameters followed by an asterisk (*) are for developer use only.

Table 18. NetServerDnafConfig.txt Parameters

PARAMETER	VALUE
DNAF SERVER	
DNAF_SERVER_RESULTS_PATH	The path to the directory that the DNAF server uses for writing the DNAFSRV files (for transferring application results from the DNAF server to the Main server). The LST files, which are loaded by the DNAF server on start-up, are also located in this directory.

PARAMETER	VALUE
SEARCH	This parameter provides the file path and name that the Main server uses for online updating of certain files including the DNAF_PARAMS_DNAF.xml file (updated using the DNAF Configuration Editor application). The default value is as follows: <code>\$DNAF_INPUT_PATH\$DNAF_PARAMS_DNAF.xml,\$MODEL_PATH\$.sm,\$DNAF_INPUT_PATH\$Manual_Entry.dat</code>
MAIN_SERVER_DYN_FILE	 Note This parameter is absent in the DNAF server configuration file and can be added manually, as needed. The default value is <code>mainserverdynamic.sav</code>
DNAF_OUTPUT_PATH	The path to the directory where the DNAF result files, which contain solution results and debugging results, are placed.
MAIN_SERVER_HOST	Host name of the main server.
MAIN_SERVER_PORT	TCP/IP port used by the main server.
RETURN_TO_NORMAL_BY_RESTART	A flag to enable (=1) or disable (=0) the DNAF server to reinitialize itself (restart) after the midnight dynamics compression.  Note This parameter is absent in the DNAF server configuration file and can be added manually, as needed. The default value is enabled (=1).
SERVICE MODULE LISTENING SOCKET	
SERVICE_PORT	TCP/IP port that is used for the DMS service module. The DMS service module on the DNAF server is used by the DNOM2HDB application to retrieve DNAF data from the DNAF server.
SERVICE_SOCKET_OUTPUTBUF_MAX_SIZE	The maximum output buffer size. The default value is 6000000.
SERVICE_SOCKET_WOULDBLOCK_RETRY	The default value is 2.
SERVICE_SOCKET_MAX_CLIENT_MES_SIZE	The maximum client message size in bytes. The default value is 640000.
DYNAMIC SCHEDULES	
DYNAMIC_SCHEDULE_PATH	The path to the directory where the dynamic schedule data is placed. The dynamic schedule data for all devices in the system for a given period of time is stored using a file name as follows: <code>dynSchedule_<Year>_<month>_<day>_<hour>_<minute>.dsc</code> For example, <code>dynSchedule_2016_8_2_10_0.dsc</code>.
DYNAMIC_SCHEDULE_PERIOD	Time period in minutes to create dynamic schedule points. The value of this parameter must be the same for all servers that use dynamic schedules (for example, DNAF server).

PARAMETER	VALUE
DYNAMIC_SCHEDULE_CACHE_SIZE	The number of future dynamic schedule points to keep in memory (cache). Typically, only a limited number of dynamic schedule data points are loaded in the system for balancing the server memory footprint and performance. As time passes, cached data is automatically updated, unloading or loading data as needed based on the current point.
DYNAMIC_SCHEDULE_QUERY_ENABLED	A flag to enable (=1) or disable (=0) dynamic schedule data querying from the main server. If enabled, the server automatically synchronizes the cached dynamic schedule data with the main server within the specified time period. This option must be enabled when secondary servers (DNAF server, standby server, or read-only server) need dynamic schedule data. When dynamic schedule information is synchronized (schedule data appears on the DNAF server), the schedules are used for DPF/BLA calculations.
DYNAMIC_SCHEDULE_QUERY_PERIOD	Time period in minutes for the secondary server to check the main server to synchronize dynamic schedule data. To ensure that DNAF functions process the correct dynamic schedule data, this parameter should be based on the dynamic schedule data cache size and schedule period. It is recommended to maintain the following condition: $DYNAMIC_SCHEDULE_PERIOD * DYNAMIC_SCHEDULE_CACHE_SIZE > 2 * DYNAMIC_SCHEDULE_QUERY_PERIOD$ In addition, the synchronization is also automatically triggered when the secondary server receives the notification that the dynamic schedules were updated from the main server.
DYNAMIC_SCHEDULE_LINEAR_MODE	A flag to enable (=1) or disable (=0) interpolating dynamic schedule values between two points. Dynamic schedule data is typically used by DNAF server. The data is specified for discrete individual points of time. • If there is no dynamic schedule value for the current point, the lookup fails and returns no value. • If there is no dynamic schedule value of the next point, the lookup succeeds and return the value of the current point. • If there are dynamic schedule values for the current and next points, the lookup succeeds and returns the interpolated value of these two points based on the time spent in between. Note This parameter is absent in the DNAF server configuration file and can be added manually, as needed. The default value is enabled (=1).

SPECIFY DNAF FUNCTION USING DNA SERVER

PARAMETER	VALUE
DNA_SCHEDULER_HOST	Host name or IP address of the DNA Scheduler server.
DNA_SCHEDULER_PORT	TCP/IP port used by the DNA Scheduler server.
DNAF_USE_DNA_DPF	Specify whether the DNAF application should be sent to the DNA server to execute. A flag to enable(=1) or disable(=0) sending a DPF request message to the DNA server.
DNAF_USE_DNA_PRV	Specify whether the DNAF application should be sent to the DNA server to execute. A flag to enable(=1) or disable(=0) sending a PRV request message to the DNA server.
DNAF_USE_DNA_SCC	Specify whether the DNAF application should be sent to the DNA server to execute. A flag to enable(=1) or disable(=0) sending a SCC request message to the DNA server.
DNAF_USE_DNA_FISR	Specify whether the DNAF application should be sent to the DNA server to execute. A flag to enable(=1) or disable(=0) sending a FISR request message to the DNA server.
DNAF_USE_DNA_LVM	Specify whether the DNAF application should be sent to the DNA server to execute. A flag to enable(=1) or disable(=0) sending a LVM request message to the DNA server.
DNAF_USE_DNA_AFR	Specify whether the DNAF application should be sent to the DNA server to execute. A flag to enable(=1) or disable(=0) sending a AFR request message to the DNA server.
DNAF_USE_DNA_PLOS	Specify whether the DNAF application should be sent to the DNA server to execute. A flag to enable(=1) or disable(=0) sending a PLOS request message to the DNA server.
DNAF_USE_DNA_FL	Specify whether the DNAF application should be sent to the DNA server to execute. A flag to enable(=1) or disable(=0) sending a radial FL request message to the DNA server.
DERM Kafka Adapter Server	
DERM_ADAPTER_HOST	Host name or ID address of the DERM Kafka Adapter.
DERM_ADAPTER_PORT	TCP/IP port used by the DERM Kafka Adapter.
DMS API SERVER RELATED SERVICES OPTIONS	
GRPC_SERVICE_PORT	TCP/IP port used by the DMS API gRPC server.
GRPC_SERVICE_SSL_CERT_FILE	The path to the directory where the SSL certificate file is placed.
GRPC_SERVICE_SSL_CERT_KEY_FILE	The path to the directory where the SSL key file is placed.
API_PLAN_EXECUTION_MODE	Plan execution mode to use for control plans created using DMS API Server (0=semi-automatic, 1=closed-loop)
API_PLAN_OPERATION_MODE_ID	Operation Mode Id to use for plans created using DMS API Server. The operation mode must be defined in DCE.
API_PLAN_LIFE_REAL_TIME	Specifies how many hours a control plan created via the DMS API Server will be kept in the DMS before it is automatically deleted.
DOTS DPF TRIGGERING FOR SPECIFIC EVALUATION TIME	
SPECIFIC_TIME_FOR_DPF_EVALUATION_ENABLED	A flag to enable (=1) or disable (=0) the ability to run the DPF solution with a specific evaluation time.
DOTS_DPF_EVALUATION_TIME_DEFINITION_FILE	The name of the file that contains the DPF evaluation time offset setting.

PARAMETER	VALUE
MANUAL TRIGGERING	
DNAF_MANUAL_TRIGGER	A flag to enable (=1) or disable (=0) manual triggering of DNAF applications.
PERIODIC TRIGGERING	
DNAF_PERIODIC_BURST_MODE	A flag to enable (=1) or disable (=0) the periodic event trigger burst mode. Burst mode determines how the stations are triggered throughout the DNAF_DPF_MIN_PERIOD time window. If burst mode is enabled, solution requests for all the stations/feeders are fired in a single burst at the beginning of each time period. If disabled, solution requests are distributed evenly throughout the time window. For example, with a system of 300stations and a DNAF_DPF_MIN_PERIOD of five minutes, burst mode requests the solution of all 300stations at the beginning of each five minute period. If burst mode is disabled, one station is solved each second throughout the five minute time window. Burst mode is an especially useful tool when doing performance testing to determine the true solution throughput of the system.
DNAF_DPF_PERIODIC_TRIGGER	A flag to enable (=1) or disable (=0) periodic triggering for the Distribution Power Flow (DPF) application.
DNAF_LVM_PERIODIC_TRIGGER	A flag to enable (=1) or disable (=0) periodic triggering for the Load and Voltage/VAR Management application.
DNAF_PRV_PERIODIC_TRIGGER	A flag to enable (=1) or disable (=0) periodic triggering for the Protection Validation application.
PERIODIC TRIGGERING INTERVALS	
DNAF_DPF_MIN_PERIOD	Time period in minutes at which each station/feeder is checked to determine whether a periodic DPF solution should be executed. If the load has changed by more than the maximum load change percentage from the previous solution, specified by the Triggering Load Percentage (%) parameter in the DNAF Configuration Editor, a solution is executed for the station.
DNAF_DPF_MAX_PERIOD	Time period in minutes at which a periodic execution of DPF is forced to run. If a solution has not been performed within the maximum periodic time interval, a forced solution is executed for the station.
DNAF_LVM_MAX_PERIOD	Time period in minutes at which a periodic execution of LVM is forced to run.
DNAF_PRV_MAX_PERIOD	Time period in minutes at which a periodic execution of PRV is forced to run. If the period is set to 0, the PRV periodic trigger is disabled.
DNAF_LVM_MIN_TIME_INTERVAL_VOLTQUALITY	Time period in minutes to check for an LVM solution. If LVM was already run within the specified time period, it does not run again, even if there are voltage violations for the measured voltages.
DNAF_TIMER_INITIAL_DELAY	Time period in seconds to wait after NTP startup before initiating DNAF triggering. This allows the user to tune the startup sequence so that DNAF solutions do not start until SCADA data has been received by ADMS (for example, 20seconds).

PARAMETER	VALUE
DNAF_TIMER_ADV_APPS_ADDITIONAL_DELAY	Time period in seconds to wait in addition to DNAF_TIMER_INITIAL_DELAY. This parameter is applicable to DNAF applications that issue controls to SCADA (for example, LVM). The objective is to allow sufficient time for the device status used in the optimization algorithms to be synchronized with the MAIN server. Note that this parameter affects DNAF periodic timers only. This parameter does not block user requests and event triggers.
DNAF_PERIODIC_DEADBAND	Time period in seconds, within which periodic application triggering for a station is skipped. The time period counts after an application is solved for a station or after an application is triggered for an adjacent station in the network selection. This prevents unnecessary periodic solutions to occur shortly after manual or event-triggered solutions. This option is currently supported for DPF and LVM only.
LOAD_AGGREGATION_PERIOD	Time period in seconds for refreshing the Demand Aggregation Summary display.
DNAF_ENERGY_SOURCE_TARGET_UPDATE_PERIOD	Time period in seconds for pre-calculating energy resource real and reactive power target values for grid tabular displays.
EVENT TRIGGERING	
DNAF_DPF_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) event triggering for the Distribution Power Flow application.
DNAF_PRV_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) event triggering for the Protection Validation (PRV) application.
DNAF_LVM_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) event triggering for the Load and Voltage/VAR Management (LVM) application.
DNAF_LVM_EVENT_VOLTQUALITY_TRIGGER	If set to 1, LVM triggers if any measured voltage is beyond the voltage high/low limit.
DNAF_LVM_EVENT_PLANEXECFAILURE_TRIGGER	A flag to enable (=1) or disable (=0) re-triggering LVM if a plan execution fails due to a device control failure.  Note LVM retriggering for a control failure also depends on the "Automatically Re-Execute for a Control Failure" setting in the DNAF Configuration Editor.
DNAF_AFR_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) event triggering for the Automatic Feeder Reconfiguration (AFR) application. After DPF runs for a station, the results are analyzed for any overloads on feeder main lines or power transformers. If any are found, AFR is executed to eliminate these violations.
DNAF_FISR_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) event triggering for the Fault Isolation and Service Restoration (FISR) application.
DNAF_FISR_FAULTIND_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) the fault indicator status change event trigger for FISR. This event trigger may also require FL application results (this option is configured by selecting the FISR FL Event Trigger option for the required stations in the DNAF System Configuration display).

PARAMETER	VALUE
DNAF_FISR_FL_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) the fault location event trigger for FISR.
DNAF_FISR_LOV_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) the loss of voltage event trigger for FISR.
DNAF_LOV_DELAY	Time period in seconds to wait before triggering the FISR application after a loss of voltage is detected.
DNAF_LOV_2ND_DELAY	Time period in seconds to wait after the initial DNAF_LOV_DELAY before triggering the FISR application if a loss of voltage is detected inside a substation.
DNAF_FL_EVENT_TRIGGER	A flag to enable (=1) or disable (=0) event triggering for the Fault Location application.
DNAF_FL_CHECK_MEAS_CHANGE	A flag that enables (=1) or disables (=0) two checks for Fault Location (FL) triggering: • Enabled (=1): FL triggers only if at least one fault current analog value changed since the previous FL run. It also ignores (does not trigger on) measurements whose timestamp age exceeds the DCE setting "Maximum Age of Fault Inputs" (stale measurements). • Disabled (=0): FL may retrigger even when fault current values are unchanged or stale. Set to 0 for demo systems with static fault currents (allows repeated FL retriggering at the same position). Set to 1 for production systems to prevent unnecessary retriggers on unchanged or stale fault current data. If this parameter is not set, the enabled flag is considered.
DNAF_FISRLOV_CHECK_MEAS_CHANGE	A flag that enables (=1) or disables (=0) measurement value change and timestamp age checks for FISR Loss of Voltage (LOV) event triggering: • Enabled (=1): Before triggering a FISR Loss of Voltage event, the system checks whether the voltage measurement timestamp is recent (within the age threshold defined by DNAF_MAXTIME_CHECK_MEAS_CHANGE). If the measurement is outdated (stale), the FISR LOV event trigger is skipped. • Disabled (=0): FISR Loss of Voltage event triggering does not check measurement staleness, allowing LOV triggers even on stale voltage measurements. Set to 0 for demo systems or environments where measurement timestamps are not reliably updated. Set to 1 for production systems to prevent unnecessary FISR LOV triggers on stale voltage data. The default value is enabled (=1).

PARAMETER	VALUE
DNAF_DPF_CHECK_MEAS_CHANGE	A flag that enables (=1) or disables (=0) measurement value change and timestamp age checks for DPF event triggering, specifically for sudden measurement change and tap position change triggers: • Enabled (=1): Before triggering a DPF event due to a sudden analog measurement change or a tap position change, the system checks whether the measurement timestamp is recent (within the age threshold defined by DNAF_MAXTIME_CHECK_MEAS_CHANGE). If the measurement is outdated (stale), the DPF event trigger is skipped. • Disabled (=0): DPF event triggering does not check measurement staleness, allowing DPF triggers even on stale measurements. Set to 0 for demo systems where measurement timestamps are static. Set to 1 for production systems to prevent unnecessary DPF retriggers on stale SCADA data. The default value is enabled (=1).
DNAF_LVM_CHECK_MEAS_CHANGE	A flag that enables (=1) or disables (=0) measurement value change and timestamp age checks for LVM voltage quality event triggering: • Enabled (=1): Before triggering an LVM voltage quality event, the system performs two checks: (1) whether the voltage measurement value has changed significantly compared to the previous value greater than a threshold (0.1 V) and (2) whether the measurement timestamp is recent (within the age threshold defined by DNAF_MAXTIME_CHECK_MEAS_CHANGE). If the value has not changed or the measurement is outdated (stale), the LVM event trigger is skipped. • Disabled (=0): LVM voltage quality event triggering does not check for measurement value changes or staleness, allowing LVM triggers even on unchanged or stale voltage measurements. Set to 0 for demo systems where measurement timestamps or values are static. Set to 1 for production systems to prevent unnecessary LVM triggers on unchanged or stale voltage data. The default value is enabled (=1).

PARAMETER	VALUE
DNAF_MAXTIME_CHECK_MEAS_CHANGE	The maximum measurement age, in seconds, for DNAF event triggering. When any of the measurement change check parameters (DNAF_FISRLOV_CHECK_MEAS_CHANGE, DNAF_DPF_CHECK_MEAS_CHANGE, or DNAF_LVM_CHECK_MEAS_CHANGE) are enabled, this threshold is used to determine whether a measurement is considered stale. If the difference between the current time and the measurement timestamp exceeds this value, the measurement is considered outdated and the corresponding DNAF event trigger is skipped.  Note For the Fault Location (FL) application, the maximum age is determined by the DCE setting "Maximum Age of Fault Inputs" rather than this parameter. This parameter applies to the FISR LOV, DPF, and LVM applications. The default value is 300 seconds (5 minutes).
DNAF_EVENT_DELAY	Time period in seconds to wait after an event trigger is received until a solution is initiated. This allows time for the system to settle before a solution is executed (for example, 10seconds). Continuing receipt of events within this delay period extends the delay with another DNAF_EVENT_DELAY period.
SPECIFIC TRIGGERING EVENTS	
DNAF_EVENT_TRIGGER_TOPOLOGY_BACKBONE	A flag to enable (=1) or disable (=0) a topology backbone switch, cut, temporary ground, or jumper status change as an event trigger. This trigger applies to DPF, PRV, and LVM.
DNAF_EVENT_TRIGGER_MEAS_CHANGE	A flag to enable (=1) or disable (=0) an event trigger for a SCADA sudden measurement change. This trigger applies to DPF only.
DNAF_EVENT_TRIGGER_TOPOLOGY_LATERAL	A flag to enable (=1) or disable (=0) a topology lateral switch, cut, temporary ground, or jumper status change as an event trigger. This trigger applies to DPF and PRV.
DPF_EVENT_TRIGGER_TAP_CHANGE	A flag to enable (=1) or disable (=0) a tap changer status change as an event trigger. This option applies to the Distribution Power Flow application.
DPF_EVENT_TRIGGER_CAP_STEP_CHANGE	A flag to enable (=1) or disable (=0) capacitor status change as event trigger. This option applies to the Distribution Power Flow application.
DPF_EVENT_TRIGGER_ENERGYSOURCE_STATUS_CHANGE	A flag to enable (=1) or disable (=0) generator or distributed energy resource status change as event trigger. This option applies to the Distribution Power Flow application.
BLOCKING EVENT TRIGGERS	
DNAF_AFR_BLOCK_SOLVE_TIMER	Do not run event-triggered AFR if the time elapsed since the last AFR solution is less than this value (in minutes).
DNAF_FL_BLOCK_SOLVE_TIMER	Do not trigger the Fault Location application by new analog fault current measurements if the time period since the last application execution is less than this value (in minutes).

PARAMETER	VALUE
DNAF_SCADA_ROC_BLOCK_TIME	If a SCADA Volt/Amp sudden measurement change (previously - Rate of Change) event for a particular station has triggered a DNAF application (for example, DPF), subsequent Sudden Measurement Change events for that station will be ignored for the period defined in this value (in minutes).
DNAF_LVM_BLOCK_TIMER	The LVM block timer indicates the time (in minutes) required for the network system conditions for any given station to stabilize after the plan manager has finished executing LVM plans for the station. This stabilization period prevents the LVM application on the DNAF server from reaching an incorrect failsafe decision and/or generating plan steps during the "station transition period" for the station.
FISR FAULT PLACEMENT LOCKOUT SETTINGS	
DNAF_FISR_MAX_LOCKOUT_WAITTIME	Maximum time period in seconds FISR waits for receiving a lockout status measurement for a tripped breaker or recloser.
DNAF_FISR_LOCKOUT_FAULTPLACEMENT	If enabled (=1), fault placement requires a valid lock out status on a tripped breaker/recloser. If the lock out is not available or suspect, the fault is not placed. If disabled (=0), FISR places a fault even with an invalid lock out status measurement on a tripped device.
DNAF_FISR_NO_LOCKOUT_WAITTIME	Time period in seconds to terminate the FISR triggering logic and place no fault if the lockout status has not been received.
DNAF_FISR_CHECK_LOCKOUT_PERIOD	Time period in seconds FISR checks for breaker or recloser lockout.
LVM	
DNAF_LVM_FAST_TRIGGER	If enabled (=1), LVM operates in fast trigger mode.
DNAF_LVM_FAST_TRIGGER_PERIOD	Time period in minutes for Optimization LVM fast trigger.
DNAF_HLVM_CAP_FAST_TRIGGER_PERIOD	Time period in minutes for fast triggering capacitors in capacitor control logic for Heuristic LVM.
DNAF_HLVM_TAP_FAST_TRIGGER_PERIOD	Periodic interval in minutes for fast triggering regulators in tap changer control logic for Heuristic LVM.
DNAF_HLVM_CAP_TAP_TRIGGER_PERIOD	Time period in minutes for switching between capacitor control and tap changer control logic in Heuristic LVM.
DNAF_LVM_MAX_CONSECUTIVE_FAST_TRIGGERS	The maximum number of fast triggers allowed. If fast-triggering is enabled (DNAF_LVM_FAST_TRIGGER = 1) and the number of consecutive fast triggers exceeds this limit before reaching the end of a Heuristic LVM execution cycle (capacitor control function, followed by LTC/regulator control function), then fast-triggering is internally disabled by the DNAF server. The subsequent LVM periodic triggers will follow the regular LVM periodicity (DNAF_LVM_MAX_PERIOD) until the end of the Heuristic LVM execution cycle. At the end of the cycle, the number of consecutive fast triggers is reset to zero. The next cycle, beginning with the capacitor control function, will follow the fast trigger timers.

PARAMETER	VALUE
DNAF_LVM_RESET_CAP_TAP_COUNT	Resets the capacitor/tap count to zero at midnight each day. This parameter is enabled by default.  Note To disable, add this parameter to the DNAF Server configuration file and set its value to 0.
DNAF_AMI_VOLTAGE_EXCEPTION_MINIMUM_CUSTOMERS	The minimum number of customers on LVM-enabled feeder required to generate the voltage quality trigger from the AMI source.
MANUAL ENABLE STATUS OVERRIDE	
LVM_USER_OVERRIDE	Determines if SCADA Enable and/or Auto Enable values override the Manual Enable values for substations, power transformers, or feeders for the LVM application in the Real-Time DNAF System Configuration. Up to two of the following options may be set: • SCADA_ON: When SCADA Enable is enabled, Manual Enable is set to enabled and cannot be edited by the user. • SCADA_OFF: When SCADA Enable is disabled, Manual Enable is set to disabled and cannot be edited by the user. • AUTO_ON: When Auto Enable is enabled, Manual Enable is set to enabled and cannot be edited by the user. • AUTO_OFF: When Auto Enable is disabled, Manual Enable is set to disabled and cannot be edited by the user.  Note This parameter must be set with the same value in the DNAF server configuration file.
FISR_USER_OVERRIDE	Determines if SCADA Enable and/or Auto Enable values override the Manual Enable values for substations, power transformers, or feeders for the FISR application in the Real-Time DNAF System Configuration. Up to two of the following options may be set: • SCADA_ON: When SCADA Enable is enabled, Manual Enable is set to enabled and cannot be edited by the user. • SCADA_OFF: When SCADA Enable is disabled, Manual Enable is set to disabled and cannot be edited by the user. • AUTO_ON: When Auto Enable is enabled, Manual Enable is set to enabled and cannot be edited by the user. • AUTO_OFF: When Auto Enable is disabled, Manual Enable is set to disabled and cannot be edited by the user.  Note This parameter must be set with the same value in the DNAF server configuration file.

PARAMETER	VALUE
AFR_USER_OVERRIDE	Determines if SCADA Enable and/or Auto Enable values override the Manual Enable values for substations, power transformers, or feeders for the AFR application in the Real-Time DNAF System Configuration. Up to two of the following options may be set: • SCADA_ON: When SCADA Enable is enabled, Manual Enable is set to enabled and cannot be edited by the user. • SCADA_OFF: When SCADA Enable is disabled, Manual Enable is set to disabled and cannot be edited by the user. • AUTO_ON: When Auto Enable is enabled, Manual Enable is set to enabled and cannot be edited by the user. • AUTO_OFF: When Auto Enable is disabled, Manual Enable is set to disabled and cannot be edited by the user.

 **Note**
This parameter must be set with the same value in the DNAF server configuration file.

DNAF DEBUG PARAMETERS

TRACE_LEVEL	The trace level controls the level of detail that is shown on the Analyst Message tab of the Net Server RT window. The following levels are currently supported: 0 = some server messages 1 = triggering messages 2 = summary iteration count and alarms 3 = internal DNAF performance 4 = additional, less-important internal DNAF performance messages 5 = configurator messages 6 = show all messages
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 **Note**
Each trace level shows messages for lower levels as well (for example, level 2 shows messages for 0, 1, and 2 levels).

This parameter can be changed at run-time via the following URL command on the client side:

NETVIEW://RT/SERVERDEBUG/LOG?LEVEL=5

The value of -1 is ignored as it is reserved by the system. Other values beyond the 0-6 range result in an error message.

PARAMETER	VALUE
TRACE_TYPE	This parameter is used to filter messages on the Analyst Message tab of the Net Server RT window by type. The following debug types are currently supported: 0 = TT_NONE 1 = TT_REGULAR 6 = TT_DNAF Types 2, 3, 4, and 5 can be entered, but no message types are currently assigned to them. This parameter can be changed at run-time via the URL commands on the client side. To define several trace types, use the following command. Trace types must be separated by commas. NETVIEW://RT/SERVERDEBUG/LOG?TYPES=1,6 To define only one trace type: NETVIEW://RT/SERVERDEBUG/LOG?TYPE=1 In the commands above, the value of -1 is ignored as it is reserved by the system. Other values beyond the 0-6 range result in an error message.

DNAF THREADS PARAMETERS	
THREADS_DNAF	The number of threads used for DNAF.
THREADS_DNAF_CBO	The number of threads used for the DNAF check-before-operate (CBO) function.
THREADS_NET	Number of network threads to handle incoming connections. The minimum is 1. The size of this pool must not be larger than the hardware core count of the system. For a system with a small number of concurrent client connections, setting this pool size to 1 is sufficient. Default is 1.
THREADS_DNAF_PRIORITY (*)	Operating system priority of DNAF worker thread. Values can range from -2 (highest priority) to 2 (lowest priority). Default is zero. This parameter is especially useful on small demo machines where the CPU may get overburdened with DNAF processing.
ENABLE_PROACTIVE_WORKER_THREAD	If enabled (=1), the DNAF and Query module worker threads actively pull messages from the Dispatcher module. If disabled (=0), the DNAF and Query module worker threads passively wait for the Dispatcher module to post messages to the DNAF and Query modules. Enabling this option evenly distributes tasks to each available worker thread, reduces the task queue, and improves system performance.

PARAMETER	VALUE
DNAF_ENABLE_DPF_CBO	If enabled (=1), the Distribution Power Flow Check Before Operate (DPF CBO) functionality is enabled. Client DPF CBO check requests are sent to the DNAF module. The Distribution Power Flow Check Before Operate (DPF CBO) functionality enables to run a DPF solution and identify abnormal power flow conditions before certain actions, such as implementing a switching step in the Switching Order.
DNAF FAST DYNAMICS PARAMETERS	
FD_FAULT_LOCATION_PERIOD	Time period in seconds for querying fault location results. A value of zero means this feature is disabled.
FD_POWER_FLOW_PERIOD	Time period in seconds for querying power flow data. A value of zero means this feature is disabled.
FD_DNAF_INPUTS_FILES	This parameter specifies files (with paths), separated by commas, that are needed when initializing a study from real time. The default files are: • \$DYNAMIC_PATH\$manuals.sav • \$DNAF_INPUT_PATH\$DNAF_PARAMS_DNAF.xml • \$DNAF_INPUT_PATH\$Manual_Entry.dat • \$DNAF_INPUT_PATH\$tftype.dat • \$DNAF_INPUT_PATH\$alg.dat • \$DNAF_INPUT_PATH\$optalg.dat • \$DNAF_INPUT_PATH\$msgdef.dat • \$DNAF_INPUT_PATH\$phid.dat The values inside the "\$" symbols specify paths that are provided by other configuration parameters.
FD_POWER_MEASUREMENT_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the customer meter power measurement query. This option is enabled (=1) by default.
FD_DER_MEASUREMENT_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the DER data query for the 2030.5 Gateway fast dynamics replication from the DMS server to DNAF server. This option is enabled (=1) by default.
RECLOSING DISABLED MEASUREMENT POINT PARAMETERS	

PARAMETER	VALUE
RECLOSING_DISABLED_POINT_ENABLED	A flag to enable (=1) or disable (=0) automatic "disarming" of all automation scheme team switching devices located on a feeder when a feeder breaker status is DA disabled, no restoration allowed, or reclosing disabled. This option is used to automatically disable normal load restoration operations for devices on a feeder to be shed so that the load that is shed by the Load Shed function is not be restored automatically by an automated switching scheme function or an ADMS automatic load restoration function. The "disarming" event is triggered by at least one of the following statuses on a breaker: StatusDAEnabled = 0, DAProhibit = 1, or DeviceReclosingEnabled = 0. "Disarming" issues the following controls to the DeviceReclosingEnabled, ProhibitRestoration, and GroundTripEnabled measurement points: • DeviceReclosingEnabled change from 1 to 0 (reclosing is disabled) • ProhibitRestoration change from 0 to 1 (no restoration allowed) • GroundTripEnabled change from 1 to 0 (ground trip protection turned off) (Reclosers have the DeviceReclosingEnabled point modeled; switches have the ProhibitRestoration point modeled.) In addition, this option specifies whether GroundTripEnabled is set to 0 for all automated switching scheme devices on a feeder if GroundTripEnabled =0 on a feeder breaker. This option is disabled (=0) by default.
ARM_DEACTIVATED_DEVICE_ENABLED	A flag to enable (=1) or disable (=0) automatic "rearming" of all "disarmed" automation scheme team switching devices located on a feeder when a feeder breaker has the following statuses: closed, DA enabled, restoration allowed, and reclosing enabled. This option is used to automatically enable normal load restoration operations for devices on a feeder after load shedding is finished. The "rearming" event is triggered when a breaker has all of the following statuses: Closed = 1, StatusDAEnabled = 1, DAProhibit = 0, and DeviceReclosingEnabled = 1. "Rearming" issues the following controls to the DeviceReclosingEnabled, ProhibitRestoration, and GroundTripEnabled measurement points: • DeviceReclosingEnabled change from 0 to 1 (reclosing is enabled) • ProhibitRestoration change from 1 to 0 (restoration is allowed) • GroundTripEnabled change from 0 to 1 (ground trip protection turned on) In addition, this option specifies whether GroundTripEnabled is set to 1 for all automated switching scheme devices on a feeder if GroundTripEnabled =1 on a feeder breaker. This option is disabled (=0) by default.
PLAN MANAGER	

PARAMETER	VALUE
PLAN_MANAGER_ENABLED	A flag to enable (=1) or disable (=0) writing the Plan Manager log. This option is enabled (=1) by default.
PLAN_MANAGER_CONSOLE	The default value is DMS.
PLAN_MANAGER_LOG_PATH	The path to the directory where the Plan Manager log file is placed.
PLAN_MANAGER_LOG_FILE	The name of the Plan Manager log file.
PLAN_MANAGER_PENDING_MANUAL_DER_PLAN_TIMEOUT	Time period in minutes for the DMS Interface server to send manual Distributed Energy Resource (DER) LVM plan steps to the 2030.5 Gateway. After this time period, the controls are considered not dispatched.

PLUGIN APPLICATIONS

APP_LVMMONITOR_ENABLED	A flag to enable (=1) or disable (=0) the LVM Monitor application, which provides a summary for monitoring system-wide LVM control events. Disabling this option prevents the application controller from trying to find and load the DLL on each "notify" request when the DLL has been deleted.
APP_LVMREADONLYMONITOR_ENABLED	A flag to enable (=1) or disable (=0) the LVM Monitor application in read-only mode, where you can create plan forecasts, but cannot implement them. Disabling this option prevents the application controller from trying to find and load the DLL on each "notify" request when the DLL has been deleted.
APP_LOADSHED_ENABLED	A flag to enable (=1) or disable (=0) the Load Shed application, which prioritizes and sheds circuits to obtain a desired target reduction in system load. Disabling this option prevents the application controller from trying to find and load the DLL on each "notify" request when the DLL has been deleted.
APP_FSD_ENABLED	A flag to enable (=1) or disable (=0) the Faulted Section Determination (FSD) application, which provides switching plans to isolate faulted sections and restore service to the non-faulted sections using a switching-based approach. Disabling this option prevents the application controller from trying to find and load the DLL on each "notify" request when the DLL has been deleted.

FAULTED SECTION DETERMINATION APPLICATION

FSD_CONTROL_COMPLETION_TIMER	Time period in seconds between implementing switching steps.
FSD_CHECK_FAULT_TIMER	Time period in seconds before checking for a fault after implementing a switching step.

DISTRIBUTION POWER FLOW CHECK BEFORE OPERATE (DPF CBO)

DPF_CBO_LOOK_AHEAD_DURATION	The look ahead duration, in hours, to find the peak loading time for the involved substations. The supported range is 0-24 hours.
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PARAMETER	VALUE
DPF_CBO_MAX_NUMBER_HIGH_VOLTAGE_REPORTED	Maximum number of top-ranking high load and bus voltages before and after operation to be reported by DPF CBO. Depending on the power flow solution, the maximum load and bus voltage locations reported before the operation may be entirely different than the maximum load and bus voltage locations reported after the device operation.
DPF_CBO_MAX_NUMBER_LOW_VOLTAGE_REPORTED	Maximum number of top-ranking low load and bus voltages before and after operation to be reported by DPF CBO. Depending on the power flow solution, the minimum load and bus voltage locations reported before the operation may be entirely different than the minimum load and bus voltage locations reported after the device operation.
DPF_CBO_MAX_NUMBER_BRANCH_LOADING_REPORTED	Maximum number of top-ranking branch loadings before and after operation to be reported by DPF CBO. Depending on the power flow solution, the maximum branch loading locations reported before the operation may be entirely different than the maximum branch loading locations reported after the device operation.

TEAM AND SCHEME	
TAS_ENABLE_FISR_INTEGRATION_WITH_LOCAL_SCHEMES	This parameter defines whether integration between ADMS switching reconfiguration applications (AFR and FISR) and local peer to peer schemes is enabled. If enabled (=1), FISR and AFR recognize the presence of local schemes and the FISR and AFR operation is delayed until the local schemes complete operation in the field and the new reconfigured topology is updated within ADMS. If disabled (=0 or not defined), FISR and AFR operation does not consider the presence of switching schemes, and the applications have the default behavior.
TAS_FISR_WAIT_FOR_LOCAL_SCHEMES_DELAY	Time delay in seconds for FISR to wait until local schemes complete their operation. This parameter only comes into effect after the MeasTransferInProgress SCADA measurement that indicates that the scheme completion in the field is received.
TAS_FISR_CAN_USE_TEAM_MEMBERS	Indicates whether FISR can utilize and control switching devices participating in teams. If enabled (=1), switches that are part of teams are eligible for FISR, after the switching schemes complete their operation. If disabled (=0 or not defined), switches that are part of teams are not eligible for FISR.

FORECAST SERVER (LVM ANALYSIS SERVER)	
APP_LVMMONITFORECAST_SERVER_MODE	Set to 0 for the ADMS server application running as a main server (NetServerMainConfig.txt). Set to 1 for a server running in the evaluation mode.
DNAF_SERVER_HOST	The host(s) where the DNAF server process is running.
DNAF_SERVER_PORTOR_ENABLED	The TCP/IP port that DNAF server listens on. This is the port to which the client connects.
DNAF_OUTPUT_PATH	The path to the directory where the LVM result files generated by the Forecast server (LVM Analysis server) are placed.

16.4. NetServerDnaConfig.txt

The following table describes the parameters that are specific to the Distribution Network Analysis Server (DNA Server). All other parameters are identical to the ADMS main server and ADMS DNAF server parameters described in [NetServerMainConfig.txt](#).

Table 19. NetServerDnaConfig.txt Parameters

PARAMETER	VALUE
DNA SERVER	
DNAF_DNA_WRITE_LOG_RESULT	Set to 0 to disable writing all application solution results *.dat files in the log folder. Set to 1 to write all application solution result files *.dat in the log folder.
DNA_WRITE_DNAF_RESULTS_ODS	This parameter defines whether DNAF results solved in DNA will be sent back to the DNAF Server or Ampere/ODS (Operational Data Store). By default, this parameter is set to 0, and DNAF results solved in DNA will be sent back to the DNAF Server. If it is set to 1, the DNAF result will be sent to Ampere/ODS. If it is set to 2, the DNAF result will be sent to Ampere/ODS and the DNAF Server.
ODS_DNAF_RESULT_DBNAME	DNAF result database name in Ampere/ODS.
DNA_ODS_CONNECTION_TIMEOUT	Time period in seconds before the connection between DNA server and Ampere/ODS server expires. By default, the parameter is commented and internally one second time out value is used.
DNA_ODS_DOPUT_RETRY_LIMIT	The number of retries for a DoPut operation of a DNA server, that a DNA server writes DNAF results to Ampere/ODS.
DNAF_SERVER_RESULTS_PATH	The path to the directory that the DNA server uses for writing the DNAFSRV files. For example, %IDMS_ROOT%\dynamics\Server\DNA\
DNAF_OUTPUT_PATH	The path to the directory where the application solution result files are placed. For example, %IDMS_ROOT%\log\DNA
SERVICE MODULE LISTENING SOCKET	
DNAF_SERVER_HOST	Host name or IP address of the DNAF server.
DNAF_SREVER_PORT	TCP/IP port used by the DNAF server.
DNA_SCHEDULER_HOST	Host name or IP address of the DNA Scheduler server.
DNA_SCHEDULER_PORT	TCP/IP port used by the DNA Scheduler server.
ODS_SERVER_HOST	Host name or IP address of the Ampere/ODS server.
ODS_SERVER_PORT	TCP/IP port used by the Ampere/ODS server.

16.5. NetServerSimConfig.txt

The following table describes the parameters that are specific to the ADMS Simulator (SIM) server. All other parameters are identical to the ADMS main server parameters described in [NetServerMainConfig.txt](#).

Table 20. NetServerSimConfig.txt Parameters

PARAMETER	VALUE
SIMULATOR SERVER	
THREADS_DYN	The number of worker threads in a thread pool in the dynamics module.
RELAYSERVER_CONNECTION_DEBUG	A flag to enable (=1) or disable (=0) the Relay server connection debug dynamics. The default value is 0.
DNAF_SERVER_FILE_SUFFIX	DNAF server file suffix (.DNAFSRV).
SCADA CONTROL	
CONTROL_VALUE_1 = RAISE,CLOSE,BLOCK, ON,ENABLE,AUTO,CPON, RESET,IN	The Simulator server CONTROL_VALUE parameters mimic the Control Function in SCADA modeling for a specific control.
CONTROL_VALUE_0 = LOWER,TRIP,OPEN, UNBLOCK,DISABLE,OFF, MANUAL,CPOF,OUT	For more details, refer to the <i>Distribution Operator Training Simulator (DOTS) User Guide</i> .
MULTISTATE_MEAS_TRANSITION_CONFIG	Path to the file which contains state transition definitions for multi-state SCADA switches.

Note

This file is optional. If specified, the SIM Server identifies switch state based on the provided definitions.

For each multi-state switch, the file can contain possible state transition entries in the following format:

```
<New_Switch_State> <Current_SDIS ><Current_XDIS> <New_SDIS><New_XDIS>
```

For example:

```
AHL
OPEN 10 00
OPEN 00 01
CLOSE 01 00
CLOSE 00 10

3STS
OPEN 10 11
OPEN 11 00
CLOSE 00 01
CLOSE 01 10
OPEN 10 00
```

For example, the "OPEN 10 00" entry specifies that when the switch status measurement changes from "SDIS = 1, XDIS = 0" to "SDIS = 0, XDIS = 0", switch status changes to Open. Alternatively, the "CLOSE 10 00" entry would specify that the switch status changes to Close when the switch status measurement changes similarly from "SDIS = 1, XDIS = 0" to "SDIS = 0, XDIS = 0".

MEASUREMENTS

PARAMETER	VALUE
SIM_MEAS_CALC_MODE	This parameter sets a measurement calculation mode, where the simulator server calculates and sends measurements to the SCADA RT server. Three modes are available: • Set to 1 for DPF-only (Distribution Power Flow) mode, where a measurement value is assigned based on the previous DPF result. • Set to 2 for Simplified mode, where the simulator server calculates measurements based on the nominal load, scale factors, and flow voltage curves. The results are then sent to SCADA RT server. • Set to 3 for Mixed mode. In Mixed mode, if the station model is detected as DNAF-incapable, the simulator server performs in Simplified mode. If the station model is DNAF-capable, the simulator server assigns the measurement value using the previous DPF results.
MEASUREMENT_MULTISTATE_FALSE_01_PNTNAM	This parameter specifies the tri-state SCADA measurements for multistate devices, values of which to be inverted. It is used to transform the measurement values to handle the difference between the SCADA and DNOM values. The default value is DAS2. By default, measurement point SCADA values for different device positions are: • Open: SDIS=0,XDIS=0 • Closed: SDIS=1,XDIS=0 • In Transit: SDIS=0,XDIS=1 • Illegal: SDIS=1,XDIS=1 When this parameter is enabled for a particular measurement, the measurement point SCADA values for different device positions are: • Open: SDIS=0,XDIS=1 • Closed: SDIS=1,XDIS=0 • In Transit: SDIS=0,XDIS=0 • Illegal: SDIS=1,XDIS=1

PARAMETER	VALUE
MEASUREMENT_INVERT_PNTNAM	This parameter specifies the measurement point types whose Scada values need to be inverted in ADMS (for example, when showing values on ADMS grid tabular displays). The parameter value is a list of measurement point types, separated by commas. Example: To invert measurement values for the point types MAN, SC, HT, and A: MEASUREMENT_INVERT_PNTNAM = MAN,SC,HT,A The default value is ARE,GTE. Note This parameter does not apply to the simulate-only status measurements (for example, a low-battery alarm). However, it does apply to SCADA controls sent for these simulate-only measurements. This parameter value must match the parameter values in NetServerOmsConfig.txt, NetServerMainConfig.txt, and NetServerStudyConfig.txt.
DISTRIBUTION OPERATOR TRAINING SIMULATOR (DOTS)	
NTP_FAULTIND_RESET_TIME	Time period in seconds to reset a telemetered fault indicator and clear the fault indicator halo. The default value is 300 (5 minutes). If both telemetered and manual fault indicators are utilized in the network section, the halo will remain until the period specified in parameter NTP_MANUAL_FAULTIND_RESET_TIME has expired.
NTP_MANUAL_FAULTIND_RESET_TIME	Time period in seconds to reset a manual fault indicator and clear the fault indicator halo. The default value is 300 (5 minutes). If both telemetered and manual fault indicators are utilized in the network section, the halo will remain until the period specified in parameter NTP_MANUAL_FAULTIND_RESET_TIME has expired.
ENABLE_RECLOSING	A flag to enable (=1) or disable (=0) the reclosing capability. For the instructor to create switch events that use a recloser or breaker, the breaker or recloser must be "reclosing enabled". Note When reclosing is disabled, the breaker/recloser lockout status measurements are set to True for the protective device upstream of a fault. This allows a downstream fault to trigger FISR without waiting for device reclosing in the real-time environment.
BREAKER_RECLOSE_SCHEDULE	Breaker reclosing schedule in seconds. VALUE1 is the time when the first breaker reclosing takes place after the initial fault. VALUE2 is the second reclosing time, counted after the first reclosing. VALUE3 is the third reclosing time, counted after the first reclosing.

PARAMETER	VALUE
RECLOSER_RECLOSE_SCHEDULE	Recloser reclosing schedule in seconds. VALUE1 is the time when the first breaker reclosing takes place after the initial fault. VALUE2 is the second reclosing time, counted after the first reclosing. VALUE3 is the third reclosing time, counted after the first reclosing.
ANALOG_MEASUREMENT_DEADBAND	The deadband for analog measurements.
ENABLE_BREAKER_OVERCURRENT_TRIP	A flag to enable (=1) or disable (=0) the feeder breaker overcurrent tripping in simulator mode. Note that to get the feeder breaker tripped, the phase trip limit in the model must be decreased. By default, the overcurrent tripping is disabled.
ENABLE_MANUAL_TRIP_LOCKOUT	If enabled (=1) the lockout status is set to "Locked Out" for a breaker or recloser when the device is manually tripped in simulator mode. This option should be disabled (=0) when changing the lockout status using the Event Scripter. Note In real-time mode, the lockout status is always set to "Locked Out" for a breaker or recloser when the device is manually tripped by the operator.
NTP_LOCKOUT_RESET_TIME	Time period in seconds to reset a breaker/recloser lockout status after close operation. The default value is 300 (5 minutes).
NTP_TEMPGROUND_REMOVE_TIME	Time period in seconds to remove a temporary fault after it is energized. The default value is 30.
NTP_THROW_TO_PREFER_TIME	Time period in seconds to set a throwover switch to the "preferred" state after topology processing. The default value is 120.
NTP_THROW_TO_ALTERNATE_TIME	Time period in seconds to set a throwover switch to the "alternate" state after topology processing. The default value is 3.
DOTS DPF TRIGGERING FOR SPECIFIC EVALUATION TIME	
SPECIFIC_TIME_FOR_DPF_EVALUATION_ENABLED	A flag to enable (=1) or disable (=0) the ability to run the DPF solution with a specific evaluation time in simulator mode. This feature enables to calculate DPF for the current network conditions for different dates with different loads schedule values. To specify the DPF evaluation time offset, use the DOTS Evaluation Time pop-up display. This display can be opened from the SIM menu.

PARAMETER	VALUE
DOTS_DPF_EVALUATION_TIME_DEFINITION_FILE	The name of the file that contains the SIM DPF evaluation time offset setting. The evaluation time file format is a one-line txt file. This file is automatically populated when the time offset is specified on the DOTS Evaluation Time pop-up display. The file can contain one of the following entries: • <code>offset=<Offset_number_of_seconds>; <State> <Description></code> • <code>paused=<Unix_Timestamp>; <State> <Description></code> Where: • offset: Specifies that DPF is run with the offset solution time. The offset is specified in seconds; positive and negative values are supported. • paused: Specifies that DPF is run once with the fixed solution time. After that, DPF is not triggered automatically. The solution time is specified in the Unix Timestamp format; future and past dates are supported. The file is updated each time the offset setting is updated on the DOTS Evaluation Time pop-up display. For example, the following entry appears in the file if SIM DPF is configured to run with a two-hour offset from the current time: <pre>offset = 7200:Started. Diff between DOTS and System time in seconds.</pre> The following entry appears in the file if SIM DPF is paused on January 27, 2021 05:43:00 AM: <pre>paused = 1611726180:Pause DateTime in seconds.</pre>

SERVER COUNTERS	
COUNTER_LOG_ENABLED	A flag to enable (=1) or disable (=0) the counter logging.
COUNTER_PERIOD	Counter period.
COUNTER_FILE_NAME	The name of the file that is used for the counter logging.

16.6. NetServerStudyConfig.txt

The following table describes the parameters that are specific to the ADMS Study server. All other parameters are identical to the ADMS main server parameters described in [NetServerMainConfig.txt](#), ADMS DNAF server parameters described in [NetServerDnafConfig.txt](#), and ADMS Simulator server parameter described in [NetServerSimConfig.txt](#).

Table 21. NetServerStudyConfig.txt Parameters

PARAMETER	VALUE
STUDY SERVERMODE	

PARAMETER	VALUE
REALTIME	This flag is used to distinguish between real-time server mode and study mode. The difference is in how the ADMS server application obtains energization information. In real-time mode, it uses SCADA. In study mode, it creates a fictitious high-side energy source for each station. For a study server, set the value to 0.
MAIN PATHS	
MODEL_PATH	The path to the directory where the model files are placed. For example, %IDMS_ROOT%\models\StudyServer\
DYNAMIC_PATH	The path to the directory where the dynamic data are placed. For example, %IDMS_ROOT%\dynamics\client\StudyServer\
SAVECASE_PATH	The path to the directory where the savecases are placed. Used on the study server only.
FD_PATH	The path to the directory where the fast dynamic data are placed. For example, %IDMS_ROOT%\dynamics\client\StudyServer\FD
LOG_FILE_NAME	The full path to the file that is used for studysrv logging purposes.
FD_DNAF_INPUTS_FILES	This parameter specifies files (with paths), separated by commas, which are needed when creating a study save case and retrieving save case. The default files are: • \$DYNAMIC_PATH\$manuals.sav • \$MODEL_PATH\$DNAF_PARAMS_DNAF.xml • \$MODEL_PATH\$Manual_Entry.dat • \$MODEL_PATH\$tftype.dat • \$MODEL_PATH\$alg.dat • \$MODEL_PATH\$optalg.dat • \$MODEL_PATH\$msgdef.dat • \$MODEL_PATH\$phid.dat • \$MODEL_PATH\$Manual_Entry.dat The values inside the "\$" symbols specify paths that are provided by other configuration parameters.
DNAF_SERVER_RESULTS_PATH	The path to the directory that the Study server uses for writing the DNAFSRV files. For example, %IDMS_ROOT%\dynamics\client\StudyServer\
DNAF_OUTPUT_PATH	The path to the directory where the DNAF result files, which contain solution results and debugging results, are placed. For example, %IDMS_ROOT%\dynamics\client\StudyServer\Output

16.7. NetServerReadOnlyConfig.txt

The following table describes the parameters that are specific to the ADMS Read-Only (RO) server. All other parameters are identical to the ADMS main server parameters described in [NetServerMainConfig.txt](#).

Table 22. NetServerReadOnlyConfig.txt Parameters

PARAMETER	VALUE
READ-ONLY SERVER	
MAIN_SERVER_HOST	The host(s) where the production NetSrv process is running.
MAIN_SERVER_PORT	The TCP/IP listening port for the production system's NetSrv process.
SERVER_PORT	The TCP/IP port that the NetServerRO listens on. This is the port to which the clients connect, such as OMS Interface server.
DNAF_RESULTS_QUERY_ENABLED	If enabled (=1), DNAF results are replicated from the DNAF server to read-only servers. If disabled (=0), DNAF results querying and replicating is disabled. Server performance slows down when sending DNAF results to multiple read-only servers. This option helps improve server performance by disabling sending DNAF results to those read-only servers that do not require them.
DNAF_SERVER_HOST	Host name of the DNAF server. This parameter is used for querying DNAF results from the DNAF server.
DNAF_SERVER_PORT	TCP/IP port used by the DNAF server. This parameter is used for querying DNAF results from the DNAF server.
PRESERVE_ALL_DYNAMICS	A flag to enable (=1) or disable (=0) creating new entries for device status changes in the dynamics.sav file. By default, this option is disabled and existing entries are updated with new device statuses. This option is used on read-only servers to keep all dynamics entries. This allows creating complete Event Scripter scripts by the Snapshot Converter tool.  Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
DISABLE_MID_NIGHT_DYNAMIC_COMPRESSION	A flag to enable (=0) or disable (=1) midnight dynamics compression, which removes all duplicated actions in the dynamics.sav file at midnight to optimize file size. (The midnight compression starts within 1-2 minutes after a date changes.) This option is used on read-only servers to keep all dynamics entries. This allows creating complete Event Scripter scripts by the Snapshot Converter tool.  Note This parameter is absent in the NetServer configuration file and can be added manually, as needed.
OMS_SERVER_PORT	The TCP/IP port that the NetServerRO listens on to support client OMS capabilities. (Managed connection port.) This is the port to which the clients connect, such as OMS Interface server, Netviewer Adapter, and Net Data Replicator.
OMS_DISABLE_LOADING_OF_NEW_DAILY_SNAPSHOT_FILE	When the value is set to true, NetServerRO will not read the daily snapshot file.

16.8. NetServerMainCustomerMode.txt

The following table describes the parameters that are specific to the ADMS Customer-Only (CM) server. All other parameters are identical to the ADMS server parameters described in [NetServerMainConfig.txt](#), [NetServerOmsConfig.txt](#), and [NetServerDnafConfig.txt](#).

Note

Parameters followed by an asterisk (*) are for developer use only.

Table 23. NetServerMainCustomerMode.txt Parameters

PARAMETER	VALUE
CUSTOMER ONLY MODE PARAMETERS	
CUSTOMERONLY_MODE	If set to 1, the server runs in customer-only mode.
OMS_METER_VOLT_QUERY_INTERVAL_SECONDS	Time period in seconds for querying meters for voltage measurements.
CALL_STATUSES_TIMEOUT	Time period in seconds for querying call status changes.
FD_POWER_MEASUREMENT_QUERY_ENABLED	A flag to enable (=1) or disable (=0) the customer meter power measurement query.
FD_POWER_MEASUREMENT_PERIOD	Time period in seconds for the server to query the customer meter power measurement data from the main server.
OMS_EXCLUDE_SWITCHING_AND_RESTORATION_STEP_PREDICTED_INCIDENT (*)	If enabled (=1), creating switching and restoration steps for predicted incidents is disabled.
OMS_INCIDENT_POS_MODE	Deprecated parameter. The default value is 0.

16.9. CallServerConfig.xml

The Call Server is configured by [CallServerConfig.xml](#), which includes the following parameters:

Warning

XML files are case-sensitive. Use all lower-case letters when configuring Boolean settings.

- **ApplicationParameters:** Call server application parameters.
 - **CallServerCallJournal:** Path to the Call server journal files location.
 - **EncryptionKeyScope:** Encryption key scope.
 - **EncryptionKeyVersion:** Encryption key version.
 - **EncryptionKeyStorageType:** Encryption key storage type.

Note

This parameter is absent in the configuration file and can be added manually. If not

defined, the system uses the EnvironmentVariableDpApi storage type.

- **EncryptJournal:** If set to true, Call server journal files are encrypted using the encryption key specified by the key scope and version parameters.
- **RetirementJournalFilePath:** Path to the folder the journal files will be retired to.
- **CallServerAreaNoteJournal:** Path to the folder the customer corrections XML files will be written to.
- **ServerCallJournalArchiveAgeHours:** The number of hours before the server call journal files are retired (the default value is 10 days, which is equal to 240 hours).
- **CustomerFile:** Path to the customer file.
- **ETRmatrixFile:** Path to the ETR matrix file. The Call server uses the ETR matrix file to provide the client (customer or Call Entry tool) the Initial Estimated Time for Restoration (ITR) when the first customer trouble call (when the customer is not part of any incident) is received. If the path to the ETR matrix file is missing or invalid, the default ITR (2.5 hours) is provided.
- **ListeningHost:** A host where the application is installed and listening to other interfaces.
- **ListeningPort:** A TCP/IP port that the application is listening to for incoming messages.
- **CheckMessageSecurity:** If set to true, this parameter enables checking for security token authenticity/expiration.
- **CheckMessageSequence:** If set to true, this parameter enables checking the sequence number of server messages.
- **UseCredentialFile:** If set to true, this parameter enables the use of the Credential file for defining the private key and user SIDs.
- **AllowUnregisteredSids:** If set to true, this parameter enables any user SID to be valid.
- **Relay:** True only for the Relay server. For the Call server, this parameter always must be false. Do not change it.
- **LogPath:** Path to the folder where the log file of the Call server is saved.
- **LogDetails:** Controls whether detailed (debug) log messages are written to the log.
- **IsServiceMode:** True only for the AMI server. For the Call server, this parameter always must be false. Do not change it.
- **DisableItronInterface:** If set to false, this parameter enables interfacing with ITRON Open Way system.
- **EnableMultiSpeakInterface:** If set to true, this parameter enables the Multispeak Interface.
- **MaxCallTypeValue:** The maximum integer value for TroubleCallCivV1.CallType.
- **MaxCondition_1Value:** The maximum integer value for TroubleCallCivV1.Condition_1.
- **MaxCondition_2Value:** The maximum integer value for TroubleCallCivV1.Condition_2.
- **MaxDescriptionValue:** The maximum integer value for TroubleCallCivV1.Description.

- **MaxPriorityCodeValue:** The maximum integer value for TroubleCallCivV1.PriorityCode.
- **CredentialInfoFile:** The file name and path of the Credential Info file that stores the valid SIDs and secret shared key.
- **TimeToLive:** The time in seconds. This is the time to live for the IDDS Messages on the network.
- **ConnectionDebug:** Controls whether debug files are created for TCP connections. Debug files are created in the ".\Server\log" folder and started with "ConnectionDebug_".
- **HelloIntervalSeconds:** The interval between automatic hello messages.
- **CallQueryIntervalSeconds:** The interval between automatic call queries.
- **MeterEventQueryIntervalSeconds:** The interval between automatic meter event queries.
- **MeterVoltEventQueryIntervalSeconds:** The interval between automatic meter volt event queries.
- **CaseNoteQueryIntervalSeconds:** The interval between automatic case note queries.
- **AreaNoteQueryIntervalSeconds:** The interval between automatic area setting queries.
- **ServicePoint2FeederQueryIntervalSeconds:** The interval between automatic service point to feeder initialization queries.
- **EnableTcpiTls:** If set to true, the TLS protocol is enabled for the TCP/IP connection.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **CfgProxyHost:** The host of the CfgProxy application.
- **CfgProxyPort:** The TCP/IP listening port of the CfgProxy application.
- **CfgProxyTimerPeriod:** Time period in seconds for CfgProxy to confirm a process role (Enabled or Standby) before rejecting the process role assignment.
- **UseRoleFromCfgProxy:** Indicates whether to obtain the role (Enabled, Standby, or Disabled) from the CfgProxy server. If false, then the role is always Enabled. This option is used for testing in a nonredundant configuration.
- **CfgmanProcessName:** The name of the process as defined in the Configuration Manager database.
- **ServerLogFileName:** The name and path to the Call server log file.
- **ServerLogEnabled:** If set to true, Call server logging is enabled.
- **CompressMessage:** If set to True, Call server messages are compressed before sending. This decreases message size and improves performance. The CompressMessage parameter must be set with the same value as the CompressMessage, OMS_SEND_COMPRESSED_MESSAGE, and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
- **ServerLogDailyEnabled:** If set to true, a new Call server log is started each new day.
- **ServerLogSizeLimit:** The maximum number of log messages written to the log file.
-

EnableFullMessageLogging: If set to true, entire message payloads are written to the log file.

- **CodedTranslationConfigFileName:** Path to the Coded Translation configuration file.
- **AmiOperationTimeout:** Time period (in seconds) after which an AMI operation is failed.

Note

This parameter is absent in the Call server configuration file and can be added manually, as needed.

- **OmsServerHost:** OMS server (NetServer) host. This parameter is required for the Site Overview display. (The Call server must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
- **OmsServerPort:** OMS server (NetServer) managed connection port. This parameter is required for the Site Overview display. (The Call server must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
- **OmsServerUseStreams:** If set to true, the OMS server uses the stream interface.
- **CustomerPhoneNumberFormat:** Specifies the customer phone number format. The format can include the AREACODE, PHONENUMBER, PHONEPREFIX, and PHONESUFFIX query fields. The default format is (%AREACODE)%PHONEPREFIX-%PHONESUFFIX. For example, (555)589-0773. The formatted phone number can be queried using the AREACODEPHONENUMBER query field.
- **Plugin URI Parameters:** The URI parameters in this group apply to all Call server web service plugins.
 - **UriScheme:** URI scheme: HTTP or HTTPS.
 - **UriHost:** URI host.
 - **UriPort:** URI port.

Note

Use different ports for Call, Relay, AMI, and Crew server web services.

- **UriPath:** URI path.
- **FullMessages:** If set to true, entire message payloads are written to the log file.
- **Log Service Plugin Parameters:** This group includes parameters for the Call server logging service plugin.
 - **Enabled:** If set to true, the Call server logging service is enabled.
 - **Type:** The type that implements the Call server logging service plugin.
 - **Name:** Call server logging service plugin name.

- **LogMessageTypes:**
 - **Type:** Application for which the log is created.
 - **SeparateLogToDisk:** If set to true, a separate log is created for this type.
 - **Color:** Color of log messages in the log (for the list of available colors, see <https://docs.microsoft.com/en-us/dotnet/api/system.drawing.color?view=netframework-4.8>).
- **LogToDisk:**
 - **Name:** Name of the log file.
 - **EnableAppendLog:** If set to true, a log appender is added.
 - **EnableDailyLog:** If set to true, a separate log file is created daily.
 - **MaxEntries:** Specifies the maximum number of entries in the log file.
- **Swagger Docs Plugin Parameters:** This group includes Swagger Docs plugin parameters.
 - **Enabled:** If set to true, the Swagger Docs plugin is enabled.
 - **Type:** The type that implements the Swagger Docs plugin.
 - **Name:** Swagger Docs plugin name.
- **OMS Logger Plugin Parameters:** This group includes OMS Logger plugin parameters for Call server web service logging.
 - **Enabled:** If set to true, the OMS Logger plugin is enabled.
 - **Type:** The type that implements the OMS Logger plugin.
 - **Name:** OMS Logger plugin name.
 - **BasePath:** OMS Logger web service root URL.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **Name:** OMS Logger web service name to be shown on the user interface.
 - **Version:** OMS Logger web service message version.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **Call Server Web Service Plugin Parameters:** This group includes Call server web service plugin parameters.
 - **Enabled:** If set to true, the Call server web service is enabled.
 - **Type:** The type that implements the Call server web service plugin.
 - **Name:** Call server web service plugin name.
 - **EnableCORS:** If set to True, Cross-Origin Resource Sharing (CORS) is enabled. (Required for Internal Notifications.)

- **BasePath:** Call server web service root URL.
- **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
- **Name:** Call server web service name to be shown on the user interface.
- **Version:** Call server web service message version.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **UseHeaderAuthorization:** If set to True, "authorization" header checking is enabled for incoming web service calls.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
- **Call Interface Web Service Client Plugin Parameters:** This group includes parameters for the Call Interface web service client plugin.
 - **Enabled:** If set to true, the Call Interface web service client is enabled.
 - **Type:** The type that implements the Call Interface web service client plugin.
 - **Name:** Call Interface web service client plugin name.
 - **ID:** Call Interface web service client plugin ID. This is an optional field. By default, plugin ID is set to the plugin name. Use this field to have multiple instances of a plugin.
 - **BaseUri:** Call Interface web service client root URL. For the communication with the Relay server Call web service, set to the URL defined in the Relay server Call Web Service Plugin Parameters group (for example, http://localhost:8443/api/adms/relay/call/v2).
 - **MessagesUri:** URL for CallMessages messages.
 - **StatusUri:** URL for Status messages.
 - **CallsUri:** URL for Calls messages.
 - **CallbackUri:** URL for Callback messages.
 - **CaseNoteResponseUri:** URL for CaseNoteResponse messages.
 - **CaseNotesUri:** URL for CaseNotes messages.
 - **CustomerMoveNotificationEnabled:** If set to true, notifications for moving a customer are enabled.
 - **CustomerMoveUri:** URL for CustomerMove messages.

- **CallMessagesInterval:** Polling time for Call messages, in seconds.
- **PostingRetryInterval:** Time period, in seconds, to retry posting messages to the remote endpoint when posting failed (for example, due to connection issues).
- **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier that is equal to "yyyy-MM-ddTHH:mm:ss.fffffffK".
- **UseHeaderAuthorization:** If set to True, "authorization" header checking is enabled for incoming web service calls.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
- **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or retrieving a valid UAA authentication token (for clients).
- **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.
- **BlockPostMessageTypes:** Call server Web Service message types to block from being sent to the Call server web service client (for example, CallNotificationV2 and CaseNotesNotificationV2).
- **RejectedHttpResponseStatusCodes:** HTTP response codes that cause the associated message to be rejected and eliminated from the retry logic.
- **RetryEnabled:** Set to true to enable retry logic.
- **MaxPostTryCnt:** The number of times to try to deliver a message. If this count is reached for a message, the message is rejected and eliminated from the retry logic. To disable limiting the number of retries, set this parameter to "-1".
- **HttpRequestTimeoutSeconds:** Time period to wait for the client response for an HTTP request. This parameter is used for testing and debugging purposes.
- **HttpDefaultRequestHeadersExpectContinue:** Specifies whether the HttpClient "ExpectContinue" behavior is enabled.

By default, DotNet enables the "ExpectContinue" behavior. This behavior results in HTTP clients posting the "Expect: 100-Continue" request header first, and then waiting for a "100-Continue" response from the server before posting the request payload. Disable this functionality when using a load balancer software which has issues supporting this functionality. Disabling this functionality eliminates one extra client server exchange for each message posted.

- **HttpKeepAlive:** Specifies whether the HttpClient "KeepAlive" behavior is enabled.

By default, DotNet enables the "KeepAlive" behavior. This behavior specifies that HTTP clients maintain the existing HTTP connection when sending messages. When this functionality is disabled, every message is sent via a new HTTP connection.

Note

Multiple instances of the Call server Web Service Client plugin and other plugins are supported.

- **AMI Interface Web Service Client Plugin Parameters:** This group includes parameters for the AMI server web service client plugin.
 - **Enabled:** If set to true, the AMI Interface web service client is enabled.
 - **Type:** The type that implements the AMI Interface web service client plugin.
 - **Name:** AMI Interface web service client plugin name.
 - **BaseUri:** AMI Interface web service client root URL. For the communication with the Relay server AMI web service, set to the URL defined in the Relay server AMI Web Service Plugin Parameters group (for example, `http://localhost:8443/api/adms/relay/ami/v2`).
 - **MessagesUri:** URL for AmiMessages messages.
 - **StatusUri:** URL for Status messages.
 - **PingUri:** URL for Ping messages.
 - **VoltPingUri:** URL for VoltPing messages.
 - **ControlUri:** URL for Control messages.
 - **ReadUri:** URL for Read messages.
 - **AmiMessagesInterval:** Polling time for AMI messages, in seconds.
 - **PostingRetryInterval:** Time period, in seconds, to retry posting messages to the remote endpoint when posting failed (for example, due to connection issues).
 - **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier that is equal to "yyyy-MM-ddTHH:mm:ss.fffffffK".
 - **UseHeaderAuthorization:** If set to True, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
 - **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
 - **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
 - **CertSubjectName:** The TLS certificate subject name's CN value.
 - **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA

authentication token (for clients).

- **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.
- **OMS Interface Server V1 Web Service Client Plugin Parameters:** This group includes OMS Interface Server V1 web service client plugin parameters. This plugin interacts with the OMS Interface Server's "oms/services/rest" endpoint.
 - **Enabled:** If set to true, the OMS Interface Server V1 web service client is enabled.
 - **Type:** The type that implements the OMS Interface Server V1 web service client plugin.
 - **Name:** OMS Interface Server V1 web service client plugin name.
 - **Config > Name:** OMS Interface Server V1 web service client to be shown on the user interface.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **UpdateAreaNoteUri:** The endpoint for update area settings queries.
 - **PostingRetryInterval:** Time period, in seconds, to retry posting messages to the remote endpoint when posting failed (for example, due to connection issues).
 - **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier that is equal to "yyyy-MM-ddTHH:mm:ss.fffffffK".
 - **HttpDefaultRequestHeadersExpectContinue:** Specifies whether the HttpClient "ExpectContinue" behavior is enabled.

By default, DotNet enables the "ExpectContinue" behavior. This behavior results in HTTP clients posting the "Expect: 100-Continue" request header first, and then waiting for a "100-Continue" response from the server before posting the request payload. Disable this functionality when using load balancer software that has issues supporting this functionality. Disabling this functionality eliminates one extra client server exchange for each message posted.
 - **HttpKeepAlive:** Specifies whether the HttpClient "KeepAlive" behavior is enabled.

By default, DotNet enables the "KeepAlive" behavior. This behavior specifies that HTTP clients maintain the existing HTTP connection when sending messages. When this functionality is disabled, every message is sent via a new HTTP connection.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **RemoteProcessCapabilities:** This group includes Call server external application parameters.
 - **ID:** ID of the server process that the Call server connects to (for example, Relay server).
 - **SupportsCaseNoteQuery:** True, if the system supports case note queries.
 - **SupportsMeterEventQuery:** True, if the system supports meter event queries.

- **SupportsMeterVoltEventQuery:** True, if the system supports meter voltage event queries.
- **SupportsMeterPingRequest:** True, if the system supports meter ping requests.
- **SupportsMeterVoltPingRequest:** True, if the system supports meter voltage ping requests.
- **SupportsCallQuery:** Deprecated functionality. Set to False to avoid having duplicate call queries (over TCP and HTTP).
- **SupportsAutomatedCallbackRequest:** True, if the system supports automatic callback requests via TCP/IP (legacy functionality). False, if the system supports automatic callback requests via HTTP (CallInterfaceWebServiceClient must be enabled).
- **SupportsManualCallbackRequest:** True, if the system supports manual callback requests via TCP/IP (legacy functionality). False, if the system supports manual callback requests via HTTP (CallInterfaceWebServiceClient must be enabled).
- **ItronConfig:** This group includes parameters for the Itron Interface.
 - **DemoMeterMode:** If set to true, demo meter mode is enabled. Demo meter mode maps external meter numbers to internal meter numbers. Alternatively, use the `AmiServiceMeters.xml` file to map meter IDs to electronic serial numbers.

16.10. RelayServerConfig.xml

The Relay Server is configured by `RelayServerConfig.xml`, which includes the following parameters:

Warning

XML files are case-sensitive. Use all lower-case letters when configuring Boolean settings.

- **ApplicationParameters:** Relay server application parameters.
 - **ProcessID:** The name of the Relay server process that will be used by the Call server for the remote connection.
 - **RelayServerCallJournal:** Path to the Relay server journal files location.
 - **EncryptionKeyScope:** Encryption key scope.
 - **EncryptionKeyVersion:** Encryption key version.
 - **EncryptionKeyStorageType:** Encryption key storage type.
-  **Note**

This parameter is absent in the configuration file and can be added manually. If not defined, the system uses the `EnvironmentVariableDpApi` storage type.
- **EncryptJournal:** If set to true, Relay server journal files are encrypted using the encryption key specified by the key scope and version parameters.
 - **RetirementJournalFilePath:** Path to the folder the journal files will be retired to.

- **ServerCallJournalArchiveAgeHours:** The number of hours before the server call journal files are retired (the default value is 10 days, which is equal to 240 hours).
- **RelayServerAreaNoteJournal:** Path to the Relay server Area Note Journal file location.
- **CustomerFile:** Path to the customer file.
- **ListeningHost:** The host where the application is installed and listening to other interfaces.
- **ListeningPort:** The TCP/IP port that the application is listening to for incoming messages.
- **CheckMessageSecurity:** If set to true, this parameter enables checking for security token authenticity/expiration.

Note

Call server does not secure DOTS and Call Entry connections while in Relay mode.

- **CheckMessageSequence:** If set to true, this parameter enables checking the sequence number of server messages.
- **UseCredentialFile:** If set to true, this parameter enables using the Credential file for defining the private key and user SIDs.
- **CompressMessage:** If set to True, Relay server messages are sent compressed. This decreases message size and improves performance. This parameter must be set with the same value with the CompressMessage, OMS_SEND_COMPRESSED_MESSAGE, and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
- **AllowUnregisteredSids:** If set to true, this parameter enables any user SID to be valid.
- **Relay:** True only for the Relay server and AMI server. For the Call server, this parameter always must be false. Do not change it.
- **CallServerTcpIpEnabled:** Specifies whether the communication with Call server via TCP is enabled. Set to False when the Relay server web service is enabled and HTTP message exchange with the Call server is supported.
- **EnableTcpiptls:** If set to true, the TLS protocol is enabled for the TCP/IP connection.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **JournalAmiMessages:** If set to true, AMI messages are journaled for future queries. If set to false, AMI messages are immediately sent to Call server.
- **CallServerHost:** The host machine name(s) of the Call server. Can list one or more comma delimited hosts. For example, host1,host2,host3.
- **CallServerPort:** The TCP/IP listening port number of the Call server.
- **LogPath:** Path where the log file of the Relay server is saved.
- **LogDetails:** Controls whether detailed (debug) log messages are written to the log.
- **IsServiceMode:** True only for the AMI server. For the Call server, this parameter must always be false. Do not change it.

- **DisableItronInterface:** If set to false, this parameter enables interfacing with ITRON Open Way System.
- **EnableMultiSpeakInterface:** If set to true, this parameter enables the Multispeak Interface.
- **CredentialInfoFile:** The file name and path of the Credential Info file that stores the valid SIDs and secret shared key.
- **TimeToLive:** Time in seconds. This is the time to live for the IDDS Messages on the network.
- **ConnectionDebug:** Controls whether debug files are created for TCP connections. Debug files are created in the ".\Server\Log" folder and started with "ConnectionDebug_".
- **HelloIntervalSeconds:** The interval between automatic hello messages.
- **CaseNoteQueryIntervalSeconds:** The interval between automatic case note queries.
- **AreaNoteQueryIntervalSeconds:** The interval between automatic area settings queries.
- **CallEntryCallResponseDays:** Maximum number of days of calls to include in response to call queries from Call Entry.
- **ServerLogFileName:** The name and path to the Relay server log file.
- **ServerLogEnabled:** If set to true, Relay server logging is enabled.
- **ServerLogDailyEnabled:** If set to true, a new Relay server log is started each new day.
- **ServerLogSizeLimit:** The maximum number of log messages written to the log file.
- **CodedTranslationConfigFileName:** Path to the Coded Translation configuration file.
- **Plugin URI Parameters:** The URI parameters in this group apply to all Relay server web service plugins.
 - **UriScheme:** URI scheme, HTTP or HTTPS.
 - **UriHost:** URI host.
 - **UriPort:** URI port.

Note

Use different ports for Call, Relay, AMI, and Crew server web services.

- **UriPath:** URI path.
- **FullMessages:** If set to true, entire message payloads are written to the log file.
- **Log Service Plugin Parameters:** This group includes parameters for the Relay server logging service plugin.
 - **Enabled:** If set to true, the Relay server logging service is enabled.
 - **Type:** The type that implements the Relay server logging service plugin.
 - **Name:** Relay server logging service plugin name.
 - **LogMessageTypes:**

- **Type:** Application for which the log is created.
- **SeparateLogToDisk:** If set to true, a separate log is created for this type.
- **Color:** Color of log messages in the log (for the list of available colors, see <https://docs.microsoft.com/en-us/dotnet/api/system.drawing.color?view=netframework-4.8>).
- **LogToDisk:**
 - **Name:** Name of the log file.
 - **EnableAppendLog:** If set to true, a log appender is added.
 - **EnableDailyLog:** If set to true, a separate log file is created daily.
 - **MaxEntries:** Specifies the maximum number of entries in the log file.
- **Swagger Docs Plugin Parameters:** This group includes Swagger Docs plugin parameters.
 - **Enabled:** If set to true, the Swagger Docs plugin is enabled.
 - **Type:** The type that implements the Swagger Docs plugin.
 - **Name:** Swagger Docs plugin name.
- **Relay Server Call Web Service Plugin Parameters:** This group includes Relay server Call web service plugin parameters.
 - **Enabled:** If set to true, the Relay server Call web service is enabled.
 - **Type:** The type that implements the Relay server Call web service plugin.
 - **Name:** The Relay server web service plugin name.
 - **BasePath:** The Relay server Call web service root URL.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **Name:** The Relay server Call web service name to show on the user interface.
 - **Version:** The Relay server Call web service message version.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
 - **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
 - **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
 - **SecureWebService:** Configures whether a secure web service connection is used (the web

service runs in HTTPS mode). The default value is false.

- **Relay Server AMI Web Service Plugin Parameters:** This group includes Relay server AMI web service plugin parameters.
 - **Enabled:** If set to true, the Relay server AMI web service is enabled.
 - **Type:** The type that implements the Relay server AMI web service plugin.
 - **Name:** The Relay server AMI web service plugin name.
 - **BasePath:** The Relay server AMI web service root URL.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **Name:** The Relay server AMI web service name to show on the user interface.
 - **Version:** The Relay server AMI web service message version.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
 - **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
 - **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
 - **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
- **OMS Logger Web Service Plugin Parameters:** This group includes OMS Logger web service plugin parameters for Relay server web service logging.
 - **Enabled:** If set to true, the OMS Logger web service plugin is enabled.
 - **Type:** The type that implements the OMS Logger web service plugin.
 - **Name:** OMS Logger web service plugin name.
 - **BasePath:** OMS Logger web service root URL.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **Name:** OMS Logger web service name to be shown on the user interface.
 - **Version:** OMS Logger web service message version.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are

ignored for the web service plugin.

- **EnableMessageLogging:** If set to true, message logging is enabled.
- **RemoteProcessCapabilities:** This group includes Relay server external application parameters.
 - **ID:** ID of the server process that the Relay server connects to (for example, Call server).
 - **SupportsCaseNoteQuery:** True, if the system supports case note queries.
 - **SupportsMeterEventQuery:** True, if the system supports meter event queries.
 - **SupportsMeterPingRequest:** True, if the system supports meter ping requests.
 - **SupportsCallQuery:** Deprecated functionality. Set to False to avoid having duplicate call queries (over TCP and HTTP).
 - **SupportsAutomatedCallbackRequest:** True, if the system supports automatic callback requests via TCP/IP (legacy functionality, `CallServerTcpIpEnabled` must be enabled). False, if the system supports automatic callback requests via HTTP (`RelayServerCallWebService` must be enabled).
 - **SupportsManualCallbackRequest:** True, if the system supports manual callback requests via TCP/IP (legacy functionality, `CallServerTcpIpEnabled` must be enabled). False, if the system supports manual callback requests via HTTP (`RelayServerCallWebService` must be enabled).

16.11. AmiServerConfig.xml

The AMI Server is configured by `AmiServerConfig.xml`, which includes the following parameters:

Warning

XML files are case-sensitive. Use all lowercase letters when configuring Boolean settings.

ApplicationParameters: This group includes general AMI server parameters:

- **CustomerFile:** Path to the customer file.
- **CheckMessageSecurity:** If set to true, this parameter enables checking for security token authenticity/expiration.

Note

Call server does not secure DOTS and Call Entry connections while in AMI mode.

- **UseCredentialFile:** If set to true, this parameter enables using a credential file for defining the private key and user SIDs.
- **AllowUnregisteredSids:** If set to true, this parameter enables any user SID to be valid.
- **Relay:** True only for the AMI server and the Relay server. For the Call server, this parameter always must be false. Do not change it.

- **CallServerPort:** The TCP/IP listening port number of the Call server.
- **CallServerHost:** The host machine name(s) of the Call server. Can list one or more comma delimited hosts. For example, host1,host2,host3.
- **EnableTcpipTls:** If set to true, the TLS protocol is enabled for the TCP/IP connection.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **CompressMessage:** If set to True, AMI server messages are sent compressed. This decreases message size and improves performance. This parameter must be set with the same value with the CompressMessage, OMS_SEND_COMPRESSED_MESSAGE, and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
- **LogPath:** Path where the log file of the Call server is saved. This parameter should be blank for the AMI server
- **LogDetails:** Controls whether detailed (debug) log messages are written to the log.
- **ServerLogFileNames:** Path where the log file of the AMI server is saved.
- **ServerLogEnabled:** If set to true, AMI server logging is enabled.
- **IsServiceMode:** True only for the AMI server. For the Call server, this parameter must always be false. Do not change it.
- **CIMInteropEnabled:** If set to true, the CIM Interop web service is enabled.
- **AmiServerJournal:** Path to the file where the AMI server journal is saved.
- **ItronPowerOffString:** The "power off" message text for the ITRON Open Way System interface (ITRON Interface).
- **ItronPowerOnString:** The "power on" message text for the ITRON Interface.
- **ItronGroupName:** Name of the ITRON Interface group.
- **DisableItronInterface:** If set to false, this parameter enables interfacing with the ITRON Open Way System.
- **CredentialInfoFile:** The file name and path of the Credential Info file that stores valid SIDs and the secret shared key.
- **EnableAmiInterface:** If set to true, this parameter enables the AMI Interface.
- **AmiInterfaceName:** The AMI Interface name to show on the user interface.
- **AmiInterfaceDllPath:** Path to the plugin .DLL file.
- **AmiInterfaceDllTypeName:** The type that implements the Call server AMI Interface in the plugin DLL file.
- **EnableAmiWebService:** If set to true, the AMI server web service is enabled.
- **AmiWebServiceName:** The name of the AMI server web service.

SystemCapabilities: This group includes system parameters.

- **SupportsCaseNoteQuery:** True, if the system supports case note queries.

- **SupportsMeterEventQuery:** True, if the system supports meter event queries.
- **SupportsMeterPingRequest:** True, if the system supports meter ping requests.
- **SupportsCallQuery:** Deprecated functionality. Set to False to avoid having duplicate call queries (over TCP and HTTP).
- **SupportsAutomatedCallbackRequest:** True, if the system supports automatic callback requests via TCP/IP (legacy functionality, CallServerTcpIpEnabled must be enabled). False, if the system supports automatic callback requests via HTTP (AmiInterfaceWebServiceClient must be enabled).
- **SupportsManualCallbackRequest:** True, if the system supports manual callback requests via TCP/IP (legacy functionality, CallServerTcpIpEnabled must be enabled). False, if the system supports manual callback requests via HTTP (AmiInterfaceWebServiceClient must be enabled).

ItronConfig: This group includes parameters for the Itron Interface.

- **Subscription Parameters:**
 - **SystemName:** System name.
 - **MeterPrefix:** Meter prefix.
 - **DemoMeterMode:** If set to true, demo meter mode is enabled. Demo meter mode maps external meter numbers to internal meter numbers.
 - **SubscribeOnStartup:** If set to true, the interface is subscribed at system startup.
 - **ExceptionSubscriptionEnabled:** If set to true, the exception subscription is enabled.
 - **ExceptionSubscriptionGroupNames:** Names of exception subscription groups.
 - **ExceptionSubscriptionCategoryConstraints:** Exception subscription category constraints.
 - **DataSubscriptionEnabled:** If set to true, data subscription is enabled.
 - **DataSubscriptionGroupNames:** Data subscription group names.
- **Message Strings Parameters:**
 - **Whd:** Name of the "Watt-hour (d)" message shown on the user interface.
 - **Whr:** Name of the "Watt-hour (r)" message shown on the user interface.
 - **VARhd:** Name of the "VAR-hour (d)" message shown on the user interface.
 - **VARhr:** Name of the "VAR-hour (r)" message shown on the user interface.
 - **Vha:** Name of the "Volt-hour phase A" message shown on the user interface.
 - **Vhb:** Name of the "Volt-hour phase B" message shown on the user interface.
 - **Vhc:** Name of the "Volt-hour phase C" message shown on the user interface.
 - **PrimaryPowerDown:** Name of the "Primary Meter Down" message shown on the user interface.
 - **PrimaryPowerUp:** Name of the "Primary Meter Up" message shown on the user interface.
 - **VoltHourAboveThreshold:** Name of the "Volt-hour above threshold" message shown on

the user interface.

- **VoltHourBelowThreshold:** Name of the "Volt-hour below threshold" message shown on the user interface.
- **VoltHourReturnToNormal:** Name of the "Volt-hour return to normal" message shown on the user interface.
- **MeterReadInterval:** Meter reading interval.
- **EnableDetailedLogging:** If set to true, detailed logging is enabled.
- **Data Conversion:**
 - **PolyPhaseMeters:** Defines poly-phase meters.
 - **DataConversionDll:** Name of the assembly (DLL) containing a type that converts data.
 - **DataConversionType:** Name of the type that performs data conversion in the format "DataConversionDll.DataConversionType".
 - **VoltHighUnreasonable:** Maximum valid reading value for Volts.
 - **VoltLowUnreasonable:** Minimum valid reading value for Volts.
 - **FlowUnreasonable:** Maximum valid reading value for Watt and VAR.

AmiConfig: This group includes AMI server interface parameters:

- **EnableEventListenerServiceCIM1stEdition:** If set to true, the Landis and Gyr (L+G) CIM interface first edition (CIM1 Interface) is enabled.
- **EventListenerServiceAddress1stEdition:** Base URL for the CIM1 Interface.
- **EnableEventListenerServiceCIM2ndEdition:** If set to true, the L+G CIM interface second edition (CIM2 Interface) is enabled.
- **EventListenerServiceAddress2ndEdition:** Base URL for the CIM1 Interface.
- **DemoMeterMode:** If set to true, demo meter mode is enabled. Demo meter mode maps external meter numbers to internal meter numbers.
- **DemoMeterFilePath:** Path to the demo meter mode meter mapping file.
- **SystemName:** System name to include in messages.
- **MeterPrefix:** Meter prefix to add to all external communications.
- **AutomaticStartup:** If set to true, automatic connection at system startup is enabled.
- **EnableDetailedLogging:** If set to true, all messages are logged.
- **OperationsCIM1stEdition:** This group includes parameters that define CIM1 Interface client operations.
 - **BaseAddress:** Base URL for the CIM1 Interface.
 - **UserName:** User name for the CIM1 Interface.
 - **Password:** User password for the CIM1 Interface.
 - **PingReadingType:** Ping reading type for the CIM1 Interface.

- **OnDemandReadingTypes:** On-demand reading types for the CIM1 Interface.
- **ReplyAddress:** Reply address for the CIM1 Interface. The reply address should point to the value specified in the EventListenerServiceAddress1stEdition parameter.
- **UseAppConfig:** If set to true, Callserver.exe.config is used.
- **CertStoreLocation:** Certificate store location.
- **CertStoreName:** Certificate store name.
- **CertFindType:** Certificate find type.
- **CertFindValue:** Certificate find value.
- **OperationsCIM2stEdition:** This group includes parameters that define CIM2 Interface client operations.
- **UserName:** User name.
- **Password:** User password.
- **EnableNominalVoltageEvent:** If set to true, nominal voltage events are enabled. If set to false, return to normal voltage events are enabled.
- **VoltageEventTimeoutMinutes:** Time period (in minutes) before creating an automatic nominal voltage event if no events are received for a meter. Set to 0 to disable automatic nominal voltage events.
- **VoltHighUnreasonable:** Maximum valid reading value for Volts.
- **VoltLowUnreasonable:** Minimum valid reading value for Volts.
- **FlowUnreasonable:** Maximum valid reading value for Watt and VAR.
- **FilterMessages:** If set to true, the entire message is discarded if any value is unreasonable.
- **FilterValues:** If set to true, unreasonable values are discarded.
- **EventValidTime:** Time period (in minutes) for processing incoming events.
- **ReadingMultiplierColumnName:** The name of the column in the customer file that is used for reading multiplier calculations.
- **TestMeterNumber:** Default meter number to use on the LGCIM tab for testing.
- **CalculateVar:** If set to true, reactive power (in VAR) is calculated based on the specified power factor.
- **PowerFactor:** Power factor that is used for reactive power calculation.
- **EventCacheEnabled:** If set to true, caching events are grouped within the AMI server.
- **EventCacheTime:** Time period (in seconds) to wait before processing "grouped" messages.
- **ProcessMeterReadingWithIndex:** If set to true, meter reading mappings are enabled. A meter reading is mapped to a different reading type than in the message based on the meter reading's order in the message.
- **MeterReadingIndexMapping:** Use these parameters to map meter readings to different reading types than the message based on the meter reading's order.

- **Index:** Meter reading index.
- **ReadingType:** Meter reading type.
- **Event Type References:**
 - **PowerOffEventType:** "Power Off" event type reference.
 - **PowerOnEventType:** "Power On" event type reference.
 - **SagStartedPhaseAEventType:** "Sag Started Phase A" event type reference.
 - **SagStartedPhaseBEventType:** "Sag Started Phase B" event type reference.
 - **SagStartedPhaseCEntityType:** "Sag Started Phase C" event type reference.
 - **SagStoppedPhaseAEventType:** "Sag Stopped Phase A" event type reference.
 - **SagStoppedPhaseBEventType:** "Sag Stopped Phase B" event type reference.
 - **SagStoppedPhaseCEntityType:** "Sag Stopped Phase C" event type reference.
 - **SwellStartedPhaseAEventType:** "Swell Started Phase A" event type reference.
 - **SwellStartedPhaseBEventType:** "Swell Started Phase B" event type reference.
 - **SwellStartedPhaseCEntityType:** "Swell Started Phase C" event type reference.
 - **SwellStoppedPhaseAEventType:** "Swell Stopped Phase A" event type reference.
 - **SwellStoppedPhaseBEventType:** "Swell Stopped Phase B" event type reference.
 - **SwellStoppedPhaseCEntityType:** "Swell Stopped Phase C" event type reference.

16.12. CrewServerConfig.xml

The Crew Server is configured by `CrewServerConfig.xml`, which includes the following parameters:



Warning

XML files are case-sensitive. Use all lower-case letters when configuring Boolean settings.

- **ListeningHost:** The host where the application is installed and listening to other interfaces.
- **ListeningPort:** The TCP/IP port the application is listening to for incoming messages.
- **LogPath:** Path where the log file of the crew server is saved.
- **CheckMessageSecurity:** If set to true, this parameter enables checking for security token authenticity/expiration.
- **CheckMessageSequence:** If set to true, this parameter enables checking the sequence number of server messages.
- **UseCredentialFile:** If set to true, this parameter enables using the credential file for defining the private key and user SIDs.
- **AllowUnregisteredSids:** If set to true, this parameter enables any user SID to be valid.
- **CredentialInfoFile:** The file name and path to the Credential Info file that stores valid SIDs and

the secret shared key.

- **TimeToLive:** Time in seconds. This is the time to live for the IDDS Messages on the network.
- **CompressMessage:** If set to True, Crew server messages are sent compressed. This decreases message size and improves performance. This parameter must be set with the same value with the CompressMessage, OMS_SEND_COMPRESSED_MESSAGE, and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
- **ConnectionDebug:** Controls whether debug files are created for TCP connections. Debug files are created in the %IDMS_Root%\log folder and start with "ConnectionDebug_".
- **ConnectionMaxKeepAliveWaitTime:** Time period in seconds to wait for a communication-keep-alive "Hello" response to be returned from a client before assuming that the connection is "bad" and terminating the connection. This value cannot be less than 30 seconds.
- **CfgProxyHost:** The host of the CfgProxy application.
- **CfgProxyPort:** The TCP/IP listening port of the CfgProxy application.
- **CfgProxyTimerPeriod:** Time period in seconds for CfgProxy to confirm a process role (Enabled or Standby) before rejecting the process role assignment.
- **UseRoleFromCfgProxy:** Indicates whether to obtain the role (Enabled, Standby, or Disabled) from the CfgProxy server. If false, then the role is always Enabled. This option is used for testing in a nonredundant configuration.
- **CfgmanProcessName:** Name of the process as defined in the Configuration Manager database.
- **ServerLogFileNames:** The name and path to the Crew server log file.
- **ServerLogEnabled:** If set to true, Crew server logging is enabled.
- **ServerLogDailyEnabled:** If set to true, a new Crew server log is started each new day.
- **ServerLogSizeLimit:** The maximum number of log messages written to the log file.
- **EnableFullMessageLogging:** If set to true, entire message payloads are written to the log file.
- **CoordinatesConverterAssemblyName:** Name of the assembly (DLL) that contains a class that converts between latitude/longitude and system X/Y coordinates.
- **CoordinatesConverterClassName:** Name of the class that converts between latitude/longitude and X/Y coordinates.

Use the coordinate converter options to configure navigation from the network view to a URL with certain latitude and longitude (for example, Google Maps).

- **OmsServerHost:** OMS server (NetServer) host. This parameter is required for the Site Overview display. (The Crew server must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
- **OmsServerPort:** OMS server (NetServer) managed connection port. This parameter is required for the Site Overview display. (The Crew server must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display

queries.)

- **OmsServerUseStreams:** If set to true, the OMS server uses the stream interface.
- **EnableTcpipTls:** If set to true, the TLS protocol is enabled for the TCP/IP connection.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **WebServiceBaseURL:** Base URL for Crew server web services. Each service adds a specific ending to this base. The default value is `http://localhost:8802/crew`.
- **UseUAATokenForIntrospect:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token. The Crew server extracts a UAA authentication token from the authorization header of incoming requests and sends the token to the UAA server for validation.
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **EnableCrewWebServices:** If set to True, Crew server runs as a web service. If set to False, Crew server runs in the regular "adapter" mode.
- **SecureWebService:** If set to True, Crew server runs in HTTPS mode.
- **DisableServiceAuthCheck:** If set to True, the "authorization" header checking is disabled for incoming web service calls.
- **CrewWebServiceRequestTimeout:** Time period, in seconds, to wait for the client response after the Crew Server web service request.
- **Plugin URI Parameters:** The URI parameters in this group apply to all Crew server modeling web service plugins.
 - **UriScheme:** URI scheme, HTTP or HTTPS.
 - **UriHost:** URI host.
 - **UriPort:** URI port.

Note

Use different ports for Call, Relay, AMI, and Crew server web services.

- **UriPath:** URI path.
- **FullMessages:** If set to true, entire message payloads are written to the log file.
- **Log Service Plugin Parameters:** This group includes parameters for the Crew server logging service plugin.
 - **Enabled:** If set to true, the Crew server logging service is enabled.
 - **Type:** The type that implements the Crew server logging service plugin.
 - **Name:** Crew server logging service plugin name.
 - **LogMessageTypes:**
 - **Type:** Application for which the log is created.
 - **SeparateLogToDisk:** If set to true, a separate log is created for this type.

- **Color:** Color of log messages in the log (for the list of available colors, see <https://docs.microsoft.com/en-us/dotnet/api/system.drawing.color?view=netframework-4.8>).
- **LogToDisk:**
 - **Name:** Name of the log file.
 - **EnableAppendLog:** If set to true, a log appender is added.
 - **EnableDailyLog:** If set to true, a separate log file is created daily.
 - **MaxEntries:** Specifies the maximum number of entries in the log file.
- **Swagger Docs Plugin Parameters:** This group includes Swagger Docs plugin parameters.
 - **Enabled:** If set to true, the Swagger Docs plugin is enabled.
 - **Type:** The type that implements the Swagger Docs plugin.
 - **Name:** Swagger Docs plugin name.
- **OMS Logger Web Service Plugin Parameters:** This group includes OMS Logger web service plugin parameters for Crew server web service logging.
 - **Enabled:** If set to true, the OMS Logger web service plugin is enabled.
 - **Type:** The type that implements the OMS Logger web service plugin.
 - **Name:** OMS Logger web service plugin name.
 - **BasePath:** OMS Logger web service root URL.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **Name:** OMS Logger web service name to be shown on the user interface.
 - **Version:** OMS Logger web service message version.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
 - **EnableMessageLogging:** If set to true, message logging is enabled.
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **Crew Server Web Service Plugin Parameters:** This group includes Crew server web service plugin parameters.
 - **Enabled:** If set to true, the Crew server web service is enabled.
 - **Type:** The type that implements the Crew server web service plugin.
 - **Name:** Crew server web service plugin name.

- **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
- **BasePath:** Crew server web service root URL.
- **Name:** Crew server web service name to be shown on the user interface.
- **Version:** Crew server web service message version.
- **Description:** Crew server web service description.
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
- **Crew Server WMS Web Service Client Plugin Parameters:** This group includes Crew server WMS web service client plugin parameters.
 - **Enabled:** If set to true, the Crew server WMS web service client is enabled.
 - **Type:** The type that implements the Crew server WMS web service client plugin.
 - **Name:** Crew server WMS web service client plugin name.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **BasePath:** Crew server WMS web service client root URL.
 - **Name:** Crew server WMS web service client name to be shown on the user interface.
 - **Version:** Crew server WMS web service client message version.
 - **Description:** Crew server WMS web service client description.
 - **MessagesGetUri:** URL for WMSModelMessages messages.
 - **MessageGetPollingInterval:** Polling time for WMSModelMessages messages, in seconds.
 - **MessagesStatusPostUri:** URL for ModelState messages.
 - **IncidentUpdatePostUri:** URL for IncidentUpdate messages.
 - **CrewUpdatePostUri:** URL for CrewUpdate messages.

- **OrderUpdateUri:** URL for OrderUpdate messages.
- **PlannedOutageUpdatePostUri:** URL for PlannedOutage messages.
- **SynchronizeCrewsUri:** URL for CrewSynch messages.
- **PostIncidentIdOrName:** Defines whether ADMS sends the incident ID or name in the IncidentUpdate.IncidentId and CrewUpdate.IncidentId message fields. The parameter value can be "name" or "id" (case insensitive). The default value is "name".
- **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier that is equal to "yyyy-MM-ddTHH:mm:ss.fffffffK".
- **Description:** The Crew server modeling web service client description.
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
- **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.
- **BlockPostMessageTypes:** Crew server Web Service message types to block from being sent to the Crew server WMS web service client (for example, CrewUpdateV2 and IncidentUpdateV2).
- **RejectedHttpResponseStatusCodes:** HTTP response codes that cause the associated message to be rejected and eliminated from the retry logic.
- **RetryEnabled:** Set to true to enable retry logic.
- **MaxPostTryCnt:** The number of times to try to deliver a message. If this count is reached for a message, the message is rejected and eliminated from the retry logic. To disable limiting the number of retries, set this parameter to "-1".
- **HttpRequestTimeoutSeconds:** Time period to wait for the client response for an HTTP request. This parameter is used for testing and debug purposes.
- **HttpDefaultRequestHeadersExpectContinue:** Specifies whether the HttpClient "ExpectContinue" behavior is enabled.

By default, DotNet enables the "ExpectContinue" behavior. This behavior results in HTTP clients posting the "Expect: 100-Continue" request header first, and then waiting for a "100-Continue" response from the server before posting the request payload. Disable this functionality when using a load balancer software which has issues supporting this functionality. Disabling this functionality eliminates one extra client server exchange for each message posted.

- **HttpKeepAlive:** Specifies whether the HttpClient "KeepAlive" behavior is enabled.

By default, DotNet enables the "KeepAlive" behavior. This behavior specifies that HTTP clients maintain the existing HTTP connection when sending messages. When this functionality is disabled, every message is sent via a new HTTP connection.

- **Crew Modeling Web Service Plugin Parameters:** This group includes Crew server modeling web service plugin parameters.

- **Enabled:** If set to true, the Crew server modeling web service is enabled.
- **Type:** The type that implements the Crew server modeling web service plugin.
- **Name:** Crew server modeling web service plugin name.
- **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
- **BasePath:** Crew server modeling web service root URL.
- **Name:** Crew server modeling web service name to be shown on the user interface.
- **Version:** Crew server modeling web service message version.
- **Description:** Crew server modeling web service description.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **ResponseBaseUri:** The remote client root URI for to receive status updates after model change requests.
- **ResponseStatusMessagesUri:** The remote client URI to receive status updates after model change requests.
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.

- **Crew Modeling Web Service Client Plugin Parameters:** This group includes parameters for the Crew Modeling web service client plugin.
 - **Enabled:** If set to true, the Crew Modeling web service client is enabled.
 - **Type:** The type that implements the Crew Modeling web service client plugin.
 - **Name:** Crew Modeling web service client plugin name.
 - **BaseUri:** Crew Modeling web service client plugin root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **BasePath:** Crew Modeling web service client plugin root URL.
 - **Version:** The Crew Modeling web service client plugin message version.
 - **MessagesGetUri:** URL for WSMessages messages.
 - **MessageGetPollingInterval:** Polling time for WSMessages messages, in seconds.
 - **MessagesStatusPostUri:** URL for Status messages.
 - **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier that is equal to "yyyy-MM-ddTHH:mm:ss.fffffffK".
 - **Description:** The Crew Modeling web service client plugin description.
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
 - **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
 - **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
 - **CertSubjectName:** The TLS certificate subject name's CN value.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
 - **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
 - **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.
- **OMS Interface Query Service Plugin Parameters:** This group includes parameters for the OMS Interface Query Service plugin.
 - **Enabled:** If set to true, the OMS Interface Query Service is enabled.
 - **Type:** The type that implements the OMS Interface Query Service plugin.
 - **Name:** The OMS Interface Query Service plugin name.

- **BaseUri:** The OMS Interface Query Service root URI.
- **OmsQueryUri:** URI for OmsQuery messages.
- **IncidentUri:** URI for Incident messages.
- **Version:** Message version.
- **Description:** Plugin description.
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **PW:** The password value that is used to hash and place in the request authorization header. Use the Encryption tool to encrypt the password.
- **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
- **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **LogService:** The name of the logging service.
- **TrustedClientList and TrustedClient:** These elements are used to set up trusted application lists. Each application can have the following settings:
 - **Name:** Used to identify the application. For example, Notification Publisher.
 - **Key:** Used to calculate the hash value for the authorization header.

The Name and Key fields are used to calculate the authorization header if DisableServiceAuthCheck is set to False.

 - **NotificationBaseURL:** The base URL for trusted applications to receive notifications. The default value is http://localhost:9090/dms/event/. The Crew server uses this base URL as an address where the POST web request for the notification will be sent.
 - **NotificationTypes:** The notification types that the application is interested in, separated by commas. The default value is INC-UPDATE,CREW-UPDATE.
- **System Capabilities:**
 - **IsAVLCapable:** If set to true, this parameter enables querying the Vehicle Location to the interface.
 - **IsSyncCapable:** If set to true, this parameter enables synchronization queries.

16.13. NetDataServerConfig.txt

Note

Boolean parameters in the NetDataServer configuration file can be set to 1/0 or True/False (case insensitive). If a parameter is removed or the parameter value is not defined, the default parameter value is used.

The Net Data Server is configured by NetDataServerConfig.txt, which includes the following parameters:

- **LOG_FILE_NAME:** Path to the Net Data Server log file.
- **CUSTOMERMODEL_CHECK_TIMER:** Time period (in seconds) for the Net Data Server to check the customer model file for changes.
- **JOURNAL_CHECK_TIMER:** Time period (in seconds) for the Net Data Server to check the journal files for changes.
- **OMS_SERVER_HOST:** OMS server (NetServer) host. This parameter is required for the Site Overview display. (The Net Data Server must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
- **OMS_SERVER_PORT:** OMS server (NetServer) managed connection port. This is the port to which the Net Data Server connects. This parameter is required for the Site Overview display. (The Net Data Server must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
- **OMS_SERVER_USE_STREAMS:** If set to true, the OMS Server uses the stream interface.
- **OMS_SERVER_USE_TLS:** If set to true, the OMS Server uses the TLS security protocol.
- **OMS_SERVER_TLS_SUBJECTNAME:** The TLS certificate subject name's CN value.
- **OMS_SERVER_CHECK_MESSAGE_SECURITY:** If set to true, the OMS Server checks for security token authenticity/expiry.
- **OMS_SERVER_CHECK_MESSAGE_SEQUENCE:** If set to true, the OMS Server checks message sequence numbers.
- **OMS_SERVER_USE_CREDENTIAL_FILE:** If set to true, the OMS Server uses a credential file.
- **OMS_SERVER_CREDENTIAL_INFO_FILE:** Path to the credential file.
- **OMS_SERVER_TIME_TO_LIVE:** Time to live (in seconds) for OMS Server connections.
- **OMS_SERVER_COMPRESSED_MESSAGE:** If set to true, the OMS Server messages are compressed.
- **OMS_POLYGON_GROUPS:** Path to the polygon group file.
- **OMS_CUSTOMER_FILE:** Path to the customer model.
- **OMS_CALL_JOURNAL_FILE_PATH:** Path to the folder that contains Call journals.
- **OMS_ARCHIVE_JOURNAL_FILE_PATH:** Path to the folder that contains Archive journals.
- **OMS_CREW_JOURNAL_FILE_PATH:** Path to the folder that contains Crew journals.
- **OMS_ORDER_JOURNAL_FILE_PATH:** Path to the folder that contains Follow-Up Work Model journals.
- **OMS_REALTIME_INCIDENT_JOURNAL_FILE_PATH:** The path to the directory where Real Time

Incident Journal files with near real-time (NRT) data are placed.

Note

Net Server can also write Real Time Incident Journal files. However, for performance issues, it is recommended that these files are written by Net Data Server.

- **OMS_MANAGED_CONFIGFILE:** Path to the OMS Server configuration file.
- **OMS_CODED_TRANSLATION_CONFIGFILE:** Path to the Coded Translation configuration file.
- **OMS_SNAPSHOT_FILE_PATH:** Path to the folder that contains snapshots.
- **OMS_CREW_EMPLOYEE_FILE:** Path to the crew model file.
- **OMS_WRITE_REALTIME_INCIDENT_JOURNAL:** If set to "true", Net Data Server writes Real Time Incident Journal files. This option is enabled (true) by default.
- **OMS_ENCRYPT_REALTIMEINCIDENT_JOURNAL:** Specifies whether all Real Time Incident journals are encrypted using the encryption key specified by the key scope and version parameters.
- **OMS_ENCRYPTION_KEY_SCOPE:** Encryption key scope.
- **OMS_ENCRYPTION_KEY_VERSION:** Encryption key version.
- **OMS_ENCRYPTION_KEY_STORAGE_TYPE:** Encryption key storage type.

Note

This parameter is absent in the configuration file and can be added manually. If not defined, the system uses the EnvironmentVariableDpApi storage type.

- **OMS_WRITE_CHRONOLOGY_REALTIME_INCIDENT_JOURNAL:** If set to "true", chronology entries are written to the Real Time Incident journal. This setting is designed for testing purposes. In production, this setting should be set to "false" to avoid having duplicate entries in the INCIDENT_CHRONOLOGY table in the ARCHIVE database, because chronology entries are written to the Archive journal.
- **OMS_SERVER_LOG_ENABLED:** If set to "1", the OMS server log is enabled.
- **OMS_SERVER_LOGFILENAME:** Path to the OMS server log file.
- **OMS_SERVER_LOG_DAILY:** If set to "1", the OMS server daily logging is enabled.
- **OMS_SERVER_LOGSIZE_LIMIT:** Specifies the maximum number of entries that can be written to an OMS Server log file. When a log file reaches the specified size, it is saved with a date timestamp, and a new log file is created.
- **OMS_SERVER_SHOW_LOG_ITEMS:** If set to "1", log records are shown in the Messages tab of the Net Data Server application window.
- **OMS_SNAPSHOT_ENTRIES_THRESHOLD:** Defines the maximum number of journal entries in the daily snapshot file. When a snapshot is written and the file reaches the maximum number of entries, the file is saved with the .snapshot_Part<N> suffix, and the next entries are written to

the new file with the .snapshot_Part<N+1> suffix.

The snapshot chunking threshold should be set to a reasonably high value to avoid overburdening the system with too many chunks at normal conditions and only use snapshot chunking when a very large number of incidents appear in the system (for example, during a storm). The default value is 1,000,000,000, which is equal to around 30,000 new incidents on the demo model (Fantasy Island).

- **OMS_NUMBER_OF_DAILY_SNAPHOTS_REQUIRED:** Configures the number of daily snapshots required in the system.

The number of daily snapshots may be equal to the number of days the history data is kept in the system. This number should be the same in all server configuration files.

- **OMS_MOMENTARY_OUTAGE_DURATION:** Time period (in seconds) of the momentary outage duration. After this period, the outage (incident) becomes confirmed. Momentary outage duration is also used for indices calculation.
- **USER_ROLE_FROM_CFGPROXY:** If set to "1", the Net Data Server uses the role from CfgProxy.
- **CFGPROXY_HOST:** Host name or IP address where the CfgProxy server is installed and listening to other interfaces.
- **CFGPROXY_PORT:** TCP/IP port used for the CfgProxy server.
- **CFGPROXY_TIMER_PERIOD:** Time period in seconds for CfgProxy to confirm a process role (Enabled or Standby) before rejecting the process role assignment.
- **CFGMAN_PROCESS_NAME:** Name of the Net Data Server process in CFGMAN.
- **WEBSERVICE_SERVER_NAME:** A host where web services are installed and listening to other interfaces.
- **WEBSERVICE_PORT_NUMBER:** TCP/IP port used for web services.
- **WEBSERVICE_PAGE_SIZE:** The maximum number of records to return per query. Set to "0" for unlimited.
- **WESERVICE_CACHE_SECONDS:** The minimum time before refreshing web service results. Must be not less than 5 seconds.
- **CUSTOMERFILE_CUSTOM_COLUMNS:** Comma-separated list of customer columns to add to the query response.
- **CUSTOMERCACHE_EXPIRE_MINUTES:** Time period (in minutes) before customer cache expires.
- **WEBSERVICE_SECURE:** Set to "true" to enable an HTTPS connection for the Net Data Server web service.
- **OMSQUERY_WEBSERVICE_URI:** The Net Data Server web API URI to send canceled and cleared incidents to the Data Correction Interface application.
- **OMS_SNAPSHOT_TIMER_INTERVAL:** The time period (in minutes) between each time a snapshot file is created.
- **OMS_DEMO_HISTORY_DATA_MODE:** Set to "true" to enable Demo History Data mode. In this mode, the system at startup creates a snapshot with history data for the previous two weeks.

- **OMS_PENDING_CALLBACK_PERIOD:** The number of seconds for an incident to have a pending callback status and be available in the Completed Incidents list. After this period the incident is purged from the list. The default is a two-week period, which is equal to 1209600 seconds.
- **OMS_CREWATE_HISTORY_DATA:** Set to "true" to enable Create History Data mode. When running in this mode, the Net Data Server creates System History Snapshot files with history data for each day.
- **OMS_CREATE_HISTORY_DATA_FILE_PATH:** The path to a folder containing sub folders with daily history data.
- **OMS_DISABLE_SNAPSHOT_CREATION:** Set to "true" to disable creating daily snapshots. This may be useful to save resources, as it can take a long time to create daily snapshots, especially on a cold start where no snapshots currently exist. The side effect of this configuration is that a replicated standby server will never complete its initialization, thus it will not be able to confirm its standby role.
- **OMS_DISABLE_HIERARCHYGROUP_CALCULATION:** Set to "true" to disable calculating hierarchy groups. Normally, hierarchy groups are calculated from midnight up to the current hour. The side effect of this configuration is that the Net Data Server will not be able to feed hierarchy group information for previous hours to dashboards.
- **OMS_ENABLE_RECENT_HISTORY:** Optional configuration parameter used to turn on hierarchy groups for recent history. If enabled, a hierarchy group is added for each hour of each day. These hierarchy groups will be included in the daily snapshots, so when loading X number of Daily Snapshots, there will be X times 24 hierarchy groups. In addition, this parameter must be enabled to include cleared incidents in daily snapshots.

By default, recent history is disabled. The creation of Daily Snapshots is much slower if this option is enabled.

Note that if OMS_DISABLE_HIERARCHYGROUP_CALCULATION is set to "true", hierarchy groups are disabled.

- **READ_ONLY_MODE:** Set to "true" to run Net Data Server in read-only mode. In readonly mode, Net Data Server does not create daily snapshots. Net Data Replicator copies the snapshots from the enabled server.
- **OMS_ENABLE_DASHBOARD_SUPPORT:** Set to "false" to disable the dashboard and history correction support. If the Net Data Server is not a server to any dashboards or history correction, then this configuration can be used to disable any processing related to dashboards and history correction support.
- **OMS_ENABLE_SERVER_DEBUG_VIEWS:** Set to "true" to enable refresh of server "debug tabs". The Net Data Server's form has various tabs where a user can view OMS data, such as active and cleared incidents. Other tabs include hierarchy group data, as well as dashboard data. All of these tabs can be used to debug the OMS data available in the Net Data Server.
- **OMS_NUMBER_OF_DAYS_BEFORE_UNASSOCIATED_CALLS_EXPIRE:** Specifies the number of days before removing unassociated calls when creating a daily snapshot.
- **OMS_NUMBER_OF_DAYS_BEFORE_CALLS_EXPIRE:** Specifies the number of days before removing calls when creating a daily snapshot.

- **OMS_NUMBER_OF_DAYS_BEFORE_METER_EVENTS_EXPIRE:** Specifies the number of days before removing meter events when creating a daily snapshot.
- **KEEP_CUSTOMER_CALL_HISTORY:** Specifies whether customer call history is kept.
- **RETAIN_CUSTOMER_CALL_TIME:** Specifies whether customer call time is retained without resetting to the minimum value.
- **OMS_IGNORE_AREA_NOTE_START_TIME:** Specifies whether to ignore the AreaNote start time and apply the AreaNote to all incidents in that area, regardless of when the incident was created. Disabled (= 0) by default and used for backward compatibility.
- **ALLOWED_NUMBER_OF_PLANNED_OUTAGES:** Specifies the allowed number of planned outages for customer. The default value is 1. A value of -1 disables the validation check.

Note

Ensure that `ALLOWED_NUMBER_OF_PLANNED_OUTAGES` is set to same value as in `NetServerOmsConfig.txt`.

- **NUMBER_OF_DAYS_PRIOR_TO_PLANNED_OUTAGE_CREATION:** Specifies the number of days prior to start time for creating a Planned Outage. The default value is 30 days.

Note

Ensure that `NUMBER_OF_DAYS_PRIOR_TO_PLANNED_OUTAGE_CREATION` is set to same value as in `NetServerOmsConfig.txt`.

16.14. NetDataReplicatorConfig.xml

The Net Data Replicator (OMS Replicator) is configured by `NetDataReplicatorConfig.xml`, which includes the following parameters:

- **ConnectionDebug:** If set to "true", debug files are created for TCP connections. The files are created in the `%IDMS_ROOT%\Server\log` folder and start with `ConnectionDebug`.
- **UseRoleFromCfgProxy:** Indicates whether to obtain role (Enabled, Standby, or Disabled) from CfgProxy server. If false, then the role is always Enabled. This option is used for testing in a nonredundant configuration.
- **CfgProxyHost:** Host name or IP address of the CfgProxy server listener.
- **CfgProxyPort:** Port number of the CfgProxy server listener.
- **CfgProxyTimerPeriod:** Time period in seconds for CfgProxy to confirm a process role (Enabled or Standby) before rejecting the process role assignment.
- **CfgmanProcessName:** Name of the process as defined in the Configuration Manager database.
- **LocalListenerAddress:** TCP/IP address, in `<Host>:<Port>` format, at which the local Net Data Replicator is listening for replication queries from the remote standby Net Data Replicator when

the local replicator's role is enabled (Primary).

- **RemoteListenerAddresses:** Comma-separated list of TCP/IP addresses, in <Host>:<Port> format, to which the local OMS replicator sends replication queries when its role is standby.
- **ReplicationQueryIntervalSeconds:** Interval between automatic replication queries.
- **OmsJournalFilePath:** Path to journal folders. Used to access journals and snapshots.
- **ReplicateCallJournal:** Indicates whether to replicate Call journal files.
- **ReplicateIncidentJournal:** Indicates whether to replicate Incident journal files.
- **ReplicateOrderJournal:** Indicates whether to replicate Follow-Up Work Order journal files.
- **ReplicateRealTimeIncidentJournal:** Indicates whether to replicate Real Time Incident journal files.
- **ReplicateCrewJournal:** Indicates whether to replicate Crew journal files.
- **ReplicateArchiveJournal:** Indicates whether to replicate Archive journal files.
- **CallServerCallJournal:** Path to the folder that contains Call server Call journal files.
- **ReplicateCallServerJournal:** Indicates whether to replicate the Call server journal files.
- **RelayServerCallJournal:** Path to the folder that contains Relay server Call journal files.
- **ReplicateRelayServerJournal:** Indicates whether to replicate the Relay server journal files.
- **ReplicateDeenergizedLineJournal:** Indicates whether to replicate the Deenergized Line journal files.
- **ReplicateWorkModelJournal:** Indicates whether to replicate the Work Model journal files.
- **WorkModelJournalFilePath:** Path to the folder that contains Work Model journal files.
- **ReplicateDailySnapshots:** Indicates whether to replicate daily snapshot files.
- **ReplicationDailySnapshotDaysRequired:** The maximum number of daily snapshot files to be replicated.
- **ReplicateSwitchOrderServerFiles:** Indicates whether to replicate Switch Order server files.
- **SwitchOrderServerFilesPath:** Path to the folder that contains Switch Order server files.
- **ReplicateOmsAdapterStorageFiles:** Indicates whether to replicate OMS Adapter storage files.
- **OmsAdapterStorageFilesPath:** Path to the folder that contains OMS Adapter storage files.
- **DmsOnlyMode:** If set to "true", DMSonly mode is enabled. In this mode the OmsReplicator only supports copying of Switch Order files, and it does not wait for snapshot files before it confirms the standby role.
- **ReadOnlyMode:** If set to "true", read-only mode is enabled. In this mode the OmsReplicator has no interface to CfgProxy and starts in standby mode to get data from the enabled machine.
- **ServerLogFileName:** Path to the server log file.
- **ServerLogEnabled:** Indicates whether messages should be written to a log file.
- **ServerLogDailyEnabled:** Indicates whether a new log file should be created each new day.
- **ServerLogSizeLimit:** Indicates the maximum number of log messages written to the log file.

- **MaxCallJournalEntries:** Maximum number of Call journal entries per query.
- **MaxIncidentJournalEntries:** Maximum number of Incident journal entries per query.
- **MaxOrderJournalEntries:** Maximum number of Follow-Up Work Model journal entries per query.
- **MaxCrewJournalEntries:** Maximum number of Crew journal entries per query.
- **MaxArchiveJournalEntries:** Maximum number of Archive journal entries per query.
- **MaxCallServerJournalEntries:** Maximum number of Call server journal entries per query.
- **MaxRelayServerJournalEntries:** Maximum number of Relay server journal entries per query.
- **MaxDeenergizedLineJournalEntries:** Maximum number of Deenergized Line journal entries per query.
- **MaxWorkModelJournalEntries:** Maximum number of Work Model journal entries per query.
- **OmsServerHost:** OMS server (NetServer) host. This parameter is required for the Site Overview display. (The Net Data Replicator must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
- **OmsServerPort:** OMS server (NetServer) managed connection port. This parameter is required for the Site Overview display. (The Net Data Replicator must connect to the OMS server managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.)
- **OmsServerUseStreams:** If set to true, the OMS server uses the streams OmsConnection interface. (This setting must be set to the same value in all configuration files.)
- **OmsServerUseTLS:** If set to true, the OMS server uses the TLS security protocol. (This setting must be set to the same value in all configuration files.)
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **OmsServerCheckMessageSecurity:** If set to true, the OMS server checks for security token authenticity/expiry. (This setting must be set to the same value in all configuration files.)
- **OmsServerCheckMessageSequence:** If set to true, the OMS server checks message sequence numbers. (This setting must be set to the same value in all configuration files.)
- **OmsServerUseCredentialFile:** If set to true, the OMS server uses a credential file for defining the private key and user SIDs. (This setting must be set to the same value in all configuration files.)
- **OmsServerCredentialInfoFile:** Path to the credential file. (This setting must be set to the same value in all configuration files.)
- **OmsServerTimeToLive:** Time to live for OMS server connections in seconds.
- **DeenergizedLinesFileTimeoutMin:** Time period in minutes before deleting Deenergized Line Journal files.
- **EncryptionKeyScope:** Encryption key scope.
- **EncryptionKeyVersion:** Encryption key version.
- **EncryptionKeyStorageType:** Encryption key storage type.

Note

This parameter is absent in the configuration file and can be added manually. If not defined, the system uses the EnvironmentVariableDpApi storage type.

- **EncryptAllJournals:** Specifies whether all journals (Incident, Crew, Call, Order, RealTimeIncident, Archive, Deenergized Lines, Call server, and Relay server journals) are encrypted using the encryption key specified by the key scope and version parameters.
- **EncryptIncidentJournal:** If set to true, Incident journals are encrypted.
- **EncryptCrewJournal:** If set to true, Crew journals are encrypted.
- **EncryptCallJournal:** If set to true, Call journals are encrypted.
- **EncryptCallServerJournal:** If set to true, Call server journals are encrypted.
- **EncryptOrderJournal:** If set to true, Order journals are encrypted.
- **EncryptRealtimeIncidentJournal:** If set to true, Archive journals are encrypted.
- **EncryptArchiveJournal:** If set to true, Incident journals are encrypted.
- **EncryptRelayServerJournal:** If set to true, Relay server journals are encrypted.
- **EncryptDeenergizedLineJournal:** If set to true, Deenergized Line journals are encrypted.
- **EncryptWorkModelJournal:** If set to true, Work Model journals are encrypted.

16.15. DataCaptureServerConfig.xml

The Core Data Capture Server is configured by `DataCaptureServerConfig.xml`, which includes the following parameters:

- **WebServiceBaseUrl:** Core Data Capture server URL.
- **DmsInterfaceServerUrl:** DMS Interface server URL.
- **TimeseriesDbUrl:** Time-series database URL.
- **DnomDictionaryFile:** The path to the DNOM dictionary file with DNOM objects and fields.
- **DpfCaptureParamsFilePollSeconds:** Time period for polling for DPF results, in seconds.
- **RetryLimit:** The number of retries for querying DPF results.
- **RetryIntervalSeconds:** Time period between retries, in seconds.
- **ExportDirectory:** The path to the folder with Core Data Capture server log files.
- **DnomQueries:** The path to the file that defines DNOM queries.
- **DpfCaptureParamsFile:** The path to the file with DPF capture parameters.
- **TriggerByParamsFileEnabled:** If set to true, triggering by the DPF capture parameters file is enabled.
- **Mode:** The application output mode. For example, file or timeseriesDB.

- **LogLevel:** The amount of detail to include in the log. The options are:
 - **1:** Basic information, such as system errors.
 - **2:** More detailed information (errors and warnings).
 - **3:** Very detailed information (errors, warnings, and information messages).
- **LogStatistics:** If set to true, application statistics are logged.
- **DisQueriesToFile:** If set to true, DMS Interface server queries are recorded to a file.
- **DisDataToFile:** If set to true, DMS Interface server responses are recorded to a file.
- **TsjsonToFile:** If set to true, time-series data in JSON is recorded logged to a file.

16.16. NetViewer Adapter Configuration File

The NetViewer Adapter reads the NetViewerAdapterConfig.xml configuration file, which is located in the %IDMS_ROOT%\config\interfaces folder.

This section provides details for NetViewerAdapterConfig.xml parameters.

General

The general parameters are as follows:

- **SecureWS:** Set to True to run NetViewer Adapter in HTTPS mode. Set to False to run NetViewer Adapter in HTTP mode.
- **CertificateSubject:** The certificate subject (name) for HTTPS mode. The certificate must be placed locally in the Certificates > Trusted Root Certification Authorities > Certificates folder.

To use different certificate subjects for different hosts in a multiple standby host configuration, you can define subjects using the "certsubject" system environment variable and set the CertificateSubject value to %certsubject%.

- **Port:** Port number that is used by the NetViewer Adapter.
- **ConfigPath:** Path to the folder with ADMS client configurations. The NetViewer Adapter reads client configuration files for styles, symbols, colors, zoom levels, and other user interface data so that the network view in Outage Assist looks similar to the network view in the ADMS client.
- **DynamicsPath:** Location of the dynamics.sav file for Outage Assist. Specify a unique folder, so that Outage Assist does not conflict with another server writing its dynamics.
- **NetConfigFileName:** Name of the Net Dynamics configuration file to be read by the NetViewer Adapter. The file is stored in the folder with ADMS client configurations.
- **DmsInterfaceServerUrl:** DMS Interface Server URL. DMS Interface Server is used to query for DNAF results, specifically, potential fault locations.
- **EnableDNAFQueries:** Specifies whether DNAF result querying is enabled (specifically, queries for potential fault locations). Set to true if NetViewer Adapter is used for generating tiles. Set to false if the NetViewer Adapter is used for DNOM device searching and data querying in a DMS-

only environment.

- **EnableFaultLocationHalos:** Specifies whether fault location halo processing is enabled.
- **DNOMQueriesInterfaceServer:** Specifies whether DNOM device searching and data querying is enabled. Set to true if the NetViewer Adapter is used for DNOM device searching and data querying in a DMS-only environment and needs device data for GraphQL queries.
- **CustomSecretStore:** The NetViewer Adapter can retrieve UAA secret values from the CyberArk Enterprise Password Vault so that you do not need to provide the value in the configuration file (as a plain text or encrypted string). For more information, refer to the *ADMS Software Installation and Maintenance Guide*.
 - **AssemblyName:** Name of the assembly (DLL) containing a class that implements the CyberArk secret store.
 - **ClassName:** Name of the class that implements the CyberArk secret store. This class is implemented based on the CyberArk software development kit (SDK) and supports connecting to CyberArk API.
 - **OtherAssemblyNames:** Name of the assembly (DLL) that implements the .NET Standard Password SDK.
 - **UaaIdObjectKey:** CyberArk object key, which contains the CyberArk application ID, safe, database (folder), and object definitions of the NetViewer Adapter UAA ID defined in the Enterprise Vault Server. The format is
<DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>.
 - **UaaSecretObjectKey:** CyberArk object key, which contains the CyberArk application ID, safe, database (folder), and object definitions of the NetViewer Adapter UAA secret defined in the Enterprise Vault Server.
- **DisableCnf:** Common Notification Framework (CNF), also known as the Notification Service, sends notifications about tile and dynamics updates to clients. This option specifies whether the NetViewer Adapter sends notifications to CNF. Typically, the NetViewer Adapter is running with the Tile Service, which itself sends notifications to CNF. Therefore, disable CNF (set this parameter to True) if the NetViewer Adapter is running with the Tile Service and EnablePublishingToTileService is set to True.
- **CnfUrl:** Notification service URL. By default, http://localhost:9090/notification-service/notify.
- **EnablePublishingToTileService:** If set to True, the adapter publishes DNOM and customer tiles to the Tile Service.
- **TileServiceUrl:** The URL of the web socket Tile Service endpoint. By default, ws://localhost:24467/tilecacheservice.
- **TileServiceCompression:** The compression type to apply to published tiles and configurations. Can be one of the following: "gzip" (Gzip compression), "br" (Brotli compression), or "none" (no compression).

Note

Using gzip compression is recommended for better client performance.

- **UaaTokenServer:** The name of the server that hosts UAA services. For security reasons, the NetViewer Adapter does not directly connect to a service in the Kubernetes cluster. The NetViewer Adapter must connect to a UAA server to get a token for authorizing the connection with a service in the cluster.
- **DnomDynamicLoopTime:** Time period, in milliseconds, between checks for dynamic updates to DNOM data. The default value is 1000.
- **CustomerDynamicLoopTime:** Time period, in milliseconds, between checks for dynamic updates to customer data. The default value is 1000.
- **DocumentDynamicLoopTime:** Time period, in milliseconds, between checks for dynamic updates to document/annotation data. The default value is 1000.
- **CustomerQueryRequestTime:** Time period, in milliseconds, between queries for customer data for updates. The NetViewer Adapter uses this time period to wait for more customer data changes after a change occurs, to avoid sending a number of notifications and updating a number of tiles. (For example, a number of successive customer events are expected after a device outage, so the NetViewer Adapter waits for those to settle.) The default value is 5000.
- **FormatJsonOutput:** If set to True, the JSON output is formatted to human-readable text. Formatted JSON files are larger than unformatted files.
- **DisableSearchService:** If set to True, the search service is disabled and returns empty results. Use for debugging purposes.
- **DisableQueryService:** If set to True, the query service is disabled and returns empty results. Use for debugging purposes.

Modes

This section describes parameters in the Modes section that define how the NetViewer Adapter processes the network view tile and configuration data.

A mode parameter value can be one of the following:

- **Normal:** Get data from ADMS and publish the data to the Tile Service. If all modules are loaded in Normal mode, the NetViewer Adapter consumes around 3GB of RAM.
- **File:** Read data from files.
- **Write:** Get data from ADMS, save the data to files, and publish the data to the Tile Service.
- **Disabled:** Disable the mode. This option only supports GraphQL queries. If all modules are disabled, the NetViewer Adapter consumes around 70MB of RAM.
- **QueryOnly:** Support DNOM, customer, and GraphQL queries. If the ConfigMode, CustomerMode, and DnomMode modules are loaded in the QueryOnly mode, the NetViewer Adapter consumes around 1.7GB of RAM.

The Modes section includes the following parameters:

- **ConfigMode:** Indicates whether to generate or read a display configuration for the Common

Network Viewer.

- **CustomerMode:** Indicates whether to generate or read customer data for the Common Network Viewer.
- **DnomMode:** Indicates whether to generate or read DNOM data for the Common Network Viewer.
- **DocumentMode:** Indicates whether to generate or read documents data, such as annotations and safety docs for the Common Network Viewer.
- **LandbaseMode:** Indicates whether to generate or read landbase data for the Common Network Viewer.
- **OverviewMode:** Indicates whether to generate or read overview data for the Common Network Viewer.

Log File

The log file parameters are as follows:

- **EnableDebugLog:** Write a debug log file.
- **LogFile:** The name and location of the debug log file.
- **MaxMessagesInLogFile:** Maximum number of log entries to retain.

Selection Configuration (SelectionConfig)

The selection parameters define the appearance of selected objects in the client display.

- **SelectionDrawStyle:** The appearance of the line selection indicator. Can be one of the following: PlainColorBorder, Pattern, BlueWhiteDiagonalChecks5, BlueWhiteDiagonalChecks7, BlueWhiteDiagonalChecks9, BlueWhiteDiagonalChecks11, BlueWhiteDiagonalChecks13, BlueWhiteDiagonalChecks15, BlueWhiteTriangle5, BlueWhiteTriangle7, BlueWhiteTriangle9, or BlueWhiteTriangle11.

Note

PlainColorBorder draws a solid color border around the selection; use SelectionColor to set the color. Pattern specifies a custom image to use for a pattern with the SelectionPattern property.

- **SelectionLineWidth:** The width of the line selection indicator.
- **SelectionAnimationStyle:** The animation for the line selection indicator: Can be one of the following:
 - **None:** No animation.
 - **Flash:** Line selection flashes.
 - **Expand:** Line selection expands/contracts on a half-second interval.
 - **ExpandSlow:** Line selection expands/contracts on a one-second interval.
 -

PatternCrawl: When used with a pattern, the pattern crawls to draw attention to the selection.

- **SelectionColor:** The color of the line selection in hex format, such as "#0088AA". Applies to the PlainColorBorder draw style only.
- **SelectionPattern:** A custom image to be used for the line selection pattern. (For the Pattern draw style only).

The SelectionPattern animation style contains the following sub elements:

- **name:** Can include any name that is unique in the configuration (CustomSelectionPattern is recommended).
- **imageData:** Image data, in data URL format.

For more information, see https://developer.mozilla.org/en-US/docs/Web/HTTP/Basics_of_HTTP/Data_URIs.

- **SelectionPatternReplacementColors:** Use this option to programmatically replace colors in the SelectionPattern image. This option should include a list of ReplacementColors elements for replacing several colors. Each ReplacementColors element includes the following sub-elements:
 - **orig:** The color to replace in the source image, in hex format.
 - **target:** The color to use to replace the "orig" color in the source image, in hex format.

Drawing Array

The drawing array configuration parameters are as follows:

- **EnableDrawingArrayExport:** Specifies whether the NetViewer Adapter automatically exports drawing array files with the station model and annotation data to XML files. Drawing arrays are an intermediate format. Data goes from in-memory DNOM to drawing arrays, from drawing arrays to tiles, and then from tiles to JSON string representing tiles. Exporting drawing array files to XML files is used for debugging purposes.
- **DrawingArrayPath:** Path to the folder with drawing array files.
- **EnableCustomerDrawingExport:** Specifies whether the NetViewer Adapter exports drawing array files with customer data to XML files.
- **EnableLandbaseDrawingExport:** Specifies whether the NetViewer Adapter exports drawing array files with landbase data to XML files.
- **EnableIncidentDrawingExport:** Specifies whether the NetViewer Adapter exports drawing array files with incident data to XML files.
- **EnableCrewDrawingExport:** Specifies whether the NetViewer Adapter exports drawing array files with crew data to XML files.
- **LandbaseBoundary:** Boundaries for landbase geo files. The boundaries are defined using the X1, Y1, X2, and Y2 coordinates. Any geo file out of boundaries is ignored.

Tiles

The tile configuration parameters are as follows:

- **LoadTilesDuringNavigation:** Specifies whether tiles are loaded in the NetViewer Adapter during navigation (panning). If set to false, tile loading is delayed until panning is complete.
- **Boundary:** Limits the tile output area. Any customer and landbase tiles outside of the configured area will not be processed.
- **RotationType:** Indicates whether rotation angles in display tiles are interpreted clockwise (cw) or counter-clockwise (ccw).
- **InitialScale:** Initial OpenLayers scaling factor for displays.
- **MinZoom:** Minimum zoom level for displays.
- **MaxZoom:** Maximum zoom level for displays.
- **SmallDisplaySize:** Size (width or height) under which a display is considered "small". This setting triggers using the SmallDisplayInitialScale setting.
- **SmallDisplayInitialScale:** Initial scaling factor for small displays.
- **DocTileWidth:** Width of document (annotation) tiles, in display units.
- **DocTileHeight:** Height of document (annotation) tiles, in display units.
- **CustomerTileWidth:** Width of customer tiles, in display units.
- **CustomerTileHeight:** Height of customer tiles, in display units.
- **DnomPatchWidth:** Width of station model (DNOM) tiles, in display units.
- **DnomPatchHeight:** Height of station model (DNOM) tile patches, in display units.

Note

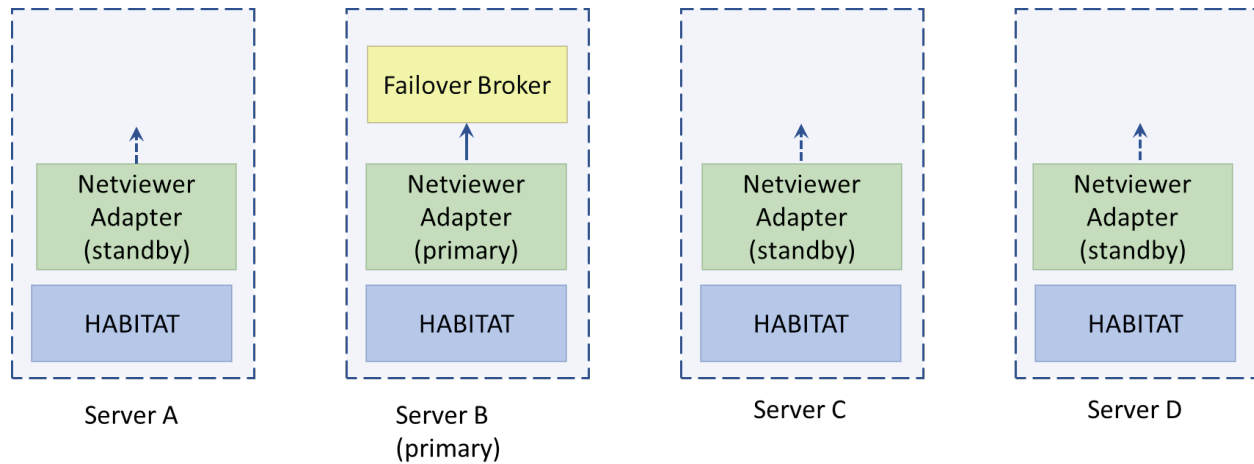
For better performance, it is recommended to adjust tile sizes on both the server and the client. Tiles that are too large take longer to generate and therefore perform poorly in the client. Tiles that are too small result in a large number of requests for tiles to the server, which may result in slower response times. To disable the tile patching functionality, remove these parameters or set their values to 0.

Also note that tile sizes are in the units of measure of the geographic display; therefore, if the coordinates are in meters, the width/height sizes are also in meters.

- **LandbaseTileLabelMinZoom:** Minimum zoom level for landbase labels. Use this parameter to declutter labels.
- **DisableIndividualDynamicUpdate:** If set to True, customer tiles are not updated for individual customer dynamics updates; customer tiles are only updated at the timeout specified by the TileStaleTimeout parameter.
- **TileStaleTimeout:** Timeout, in seconds, for recreating a tile affected by dynamics. When the last dynamic that affects a tile is older than this timeout, the tile is recreated.

Failover

The Netviewer Adapter does not use the same failover mechanism as other applications, that is, it does not use the CFGMONI interface to get a role. Rather, it uses the Failover Broker application to get its role.



Netviewer Adapter Set up in a 4 CPU configuration

The Netviewer Adapter and Failover Broker will be started by PROCMAN on the local Habitat. In the example above, the Netviewer Adapter configurations are set to connect to localhost, and the Netviewer Adapter will follow the Habitat primary system. It is possible to set the Netviewer Adapter configurations to connect to any of the 4 cpus, and this will yield faster failover times, but the primary Netviewer Adapter may not be on the primary Habitat server.

The Netviewer Adapter pushes tiles to the tile service, and only one Netviewer Adapter can connect to the tile service at a time. For read-only instances of the Netviewer Adapter, there is no restriction on running many of these at the same time, and they should be configured to not need a failover role.

! Important

Only one instance of Failover Broker can be active. This task should be running on the primary Habitat server only.

The Netviewer Adapters query the Failover Broker for a role, periodically, based on a health timer (15 seconds by default). The Failover Broker will assign the primary role to one Netviewer Adapter, and it will give this role to another Netviewer Adapter if it fails to respond.

To make the Netviewer Adapter follow the active CFGMONI server, the PROCMAN model must perform this logic based on the transitions:

- **Transition from the Enabled role to the Disabled role:**
 - Stop the Failover Broker.
 - Stop the Netviewer Adapter.
- **Transition from the Disabled role to the Standby role:**

- Start the Netviewer Adapter.
- The Failover Broker remains stopped.
- **Transition from the Standby role to the Enabled role:**
 - The Netviewer Adapter remains running.
 - Start the Failover Broker.

There is no configuration file for the Failover Broker, but parameters can be passed into the command line of the Failover Broker to override these settings:

- **port=newportnum**

By default, the Failover Broker listens on port 24470.

- **enablewaittime=newwaittime**

The wait time is used to wait before assigning the enabled role, in case another Netviewer Adapter is already enabled and has yet to poll the Failover broker. The default wait time is 15 seconds.

- **healthcheckinterval=newinterval**

The health check interval is used to check that the currently enabled Netviewer Adapter is still active, before assigning the enabled role to another Netviewer Adapter. The default health check interval is 15 seconds.

The failover configuration parameters for the NetViewer Adapter are as follows:

- **EnableFailoverRoleAssignment:** If set to True, the adapter queries the Failover Broker for its role (ENABLED or STANDBY).
- **FailOverBrokerUrl:** A comma-separated list of Failover Broker base URLs. For example: `http://hostname_a:24470/api/role,http://hostname_b:24479/api/role,http://hostname_c:27832/api/role`.
- **HealthCheckInterval:** Time period, in seconds, for sending health check messages to the Failover Broker. This interval should be less than the health check interval that is applied to the Failover Broker application.

Geographic Background

The geographic configuration parameters are as follows:

- **MapSource:** The source URL template for the background base map. The source URL template must include placeholders for x, y, and z coordinates, although their placement in the URL is up to the service provider.

Examples:

- `http://{1-4}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}.png`
- `http://mt0.google.com/vt/lyrs=y&hl=en&x={x}&y={y}&z={z}&s=Ga`



Note

You can also define a URL for an internally-hosted resource.

Note

The MapSource configuration is a legacy configuration, which supports only one map source. To use multiple map sources for background and overlay map layers, use the MapBackgroundLayerSources parameter group.

- **Proj4Def:** Projection definition in "PROJ4" format. Use the same projection that is used to translate DNOM/customer data into x/y coordinates in which the data is expressed. In particular, the "units" setting must match the units of measure of the data. For more information, see <https://proj4.org/>.

For example:

```
+proj=tmerc +lat_0=24.3333333333333 +lon_0=-81 +k=0.999941177 +x_0=200000.0001016002  
+y_0=0 +ellps=GRS80 +towgs84=0,0,0,0,0,0 +units=us-ft +no_defs
```

- **Proj4Name:** Spatial reference name used internally by OpenLayers. You can choose any value, not necessarily a standard name. For more information, see <http://spatialreference.org/ref/?search=oregon&srtext=Search>.
- **Proj4Extent1:** Minimum longitude value for geographic data, expressed in decimal degrees.
- **Proj4Extent2:** Minimum latitude value for geographic data, expressed in decimal degrees.
- **Proj4Extent3:** Maximum longitude value for geographic data, expressed in decimal degrees.
- **Proj4Extent4:** Maximum latitude value for geographic data, expressed in decimal degrees.

Note

Proj4Extent settings are optional. They indicate the maximum bounds at which the projection string is reliable. If bounds are specified, data beyond the bounds may be ignored by OpenLayers; this may or may not be desirable.

- **StartCenterX:** X-coordinate for the initial map position.
- **StartCenterY:** Y-coordinate for the initial map position.
- **MapBackgroundLayerSources/MapOverlayLayerSources:** Configurations for multiple background and overlay map sources.

Note

Ampersand characters ("&") must be escaped with ' to pass the XML parser validation.

Multiple map sources can be defined using multiple MapTileSource parameter groups. Each MapTileSource parameter group includes the following parameters:

- **name:** Map source unique identifier.
- **type:** Map source type. Only the "xyz" and "tilewms" types are supported.

XYZ tiles come from compliant web data sources that expect the x, y, and z position of the display to be embedded in the URL when fetching tiles. WMS (web map service) tiles are returned as images or tiled data.

- **declutter:** The name of the associated declutter layer. This parameter is optional. It should be used for layers that need to be visible only within certain zoom ranges.
- **label:** Map source layer label to be shown on the user interface.
- **url:** Map source URL.
- **opacity:** Layer opacity. This parameter is optional. The allowed range is 0-1.

For example, 0.5 is 50% opacity so that the layer does not completely overlay the background.

- **MapSourceParameters:** Map source parameters only apply to the "tilewms" map source types. This is an "any" object allowing the specifying of params understood for this map type.
 - **ParamName:** Parameter name. For example, LAYERS.
 - **ParamValue:** Parameter value. For example, 1.
- **defaultVisible:** Specifies whether the map source layer is visible by default.

Oms Server Connection

The NetViewer Adapter sends query and search requests that cannot be resolved locally on the OMS Server. The OMS server configuration parameters are as follows:

- **OmsServerIPs:** OMS server IPs.
- **OmsServerPort:** OMS server port number.
- **DmsQueryUserId:** DMS query user ID. The default value is {8599AE31-2409-4875-9ECF-0698FAE23EDA}.
- **RequestTimeoutPeriod:** Request timeout period, in seconds.
- **UseBinarySerialization:** Specifies whether binary serialization is used.
- **SendCompressedMessage:** Specifies whether OMS message data is compressed.

Work Model Configuration

The Work Model (Enterprise OMS) configuration parameters are as follows:

- **WorkModelWebDbEnabled:** If enabled (=1), NetViewer Adapter queries the Enterprise OMS REST API for the order data.
- **WorkModelServiceApi:** Work Model Event Handler Service endpoint.
- **WorkModelEventApi:** Work Model Notification Service events endpoint.
- **WorkModelTlsEnabled:** Specifies whether HTTPS TLS security is enabled for the Work Model Event Handler Service connection for web clients and web socket clients.
- **WorkModelTlsSubjectName:** Specifies the certificate subject name that is used when

searching for client certificates to send to the Work Model Event Handler Service.

- **WorkModelTlsIgnoreRemoteNameMismatch:** Specifies whether the remote server certificate name mismatch is ignored. If enabled (=1), the remote server certificate name can be different from the URL host name that is used to connect to the server.

DMS Server Connection

The DMS server configuration parameters are as follows:

- **HostName:** DMS server name.
- **HostPort:** DMS server port number.
- **Options:** Hello period and connection retry period. For example:

```
HELLO_PERIOD:45;CONNECT_RETRY_PERIOD:45
```

Tags

The tag configuration parameters are as follows:

- **Name:** Name of the tag image.
- **ImageData:** Tag image, in data URL format. For more information, see https://developer.mozilla.org/en-US/docs/Web/HTTP/Basics_of_HTTP/Data_URIs.
- **Width:** Tag image width.
- **Height:** Tag image height.
- **InsertionPoint:** X and Y tag offset relative to the associated device.
- **ColorToReplace:** Tag image color to replace to indicate the tag type.
- **TagPriority:** Collection of tag type colors, where the first entry is the highest priority.

Customer Search

The customer search configuration parameters are as follows:

- **SearchableFields:** List of customer fields that are searched by a customer query/search.
- **ResultFields:** List of customer fields that are returned by the server as the result of a customer query/search.

Device Search

The device search configuration parameters are as follows:

- **SearchType:** Type of search. Can be one of the following: CONTAINS, EXACTMATCH, BEGINSWITH, ENDSWITH, or DYNAMIC. If not specified, the CONTAINS type is used.
- **ResultFields:** List of device fields that are returned by the server as the result of a device query/search.

Crew Search

The crew search configuration parameters are as follows:

- **SearchableFields:** List of crew fields that are searched by a crew query/search.
- **ResultFields:** List of crew fields that are returned by the server as the result of a crew query/search.

Display Parameters

The display configuration parameters define which filters and layers apply to primary (geographic) and non-primary (non-geographic) displays:

- **Zoom:** Default zoom level for the map.
- **Filters:** Default filters for the map.
- **Layers:** Default layers to be shown on the map.

16.17. DnaSchedulerConfig.xml

The DNA Scheduler is configured by the DnaSchedulerConfig.xml file, which includes the following parameters:

- **AppPort:** The local machine port that this service is listening for gRPC connection.
- **RESTfulAppPort:** The local machine port that this service is listening for RESTful connection.
- **DNAFParamsConfigPath:** The path to the directory where the DNAF parameter files are placed.
- **Msgdef:** Message definition file name.
- **Phid:** Phase identifiers file name.
- **Alg:** algorithm control parameters file name.
- **TfType:** Transformer types file name.
- **OptAlg:** Algorithm options for optimization file name.
- **DnaRequestRecordFileName:** The local file name where request messages are saved.
- **LogFileName:** DNA Scheduler log file name.
- **LogLevel:** log level.
- **MaxMessagesInLogFile:** Maximum lines of log in one log file. When this number is reached, current log file is re-named and a new log file is created using the LogFileName.
- **MaxRequestStatusCountForEachDna:** Maximum request status of individual DNA server kept for DNA Scheduler monitor.
- **DnaRequestTimeoutInSecond:** Time out for individual request.
- **MaxReceiveMessageSize:** Maximum receive message size in MB. Default is 4MB.
- **MaxSendMessageSize:** Maximum send message size in MB. Default is unlimited.
- **EnableDetailedErrors:** If true, detailed exception messages are returned to clients when an

exception is thrown in a service method. The default is false.

Caution

Setting `EnableDetailedErrors` to true can leak sensitive information.

16.18. Structure Server Configuration File

The Structure Server reads the `StructureServerConfig.xml` configuration file, which is located in the `%IDMS_ROOT%\config\interfaces` folder.

`StructureServerConfig.xml` contains the following parameters:

- **WebServiceBaseUrl:** The base URL to specify the domain and whether to use HTTP or HTTPS.
Example: `https://localhost:17642`
- **CertificateSubject:** Specifies the certificate to be trusted in the Trusted Root Certification Authorities.
- **LogFileName:** The path including the name for the log file for StructureServer.
- **MaxMessagesInLogFile:** The maximum number of messages a log file can contain before it creates a new log file.
- **LogLevel:** The log level for the StructureServer:
`OFF = 1, FATAL = 2, ERROR = 3, WARN = 4, INFO = 5, DEBUG = 6, ALL = 7`

16.19. DermKafkaAdapterConfig.xml

The DERM Kafka Adapter is configured by the `DermKafkaAdapterConfig.xml` file, which includes the following parameters:

- **GrpcServerPort:** The local machine port that this service is listening to for a gRPC connection.
- **RESTfulAppPort:** The local machine port that this service is listening to for a RESTful API connection.
- **DerEnergizationStatusTimeoutMs:** Main server timeout in milliseconds when calling for a DER energization status.
- **AppId:** The Id of this running DERM Kafka Adapter.
- **LogFileName:** The log file name.
- **LogLevel:** The log level. 0 → Trace/All, 1 → Debug, 2 → Information, 3 → Warning, 4 → Error, 5 → Critical/Fatal.
- **MaxMessagesInLogFile:** The maximum number of lines in a single log file. When this number is reached, the current log file is renamed and a new log file is created using the `LogFileName` parameter.

- **MaxReceiveMessageSize:** The maximum size in MB for an incoming gRPC message. The default is 4 MB.
- **MaxSendMessageSize:** The maximum size in MB for an outgoing gRPC message. Default is unlimited.
- **EnableDetailedErrors:** If enabled, detailed exception messages are returned to the clients when an exception is thrown in a service method for gRPC messages. The default is disabled (false).
- **GrpcSslCertificatePfxFile:** The Pfx certificate file name for the gRPC connection.
- **RESTfulSslCertificatePfxFile:** The Pfx certificate file name for the RESTful connection.
- **BootstrapServers:** The list of comma separated host and port pairs ('Host Name:Port Number') that provides the path of the bootstrap nodes (aka 'brokers') in a Kafka cluster.
- **UseSecurityProtocol:** If security protocol is used to connect to the Kafka cluster, this flag should be enabled (true). The default value is disabled (false).
- **CertificateAuthorityLocation:** The security certificate authority is stored if the UseSecurityProtocol parameter is enabled.
- **UAAEnabled:** The flag to enable (true) or disable (false) the UAA Token configuration for accessing the Kafka cluster. The default value is disabled (false).
- **UAAEndpoint:** UAA Endpoint.
- **ConsumerPollingDelayInMs:** The sleep/pause time In milliseconds for the consumer thread processing of a group of messages from Kafka.

Selection Configuration (DerKafkaTopics)

The desired set of topics to consume/produce (subscribe/publish). These names must match across all consumers/producers who plan to exchange data on the topic.

- **DerPlan:** Plan topic name.
- **DerStatus:** Status topic name.
- **DerLimit:** Limit topic name.
- **DerAnalog:** Analog topic name.
- **DerControlStatus:** Plan response topic name.
- **DerEnergizationUpdate:** DER energization update topic name.

Selection Configuration (DerKafkaConsumerConfig)

The parameters that effect the behavior of the DERM Kafka Adapter consumer thread.

- **GroupId:** The group id of this consumer, which identifies the consumer to Kafka.
- **AutoOffsetReset:** The offset is the position in a topic stream that marks the point in the stream for subsequent I/O.
- **EnableAutoCommit:** Enables setting the consumed offset automatically at the defined AutoCommitIntervalMs.

- **AutoCommitIntervalMs:** The frequency in milliseconds that the consumed offsets are auto-committed to Kafka if EnableAutoCommit is set to true.
- **EnableAutoOffsetStore:** Enables automatic offset storing by the Kafka cluster. Should be disabled if EnableAutoCommit == true.
- **EnablePartitionEOF:** Notify the client when the end of the topic stream is reached.

Selection Configuration (DerKafkaProducerConfig)

The parameters that effect the behavior of the adapter producer thread.

- **MessageTimeoutMs:** This value, in milliseconds, is enforced locally and limits the time a produced message waits for successful delivery.

17. Command-Line Parameters

Note

- Command-line parameter names are not case sensitive.
- Command-line parameters can be used with any of the two prefixes: "/" or "-". For example, you can use /CONFIG or -CONFIG to specify the path to a configuration file.
- Paths to files can contain environment variables, which should be separated by the "%" symbol. For example, CallEntry.exe /CONFIG %IDMS_ROOT%\config\client\CallEntryConfig.xml

Executable file name: ArchiveApp.exe

Startup menu shortcut name: Archive

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application automatically tries to find the configuration file "ArchiveAppConfig.xml" in the directory where the executable file is located.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: ArchiveApp.exe /CONFIG %IDMS_ROOT%\config\server\ArchiveAppConfig.xml
3	/NOCOREDUMP	Used to disable creating a crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: CallEntry.exe

Startup menu shortcut name: Call Entry

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: CallEntry.exe /CONFIG %IDMS_ROOT%\config\client\CallEntryConfig.xml
3	/PERFORMANCE [PATH TO FILE]	Used to specify the path to the performance scripiter file. Tip: This parameter is optional.
4	/AUTOSTART	If specified, the performance scripiter file is executed automatically. Tip: This parameter is optional.
5	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: CallServer.exe

Startup menu shortcut name: Call server

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: CallServer.exe /CONFIG %IDMS_ROOT%\config\server\CallServerConfig.xml
3	/USENEWCOMMS [PATH TO FILE]	This flag specifies new communication code. Tip: This parameter is optional.
4	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: CodedTranslationEditor.exe

Startup menu shortcut name: Coded Translation Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/BPATH [PATH TO FOLDER]	Used to specify the path to the bitmap directory. Example: CodedTranslationEditor.exe /CONFIG %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /BPATH %IDMS_ROOT%\bitmaps Tip: This parameter is required to execute the application.
3	/CONFIG [PATH TO FILE] or /X	Used to specify the path to a configuration file.
4	/HPATH [PATH TO FILE]	Used to specify the path to the help file.

Executable file name: Converter.exe

Startup menu shortcut name: Converter

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: Converter.exe /CONFIG %IDMS_ROOT%\ToolsWorkspace\Converter\converter.xml
3	/DNAFINPUT [PATH TO FOLDER]	Defines the folder that contains DNAF input files.

#	Parameter	Description
4	/CONVINPUT [PATH TO FILE]	Defines the input file that should be converted.
5	/CUSTFILE [PATH TO FILE]	Defines the full path to the customer "*.bpcl" file.
6	/INPUT [PATH TO FOLDER]	Defines the full path to the converter input files.
7	/OUTPUT [PATH TO FOLDER]	Defines the full path to the converter output files.
8	/MAPPERFILE [PATH TO FILE]	Defines the full path to the map file.
9	/BATCH or /B	Defines when to run the Converter in batch mode.
10	/HELP or /H	Used to output help about command-line argument usage.
11	-TARGET ["stationName1,stationName2,stationName3"]	Specifies station names to be processed by the Converter in one string. Station names are separated with commas. Example: -Target "3LIONS,BACHELOR,RAVEN,PEBBLE BEACH"
12	-TARGET *	Specifies all stations in the input folder for conversion.
13	-TARGET [TXTFilePath]	Specifies all stations in the TXT file for conversion. It provides the path to the TXT file, including the file's name and the .txt suffix. Example: -Target %IDMS_ROOT%\ToolsWorkspace\Converter\input\StationNames.txt The TXT file lists all the stations to be processed by Converter, one station per line.
14	-MMAMODELREVIEWCONFIG	Specifies that the Model Review feature is enabled in the Model Manager application. This option is used when the Model Manager runs the Converter in silent mode.

Executable file name: CrewModelingTool.exe

Startup menu shortcut name: Crew Modeling Tool

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/BPATH [PATH TO FOLDER]	Used to specify the path to the bitmap directory. Note: This parameter is required to execute the application.

#	Parameter	Description
3	/CODEDTRANS [PATH TO FILE]	Used to specify the path to a configuration file. Example: CrewModelingTool.exe /CODEDTRANS %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /BPATH %IDMS_ROOT%\bitmaps Note: This parameter is required to execute the application.
4	/IMPORTXML [PATH TO FILE]	Imports crew data from an XML file.
5	/SAVE [PATH TO FILE]	Saves the imported crew data as a CREW file. Example: CrewModelingTool.exe /CODEDTRANS %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /BPATH %IDMS_ROOT%\bitmaps /IMPORTXML %IDMS_ROOT%\models\crew\CrewFile.XML /SAVE %IDMS_ROOT%\models\crew\CrewFile.Crew
6	/OVERWRITEIFEXISTS	Allows saving a crew file with the name of an already existing file.
7	/LOG [PATH TO FILE]	Writes all log messages in the specified text file. Example: CrewModelingTool.exe /CODEDTRANS %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /BPATH %IDMS_ROOT%\bitmaps /LOG %IDMS_ROOT%\Log\CrewToolLog.txt
8	/SILENT	If this parameter is used, the Crew Modeling Tools is working in silent mode (without user interface). Example: CrewModelingTool.exe /CODEDTRANS %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /BPATH %IDMS_ROOT%\bitmaps /LOG %IDMS_ROOT%\Log\CrewToolLog.txt /SILENT

Executable file name: CrewServer.exe

Startup menu shortcut name: Crew Server

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: CrewServer.exe /CONFIG %IDMS_ROOT%\config\server\CrewServerConfig.xml
3	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: CustomerCompiler.exe

Startup menu shortcut name: Customer List Compiler

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONVERT	Used to convert a customer file from a .txt file to a .bpcl file.
3	/SCHEMA [PATH TO FILE]	Used to specify the path to a schema file.
4	/INPUT [PATH TO FILE]	Used to specify the path to the input text file.
5	/OUTPUT [PATH TO FILE]	Used to specify the path to the compiled binary file.
6	/GENERATESQLTABLE	Used to generate an SQL script for creating the CUSTOMER table with dedicated fields from the schema file in the Archive database.
7	-DBMS	Used to specify the relational database management system (MSSQL or ORCL) for which to create an SQL script for creating the CUSTOMER table.
8	/HELP or /H	Used to output help about command-line argument usage.

Executable file name: DmsInterfaceServer.exe

Startup menu shortcut name: DMS Interface Server

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: DmsInterfaceServer.exe /CONFIG "%IDMS_ROOT%\config\client\netconf_dataservice.xml" "%IDMS_ROOT%\config\client\DmsInterfaceServerConfig.xml"
3	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.
4	/INSTALL	Used to install the DMSInterfaceServerService Windows service.
5	/UNINSTALL	Used to uninstall the DMSInterfaceServerService Windows service.

Executable file name: DNAFConfiguration.Editor.exe

Startup menu shortcut name: DNAF Configuration Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	[PATH TO FILE]	Used to specify the path to a configuration file. Example: DNAFConfiguration.Editor.exe D:\Eterra\Distribution\FantasyIsland\models\DNAFParams\DNAFParams.xml

#	Parameter	Description
3	/BATCH or /B	Used to specify that the application runs in command-line mode.
4	/EXPORTFILE	Used to specify the full path of the exported file.
5	/SAVE	Resaves the DNAF parameters file.
6	/HELP or /H	Used to output help about command-line argument usage.

Executable file name: DnomDBLoaderApplication.exe

Startup menu shortcut name: DNOM DB Loader Application

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: DnomDBLoaderApplication.exe /CONFIG D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\DNOMDatabaseLoader\DnomDBLoaderConfig.xml

Executable file name: DnomDBLoaderConfigEditor.exe

Startup menu shortcut name: DNOM DB Loader Configuration Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: DnomDBLoaderConfigEditor.exe /CONFIG D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\DNOMDatabaseLoader\DnomDBLoaderConfig.xml

Executable file name: DnomDBLoaderService.exe

Startup menu shortcut name: DNOM DB Loader Service Manager

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG=[PATH TO FILE]	Used to specify the path to a configuration file.
3	/INSTALL [NEW SERVICE NAME]	Used to specify the path to the InstallUtility.exe file.

#	Parameter	Description
4	/SERVICENAME=[NEW SERVICE NAME]	Used to specify the path to the InstallUtility.exe, which is required to run the application from the command line. Example: DnomDBLoaderService.exe /INSTALL /SERVICENAME=DNOMDBLoaderService /CONFIG="D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\DNOM DatabaseLoader\DnomDBLoaderConfig.xml"

Executable file name: DnomUniqueIDChecker.exe

Startup menu shortcut name: DNOM Unique ID Checker

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/BATCH	Used to flag that the program is running in a command-line background without the user interface.
3	/INPUT [PATH TO FOLDER]	Used to specify the path to the folder with the MOD files.
4	/OUTPUT	Used to specify the path to the folder for the output file, or the complete file path with file name and suffix. For example: DnomUniqueIDChecker.exe /BATCH /INPUT D:\Eterra\Distribution\FantasyIsland\models\RT /OUTPUT D:\Eterra\Distribution\FantasyIsland\log\CheckResult.txt
5	/HELP or /H	Used to output help about command-line argument usage.

Executable file name: EventScripter.exe

Startup menu shortcut name: Event Scripter

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/NETCONF [PATH TO FILE]	Used to specify the path to the network configuration file.
3	/SCRIPT [PATH TO FILE]	Used to specify the path to a script file.
4	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.
5	/BATCH	Used to flag that the program is running in a command-line background without the user interface. Example: EventScripter.exe /NETCONF "%IDMS_ROOT%\config\client\netconfes.xml" /SCRIPT "%IDMS_ROOT%\..\FantasyIsland\misc\EventScripts\Afternoon_storm_s cript.xml" /BATCH

Executable file name: GeographicBackgroundCompiler.exe

Startup menu shortcut name: Geographic Background Compiler

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file.
3	/BATCH	Used to flag that the program is running in a command-line background without the user interface.
4	/BPATH [PATH TO FOLDER] or /BITMAPPATH [PATH TO FOLDER]	Used to specify the path to the bitmap directory. Example: GeographicBackgroundCompiler.exe /CONFIGFILE %IDMS_ROOT%\ToolsWorkspace\GeographicBackground_Shape\GeographicBackgroundCompilerConfiguration.xml /BPATH %IDMS_ROOT%\bitmaps Tip: This parameter is required to execute the application.
5	/INPUT [PATH TO FOLDER] or /INPUTFOLDER [PATH TO FOLDER]	Used to specify the path to the input folder. This is an optional argument that overrides the input folder configuration specified in the Geographic Background Compiler configuration file.
6	/OUTPUT [PATH TO FOLDER] or /OUTPUTFOLDER [PATH TO FOLDER]	Used to specify the path to the output folder. This is an optional argument that, if presented, overrides the output folder configuration specified in the Geographic Background Compiler configuration file.
7	/FOLDER [PATH TO FOLDER] or -FOLDER [PATH TO FOLDER]	An optional argument used to specify the name of the landbase tile to compile. In this context, the tile name is equivalent to the name of the folder name describing the tile. If this argument is not provided, all tiles are compiled.
8	/HELP or /H	Used to output help about command-line argument usage.

Executable file name: GridTabularBuilder.exe

Startup menu shortcut name: Grid Tabular Builder

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/BPATH [PATH TO FOLDER]	Used to specify the path to the bitmap directory. Example: GridTabularBuilder.exe /BPATH %IDMS_ROOT%\bitmaps Tip: This parameter is required to execute the application.
3	/HPATH [PATH TO FILE]	Used to specify the path to the help file.

Executable file name: ModelExport.exe

Startup menu shortcut name: Model Export

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: ModelExport.exe /CONFIG D:\Eterra\Distribution\ModelExport\Bin\ModelExportConfig.xml
3	-TARGET ["stationName1,stationName2,stationName3"]	Specifies station names to be processed by the Model Export in one string. Station names are separated with commas. Example: -Target "3LIONS,BACHELOR,RAVEN,PEBBLE BEACH"
4	-TARGET *	Specifies all stations in the input folder for conversion.
5	-TARGET [TXTFilePath]	Specifies all stations in the TXT file for conversion. It provides the path to the TXT file, including the file's name and the .txt suffix. Example: -Target D:\Eterra\Distribution\ModelExport\Bin\StationNames.txt The TXT file lists all stations to be processed by the Model Export, one station per line.
6	/BATCH or /B	Used to flag that the program is running in a command-line background without the user interface.
7	/HELP or /H	Used to output help about command-line argument usage.

Executable file name: ModelManager.ConfigurationEditor.exe

Startup menu shortcut name: Model Manager Configuration Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	[PATH TO FILE] or /CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: ModelManager.ConfigurationEditor.exe /CONFIG %IDMS_ROOT%\ToolsWorkspace\ModelManagerFolders\ModelManager.xml

Executable file name: ModelManagerApplication.exe

Startup menu shortcut name: Model Manager Application

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	[PATH TO FILE] or /CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: ModelManagerApplication.exe /CONFIG %IDMS_ROOT%\ToolsWorkspace\ModelManagerFolders\ModelManager.xml

Executable file name: ModelManagerService.Client.exe

Startup menu shortcut name: Model Manager Service Client

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	[URL TO SERVICE] or /SERVICE [URL TO SERVICE]	Used to specify URL to service. Example: ModelManagerService.Client.exe /SERVICE http://localhost:8085/ModelManagerService

Executable file name: ModeFileExporter.exe

Startup menu shortcut name: N/A

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/MODFILE [PATH TO FILE]	Used to specify URL to service. Example: ModelManagerService.Client.exe /SERVICE http://localhost:8085/ModelManagerService
3	/DBUSER [USER NAME]	Used to specify the database (DB) user name.
4	/DBPASSWORD [PASSWORD]	Used to specify the DB password.
5	/DBCONNECTSTR [CONNECTION STRING]	Used to specify the DB connection string.
6	/CONFIG [PATH TO FILE]	Used to specify the path to the configuration file.
7	/DB [DBSERVER]	Used to specify the DB server type. Can be ORACLE or MSSQL. Example: ModFileExporter.exe /MODFILE "D:\Eterra\distribution\FantasyIsland\models\RT\RAVEN.mod" /DB ORACLE /DBUSER scott /DBPASSWORD tiger /DBCONNECTSTR //asterix.empgrid.esca.com:1521/idms.esca.com /CONFIG "D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\DnomDatabaseLoader\DnomExporterConfig.txt"

#	Parameter	Description
8	/TOCSV [PATH TO FILE]	Used to export a model to a CSV file. Optionally, you can specify a path to the CSV file. If a file path is not specified, the file will be created in the ModFileExporter.exe directory. Example: ModFileExporter.exe /ModFile "D:\Eterra\Distribution\FantasyIsland\Models\RT\RAVEN.mod" /TOCSV "D:\RAVEN.csv"
9	/HELP or /H	Used to output help about command-line argument usage.

Executable file name: NetDataReplicator.exe

Startup menu shortcut name: NetData Replicator

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: NetDataReplicator.exe /CONFIG %IDMS_ROOT%\config\server\NetDataReplicatorConfig.xml
3	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: NetDataServer.exe

Startup menu shortcut name: Net Data Server

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	[PATH TO FILE] or /CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: NetDataServer.exe /CONFIG %IDMS_ROOT%\config\server\NetDataServerConfig_localhost.txt
3	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: NetDynamicsConfigEditor.exe

Startup menu shortcut name: Net Dynamics Configuration Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/DEFAULTFILE [PATH TO FILE]	Used to specify the path to the bitmap directory. Example: NetDynamicsConfigEditor.exe /DEFAULTFILE %IDMS_ROOT%\config\client\netconf_localhost.xml Tip: This parameter is required to execute the application.
3	/HPATH [PATH TO FILE]	Used to specify the path to the help file.

Executable file name: NetServer.exe

Startup menu shortcut name: Net Server

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: NetServer.exe /CONFIG %IDMS_ROOT%\config\server\NetServerOmsConfig_localhost.txt
3	/SERVER [PATH TO FILE]	Used to specify the path to the OMS server configuration file.
4	/CREW [PATH TO FILE]	Used to specify the path to the OMS Crew Employee file.
5	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: netsrv.exe

Startup menu shortcut name: DMS Main Server

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	[PATH TO FILE] or /CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: netsrv.exe /CONFIG %IDMS_ROOT%\config\server\NetServerMainConfig_localhost.txt
3	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.
4	/FULLCOREDUMP	Used to specify the necessity of full memory dump output.

Executable file name: OmsConfigClient.exe

Startup menu shortcut name: OMS Client Configuration Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/BPATH [PATH TO FOLDER]	Used to specify the path to the bitmap directory. Tip: This parameter is required to execute the application.
3	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: OmsConfigClient.exe /BPATH "%IDMS_ROOT%\bitmaps" /CONFIG %IDMS_ROOT%\config\client\OmsClientConfig.xml Tip: This parameter is required to execute the application.
4	/HPATH [PATH TO FILE]	Used to specify the path to the help file.

Executable file name: OmsConfigServer.exe

Startup menu shortcut name: OMS Server Configuration Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CODEDTRANS [PATH TO FILE]	Used to specify the path to the OMS Coded Translation file. Tip: This parameter is required to execute the application.
3	/CONFIG [PATH TO FILE]	Used to specify the path to the OMS server Configuration file. Example: OmsConfigServer.exe /CODEDTRANS %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /CONFIG %IDMS_ROOT%\config\server\OmsServerConfig.xml Tip: This parameter is required to execute the application.
4	/HPATH [PATH TO FILE]	Used to specify the path to the help file.

Executable file name: PermitBuilder.exe

Startup menu shortcut name: Permit Builder

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Specifies the path to the Permission Management plug-in configuration file. If Permit Builder is run with this option, it will have the same colors and settings as the Permission Management plug-in.
3	/INPUT [PATH TO FILE]	Specifies the path to the AORRef hierarchy model.
4	/OUTPUTPATH [PATH TO FOLDER]	Specifies the path to the output folder for hdbimport and intersection files. If this option is not specified, the default location is used (D:\Eterra\Distribution\Tools\bin).
5	/OUTPUT [NAME OF THE OUTPUT HDBIMPORT FILE]	Specifies the name of the output hdbimport file. If this option is not specified, the default file name is used (hdbimport.txt).

#	Parameter	Description
6	/INTERSECTION [PATH TO THE OUTPUT INTERSECTION FILE]	Specifies the path to the output intersection model file. If this option is not specified, the file is exported to the OUTPUTPATH folder, if specified, and the default file name is used (Intersections.txt).
7	/RECORD [FILE TYPE]	Specifies the type of output file to be created. HMODE: Use this argument to create an hdbimport file with the HMODE records (hdbimport_hmode.txt). AREA: Use this argument to create an hdbimport file with the AREA records (hdbimport_area.txt). UWMODE: Use this argument to create an intersection model file (Intersections.txt). ALL: Use this argument to create an hdbimport file with both AREA and HMODE records (hdbimport.txt) and an intersection model file (Intersections.txt).
8	/VERBOSE	Add this option to an import or export command to show the command processing status in the command line.
9	/HELP or /H	Used to output help about command-line argument usage.

Executable file name: RelayServer.exe

Startup menu shortcut name: Relay Server

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: RelayServer.exe /CONFIG D:\Eterra\Distribution\Server\bin\CallServer.exe /CONFIG "%IDMS_ROOT%\config\server\RelayServerConfig.xml"
3	/NOCOREDUMP	Used to disable creating crash dump and crash log files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: ResourceEditor.exe

Startup menu shortcut name: Resource Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.

#	Parameter	Description
2	/CONFIG [PATH TO FILE] or /CONFIGFILE [PATH TO FILE]	Used to specify the path to a configuration file. Example: ResourceEditor.exe /CONFIG %IDMS_ROOT%\ToolsWorkspace\ResourceEditor\ResourceEditorConfig.xml
3	/RESOURCEDLLDIR [PATH TO FILE] or /RESOURCE_DLL_DIR [PATH TO FILE]	Used to specify the path to the folder with default resources.
4	/RESOURCEDATADIR [PATH TO FILE] or /RESOURCE_DATA_DIR [PATH TO FILE]	Used to specify the path to the folder with pictures.

Executable file name: RoboCrew.exe

Startup menu shortcut name: Robo Crew

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: RoboCrew.exe /CONFIG %IDMS_ROOT%\config\server\CrewSimulatorConfig.xml Tip: This parameter is required to execute the application.
3	/SERVER [PATH TO FILE]	Used to specify the path to the server configuration file.
4	/NETCONF [PATH TO FILE]	Used to specify the path to the net-server configuration file.
5	/LOG [PATH TO FILE] or /LOG_PATH [PATH TO FILE]	Used to specify the path to the log file.

Executable file name: RoboDispatch.exe

Startup menu shortcut name: Robo Dispatcher

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/NETCONF [PATH TO FILE]	Used to specify the path to a configuration file. Example: RoboDispatch.exe /NETCONF %IDMS_ROOT%\config\server\NetconfRoboDispatch.xml

Executable file name: SnapshotViewer.exe

Startup menu shortcut name: Snapshot Viewer

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: SnapshotViewer.exe /CONFIG %IDMS_ROOT%\ToolsWorkspace\SnapshotViewerConfig.xml
		 Tip This parameter is required to execute the application.
3	/HPATH [PATH TO FILE]	Used to specify the path to the help file.

Executable file name: Stat-Logger.exe

Startup menu shortcut name: Model Quality Stat Logger

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/PATH [PATH TO FOLDER]	Used to specify the path to the Converter logs. Typically, this is the output folder path of the Converter. The Model Quality Stat Logger writes output files to this folder.
3	/DNOM	Used to create a file with DNOM statistics.
4	/F	Used to create a file with Fatal statistics.
5	/W	Used to create a file with Warning statistics.
6	/E	Used to create a file with Error statistics.
7	/I	Used to create a file with Info statistics.
8	/HELP	Used to output help about command-line argument usage.
	or	
	/H	

Executable file name: StructureCompiler.exe

Startup menu shortcut name: N/A

#	Parameter	Description
1	/INPUT	The path to the directory containing one or more *.geojson files, with the structure object and the associated equipment objects to create the equipment and structure index. The index is used by the Converter tool as well as the StructureServer.
2	/OUTPUT	The filename of the equipment and structure index file that will be created.
3	/ENCMODE	Example: StructureCompiler /INPUT D:\Structures /OUTPUT D:\Structures\EquipmentAndStructures.index

#	Parameter	Description
4	/ENCKEYSCOPE	Defines if the resulting index must be encrypted or not. The value can be Encrypted or None. By default, this is assumed to be None. This parameter is only required if the index is required to be encrypted.
5	/ENCKEYVERSION	The encryption scope. The format for this value is an uppercase text string, with a maximum length of 11-characters. This parameter is required if the index is required to be encrypted.
6	/ENCKEYSTORAGE	The encryption key storage. This value can be any one of the following: Default: The value internally resolved to EnviromentVariableDpApi. EnvironmentVariable: Specifies that the encryption key is defined in the system using the environment variable and that the key value is not encrypted. EnvironmentVariableDpApi: Specifies that the encryption key is defined in the system using an environment variable and that the key is encrypted using the DPAPI encryption method. This parameter is required if the index is required to be encrypted.

Executable file name: StructureIndexer.exe

Startup menu shortcut name: N/A

#	Parameter	Description
1	/IMPORT [PATH TO FOLDER]	Path to a directory containing one or more "*.geojson" files with structure objects to create index from. The index is used for searching structure objects.
2	/ID	Optional parameter. Specifies which structure object property to use as the "ID" property for index (by default, "id").
3	/NAME	Optional parameter. Specifies which structure object property to use as the "Name" property for index (by default, "name").
4	/OUTPUT	Optional parameter. Path to a directory where the Structure.index file is saved. By default, the file is saved to the Import directory. Example: StructureIndexer.exe /INPUT D:/structures /ID OBJECTID /NAME ID_NO /OUTPUT D:/structures

Executable file name: SubstationEditor.exe

Startup menu shortcut name: Substation Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file.
	or	
	/CONFIGFILE	

#	Parameter	Description
3	/EXPORT [PATH TO FOLDER] or /EXPORTDIR	A directory containing one or more "*_subeditor.xml" files to be used for the export process. The "-EXPORTTYPE" argument determines the files to be exported. This argument overrides (but does not overwrite) the path in the preferences file.
4	/EXPORTFILE [PATH TO FILE]	The name of the Substation Editor file to be used for the export process. The "-EXPORTTYPE" argument determines the files to be exported.
5	/EXPORTTYPE [TYPE]	The type of export to perform. The arguments can be: ALL: Export the station internals and placements, the Browser displays, and the Control and Scada databases. SMP: Export a Control database. DDL: Export the Browser display. INTERNALS: Export the station internals and placements. HDB: Export a Scada database. Example: SubstationEditor.exe /CONFIG "%IDMS_ROOT%\ToolsWorkspace\SubstationEditor\SubstationEditorConfiguration.xml" /EXPORTFILE " %IDMS_ROOT%\ToolsWorkspace\SubstationEditor\data\PEBBLE BEACH_subeditor.xml" /OUTPUT %IDMS_ROOT%\ToolsWorkspace\Converter\input /EXPORTTYPE ALL
6	/EXPORTSCOPE [SCOPE]	This parameter specifies the scope of the export, which can be equal to one of the following states: AllStations: Export all stations. DnomstationOnly: Export only DNOM stations.
7	/MMAUTOPROCESSFLAG	Specify either "true" or "false" to use the <AutoProcess> element in the ISSG file. The configuration value is used if it is not defined by the command-line parameter.
8	/OUTPUT [PATH TO FOLDER] or /OUT [PATH TO FOLDER]	Specifies the directory where the exported file is saved. If this is not specified, the file is exported to the same directory as the input file.
9	/CULTURE [CULTURE]	The culture name to use when the user interface is presented in a locale other than U.S.English.
10	/EMPSCADADATASOURCE [PATH TO FILE]	The SCADA datasource to use for all SCADA datasource property values that are blank in the substation model. This is used only when the EXPORTTYPEvalue is specified as HDB or ALL.

#	Parameter	Description
11	/MULTIFILE [VALUE]	Parameter that can be used with the following values: ASSOC: The command to reapply associations for multi-file processing to all the substations in INPUTDIR. TEMPLATE: The command to reapply template values for multi-file processing to all the substations in INPUTDIR. CORRECTMEAS: The command to correct measurement associations. Example: SubstationEditor.exe /CONFIG %SSECFG%\SubstationEditorConfiguration.xml /MULTIFILE ASSOC /INPUTDIR %IDMS_ROOT%\ToolsWorkspace\SubstationEditor\data Example: SubstationEditor.exe /CONFIG "%IDMS_ROOT%\ToolsWorkspace\SubstationEditor\SubstationEditorConfiguration.xml" /EXPORTFILE " %IDMS_ROOT%\ToolsWorkspace\SubstationEditor\data\PEBBLE BEACH_subeditor.xml" /INPUTDIR %IDMS_ROOT%\ToolsWorkspace\SubstationEditor\data /EXTDIR %IDMS_ROOT%\ToolsWorkspace\Converter\input -MULTIFILE CORRECTMEAS
12	/IMPORT [PATH TO FOLDER] or /INPUTDIR [PATH TO FOLDER]	A directory containing one or more "*.subeditor.xml" files on which a MULTIFILE operation is performed.
13	/EXTDIR [PATH TO FOLDER]	A directory containing station external files (*.xml).

Executable file name: SwitchOrderConfiguration.exe

Startup menu shortcut name: Switching Order Configuration Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.
2	/CONFIG [PATH TO FILE] or /X	Used to specify the path to a configuration file. Example: SwitchOrderConfiguration.exe /CONFIG %IDMS_ROOT%\config\client\SwitchOrderClientConfig.xml Tip: This parameter is required to execute the application.
3	/HPATH [PATH TO FILE]	Used to specify the path to the help file.

Executable file name: SwitchOrderServer.exe

Startup menu shortcut name: Switch Order Server

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is executed in its default state.

#	Parameter	Description
2	[PATH TO FILE] or /CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Example: SwitchOrderServer.exe /CONFIG %IDMS_ROOT%\config\server\SwitchOrderServerConfig.txt
3	/NOCOREDUMP	Used to disable creating crash dump and crash files when the application crashes. By default, the core dump is output to the application file's location.

Executable file name: ViewerConfigEditor.exe

Startup menu shortcut name: Viewer Configuration Editor

#	Parameter	Description
1	[NOT SPECIFIED]	If command-line parameters are not specified, the application is not executed.
2	/CONFIG [PATH TO FILE]	Used to specify the path to a configuration file. Tip: This parameter is required to execute the application.
3	/BPATH [PATH TO FOLDER]	Used to specify the path to the bitmap directory. Example: ViewerConfigEditor.exe /CONFIG %IDMS_ROOT%\config\client\ViewerConfig.xml /BPATH %IDMS_ROOT%\bitmaps  Tip This parameter is required to execute the application.
4	/HPATH [PATH TO FILE]	Used to specify the path to the help file.

18. Configuration Changes in ADMS 3.16.0.0

18.1. Server Configuration Changes

This chapter lists changes that occurred in the server configuration files for ADMS3.16.0.0.

Note

Parameters followed by an asterisk (*) are optional and are not included in the configuration file by default.

ANPSAdapterConfig.xml

ANPSAdapterConfig.xml Parameter

New Parameters

```
<Field name="UseUAAToken">false</Field>
```

```
<Field name="UaaEndPoint">https://ingress.local/uaa/oauth/token</Field>
```

ArchiveAppConfig.xml

ArchiveAppConfig.xml Parameter

New Parameters

```
<DataArchivedIntervalDays>1</DataArchivedIntervalDays>
```

ArchiveAppConfig_ool.xml

This configuration file is deleted.

CallServerConfig.xml

CallServerConfig.xml Parameter

New Parameters

```
<LogMessageTypes json:Array='true'>
```

```
<Type>AmiServerWebService</Type>
```

```
<SeparateLogToDisk>true</SeparateLogToDisk>
```

```
<EnableAppendLog>false</EnableAppendLog>
```

```
<Color>LightBlue</Color>
```

```
</LogMessageTypes>
```

```
<UseCustomHeader>false</UseCustomHeader>
```

```
<CustomHeaderKey>myCustomCallHeader</CustomHeaderKey>
```

```
<CustomHeaderValue>myCustomCallCompany</CustomHeaderValue>
```

CallServerConfig.xml Parameter

```
<UseCustomHeader>false</UseCustomHeader>

<CustomHeaderKey>myCustomAMIHeader</CustomHeaderKey>

<CustomHeaderValue>myCustomAMISCompany</CustomHeaderValue>

<PluginDefinition>

<Type>WebService</Type>

<Name>AmiServerWebService</Name>

<Enabled>true</Enabled>

...
```

CrewServerConfig.xml

CrewServerConfig.xml Parameter

New Parameters

```
<LogMessageTypes json:Array='true'>

<Type>MobilityGatewayCrewWebClient</Type>

<SeparateLogToDisk>true</SeparateLogToDisk>

<Color>Pink</Color>

</LogMessageTypes>

<UseCustomHeader>false</UseCustomHeader>

<CustomHeaderKey>myCustomWMSWebHeader</CustomHeaderKey>

<CustomHeaderValue>myCustomWMSCompany</CustomHeaderValue>

<UseCustomHeader>false</UseCustomHeader>

<CustomHeaderKey>myCustomCrewModelingHeader</CustomHeaderKey>

<CustomHeaderValue>myCustomCrewModelingCompany</CustomHeaderValue>

<PluginDefinition>

<Enabled>true</Enabled>

<Type>WebServiceClient</Type>

<Name>MobilityGatewayCrewWebClient</Name>

...
```

DnaSchedulerConfig.xml

DnaSchedulerConfig.xml Parameter

New Parameters

```
<MaxRequestStatusCountForEachDna>10</MaxRequestStatusCountForEachDna>

<DnaRequestTimeoutInSeconds>300</DnaRequestTimeoutInSeconds>
```

DnaSchedulerConfig.xml Parameter

<MaxReceiveMessageSize>10</MaxReceiveMessageSize>

<MaxSendMessageSize>10</MaxSendMessageSize>

<EnableDetailedErrors>>false</EnableDetailedErrors>

<RESTfulAppPort>7168</RESTfulAppPort>

<DNAFParamsConfigPath>%IDMS_ROOT%\models\DNAFParams\</DNAFParamsConfigPath>

<Msgdef>msgdef.dat</Msgdef>

<Phid>phid.dat</Phid>

<Alg>alg.dat</Alg>

<TfType>tftype.dat</TfType>

<OptAlg>optalg.dat</OptAlg>

Changed Parameters

<LogFileName>%IDMS_ROOT%\log\DnaScheduler\DnaScheduler.log</LogFileName>

0 - Trace/All

1 - Debug

2 - Information

3 - warning

4 - Error

5 - Critical/Fatal

<LogLevel>2</LogLevel>

Removed Parameters

<DnaInstanceCount>2</DnaInstanceCount>

<DMS_BINDIR>%DMS_BINDIR%</DMS_BINDIR>

<DnaExecutableName>DnaSrv.exe</DnaExecutableName>

<DnaConfig>%IDMS_ROOT%\config\server\NetServerDnaConfig_localhost.txt</DnaConfig>

<DnaInstanceStatusValidSeconds>2</DnaInstanceStatusValidSeconds>

NetServerDnaConfig_localhost.txt

NetServerDnaConfig_localhost.txt Parameter

New Parameters

DNA_WRITE_DNAF_RESULTS_ODS = 0

ODS_DNAF_RESULT_DBNAME = DNAF

ODS_SERVER_HOST = 127.0.0.1

ODS_SERVER_PORT = 5000

DNA_ODS_CONNECTION_TIMEOUT = 3600

Changed Parameters

NetServerDnaConfig_localhost.txt Parameter

Logging

LOG_ADD_FILENAME_PREFIX = 1

LOG_FILE_NAME = %IDMS_ROOT%\log\dna\netsrv_dna.log

Thread pools

THREADS_CLIENT = 1

Removed Parameters

Logging

ISD_LOG_FILE_NAME = %IDMS_ROOT%\log\isd_dna.log

Server Mode

REALTIME = 1

Debug Parameters

NTP_OUTPUT_DETAIL = 1

STATION_MAP_PUBLISH_PERIOD = 30

Client connection parameters

CLIENT_HELLO_TIMEOUT = 180

DNAF_RESULTS_BY_TCP = 1

DNAF_DPF_ENTRY_SUB_ONLY = 1

DNA Server Specific

SEARCH = \$DNAF_INPUT_PATH\$DNAF_PARAMS_DNAF.xml,\$MODEL_PATH\$*.sm,\$DNAF_INPUT_PATH\$Manual_Entry.dat

GRPC_CLIENT_TIMEOUT = 2000 millionseconds

Localization

ISD_REMOTE_ENCODING = MBCS

ISD_REMOTE_LOCALE = <language code>_<variant/locale>

PERMIT_REMOTE_ENCODING = MBCS

PERMIT_REMOTE_LOCALE = <language code>_<variant/locale>

NetServerDnafConfig_localhost.txt

NetServerDnafConfig_localhost.txt Parameter

New Parameters

e-terrahabitat PERMIT Server

PERMIT_CHECK_ENABLED = 0

PERMIT_SERVER_IP = 127.0.0.1

PERMIT_SERVER_PORT = 5345

PERMIT_SERVER_TIMER_PERIOD = 9

NetServerDnafConfig_localhost.txt Parameter

0 - No permit data

1 - Get Permit data from Main server

2 - Connect permit server to get permit data, the above permit server info has to be provided

PERMIT_DATA_OPTION = 2

Use External DNA server

Connecting to Dna Scheduler

NA_SCHEDULER_HOST = 127.0.0.1

DNA_SCHEDULER_PORT = 5028

Specify which DNAF app should be sent to DNA server to evaluate

DNAF_USE_DNA_DPF = 1

DNAF_USE_DNA_PRV = 1

DNAF_USE_DNA_SCC = 1

DNAF_USE_DNA_LVM = 1

DNAF_USE_DNA_FISR = 1

DNAF_USE_DNA_AFR = 1

DNAF_USE_DNA_PLOS = 1

Changed Parameters

FD_DNAF_INPUTS_FILES = \$DYNAMIC_PATH\$manuals.sav,\$DNAF_INPUT_PATH\$DNAF_PARAMS_DNAF.xml
,\$DNAF_INPUT_PATH\$Manual_Entry.dat,\$DNAF_INPUT_PATH\$tftype.dat,\$DNAF_INPUT_PATH\$alg.dat,
\$DNAF_INPUT_PATH\$optalg.dat,\$DNAF_INPUT_PATH\$msggdef.dat,\$DNAF_INPUT_PATH\$phid.dat

Removed Parameters

DNAF server

DNA_SCHEDULER_HOST = 127.0.0.1

DNA_SCHEDULER_PORT = 5028

GRPC_CLIENT_TIMEOUT = 2000 millionseconds

NetServerFollowerConfig_localhost.txt

NetServerFollowerConfig_localhost.txt Parameter

New Parameters

Follower server mode

DMS_MANUAL_DYN_KEY_CONTAINS_STR = /RFBSIF=,/FLTRIGGER=

Sample CyberArk Secret store configuration

TRUSTED_USER_ID_IFOLLOWER = {6924103C-446A-4636-B0E6-C91673E74899}

TRUSTED_USER_NAME_IFOLLOWER = INDEPENDENT_FOLLOWER

NetServerMainConfig_localhost.txt

NetServerMainConfig_localhost.txt Parameter

New Parameters

Sample CyberArk Secret store configuration

TRUSTED_USER_ID_IFOLLOWER = {6924103C-446A-4636-B0E6-C91673E74899}

TRUSTED_USER_NAME_IFOLLOWER = INDEPENDENT_FOLLOWER

NetServerOmsConfig_localhost.txt

NetServerOmsConfig_localhost.txt Parameter

New Parameters

OMS Server

If UPDATE_INCIDENT_ETR_WHEN_AREA_NOTE_DELETED = 1, Incident ETR will be cleared when the Area Settings is deleted.

UPDATE_INCIDENT_ETR_WHEN_AREA_NOTE_DELETED = 0

OMS Incident Clearing

FIELD_CLEAR_DEVICE_LEVEL = customer

AMI_PRN_CLEAR_DEVICE_LEVEL = recloser

Momentary Incidents

OMS_MOMENTARY_OUTAGE_DURATION = 120

Incident ITR/ETR

DEFAULT_ETR_OPTION_FOR_SPLIT_INCIDENT_BY_CUSTOMER = KEEP

DEFAULT_ETR_OPTION_FOR_SPLIT_INCIDENT_BY_DEVICE = NEW

Other Incident Parameters

AUTOMATICALLY_CONVERT_PLANNED_OUTAGE_TO_INCIDENT_WHEN_ASSOCIATED_DEVICE_IS_OPEN = 0

Other Miscellaneous Parameters

OMS_RELATED_MAINTENANCE_ORDER_ENABLED = 1

Changed Parameters

AOR Permission Manager

PERMISSION_UAA_END_POINT = https://k8s-idmst-wmvuvrgx-ing-86537516d61f7b14.elb.us-east-1.amazonaws.com/uaa/oauth/token

OMS Incident Clearing

OMS_AUTOCLEAR_MOMENTARY_INCIDENTS = 0

NetServerReadOnlyConfig_localhost.txt

NetServerReadOnlyConfig_localhost.txt Parameter

New Parameters

NetServerReadOnlyConfig_localhost.txt Parameter

Oms Server Specific Parameters

OMS_ETR_EXPIRY_TIME = 30

OMS_ETR_EXPIRY_SOON = 15

AOR Permission Manager

PERMISSION_UAA_END_POINT = https://k8s-idmst-wmvuvrgx-ing-86537516d61f7b14.elb.us-east-1.amazonaws.com/uaa/oauth/token

PERMISSION_READ_VALUE = oms.Read

PERMISSION_WRITE_VALUE = oms.Write

PERMISSION_API_VERSION = v2

PERMISSION_REFRESH_INTERVAL = 10

UAA Token Service

NETSERVER_UAA_AUTH_CLIENTID, NETSERVER_UAA_AUTH_CLIENTSECRET

NETSERVER_UAA_AUTH_USERNAME, NETSERVER_UAA_AUTH_PASSWORD

OmsAdapterConfig.xml

This configuration file is deleted.

OmsServerConfig.xml

Multiple additions, deletions, and changes.

SwitchOrderServerConfig.xml

Multiple additions, deletions, and changes.

18.2. Client Configuration Changes

This chapter lists changes that occurred in the client configuration files for ADMS3.16.0.0.

Note

The following tables exclude some configuration parameters. These cases are indicated with an ellipsis (...).

DermKafkaAdapterConfig.xml

DermKafkaAdapterConfig.xml Parameter

New Parameters

<UAAEnabled>>false</UAAEnabled>

<UAAEndpoint></UAAEndpoint>

DermKafkaAdapterConfig.xml Parameter

DER_KAFKA_UAA_AUTH_CLIENTID

DER_KAFKA_UAA_AUTH_CLIENTSECRET

<TagRestSettings>

<PermitConsole>UTDMSA00</PermitConsole>

<EnableDualSSL>>false</EnableDualSSL>

<CertStore>Personal</CertStore>

<CertSubjectName>GC89Z3D3E.logon.ds.ge.com</CertSubjectName>

Changed Parameters

<QueryIntervalInSeconds>600</QueryIntervalInSeconds>

<DisableAllArchive>true</DisableAllArchive>

<DisableFISRArchive>true</DisableFISRArchive>

<DisableVVCArchive>true</DisableVVCArchive>

<DisableAFRArchive>true</DisableAFRArchive>

<DisableFLArchive>true</DisableFLArchive>

<DisableDPFArchive>true</DisableDPFArchive>

<DisableStationFilesArchive>true</DisableStationFilesArchive>

<DisableStationExceptionArchive>true</DisableStationExceptionArchive>

<DisableFeederExceptionArchive>true</DisableFeederExceptionArchive>

<DisableAllDpfSubQueries>true</DisableAllDpfSubQueries>

<DisableDPFTfLossResultsQuery>true</DisableDPFTfLossResultsQuery>

<DisableDPFMeasIslandQuery>true</DisableDPFMeasIslandQuery>

<DisableDPFFeederLossQuery>true</DisableDPFFeederLossQuery>

<DisableFeederExceptionQuery>true</DisableFeederExceptionQuery>

<DisableStationExceptionQuery>true</DisableStationExceptionQuery>

<EnableMessageLog>true</EnableMessageLog>

OFF = 1,

FATAL = 2,

ERROR = 3,

WARN = 4,

INFO = 5,

DEBUG = 6,

ALL = 7

<LogLevel>7</LogLevel>

DermKafkaAdapterConfig.xml Parameter

<Url>http://localhost:38669/tagrest/api/V3?requestType=tag_action</Url>

DnafDisplayPluginConfig.xml

DnafDisplayPluginConfig.xml Parameter

Changed Parameters

<StartupTabs>

<string>DNAFAGGREGATION</string>

netconf_localhost.xml

netconf_localhost.xml Parameter

New Parameters

<NetDynCtrl>

<ConnectingTimeoutSeconds>0</ConnectingTimeoutSeconds>

<MaxAutoCheckCrcCount>5</MaxAutoCheckCrcCount>

<EnableStrictSecurity>>false</EnableStrictSecurity>

<MaxConnectionRefuseLog>5</MaxConnectionRefuseLog>

<CheckPersistTraceBeforeClear>>false</CheckPersistTraceBeforeClear>

<ParallelModelLoad>>true</ParallelModelLoad>

<DisableFireDynChangedEvent>>true</DisableFireDynChangedEvent>

Changed Parameters

<MaxZoom>500</MaxZoom>

<MaxAllowedMessageSizeMB>128</MaxAllowedMessageSizeMB>

NetServerStudyConfig.txt

NetServerStudyConfig.txt Parameter

Changed Parameters

DPF CBO Parameters

DPF_CBO_LOOK_AHEAD_DURATION = 0

NetServerStudyDnaConfig.txt

Multiple additions, deletions, and changes.

OmsDisplayPluginConfig.xml

OmsDisplayPluginConfig.xml Parameter

New Parameters

OmsDisplayPluginConfig.xml Parameter

<OverviewColumnConfig>

<Name>New</Name>

<Field>SummaryCount1</Field>

...

Changed Parameter

<NewOrderWorksheetMode>Tab</NewOrderWorksheetMode>

<Select>.....ISRELATEDMAINTENANCEORDERENABLED....</Select>

OmsInterfaceServerConfig_UAA.xml

OmsInterfaceServerConfig_UAA.xml Parameter

Changed Parameters

<UaaEndpoints>https://k8s-idmst-wmvuvrgx-ing-86537516d61f7b14.elb.us-east-1.amazonaws.com/uaa/oauth/token</UaaEndpoints>

SwitchOrderClientConfig.xml

SwitchOrderClientConfig.xml Parameter

Changed Parameters

<UiSwoPropColapsed>false</UiSwoPropColapsed>

<UiDIPropColapsed>false</UiDIPropColapsed>

<UiSwoStateColapsed>false</UiSwoStateColapsed>

<UiDIStateColapsed>false</UiDIStateColapsed>

ViewerConfig.xml

Multiple additions, deletions, and changes.

18.3. Interfaces Configuration Changes

This chapter lists changes that occurred in the interfaces configuration files for ADMS3.16.0.0.

Note

The following tables exclude some configuration parameters. These cases are indicated with an ellipsis (...).

interface_CallServerConfig.xml

This configuration file is added newly.

interface_CertificateLoaderAppConfig.xml

This configuration file is added newly.

interface_CrewServerConfig.xml

This configuration file is added newly.

interface_DmsInterfaceServerConfig.xml

This configuration file is added newly.

interface_NetDataReplicatorConfig.xml

This configuration file is added newly.

interface_OmsInterfaceServerConfig.xml

This configuration file is added newly.

interface_OmsInterfaceServerConfig_UAA

This configuration file is added newly.

MobileEnterpriseSimulatorConfig.xml

MobileEnterpriseSimulatorConfig.xml Parameter
New Parameters
<LogMessageTypes json:Array='true'>
<Type>OmsInterfaceV1WebServiceClient</Type>
<SeparateLogToDisk>true</SeparateLogToDisk>
</LogMessageTypes>
<PluginDefinition>
<Type>WebServiceClient</Type>
<Name>OmsInterfaceV1WebServiceClient</Name>
...

MobilityGatewayAppConfig.xml

Multiple additions, deletions, and changes.

NetviewerAdapterConfig-24461.xml

This configuration file is added newly.

OmsAdapter_OmsServerConfig.xml

This configuration file is added newly.

ProcessAutoConfig.xml

This configuration file is deleted.

18.4. DNAF Application Configuration Changes

This chapter lists changes that occurred in the DNAF_PARAMS_DNAF.xml configuration file for ADMS3.16.0.0.

DNAF_PARAMS.xml

DNAF_PARAMS.xml Parameter
New Parameters
<Params xsi:type="POSParams">
<RealTime>true</RealTime>
Changed Parameters
<Params xsi:type="POSParams">
<DefaultProbIndex>8</DefaultProbIndex>

DNAF_PARAMS_DNAF.xml

DNAF_PARAMS_DNAF.xml Parameter
New Parameters
<Params xsi:type="POSParams">
<RealTime>true</RealTime>
Changed Parameters
<Params xsi:type="POSParams">
<DefaultProbIndex>8</DefaultProbIndex>

DNAF_PARAMS_SIM.xml

DNAF_PARAMS_SIM.xml Parameter
New Parameters
<Params xsi:type="POSParams">
<RealTime>true</RealTime>
Changed Parameters
<Params xsi:type="POSParams">
<DefaultProbIndex>8</DefaultProbIndex>